



Coronavirus Disease 2019 (COVID-19) Treatment Guidelines

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How to Cite the COVID-19 Treatment Guidelines:

COVID-19 Treatment Guidelines Panel. Coronavirus Disease 2019 (COVID-19) Treatment Guidelines. National Institutes of Health. Available at <https://www.covid19treatmentguidelines.nih.gov/>. Accessed [insert date].

The COVID-19 Treatment Guidelines Panel regularly updates the recommendations in these guidelines as new information on the management of COVID-19 becomes available. The most recent version of the guidelines can be found on the COVID-19 Treatment Guidelines website (<https://www.covid19treatmentguidelines.nih.gov/>).

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What's New in the Guidelines

Last Updated: February 29, 2024

In response to the rapidly evolving COVID-19 pandemic, the National Institutes of Health assembled a panel of experts to provide practical recommendations for health care providers and issued the first version of the *Coronavirus Disease 2019 (COVID-19) Treatment Guidelines* on April 21, 2020. For close to 4 years, the COVID-19 Treatment Guidelines Panel (the Panel) has critically reviewed the growing body of research data on COVID-19 and used that information to develop and revise their recommendations for treating patients with this disease. The Panel has released a total of 72 versions of the Guidelines.

The federal COVID-19 Public Health Emergency ended in May 2023, and several professional societies currently provide COVID-19 treatment guidelines for their medical specialties or subspecialties. Accordingly, this will be the final update of the COVID-19 Treatment Guidelines.

The Panel members hope these Guidelines have been of value to health care providers, and they appreciate the support and input they have received over the past 4 years.

The COVID-19 Treatment Guidelines website will remain available until August 16, 2024, and will provide a [downloadable PDF](#) of the final version of the Guidelines.

February 29, 2024

In preparation for this final version of the Guidelines, the Panel reviewed all the sections that were not updated on December 20, 2023. The information in these sections is current as of February 2024.

The Viral Rebound and Symptom Recurrence subsections in [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) and [Ritonavir-Boosted Nirmatrelvir \(Paxlovid\)](#) have been updated with new references. The Panel noted that concerns about the recurrence of symptoms or viral rebound should not be a reason to avoid using antiviral therapy when indicated.

The Panel updated the discussion on the role of remdesivir in adults with COVID-19 who require mechanical ventilation or extracorporeal membrane oxygenation in [Therapeutic Management of Hospitalized Adults With COVID-19](#).

In [Therapeutic Management of Nonhospitalized Children With COVID-19](#), the vaccination status categories that determine a child's risk level for progression to severe disease have been changed from "Unvaccinated," "Primary Series," and "Up to Date" to "Not Up to Date" and "Up to Date." Chronic kidney disease and pregnancy were added to the list of risk factors that are associated with progression to severe COVID-19.

Other sections that were reviewed for this final version of the Guidelines can be found in:

- [Clinical Management of Adults](#)
- [Clinical Management of Children](#)
- [Critical Care for Adults](#)
- [Critical Care for Children](#)
- [Antivirals, Including Antibody Products](#)
- [Immunomodulators](#)
- [Special Populations](#)

Guidelines Development

Last Updated: February 29, 2024

The COVID-19 Treatment Guidelines were developed in response to the COVID-19 Public Health Emergency declared by the U.S. Department of Health and Human Services in late January 2020. The goal of the Guidelines was to provide clinicians with guidance on caring for patients with COVID-19. Because clinical information about the optimal management of COVID-19 evolved quickly, a multidisciplinary panel of experts frequently updated the Guidelines based on their assessments of the emerging evidence on treatments for this disease.

Panel Composition

The COVID-19 Treatment Guidelines Panel (the Panel) co-chairs appointed Panel members with clinical experience and expertise in adult or pediatric patient management, translational and clinical science, or the development of treatment guidelines. Panel members included representatives from federal agencies, health care organizations, academic institutions, professional societies, and the community. Federal agencies and professional societies represented on the Panel include:

- American Association for Respiratory Care
- American Association of Critical-Care Nurses
- American College of Chest Physicians
- American College of Emergency Physicians
- American College of Obstetricians and Gynecologists
- American Society of Hematology
- American Thoracic Society
- Biomedical Advanced Research and Development Authority
- Centers for Disease Control and Prevention
- Department of Defense
- Department of Veterans Affairs
- Food and Drug Administration
- Infectious Diseases Society of America
- National Institutes of Health
- Pediatric Infectious Diseases Society
- Society of Critical Care Medicine
- Society of Infectious Diseases Pharmacists

The inclusion of representatives from professional societies does not imply that these societies endorsed all elements of the Guidelines.

[Appendix A, Table 1](#), provides the names and affiliations of the Panel members, ex officio members, consultants, and support team members on the Panel roster as of the final update of the Guidelines. Financial disclosures for the Panel members can be found in [Appendix A, Table 2](#).

Development of the Guidelines

Each section of the Guidelines was developed by a working group of Panel members who had expertise in the area addressed in that section. Each working group was responsible for identifying relevant information and published scientific literature and for conducting a systematic, comprehensive review of that information and literature. The working groups proposed updates to the Guidelines based on the latest published research findings and clinical information.

Voting members of the Panel reviewed and voted on new Guidelines sections and recommendations. A majority of voting members endorsed each recommendation statement before it was included in the Guidelines. This requirement applied to recommendations for and against treatments and in cases when there was insufficient evidence to recommend either for or against treatments. Section updates that did not affect rated recommendations were approved by the Panel co-chairs without a Panel vote. During the development of the Guidelines, Panel members were required to keep all Panel deliberations and evaluations of unpublished data confidential.

Method of Synthesizing Data and Formulating Recommendations

The working groups critically reviewed and synthesized the available data to develop recommendations. During this process, the Panel evaluated the data, including the source of the data, the type of study (e.g., randomized controlled trial, prospective or retrospective cohort study, case series, in vitro study), the quality and suitability of the methods, the number of participants, and the effect sizes observed. In addition to evaluating data and reviewing clinical research on COVID-19, Panel members used clinical experiences with COVID-19 and other diseases to develop recommendations.

The recommendations in these Guidelines are based on scientific evidence and expert opinion. Each recommendation includes 2 ratings: an uppercase letter (**A**, **B**, or **C**) that indicates the strength of the recommendation and a Roman numeral with or without a lowercase letter (**I**, **IIa**, **IIb**, or **III**) that indicates the quality of the evidence that supports the recommendation (see Table 1).

The ratings for the quality of the evidence reflect both the likelihood of bias in the treatment effect estimate and the precision of the estimate. A rating of **I** corresponds to a low likelihood of bias and a high precision, a rating of **IIa** (for randomized trials) or **IIb** (for observational studies) corresponds to a moderate likelihood of bias and a moderate or high precision, and a rating of **III** corresponds to a high likelihood of bias (for any type of study).

Table 1. Recommendation Rating Scheme

Strength of Recommendation	Evidence for Recommendation
A: Strong recommendation for the statement B: Moderate recommendation for the statement C: Weak recommendation for the statement	I: <i>High quality of evidence:</i> 1 or more randomized trials without major limitations, ^a well-powered subgroup analyses of such trials, or meta-analyses without major limitations IIa: <i>Moderate quality of evidence:</i> Randomized trials and subgroup analyses of randomized trials that do not meet the criteria for a I rating IIb: <i>Moderate quality of evidence:</i> Observational studies without major limitations ^b III: Expert opinion
^a The rating may be lower than I in cases where trials have produced conflicting results.	
^b This category also includes meta-analyses of observational studies.	

In general, the recommendations in these Guidelines fall into the following categories:

- **The Panel recommends using [blank] for the treatment of COVID-19 (rating).**
Recommendations in this category are based on evidence that the potential benefits of using the intervention outweigh the potential risks.
- **There is insufficient evidence for the Panel to recommend either for or against the use of [blank] for the treatment of COVID-19 (no rating).** This statement is used when data are not sufficient to support a recommendation or when the available data are conflicting.
- **The Panel recommends against the use of [blank] for the treatment of COVID-19, except in a clinical trial (rating).** This recommendation is used in cases when the available data have shown no benefit from using the intervention for the treatment of COVID-19, or the intervention has demonstrated safety concerns. More results from clinical trials are needed to further define the role of the intervention in treating COVID-19.
- **The Panel recommends against the use of [blank] for the treatment of COVID-19 (rating).**
This recommendation is used in cases when the available data show no benefit from using the intervention to treat COVID-19, or the safety concerns for the intervention outweigh any potential benefits.

Evolving Knowledge on Treatments for COVID-19

The Food and Drug Administration approved several agents (e.g., baricitinib, ritonavir-boosted nirmatrelvir [Paxlovid], remdesivir, tocilizumab) for the treatment of COVID-19, and a number of other agents have received Emergency Use Authorizations. An array of drugs approved for other indications and multiple investigational agents are being studied for the treatment of COVID-19 in clinical trials around the globe. Information about these trials can be found at [ClinicalTrials.gov](https://clinicaltrials.gov).

Whenever possible, the Panel recommends that unapproved or unlicensed treatments for COVID-19 be studied in well-designed, controlled clinical trials. This recommendation also applies to drugs that have been approved or licensed for indications other than the treatment of COVID-19. Clinical research is critically important to generating evidence that can be used to answer questions about the safety and efficacy of potential treatments for COVID-19.

Finally, it is important to stress that the rated treatment recommendations in these Guidelines should not be considered mandates. Ultimately, patients and their health care providers should use a shared decision-making process when considering treatments for COVID-19.

Overview of COVID-19

Last Updated: December 20, 2023

Epidemiology

Individuals of all ages are at risk of SARS-CoV-2 infection. However, the probability of severe COVID-19 is higher in people aged ≥ 65 years, those living in nursing homes or long-term care facilities, those who are not vaccinated against COVID-19 or who have poor responses to COVID-19 vaccines, and those with certain chronic medical conditions. Data on comorbid health conditions among patients with COVID-19 indicate that patients with cardiovascular disease, chronic kidney disease, chronic obstructive pulmonary disease, diabetes with complications, neurocognitive disorders, and obesity are at increased risk of severe COVID-19. The risk appears to be higher in patients with multiple comorbid conditions. Other conditions that may lead to a high risk of severe COVID-19 include cancer, cystic fibrosis, immunocompromising conditions, liver disease (especially in patients with cirrhosis), pregnancy, and sickle cell disease. Transplant recipients and people who are taking immunosuppressive medications are also at high risk of severe COVID-19.¹ See [Clinical Spectrum of SARS-CoV-2 Infection](#) for a description of the clinical manifestations of SARS-CoV-2 infection and a discussion of the spectrum of disease.

Although COVID-19 vaccination does not eliminate the risk of SARS-CoV-2 infection, vaccination does significantly reduce the risk of COVID-19–related morbidity and mortality, particularly in individuals who are at high risk of progressing to severe disease.^{2,3}

Racial and Ethnic Minorities and Other Marginalized Groups

Communities that have been historically marginalized or made socially vulnerable due to a lack of access to health care or an inability to socially isolate are at increased risk of SARS-CoV-2 acquisition, COVID-19–related hospitalization, and death. These communities include racial and ethnic minorities, essential non-health care workers, and some people with disabilities.

Key Considerations

- The COVID-19 Treatment Guidelines Panel recommends that health care providers, health care systems, and payers ensure equitable access to high-quality care and treatment for all patients, regardless of race, ethnic identity, or other minoritized identity or social status (**AIII**). “Minoritized” refers to social groups that have been deprived of power and status by the dominant culture in society and encompasses not just racial identities but other identities as well, including gender identity and sexual orientation.⁴
- Promoting equitable care for these groups must include considering the full range of medical, demographic, and social factors that may negatively impact health outcomes.
- Clinicians should be aware that pulse oximeters may not accurately detect hypoxemia in people with darker skin pigmentation.^{5,6} This may delay treatment and lead to worse clinical outcomes in patients with COVID-19.⁷ See [Clinical Spectrum of SARS-CoV-2 Infection](#) for more information.
- Supporting equitable access to high-quality care and treatment for all patients is now an imperative for all health care organizations accredited by the Joint Commission, as well as a priority for the Centers for Disease Control and Prevention (CDC) and other public health agencies.

COVID-19–Related Health Outcomes

Historical structural inequities significantly contribute to the health disparities experienced by racial and ethnic minority groups (e.g., Black/African American people, Hispanic people, American Indian/Alaska Native people).⁸ Some data have highlighted that select racial and ethnic minority groups experience higher rates of COVID-19, subsequent hospitalization, and death in relation to their share of the total U.S. population. Black/African American people, Hispanic people, and American Indian/Alaska Native people also experience rates of hospitalization that are more than 2 times higher and rates of COVID-19–related death that are approximately 2 times higher than those experienced by White people. The largest disparities were observed among American Indian/Alaska Native people, who experienced a rate of hospitalization almost 3 times higher and a rate of death 2.1 times higher than White people.⁹

The increased risk of severe COVID-19 among racial and ethnic minority groups may be partly attributed to higher rates of comorbid conditions in these populations (e.g., cardiovascular disease, diabetes, chronic kidney disease, hypertension, obesity, pulmonary disease).⁹

Disparities in Access to Care

Members of racial and ethnic minority groups have an increased risk of exposure to COVID-19 and decreased access to care. Large-scale mobility data reveals that people living in lower-income communities were less able to physically isolate during COVID-19 emergency declarations,¹⁰ as members of these communities were frequently unable to work from home.¹¹ A 2020 study evaluating access to health care resources in New York City found that in areas of the city where the majority of the population was Black/African American and Hispanic, there were higher COVID-19 positivity rates and fewer licensed hospital beds and intensive care unit beds than in areas where the majority of the population was White.¹²

Disparities in Access to COVID-19 Treatments

Data from 41 U.S. health care systems reveal racial and ethnic disparities in the use of anti-SARS-CoV-2 monoclonal antibodies (mAbs) for the treatment of COVID-19.¹³ Black/African American patients, Asian patients, and patients of other races were, respectively, 22.4%, 48.3%, and 46.5% less likely to receive anti-SARS-CoV-2 mAbs for the treatment of COVID-19 than White patients.^{13,14} Disparities have also been observed in the dispensing rates for ritonavir-boosted nirmatrelvir (Paxlovid) and molnupiravir. One study reported that between April and July 2022, Black/African American patients were prescribed ritonavir-boosted nirmatrelvir 35.8% less often than White patients, and Hispanic patients were prescribed this drug 29.9% less often than White patients.¹⁵ Despite a greater number of dispensing sites in neighborhoods with a higher social vulnerability, oral antivirals were prescribed at a lower rate for patients with COVID-19 who were living in these areas than in those with a lesser degree of social vulnerability.¹⁶ These disparities are not limited to outpatient settings. One retrospective cohort study of veterans hospitalized with COVID-19 reported that Black veterans had lower odds of receiving COVID-19–specific treatments, including systemic steroids, remdesivir, and immunomodulators, than White veterans.¹⁷

Other Marginalized Groups

Other marginalized groups also experience worse outcomes for COVID-19. Hospitalization rates for COVID-19 among Medicare beneficiaries who were eligible for disability were approximately 50% higher than those among people who were eligible for Medicare based on age alone, and this discrepancy disproportionately affected Black/African American people, Hispanic people, and American Indian/Alaska Native people.¹⁸

Migrants, refugees, and essential non-health care workers (e.g., food supply, food service, public transportation, and agricultural workers) also have disproportionately high rates of COVID-19 cases and deaths. These high rates can be attributed to overcrowding, an inability to physically isolate, and inadequate access to health care.¹⁹⁻²¹

Given the pervasiveness of disparities in access to care and provision of treatment, it is imperative for clinicians, working with others on the patient care team, to assess the social factors that contribute to access and quality gaps and to strive to provide equitable treatment to all patients. These issues have been identified as a strategic priority by the [Joint Commission](#) and the [CDC](#).

SARS-CoV-2 Variants

Like other RNA viruses, SARS-CoV-2 is constantly evolving through random mutations. New mutations can potentially increase or decrease infectiousness and virulence. In addition, mutations can increase the virus' ability to evade adaptive immune responses from previous SARS-CoV-2 infections or vaccination. This viral evolution may increase the risk of reinfection or decrease the efficacy of vaccines.²² There is evidence that some SARS-CoV-2 variants have reduced susceptibility to plasma from people who were previously infected or immunized, as well as to anti-SARS-CoV-2 mAbs.²³⁻²⁵

Since December 2020, the World Health Organization has assigned Greek letter designations to several identified variants. A SARS-CoV-2 variant designated as a variant of concern displays certain characteristics, such as increased transmissibility or virulence. In addition, vaccines and therapeutics may have decreased effectiveness against variants of concern, and the mutations found in these variants may interfere with the targets of diagnostic tests. The variant of interest designation has been used for important variants that are not fully characterized; however, organizations do not use the same variant designations, and they may define their variant designations differently.^{26,27}

In September 2021, the CDC added a new designation for variants: [variants being monitored](#). The data on these variants indicate a potential or clear impact on approved or authorized medical countermeasures, or these variants are associated with cases of more severe disease or increased transmission rates. However, these variants are either no longer detected or are circulating at very low levels in the United States; therefore, they do not pose a significant and imminent risk to public health in the United States.

The Omicron variant was designated as a variant of concern in November 2021 and rapidly became the dominant variant across the globe. The Omicron subvariants BA.1, BA.1.1, and BA.2 emerged in early to mid-2022, followed by the subvariants BA.4, BA.5, BQ.1, BQ.1.1, XBB, EG.5, HV.1, and FL.1.5.1. The newer Omicron subvariants are generally more transmissible than previous variants and are not susceptible to any of the anti-SARS-CoV-2 mAbs that were previously authorized for the treatment and prevention of COVID-19.^{24,25,28,29}

Data on the emergence, transmission, and clinical relevance of these new variants are rapidly evolving; this is especially true for research on how variants might affect transmission rates, disease progression, vaccine development, and the efficacy of current therapeutics. Because the research on variants is moving quickly and the classification of the different variants may change over time, websites such as the CDC's [COVID Data Tracker](#), [CoVariants.org](#), and the World Health Organization's [Tracking SARS-CoV-2 Variants](#) provide regular updates on data for SARS-CoV-2 variants.

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Testing for SARS-CoV-2 Infection

Last Updated: December 20, 2023

Summary of Testing for SARS-CoV-2 Infection

The COVID-19 Treatment Guidelines Panel (the Panel) defers to the Centers for Disease Control and Prevention (CDC) for recommendations on diagnostic testing for SARS-CoV-2 infection. The Panel also defers to the CDC for recommendations on the use of testing for screening purposes, such as for screening among people who are asymptomatic but have had recent known or suspected exposure to SARS-CoV-2. Some key CDC recommendations include:

- For diagnosing current SARS-CoV-2 infection, the CDC recommends using either a nucleic acid amplification test (NAAT) or an antigen test and using a specimen from the upper respiratory tract (e.g., nasal, nasopharyngeal).
- There may be a window period of up to 5 days after exposure before viral antigens or nucleic acids can be detected.
- NAATs are the most sensitive tests for detecting current SARS-CoV-2 infection. Because antigen tests are less sensitive than NAATs, the Food and Drug Administration recommends repeating antigen tests that produce negative results in certain circumstances, such as when clinical suspicion of COVID-19 is high in people who are symptomatic or when people who are asymptomatic have had known or suspected exposure to SARS-CoV-2.
- Antibody tests should not be used to diagnose current SARS-CoV-2 infection.
- Currently, antibody tests are not recommended for assessing SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated.

Diagnostic Testing for SARS-CoV-2 Infection

For diagnosing current SARS-CoV-2 infection, the Centers for Disease Control and Prevention (CDC) recommends using either a nucleic acid amplification test (NAAT) or an antigen test.¹ Testing may also be used for screening and to determine the length of a patient's isolation period.² There may be a window period of up to 5 days after exposure before viral antigens or nucleic acids can be detected.

A number of diagnostic tests for SARS-CoV-2 infection (e.g., NAATs, antigen tests) have received Food and Drug Administration (FDA) Emergency Use Authorizations (EUAs) for use in laboratories and points of care (e.g., physician offices, pharmacies, long-term care facilities, school clinics) and for self-administered testing.³ An influenza and SARS-CoV-2 multiplex NAAT that can simultaneously detect and differentiate between influenza A, influenza B, and SARS-CoV-2 also received an EUA from the FDA.⁴ The FDA also granted authorization to market the first over-the-counter, at-home, molecular NAAT (i.e., Cue COVID-19) and antigen test (i.e., Flowflex COVID-19) for use in people with symptomatic COVID-19.

For diagnosing current SARS-CoV-2 infection, the CDC recommends using a specimen from the upper respiratory tract (e.g., nasal, nasopharyngeal).⁵ Testing lower respiratory tract specimens is also an option in certain circumstances (e.g., in those receiving mechanical ventilation). For details about collecting and handling specimens for COVID-19 testing, please refer to the CDC's recommendations.

Antigen Testing for SARS-CoV-2 Infection

Antigen-based diagnostic tests are widely used at home, at the point of care, and in the laboratory because of their low cost, rapid turnaround time, and availability. Antigen tests and laboratory-based NAATs have similar high specificity. False positive test results can occur with antigen tests, although they are unlikely when the tests are used correctly.⁶ The likelihood of a false positive antigen test result is higher when the expected probability of SARS-CoV-2 infection is low. Because antigen tests are less sensitive than NAATs, the FDA recommends repeating antigen tests that produce negative results in certain

circumstances, such as when clinical suspicion of COVID-19 is high in people who are symptomatic or when people who are asymptomatic have had known or suspected exposure to SARS-CoV-2.

Nucleic Acid Amplification Testing for SARS-CoV-2 Infection

NAATs, such as reverse transcription polymerase chain reaction–based diagnostic tests, which detect viral nucleic acids,⁷ are the most sensitive tests for detecting current SARS-CoV-2 infection.

Diagnostically, some NAATs may produce false negative results if a mutation occurs in the part of the virus's genome that is assessed by that test.⁸ The FDA monitors the potential effects of SARS-CoV-2 genetic variations on NAAT results and issues updates when specific variations could affect the performance of NAATs that have received EUAs.⁹ A single negative test result does not exclude the possibility of SARS-CoV-2 infection in people who have a high likelihood of infection based on their exposure history or clinical presentation.¹⁰

Reinfection

Reinfection has been reported in people after an initial diagnosis of SARS-CoV-2 infection. Because reinfection can be difficult to distinguish from persistent shedding (i.e., positive NAAT results persisting for weeks or months), the CDC recommends using an antigen test instead of a NAAT in patients who have symptoms compatible with SARS-CoV-2 infection who are within 90 days of recovering from a previous SARS-CoV-2 infection. Because intermittent detection of viral RNA can occur, a negative result on an initial NAAT followed by a positive result on a subsequent test does not necessarily mean a person has been reinfected.¹¹ When the results for an initial and subsequent test are positive, comparative viral sequence data from both tests are needed to distinguish between the persistent presence of viral fragments and reinfection. In the absence of viral sequence data, the cycle threshold (Ct) value from a positive NAAT result may provide information about whether a newly detected infection is related to the persistence of viral fragments or to reinfection. The Ct value is the number of PCR cycles at which the nucleic acid target in the sample becomes detectable. In general, the Ct value is inversely related to the SARS-CoV-2 viral load. Because the clinical utility of Ct values is unclear, an expert should be consulted if these values are used to guide clinical decisions.

Serologic or Antibody Testing for Diagnosis of SARS-CoV-2 Infection

Unlike NAATs and antigen tests, which detect the presence of SARS-CoV-2, serologic or antibody tests can detect recent or prior SARS-CoV-2 infection or vaccination. The CDC recommends that antibody tests should not be used to diagnose current SARS-CoV-2 infection.¹² It may take 21 days or longer after symptom onset for seroconversion to occur (i.e., the development of detectable immunoglobulin M or immunoglobulin G antibodies to SARS-CoV-2).¹³⁻¹⁸

No serologic tests for SARS-CoV-2 have been approved by the FDA. Some, but not all, commercially available serologic tests for SARS-CoV-2 have received EUAs from the FDA.¹⁹ Several professional societies and federal agencies, including the [Infectious Diseases Society of America](#), the [CDC](#), and the [FDA](#), provide guidance on the use of serologic testing for SARS-CoV-2.

Serologic Testing and Immunity to SARS-CoV-2 Infection

Currently, antibody tests are not recommended for assessing SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated.

The FDA has issued EUAs for more than 80 SARS-CoV-2 serologic tests since the beginning of the pandemic. However, these tests are not currently authorized for routine use in making individual medical decisions.¹⁹ SARS-CoV-2 serologic tests are authorized for detecting antibodies, but their

ability to predict protective immunity has not been validated. Most of these tests are not standardized. Furthermore, as SARS-CoV-2 is not a well-conserved virus, mutations in the receptor binding domain of the virus could lead to decreased binding affinity between antibodies and SARS-CoV-2–specific antigens.

If a serologic test is performed, the result should be interpreted with caution. First, it remains unclear how long SARS-CoV-2 antibodies persist following infection or vaccination. A negative serologic test result also does not preclude prior SARS-CoV-2 infection or vaccination against COVID-19. Second, some people who are infected with SARS-CoV-2 or who are vaccinated against COVID-19 (e.g., those who are immunocompromised) may not develop measurable levels of antibodies. It is presumed that those who do not have measurable antibodies after vaccination are at higher risk of SARS-CoV-2 infection than those who have measurable antibodies. Third, because nucleocapsid proteins are not a constituent of the vaccines that are currently approved by the FDA, available through EUAs, or in late-stage clinical trials, serologic tests that detect antibodies by recognizing nucleocapsid proteins should be used to distinguish between antibody responses to natural infection and vaccine-induced antibody responses to the SARS-CoV-2 spike protein antigen.

Assuming that the test is reliable, serologic tests that identify recent or prior SARS-CoV-2 infection may be used to determine who may be eligible to donate COVID-19 convalescent plasma and may aid in diagnosing multisystem inflammatory syndrome in children (MIS-C) and multisystem inflammatory syndrome in adults (MIS-A).

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Prevention of SARS-CoV-2 Infection

Last Updated: December 20, 2023

Summary Recommendation
The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible according to the Centers for Disease Control and Prevention's Advisory Committee on Immunization Practices (AI).
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

General Prevention Measures

Transmission of SARS-CoV-2 occurs primarily through exposure to respiratory droplets.¹ Exposure can occur when individuals inhale droplets or particles that contain the virus or touch mucous membranes with hands that have been contaminated with the virus. Exhaled droplets or particles can also deposit the virus onto exposed mucous membranes.

The risk of SARS-CoV-2 transmission can be reduced by covering coughs and sneezes, wearing a well-fitted mask around others, and isolating when experiencing symptoms. Frequent handwashing also effectively reduces the risk of infection.² Health care providers should follow the Centers for Disease Control and Prevention (CDC) recommendations for infection control and the appropriate use of personal protective equipment.³

COVID-19 Vaccines

Recommendation

- The Panel recommends COVID-19 vaccination for everyone who is eligible according to the CDC's Advisory Committee on Immunization Practices (AI).

Rationale

Vaccination is the most effective way to prevent COVID-19. Two 2023–2024 mRNA vaccines, BNT162b2 (Pfizer-BioNTech) and mRNA-1273 (Moderna), and the 2023–2024 recombinant spike protein with adjuvant vaccine NVX-CoV2373 (Novavax)⁴ are currently available in the United States. The adenovirus vector vaccine Ad26.COV2.S (Johnson & Johnson/Janssen) is no longer available in the United States.

COVID-19 vaccination is recommended for everyone aged ≥ 6 months in the United States. The Food and Drug Administration (FDA) Emergency Use Authorization fact sheet and the product label for each vaccine provide detailed information on the vaccination schedule and the doses that are approved or authorized for that vaccine. The type and dose of vaccine and the timing of the doses depend on the recipient's age and underlying medical conditions. The CDC regularly updates the clinical considerations for the COVID-19 vaccines currently approved by the FDA or authorized for use in the United States.⁵

Adverse Events

COVID-19 vaccines are safe and effective. Local and systemic adverse events are relatively common with these vaccines. Most of the adverse events that occurred during vaccine trials were mild or

moderate in severity (i.e., they did not prevent vaccinated people from engaging in daily activities) and resolved after 1 or 2 days. There have been a few reports of severe allergic reactions following COVID-19 vaccination, including rare reports of patients who experienced anaphylaxis after receiving an mRNA vaccine.^{6,7}

Thrombosis with thrombocytopenia syndrome is a serious condition characterized by blood clots in large blood vessels and low platelet levels. The prevalence of the syndrome was approximately 4 per million among people who received the Johnson & Johnson/Janssen vaccine.^{8,9} That vaccine is no longer available in the United States. If a patient experiences thrombosis and thrombocytopenia syndrome after receiving a COVID-19 vaccine outside of the United States, a hematologist should be consulted about evaluation and management.

Myocarditis and pericarditis after COVID-19 vaccination are rare, and most of the reported cases were very mild and self-limiting.¹⁰ These conditions have occurred most often in male adolescents, young adults, and people who have received mRNA vaccines.

The results of recent studies suggest that adults aged ≥ 18 years who received the Johnson & Johnson/Janssen vaccine have an increased risk of Guillain-Barré syndrome.¹¹ In contrast, people who received mRNA vaccines do not have an increased risk of Guillain-Barré syndrome.¹²

The CDC monitors severe adverse events, such as strokes, and provides regular updates on [selected adverse events of COVID-19 vaccines](#).

Vaccination in Pregnant and Lactating People

Pregnant and lactating individuals were not included in the initial COVID-19 vaccine trials. However, the CDC and the American College of Obstetricians and Gynecologists recommend vaccination for pregnant and lactating people. This recommendation is based on the accumulated safety and efficacy data on the use of these vaccines in pregnant people, as well on as the increased risk of severe disease in pregnant individuals with COVID-19.¹³⁻¹⁷ These organizations also recommend vaccination for people who are trying to become pregnant or who may become pregnant in the future. The American College of Obstetricians and Gynecologists provides guidance for clinicians on counseling pregnant patients about COVID-19 vaccination.¹⁸

Pre-Exposure Prophylaxis

As of January 2024, no biomedical intervention other than vaccines prevents COVID-19 disease. Previously, the FDA authorized the use of the anti-SARS-CoV-2 monoclonal antibodies tixagevimab plus cilgavimab (Evusheld) as pre-exposure prophylaxis (PrEP) of COVID-19 in people who were not expected to mount an adequate immune response to COVID-19 vaccination and in people with COVID-19 vaccine contraindications.¹⁹ Due to the increased prevalence of Omicron subvariants that are not susceptible to tixagevimab plus cilgavimab, this combination is not currently authorized by the FDA for use as PrEP of COVID-19.²⁰ It remains critical that these individuals:

- Keep up to date with COVID-19 vaccination unless a contraindication exists.
- Take precautions to avoid infection. The CDC provides [information on the prevention of COVID-19](#) in people who are immunocompromised.
- Be tested for SARS-CoV-2 infection if they experience signs and symptoms consistent with COVID-19 and, if infected, promptly seek medical attention.

Post-Exposure Prophylaxis

As of January 2024, no biomedical intervention other than vaccines prevents disease after exposure to SARS-CoV-2. Previously, the FDA authorized the use of the anti-SARS-CoV-2 monoclonal antibody products bamlanivimab plus etesevimab and casirivimab plus imdevimab as post-exposure prophylaxis (PEP) in certain people at high risk of progression to severe COVID-19. However, the Omicron subvariants are not susceptible to these products; therefore, their use as SARS-CoV-2 PEP is not recommended.

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Clinical Spectrum of SARS-CoV-2 Infection

Last Updated: February 29, 2024

Patients with SARS-CoV-2 infection can experience a range of clinical manifestations, from no symptoms to critical illness. In general, adults with SARS-CoV-2 infection can be grouped into the following severity of illness categories; however, the criteria for each category may overlap or vary across clinical guidelines and clinical trials, and a patient's clinical status may change over time.

- *Asymptomatic or presymptomatic infection:* Individuals who test positive for SARS-CoV-2 using a virologic test (i.e., a nucleic acid amplification test [NAAT] or an antigen test) but have no symptoms consistent with COVID-19.
- *Mild illness:* Individuals who have any of the various signs and symptoms of COVID-19 (e.g., fever, cough, sore throat, malaise, headache, muscle pain, nausea, vomiting, diarrhea, loss of taste and smell) but do not have shortness of breath, dyspnea, or abnormal chest imaging.
- *Moderate illness:* Individuals who show evidence of lower respiratory disease during clinical assessment or imaging and who have an oxygen saturation measured by pulse oximetry (SpO_2) $\geq 94\%$ on room air at sea level.
- *Severe illness:* Individuals who have an $\text{SpO}_2 < 94\%$ on room air at sea level, a ratio of arterial partial pressure of oxygen to fraction of inspired oxygen ($\text{PaO}_2/\text{FiO}_2$) < 300 mm Hg, a respiratory rate > 30 breaths/min, or lung infiltrates $> 50\%$.
- *Critical illness:* Individuals who have respiratory failure, septic shock, or multiple organ dysfunction.

SpO_2 is a key parameter for defining the illness categories listed above. However, pulse oximetry has important limitations (discussed in more detail below). Clinicians who use SpO_2 when assessing a patient must be aware of those limitations and conduct the assessment in the context of that patient's clinical status.

The risk of progressing to severe disease increases with age and the number of underlying conditions. Patients aged ≥ 50 years, especially those aged ≥ 65 years, and patients who are immunosuppressed, unvaccinated, or not up to date with COVID-19 vaccinations are at a higher risk of progressing to severe COVID-19.^{1,2} Certain underlying conditions are also associated with a higher risk of severe COVID-19, including cancer, cardiovascular disease, chronic kidney disease, chronic liver disease, chronic lung disease, diabetes, advanced or untreated HIV infection, obesity, pregnancy, cigarette smoking, and being a recipient of immunosuppressive therapy or a transplant.³ Health care providers should closely monitor patients who have COVID-19 and any of these conditions until clinical recovery is achieved.

The initial evaluation for patients may include chest imaging (e.g., X-ray, ultrasound or computed tomography scan) and an electrocardiogram, if indicated. Laboratory testing should include a complete blood count with differential and a metabolic profile, including liver and renal function tests. Although inflammatory markers such as C-reactive protein (CRP), D-dimer, and ferritin are not routinely measured as part of standard care, results from such measurements may have prognostic value.⁴⁻⁷

The definitions for the severity of illness categories also apply to pregnant patients. However, the threshold for certain interventions is different for pregnant and nonpregnant patients. For example, oxygen supplementation for pregnant patients is generally used when SpO_2 falls below 95% on room air at sea level to accommodate the physiologic changes in oxygen demand during pregnancy and to ensure

adequate oxygen delivery to the fetus.⁸

If laboratory parameters are used for monitoring pregnant patients and making decisions about interventions, clinicians should be aware that normal physiologic changes during pregnancy can alter several laboratory values. In general, leukocyte cell count increases throughout gestation and delivery and peaks during the immediate postpartum period. This increase is mainly due to neutrophilia.⁹ D-dimer and CRP levels also increase during pregnancy and are often higher in pregnant patients than in nonpregnant patients.¹⁰ Detailed information on treating COVID-19 in pregnant patients can be found in [Special Considerations During Pregnancy and After Delivery](#) and in the pregnancy considerations subsections in the Guidelines.

In children with COVID-19, radiographic abnormalities are common and, for the most part, should not be the only criteria used to determine the severity of illness. The normal values for respiratory rate also vary with age in children. Therefore, hypoxemia should be the primary criterion used to define severe COVID-19, especially in younger children. In a small subset of children and young adults, SARS-CoV-2 infection may be followed by the severe inflammatory condition multisystem inflammatory syndrome in children (MIS-C).^{11,12} This syndrome is discussed in detail in [Special Considerations in Children](#).

Clinical Considerations for the Use of Pulse Oximetry

During the COVID-19 pandemic, the use of pulse oximetry to assess and monitor patients' oxygenation status increased in hospital, outpatient health care facility, and home settings. Although pulse oximetry is useful for estimating blood oxygen levels, pulse oximeters may not accurately detect hypoxemia under certain circumstances. To avoid delays in recognizing hypoxemia, clinicians who use pulse oximetry to assist with clinical decisions should keep these limitations in mind.

Pulse oximetry results can be affected by skin pigmentation, thickness, or temperature. Poor blood circulation or the use of tobacco or fingernail polish also may affect results. The Food and Drug Administration (FDA) advises clinicians to refer to the label or manufacturer website of a pulse oximeter or sensor to ascertain its accuracy.¹³ The FDA also advises using pulse oximetry only as an estimate of blood oxygen saturation, because an SpO₂ reading represents a range of arterial oxygen saturation (SaO₂). For example, an SpO₂ reading of 90% may represent a range of SaO₂ from 86% to 94%. Studies that compared SpO₂ and SaO₂ levels measured before the pandemic found that pulse oximeters overestimated oxygen saturation in people who were classified as having darker skin pigmentation and in people whose race or ethnic origin was reported as non-Hispanic Black, Black, or African American.^{14,15}

Several published reports have compared SpO₂ and SaO₂ measurements in patients with COVID-19, including children.^{14,16-18} The studies demonstrated that occult hypoxemia (defined as an SaO₂ <88% despite an SpO₂ >92%) was more common in patients with darker skin pigmentation, which may result in adverse consequences. The likelihood of error was greater in the lower ranges of SpO₂. In 1 of these studies, occult hypoxemia was associated with more organ dysfunction and hospital mortality.¹⁷ These studies did not specify the specific devices used to assess SpO₂ levels. The FDA has recognized the need for better real-world evidence to address ongoing concerns about the accuracy of pulse oximeters when they are used to measure oxygen saturation in people with darker skin pigmentation.¹⁹

A 5-hospital registry study of patients evaluated in the emergency department or hospitalized for COVID-19 found that 24% were not identified as eligible for treatment due to overestimation of SaO₂.²⁰ The majority of patients (55%) who were not identified as eligible were Black. The study also examined the amount of time delay patients experienced before being identified as eligible for treatment. The median delay for patients who were Black was 1 hour longer than the delay for patients who were White.

In pulse oximetry, skin tone is an important variable, but accurately measuring oxygen saturation is a complex process. One observational study in adults was unable to identify a consistently predictable difference between SaO_2 and SpO_2 over time for individual patients.¹⁶ Factors other than skin pigmentation (e.g., peripheral perfusion, pulse oximeter sensor placement) are likely involved.

Despite the limitations of pulse oximetry, an FDA-cleared pulse oximeter for home use can contribute to an assessment of a patient's overall clinical status. Practitioners should advise patients to follow the manufacturer's instructions for use, place the oximeter on the index or ring finger, and ensure the hand is warm, relaxed, and held below the level of the heart. Fingernail polish should be removed before testing. Patients should be at rest, indoors, and breathing quietly without talking for several minutes before testing. Rather than accepting the first reading, patients or caretakers should observe the readings on the pulse oximeter for ≥ 30 seconds until a steady number is displayed and inform their health care provider if the reading is repeatedly below a previously specified value (generally 95% on room air at sea level).^{13,21} Pulse oximetry has been widely adopted as a remote patient monitoring tool, but when the use of pulse oximeters is compared with close monitoring of clinical progress via video consultation, telephone calls, text messaging, or home visits, there is insufficient evidence that it improves clinical outcomes.^{22,23}

Not all commercially available pulse oximeters have been cleared by the FDA. SpO_2 readings obtained through devices not cleared by the FDA, such as over-the-counter sports oximeters or mobile phone applications, lack sufficient accuracy for clinical use. Abnormal readings on these devices should be confirmed with an FDA-cleared device or an arterial blood gas analysis.^{24,25}

Regardless of the setting, SpO_2 should always be interpreted within the context of a patient's entire clinical presentation. Regardless of a pulse oximeter reading, a patient's signs and symptoms (e.g., dyspnea, tachypnea, chest pain, changes in cognition or attentional state, cyanosis) should be evaluated.

Asymptomatic or Presymptomatic Infection

Asymptomatic SARS-CoV-2 infection can occur, although the percentage of patients who remain truly asymptomatic throughout the course of infection is variable and incompletely defined. The percentage of individuals who present with asymptomatic infection and progress to clinical disease is unclear. Some asymptomatic individuals have been reported to have objective radiographic findings consistent with COVID-19 pneumonia.^{26,27}

Mild Illness

Patients with mild illness may exhibit a variety of signs and symptoms (e.g., fever, cough, sore throat, malaise, headache, muscle pain, nausea, vomiting, diarrhea, loss of taste and smell). They do not have shortness of breath, dyspnea on exertion, or abnormal imaging. Most patients who are mildly ill can be managed in an ambulatory setting or at home. No imaging or specific laboratory evaluations are routinely indicated in otherwise healthy patients with mild COVID-19. Patients aged ≥ 50 years, especially those aged ≥ 65 years, patients with certain underlying comorbidities, and patients who are immunosuppressed, unvaccinated, or not up to date with COVID-19 vaccinations are at [higher risk of disease progression](#) and are candidates for antiviral therapy.^{1,2} See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for recommendations regarding anti-SARS-CoV-2 therapies.

Moderate Illness

Moderate illness is defined as evidence of lower respiratory disease during clinical assessment or imaging, with an $\text{SpO}_2 \geq 94\%$ on room air at sea level. Given that pulmonary disease can progress rapidly in patients with COVID-19, patients with moderate disease should be closely monitored. See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for recommendations regarding

anti-SARS-CoV-2 therapies in patients at high risk of progression to severe disease.

Severe Illness

Patients with COVID-19 are considered to have severe illness if they have an $\text{SpO}_2 < 94\%$ on room air at sea level, $\text{PaO}_2/\text{FiO}_2 < 300$ mm Hg, a respiratory rate > 30 breaths/min, or lung infiltrates $> 50\%$. These patients may experience rapid clinical deterioration and should be given oxygen therapy and hospitalized. See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for treatment recommendations.

Critical Illness

SARS-CoV-2 infection can cause acute respiratory distress syndrome, virus-induced distributive (septic) shock, cardiac shock, an exaggerated inflammatory response, thrombotic disease, and exacerbation of underlying comorbidities.

The clinical management of patients with COVID-19 who are in the intensive care unit should include treatment with immunomodulators and, in some cases, the addition of remdesivir. These patients should also receive treatment for any comorbid conditions and nosocomial complications. For more information, see [Critical Care for Adults](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#).

Infectious Complications in Patients With COVID-19

Some patients with COVID-19 may have additional infections when they present for care or that develop during the course of treatment. These coinfections may complicate treatment and recovery. Older patients or those with certain comorbidities or immunocompromising conditions may be at higher risk for these infections. The use of immunomodulators such as dexamethasone, Janus kinase inhibitors (e.g., baricitinib, tofacitinib), interleukin-6 inhibitors (e.g., tocilizumab, sarilumab), tumor necrosis factor inhibitors (e.g., infliximab), or abatacept to treat COVID-19 may also increase the risk of infectious complications. However, when these therapies are used appropriately, the benefits outweigh the risks.

Infectious complications in patients with COVID-19 can be categorized as follows:

- *Coinfections at presentation:* Although most individuals present with only SARS-CoV-2 infection, concomitant viral infections, including influenza and other respiratory viruses, have been reported.²⁸⁻³⁰ Community-acquired bacterial pneumonia has also been reported, but it is uncommon.^{28,31,32} Antibacterial therapy is generally not recommended unless additional evidence for bacterial pneumonia is present (e.g., leukocytosis, the presence of a focal infiltrate on imaging).
- *Reactivation of latent infections:* There are case reports of underlying chronic hepatitis B virus and latent tuberculosis infections reactivating in patients with COVID-19 who receive immunomodulators as treatment, although the data are currently limited.³³⁻³⁵ Reactivation of herpes simplex virus and varicella zoster virus infections have also been reported.³⁶ Cases of severe and disseminated strongyloidiasis have been reported in patients with COVID-19 during treatment with tocilizumab and corticosteroids.^{37,38} Many clinicians would initiate empiric treatment (e.g., with the antiparasitic drug ivermectin), with or without serologic testing, in patients who require immunomodulators for the treatment of COVID-19 and have come from areas where *Strongyloides* is endemic (i.e., tropical, subtropical, or warm temperate areas).^{39,40}
- *Nosocomial infections:* Hospitalized patients with COVID-19 may acquire common nosocomial infections, such as hospital-acquired pneumonia (including ventilator-associated pneumonia), line-

related bacteremia or fungemia, catheter-associated urinary tract infection, and diarrhea associated with *Clostridioides difficile*.^{41,42} Early diagnosis and treatment of these infections are important for improving outcomes in these patients.

- *Opportunistic fungal infections:* Invasive fungal infections, including aspergillosis and mucormycosis, have been reported in hospitalized patients with COVID-19.⁴³⁻⁴⁶ Although these infections are relatively rare, they can be fatal, and they may be seen more commonly in patients who are immunocompromised or receiving mechanical ventilation. The majority of mucormycosis cases have been reported in India and are associated with diabetes mellitus or the use of corticosteroids.^{47,48} The approach for managing these fungal infections should be the same as the approach for managing invasive fungal infections in other settings.

SARS-CoV-2 Reinfection and Breakthrough Infection

As seen with other respiratory viral infections, reinfection after recovery from prior infection has been reported for SARS-CoV-2.⁴⁹ Reinfection may occur as initial immune responses to the primary infection wane over time. Data regarding the prevalence, risk factors, timing, and severity of reinfection are evolving and vary depending on the circulating variants. Breakthrough SARS-CoV-2 infections (i.e., infection in people who are up to date with COVID-19 vaccinations) also occur.⁵⁰ When compared with infection in people who are unvaccinated, breakthrough infections in vaccinated individuals appear less likely to lead to severe illness or symptoms that persist ≥ 28 days.⁵⁰⁻⁵³ The time to breakthrough infection has been reported to be shorter for patients with immunocompromising conditions (i.e., solid organ or bone marrow transplant recipients or people with HIV) than for those with no immunocompromising conditions.⁵⁰

Although data are limited, no evidence suggests that the treatment of suspected or documented SARS-CoV-2 reinfection or breakthrough infection should be different from the treatment used during the initial infection, as outlined in [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#).

Prolonged Viral Shedding, Persistent Symptoms, and Other Conditions After SARS-CoV-2 Infection

Symptomatic SARS-CoV-2 infection is typically associated with a decline in viral shedding and resolution of COVID-19 symptoms over days to weeks. However, in some cases, reduced viral shedding and symptom resolution are followed by viral or symptom rebound. People who are immunocompromised may experience viral shedding for many weeks. Some people experience symptoms that develop or persist for more than 4 weeks after the initial COVID-19 diagnosis.

Viral or Symptom Rebound Soon After COVID-19

Observational studies and results from clinical trials of therapeutic agents have described SARS-CoV-2 viral or COVID-19 symptom rebound in patients who have completed treatment for COVID-19.⁵⁴⁻⁵⁶ Viral and symptom rebounds have also occurred when anti-SARS-CoV-2 therapies were not used.^{56,57} Typically, this phenomenon has not been associated with progression to severe COVID-19.

Prolonged Viral Shedding in Patients Who Are Immunocompromised

Patients who are immunocompromised may experience prolonged shedding of SARS-CoV-2 with or without COVID-19 symptoms.^{58,59} Prolonged viral shedding may affect SARS-CoV-2 testing strategies and isolation duration for these patients. In some cases, the prolonged shedding may be associated with persistent COVID-19 symptoms. See [Special Considerations in People Who Are Immunocompromised](#) for more information on the clinical management of people who are immunocompromised.

Persistent, New, or Recurrent Symptoms More Than 4 Weeks After SARS-CoV-2 Infection

Some patients report persistent, new, or recurrent symptoms and conditions (e.g., cardiopulmonary injury, neurocognitive impairment, new-onset diabetes, gastrointestinal and dermatologic manifestations) more than 4 weeks after the initial COVID-19 diagnosis.⁶⁰ The nomenclature for this phenomenon is evolving; no clinical terminology has been established. The terminology used includes long COVID, post-COVID condition, post-COVID-19 syndrome, and post-acute sequelae of SARS-CoV-2. Patients who have these symptoms or conditions have been called “long haulers.”

Data on the incidence, natural history, and etiology of these symptoms are emerging. However, reports on these syndromes have several limitations, such as differing case definitions, a lack of comparator groups, and overlap between the reported symptoms and the symptoms of post-intensive care syndrome that have been described in patients without COVID-19. In addition, many reports only included patients who attended post-COVID clinics. Details on the pathogenesis, clinical presentation, and treatment for these conditions are beyond the scope of these Guidelines. The [Centers for Disease Control and Prevention](#) provides information about the timeframes, presentation of symptoms, and management strategies for post-COVID conditions. Research on the prevalence, characteristics, and pathophysiology of persistent symptoms and conditions after COVID-19 is ongoing, including research through the National Institutes of Health’s [RECOVER Initiative](#), which aims to inform potential therapeutic strategies.

MIS-C and multisystem inflammatory syndrome in adults (MIS-A) are serious postinfectious complications of SARS-CoV-2 infection. For more information on these syndromes, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

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Prioritization of Anti-SARS-CoV-2 Therapies for the Treatment of COVID-19 in Nonhospitalized Patients When There Are Logistical Constraints

Last Updated: December 20, 2023

The prioritization guidance in this section should be used **only** when logistical constraints limit the availability of therapies. When there are no logistical constraints, clinicians can prescribe therapies to treat SARS-CoV-2 infection to any eligible individual following the recommendations in these Guidelines.

If it is necessary to triage patients for receipt of anti-SARS-CoV-2 therapies, the COVID-19 Treatment Guidelines Panel (the Panel) suggests prioritizing patients based on their clinical risk factors for severe disease, their vaccination status, and their ability to mount an adequate immune response to COVID-19 vaccination or SARS-CoV-2 infection.

Prioritization schemes should include a plan for equitable distribution of scarce resources to individuals who may have less knowledge of or access to these therapies. The availability and distribution of recommended therapies should be monitored to ensure that access to products is equitable.

Patient Prioritization for Treatment

The Panel recommends using **ritonavir-boosted nirmatrelvir (Paxlovid)** to treat nonhospitalized adults (**AIIa**) and adolescents (**BIII**) who have mild to moderate COVID-19 and are at high risk of progressing to severe disease.

Remdesivir is a recommended option if ritonavir-boosted nirmatrelvir cannot be used. However, some treatment facilities may not have the ability to provide a 3-day course of remdesivir intravenous infusions to all eligible patients. In these situations, prioritizing patients who will benefit the most from the therapy becomes necessary. If administration of remdesivir is not feasible, clinicians should review the Panel's recommendations in [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for alternative treatment options.

The prioritization scheme below is based on 4 key elements: age, vaccination status, immune status, and clinical risk factors. For a list of risk factors, see the Centers for Disease Control and Prevention (CDC) webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#). The groups are listed by tier in descending order of priority.

Tier	Risk Group
1	• People who are immunocompromised and are not expected to mount an adequate immune response to COVID-19 vaccination or SARS-CoV-2 infection due to their underlying conditions, regardless of vaccine status (see Immunocompromising Conditions below); <i>or</i> • Unvaccinated individuals at the highest risk of severe disease (anyone aged ≥75 years or anyone aged ≥65 years with additional risk factors)
2	• Unvaccinated individuals not included in Tier 1 who are at risk of severe disease (anyone aged ≥65 years or anyone aged <65 years with clinical risk factors)
3	• Vaccinated individuals at risk of severe disease (anyone aged ≥65 years or anyone aged <65 years with clinical risk factors) • Vaccinated individuals who are not up to date with their immunizations are likely at higher risk for severe disease; patients within this tier who are in this situation should be prioritized for treatment.

Immunocompromising Conditions

The CDC website [COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised](#) provides a list of moderate or severe immunocompromising conditions.

If there are logistical constraints to providing the Panel's recommended therapies to all individuals who are moderately to severely immunocompromised, the Panel suggests prioritizing patients who are least likely to mount an adequate response to COVID-19 vaccination or SARS-CoV-2 infection and are at risk for severe outcomes. This includes, but is not limited to, patients who:

- Are receiving active treatment for solid tumor and hematologic malignancies.
- Have hematologic malignancies (e.g., chronic lymphocytic lymphoma, non-Hodgkin lymphoma, multiple myeloma, acute leukemia) and are known to have poor responses to COVID-19 vaccines, regardless of the treatment status for the hematologic malignancy.
- Received a solid organ or islet transplant and are receiving immunosuppressive therapy.
- Received chimeric antigen receptor T cell therapy or a hematopoietic cell transplant and are within 2 years of transplantation or are receiving immunosuppressive therapy.
- Have a moderate or severe primary immunodeficiency (e.g., severe combined immunodeficiency, DiGeorge syndrome, Wiskott-Aldrich syndrome, common variable immunodeficiency disease).
- Have advanced or untreated HIV infection (defined as people with HIV and CD4 T lymphocyte cell counts <200 cells/mm³, a history of an AIDS-defining illness without immune reconstitution, or clinical manifestations of symptomatic HIV).
- Are receiving active treatment with high-dose corticosteroids (i.e., ≥ 20 mg prednisone or equivalent per day for ≥ 2 weeks), alkylating agents, antimetabolites, transplant-related immunosuppressive drugs, cancer chemotherapeutic agents classified as severely immunosuppressive, or immunosuppressive or immunomodulatory biologic agents (e.g., B cell-depleting agents).

If logistical constraints preclude administration of remdesivir to all prioritized patients, the Panel suggests further prioritizing patients who are more severely immunocompromised and have additional risk factors for severe disease.

Clinical Risk Factors

Some of the most important risk factors for severe COVID-19 include age (risk increases with each decade after age 50),¹ receipt of cancer treatment, immunocompromising conditions or receipt of immunosuppressive medications, cardiovascular disease, chronic kidney disease, chronic lung disease, diabetes, obesity (i.e., body mass index ≥ 30), and pregnancy. For a complete list of risk factors, including information on the relative risk of severe disease, see the CDC webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#). Of note, the likelihood of developing severe COVID-19 increases when a person has multiple comorbidities.² For people who are not immunocompromised, vaccination with a primary COVID-19 vaccine series and booster doses dramatically reduces the risk of progressing to severe disease.

Although children with COVID-19 have substantially lower mortality than adults with COVID-19, severe disease can occur, especially in those with risk factors. See Table 3b for the Panel's framework for assessing the risk of progression to severe COVID-19 in children.

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Clinical Management of Adults Summary

Last Updated: February 29, 2024

Two main processes are thought to drive the pathogenesis of COVID-19. Early in the clinical course, the disease is primarily driven by the replication of SARS-CoV-2. Later in the clinical course, the disease is driven by a dysregulated immune/inflammatory response to SARS-CoV-2 infection that may lead to further tissue damage and thrombosis. Based on this understanding, therapies that directly target SARS-CoV-2 are anticipated to have the greatest effect early in the course of the disease, whereas immunosuppressive, anti-inflammatory, and antithrombotic therapies are likely to be more beneficial after COVID-19 has progressed to stages characterized by hypoxemia, endothelial dysfunction, and immunothrombosis.

The clinical spectrum of SARS-CoV-2 infection includes asymptomatic or presymptomatic infection and mild, moderate, severe, and critical illness. Table 2a provides guidance for clinicians on the therapeutic management of nonhospitalized adults. This includes patients who do not require hospitalization or supplemental oxygen and those who have been discharged from an emergency department or a hospital. Table 2c provides guidance on the therapeutic management of hospitalized adults according to their disease severity and oxygen requirements.

Table 2a. Therapeutic Management of Nonhospitalized Adults With Mild to Moderate COVID-19 Who Do Not Require Supplemental Oxygen

Patient Disposition	Panel's Recommendations
All Patients	• Symptom management should be initiated for all patients (AIII). • The Panel recommends against the use of dexamethasone^a or other systemic corticosteroids (AIIb), unless these agents are being used to treat an underlying condition (AIII).
Patients Who Are at High Risk of Progressing to Severe COVID-19 ^{b,c,d}	<i>Preferred therapies. Listed in order of preference:</i> • Ritonavir-boosted nirmatrelvir (Paxlovid)^e (AIIa). Start as soon as possible and within 5 days of symptom onset. See footnote on drug-drug interactions.^f • Remdesivir^{e,g} (BIIa). Start as soon as possible and within 7 days of symptom onset. <i>Alternative therapy. For use when the preferred therapies are not available, feasible to use, or clinically appropriate:^h</i> • Molnupiravir^{e,i} (CIIa). Start as soon as possible and within 5 days of symptom onset. There is insufficient evidence for the Panel to recommend either for or against initiating these antiviral agents after the timeframes listed above.
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.	

^a There is currently a lack of safety and efficacy data on the use of dexamethasone in outpatients with COVID-19. Using systemic glucocorticoids in outpatients with COVID-19 may cause harm.

^b For a list of risk factors, see the CDC webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#). When deciding whether to prescribe an antiviral agent to a patient who has been vaccinated, clinicians should be aware of the conditions associated with a high risk of disease progression. These conditions include older age, a prolonged amount of time since the most recent vaccine dose (e.g., >6 months), and a decreased likelihood of an adequate immune response to vaccination due to a moderate to severe immunocompromising condition or the receipt of immunosuppressive medications. The number and severity of risk factors also affects the level of risk.

^c For a discussion of potential treatment options for patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication, see [Special Considerations in People Who Are Immunocompromised](#).

^d Concerns about viral rebound or the recurrence of symptoms should not be a reason to avoid using antiviral therapies when their use is indicated.

^e If a patient requires hospitalization after starting treatment, the full treatment course can be completed at the health care provider's discretion.

^f Ritonavir-boosted nirmatrelvir has significant drug-drug interactions. Clinicians should carefully review a patient's concomitant medications and evaluate potential drug-drug interactions. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

^g Administration of remdesivir requires an IV infusion once daily for 3 days.

^h Molnupiravir appears to have lower efficacy than the other options recommended by the Panel.

ⁱ The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated **(AIII)**.

Key: CDC = Centers for Disease Control and Prevention; IV = intravenous; the Panel = the COVID-19 Treatment Guidelines Panel

Table 2c. Therapeutic Management of Hospitalized Adults With COVID-19

Disease Severity	Recommendations for Antiviral or Immunomodulator Therapy		Recommendations for Anticoagulant Therapy
	Clinical Scenario	Recommendation	
Hospitalized for Reasons Other Than COVID-19	Patients with mild to moderate COVID-19 who are at high risk of progressing to severe COVID-19 ^a	See Therapeutic Management of Nonhospitalized Adults With COVID-19 . ^b	For patients without an indication for therapeutic anticoagulation: • Prophylactic dose of heparin, unless contraindicated (AI); (BIII) for pregnant patients
Hospitalized but Does Not Require Supplemental Oxygen	All patients	The Panel recommends against the use of dexamethasone (AIIa) or other systemic corticosteroids (AIII) for the treatment of COVID-19. ^c	
	Patients who are at high risk of progressing to severe COVID-19 ^a	Remdesivir^d (BIIb) for patients who are immunocompromised; (BIII) for other high-risk patients	
Hospitalized and Requires Conventional Oxygen^e	Patients who require minimal conventional oxygen	Remdesivir^{d,f} (BIIa)	For nonpregnant patients with D-dimer levels above the ULN who do not have an increased bleeding risk: • Therapeutic dose of heparin^h (CIIa)
	Most patients	Use dexamethasone plus remdesivir^f (BIIa) . If remdesivir cannot be obtained, use dexamethasone (BI) .	For other patients: • Prophylactic dose of heparin, unless contraindicated (AI); (BIII) for pregnant patients
	Patients who are receiving dexamethasone and who have rapidly increasing oxygen needs and systemic inflammation	Add 1 of the following immunomodulators: ^g <i>Preferred</i> • PO baricitinib (BIIa) • IV tocilizumab (BIIa) <i>Alternatives (Listed in Alphabetical Order)</i> • IV abatacept (CIIa) • IV infliximab (CIIa)	
Hospitalized and Requires HFNC Oxygen or NIV	All patients	Dexamethasone should be administered to all patients (AI). If not already initiated, promptly add 1 of the following immunomodulators: ^{g,i} <i>Preferred</i> • PO baricitinib (AI) <i>Preferred Alternative</i> • IV tocilizumab (BIIa) <i>Additional Alternatives (Listed in Alphabetical Order)</i> • IV abatacept (CIIa) • IV infliximab (CIIa) Add remdesivir to 1 of the options above in certain patients (for examples, see footnote). ^j	For patients without an indication for therapeutic anticoagulation: • Prophylactic dose of heparin, unless contraindicated (AI); (BIII) for pregnant patients For patients who start on a therapeutic dose of heparin in a non-ICU setting and then transfer to the ICU, the Panel recommends switching to a prophylactic dose of heparin , unless there is another indication for therapeutic anticoagulation (BIII).
Hospitalized and Requires MV or ECMO	All patients	Dexamethasone should be administered to all patients (AI). If the patient has not already received a second immunomodulator, promptly add 1 of the following (listed in alphabetical order): ^{i,k} • PO baricitinib^l (BIIa) • IV tocilizumab^l (BIIa) See footnote k for a discussion on the use of remdesivir.	

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

- ^a For a list of risk factors, see the CDC webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#).
- ^b If the patient is hospitalized for reasons other than COVID-19, the treatment duration for remdesivir is 3 days.
- ^c Corticosteroids that are prescribed for an underlying condition should be continued.
- ^d Evidence suggests that the benefit of remdesivir is greatest when the drug is given early in the course of COVID-19 (e.g., within 10 days of symptom onset).
- ^e Conventional oxygen refers to oxygen supplementation that is not HFNC oxygen, NIV, MV, or ECMO.
- ^f If these patients progress to requiring HFNC oxygen, NIV, MV, or ECMO, the full course of remdesivir should still be completed.
- ^g If none of the preferred or alternative options are available or feasible to use, the JAK inhibitor **PO tofacitinib (CIIa)** or the IL-6 inhibitor **IV sarilumab (CIIa)** can be used in combination with dexamethasone. Sarilumab is only commercially available as a SUBQ injection; see [Table 5e](#) for information regarding the preparation of an IV infusion using the SUBQ product.
- ^h Contraindications for the use of therapeutic anticoagulation in patients with COVID-19 include a PLT <50,000 cells/μL, Hgb <8 g/dL, the need for dual antiplatelet therapy, bleeding within the past 30 days that required an ED visit or hospitalization, a history of a bleeding disorder, or an inherited or active acquired bleeding disorder.
- ⁱ Dexamethasone should be initiated immediately, without waiting until the second immunomodulator can be acquired. If other immunomodulators cannot be obtained or are contraindicated, use dexamethasone alone (**AI**).
- ^j Examples of patients who may benefit most from remdesivir include patients who are immunocompromised (**BIIB**); patients with evidence of ongoing viral replication (e.g., those with a low Ct value, as measured by an RT-PCR result or with a positive rapid antigen test result) (**BIII**); or patients who are within 10 days of symptom onset (**CIIa**). For more information on using remdesivir in people with immunocompromising conditions, see [Special Considerations in People Who Are Immunocompromised](#).
- ^k There is insufficient evidence for the Panel to recommend either for or against the use of remdesivir in hospitalized patients with COVID-19 who require MV or ECMO. Some Panel members would add remdesivir to immunomodulator therapy in patients who have recently been placed on MV or ECMO, who are immunocompromised, who have evidence of ongoing viral replication, or who are within 10 days of symptom onset. See text for more information.
- ^l If PO baricitinib and IV tocilizumab are not available or feasible to use, **PO tofacitinib** can be used instead of PO baricitinib (**CIIa**), and **IV sarilumab** can be used instead of IV tocilizumab (**CIIa**).

Key: CDC = Centers for Disease Control and Prevention; Ct = cycle threshold; ECMO = extracorporeal membrane oxygenation; ED = emergency department; HFNC = high-flow nasal cannula; Hgb = hemoglobin; ICU = intensive care unit; IL = interleukin; IV = intravenous; JAK = Janus kinase; MV = mechanical ventilation; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel; PLT = platelet count; PO = oral; RT-PCR = reverse transcription polymerase chain reaction; SUBQ = subcutaneous; ULN = upper limit of normal

General Management of Nonhospitalized Adults With Acute COVID-19

Last Updated: February 29, 2024

Summary Recommendations
• Management of nonhospitalized patients with acute COVID-19 should include providing supportive care, taking steps to reduce the risk of SARS-CoV-2 transmission (including isolating the patient), and advising patients on when to contact a health care provider and seek an in-person evaluation (AIII). • Patients who are at high risk of progressing to severe COVID-19 may be eligible for pharmacologic therapy. See Therapeutic Management of Nonhospitalized Adults With COVID-19 for specific recommendations. • Patients with persistent or progressive dyspnea, especially those who have an oxygen saturation measured by pulse oximetry (SpO₂) ≤94% on room air at sea level or have symptoms that suggest high acuity (e.g., chest pain or tightness, dizziness, confusion, other mental status changes), should be referred to a health care provider for an in-person evaluation (AIII). • Clinicians should be aware that using pulse oximeters to measure oxygen saturation has important limitations. Therefore, SpO₂ results should be considered in the context of the patient's clinical condition. See Clinical Spectrum of SARS-CoV-2 Infection for more information.
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

This section of the Guidelines is intended to provide general information to health care providers who are caring for nonhospitalized adults with COVID-19. The COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for pharmacologic management can be found in [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

This section focuses on the evaluation and management of:

- Adults with COVID-19 in an ambulatory care setting
- Adults with COVID-19 following discharge from the emergency department (ED)
- Adults with COVID-19 following inpatient discharge

Outpatient evaluation and management in each of these settings may include some or all of the following: in-person visits, telemedicine, remote monitoring, and home visits by nurses or other health care providers.

Data from studies in the United States show that certain racial and ethnic minorities experience higher rates of COVID-19, hospitalization, and death in relation to their share of the total U.S. population.¹⁻⁵ In addition, some studies have found that members of certain racial, ethnic, and socioeconomic groups were less likely to receive COVID-19 treatments.⁶⁻⁸ The underlying causes of these observed disparities may include inadequate insurance coverage, a lack of primary care providers, hesitancy about receiving treatment, barriers to telehealth visits, and transportation challenges. To reduce COVID-19 treatment disparities, providers must be aware of the problem and provide patient-centered care. The Panel recommends that health care providers, health care systems, and payers ensure equitable access to high-quality care and treatment for all patients, regardless of race, ethnic identity, or other minoritized identity or social status (AIII).

Managing Patients With COVID-19 in an Ambulatory Care Setting

Approximately 80% of patients with COVID-19 who are unvaccinated have mild illness that does not

require medical intervention or hospitalization,⁹ and the proportion is likely higher in patients who are up to date with COVID-19 vaccinations.^{10,11} Most patients with mild COVID-19 (defined as the absence of viral pneumonia and hypoxemia) can be managed in an ambulatory care setting or at home. Patients with moderate COVID-19 (e.g., those with viral pneumonia but without hypoxemia) or severe COVID-19 (e.g., those with dyspnea, hypoxemia, or lung infiltrates >50%) need in-person evaluation and close monitoring, as pulmonary disease can progress rapidly and require hospitalization.

When managing outpatients with COVID-19, clinicians should provide supportive care, take steps to reduce the risk of SARS-CoV-2 transmission as recommended by the Centers for Disease Control and Prevention,^{12,13} and advise patients on when to seek an in-person evaluation.¹⁴ Supportive care includes managing symptoms (as described below), ensuring that patients are receiving the proper nutrition, and being cognizant of the risks of social isolation, particularly for older adults.¹⁵ Health care providers should identify patients who are at high risk of progressing to severe COVID-19.¹⁶ These patients are candidates for antiviral therapy. See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for more information.

Older patients and those with chronic medical conditions, especially those who are not up to date with COVID-19 vaccinations, have a higher risk of hospitalization and death.¹⁷ However, SARS-CoV-2 infection may cause severe disease and death in patients of any age, even in the absence of risk factors. When managing older adults with COVID-19, factors such as cognitive impairment, frailty, the risk of falls, and polypharmacy should be considered. The decision to monitor a patient in the outpatient setting should be made on a case-by-case basis.

Clinical Considerations When Managing Patients in an Ambulatory Care Setting

Patients with SARS-CoV-2 infection may be asymptomatic or experience symptoms that are indistinguishable from other acute viral or bacterial infections (e.g., fever, cough, sore throat, malaise, muscle pain, headache, gastrointestinal symptoms). People who have symptoms compatible with COVID-19 should undergo diagnostic SARS-CoV-2 testing (see [Testing for SARS-CoV-2 Infection](#)). Considering other possible etiologies of symptoms is important, including other respiratory viral infections (e.g., influenza), community-acquired pneumonia, congestive heart failure, asthma or chronic obstructive pulmonary disease exacerbations, and streptococcal pharyngitis.

Although mild dyspnea is common, worsening dyspnea and severe chest pain or tightness suggest the development or progression of pulmonary involvement. In earlier studies of patients with COVID-19 who developed acute respiratory distress syndrome, progression occurred a median of 2.5 days after the onset of dyspnea.¹⁸⁻²⁰ Patients with persistent or progressive dyspnea, especially those who have an oxygen saturation measured by pulse oximetry (SpO_2) $\leq 94\%$ on room air at sea level or have symptoms that suggest high acuity (e.g., chest pain or tightness, dizziness, confusion, other mental status changes), should be referred to a health care provider for an in-person evaluation (**AIII**).

If an adult patient has access to a pulse oximeter at home, SpO_2 measurements can be used to help assess overall clinical status. Patients should be advised to use pulse oximeters on warm fingers rather than cold fingers for better accuracy. Patients should inform their health care providers if the value is repeatedly below 95% on room air at sea level. Pulse oximetry may not accurately detect hypoxemia, especially in patients who have dark skin pigmentation.^{21,22}

Not all commercially available pulse oximeters have been cleared by the Food and Drug Administration (FDA). SpO_2 readings obtained through non-FDA-cleared devices, such as over-the-counter sports oximeters or mobile phone applications, lack sufficient accuracy for clinical use.^{23,24} Abnormal readings on these devices should be confirmed with an FDA-cleared device or an arterial blood gas analysis.

Importantly, SpO₂ readings should only be interpreted within the context of a patient's entire clinical presentation (i.e., results should be disregarded if a patient is complaining of increasing dyspnea). See [Clinical Spectrum of SARS-CoV-2 Infection](#) for more information regarding the limitations of pulse oximetry.

Assessing the Need for In-Person Evaluation

When possible, patients with symptoms of COVID-19 may be triaged via telehealth visits to determine whether they require COVID-19–specific therapy and in-person care. Outpatient management may include the use of patient self-assessment tools. During the initial triage, clinic staff should determine which patients are eligible to receive supportive care at home and which patients warrant an in-person evaluation.²⁵

Patients with persistent or progressive dyspnea, especially those who have an SpO₂ ≤94% on room air at sea level or have symptoms that suggest high acuity (e.g., chest pain or tightness, dizziness, confusion, other mental status changes), should be evaluated by a health care provider (AIII).

Clinicians who use SpO₂ results to assess patients must be aware of the important limitations of pulse oximeters and conduct assessments in the context of a patient's clinical condition. See [Clinical Spectrum of SARS-CoV-2 Infection](#) for more information.

The criteria used to determine the appropriate clinical setting for an in-person evaluation may vary by location and institution. There should be a low threshold for in-person evaluation of older people and those with medical conditions associated with an increased risk of progressing to severe COVID-19. The individuals who perform the initial triage should use their clinical judgment to determine whether patients require ambulance transport.

In some settings where clinical evaluation is challenged by geography, home visits from a health care provider may be used to evaluate patients.²⁶ Patients who are homeless should be provided with housing where they can adequately self-isolate. All outpatients should receive instructions regarding self-care, isolation, and follow-up, and they should be advised to contact a health care provider or a local ED about any worsening symptoms.²⁷

Counseling Regarding the Need for Follow-Up

Clinicians should ensure that these patients receive adequate medical follow-up. The frequency and duration of follow-up will depend on the risk for severe disease, the severity of symptoms, and the patient's ability to self-report worsening symptoms. Health care providers should determine whether a patient has adequate transportation for clinic visits; whether they have access to a phone, computer, or tablet for telehealth visits; and whether they have regular access to food. The clinician should also confirm that the patient has a caregiver who can assist with daily activities if needed.

All patients and/or their family members or caregivers should be counseled about the warning symptoms that should prompt re-evaluation through a telehealth visit or an in-person evaluation in an ambulatory care setting or ED. These symptoms include new onset of dyspnea; worsening dyspnea (particularly if dyspnea occurs while resting or if it interferes with daily activities); dizziness; and mental status changes, such as confusion.

Managing Adults With COVID-19 Following Discharge From the Emergency Department

Clinicians who work in EDs should assess whether a patient requires hospital admission for the management of COVID-19 or whether the symptoms can be managed in the outpatient setting.

Treatment with an antiviral agent is recommended for outpatients with mild to moderate COVID-19 who are not on supplemental oxygen and are at high risk of clinical progression (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)).

If a patient is not admitted to the hospital, the Panel **recommends against** the use of anticoagulants and antiplatelet therapy in the ED for the prevention of venous thromboembolism or arterial thrombosis, except in a clinical trial (**AIIa**). This recommendation does not apply to patients with other indications for antithrombotic therapy. For more information, see [Antithrombotic Therapy in Patients With COVID-19](#). Patients should be encouraged to ambulate, and activity should be increased according to the patient's tolerance.

Managing Adults With COVID-19 Following Hospital Discharge

Most patients who are discharged from the hospital setting should have a follow-up visit with a health care provider soon after discharge. Whether an in-person or telehealth visit is most appropriate depends on the clinical and social situation. In some cases, adult patients are deemed to be stable for discharge from the inpatient setting while they still require supplemental oxygen. When possible, these individuals should receive oximetry monitoring and close follow-up through telehealth visits, visiting nurse services, or in-person clinic visits.

The Panel **recommends against** routinely continuing venous thromboembolism prophylaxis in patients with COVID-19 after hospital discharge unless they have another indication for anticoagulation (**AIIa**). For more information, see [Antithrombotic Therapy in Patients With COVID-19](#). Patients should be encouraged to ambulate, and activity should be increased according to the patient's tolerance.

Considerations in Pregnant People

Managing pregnant outpatients with COVID-19 is similar to managing nonpregnant patients. Clinicians should offer supportive care and therapeutics as indicated, take steps to reduce the risk of SARS-CoV-2 transmission, and provide guidance on when to seek an in-person evaluation. Pregnant patients who have COVID-19 are at higher risk of severe disease, including a higher risk of death, intensive care unit admission, and mechanical ventilation, than nonpregnant people with COVID-19.^{28,29} Pregnant patients with COVID-19 also have an increased risk of poor obstetric and neonatal outcomes, and this risk may be even higher in people with comorbidities such as obesity, diabetes, hypertension, and lung disease.³⁰⁻³⁴ It is important for health care providers to thoroughly assess pregnant patients for potential risks of severe COVID-19 and offer antiviral therapy when indicated. Please see [Special Considerations During Pregnancy and After Delivery](#) for more information on managing pregnant patients with COVID-19.

In pregnant patients, SpO₂ should be maintained at ≥95% on room air at sea level; therefore, the threshold for monitoring pregnant patients in an inpatient setting may be lower than in nonpregnant patients. At this time, there are no changes to fetal monitoring recommendations in the outpatient setting, and fetal surveillance and management should be similar to the fetal surveillance and management used for pregnant patients with medical illness.^{35,36} However, these monitoring strategies can be discussed on a case-by-case basis with an obstetrician. Pregnant and lactating patients should be given the opportunity to participate in clinical trials of outpatients with COVID-19 to help inform decision making in this population.

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Therapeutic Management of Nonhospitalized Adults With COVID-19

Last Updated: February 29, 2024

Symptom management should be initiated for all nonhospitalized adults with mild to moderate COVID-19. For adults who are at high risk of progressing to severe disease, several antiviral therapeutic options are available to reduce the risk of hospitalization or death. The COVID-19 Treatment Guidelines Panel's (the Panel) recommendations on the use of these drugs for the treatment of COVID-19 are outlined in this section.

The main goal of therapeutic management for nonhospitalized patients is to prevent progression to severe disease, hospitalization, or death. Other goals may include accelerating symptom recovery and viral clearance and preventing long-term sequelae. Current data on the impact of therapy on these secondary goals are limited.

Several factors affect the selection of the best treatment option for a specific patient. These factors include the clinical efficacy and availability of the treatment option, the feasibility of administering parenteral medications, the potential for significant drug-drug interactions, the patient's pregnancy status, the time from symptom onset, and the *in vitro* activities of the available products against the currently circulating SARS-CoV-2 variants and subvariants.

Most of the data that support the use of the recommended treatment options come from clinical trials that enrolled individuals who were at high risk of disease progression and who had no pre-existing immunity from COVID-19 vaccination or prior SARS-CoV-2 infection. Accordingly, the proportion of hospitalizations and deaths in the placebo arms of these trials was high compared to what has been seen more recently in populations where most people are vaccinated or have had prior SARS-CoV-2 infection. Although these trials demonstrated the efficacy of using antiviral drugs in high-risk populations, it is difficult to know their precise effectiveness in the current setting because of the low rates of hospitalization and death among those who have been vaccinated.

Nevertheless, some patients continue to have an increased risk of disease progression, and it is in those people that therapies are most likely to be beneficial. Patients who are at the highest risk are older patients (i.e., those aged >50 years and especially those aged ≥65 years) and patients who are unlikely to have an adequate immune response to COVID-19 vaccines due to a moderate to severe immunocompromising condition or the receipt of immunosuppressive medications. Other risk factors include lack of vaccination or incomplete vaccination; a prolonged amount of time since the most recent vaccine dose (e.g., >6 months); and conditions such as obesity, diabetes, and chronic pulmonary, cardiac, or renal disease.¹

People who are members of racial and ethnic minority groups have higher rates of hospitalization and death from COVID-19 than people who are White.² Disparities in the use of antiviral treatments in patients who are not White have been reported; therefore, attention to equitable access is critical.^{3,4}

The Panel's recommendations reflect the available data on the benefits of using antiviral therapies to prevent progression to severe COVID-19. Table 2a outlines the Panel's recommendations for the therapeutic management of nonhospitalized adults with COVID-19. For the recommended doses for the agents listed in Table 2a, refer to Table 2b below.

Table 2a. Therapeutic Management of Nonhospitalized Adults With Mild to Moderate COVID-19 Who Do Not Require Supplemental Oxygen

Patient Disposition	Panel's Recommendations
All Patients	- Symptom management should be initiated for all patients (AIII). - The Panel recommends against the use of dexamethasone^a or other systemic corticosteroids (AIIb), unless these agents are being used to treat an underlying condition (AIII).
Patients Who Are at High Risk of Progressing to Severe COVID-19 ^{b,c,d}	<i>Preferred therapies. Listed in order of preference:</i> Ritonavir-boosted nirmatrelvir (Paxlovid)^e (AIIa). Start as soon as possible and within 5 days of symptom onset. See footnote on drug-drug interactions.^f Remdesivir^{e,g} (BIIa). Start as soon as possible and within 7 days of symptom onset. <i>Alternative therapy. For use when the preferred therapies are not available, feasible to use, or clinically appropriate.^h</i> Molnupiravir^{e,i} (CIIa). Start as soon as possible and within 5 days of symptom onset. There is insufficient evidence for the Panel to recommend either for or against initiating these antiviral agents after the timeframes listed above.
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.	

^a There is currently a lack of safety and efficacy data on the use of dexamethasone in outpatients with COVID-19. Using systemic glucocorticoids in outpatients with COVID-19 may cause harm.

^b For a list of risk factors, see the CDC webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#). When deciding whether to prescribe an antiviral agent to a patient who has been vaccinated, clinicians should be aware of the conditions associated with a high risk of disease progression. These conditions include older age, a prolonged amount of time since the most recent vaccine dose (e.g., >6 months), and a decreased likelihood of an adequate immune response to vaccination due to a moderate to severe immunocompromising condition or the receipt of immunosuppressive medications. The number and severity of risk factors also affects the level of risk.

^c For a discussion of potential treatment options for patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication, see below and [Special Considerations in People Who Are Immunocompromised](#).

^d Concerns about viral rebound or the recurrence of symptoms should not be a reason to avoid using antiviral therapies when their use is indicated. See Viral Rebound and Symptom Recurrence below for details.

^e If a patient requires hospitalization after starting treatment, the full treatment course can be completed at the health care provider's discretion.

^f Ritonavir-boosted nirmatrelvir has significant drug-drug interactions. Clinicians should carefully review a patient's concomitant medications and evaluate potential drug-drug interactions. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

^g Administration of remdesivir requires an IV infusion once daily for 3 days.

^h Molnupiravir appears to have lower efficacy than the other options recommended by the Panel.

ⁱ The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (**AIII**).

Key: CDC = Centers for Disease Control and Prevention; IV = intravenous; the Panel = the COVID-19 Treatment Guidelines Panel

Symptom Management

Treatment of symptoms includes using over-the-counter antipyretics, analgesics, or antitussives for fever, headache, myalgias, and cough. Patients should be advised to drink fluids regularly to avoid dehydration. Rest is recommended as needed during the acute phase of COVID-19, and ambulation and other forms of activity should be increased according to the patient's tolerance. Patients should be educated about the variability in time to symptom resolution and complete recovery. When possible, patients with symptoms of COVID-19 may be triaged via telehealth visits to determine whether they require COVID-19-specific therapy and in-person care.

Patients with persistent or progressive dyspnea, especially those who have an oxygen saturation measured by pulse oximetry $\leq 94\%$ on room air at sea level or have symptoms that suggest high acuity (e.g., chest pain or tightness, dizziness, confusion, other mental status changes), should be evaluated by a health care provider (**AIII**). For more information, see [General Management of Nonhospitalized Adults With Acute COVID-19](#).

Rationale for the Panel's Recommendations

The Panel's recommendations for the antiviral agents that are used to treat nonhospitalized patients with mild to moderate COVID-19 who are at high risk of disease progression are based on the results of clinical trials.

The Panel favors the use of ritonavir-boosted nirmatrelvir (Paxlovid) in most high-risk, nonhospitalized patients with mild to moderate COVID-19. When ritonavir-boosted nirmatrelvir is not clinically appropriate (e.g., because of significant drug-drug interactions), the Panel recommends using remdesivir. Ritonavir-boosted nirmatrelvir has high efficacy; has been shown to reduce hospitalization and death when administered to high-risk, unvaccinated, nonhospitalized patients within 5 days of symptom onset;⁵ and is an oral medication, whereas remdesivir requires intravenous (IV) administration.

The Panel's recommendation for remdesivir is based on the results from a Phase 3, randomized, placebo-controlled trial that reported high clinical efficacy for remdesivir in high-risk, nonhospitalized patients with COVID-19 who were unvaccinated.⁶ However, in some settings, daily IV administration of remdesivir for 3 days may be a logistical challenge.

The Panel recommends **molnupiravir** as a therapeutic option when the other recommended antiviral treatment options are not available, feasible to use, or clinically appropriate (**CIIa**). Molnupiravir appears to have lower clinical efficacy than the other treatment options, although no randomized studies have compared these therapies directly.

There is insufficient evidence for the Panel to recommend either for or against initiating these antiviral agents after the recommended timeframes. Because the drugs listed above were studied soon after symptom onset (i.e., within 5 days for molnupiravir and ritonavir-boosted nirmatrelvir and within 7 days for remdesivir), it is not known whether they would confer a clinical benefit if they were started beyond the recommended timeframe.

Currently, data on the use of combinations of antiviral agents for the treatment of COVID-19 are limited. Clinical trials are needed to determine whether combination therapy has a role in the treatment of COVID-19.

Strategies for the Use of Ritonavir-Boosted Nirmatrelvir

Because ritonavir is a strong cytochrome P450 3A4 inhibitor and a P-glycoprotein inhibitor, it may increase blood concentrations of certain concomitant medications and increase the potential for serious drug toxicities. Therefore, the Food and Drug Administration (FDA) prescribing information includes a

boxed warning for significant drug-drug interactions with ritonavir-boosted nirmatrelvir.⁷ Clinicians should consider both the potential benefits of treatment with ritonavir-boosted nirmatrelvir and the potential risks related to drug-drug interactions.

Many drug-drug interactions between ritonavir-boosted nirmatrelvir and concomitant medications (e.g., certain statins, calcium channel blockers, or direct oral anticoagulants) **can be safely managed**. If a significant drug-drug interaction is identified, prescribers should consider consulting with a pharmacist.

The following resources provide information on identifying and managing drug-drug interactions:

- [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#)
- The [Liverpool COVID-19 Drug Interactions website](#)
- The [University of Waterloo/University of Toronto drug interaction guide](#) for ritonavir-boosted nirmatrelvir
- The FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir

The use of ritonavir-boosted nirmatrelvir may be challenging in patients with severe renal impairment. The FDA prescribing information states that ritonavir-boosted nirmatrelvir is not recommended for patients with an estimated glomerular filtration rate (eGFR) of <30 mL/min until more data are available to establish appropriate dosing.⁷ Although data on dose adjustments are limited, some groups have proposed dosing adjustments for ritonavir-boosted nirmatrelvir in patients with an eGFR of <30 mL/min or in patients who require hemodialysis.⁸⁻¹¹

The decision to prescribe ritonavir-boosted nirmatrelvir to patients who have received transplants and are taking calcineurin and mammalian target of rapamycin inhibitors should always be made in consultation with the patient's specialist providers. Among reports submitted to the FDA Adverse Event Reporting System, the most commonly reported concomitant medications that resulted in serious adverse reactions, including fatal events, were calcineurin inhibitors (e.g., tacrolimus).¹² Ritonavir-boosted nirmatrelvir may be prescribed to select patients if an expert in managing the interaction is available and close therapeutic drug monitoring is logistically feasible. Otherwise, an alternative therapy for COVID-19 should be considered. See the [American Society of Transplantation statement](#) for additional information.

Interactions between ritonavir-boosted nirmatrelvir and cancer chemotherapeutic agents should also be managed in consultation with the patient's specialist providers. For guidance on managing these interactions, refer to the FDA prescribing information for ritonavir-boosted nirmatrelvir and the prescribing information for the chemotherapeutic agent. The [University Health Network/Kingston Health Sciences Centre](#) provides an additional resource for evaluating drug-drug interactions between ritonavir-boosted nirmatrelvir and chemotherapeutic agents.

Strategies for the Use of Remdesivir

Advanced planning (e.g., reserving infusion slots, identifying alternative infusion sites) may be needed to increase access to IV remdesivir. IV remdesivir can be administered in skilled nursing facilities, home health care settings, and outpatient facilities such as infusion centers. If treatment facilities cannot provide a 3-day course of remdesivir IV infusions to all eligible patients, prioritizing patients who will benefit the most from the therapy becomes necessary. The prioritization scheme below is based on 4 key elements: age, vaccination status, immune status, and clinical risk factors. For a list of risk factors, see the Centers for Disease Control and Prevention (CDC) webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#). The groups are listed by tier in descending order of priority.

Tier	Risk Group
1	- Immunocompromised individuals not expected to mount an adequate immune response to COVID-19 vaccination or SARS-CoV-2 infection due to their underlying conditions,^a regardless of vaccine status; <i>or</i> - Unvaccinated individuals at the highest risk of severe disease (anyone aged ≥75 years or anyone aged ≥65 years with additional risk factors)
2	Unvaccinated individuals not included in Tier 1 who are at risk of severe disease (anyone aged ≥65 years or anyone aged <65 years with clinical risk factors)
3	Vaccinated individuals at risk of severe disease (anyone aged ≥65 years or anyone aged <65 years with clinical risk factors)^b

^a See the CDC webpage [COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised](#) for a discussion of immunocompromising conditions.

^b Vaccinated individuals who are not up to date with their immunizations are likely at higher risk of severe disease; patients within this tier who are in this situation should be prioritized for treatment. See the CDC webpage [Stay Up to Date With COVID-19 Vaccines](#) for more information.

See [Prioritization of Anti-SARS-CoV-2 Therapies for the Treatment of COVID-19 in Nonhospitalized Patients When There Are Logistical Constraints](#) for more information.

Patients Who Are Immunocompromised and Have Prolonged COVID-19 Symptoms and Evidence of Ongoing Viral Replication

For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication (e.g., those with a low cycle threshold value, as measured by a reverse transcription polymerase chain reaction result or with a positive rapid antigen test result) despite receiving a course of antiviral therapy, the optimal management is unknown. Case reports and case series have described the treatment of these patients with additional antiviral treatments, prolonged courses of antiviral treatments, high-titer COVID-19 convalescent plasma (CCP), or combination therapy.¹³⁻¹⁹ The data for these approaches are not definitive, but some Panel members would use 1 or more of the following treatment options:

- Longer and/or additional courses of ritonavir-boosted nirmatrelvir
- Longer and/or additional courses of remdesivir
- High-titer CCP from a vaccinated donor who recently recovered from COVID-19 likely caused by a SARS-CoV-2 variant similar to the variant causing the patient's illness

For further discussion of these potential treatment options, see [Special Considerations in People Who Are Immunocompromised](#).

Additional Information on Ritonavir-Boosted Nirmatrelvir

Nirmatrelvir is an orally bioavailable protease inhibitor that is active against M^{PRO}, a viral protease that plays an essential role in viral replication.²⁰ The FDA has approved ritonavir-boosted nirmatrelvir for the treatment of mild to moderate COVID-19 in nonhospitalized adults who are at high risk of progressing to severe COVID-19.⁷

Patients should complete the 5-day treatment course of ritonavir-boosted nirmatrelvir, which was shown to be efficacious in the EPIC-HR trial.⁵ If a patient requires hospitalization after starting treatment, the full 5-day treatment course of ritonavir-boosted nirmatrelvir should be completed unless there are drug-drug interactions that preclude its use.

In the EPIC-HR trial, ritonavir-boosted nirmatrelvir reduced the risk of hospitalization or death by 89%

compared to placebo in unvaccinated, nonhospitalized adults with laboratory-confirmed SARS-CoV-2 infection.⁵ This efficacy is comparable to the efficacies reported in similar patient populations for remdesivir (87% relative reduction)⁶ and greater than the efficacy reported for molnupiravir in this setting (31% relative reduction).²¹

Because ritonavir-boosted nirmatrelvir has the potential for significant drug-drug interactions with concomitant medications, this regimen may not be the optimal choice for all patients (see [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \[Paxlovid\] and Concomitant Medications](#)). However, because ritonavir-boosted nirmatrelvir is the only highly effective oral antiviral available for the treatment of COVID-19, drug-drug interactions that can be safely managed should not preclude the use of this medication.

For more information on the use of ritonavir-boosted nirmatrelvir, see [Ritonavir-Boosted Nirmatrelvir \(Paxlovid\)](#). See Viral Rebound and Symptom Recurrence below for information regarding SARS-CoV-2 viral rebound in patients who have completed treatment with ritonavir-boosted nirmatrelvir.

Additional Information on Remdesivir

Remdesivir is a nucleotide prodrug of an adenosine analog that inhibits SARS-CoV-2 replication. It is approved by the FDA for the treatment of COVID-19 in adults and children aged ≥ 28 days and weighing ≥ 3 kg who are hospitalized with COVID-19 and for those with mild to moderate COVID-19 who are not hospitalized and are at high risk of progressing to severe disease. In the PINETREE trial, nonhospitalized patients with mild to moderate COVID-19 who were unvaccinated and at high risk of progressing to severe disease received 3 days of IV remdesivir or placebo. The use of remdesivir resulted in an 87% relative reduction in the risk of hospitalization or death.⁶

Remdesivir should be administered in a setting where severe hypersensitivity reactions, such as anaphylaxis, can be managed. Patients should be monitored during the infusion and observed for at least 1 hour after the infusion as clinically appropriate.

For more information, see [Remdesivir](#).

Additional Information on Molnupiravir

Molnupiravir is the oral prodrug of beta-D-N4-hydroxycytidine, a ribonucleoside that has shown antiviral activity against SARS-CoV-2 in vitro and in clinical trials.²²⁻²⁴ The FDA issued an Emergency Use Authorization for molnupiravir for the treatment of mild to moderate COVID-19 in nonhospitalized patients aged ≥ 18 years who are at high risk of disease progression and for whom other antiviral treatment options are not accessible or clinically appropriate.

The MOVE-OUT trial enrolled high-risk, unvaccinated, nonhospitalized adults in the pre-Omicron era.²¹ The study found that molnupiravir reduced the rate of hospitalization or death by 31% compared to placebo. A secondary analysis of patients who required hospitalization during the trial found a reduced need for respiratory interventions among those who received molnupiravir compared to those who received placebo.²⁵

The PANORAMIC trial enrolled nonhospitalized adults with COVID-19 who were at high risk of severe disease during a period when the Omicron variant was circulating.²⁶ Ninety-four percent of the patients had received at least 3 doses of a COVID-19 vaccine. The study found that the use of molnupiravir plus usual care did not reduce the occurrence of the primary composite outcome of hospitalization or death compared to usual care alone. The proportion of patients who met this composite outcome was 1% in both arms. Molnupiravir plus usual care was superior to usual care alone for several secondary clinical endpoints. For example, the time to self-reported recovery was substantially shorter in people who

received molnupiravir plus usual care than in people who received usual care alone (median of 9 days vs. 15 days). Because the PANORAMIC trial was an open-label study with self-reported symptoms, the findings are less reliable than those from a placebo-controlled trial.

Although the different COVID-19 treatment options have not been directly compared in clinical trials, the Panel recommends using molnupiravir as an alternative therapy only when ritonavir-boosted nirmatrelvir and remdesivir are not available, feasible to use, or clinically appropriate, because molnupiravir appears to have lower clinical efficacy than these other options.

There is a theoretical risk that the molnupiravir metabolite beta-D-N4-hydroxycytidine could be incorporated into host DNA, leading to mutations. The available genotoxicity data and the 5-day duration of treatment led the FDA to conclude that molnupiravir has a low risk for genotoxicity.²⁷

The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (**AIII**). People who engage in sexual activity that may result in conception should use effective contraception during and following treatment with molnupiravir.

Fetal toxicity has been reported in animal studies of molnupiravir.²⁷ However, when other therapies are not available, pregnant patients with COVID-19 who are at high risk of progressing to severe disease may reasonably choose molnupiravir after being fully informed of the risks, particularly if they are beyond the time of embryogenesis (i.e., >10 weeks' gestation). See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for more information.

For more information, see [Molnupiravir](#).

Viral Rebound and Symptom Recurrence

Observational studies and the EPIC-HR and MOVE-OUT trials have described SARS-CoV-2 viral rebound and the recurrence of COVID-19 symptoms in some patients who have completed treatment with ritonavir-boosted nirmatrelvir or molnupiravir.^{7,27-30} However, viral rebound can also occur in patients who have not received treatment for COVID-19.³¹ Some observational studies have reported that patients who were treated with ritonavir-boosted nirmatrelvir had a higher frequency of viral rebound and symptom recurrence than those who did not receive treatment.^{32,33} The re-emergence of culturable SARS CoV-2 has been reported in some individuals with viral rebound.

To date, virus detection and the recurrence of COVID-19 symptoms following the use of antiviral therapies have not been associated with progression to severe COVID-19.^{34,35} Therefore, concerns about viral rebound or the recurrence of symptoms should not be a reason to avoid using antiviral therapies when their use is indicated.^{27,36-38} There are insufficient data on whether a longer course of ritonavir-boosted nirmatrelvir or molnupiravir will prevent viral rebound or symptom recurrence. There also are insufficient data on the efficacy of administering a second course of antiviral therapy to treat viral rebound or symptom recurrence. However, a clinical trial that is evaluating the use of a second course of ritonavir-boosted nirmatrelvir to treat patients with viral rebound and symptom recurrence is underway (ClinicalTrials.gov Identifier [NCT05567952](#)).

Immunomodulators

The Panel **recommends against** the use of **dexamethasone** or other systemic corticosteroids to treat outpatients with mild to moderate COVID-19 who do not require hospitalization or supplemental oxygen (**AIIb**). Patients with COVID-19 who are receiving **dexamethasone** or another corticosteroid for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Medicare and FDA data show a significant increase in the number of prescriptions for systemic

corticosteroids among nonhospitalized patients with COVID-19³⁹ despite a lack of safety and efficacy data on the use of systemic corticosteroids in this setting. Systemic glucocorticoids may cause harm in nonhospitalized patients with COVID-19. Results from 1 randomized controlled trial and 1 observational cohort study did not demonstrate a clinical benefit of dexamethasone among hospitalized patients who did not require supplemental oxygen,⁴⁰ and dexamethasone may potentially cause harm in these patients.⁴¹

In the RECOVERY trial, the use of dexamethasone had no effect on mortality among hospitalized patients with COVID-19 who did not require supplemental oxygen (rate ratio 1.19; 95% CI, 0.91–1.55).⁴¹ A large observational study of patients at Veterans Affairs hospitals showed that patients with COVID-19 who did not require supplemental oxygen and received dexamethasone had an increased risk of 90-day mortality (HR 1.76; 95% CI, 1.47–2.12).⁴¹

Concomitant Medication Management

In general, a patient's usual medication and/or supplement regimen should be continued after the diagnosis of COVID-19 (see [Considerations for Using Concomitant Medications in Patients With COVID-19](#)). Patients should be advised to avoid the use of nebulized medications in the presence of others to avoid potential aerosolization of SARS-CoV-2.⁴² In patients with HIV, an antiretroviral regimen should not be modified for the purpose of preventing or treating SARS-CoV-2 infection. For more information, see [Special Considerations in People With HIV](#).

When a patient is receiving an immunomodulating medication, the prescribing clinician or an expert in the subspecialty should be consulted about the risks and benefits associated with a temporary dose reduction or discontinuation. These risks and benefits will depend on the medication's indication and the severity of the underlying condition (see [Special Considerations in People Who Are Immunocompromised](#)).

Before prescribing ritonavir-boosted nirmatrelvir, clinicians should carefully review the patient's concomitant medications, including over-the-counter medications, herbal supplements, and recreational drugs, to evaluate potential drug-drug interactions.

Table 2b. Dosing Regimens for the Drugs Recommended in Table 2a

Drug Name	Dosing Regimen	Comments
Ritonavir-Boosted Nirmatrelvir (Paxlovid)	eGFR ≥60 mL/min - Nirmatrelvir 300 mg with RTV 100 mg PO twice daily for 5 days eGFR ≥30 to <60 mL/min - Nirmatrelvir 150 mg with RTV 100 mg PO twice daily for 5 days eGFR <30 mL/min Not recommended - For more information on the use of this agent in patients with eGFR <30 mL/min, see Ritonavir-Boosted Nirmatrelvir (Paxlovid). Severe Hepatic Impairment (Child-Pugh Class C) Not recommended	Clinicians should evaluate potential drug-drug interactions. See Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir (Paxlovid) and Concomitant Medications for more information.
Remdesivir	RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily on Days 2 and 3. Administer each infusion over 30–120 minutes.	Patients should be monitored for ≥1 hour after the infusion as clinically appropriate.
Molnupiravir	MOV 800 mg PO every 12 hours for 5 days	Before initiating MOV, assess the patient's pregnancy status as clinically indicated. See Molnupiravir for more information.

Key: eGFR = estimated glomerular filtration rate; IV = intravenous; MOV = molnupiravir; PO = oral; RDV = remdesivir; RTV = ritonavir

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Therapeutic Management of Hospitalized Adults With COVID-19

Last Updated: February 29, 2024

Table 2c. Therapeutic Management of Hospitalized Adults With COVID-19

Disease Severity	Recommendations for Antiviral or Immunomodulator Therapy		Recommendations for Anticoagulant Therapy
	Clinical Scenario	Recommendation	
Hospitalized for Reasons Other Than COVID-19	Patients with mild to moderate COVID-19 who are at high risk of progressing to severe COVID-19 ^a	See Therapeutic Management of Nonhospitalized Adults With COVID-19 . ^b	For patients without an indication for therapeutic anticoagulation: • Prophylactic dose of heparin , unless contraindicated (AI); (BIII) for pregnant patients
Hospitalized but Does Not Require Supplemental Oxygen	All patients	The Panel recommends against the use of dexamethasone (AIIa) or other systemic corticosteroids (AIII) for the treatment of COVID-19. ^c	
	Patients who are at high risk of progressing to severe COVID-19 ^a	Remdesivir^d (BIIb) for patients who are immunocompromised; (BIII) for other high-risk patients	
Hospitalized and Requires Conventional Oxygen^e	Patients who require minimal conventional oxygen	Remdesivir^{d,f} (BIIa)	For nonpregnant patients with D-dimer levels above the ULN who do not have an increased bleeding risk: • Therapeutic dose of heparin^h (CIIa)
	Most patients	Use dexamethasone plus remdesivirⁱ (BIIa) . If remdesivir cannot be obtained, use dexamethasone (BI) .	For other patients: • Prophylactic dose of heparin , unless contraindicated (AI); (BIII) for pregnant patients
	Patients who are receiving dexamethasone and who have rapidly increasing oxygen needs and systemic inflammation	Add 1 of the following immunomodulators: ^g <i>Preferred</i> • PO baricitinib (BIIa) • IV tocilizumab (BIIa) <i>Alternatives (Listed in Alphabetical Order)</i> • IV abatacept (CIIa) • IV infliximab (CIIa)	

Disease Severity	Recommendations for Antiviral or Immunomodulator Therapy		Recommendations for Anticoagulant Therapy
	Clinical Scenario	Recommendation	
Hospitalized and Requires HFNC Oxygen or NIV	All patients	Dexamethasone should be administered to all patients (AI). If not already initiated, promptly add 1 of the following immunomodulators:^{g,i} <i>Preferred</i> • PO baricitinib (AI) <i>Preferred Alternative</i> • IV tocilizumab (BIIa) <i>Additional Alternatives (Listed in Alphabetical Order)</i> • IV abatacept (CIIa) • IV infliximab (CIIa) Add remdesivir to 1 of the options above in certain patients (for examples, see footnote).^j	For patients without an indication for therapeutic anticoagulation: • Prophylactic dose of heparin, unless contraindicated (AI); (BIII) for pregnant patients For patients who start on a therapeutic dose of heparin in a non-ICU setting and then transfer to the ICU, the Panel recommends switching to a prophylactic dose of heparin, unless there is another indication for therapeutic anticoagulation (BIII).
	All patients	Dexamethasone should be administered to all patients (AI). If the patient has not already received a second immunomodulator, promptly add 1 of the following (listed in alphabetical order):^{i,k} • PO baricitinib^l (BIIa) • IV tocilizumab^l (BIIa) See footnote k for a discussion on the use of remdesivir.	

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

^a For a list of risk factors, see the CDC webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#).

^b If the patient is hospitalized for reasons other than COVID-19, the treatment duration for remdesivir is 3 days.

^c Corticosteroids that are prescribed for an underlying condition should be continued.

^d Evidence suggests that the benefit of remdesivir is greatest when the drug is given early in the course of COVID-19 (e.g., within 10 days of symptom onset).

^e Conventional oxygen refers to oxygen supplementation that is not HFNC oxygen, NIV, MV, or ECMO.

^f If these patients progress to requiring HFNC oxygen, NIV, MV, or ECMO, the full course of remdesivir should still be completed.

^g If none of the preferred or alternative options are available or feasible to use, the JAK inhibitor **PO tofacitinib (CIIa)** or the IL-6 inhibitor **IV sarilumab (CIIa)** can be used in combination with dexamethasone. Sarilumab is only commercially available as a SUBQ injection; see [Table 5e](#) for information regarding the preparation of an IV infusion using the SUBQ product.

^h Contraindications for the use of therapeutic anticoagulation in patients with COVID-19 include a PLT <50,000 cells/μL, Hgb <8 g/dL, the need for dual antiplatelet therapy, bleeding within the past 30 days that required an ED visit or hospitalization, a history of a bleeding disorder, or an inherited or active acquired bleeding disorder.

ⁱ Dexamethasone should be initiated immediately, without waiting until the second immunomodulator can be acquired. If other immunomodulators cannot be obtained or are contraindicated, use dexamethasone alone **(AI)**.

^j Examples of patients who may benefit most from remdesivir include patients who are immunocompromised **(BIIb)**; patients with evidence of ongoing viral replication

(e.g., those with a low Ct value, as measured by an RT-PCR result or with a positive rapid antigen test result) **(BIII)**; or patients who are within 10 days of symptom onset **(CIIa)**. For more information on using remdesivir in people with immunocompromising conditions, see [Special Considerations in People Who Are Immunocompromised](#).

^k There is insufficient evidence for the Panel to recommend either for or against the use of remdesivir in hospitalized patients with COVID-19 who require MV or ECMO. Some Panel members would add remdesivir to immunomodulator therapy in patients who have recently been placed on MV or ECMO, who are immunocompromised, who have evidence of ongoing viral replication, or who are within 10 days of symptom onset. See text for more information.

^l If PO baricitinib and IV tocilizumab are not available or feasible to use, **PO tofacitinib** can be used instead of PO baricitinib **(CIIa)**, and **IV sarilumab** can be used instead of IV tocilizumab **(CIIa)**.

Key: CDC = Centers for Disease Control and Prevention; Ct = cycle threshold; ECMO = extracorporeal membrane oxygenation; ED = emergency department; HFNC = high-flow nasal cannula; Hgb = hemoglobin; ICU = intensive care unit; IL = interleukin; IV = intravenous; JAK = Janus kinase; MV = mechanical ventilation; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel; PLT = platelet count; PO = oral; RT-PCR = reverse transcription polymerase chain reaction; SUBQ = subcutaneous; ULN = upper limit of normal

Two main processes are thought to drive the pathogenesis of COVID-19. Early in the clinical course, the disease is primarily driven by the replication of SARS-CoV-2. Later in the clinical course, the disease is driven by a dysregulated immune/inflammatory response to SARS-CoV-2 infection that may lead to further tissue damage and thrombosis. Based on this understanding, therapies that directly target SARS-CoV-2 are anticipated to have the greatest effect early in the course of the disease, whereas immunosuppressive, anti-inflammatory, and antithrombotic therapies are likely to be more beneficial after COVID-19 has progressed to stages characterized by hypoxemia, endothelial dysfunction, and immunothrombosis.

Currently, most people in the United States have some degree of immunity to SARS-CoV-2 due to COVID-19 vaccination or SARS-CoV-2 infection. The increase in population immunity and the change in variants have led to a decrease in the rate of severe disease caused by COVID-19. Because other co-existing diseases can cause hypoxemia in patients who test positive for SARS-CoV-2 infection, clinicians should perform the appropriate evaluations to rule out alternative diagnoses. However, the ongoing evolution of SARS-CoV-2 can lead to immune escape, allowing the virus to continue circulating in the community. Thus, COVID-19 remains a concern for public health.

Many of the patients who are hospitalized for COVID-19 are immunocompromised to some degree. For most hospitalized patients with severe or critical COVID-19 who are immunocompromised, the COVID-19 Treatment Guidelines Panel (the Panel) recommends using antiviral drugs and immunomodulatory therapies at the doses and durations recommended for the general population **(AIII)**. Some patients who are immunocompromised have prolonged COVID-19 symptoms and evidence of ongoing SARS-CoV-2 replication. See [Special Considerations in People Who Are Immunocompromised](#) for additional guidance on managing these patients.

Below is a summary of the rationale for the Panel's recommendations on the therapeutic management of hospitalized patients with COVID-19. For dosing information for each of the recommended drugs, see Table 2d below. For more information about these therapies and the evidence that supports the Panel's recommendations, please refer to the specific drug pages and clinical data tables.

Patients Who Are Hospitalized for Reasons Other Than COVID-19 and Who Do Not Require Supplemental Oxygen

Hospitalized patients with COVID-19 who do not require supplemental oxygen are a heterogeneous population. Some patients may be hospitalized for reasons other than COVID-19 but may also have mild to moderate COVID-19 (see [Clinical Spectrum of SARS-CoV-2 Infection](#)). In these cases, patients who

are at high risk of progressing to severe COVID-19 may benefit from receiving antiviral therapy.

Remdesivir is approved by the Food and Drug Administration (FDA) for the treatment of COVID-19 in adults who:

- Are hospitalized; *or*
- Are not hospitalized, have mild to moderate COVID-19, and are at high risk of progressing to severe COVID-19.

If the patient is hospitalized for reasons other than COVID-19, the treatment duration for remdesivir is 3 days.

Ritonavir-boosted nirmatrelvir (Paxlovid) is approved by the FDA and molnupiravir has an Emergency Use Authorization from the FDA for use in patients with mild to moderate COVID-19 who are at high risk of progressing to severe disease. These therapies can be used in hospitalized patients who qualify for therapy if they were admitted to the hospital for a diagnosis other than COVID-19. The Panel's recommendations for these patients are the same as those for nonhospitalized patients (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)).

Patients Who Are Hospitalized for COVID-19 and Who Do Not Require Supplemental Oxygen

Recommendations

- The Panel recommends using **remdesivir** for the treatment of COVID-19 in hospitalized patients who do not require supplemental oxygen and who are immunocompromised (**BI Ib**) and for other patients who are at high risk of progressing to severe disease (**BIII**).
- Remdesivir should be administered for 5 days or until hospital discharge, whichever comes first.

The rationale for using remdesivir in high-risk patients is based on several lines of evidence. In a trial conducted predominantly among hospitalized patients with COVID-19 who were not receiving supplemental oxygen at enrollment, a 5-day course of remdesivir was associated with greater clinical improvement when compared with standard of care.¹ Evidence from the PINETREE trial also suggests that early therapy reduces the risk of progression, although that study was performed in high-risk, unvaccinated, nonhospitalized patients with ≤ 7 days of symptoms who received a 3-day course of remdesivir.²

Other studies have not shown a clinical benefit of remdesivir in this group of patients. In the ACTT-1 trial, remdesivir showed no significant benefit in hospitalized patients with mild to moderate COVID-19; however, only 13% of the study population did not require supplemental oxygen.³ In the large Solidarity trial, the use of remdesivir was not associated with a survival benefit among the subset of hospitalized patients who did not require supplemental oxygen.⁴ See [Table 4a](#) for more information.

The aggregate data on using remdesivir to treat all high-risk patients with COVID-19 show a faster time to recovery in patients who received remdesivir but no clear evidence of a survival benefit. Therefore, the Panel recommends using **remdesivir** in hospitalized patients with COVID-19 who do not require supplemental oxygen and who are at high risk of progressing to severe disease (**BIII**).

In a large, retrospective cohort study of hospitalized patients with COVID-19 who were immunocompromised ($n = 28,338$), patients who received remdesivir had a lower risk of mortality than those who did not receive remdesivir.⁵ Forty percent of patients in this cohort were not receiving supplemental oxygen at baseline; mortality was reduced in this subset of patients. Therefore, the Panel recommends using **remdesivir** in hospitalized patients with COVID-19 who do not require supplemental oxygen and who are immunocompromised (**BI Ib**).

For information on medical conditions that confer high risk, see the Centers for Disease Control and Prevention webpage [People With Certain Medical Conditions](#).

Recommendation

- The Panel **recommends against** the use of **dexamethasone (AIIa)** or other systemic corticosteroids (**AIII**) for the treatment of COVID-19 in patients who do not require supplemental oxygen.

In the RECOVERY trial, no survival benefit was observed for dexamethasone among the subset of patients with COVID-19 who did not require supplemental oxygen at enrollment.⁶ In an observational cohort study of U.S. veterans, the use of dexamethasone was associated with higher mortality in hospitalized patients with COVID-19 who did not require supplemental oxygen.⁷

There are insufficient data to inform the use of other systemic corticosteroids in hospitalized patients with COVID-19 who do not require supplemental oxygen. See [Table 5a](#) for more information about the clinical trials that have evaluated the use of these drugs in patients with COVID-19. Patients who are receiving corticosteroid treatment for an underlying condition should continue to receive corticosteroids.

Patients Who Require Conventional Oxygen

Patients with COVID-19 who require conventional oxygen (i.e., those who do not require high-flow nasal cannula [HFNC] oxygen, noninvasive ventilation [NIV], or mechanical ventilation) are a heterogeneous population. Although all patients who require oxygen are considered to have severe disease, some of these patients will improve after a short period with or without treatment; others will develop progressive disease. There is no consensus on which clinical or laboratory parameters should be used to determine a patient's risk of progression and guide therapy.

Recommendation

- For patients with COVID-19 who only require minimal conventional oxygen, the Panel recommends using **remdesivir** without dexamethasone (**BIIa**).

In a subgroup analysis during the ACTT-1 trial, remdesivir significantly reduced the time to clinical recovery and significantly reduced mortality among the subset of patients who were receiving conventional oxygen at enrollment.³ Evidence from ACTT-1 and a pooled analysis of individual data from 9 randomized controlled trials⁸ suggest that remdesivir will have its greatest benefit when administered early in the clinical course of COVID-19 (e.g., within 10 days of symptom onset). See [Table 4a](#) for more information.

Recommendations

- For most patients with COVID-19 who require conventional oxygen, the Panel recommends using **dexamethasone plus remdesivir (BIIa)**.
- If dexamethasone is not available, an equivalent dose of another corticosteroid (e.g., **prednisone**, **methylprednisolone**, **hydrocortisone**) may be used (**BIII**).

The results of several studies suggest that the use of remdesivir plus dexamethasone improves clinical outcomes among hospitalized patients with COVID-19. In the CATCO trial, in which 87% of patients received corticosteroids and 54% were on conventional oxygen, remdesivir significantly reduced the need for mechanical ventilation among the subset of patients who did not require mechanical ventilation at enrollment when compared with standard of care.⁹ In the Solidarity trial, in which approximately two-thirds of the patients received corticosteroids, remdesivir significantly reduced mortality among the large subset of patients (n > 7,000) who were receiving conventional or HFNC oxygen at enrollment.⁴ See [Table 4a](#) for more information.

An individual patient–level meta-analysis of 8 clinical trials examined the efficacy of using remdesivir in hospitalized patients with COVID-19.⁸ This meta-analysis found that remdesivir significantly reduced the number of patients who required mechanical ventilation or who died by Day 28 in the combined subgroups of patients who did not require oxygen or who were receiving conventional oxygen at baseline. However, the effect of remdesivir was not evaluated separately in the subgroup of patients who were receiving conventional oxygen at enrollment.

Recommendation

- If remdesivir is not available, the Panel recommends using **dexamethasone alone** in patients with COVID-19 who require conventional oxygen (**BI**).

In the RECOVERY trial, the use of dexamethasone 6 mg once daily for 10 days or until hospital discharge significantly reduced mortality among the subset of patients who were receiving oxygen (defined as receiving oxygen supplementation but not mechanical ventilation or extracorporeal membrane oxygenation [ECMO]) at enrollment.⁶ Remdesivir was administered to <1% of the study participants. Results for patients who were only receiving conventional oxygen at enrollment were not available. See [Table 5a](#) for more information.

Recommendations

- For patients with COVID-19 who have rapidly increasing oxygen needs and systemic inflammation, the Panel recommends adding 1 of the following immunomodulators to dexamethasone:
 - *Preferred Second Immunomodulators*
 - **Oral (PO) baricitinib (BIIa)**
 - **Intravenous (IV) tocilizumab (BIIa)**
 - *Alternative Second Immunomodulators (Listed in Alphabetical Order)*
 - **IV abatacept (CIIa)**
 - **IV infliximab (CIIa)**

If none of these options are available or feasible to use, the Janus kinase (JAK) inhibitor **PO tofacitinib (CIIa)** or the interleukin (IL)-6 inhibitor **IV sarilumab (CIIa)** can be used in combination with dexamethasone. Sarilumab is only commercially available as a subcutaneous (SUBQ) injection; see [Table 5e](#) for information regarding the preparation of an IV infusion using the SUBQ product.

Several large randomized controlled trials have evaluated the use of dexamethasone in combination with a second immunomodulator, including:

- Abatacept, a cytotoxic T-lymphocyte–associated antigen 4 agonist¹⁰
- Baricitinib, a JAK inhibitor¹¹⁻¹³
- Infliximab, a tumor necrosis factor–alpha inhibitor¹⁰
- Tocilizumab, an IL-6 inhibitor¹⁴⁻¹⁶

These studies included patients who required conventional oxygen only, as well as those with increasing oxygen needs and/or elevated levels of inflammatory markers. Subgroup analyses in these trials have not clearly defined which patients in this heterogeneous group are most likely to benefit from adding a second immunomodulator to corticosteroid therapy. The study endpoints for these trials included progression to more severe disease, the need for mechanical ventilation, and death. Nonetheless, some trials suggest that adding a second immunomodulator provides benefits to patients who require conventional oxygen,

especially those with rapidly increasing oxygen requirements and systemic inflammation.

The Panel recommends either baricitinib or tocilizumab as the preferred second immunomodulator because both are approved by the FDA for the treatment of COVID-19, and data from multiple clinical trials have demonstrated that these agents provide a clinical benefit in patients with COVID-19 who require conventional oxygen.^{6,11-16} There is also more clinical experience with the use of these 2 agents in this setting than other potential treatment options.

The ACTIV-1 immunomodulator trial was a double-blind, multi-arm, randomized trial in moderately to severely ill adults who were hospitalized with COVID-19.¹⁰ The trial separately evaluated treatment with abatacept, cenicriviroc, and infliximab versus placebo. All arms received standard care, and the separate analyses included data from a shared placebo arm. The primary endpoint was time to recovery by Day 28. Key secondary endpoints included clinical status at Day 14 and mortality through Days 28 and 60. The majority of patients received corticosteroids (>89%) and remdesivir (>93%).

None of the study drugs had a significant effect on the time to recovery. Mortality by Day 28 was lower among patients in the abatacept and infliximab arms than among those in the shared placebo arm. Based on the results of this trial, abatacept or infliximab may be considered alternatives to baricitinib or tocilizumab. There are no studies that directly compare the use of abatacept or infliximab to the use of baricitinib or tocilizumab in people with COVID-19.

When baricitinib, tocilizumab, abatacept, or infliximab are not available or feasible to use, the JAK inhibitor tofacitinib or the IL-6 inhibitor sarilumab may be used as alternative agents. Tofacitinib decreased the risk for respiratory failure or death among hospitalized patients with COVID-19 in the STOP-COVID trial,¹⁷ and sarilumab reduced mortality and the duration of organ support to the same degree as tocilizumab in the REMAP-CAP trial.^{14,15}

Use of Anticoagulants

- The Panel recommends using a **therapeutic dose of heparin** for nonpregnant patients with D-dimer levels above the upper limit of normal who require conventional oxygen and who do not have an increased bleeding risk (**CIIa**).
- The Panel recommends the use of a **prophylactic dose of heparin** for patients who do not meet the criteria for receiving therapeutic heparin or are not receiving a therapeutic dose of heparin for other reasons, unless a contraindication exists (**AI**).
- The Panel recommends the use of a prophylactic dose of anticoagulation for pregnant patients who are hospitalized for manifestations of COVID-19, unless a contraindication exists (**BIII**).

The Panel's recommendations for the use of heparin are based on data from 3 open-label randomized controlled trials that compared the use of therapeutic doses of heparin to prophylactic or intermediate doses of heparin in hospitalized patients who did not require intensive care unit (ICU)-level care. Pooled data from the ATTACC/ACTIV-4a/REMAP-CAP trials reported more organ support-free days (i.e., days alive and free of ICU-based organ support) for patients in the therapeutic heparin arm than for those in the usual care arm, but there was no difference between the arms in in-hospital mortality or length of hospitalization.¹⁸ The RAPID trial compared a therapeutic dose of heparin to a prophylactic dose in hospitalized patients with moderate COVID-19. There was no statistically significant difference between the arms in the occurrence of the primary endpoint (which was a composite of ICU admission, NIV or mechanical ventilation, or death by Day 28), but the therapeutic dose of heparin reduced the risk of all-cause death.¹⁹ In the HEP-COVID trial, venous thromboembolism (VTE), arterial thromboembolism, and death by Day 32 occurred significantly less frequently in patients who received a therapeutic dose of heparin than in those who received a prophylactic dose of heparin, but there was no difference in mortality

by Day 32 between the arms.²⁰

Patients Who Require High-Flow Nasal Cannula Oxygen or Noninvasive Ventilation

In these patients, systemic inflammation contributes to hypoxemia, and thus these patients may benefit from receiving a second immunomodulator in addition to dexamethasone. There is no consensus on which clinical or laboratory parameters reliably predict the risk of progression to mechanical ventilation or death.

The available evidence suggests that the benefits of adding baricitinib or tocilizumab to dexamethasone treatment outweigh the potential risks in patients with COVID-19 who require HFNC oxygen or NIV. Although the combination of dexamethasone and secondary immunomodulating medications may increase the risk of opportunistic infections or the risk of reactivating latent infections, there are insufficient data to make recommendations about initiating prophylaxis against these infections.

Recommendations

- **Dexamethasone** should be administered to all patients with COVID-19 who require HFNC oxygen or NIV (AI).
- If not already initiated, promptly add 1 of the following immunomodulators to dexamethasone:
 - *Preferred Second Immunomodulator*
 - **PO baricitinib (AI)**
 - *Preferred Alternative Second Immunomodulator*
 - **IV tocilizumab (BIIa)**
 - *Additional Alternative Second Immunomodulators (Listed in Alphabetical Order)*
 - **IV abatacept (CIIa)**
 - **IV infliximab (CIIa)**

If none of these options are available or feasible to use, **PO tofacitinib (CIIa)** or **IV sarilumab (CIIa)** can be used in combination with dexamethasone. Sarilumab is only commercially available as a SUBQ injection; see [Table 5e](#) for information regarding the preparation of an IV infusion using the SUBQ product.

Clinicians should make a significant effort to obtain and administer 1 of the recommended second immunomodulators. However, dexamethasone should be initiated immediately, without waiting until the second immunomodulator can be acquired. Dexamethasone was used as a single-agent immunomodulatory strategy in the RECOVERY trial and demonstrated a survival benefit among patients who required supplemental oxygen.⁶ In this trial, the treatment effect for dexamethasone was not evaluated separately for those who required conventional oxygen and those who required HFNC oxygen or NIV.

Several large randomized controlled trials have demonstrated that patients with COVID-19 who require HFNC oxygen or NIV benefit from combining dexamethasone with an additional immunomodulator, such as a JAK inhibitor or an IL-6 inhibitor. The quality of the evidence and the totality of the data support a stronger recommendation for baricitinib than for tocilizumab.

Two large randomized controlled trials (RECOVERY and COV-BARRIER) both reported a survival benefit among hospitalized patients with COVID-19 who required HFNC oxygen or NIV and who received baricitinib plus dexamethasone.^{12,13} Data from the ACTT-2¹¹ and ACTT-4²¹ trials support the overall safety of using baricitinib in combination with remdesivir and the potential for a clinical benefit of this combination, but neither trial studied baricitinib in combination with dexamethasone as the standard of care. A retrospective analysis of data from 11 U.S. health systems suggests that the use of

baricitinib may be associated with fewer adverse effects than tocilizumab, including fewer secondary infections, thrombotic events, and cases of acute liver injury.²²

The use of tocilizumab in combination with corticosteroids reduced in-hospital mortality in patients with rapid respiratory decompensation who were admitted to the ICU in the REMAP-CAP trial.¹⁴ Similar results were reported during the RECOVERY trial, although patients were only randomized into the tocilizumab arm if they had an oxygen saturation <92% on room air and C-reactive protein levels ≥ 75 mg/L.¹⁶ Both REMAP-CAP and RECOVERY evaluated the efficacy of adding tocilizumab to standard care; in both cases, standard care included dexamethasone therapy. Other randomized trials that have evaluated the use of tocilizumab have demonstrated mixed results, including a lack of benefit when tocilizumab was administered without dexamethasone as part of standard care.²³⁻²⁶

In the ACTIV-1 trial, which evaluated the use of abatacept, cenicriviroc, and infliximab in hospitalized patients with COVID-19, neither abatacept nor infliximab demonstrated a significant effect on the primary endpoint of time to recovery.¹⁰ In the subgroup of patients who received HFNC oxygen or NIV, mortality at Day 28 (a secondary outcome) was lower in both the abatacept and the infliximab arms than in the shared placebo arm.

Combinations of 3 immunomodulators (e.g., dexamethasone plus baricitinib plus tocilizumab) have not been studied in clinical trials. Although some patients in the baricitinib arm of the RECOVERY trial also received tocilizumab, data from the study are insufficient to issue a recommendation.¹³ When both agents are used, there is a potential for greater risk of secondary infections.

The clinical trial data cited above informed the Panel's recommendations for adding a second immunomodulator to dexamethasone in hospitalized patients who require HFNC oxygen or NIV. After reviewing these clinical trial results, the Panel recommends baricitinib over tocilizumab as the second immunomodulator. See [Table 5c](#) and [Table 5d](#) for more information. Because the evidence for the use of either abatacept or infliximab in people with COVID-19 is derived from a single study while multiple trials have demonstrated a beneficial effect of using baricitinib or tocilizumab, the Panel recommends using abatacept or infliximab only when baricitinib and tocilizumab are not available or their use is contraindicated.

Recommendations

- For certain hospitalized patients who require HFNC oxygen or NIV, the Panel recommends adding **remdesivir** to 1 of the recommended immunomodulator combinations. Examples of patients who may benefit most from adding remdesivir include:
 - Patients who are immunocompromised (**BIIB**);
 - Patients with evidence of ongoing viral replication (e.g., those with a low cycle threshold [Ct] value, as measured by a reverse transcription polymerase chain reaction [RT-PCR] result or with a positive rapid antigen test result) (**BIII**); *or*
 - Patients who are within 10 days of symptom onset (**CIIa**).

Clinical trial data have not clearly established that remdesivir reduces the time to recovery or improves survival in patients who require HFNC oxygen or NIV. However, because clinical trials have found that remdesivir prevents clinical progression in patients who are not on mechanical ventilation, some patients receiving HFNC oxygen or NIV might benefit from receiving remdesivir. In the Solidarity trial, remdesivir had a modest but statistically significant effect on reducing the risk of death or progression to ventilation in patients who were receiving oxygen but who were not ventilated at baseline.⁴

However, these effects could not be evaluated separately for patients who required conventional oxygen supplementation and those who required HFNC oxygen or NIV.⁴ In the CATCO trial, among the patients

who were not receiving mechanical ventilation at baseline, 8% of patients who received remdesivir required mechanical ventilation compared to 15% of those who received standard of care (relative risk 0.53; 95% CI, 0.38–0.75).⁹ See [Table 4a](#) for more information.

The Panel's rationale for recommending remdesivir for certain patients who require HFNC oxygen or NIV is discussed below. This discussion includes examples of patients who may benefit most from receiving remdesivir. In addition, clinicians may extend the course of remdesivir beyond 5 days in this population based on clinical response.

Patients Who Are Immunocompromised

People who are immunocompromised already have difficulty achieving viral clearance. The use of immunomodulators to treat COVID-19 may further impair this process. Because SARS-CoV-2 replication may be prolonged in these patients, remdesivir may help enhance viral clearance and improve outcomes. In a large, retrospective study of a cohort of patients who were immunocompromised, patients who received remdesivir had a lower risk of mortality than those who did not receive remdesivir; however, only 19% of the patients in the study were receiving HFNC oxygen or NIV.⁵ For more information, see [Special Considerations in People Who Are Immunocompromised](#).

Patients With Evidence of Ongoing Viral Replication

Hospitalized patients who require HFNC oxygen or NIV are routinely treated with 2 immunomodulators to prevent or mitigate inflammation-mediated injury. These treatments may impair the patient's ability to achieve viral clearance; thus, directly treating the virus with remdesivir may theoretically help improve outcomes. Substantial evidence from studies of other viral diseases supports the benefits of reducing the viral burden. Ct values can be obtained from some SARS-CoV-2 RT-PCR assays, and these values may be used as a proxy for the level of ongoing viral replication (low Ct values correspond to higher viral loads). While this information is not available on all RT-PCR platforms, Ct values may be helpful in informing decisions regarding the use of remdesivir. Positive rapid antigen test results are also consistent with higher viral loads.²⁷

Patients Who Are Within 10 Days of Symptom Onset

Active viral replication occurs early in the course of SARS-CoV-2 infection. Evidence from the ACTT-1 and PINETREE trials suggests that remdesivir will have the greatest benefit when administered early in the clinical course of COVID-19.^{2,3} In the ACTT-1 trial, remdesivir demonstrated a greater benefit in patients who were enrolled within 10 days of symptom onset than in those who were enrolled later in the disease course.

Use of Anticoagulants

- The Panel recommends using a **prophylactic dose of heparin** as VTE prophylaxis, unless a contraindication exists (**AI**); (**BIII**) for pregnant patients.
- The Panel **recommends against** the use of a therapeutic dose of anticoagulation for VTE prophylaxis (**BI**).
- For patients who start on a therapeutic dose of heparin in a non-ICU setting due to COVID-19 and then transfer to the ICU, the Panel recommends switching from the therapeutic dose to a **prophylactic dose of heparin**, unless VTE is confirmed (**BIII**).

The multiplatform randomized controlled trial REMAP-CAP/ACTIV-4a/ATTACC compared the effectiveness of a therapeutic dose of heparin to standard care in critically ill patients with COVID-19.²⁸ The study did not show an increase in the number of organ support-free days or the probability of survival to hospital discharge among patients who received therapeutic doses of anticoagulation. See [Antithrombotic Therapy in Patients With COVID-19](#) for more information.

Patients Who Require Mechanical Ventilation or Extracorporeal Membrane Oxygenation

Recommendations

- **Dexamethasone** should be administered to all patients with COVID-19 who require mechanical ventilation or ECMO (AI).
- If the patient has not already received a second immunomodulator in addition to dexamethasone, promptly add 1 of the following (listed in alphabetical order):
 - **PO baricitinib (BIIa)**
 - **IV tocilizumab (BIIa)**

Dexamethasone was shown to reduce mortality in critically ill patients with COVID-19 in a meta-analysis that aggregated 7 randomized trials and included data on 1,703 critically ill patients.²⁹ The largest trial included in the meta-analysis was the RECOVERY trial, which had a subgroup of patients who were receiving mechanical ventilation (see [Systemic Corticosteroids](#) and [Table 5a](#)).⁶ Subsequent studies of immunomodulator therapy suggest that using a second immunomodulator in combination with dexamethasone is more effective than dexamethasone alone in patients with COVID-19 who require mechanical ventilation or ECMO.^{14,16,30}

Clinical trials that have evaluated combining IL-6 inhibitors or JAK inhibitors with corticosteroids for the treatment of patients with COVID-19 provide the most robust evidence for the Panel's recommendations.

Clinical trials of tocilizumab have reported an overall survival benefit in patients with hypoxemia and signs of systemic inflammation (RECOVERY)¹⁶ and in patients who are critically ill and require organ support (REMAP-CAP).¹⁴ Although these studies included patients who were receiving mechanical ventilation at randomization, the studies were not specifically powered to assess the effectiveness of IL-6 inhibitors in these patients. Other studies of tocilizumab in critically ill patients did not find a survival benefit, although the time between initiation of organ support in the ICU and study enrollment differed across the studies (see [Table 5c](#)).^{23,26} The use of corticosteroids also varied across the studies.

An extension of the COV-BARRIER trial compared the efficacy of baricitinib to placebo in 101 critically ill patients with COVID-19.³⁰ The study reported significant reductions in mortality (relative reduction of 46% at 28 days and 44% at 60 days) and no major adverse events among patients who received baricitinib. Systematic reviews of JAK inhibitors confirm the efficacy of using baricitinib in hospitalized patients with COVID-19 who require oxygen support. There is a lower certainty of evidence for patients who were receiving mechanical ventilation or ECMO, and baricitinib may have modestly attenuated efficacy in this group.³¹ Baricitinib or tocilizumab should only be administered in combination with dexamethasone or another corticosteroid.

In the ACTIV-1 trial, the use of abatacept, cenicriviroc, or infliximab did not reduce the time to recovery or mortality in patients with COVID-19 who required mechanical ventilation or ECMO. Therefore, these immunomodulators are not recommended for these patients.

Considerations for the Use of Remdesivir

There is insufficient evidence for the Panel to recommend either for or against the use of remdesivir in hospitalized patients with COVID-19 who require mechanical ventilation or ECMO. For patients who progress to requiring mechanical ventilation or ECMO after they initiate remdesivir, the Panel suggests continuing remdesivir until the treatment course is completed. Remdesivir is most effective against COVID-19 in patients who are earlier in the course of the disease.

In the ACTT-1 trial, there was no difference between the remdesivir and placebo arms in the time to recovery among patients with COVID-19 who were receiving mechanical ventilation or ECMO; however, very few patients received corticosteroids in this trial.³ The Solidarity trial reported no benefit of using remdesivir in hospitalized patients with COVID-19 who were already on mechanical ventilation (mortality rate ratio 1.13; $P = 0.32$).⁴ It is worth noting that only a few patients required mechanical ventilation in these 2 randomized trials, and there was substantial variation in the timing of remdesivir initiation. In contrast, in a propensity score-matched retrospective observational study, the use of remdesivir was associated with a reduction in mortality at 14 and 28 days among patients who required mechanical ventilation or ECMO within 48 hours of hospital admission for COVID-19 pneumonia.³² In this study, over 97% of patients received corticosteroids.

Given the results of the randomized clinical trials and the limitations of observational data, the Panel cannot recommend either for or against the use of remdesivir in this group of patients. However, some Panel members would add remdesivir to immunomodulator therapy in patients with COVID-19 who have recently been placed on mechanical ventilation or ECMO. In this case, the rationale to add remdesivir is based on the observed clinical benefit of remdesivir during earlier stages of infection, the results of the observational study, and the demonstrated safety of remdesivir. In addition, some Panel members would add remdesivir to the regimen for patients with COVID-19 who are immunocompromised, who have evidence for ongoing viral replication, or who are within 10 days of symptom onset.

Use of Anticoagulants

- The Panel recommends using a **prophylactic dose of heparin** as VTE prophylaxis, unless a contraindication exists (**AI**); (**BIII**) for pregnant patients.
- For patients who start on a therapeutic dose of heparin in a non-ICU setting due to COVID-19 and then transfer to the ICU, the Panel recommends switching from the therapeutic dose to a **prophylactic dose of heparin**, unless there is another indication for therapeutic anticoagulation (**BIII**).
- The Panel **recommends against** the use of a therapeutic dose of anticoagulation for VTE prophylaxis (**BI**).

Patients who required mechanical ventilation or ECMO were included in the multiplatform REMAP-CAP/ACTIV-4a/ATTACC trial that studied therapeutic doses of heparin.²⁸ Because these studies reported no benefits of using therapeutic doses of heparin, the recommendations for using prophylactic doses of heparin in hospitalized patients who require mechanical ventilation or ECMO are the same as those for patients who require HFNC oxygen or NIV.

Table 2d. Dosing Regimens for the Drugs Recommended in Table 2c

The drugs in this table are listed in alphabetical order.

Drug Name	Dosing Regimen	Comments
Abatacept	Abatacept 10 mg/kg actual body weight (up to 1,000 mg) administered as a single IV dose	• No adjustment based on eGFR
Baricitinib	BAR dose is dependent on eGFR; duration of therapy is up to 14 days or until hospital discharge, whichever comes first.	• eGFR ≥ 60 mL/min/1.73 m²: BAR 4 mg PO once daily • eGFR 30 to <60 mL/min/1.73 m²: BAR 2 mg PO once daily • eGFR 15 to <30 mL/min/1.73 m²: BAR 1 mg PO once daily • eGFR <15 mL/min/1.73 m²: Not recommended.
Dexamethasone	DEX 6 mg IV or PO once daily for up to 10 days or until hospital discharge, whichever comes first	• If DEX is not available, an equivalent dose of another corticosteroid may be used. • For more information, see Systemic Corticosteroids.

Drug Name	Dosing Regimen	Comments
Heparin	Therapeutic dose of SUBQ LMWH or IV UFH	• Administer for 14 days or until hospital discharge (whichever comes first) unless there is a diagnosis of VTE or another indication for therapeutic anticoagulation.
	Prophylactic dose of SUBQ LMWH or SUBQ UFH	• Administer for the duration of the hospital stay.
Infliximab	Infliximab 5 mg/kg actual body weight administered as a single IV dose	• No adjustment based on eGFR
Remdesivir	RDV 200 mg IV once, then RDV 100 mg IV once daily for 4 days or until hospital discharge, whichever comes first	• If the patient is hospitalized for reasons other than COVID-19, the treatment duration is 3 days. For more information, see Therapeutic Management of Nonhospitalized Adults With COVID-19. • If the patient progresses to more severe illness, complete the course of RDV. • For a discussion on using RDV in patients with renal insufficiency, see Remdesivir.
Sarilumab	Use the single-dose, prefilled syringe (not the prefilled pen) for SUBQ injection. Reconstitute sarilumab 400 mg in 100 cc 0.9% NaCl and administer as an IV infusion over 1 hour.	• In the United States, the currently approved route of administration for sarilumab is SUBQ injection. In the REMAP-CAP trial, the SUBQ formulation was used to prepare the IV infusion.
Tocilizumab	Tocilizumab 8 mg/kg actual body weight (up to 800 mg) administered as a single IV dose	• In clinical trials, a third of the participants received a second dose of tocilizumab 8 hours after the first dose if no clinical improvement was observed.
Tofacitinib	Tofacitinib 10 mg PO twice daily for up to 14 days or until hospital discharge, whichever comes first	• eGFR <60 mL/min/1.73 m ² : tofacitinib 5 mg PO twice daily

Key: BAR = baricitinib; DEX = dexamethasone; eGFR = estimated glomerular filtration rate; IV = intravenous; LMWH = low-molecular-weight heparin; NaCl = sodium chloride; PO = oral; RDV = remdesivir; SUBQ = subcutaneous; UFH = unfractionated heparin; VTE = venous thromboembolism

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Clinical Management of Children Summary

Last Updated: February 29, 2024

Data from the Centers for Disease Control and Prevention demonstrate that severe disease and death due to COVID-19 occur less often in children than in adults.¹⁻⁴ However, weekly hospitalization rates for children aged <6 months are high, exceeded only by the rates for adults aged ≥75 years.⁵ The overall incidence of SARS-CoV-2 infection and, by extension, COVID-19–related hospitalizations among children increased substantially with the emergence of newer variants, particularly the Omicron variant and its subvariants.^{6,7} According to the Centers for Disease Control and Prevention, by December 2022, an estimated 96% of children and adolescents had serologic evidence of prior SARS-CoV-2 infection.⁸ The high infection rates among children makes the overall burden substantial despite the low rate of severe outcomes.⁹

Data on the pathogenesis and clinical spectrum of SARS-CoV-2 infection in children are still limited compared to the data for adults. Although only a small percentage of children with COVID-19 will require medical attention, the percentage of intensive care unit admissions among hospitalized children is comparable to that for hospitalized adults with COVID-19.^{6,10-18}

Observational studies and meta-analyses have found that children with certain comorbidities are at increased risk of severe COVID-19. These comorbidities include cardiac disease, neurologic disorders, prematurity (in young infants), diabetes, obesity (particularly severe obesity), chronic lung disease, feeding tube dependence, and immunocompromised status.¹⁹⁻²² Demographic factors, such as age (<1 year and 10–14 years)²³ and non-White race/ethnicity,^{15,24-26} have also been associated with severe disease. However, many studies did not assess the relative severity of underlying medical conditions in children with severe COVID-19.

In general, COVID-19 has similar clinical manifestations and disease stages in children and adults, including an early phase driven by viral replication and a late phase that appears to be driven by a dysregulated immune/inflammatory response to SARS-CoV-2 that leads to tissue damage. Respiratory complications in young children that can occur during the early clinical phase include croup and bronchiolitis. In addition, a small number of children who have recovered from acute SARS-CoV-2 infection develop multisystem inflammatory syndrome in children (MIS-C) 2 to 6 weeks after infection. MIS-C is a postinfectious inflammatory condition that can lead to severe organ dysfunction.

Published guidance on the treatment of COVID-19 in children has been extrapolated mostly from recommendations for adults with COVID-19, recommendations for children with other viral infections, and expert opinion.²⁷⁻²⁹ Applying adult data from COVID-19 trials to children is a unique challenge because most children experience a mild course of illness with COVID-19.³⁰ Relative to adults, children with COVID-19 have substantially lower mortality and less need for hospitalization.^{10,11,31} Because of these differences in epidemiology and disease severity, the effect sizes of treatments for children are likely to be smaller than those observed in adults. Therefore, to produce a beneficial outcome in children, the number needed to treat is higher. Collectively, these factors influence the risk versus benefit balance for pharmacologic therapies in children.

In the absence of sufficient clinical trial data on the treatment of children with COVID-19, the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for the therapeutic management of hospitalized children are based largely on safety and efficacy data from clinical trials in adults, the child's risk of disease progression, and expert opinion.

In general, data from clinical trials in adults are most applicable to the treatment of older children

with severe COVID-19 and predominantly lower respiratory tract disease. Using data from clinical trials in adults to develop recommendations for children with SARS-CoV-2 infection who have clinical syndromes associated with other respiratory viruses (e.g., bronchiolitis, croup, asthma) is a challenge. No evidence suggests that these syndromes should be managed differently when caused by SARS-CoV-2 infection. Clinical judgment is needed when applying recommendations for treatment in adults to children, particularly young children, with these clinical syndromes.

The Panel's recommendations for the management of children with COVID-19 or MIS-C are summarized in the tables below. Table 3a provides recommendations for the therapeutic management of nonhospitalized children with COVID-19. The Panel's recommendations are stratified by age (per the Food and Drug Administration Emergency Use Authorizations) and risk level. See [Therapeutic Management of Nonhospitalized Children With COVID-19](#) for more information. Table 3b includes a framework to help clinicians evaluate the risk for severe COVID-19 based on patient conditions and COVID-19 vaccination status.

The recommendations for hospitalized children in Table 3c are stratified by disease severity. See [Therapeutic Management of Hospitalized Children With COVID-19](#) for more information. Table 3d summarizes the recommendations for the therapeutic management of MIS-C. For the rationale behind these recommendations and supporting data, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Table 3a. Therapeutic Management of Nonhospitalized Children With COVID-19

Risk of Severe COVID-19	Panel's Recommendations	
	Aged 12–17 Years	Aged <12 Years
Symptomatic, Regardless of Risk Factors	• Provide supportive care (AIII).	• Provide supportive care (AIII).
High Risk^{a,b}	• Use 1 of the following options (listed in order of preference):^c • Ritonavir-boosted nirmatrelvir (Paxlovid) within 5 days of symptom onset (BIII) • Remdesivir within 7 days of symptom onset (CIII)	• Ritonavir-boosted nirmatrelvir is not authorized by the FDA for use in children aged <12 years. • There is insufficient evidence to recommend either for or against the routine use of remdesivir. Consider treatment based on age and other risk factors.
Intermediate Risk^{b,d}	• There is insufficient evidence to recommend either for or against the use of any antiviral therapy. Consider treatment based on age and other risk factors.	• There is insufficient evidence to recommend either for or against the routine use of remdesivir.
Low Risk^{b,e}	• Manage with supportive care alone (AIII).	• Manage with supportive care alone (AIII).
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.		

^a Molnupiravir is not authorized by the FDA for use in children aged <18 years and should not be used.

^b See Table 3b for the Panel's framework for assessing the risk of progression to severe COVID-19 based on patient conditions and COVID-19 vaccination status.

^c Initiate treatment as soon as possible after symptom onset.

^d The relative risk of severe COVID-19 for intermediate-risk patients is lower than the risk for high-risk patients but higher than the risk for low-risk patients.

^e Low-risk patients include those with comorbid conditions that have a weak or unknown association with severe COVID-19.

Patients with no comorbidities are included in this group.

Key: FDA = Food and Drug Administration; the Panel = the COVID-19 Treatment Guidelines Panel

Table 3b. The Panel’s Framework for Assessing the Risk of Progression to Severe COVID-19 Based on Patient Conditions and COVID-19 Vaccination Status

Conditions	Risk Level by Vaccination Status	
	Not Up to Date ^{a,b}	Up to Date ^a
Strong or Consistent Association With Progression to Severe COVID-19		
• Moderately or severely immunocompromised (see Special Considerations in People Who Are Immunocompromised)	High	
• Obesity (BMI ≥95th percentile for age), especially severe obesity (BMI ≥120% of 95th percentile for age)^c • Medical complexity with dependence on respiratory technology^d • Severe neurologic, genetic, metabolic, or other disability that results in impaired airway clearance or limitations in self-care or activities of daily living • Severe asthma or other severe chronic lung disease requiring ≥2 inhaled or ≥1 systemic medications daily • Severe congenital or acquired cardiac disease • Multiple moderate to severe chronic diseases • Pregnancy	High	
Moderate or Inconsistent Association With Progression to Severe COVID-19		
• Aged <1 year • Prematurity in children aged ≤2 years • Sickle cell disease • Diabetes mellitus (poorly controlled) • Chronic kidney disease • Nonsevere cardiac, neurologic, or metabolic disease^e	Intermediate	
Weak or Unknown Association With Progression to Severe COVID-19		
• Mild asthma • Overweight • Diabetes mellitus (well controlled)	Low	

^a See the current [COVID-19 vaccination schedule](#) from the CDC.

^b Recent SARS-CoV-2 infection (i.e., within 3–6 months) may confer substantial immunity against closely related variants. A patient’s recent infection history should be factored into the risk assessment.

^c The degree of risk conferred by obesity in younger children is less clear than it is in older adolescents.

^d This includes patients with a tracheostomy and those who require NIV.

^e The data for this group are particularly limited.

Key: BMI = body mass index; CDC = Centers for Disease Control and Prevention; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel

Table 3c. Therapeutic Management of Hospitalized Children With COVID-19

Disease Severity	Panel's Recommendations
Hospitalized for COVID-19	For children aged ≥ 12 years admitted for COVID-19, use prophylactic anticoagulation unless contraindicated (BIII). ^a
Does Not Require Supplemental Oxygen	For children admitted for COVID-19 who are at the highest risk of progression to severe COVID-19 ^b (especially those who are severely immunocompromised), consider using remdesivir ^c for children aged 12–17 years (CIII). There is insufficient evidence for using remdesivir in children aged 28 days to <12 years and weighing ≥ 3 kg.
	For children admitted for reasons other than COVID-19 who have mild to moderate COVID-19 and are at the highest risk of progression, ^b refer to Therapeutic Management of Nonhospitalized Children With COVID-19 .
Requires Conventional Oxygen^d	Use 1 of the following options: • Remdesivir^c (BIII) • Dexamethasone plus remdesivir^c for children with increasing oxygen needs, particularly adolescents (BIII)
Requires Oxygen Through High-Flow Device or NIV^e	Use 1 of the following options: • Dexamethasone (BIII) • Dexamethasone plus remdesivir^c (BIII)
	For children who do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, baricitinib ^f or tocilizumab can be considered for children aged 12–17 years (BIII) and for children aged 2–11 years (CIII).
Requires MV or ECMO^g	Dexamethasone ^g (AIII)
	For children who do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, baricitinib ^f or tocilizumab may be considered for children aged 12–17 years (BIII) and for children aged 2–11 years (CIII).
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.	

^a Weighing the risk factors for thrombosis and bleeding, some Panel members would use prophylactic anticoagulation for children aged <12 years who are hospitalized for COVID-19.

^b See [Therapeutic Management of Nonhospitalized Children With COVID-19](#) for a list of conditions that will put children at highest risk for progression to severe COVID-19.

^c The clinical benefit of remdesivir is greatest if it is initiated within 10 days of symptom onset. Remdesivir should be given for 5 days or until hospital discharge, whichever comes first. Examples of patients who may benefit most from adding remdesivir >10 days from symptom onset include patients who are severely immunocompromised, particularly if they have evidence of ongoing viral replication (e.g., those with a low Ct value, as measured by an RT-PCR result or with a positive rapid antigen test result).

^d Conventional oxygen refers to oxygen supplementation that is not high-flow oxygen, NIV, MV, or ECMO.

^e Patients who are receiving NIV or MV at baseline and require a substantial increase in baseline support should be treated per the recommendations for patients requiring new NIV or MV.

^f **Tofacitinib** is an alternative if baricitinib is not available (**BIII**).

^g For children who started receiving remdesivir before admission to the ICU, the remdesivir should be continued to complete the treatment course.

Key: ECMO = extracorporeal membrane oxygenation; ICU = intensive care unit; MV = mechanical ventilation; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel; RT-PCR = reverse transcription polymerase chain reaction

Table 3d. Therapeutic Management of Hospitalized Patients With MIS-C

	Panel's Recommendations
MIS-C	Initial treatment for MIS-C includes both immunomodulatory and antithrombotic therapy. <i>Initial Immunomodulatory Therapy</i> • IVIG 2 g/kg IBW IV (up to a maximum total dose of 100 g) plus low to moderate dose methylprednisolone (1–2 mg/kg/day IV)^a or another glucocorticoid at an equivalent dose^a (AIIb). • Glucocorticoid monotherapy, only if IVIG is unavailable or contraindicated (BIIa). • IVIG monotherapy, only if glucocorticoids are contraindicated (BIIb). <i>Intensification Immunomodulatory Therapy</i> • Intensification therapy is recommended for children with refractory MIS-C who do not improve within 24 hours of receiving initial immunomodulatory therapy (AIII). One of the following can be used (listed in alphabetical order): • High-dose anakinra 5–10 mg/kg/day IV (preferred) or SUBQ in 1–4 divided doses^b (BIIb) • Higher-dose glucocorticoid (e.g., methylprednisolone 10–30 mg/kg/day IV for 1–3 days, up to a maximum of 1,000 mg/day, or equivalent glucocorticoid for 1–3 days)^{a,b} (BIIb) • Infliximab 5–10 mg/kg IV for 1 dose^{b,c} (BIIb) <i>Antithrombotic Therapy</i> • Low-dose aspirin (3–5 mg/kg PO once daily, up to a maximum dose of 81 mg) for all patients without risk factors for bleeding (AIII), AND • Anticoagulation for patients who fall under 1 of the following clinical scenarios: • Therapeutic anticoagulation for patients with large CAAs according to the American Heart Association guidelines for Kawasaki disease (AIII). • Therapeutic anticoagulation for patients with moderate to severe LV dysfunction who have no risk factors for bleeding (AIII). • For patients with MIS-C who do not have large CAAs or moderate to severe LV dysfunction, consider prophylactic or therapeutic anticoagulation on an individual basis, taking into consideration risk factors for thrombosis and bleeding. See Table 3e for additional information. Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

^a The duration of glucocorticoid therapy may vary. When a patient shows clinical improvement (e.g., resolution of fever, improvement of organ function, reduction of levels of inflammatory markers), a steroid taper should be initiated. Typically, the patient's clinical status guides the taper, and it continues for several weeks to avoid rebound inflammation. For more information, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

^b In certain patients with severe illness, intensification therapy may include dual therapy with higher-dose glucocorticoids plus anakinra (**BIII**) or higher-dose glucocorticoids plus infliximab (**BIII**). **Anakinra and infliximab should not be used in combination.**

^c **Infliximab should not be used** in patients with macrophage activation syndrome.

Key: CAA = coronary artery aneurysm; IBW = ideal body weight; IV = intravenous; IVIG = intravenous immunoglobulin; LV = left ventricular; MIS-C = multisystem inflammatory syndrome in children; PO = oral; SUBQ = subcutaneous

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Special Considerations in Children

Last Updated: February 29, 2024

Key Considerations
• SARS-CoV-2 infection is generally milder in children than in adults, and a substantial proportion of children with the infection are asymptomatic. • Most nonhospitalized children with COVID-19 will not require any specific therapy. • Children with ≥ 1 of the following comorbidities are at increased risk of severe COVID-19: cardiac disease, neurologic disorders, prematurity (in young infants), diabetes, obesity (particularly severe obesity), chronic lung disease, feeding tube dependence, and immunocompromised status. Age (<1 year and 10–14 years) and non-White race/ethnicity are also associated with severe disease. • Data on the pathogenesis and clinical spectrum of SARS-CoV-2 infection are more limited for children than for adults. • Vertical transmission of SARS-CoV-2 appears to be rare. • A small subset of children and young adults with SARS-CoV-2 infection may develop multisystem inflammatory syndrome in children (MIS-C). Many patients with MIS-C require intensive care management. The majority of children with MIS-C do not have underlying comorbidities. • Data on the prevalence of post-COVID conditions in children are limited but suggest that younger children may have fewer persistent symptoms than older children and adults.
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

This section provides an overview of the epidemiology and clinical spectrum of disease, including COVID-19, multisystem inflammatory syndrome in children (MIS-C), and post-COVID conditions. It also includes information on risk factors for severe COVID-19, vertical transmission, and infants born to a birth parent with SARS-CoV-2 infection. Throughout this section, the term “COVID-19” refers to the acute, primarily respiratory illness due to infection with SARS-CoV-2. MIS-C refers to the postinfectious inflammatory condition.

Epidemiology

Data from the Centers for Disease Control and Prevention (CDC) demonstrate that severe disease and death due to COVID-19 occur less often in children than in adults.¹⁻⁴ However, weekly hospitalization rates for children aged <6 months are high, exceeded only by the rates for adults aged ≥ 75 years.⁵ The overall incidence of SARS-CoV-2 infection and, by extension, COVID-19–related hospitalizations among children increased substantially with the emergence of newer variants, particularly the Omicron variant and its subvariants.^{6,7} According to the CDC, by December 2022, an estimated 96% of children and adolescents had serologic evidence of prior SARS-CoV-2 infection.⁸ The high infection rate among children makes the overall burden substantial despite the low rate of severe outcomes.⁹

Data on the pathogenesis and clinical spectrum of SARS-CoV-2 infection in children are still limited compared to the data for adults. Although only a small percentage of children with COVID-19 will require medical attention, the percentage of intensive care unit (ICU) admissions among hospitalized children is comparable to that for hospitalized adults with COVID-19.^{6,10-18}

Children from some racial and ethnic groups experience disproportionate rates of COVID-19–related hospitalization, which may be a result of barriers to accessing health care and economic and structural inequities. From 2020 to 2021, Black/African American children with COVID-19 in the United States

were 2 times more likely to be hospitalized and 5 times more likely to be admitted to the ICU than White children.¹⁹

A U.S. study of children with COVID-19 who were hospitalized between April and September 2020 reported an association between race/ethnicity and disease severity.²⁰ In a large United Kingdom study, admission to critical care was independently associated with hospitalized children who self-reported as being of Black ethnicity.¹⁵ A study in England reported that children who identified as Asian were more likely than children who identified as White to be hospitalized for COVID-19 and to be admitted to an ICU.²¹ The study also found that children who identified as Black or as mixed or other races/ethnicities had significantly more hospitalizations than children who identified as White.

Clinical Manifestations of COVID-19

The signs and symptoms of SARS-CoV-2 infection in symptomatic children are similar to those in adults. However, a greater proportion of children may be asymptomatic or have only mild illness when compared with adults. Although the true incidence of asymptomatic SARS-CoV-2 infection is unknown, a small study reported that 45% of children who underwent surveillance testing at the time of hospitalization for a non-COVID-19 indication had asymptomatic infection.²² The most common signs and symptoms of COVID-19 in hospitalized children are fever, nausea/vomiting, cough, shortness of breath, and upper respiratory symptoms.^{15,23} The signs and symptoms of COVID-19 may overlap significantly with those of influenza and other respiratory and enteric viral infections. Critical disease, including respiratory failure, acute respiratory distress syndrome, and, less commonly, shock, may occur in children with COVID-19.^{24,25} For more information, see [Therapeutic Management of Hospitalized Children With COVID-19](#) and [Introduction to Critical Care Management of Children With COVID-19](#).

Risk Factors for Severe COVID-19

Observational studies and meta-analyses have found that children with certain comorbidities are at increased risk of severe COVID-19. These comorbidities include cardiac disease, neurologic disorders, prematurity (in young infants), diabetes, obesity (particularly severe obesity), chronic lung disease, feeding tube dependence, and immunocompromised status.²⁶⁻²⁹ Demographic factors, such as age (<1 year and 10–14 years)³⁰ and non-White race/ethnicity,^{15,19-21} have also been associated with severe disease. However, many studies did not assess the relative severity of underlying medical conditions in children with severe COVID-19.

Many published studies reported an increased relative risk of severe disease in children with comorbidities,^{20,26-30} but the overall risk of severe COVID-19 among children remains low. Protocolized admissions for certain populations (e.g., febrile young infants) may confound the association between comorbidities and severe COVID-19. However, nearly half of the children aged 8 months to <5 years who were hospitalized for COVID-19 from September 20, 2022, to May 31, 2023, were previously healthy.³¹ Most children who have been hospitalized for severe COVID-19 have not been fully vaccinated, as many were not eligible for COVID-19 vaccination because of their age when the studies were conducted.⁵ The CDC has additional information on the underlying conditions that are [risk factors for severe COVID-19](#).

The risk of severe disease is an important factor to consider when making treatment decisions for children with COVID-19. The children most likely to benefit from antiviral treatment are those who are nonhospitalized, have mild to moderate COVID-19, and are at the highest risk of severe COVID-19 (e.g., those with severe comorbidities). For a description of children who are considered at high risk of severe COVID-19 and for the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for their treatment, see [Therapeutic Management of Nonhospitalized Children With COVID-19](#).

Age

Among all children, infants and adolescents have the highest risk of COVID-19–related hospitalization, ICU admission, or death. From March 2020 to mid-August 2021, U.S. children aged <5 years had the highest cumulative COVID-19–related hospitalization rates, followed closely by adolescents.³² Children aged 5 to 11 years had the lowest hospitalization rates. From July to August 2021, when the Delta variant was the dominant variant, 25% of 713 children admitted to 6 U.S. hospitals were aged <1 year, 17% were aged 1 to 4 years, 20% were aged 5 to 11 years, and 38% were aged 12 to 17 years.²⁶ From March 2020 to mid-June 2021, 26.5% of 3,116 U.S. children hospitalized for COVID-19 were admitted to an ICU.³² In 2023, children aged <6 months had the highest weekly COVID-19–related pediatric hospitalization rates.⁵

A meta-analysis of individual patient data showed that among hospitalized children with COVID-19, patients aged <1 year and those aged 10 to 14 years had the highest risks of ICU admission and death.³⁰ In another meta-analysis, neonates had an increased risk of severe COVID-19 when compared with other pediatric age groups, but infants aged 1 to 3 months did not.²⁷ When the original Omicron variant was the dominant circulating variant, children and adolescents had higher hospitalization rates than they did when the Delta variant was dominant, and children aged <5 years had the highest rates.^{7,33} However, the proportion of hospitalized children who required ICU admission was significantly lower when the original Omicron variant was dominant.

Comorbidities

Several chronic conditions are prevalent in hospitalized children with COVID-19. When the Delta variant was the dominant variant in the United States, 68% of hospitalized children had ≥ 1 underlying medical conditions, such as obesity (32%), asthma or reactive airway disease (16%), or feeding tube dependence (8%).²⁶ Obesity was present in approximately a third of hospitalized children aged 5 to 11 years, 60% of whom had severe obesity (i.e., a body mass index [BMI] $\geq 120\%$ of the 95th percentile). For adolescents, 61% had obesity; of those patients, 61% had a BMI $\geq 120\%$ of the 95th percentile.

Meta-analyses and observational studies identified risk factors for ICU admission, mechanical ventilation, or death among hospitalized children with COVID-19.²⁷⁻²⁹ These risk factors included prematurity in young infants, obesity, diabetes, chronic lung disease, cardiac disease, neurologic disease, and immunocompromising conditions. Another study found that having a complex chronic condition that affected ≥ 2 body systems or having a progressive chronic condition or continuous dependence on technology for ≥ 6 months (e.g., dialysis, tracheostomy with ventilator assistance) was significantly associated with an increased risk of moderate or severe COVID-19.³⁴ The study also found that children with more severe chronic diseases (e.g., active cancer treated within the previous 3 months or asthma with hospitalization within the previous 12 months) had a higher risk of critical COVID-19 or death than those with less severe conditions. The CDC has additional information on the underlying conditions that are [risk factors for severe COVID-19](#).

Having multiple comorbidities increases the risk of severe COVID-19 in children. A meta-analysis of data from children hospitalized with COVID-19 found that the risk of ICU admission was greater for children with 1 chronic condition than for those with no comorbidities, and the risk increased substantially as the number of comorbidities increased.³⁰

COVID-19 Vaccination

Staying up to date with COVID-19 vaccinations remains the most effective way to prevent severe COVID-19. See the CDC webpages [Stay Up to Date With COVID-19 Vaccines](#) and [Use of COVID-19 Vaccines in the United States](#) for more information on COVID-19 vaccination schedules. Estimates of

vaccine effectiveness vary by age group and time period. From July 2022 to September 2023 (when Omicron subvariants were dominant in the United States), 86% of vaccine-eligible U.S. children aged 6 months to <5 years who were hospitalized or sought care for acute respiratory illness in emergency departments had not received any COVID-19 vaccines.³⁵ Two or more doses of COVID-19 vaccine were 40% effective in preventing emergency department visits or hospitalization due to COVID-19 in children aged <5 years compared with unvaccinated children. In a study of U.S. children aged 8 months to <5 years who were hospitalized for COVID-19 from September 20, 2022, to May 31, 2023, only 4.5% had completed a primary COVID-19 vaccine series.³¹

The estimates for vaccine effectiveness against severe COVID-19 in adolescents aged 12 to 18 years exceeded 90% while Delta was the dominant variant in the United States.^{36,37} When Omicron was the dominant variant, vaccine effectiveness against hospitalization for noncritical COVID-19 was 20% in adolescents; vaccine effectiveness against critical illness was 79% in these patients.³⁷ In children aged 5 to 11 years, vaccine effectiveness against hospitalization was more variable, with an estimated effectiveness of 68% after Omicron became the dominant variant in the United States.

A meta-analysis of COVID-19 vaccination in adolescents aged 12 to 17 years reported a vaccine effectiveness of 88% against severe disease and 35% against nonsevere COVID-19.³⁸ An Italian study estimated that vaccine effectiveness was 38% in children aged 5 to 11 years during the Omicron period.³⁹ A Canadian study reported that the effectiveness of 2 doses of COVID-19 vaccine against symptomatic COVID-19 in children aged 5 to 11 years varied widely.⁴⁰ Vaccine effectiveness decreased over time after the last dose and decreased against Omicron subvariants that were antigenically distinct from the vaccine. See [Prevention of SARS-CoV-2 Infection](#) for more information about COVID-19 vaccines.

Mortality

Death from COVID-19 is uncommon in children. Risk factors for death include having chronic conditions, such as neurologic or cardiac disease, and having multiple comorbidities. Among children aged <21 years in the United States, the number of deaths associated with COVID-19 has been higher for children aged 10 to 20 years, especially for young adults aged 18 to 20 years, and for those who identify as Hispanic, Black, or American Indian/Alaskan Native.^{41,42}

A systematic review and meta-analysis reported that neurologic or cardiac comorbidities were associated with the greatest increase in risk of death among hospitalized children with COVID-19.³⁰ In the same study, an individual patient data meta-analysis found that the risk of death related to COVID-19 was greater for children with 1 chronic condition than for those with no comorbidities, and the risk increased substantially as the number of comorbidities increased.

Vertical Transmission and Infants Born to People With SARS-CoV-2 Infection

Systematic reviews and meta-analyses have reported that confirmed vertical transmission of SARS-CoV-2 appears to be rare, but severe maternal COVID-19 has been associated with SARS-CoV-2 infection in babies.⁴³ In 2 large, combined cohorts of pregnant individuals from the United States and United Kingdom, SARS-CoV-2 infection was reported in 1.8% and 2% of the babies born to people with SARS-CoV-2 infection.⁴⁴ A systematic review and meta-analysis of prospective observational studies from high-income countries estimated that the frequency of SARS-CoV-2 infection in infants born to people with SARS-CoV-2 infection was 2.3%.⁴⁵

Case reports have described intrauterine fetal demise during the third trimester of pregnancy in individuals with mild COVID-19 due to infection with the Delta variant.^{46,47} These individuals had evidence of placental SARS-CoV-2 infection, placental malperfusion, and placental inflammation. One case report described a person with asymptomatic SARS-CoV-2 infection and severe preeclampsia who

gave birth at 25 weeks of gestation by emergency cesarean delivery. The neonate died on Day 4, and evidence of SARS-CoV-2 infection was found in placental tissues and in the infant's lungs and vascular endothelium at autopsy.⁴⁸ Evidence of placental SARS-CoV-2 infection was reported in 5 stillbirths and for a live-born neonate in Sweden.⁴⁹

A systematic review of neonatal SARS-CoV-2 infections reported that 70% were due to postpartum transmission, and 30% were due to vertical transmission from the infected birth parent.⁵⁰ Two systematic reviews reported that newborn infants rooming-in with an infected birth parent did not have an increased risk of SARS-CoV-2 transmission when compared with newborns who were isolated from the birth parent.^{45,51}

Detection of SARS-CoV-2 RNA in the breast milk of individuals with confirmed cases of COVID-19 is very uncommon.⁵² Currently, there is no evidence of SARS-CoV-2 transmission through breast milk.⁵³ Breast milk from people with SARS-CoV-2 infection can contain antibodies to SARS-CoV-2.⁵⁴⁻⁵⁶ For information regarding the safety of feeding infants breast milk from individuals who are receiving treatment for COVID-19, see [Pregnancy, Lactation, and COVID-19 Therapeutics](#).

Multisystem Inflammatory Syndrome in Children

A small subset of children and young adults with SARS-CoV-2 infection, including those with asymptomatic infection, may develop MIS-C. This syndrome is also called pediatric inflammatory multisystem syndrome—temporally associated with SARS-CoV-2 (PIMS-TS). Although the case definitions for these syndromes differ slightly, they are likely the same disease. The syndrome was first described in Europe, where previously healthy children with severe inflammation and features similar to Kawasaki disease were identified as having current or recent infection with SARS-CoV-2.^{57,58} Subsequently, children with MIS-C were identified in the United States and in many other locations outside of Europe.⁵⁹ Most patients with MIS-C have serologic evidence of previous SARS-CoV-2 infection, but only a minority have had a positive reverse transcription polymerase chain reaction (RT-PCR) result for SARS-CoV-2 at presentation.^{60,61} The peak, population-based incidence of MIS-C lags about 4 weeks behind the peak of acute pediatric COVID-19–related hospitalizations.

Although risk factors for the development of MIS-C have not been established, an analysis of MIS-C cases in the United States found that ICU admission was more likely for patients aged 6 to 12 years than for younger children, and it was more likely for children who identified as non-Hispanic Black than for those who identified as non-Hispanic White.⁶² Unlike most children who present with severe COVID-19, the majority of children who present with MIS-C do not seem to have common underlying comorbidities other than obesity.⁶² In addition, children whose deaths were related to MIS-C were less likely to have underlying medical conditions than children who died of COVID-19.⁴²

Several studies have suggested that COVID-19 vaccination protects against the development of MIS-C.⁶³⁻⁶⁵ Following the emergence of the Omicron variant, the incidence of MIS-C and the clinical severity of MIS-C have declined.⁶⁶⁻⁷⁰ This decline may be a result of several factors. For example, more children have now received COVID-19 vaccines and have had infection with SARS-CoV-2, which may provide some protection against MIS-C.

Clinical Manifestations of Multisystem Inflammatory Syndrome in Children

The CDC and the Council of State and Territorial Epidemiologists (CSTE) issued an updated case definition for MIS-C on January 1, 2023.⁷¹ The 2023 CSTE/CDC Surveillance Case Definition for MIS-C is an individual aged <21 years who:

- Presents with fever,^a laboratory evidence of inflammation,^b and illness with a clinical severity that

requires hospitalization or results in death, with new-onset clinical manifestations in ≥ 2 categories (i.e., cardiac, shock, hematologic, gastrointestinal, dermatologic)^c; *and* does not have a more likely alternative diagnosis; *and*

- Has a positive viral test result from a molecular test that detects SARS-CoV-2 RNA or a SARS-CoV-2 antigen test up to 60 days prior to or during hospitalization or in a postmortem specimen; *or*
- Has a positive viral test result from a test that detects SARS-CoV-2–specific antibodies associated with current illness; *or*
- Has a close contact with a confirmed or probable case of COVID-19 in the 60 days prior to hospitalization; *or*
- Has a death certificate that lists MIS-C as an underlying cause of death or a significant condition contributing to death.

^a Subjective or documented fever $\geq 38.0^{\circ}\text{C}$.

^b C-reactive protein level ≥ 3.0 mg/dL (30 mg/L).

^c See Table A for a list of categories for these organ manifestations.

Table A. Clinical Manifestation Criteria for the 2023 CSTE/CDC MIS-C Surveillance Case Definition

Clinical Manifestation	Criteria
Cardiac Involvement	• Left ventricular ejection fraction $<55\%$ • Coronary artery dilatation, aneurysm, or ectasia • Troponin levels elevated above laboratory normal range or indicated as elevated in a clinical note
Shock	• Clinician diagnosis, as documented in clinical note
Hematologic Involvement	• Thrombocytopenia (i.e., platelet count $<150,000$ cells/μL) • Lymphopenia (i.e., absolute lymphocyte count $<1,000$ cells/μL)
Gastrointestinal Involvement	• Abdominal pain • Vomiting • Diarrhea
Dermatologic/Mucocutaneous Involvement	• Rash • Inflammation of the oral mucosa • Conjunctivitis or conjunctival injection • Extremity findings (e.g., erythema, edema)

Key: CDC = Centers for Disease Control and Prevention; CSTE = Council of State and Territorial Epidemiologists; MIS-C = multisystem inflammatory syndrome in children

Distinguishing MIS-C from other febrile illnesses in the community setting remains challenging, particularly with the declining incidence of MIS-C, but the presence of persistent fever, multisystem manifestations, and laboratory abnormalities could help early recognition.⁷² The clinical spectrum of hospitalized cases has included younger children with mucocutaneous manifestations that overlap with Kawasaki disease, older children with more multiorgan involvement and shock, and patients with respiratory manifestations that overlap with COVID-19.

Patients with MIS-C are often critically ill, and up to 80% of children require ICU admission; however, data collected while Omicron was the dominant variant in the United States suggest that the cases of MIS-C reported during this period were less severe than those reported when other variants were dominant.^{66,73,74} Most patients with MIS-C have markers of cardiac injury or dysfunction, including

elevated levels of troponin and brain natriuretic protein.⁶² Higher levels of these markers are associated with ICU admission, myocardial dysfunction, and shock. In these cases, echocardiographic findings may include impaired left ventricular function, coronary artery dilations, and, rarely, coronary artery aneurysms. During the period when Omicron was the dominant variant in the United States, the clinical phenotype of MIS-C appeared to be more consistent with classic Kawasaki disease.^{66,73} The reported mortality in the United States for hospitalized children with MIS-C is 1% to 2%. Longitudinal studies to examine the long-term sequelae of MIS-C are currently ongoing.

The pathogenesis of MIS-C is still being elucidated and may include a unique, distinct antibody response; excessive activation of elements of the innate immune system; or aberrant T cell responses, including a superantigen effect.⁷⁵ Several studies reported that an expansion of T cells that express the T cell receptor beta variable 11-2 (TRBV11-2) gene was detected in many children with MIS-C.⁷⁶⁻⁷⁹ This expansion of T cells was not seen in children who had conditions similar to MIS-C, including Kawasaki disease and bacterial toxic shock syndrome, which supports the hypothesis that a superantigen effect may be involved in MIS-C. Another study demonstrated that 1% of patients with MIS-C had inborn errors of immunity.⁸⁰ Other studies have demonstrated that MIS-C and typical Kawasaki disease have differences in epidemiology, cytopenias, cytokine expression, and elevation of inflammatory markers.^{81,82} Immunologic profiling has shown that MIS-C and COVID-19 in children have differences in cytokine expression (e.g., tumor necrosis factor–alpha, interleukin-10, and interferon gamma).^{78,83,84}

For the Panel’s recommendations on the treatment of MIS-C, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Post-COVID Conditions

The persistent symptoms after COVID-19 that have been described in children are similar to those seen in adults. The terminology for these collective symptoms is evolving and includes long COVID, post-COVID-19 condition, and post-acute sequelae of SARS-CoV-2 infection (PASC). The data on the incidence of post-COVID conditions in children are limited and somewhat conflicting, but the overall incidence appears to be lower in children than in adults (see [Clinical Spectrum of SARS-CoV-2 Infection](#)).⁸⁵⁻⁹²

Case definitions for post-COVID conditions vary between studies, which makes determining the true incidence of these conditions challenging. The incidence of post-COVID symptoms in children appears to increase with age. The most common symptoms reported include persistent fatigue, headache, shortness of breath, sleep disturbances, gastrointestinal symptoms, and an altered sense of smell.⁹³ Cardiopulmonary injury, neurocognitive impairment, and new-onset diabetes may occur. However, some studies did not include control groups of children who did not have SARS-CoV-2 infection, which makes assessing the relative risk of these symptoms a challenge.

Details on the pathogenesis, clinical presentation, and treatment for post-COVID conditions in children are beyond the scope of these Guidelines. The CDC provides additional information about the incidence, presentation, and management strategies for [post-COVID conditions in children](#) as well as adults. Additional research is needed to define the incidence, pathophysiology, spectrum, and severity of post-COVID conditions in children and to identify the optimal strategies for the prevention, diagnosis, and treatment of these conditions.

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Therapeutic Management of Nonhospitalized Children With COVID-19

Last Updated: February 29, 2024

This section outlines the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for the therapeutic management of nonhospitalized children (i.e., pediatric patients aged <18 years) with mild to moderate COVID-19. These recommendations are also for children who have mild to moderate COVID-19 and are hospitalized for reasons other than COVID-19. For patients aged ≥18 years, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#). Throughout this section, the term "COVID-19" refers to the acute, primarily respiratory illness due to infection with SARS-CoV-2. For the Panel's recommendations for managing multisystem inflammatory syndrome in children (MIS-C), see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Treatment Considerations

Currently, no results from pediatric clinical trials that evaluated the treatment of COVID-19 have been published. Data evaluating the use of pharmacologic therapy in children with COVID-19 are limited largely to descriptive reports.¹⁻³ Therefore, more high-quality randomized trials, observational studies, and pharmacokinetic studies are urgently needed. Whenever possible, clinical trials of therapeutics and multicenter observational cohorts should enroll children with COVID-19.

Published guidance on the treatment of COVID-19 in children has been extrapolated mostly from recommendations for adults with COVID-19, recommendations for children with other viral infections, and expert opinion.⁴⁻⁶ Applying adult data from COVID-19 trials to children is a unique challenge because most children experience a mild course of illness with COVID-19. Relative to adults, children with COVID-19 have substantially lower mortality and less need for hospitalization.⁷⁻⁹ Because of these differences in epidemiology and disease severity, the effect sizes of treatments for children are likely to be smaller than those observed in adults. Therefore, to produce a beneficial outcome in children, the number needed to treat is higher. Collectively, these factors influence the risk versus benefit balance for pharmacologic therapies in children.

Other challenges are the uncertainty about which comorbid conditions place children at the highest risk of severe COVID-19 and the uncertainty about the absolute magnitude of the increased risk from those comorbid conditions. For children with COVID-19, the type, number, and severity of comorbid conditions influence decisions about pharmacologic treatments. For more information on risk factors for children with COVID-19, see [Special Considerations in Children](#).

Recommendations

In the absence of sufficient clinical trial data on the treatment of children with COVID-19, the Panel's recommendations for the therapeutic management of nonhospitalized children are based largely on safety and efficacy data from clinical trials in adults (see Table 3a). No pediatric comparative studies have been published; therefore, all the quality of evidence ratings for the Panel's recommendations in this section are based on expert opinion (i.e., a **III** rating).

The majority of children with mild to moderate COVID-19 will not progress to more severe illness; therefore, the Panel recommends managing these patients with supportive care alone (**AIII**). The risks and benefits of therapy should be assessed based on COVID-19 disease severity, age, vaccination status,

and the presence of underlying medical conditions that may place the patient at high risk of severe COVID-19.

Table 3a. Therapeutic Management of Nonhospitalized Children With COVID-19

Risk of Severe COVID-19	Panel's Recommendations	
	Aged 12–17 Years	Aged <12 Years
Symptomatic, Regardless of Risk Factors	• Provide supportive care (AIII).	• Provide supportive care (AIII).
High Risk^{a,b}	• Use 1 of the following options (listed in order of preference):^c • Ritonavir-boosted nirmatrelvir (Paxlovid) within 5 days of symptom onset (BIII) • Remdesivir within 7 days of symptom onset (CIII)	• Ritonavir-boosted nirmatrelvir is not authorized by the FDA for use in children aged <12 years. • There is insufficient evidence to recommend either for or against the routine use of remdesivir. Consider treatment based on age and other risk factors.
Intermediate Risk^{b,d}	• There is insufficient evidence to recommend either for or against the use of any antiviral therapy. Consider treatment based on age and other risk factors.	• There is insufficient evidence to recommend either for or against the routine use of remdesivir.
Low Risk^{b,e}	• Manage with supportive care alone (AIII).	• Manage with supportive care alone (AIII).
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.		

^a Molnupiravir is not authorized by the FDA for use in children aged <18 years and should not be used.

^b See Table 3b for the Panel's framework for assessing the risk of progression to severe COVID-19 based on patient conditions and COVID-19 vaccination status.

^c Initiate treatment as soon as possible after symptom onset.

^d The relative risk of severe COVID-19 for intermediate-risk patients is lower than the risk for high-risk patients but higher than the risk for low-risk patients.

^e Low-risk patients include those with comorbid conditions that have a weak or unknown association with severe COVID-19. Patients with no comorbidities are included in this group.

Key: FDA = Food and Drug Administration; the Panel = the COVID-19 Treatment Guidelines Panel

Table 3b. The Panel’s Framework for Assessing the Risk of Progression to Severe COVID-19 Based on Patient Conditions and COVID-19 Vaccination Status

Conditions	Risk Level by Vaccination Status	
	Not Up to Date ^{a,b}	Up to Date ^a
Strong or Consistent Association With Progression to Severe COVID-19		
Moderately or severely immunocompromised (see Special Considerations in People Who Are Immunocompromised)	High	
- Obesity (BMI ≥95th percentile for age), especially severe obesity (BMI ≥120% of 95th percentile for age)^c - Medical complexity with dependence on respiratory technology^d - Severe neurologic, genetic, metabolic, or other disability that results in impaired airway clearance or limitations in self-care or activities of daily living - Severe asthma or other severe chronic lung disease requiring ≥2 inhaled or ≥1 systemic medications daily - Severe congenital or acquired cardiac disease - Multiple moderate to severe chronic diseases - Pregnancy	High	Intermediate
Moderate or Inconsistent Association With Progression to Severe COVID-19		
- Aged <1 year - Prematurity in children aged ≤2 years - Sickle cell disease - Diabetes mellitus (poorly controlled) - Chronic kidney disease - Nonsevere cardiac, neurologic, or metabolic disease^e	Intermediate	
Weak or Unknown Association With Progression to Severe COVID-19		
- Mild asthma - Overweight - Diabetes mellitus (well controlled)	Low	

^a See the current [COVID-19 vaccination schedule](#) from the CDC.

^b Recent SARS-CoV-2 infection (i.e., within 3–6 months) may confer substantial immunity against closely related variants. A patient’s recent infection history should be factored into the risk assessment.

^c The degree of risk conferred by obesity in younger children is less clear than it is in older adolescents.

^d This includes patients with a tracheostomy and those who require NIV.

^e The data for this group are particularly limited.

Key: BMI = body mass index; CDC = Centers for Disease Control and Prevention; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel

Rationale for the Panel's Framework for Assessing the Risk of Progression to Severe COVID-19

Although mortality associated with COVID-19 in children is low overall, severe disease can occur, especially in those with risk factors.¹⁰ Risk stratification for severe disease in children remains challenging. Imprecise definitions of comorbid conditions, insufficient granularity for differentiating the severity of comorbidities (e.g., mild vs. severe lung disease, poorly controlled vs. well-controlled diabetes), and small sample sizes limit the conclusions that can be drawn from individual studies and make comparing findings across studies difficult.

Furthermore, asymptomatic SARS-CoV-2 infection detected during admission screening of children who are hospitalized for reasons other than COVID-19 may affect the estimated risk of severe COVID-19, particularly for patient groups that may have protocolized admissions (e.g., children with febrile neutropenia, infants aged <90 days with fever). In addition, the published studies that have evaluated these associations in children are limited largely to case series without control groups and observational studies with methodologic limitations.

Despite these challenges, a risk-stratification framework needs to be developed that will allow clinicians to identify the patients who are most likely to benefit from receiving treatment. These patients can be prioritized in situations where supply or logistical constraints make it impossible to offer therapy to all eligible patients. Both the Pediatric Infectious Diseases Society and the American Academy of Pediatrics advocate for a risk-stratified approach to identifying the patients who are at the highest risk of progression to severe COVID-19 among those eligible for therapies under Food and Drug Administration (FDA) Emergency Use Authorizations (EUAs).^{11,12}

The Panel's approach to risk stratification and prioritization considers COVID-19 vaccination status, immune function, clinical risk factors, the strength of the evidence that demonstrates an association between each clinical risk factor and severe disease, and expert opinion.^{7-10,13-27} See [Special Considerations in Children](#) for more information on clinical risk factors. The Panel suggests that decisions regarding treatment be individualized, particularly for patients in the intermediate risk category. Clinicians should consider the type, number, and severity of comorbid conditions, the child's vaccination status, and the time since vaccination.

Comorbid conditions associated with severe COVID-19 are separated into the following categories in Table 3b:

- *Strong or Consistent Association With Progression to Severe COVID-19:* Comorbid conditions for which the published literature most consistently supports an increased risk of severe COVID-19. Patients in this category are moderately or severely immunocompromised, at risk of severe COVID-19, and not expected to develop an adequate immune response to COVID-19 vaccination.
- *Moderate or Inconsistent Association With Progression to Severe COVID-19:* Comorbid conditions and ages for which the published literature supports an association with severe COVID-19, but the association may be moderate or inconsistent across studies. In addition, the absolute risk of progression to severe disease or death is likely modest for any of the patients in this category.
- *Weak or Unknown Association With Progression to Severe COVID-19:* Comorbid conditions for which the data suggesting an association with severe COVID-19 are weak or for which an association is unknown. Patients with no comorbidities are included in this category.

Vaccination Status

Because COVID-19 vaccines are highly effective in preventing severe disease, individuals who are not immunocompromised and are up to date on their vaccines²⁸ are likely to have a low absolute risk of severe disease. Therefore, the potential benefit from antiviral treatment is less clear for these patients. Patients who are not up to date may have a lower level of protection against severe disease than patients who are up to date, but the data comparing these groups are limited. Evidence suggests that vaccine protection against severe COVID-19 wanes over time, particularly protection against the Omicron variant and its subvariants.²⁹⁻³² Several studies suggested that people who have had SARS-CoV-2 infection and were vaccinated have more durable protection than those who have had only an infection or only been vaccinated.^{33,34} Clinicians should consider the time since a child's vaccination when making treatment decisions.

Health Disparities

COVID-19–related outcomes are worse among medically underserved populations, although this factor is not strictly a comorbid condition. Some racial and ethnic minority groups experience disproportionate rates of COVID-19 hospitalization and are less likely to receive specific therapies.^{23,35-37} These factors may be relevant when making clinical decisions about treatment.^{38,39} See [Special Considerations in Children](#) for more information.

Rationale for the Panel's Recommendations

Ritonavir-Boosted Nirmatrelvir (Paxlovid)

Ritonavir-boosted nirmatrelvir has received an FDA EUA for the treatment of mild to moderate COVID-19 in nonhospitalized adolescents aged ≥ 12 years and weighing ≥ 40 kg who are at high risk of progression to severe COVID-19.⁴⁰

The EPIC-HR trial enrolled adults aged ≥ 18 years who were at high risk of severe COVID-19.⁴¹ They were randomized to receive ritonavir-boosted nirmatrelvir or placebo. The primary outcome of COVID-19–related hospitalization or all-cause mortality occurred in 8 of 1,039 patients (0.8%) who received ritonavir-boosted nirmatrelvir and in 66 of 1,046 patients (6.3%) who received placebo, an 89% relative risk reduction. No pediatric patients were included in the trial, and no pediatric safety data were made available.

Ritonavir has been used extensively in pediatric patients as a pharmacokinetic booster for the treatment of HIV and hepatitis C virus infection, and it has a known and tolerable side effect profile. In the FDA EUA, the dose of ritonavir-boosted nirmatrelvir authorized for adolescents aged ≥ 12 years and weighing ≥ 40 kg is expected to result in a drug exposure similar to that observed in adults.⁴⁰

Given the high efficacy of ritonavir-boosted nirmatrelvir in adults, its overall manageable side effect profile, the pediatric clinical experience with ritonavir, and the convenience of an oral medication, the Panel recommends the use of **ritonavir-boosted nirmatrelvir (Paxlovid)** for nonhospitalized adolescents aged ≥ 12 years and weighing ≥ 40 kg who have mild to moderate COVID-19 and are at the highest risk of progression to severe COVID-19 (**BIII**). Ritonavir-boosted nirmatrelvir is expected to be active against circulating variants, although clinical efficacy data are currently limited in children.⁴²⁻⁴⁴

Because of the potential for significant drug-drug interactions with some concomitant medications, ritonavir-boosted nirmatrelvir may not be the safest choice for some patients. See [Ritonavir-Boosted Nirmatrelvir \(Paxlovid\)](#) and [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

Remdesivir

Remdesivir is approved by the FDA for use in hospitalized and nonhospitalized pediatric patients aged ≥ 28 days and weighing ≥ 3.0 kg.⁴⁵ Remdesivir is expected to be active against circulating variants, although clinical efficacy data are currently limited in children.^{44,46-48}

In a study that included nonhospitalized adults with mild to moderate COVID-19 who were at high risk of progression to severe disease, administering an intravenous infusion of remdesivir once daily for 3 days resulted in an 87% relative reduction in the risk of hospitalization or death when compared with placebo.⁴⁹ Although adolescents aged ≥ 12 years were eligible for inclusion, the trial included only 8 patients aged < 18 years. Therefore, no conclusions regarding the efficacy of remdesivir in children can be made from this trial. In addition, clinical experience data from hospitalized children with COVID-19 who received remdesivir through a compassionate use program have been reported.^{2,50} Given the demonstrated efficacy of remdesivir in the overall study population, its overall favorable side effect profile, and clinical experience with remdesivir in hospitalized children, **remdesivir**, as an alternative to ritonavir-boosted nirmatrelvir, can be considered for children aged ≥ 12 years who are at the highest risk of progression to severe COVID-19 (CIII).

There is insufficient evidence to recommend either for or against the routine use of remdesivir for the treatment of COVID-19 in nonhospitalized children aged < 12 years who are at the highest risk of progression to severe disease or who are at intermediate risk of severe disease. Administering remdesivir requires performing an intravenous infusion once daily for 3 days, so logistical constraints may preclude the use of remdesivir in many settings.

Pharmacologic Therapies Not Recommended

Molnupiravir

The FDA EUA for molnupiravir is limited to people aged ≥ 18 years, and there are no data on the safety of using molnupiravir in children.⁵¹ The mechanism of action of molnupiravir has raised concerns about potential mutagenesis in mammalian cells. See [Molnupiravir](#) and [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for additional information.

Corticosteroids

Corticosteroids are not indicated for the treatment of COVID-19 in nonhospitalized children. However, corticosteroids should be used per usual standards of care in children with asthma and croup triggered by SARS-CoV-2 infection. Children with COVID-19 who are receiving corticosteroids for an underlying condition should continue this therapy as directed by their health care providers.

Other Therapeutic Agents

For other therapies that have been studied or are under investigation for the treatment of COVID-19, see [Therapies](#).

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Therapeutic Management of Hospitalized Children With COVID-19

Last Updated: February 29, 2024

This section outlines the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for the therapeutic management of children (i.e., pediatric patients aged <18 years) who are hospitalized for COVID-19. Throughout this section, the term "COVID-19" refers to the acute, primarily respiratory illness due to infection with SARS-CoV-2. Multisystem inflammatory syndrome in children (MIS-C) refers to the postinfectious inflammatory condition.

Treatment Considerations for Children With COVID-19

To date, no comparative clinical trials evaluating the treatment of COVID-19 in children have been published. Evaluations of pharmacologic therapies in children with COVID-19 have been limited to small pharmacokinetic studies and retrospective observational reports.¹⁻⁵ Therefore, more high-quality randomized trials, observational studies, and pharmacokinetic studies are urgently needed. Whenever possible, clinical trials of therapeutics and multicenter observational cohorts should enroll children with COVID-19.

Published guidance on the treatment of COVID-19 in children has been extrapolated mostly from recommendations for adults with COVID-19, recommendations for children with other viral infections, and expert opinion.⁶⁻⁸ Applying adult data from COVID-19 trials to children is a unique challenge because most children experience a mild course of illness with COVID-19.³ Relative to adults, children with COVID-19 have substantially lower mortality and less need for hospitalization.⁹⁻¹¹ Because of these differences in epidemiology and disease severity, the effect sizes of treatments for children are likely to be smaller than those observed in adults. Therefore, to produce a beneficial outcome in children, the number needed to treat is higher. Collectively, these factors influence the risk versus benefit balance for pharmacologic therapies in children.

Other challenges are the uncertainty about which comorbid conditions place children at the highest risk of severe COVID-19 and the uncertainty about the absolute magnitude of the increased risk from those comorbid conditions. For children with COVID-19, the type, number, and severity of comorbid conditions influence decisions about pharmacologic treatment. For more information on risk factors in children with COVID-19, see [Special Considerations in Children](#) and [Therapeutic Management of Nonhospitalized Children With COVID-19](#).

Recommendations

In the absence of sufficient clinical trial data on the treatment of children with COVID-19, the Panel's recommendations for the therapeutic management of hospitalized children are based largely on safety and efficacy data from clinical trials in adults, the child's risk of disease progression, and expert opinion (see Table 3c). For the Panel's recommendations for adults, see [Therapeutic Management of Hospitalized Adults With COVID-19](#).

In general, data from clinical trials in adults are most applicable to the treatment of older children with severe COVID-19 and predominantly lower respiratory tract disease. Using data from clinical trials in adults to develop recommendations for children with SARS-CoV-2 infection who have clinical syndromes associated with other respiratory viruses (e.g., bronchiolitis, croup, asthma) is a challenge. No evidence suggests that these syndromes should be managed differently when caused by

SARS-CoV-2 infection. Clinical judgment is needed when applying recommendations for treatment in adults to children, particularly young children, with these clinical syndromes.

Table 3c. Therapeutic Management of Hospitalized Children With COVID-19

Disease Severity	Panel's Recommendations
Hospitalized for COVID-19	For children aged ≥ 12 years admitted for COVID-19, use prophylactic anticoagulation unless contraindicated (BIII) . ^a
Does Not Require Supplemental Oxygen	For children admitted for COVID-19 who are at the highest risk of progression to severe COVID-19 ^b (especially those who are severely immunocompromised), consider using remdesivir ^c for children aged 12–17 years (CIII) . There is insufficient evidence for using remdesivir in children aged 28 days to <12 years and weighing ≥ 3 kg.
	For children admitted for reasons other than COVID-19 who have mild to moderate COVID-19 and are at the highest risk of progression, ^b refer to Therapeutic Management of Nonhospitalized Children With COVID-19 .
Requires Conventional Oxygen^d	Use 1 of the following options: • Remdesivir^c (BIII) • Dexamethasone plus remdesivir^c for children with increasing oxygen needs, particularly adolescents (BIII)
Requires Oxygen Through High-Flow Device or NIV^e	Use 1 of the following options: • Dexamethasone (BIII) • Dexamethasone plus remdesivir^c (BIII)
	For children who do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, baricitinib ^f or tocilizumab can be considered for children aged 12–17 years (BIII) and for children aged 2–11 years (CIII) .
Requires MV or ECMO^g	Dexamethasone ^g (AIII)
	For children who do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, baricitinib ^f or tocilizumab may be considered for children aged 12–17 years (BIII) and for children aged 2–11 years (CIII) .
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.	

^a Weighing the risk factors for thrombosis and bleeding, some Panel members would use prophylactic anticoagulation for children aged <12 years who are hospitalized for COVID-19.

^b See [Therapeutic Management of Nonhospitalized Children With COVID-19](#) for a list of conditions that will put children at highest risk for progression to severe COVID-19.

^c The clinical benefit of remdesivir is greatest if it is initiated within 10 days of symptom onset. Remdesivir should be given for 5 days or until hospital discharge, whichever comes first. Examples of patients who may benefit most from adding remdesivir >10 days from symptom onset include patients who are severely immunocompromised, particularly if they have evidence of ongoing viral replication (e.g., those with a low Ct value, as measured by an RT-PCR result or with a positive rapid antigen test result).

^d Conventional oxygen refers to oxygen supplementation that is not high-flow oxygen, NIV, MV, or ECMO.

^e Patients who are receiving NIV or MV at baseline and require a substantial increase in baseline support should be treated per the recommendations for patients requiring new NIV or MV.

^f **Tofacitinib** is an alternative if baricitinib is not available **(BIII)**.

^g For children who started receiving remdesivir before admission to the ICU, the remdesivir should be continued to complete the treatment course.

Key: Ct = cycle threshold; ECMO = extracorporeal membrane oxygenation; ICU = intensive care unit; MV = mechanical ventilation; NIV = noninvasive ventilation; the Panel = the COVID-19 Treatment Guidelines Panel; RT-PCR = reverse transcription polymerase chain reaction

Rationale for the Panel's Recommendations for Drug Therapies

Remdesivir

Remdesivir is approved by the Food and Drug Administration (FDA) for hospitalized and nonhospitalized pediatric patients aged ≥ 28 days and weighing ≥ 3 kg.¹² Remdesivir is expected to be active against circulating variants, although clinical efficacy data in children are currently limited.¹³⁻¹⁶ For most hospitalized patients, remdesivir should be administered for 5 days or until the patient is ready for discharge, whichever comes first. Treatment may be extended to 10 days for patients who have not clinically improved or for patients who are severely immunocompromised. See [Remdesivir](#) and [Special Considerations in People Who Are Immunocompromised](#) for more information.

In a trial conducted mostly among hospitalized adults with COVID-19 who did not receive supplemental oxygen at enrollment, a 5-day course of remdesivir was associated with greater clinical improvement when compared with the standard of care.¹⁷ Remdesivir was also studied in ACTT-1, a double-blind, placebo-controlled, randomized trial for hospitalized adults with COVID-19 who received remdesivir for 10 days (or until hospital discharge) or placebo.¹⁸ The study reported that the remdesivir arm had a shorter time to clinical recovery than the placebo arm (10 days vs. 15 days; $P < 0.001$). A subgroup analysis demonstrated that patients who received conventional oxygen therapy had the greatest benefit. No benefit was detected for patients who did not receive supplemental oxygen or for those who received noninvasive ventilation (NIV) or mechanical ventilation. No statistically significant differences in mortality or in the need for new mechanical ventilation were detected, and the benefit of remdesivir in this study was limited to patients with symptoms for < 10 days.

Three open-label trials in adults compared remdesivir to a local standard of care. The World Health Organization's Solidarity trial enrolled hospitalized adult patients with COVID-19 in 35 countries.¹⁹ In the overall cohort, no difference in hospital mortality was demonstrated (14.5% in the remdesivir arm vs. 15.6% in the usual care arm; rate ratio 0.91; 95% CI, 0.82–1.02; $P = 0.12$). However, in the large subset of patients receiving supplemental oxygen but not NIV or mechanical ventilation, remdesivir significantly reduced the risk of in-hospital mortality by 13% (14.6% vs. 16.3%; rate ratio 0.87; 95% CI, 0.76–0.99; $P = 0.03$).

The CATCO study demonstrated similar findings.²⁰ Treatment with remdesivir, when compared with standard care, reduced the need for mechanical ventilation in hospitalized adults with COVID-19 (8% vs. 15%; relative risk 0.53; 95% CI, 0.38–0.75). In this study, 87% of adult patients in both the remdesivir arm and standard of care arm received dexamethasone. In contrast to these 2 studies, the DisCoVeRy trial demonstrated no difference for any clinical outcome when the use of remdesivir plus usual care was compared to usual care alone.²¹

The efficacy of remdesivir has not been evaluated in clinical trials of hospitalized children with COVID-19. A Phase 2/3, single-arm, open-label study evaluated the safety, tolerability, and pharmacokinetics of remdesivir in 53 hospitalized children with COVID-19.²² Children weighing 3 to < 40 kg received remdesivir 5 mg/kg on Day 1, followed by remdesivir 2.5 mg/kg daily. Adverse events included acute kidney injury (11%) and an increase in alanine transaminase levels (8%). However, this study did not have a placebo group, limiting the ability to draw conclusions regarding the significance of these adverse events. Published observational data are limited to descriptive case series.^{1,2,23} Other uncontrolled case series reported similar safety profiles.^{4,23-25}

Findings from the adult trials and the pediatric pharmacokinetic study led the Panel to recommend **remdesivir** for hospitalized children who have a new or increasing need for conventional oxygen (BIII) and to recommend **dexamethasone plus remdesivir** for children who require oxygen through a high-flow device or NIV (BIII). It is not known if remdesivir offers an additional clinical benefit to

standard care in younger children with SARS-CoV-2 infection who are receiving respiratory support for bronchiolitis, asthma, or croup.

For children hospitalized for COVID-19 who do not require supplemental oxygen, the Panel recommends **remdesivir** for children aged 12 to 17 years who are at the highest risk for progression to severe disease, especially those who are severely immunocompromised (**CIII**). This recommendation was extrapolated from the findings of the PINETREE study, which demonstrated a reduction in hospitalization among high-risk, unvaccinated adults treated in the outpatient setting.²⁶ However, there is insufficient evidence for or against the use of remdesivir in children aged 28 days to <12 years and weighing ≥ 3 kg who do not require supplemental oxygen. Given the reported clinical experience with the use of remdesivir in younger patients,²² the use of remdesivir in high-risk, younger children who do not require supplemental oxygen may be considered on a case-by-case basis.

Dexamethasone

Dexamethasone was evaluated in the RECOVERY trial, which was an open-label, randomized trial conducted in the United Kingdom.²⁷ The trial compared the use of up to 10 days of dexamethasone 6 mg, administered by intravenous injection or orally, with usual care among hospitalized adults with COVID-19. The primary outcome was all-cause mortality at 28 days, which occurred in 22.9% of patients randomized to receive dexamethasone versus 25.7% of patients randomized to receive usual care (age-adjusted rate ratio 0.83; 95% CI, 0.75–0.93; $P < 0.001$). Patients who required mechanical ventilation or extracorporeal membrane oxygenation (ECMO) had the greatest effect size (29.3% vs. 41.4%; rate ratio 0.64; 95% CI, 0.51–0.81). No difference in outcomes was observed in patients who did not require supplemental oxygen (17.8% vs. 14.0%; rate ratio 1.19; 95% CI, 0.92–1.55). For the 28-day mortality outcome, a difference between arms was observed in patients who required supplemental oxygen (23.3% vs. 26.2%; rate ratio 0.82; 95% CI, 0.72–0.94). However, it should be noted that these patients were a heterogeneous group, including those who received either conventional oxygen or NIV. See [Systemic Corticosteroids](#) for detailed information.

The safety and efficacy of using dexamethasone or other corticosteroids for the treatment of COVID-19 have not been evaluated in pediatric patients. Given that the mortality for adults in the placebo arm in the RECOVERY trial was substantially greater than the mortality generally reported for children with COVID-19, caution is warranted when extrapolating from recommendations for adults and applying them to patients aged <18 years.

However, because of the effect size observed in the RECOVERY trial, the Panel recommends the use of **dexamethasone** for children who require mechanical ventilation or ECMO (**AIII**). The Panel also recommends the use of **dexamethasone**, with or without concurrent **remdesivir**, for children who require oxygen through a high-flow device or NIV (**BIII**). The Panel **does not recommend** routine use of corticosteroids for children who require only conventional oxygen, but corticosteroids can be considered in combination with remdesivir for patients with increasing oxygen needs, particularly adolescents (**BIII**).

Evidence has demonstrated that the use of corticosteroids does not benefit infants with viral bronchiolitis not related to COVID-19, and current American Academy of Pediatrics guidelines recommend against the use of corticosteroids in this population.²⁸ There are no data specific to COVID-19 that support the use of corticosteroids in children with bronchiolitis due to SARS-CoV-2 infection. Corticosteroids should be used per the usual standards of care in children with asthma or croup triggered by SARS-CoV-2.

The use of dexamethasone for the treatment of severe COVID-19 in children who are profoundly

immunocompromised has not been evaluated, and there is a potential risk of harm. Therefore, the use of corticosteroids should be considered on a case-by-case basis in consultation with relevant specialists, and the benefits and risks of the therapy should be weighed. If dexamethasone is not available, alternative glucocorticoids such as prednisone, methylprednisolone, or hydrocortisone can be considered. The dexamethasone dose for pediatric patients is 0.15 mg/kg (with a maximum dose of 6 mg) once daily for ≤ 10 days or until hospital discharge, whichever comes first.

Baricitinib

The Janus kinase inhibitor baricitinib was approved by the FDA for the treatment of COVID-19 in hospitalized adults. An FDA Emergency Use Authorization for baricitinib remains active for the treatment of COVID-19 in hospitalized children aged 2 to 17 years who require supplemental oxygen, NIV, mechanical ventilation, or ECMO.²⁹

In the COV-BARRIER trial, adults with COVID-19 pneumonia were randomized to receive baricitinib or standard care. Patients treated with baricitinib showed a reduction in mortality when compared with those who received standard care.³⁰ The reduction was greatest in patients who received high-flow oxygen or NIV. Similarly, the ACTT-2 trial in adults showed that patients who received baricitinib plus remdesivir had improved time to recovery when compared with patients who received remdesivir alone.³¹ This effect was most pronounced in patients who received high-flow oxygen or NIV. In the ACTT-4 trial, 1,010 patients were randomized 1:1 to receive baricitinib plus remdesivir or dexamethasone plus remdesivir.³² The study reported no difference between the arms for the outcome of mechanical ventilation-free survival.

In the RECOVERY trial, 8,156 patients, including 33 children aged 2 to 17 years, were randomized to receive baricitinib or usual care (95% received corticosteroids).³³ Treatment with baricitinib was associated with a 13% proportional reduction in mortality, with the greatest effect size occurring in patients who received NIV. The RECOVERY investigators included these patients in a meta-analysis and found that treatment with baricitinib was associated with a 20% relative reduction in mortality (rate ratio 0.80; 95% CI, 0.72–0.89; $P < 0.0001$). See [Janus Kinase Inhibitors](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#) for additional information. These data in adults indicate that baricitinib is likely to be most beneficial for patients receiving noninvasive forms of respiratory support.

Several open-label trials and cohort studies have evaluated baricitinib in children with autoinflammatory and rheumatic diseases, including many children aged < 5 years, and found the treatment was well tolerated; however, the pharmacokinetics of baricitinib in younger children are not well studied.³⁴⁻³⁷ Information on the safety and effectiveness of the use of baricitinib in children with COVID-19 is limited to case reports.

In contrast to the strong recommendation for its use in adults, baricitinib is not considered the standard of care for all children who require high-flow oxygen or NIV because of the low mortality in children with COVID-19 (especially young children) and the limited data on the use of baricitinib in these children.

Extrapolating from clinical trials among adults with COVID-19, the Panel recommends that:

- For children who require oxygen through a high-flow device or NIV *and* do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, **baricitinib** can be considered for children aged 12 to 17 years (**BIII**) and for children aged 2 to 11 years (**CIII**).
- For children who require mechanical ventilation or ECMO *and* do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, **baricitinib** may be

considered for children aged 12 to 17 years (**BIII**) and for children aged 2 to 11 years (**CIII**).

Clinicians should consult with specialists experienced in treating children with immunosuppression (e.g., in pediatric infectious disease, pediatric rheumatology) when considering the use of baricitinib in hospitalized children with COVID-19. Data from adults indicate that baricitinib should be initiated promptly. Ideally, it should be initiated at the onset of clinical deterioration or respiratory failure.

Tofacitinib

There are no data on the efficacy of tofacitinib in pediatric patients with COVID-19; the Panel's recommendation is extrapolated from data in adults. The STOP-COVID trial compared tofacitinib to the standard of care in adults hospitalized for COVID-19 pneumonia.³⁸ The standard of care included glucocorticoids for most patients. The study demonstrated a reduction in mortality and respiratory failure at Day 28 for the tofacitinib arm when compared with the placebo arm. Tofacitinib has been studied less extensively than baricitinib for the treatment of COVID-19. Thus, tofacitinib, as an alternative to baricitinib, is recommended to be used in combination with dexamethasone in adults with COVID-19 who require high-flow oxygen or NIV. See [Janus Kinase Inhibitors](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#) for additional information.

No trials have evaluated the safety of using tofacitinib in children with COVID-19. Overall, there has been more clinical experience with the use of tofacitinib than baricitinib in children, particularly when used in children with juvenile idiopathic arthritis (JIA) as young as 2 years of age. A Phase 1 study was conducted to define the pharmacokinetics and safety of using tofacitinib in children,³⁹ and a Phase 3, double-blind, randomized, placebo-controlled trial investigated the efficacy of using tofacitinib in children with JIA.⁴⁰ Tofacitinib is available as a liquid formulation for children.

Given the established safety of tofacitinib in the pediatric population, **tofacitinib** can be considered an alternative for children hospitalized for COVID-19 if baricitinib is not available (**BIII**). The dose of tofacitinib that should be used to treat hospitalized children with COVID-19 has not been established. As with baricitinib, the dose of tofacitinib for hospitalized children with COVID-19 likely needs to be higher than the dose typically used to treat pediatric rheumatologic diseases. Therefore, clinicians should consult with specialists experienced in treating children with immunosuppression (e.g., in pediatric infectious disease, pediatric rheumatology) when considering administering tofacitinib to hospitalized children with COVID-19.

Tocilizumab

Tocilizumab is an interleukin-6 inhibitor that has received an FDA Emergency Use Authorization for the treatment of hospitalized adults and children with COVID-19 who are aged ≥ 2 years; receiving systemic corticosteroids; and require supplemental oxygen, NIV, mechanical ventilation, or ECMO.⁴¹ Two large randomized controlled trials (REMAP-CAP and RECOVERY) conducted among hospitalized adults with COVID-19 have demonstrated reductions in mortality with the use of tocilizumab.^{42,43} See [Interleukin-6 Inhibitors](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#) for additional information.

The RECOVERY trial was an open-label study that included hospitalized adults who had an oxygen saturation of $<92\%$ on room air or were receiving supplemental oxygen therapy; patients also had C-reactive protein levels ≥ 75 mg/L.⁴³ Patients were randomized to receive tocilizumab plus usual care or usual care alone. Mortality at 28 days was significantly lower in the tocilizumab arm compared to the usual care arm. The REMAP-CAP trial included adults with suspected or confirmed COVID-19 who were admitted to an intensive care unit and received either respiratory (i.e., NIV or mechanical ventilation) or cardiovascular organ (i.e., vasopressor/inotrope) support.⁴² Patients were randomized

within 24 hours of organ failure to receive either tocilizumab or sarilumab (the majority received tocilizumab) or to receive standard care. The median number of organ support-free days was higher for those who received tocilizumab than for those who received standard care, and in-hospital mortality was lower in the combined tocilizumab or sarilumab arm than in the standard care arm. In both studies, the majority of patients received dexamethasone (82% in the RECOVERY trial and 93% in the REMAP-CAP trial).

Studies have evaluated the use of tocilizumab for the treatment of non–COVID-19 conditions in children, including JIA^{44–48} and cytokine release syndrome related to chimeric antigen receptor T cell therapy.⁴⁹ The FDA approved tocilizumab for use in children aged ≥ 2 years for these indications.^{44–48} The use of tocilizumab in children with severe cases of COVID-19 has been described only in case series.^{50–52}

Extrapolating from clinical trials among adults with COVID-19, the Panel recommends that:

- For children who require oxygen through a high-flow device or NIV *and* do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, **tocilizumab** can be considered for children aged 12 to 17 years (**BIII**) and for children aged 2 to 11 years (**CIII**).
- For children who require mechanical ventilation or ECMO *and* do not have rapid (e.g., within 24 hours) improvement in oxygenation after initiation of dexamethasone, if tocilizumab has not been started, addition of **tocilizumab** may be considered for children aged 12 to 17 years (**BIII**) and for children aged 2 to 11 years (**CIII**).

Data from REMAP-CAP and RECOVERY are most likely to be applicable to high-risk adolescent patients. Clinicians should consult with specialists experienced in treating children with immunosuppression (e.g., in pediatric infectious disease, pediatric rheumatology) when considering the use of tocilizumab in younger children with COVID-19.

Sarilumab

Sarilumab, a monoclonal antibody that blocks interleukin-6 receptors, is not authorized by the FDA for the treatment of COVID-19. Data on the efficacy of sarilumab for the treatment of COVID-19 hyperinflammation are limited, and there is a lack of pediatric dosing information. Therefore, the Panel **recommends against** the use of **sarilumab** in hospitalized children with COVID-19, except in a clinical trial (**AIII**).

Anticoagulation in Children With COVID-19

Recommendations

- The Panel recommends prophylactic anticoagulation for children aged ≥ 12 years who are hospitalized for COVID-19, unless there are contraindications (**BIII**).
- Weighing the risk factors for thrombosis and bleeding, some Panel members would use prophylactic anticoagulation for children aged < 12 years who are hospitalized for COVID-19. Institutional standards for anticoagulation should be followed.
- There is insufficient evidence for the Panel to recommend either for or against the use of therapeutic anticoagulation for venous thromboembolism prophylaxis in children of any age with COVID-19.

Limited data characterize the risk of thromboembolic disease in children with COVID-19. Among children who do not have COVID-19, most thromboembolic events occur in neonates and adolescents.^{53,54} In a multicenter, retrospective cohort study that included 814 pediatric patients with

COVID-19 or MIS-C,⁵⁵ thromboembolic events were detected in 2.1% of patients with COVID-19 and in 6.5% of patients with MIS-C. A single-center study found that 1 of 596 participants (0.2%) with COVID-19 and 2 of 94 participants (2.1%) with MIS-C experienced thrombosis.⁵⁶

Limited data inform the clinical use of anticoagulation in children with COVID-19. Only the COVAC-TP trial has evaluated the dose, safety, and efficacy of prophylactic anticoagulants in children with COVID-19 or MIS-C.⁵⁷ In this multicenter, Phase 2 clinical trial of children hospitalized with COVID-19–related illness (including MIS-C) in the United States, a starting dose of enoxaparin 0.5 mg/kg achieved targeted anticoagulant activity (as measured by antifactor Xa level) in the majority of patients with few dose changes, and no patients experienced clinically relevant bleeding as defined by the International Society on Thrombosis and Haemostasis.⁵⁸ In this trial, thromboembolic events occurred in 2 patients (5.3%; 90% CI, 1.0%–15.7%); both events were related to central venous catheters.⁵⁷ These results raise the question of whether prophylactic doses of anticoagulants sufficiently reduce thromboembolism risk in children hospitalized with COVID or MIS-C.

To date, no clinical trial has evaluated the safety and efficacy of therapeutic anticoagulation in hospitalized children with COVID-19. Therefore, the Panel has determined that there is insufficient evidence to recommend either for or against the use of therapeutic anticoagulation for venous thromboembolism prophylaxis in children of any age with COVID-19.

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Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A

Last Updated: February 29, 2024

This section outlines the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations for the therapeutic management of pediatric patients with multisystem inflammatory syndrome in children (MIS-C). The case definition for MIS-C from the Council of State and Territorial Epidemiologists (CSTE) and the Centers for Disease Control and Prevention includes individuals aged <21 years.¹ The recommendations in this section encompass this age group. There are few randomized controlled trials that directly compared different treatment approaches for MIS-C, and these trials have limitations. Because of those limitations, the Panel's recommendations for the therapeutic management of patients with MIS-C are primarily based on data from large descriptive and observational comparative effectiveness studies. For information on the clinical manifestations of MIS-C, see [Special Considerations in Children](#).

Multisystem Inflammatory Syndrome in Adults

Adults can present with a syndrome similar to MIS-C, termed multisystem inflammatory syndrome in adults (MIS-A).² The published literature on MIS-A is restricted to small case series and observational epidemiologic studies that provide little data to guide treatment decisions in patients with MIS-A.³⁻⁵ Although the therapeutic management of MIS-A has not been studied, it is reasonable to extrapolate from data on treating patients with MIS-C to aid in the management of individuals with MIS-A.

Table 3d. Therapeutic Management of Hospitalized Patients With MIS-C

	Panel's Recommendations
MIS-C	Initial treatment for MIS-C includes both immunomodulatory and antithrombotic therapy. <i>Initial Immunomodulatory Therapy</i> • IVIG 2 g/kg IBW IV (up to a maximum total dose of 100 g) plus low to moderate dose methylprednisolone (1–2 mg/kg/day IV)^a or another glucocorticoid at an equivalent dose^a (AIIb). • Glucocorticoid monotherapy, only if IVIG is unavailable or contraindicated (BIIa). • IVIG monotherapy, only if glucocorticoids are contraindicated (BIIb). <i>Intensification Immunomodulatory Therapy</i> • Intensification therapy is recommended for children with refractory MIS-C who do not improve within 24 hours of receiving initial immunomodulatory therapy (AIII). One of the following can be used (listed in alphabetical order): • High-dose anakinra 5–10 mg/kg/day IV (preferred) or SUBQ in 1–4 divided doses^b (BIIb) • Higher-dose glucocorticoid (e.g., methylprednisolone 10–30 mg/kg/day IV for 1–3 days, up to a maximum of 1,000 mg/day, or equivalent glucocorticoid for 1–3 days)^{a,b} (BIIb) • Infliximab 5–10 mg/kg IV for 1 dose^{b,c} (BIIb) <i>Antithrombotic Therapy</i> • Low-dose aspirin (3–5 mg/kg PO once daily, up to a maximum dose of 81 mg) for all patients without risk factors for bleeding (AIII), AND • Anticoagulation for patients who fall under 1 of the following clinical scenarios: • Therapeutic anticoagulation for patients with large CAAs according to the American Heart Association guidelines for Kawasaki disease (AIII). • Therapeutic anticoagulation for patients with moderate to severe LV dysfunction who have no risk factors for bleeding (AIII). • For patients with MIS-C who do not have large CAAs or moderate to severe LV dysfunction, consider prophylactic or therapeutic anticoagulation on an individual basis, taking into consideration risk factors for thrombosis and bleeding. See Table 3e for additional information.
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.	

^a The duration of glucocorticoid therapy may vary. When a patient shows clinical improvement (e.g., resolution of fever, improvement of organ function, reduction of levels of inflammatory markers), a steroid taper should be initiated. Typically, the patient's clinical status guides the taper, and it continues for several weeks to avoid rebound inflammation. See Table 3e and text below.

^b In certain patients with severe illness, intensification therapy may include dual therapy with higher-dose glucocorticoids plus anakinra (**BIII**) or higher-dose glucocorticoids plus infliximab (**BIII**). **Anakinra and infliximab should not be used in combination.**

^c **Infliximab should not be used** in patients with macrophage activation syndrome.

Key: CAA = coronary artery aneurysm; IBW = ideal body weight; IV = intravenous; IVIG = intravenous immunoglobulin; LV = left ventricular; MIS-C = multisystem inflammatory syndrome in children; PO = oral; SUBQ = subcutaneous

Table 3e. Dosing Regimens for the Drugs Recommended for the Treatment of MIS-C

	Dosing Regimens <i>For infants, children, and adolescents unless otherwise specified.</i> <i>The doses listed are for FDA-approved indications for other diseases or from reported experiences or clinical trials.</i>	Adverse Events	Monitoring Parameters
Intravenous Immunoglobulin	IVIG 2 g/kg IBW IV (up to a maximum total dose of 100 g) In the event of cardiac dysfunction or fluid overload, consider administering IVIG in divided doses (i.e., IVIG 1 g/kg IBW IV, up to 50 g daily for 2 doses).	• Hypersensitivity • Fever • Chills • Flushing • Hemolytic anemia	• Renal function • Urine output • CBC with differential • Infusion-related AEs • Anaphylaxis • Signs and symptoms of hemolysis
Methyl-prednisolone	Methylprednisolone 1–2 mg/kg IV every 12 hours Intensification immunomodulatory therapy: If the patient does not respond to 1–2 mg/kg IV every 12 hours, increase the dose to 10–30 mg/kg/day IV for 1–3 days (up to a maximum of 1,000 mg/day). Glucocorticoids should be tapered gradually after signs of clinical improvement. The duration of the taper will depend on the patient's clinical status.	• Adrenal suppression • Hyperglycemia • Sodium retention • Fluid retention • Leukocytosis • Immune suppression • Psychiatric disturbances	• Blood pressure • CBC with differential • BMP
Anakinra	Anakinra 5–10 mg/kg/day IV (preferred) or SUBQ in 1 to 4 divided doses	• Headache • Fever • Hypersensitivity • Immune suppression • Transaminase elevation • Injection site reactions (for SUBQ)	• CBC with differential • LFTs • SCr
Infliximab	Infliximab 5–10 mg/kg IV for 1 dose	• Infusion-related reaction • Headache • Immune suppression	• Monitor vital signs every 2–10 minutes during infusion. • CBC with differential
Aspirin	Aspirin 3–5 mg/kg PO once daily (up to a maximum of 81 mg)	• Gastrointestinal ulcers • Hypersensitivity	• Signs or symptoms of bleeding • Renal function
Enoxaparin	For Prophylaxis <i>Aged >2 Months to <18 Years</i> • 0.5 mg/kg SUBQ every 12 hours (up to maximum of 30 mg) For Treatment <i>Aged >2 Months to <18 Years</i> • 1 mg/kg SUBQ every 12 hours • Monitor antifactor Xa activity (treatment goal: 0.5–1 unit/mL).	• Increased risk of bleeding	• CBC with differential • Renal function

Key: AE = adverse event; BMP = basic metabolic panel; CBC = complete blood count; FDA = Food and Drug Administration; IBW = ideal body weight; IV = intravenous; IVIG = intravenous immunoglobulin; LFT = liver function test; MIS-C = multisystem inflammatory syndrome in children; PO = oral; SCr = serum creatinine; SUBQ = subcutaneous

Treatment Considerations for Children With MIS-C

Initial Immunomodulatory Therapy for MIS-C

The Panel recommends consulting with a multidisciplinary team when managing immunomodulatory therapy for children with MIS-C (**AIII**). The multidisciplinary team may include experts in cardiology, hematology, infectious disease, intensive care, and rheumatology. MIS-C is defined by multiorgan dysfunction, and input from other pediatric subspecialists may be needed depending on the presentation of the individual patient. Thus, children with MIS-C should be cared for at centers with access to these pediatric specialists.

Intravenous immunoglobulin (IVIG) and glucocorticoids are the most commonly used immunomodulatory medications in reported cohorts of children with MIS-C.⁶⁻¹⁴ The American College of Rheumatology has outlined initial diagnostic and treatment considerations for patients with MIS-C and recommends using IVIG in combination with glucocorticoids as first-tier therapy for most hospitalized children with MIS-C.¹⁵ Several nonrandomized studies suggest that the use of IVIG plus glucocorticoids is associated with less treatment failure, faster recovery of cardiac function, shorter intensive care unit (ICU) stays, and less need for treatment escalation than IVIG monotherapy.^{7,16-22} Based on these data, the Panel recommends using **IVIG** in combination with low to moderate doses of glucocorticoids for children hospitalized with MIS-C (**AIIb**).

IVIG should be administered at a dose of 2 g/kg of ideal body weight, with a maximum total dose of 100 g. The patient's cardiac function and fluid status should be monitored carefully during the IVIG infusion. IVIG can be given in divided doses of 1 g/kg of ideal body weight over 2 days if there is a concern about the patient's fluid status. Methylprednisolone 1 to 2 mg/kg/day, or another glucocorticoid at an equivalent dose, is considered low to moderate glucocorticoid dosing. When the patient has improved clinically (e.g., when the patient is afebrile, end-organ dysfunction resolves, and inflammatory markers trend downward), a steroid taper should be initiated. Typically, the patient's clinical status guides the taper, and it continues for several weeks to avoid rebound inflammation.

Glucocorticoid monotherapy is an alternative initial treatment for MIS-C. Some studies have shown that patients treated with this approach had similar outcomes to patients treated with IVIG monotherapy or IVIG plus glucocorticoids.^{18,20,23,24} However, secondary analyses indicate that patients who were initially treated with IVIG plus glucocorticoids had faster time to improvement, less need for treatment escalation, and faster time to defervescence than patients who received glucocorticoid monotherapy.²⁰ Thus, the combination of IVIG and glucocorticoids appears to provide additional benefits that are not provided by glucocorticoid monotherapy.

Initial treatment that includes IVIG is also beneficial because it reduces the frequency of coronary artery aneurysms (CAAs) in patients with Kawasaki disease.^{14,25} Kawasaki disease is increasingly difficult to differentiate from MIS-C, and more recent SARS-CoV-2 variants have resulted in MIS-C presentations that are similar to Kawasaki disease.²⁶ Distinguishing MIS-C from Kawasaki disease is further complicated by the fact that seropositivity for SARS-CoV-2 is now widespread, which makes establishing the epidemiological link required for the MIS-C diagnosis difficult. For these reasons, the Panel recommends using **IVIG** plus glucocorticoids as the initial therapy for patients with MIS-C (**AIIb**). **Glucocorticoid monotherapy** is recommended **only** if IVIG is unavailable or contraindicated (**BIIa**). **IVIG monotherapy** is recommended **only** if glucocorticoids are contraindicated (**BIIb**).

Clinical Data on Initial Immunomodulatory Therapy for MIS-C

Intravenous Immunoglobulin in Combination With Glucocorticoids

No randomized clinical trials evaluating the use of IVIG plus glucocorticoids for the treatment of MIS-C

have been completed. The comparative benefit of adding steroids to IVIG for MIS-C treatment has been estimated in observational cohort studies that used statistical techniques to adjust for confounders. The first of these studies employed observational data from a national surveillance system cohort in France and used propensity matching to compare short-term outcomes in children with MIS-C who were treated initially with IVIG 2 gm/kg alone or IVIG plus methylprednisolone (most patients received 1.6–2 mg/kg/day for 5 days).¹⁶ The study team observed a lower risk of treatment failure (defined as a fever that persisted for 2 days after treatment or recurrent fever within 7 days), less need for hemodynamic support, less severe left ventricular dysfunction, and shorter ICU stays among the children who were initially treated with the combination therapy. This was a small study, and only 32 patients treated with IVIG plus methylprednisolone and 64 patients treated with IVIG alone could be matched based on propensity score.

A larger study in the United States analyzed data from the Overcoming COVID-19 surveillance registry to evaluate immunomodulatory therapy for MIS-C.¹⁷ The study included 103 patients who received initial treatment with IVIG plus glucocorticoids and an equal number of propensity score–matched patients who received IVIG alone. The risk of cardiovascular dysfunction on or after Day 2 was measured among these patients using a composite outcome of left ventricular ejection fraction of <55% or vasopressor use. The composite outcome occurred in 17% of patients in the IVIG plus glucocorticoids arm and in 31% of patients in the IVIG alone arm (risk ratio 0.56; 95% CI, 0.34–0.94). In addition, patients treated with the combination of IVIG and glucocorticoids were less likely to require adjunctive immunomodulatory therapy than those treated with IVIG alone. Methylprednisolone, the glucocorticoid that was prescribed most often, was administered to 353 patients (68% of patients, including nonpropensity score–matched patients, in the entire cohort). Among these patients, the dosing of methylprednisolone ranged from 2 mg/kg/day in 284 patients (80%) to 10 to 30 mg/kg/day in 69 patients (20%).

A third study, the international, observational BATS study, compared patients with MIS-C who received IVIG alone (n = 246) with those who received IVIG plus glucocorticoids (n = 208).¹⁸ This study found similar rates for the composite outcome of inotropic support or mechanical ventilation by Day 2 or later or death in both treatment arms. The composite outcome occurred in 44 of 221 patients (21%) in the IVIG alone arm and in 56 of 180 patients (31%) in the IVIG plus glucocorticoids arm (OR 0.77; 95% CI, 0.33–1.82). However, escalation of immunomodulatory treatment was less common among the patients who received IVIG plus glucocorticoids than among those who received IVIG alone (OR 0.18; 95% CI, 0.10–0.33). Notably, the study allowed for the inclusion of patients who had any inflammatory illness after acute COVID-19 but who did not meet the Centers for Disease Control and Prevention or World Health Organization (WHO) criteria for MIS-C. This multicenter study included sites from 34 countries, which introduced the potential for more variability in supportive care. In addition, the overall percentage of patients with abnormal cardiac findings (12% of the 538 patients) was lower than in other cohorts.

Intravenous Immunoglobulin Monotherapy

The use of IVIG is long established for patients with Kawasaki disease, a syndrome that has overlapping manifestations with MIS-C. Thus, the product's safety profile is well understood. In patients with Kawasaki disease, IVIG reduces the development of CAAs,^{25,27} a complication also observed in some patients with MIS-C. IVIG is the most frequently used therapy for MIS-C. In a national survey of U.S. institutional protocols for managing MIS-C, IVIG was the first-line therapy in 98% of 40 participating centers.²⁸

Data on the efficacy of IVIG in patients with MIS-C is extrapolated from case series that show mostly favorable outcomes. In a series of 539 MIS-C cases, 77% of the children received IVIG.¹⁴ A sizeable proportion of these children had reduced left ventricular ejection fraction at admission (172 of 503 evaluable patients [34.2%]). The symptom resolved by Day 30 in 156 of the children (90.7%). Although these studies have not described the occurrence of specific adverse events related to IVIG use, the

dosing used (IVIG 2 g/kg) has a well-established safety profile when used for Kawasaki disease.

A limitation of all published studies on IVIG use for MIS-C is the frequent and often rapid sequential addition of other immunomodulatory therapies, such as corticosteroids. In addition, there is accumulating evidence that glucocorticoids administered in combination with IVIG are more effective as treatment for MIS-C. However, IVIG monotherapy may be a reasonable treatment option for a small subset of patients with MIS-C who are stable (i.e., not in shock or with organ-threatening disease) and have contraindications for glucocorticoid therapy. Such contraindications may include concern about the impact of corticosteroids on the diagnostic evaluation or an underlying medical condition.

Glucocorticoid Monotherapy

The observational BATS study also compared initial IVIG monotherapy treatment (n = 246) with glucocorticoid treatment (n = 99) and found no differences in primary or secondary outcomes between these 2 cohorts.¹⁸ However, in a subgroup analysis of patients who met the WHO criteria for MIS-C, the glucocorticoid arm (n = 78) had significantly fewer patients who required respiratory support by Day 2 or later or died than the IVIG arm (n = 192).

In a subsequent publication, the BATS consortium reported on additional patients with MIS-C who were enrolled in the study (over 2,000 patients in total).²⁰ The study had 2 primary outcomes. The first was a composite of the need for inotropic or ventilator support on or after Day 2 or death. The second was time to improvement by 1 level on an ordinal severity scale. In this larger study, there was once again no difference in the primary outcomes among the arms in a propensity-weighted analysis (combination therapy with IVIG plus glucocorticoids was compared to IVIG alone, and glucocorticoid monotherapy was compared to IVIG alone).

In secondary analyses, patients who received combination therapy had a lower rate of treatment escalation than those who received IVIG alone, and patients who received glucocorticoid monotherapy had a lower rate of treatment escalation than those who received IVIG alone.²⁰ The combination therapy arm had a faster time to improvement, less need for treatment escalation, and a lower rate of persistent fever on Day 2 than the glucocorticoid monotherapy arm. These treatment arms had similar frequencies of CAAs measured at hospital discharge and similar CAA severity. Among the 236 patients with documented CAAs during the initial hospitalization, 196 patients had follow-up echocardiograms. More than 90% of the CAAs resolved, with similar rates of resolution across the treatment groups.

As reported in the initial publication for the observational BATS study, the inclusion criteria were broad, and the patients did not need to meet the full WHO case definition for MIS-C.²⁰ Compared to the other treatment arms, the IVIG plus glucocorticoid arm had a greater proportion of patients that met the WHO case definition for MIS-C, were ventilated and/or treated with inotropes at Day 0, and had CAAs (even before the initiation of immunomodulators). Many patients received additional immunomodulatory agents after Day 1, including 230 of 487 patients in the initial glucocorticoids alone group who also received IVIG. Finally, COVID-19 vaccination has been associated with reduced incidence and severity of MIS-C,²⁹⁻³¹ but vaccination was not evaluated in the study.

To date, the only randomized trial that evaluated treatments in patients with MIS-C was conducted in Switzerland.²³ This open-label, multicenter study compared methylprednisolone 10 mg/kg per day for 3 days (n = 37) to a single dose of IVIG 2 gm/kg (n = 38). In this study, patients met the criteria for the case definition of pediatric inflammatory multisystem syndrome—temporally associated with SARS-CoV-2 (PIMS-TS). There was no difference between the arms for the primary outcome of length of hospital stay or death. The length of hospital stay from admission to discharge was 6 days for both arms (estimated effect size -0.037 of the log₁₀ transformed times; 95% CI, -0.13 to 0.065; *P* = 0.42). No deaths were reported in either arm.

In a secondary analysis, 27% of patients in the glucocorticoid arm required respiratory support compared to 55% of those in the IVIG arm, which was a statistically significant result.²³ There was no difference between the arms for the occurrence of coronary artery enlargement. The small sample size in this study limited the power for treatment comparisons, and many patients received additional therapies for MIS-C after randomization.

Intensification Immunomodulatory Therapy for MIS-C

Children with MIS-C typically respond briskly to immunomodulatory therapy and show clinical improvements within the first 24 hours of treatment. Treatment response is characterized by resolution of fever, improvement of organ function, and reduction of levels of inflammatory markers, particularly C-reactive protein. In contrast, refractory disease is often accompanied by persistent fever, worsening organ dysfunction, and increasing levels of inflammatory markers. Intensification therapy is recommended for children with refractory MIS-C who do not improve within 24 hours of receiving initial immunomodulatory therapy (**AIII**). Children with uncontrolled MIS-C despite treatment with IVIG and low to moderate doses of glucocorticoids will often continue to deteriorate, and this decline in clinical status can be quite rapid.

No studies have compared the different intensification therapies used for patients with MIS-C. The data on this topic are limited to results from cohort studies in patients with MIS-C, opinions from experts, and clinician experiences treating children with other hyperinflammatory syndromes, such as Kawasaki disease and macrophage activation syndrome. For children with refractory MIS-C, the Panel recommends providing additional immunomodulatory therapy (listed in alphabetical order) with **high-dose anakinra (BIIB)**, **higher-dose glucocorticoids (BIIB)**, or **infliximab (BIIB)**. Currently, there is insufficient evidence to determine which of these agents is most effective for intensification therapy in patients with refractory MIS-C. In certain patients with severe illness, intensification therapy may include dual therapy with **higher-dose glucocorticoids plus anakinra (BIII)** or **higher-dose glucocorticoids plus infliximab (BIII)**. **Anakinra and infliximab should not be used in combination.** A second dose of IVIG is not commonly reported in the literature as a strategy for intensification therapy in patients with MIS-C. This may be due to the high rates of IVIG resistance, the rapid pace of disease escalation, and the risk for fluid overload in patients with MIS-C.¹⁰ Therefore, the Panel **recommends against** a second dose of **IVIG** for intensification therapy in patients with refractory MIS-C (**BIII**).

Patients with MIS-C who receive multiple immunomodulatory agents are at risk for infection and need to be monitored carefully. Most children with MIS-C were previously healthy. In patients who have an immune disorder or are taking immunosuppression therapy, the risk of infection is greater. The risks and benefits of using immunomodulatory agents in patients with MIS-C who are immunocompromised need to be evaluated on a case-by-case basis.

Clinical Data on Intensification Immunomodulatory Therapy for MIS-C

High-Dose Glucocorticoids

The use of high-dose glucocorticoids in pediatric patients with other inflammatory conditions, such as Kawasaki disease and macrophage activation syndrome, is well established.^{32,33} High-dose glucocorticoid therapy is defined as intravenous (IV) administration of 10 to 30 mg/kg/day of methylprednisolone (or an equivalent corticosteroid). Often, this higher dose of glucocorticoids is administered for 1 to 3 days before returning to low to moderate doses (1–2 mg/kg/day). Multiple observational studies have evaluated the use of high-dose glucocorticoids in children with MIS-C.^{17,34} In addition, single-center treatment protocols for MIS-C that incorporate high-dose glucocorticoids into the treatment algorithm have been published.¹⁹ Implementation of the protocols has resulted in positive clinical outcomes in patients with MIS-C.

Anakinra

Anakinra is the most commonly used biologic medication for the treatment of MIS-C in the United States.²⁸ Multiple noncomparative, observational cohorts have reported on the use of anakinra in patients with MIS-C.^{10,11,13,35-39} This medication has been used extensively and has a good safety record in pediatric patients with other hyperinflammatory syndromes (e.g., systemic juvenile idiopathic arthritis, macrophage activation syndrome).⁴⁰⁻⁴² Anakinra has also been used successfully to treat IVIG-resistant Kawasaki disease. Anakinra has a short half-life (4–6 hours), and the medication can be stopped quickly, which many providers regard as a benefit relative to longer-acting immunomodulators. The Panel's recommendation for the use of high-dose anakinra (5–10 mg/kg/day) in patients with MIS-C is based on the demonstrated efficacy of high-dose anakinra in patients with macrophage activation syndrome. The duration of anakinra therapy varies in the literature. Some clinicians prescribe anakinra as a steroid-sparing agent to manage MIS-C for longer periods (e.g., up to 2 weeks).

Infliximab

The Panel recommends a single dose of infliximab 5 to 10 mg/kg IV as an option for intensification therapy. Infliximab has been studied for the treatment of MIS-C in a single-center retrospective study that compared patients treated with IVIG alone (n = 20) to those treated with IVIG and a single dose of infliximab 10 mg/kg IV (n = 52).⁴³ Infliximab was used as the first-line therapy in this study, and the patients were not treated with glucocorticoids. The patients who received IVIG and infliximab were more likely to be admitted to the ICU and had more severe illness than those who received IVIG alone. However, the patients who received the combination therapy were less likely to require additional therapy after 24 hours (the primary outcome). In addition, patients who received IVIG and infliximab had shorter stays in the ICU and improved cardiac outcomes. These results suggest that infliximab has a therapeutic effect in patients with MIS-C.

Infliximab is approved by the Food and Drug Administration for use in children with inflammatory bowel disease and is used widely to treat juvenile idiopathic arthritis. Infliximab has also been used to treat IVIG-resistant Kawasaki disease.^{44,45} Although the half-life of infliximab in patients with MIS-C is unknown, it likely has effects that persist for several weeks. This extended period of drug activity may provide a steroid-sparing effect in patients with MIS-C.

Antithrombotic Therapy for MIS-C

The Panel recommends the use of **low-dose aspirin** for patients with MIS-C who do not have risk factors for bleeding (**AIII**). This recommendation is largely derived from experience treating children with Kawasaki disease and the increased likelihood of analogous platelet activation and endothelial dysfunction in children with MIS-C.⁴⁶ Children treated with aspirin and steroids should also receive prophylactic H2 blockers or proton pump inhibitors. Patients with MIS-C who have large CAAs (Z-score ≥ 10) should receive **therapeutic anticoagulation** according to the [American Heart Association guidelines for Kawasaki disease](#) (**AIII**). Children with left ventricular dysfunction are at risk for intracardiac thrombosis. Patients with MIS-C and moderate to severe left ventricular dysfunction who have no risk factors for bleeding should receive **therapeutic anticoagulation** (**AIII**).

There is less consensus on the use of either prophylactic or therapeutic anticoagulation in patients with MIS-C who do not have large CAAs or moderate to severe left ventricular dysfunction. Children with MIS-C have marked elevations in D-dimer levels and other abnormalities of coagulation, which suggests that they may be at increased risk for thrombosis.⁴⁷ In a multicenter retrospective study of children with acute COVID-19 and MIS-C, the independent risk factors for thrombosis included indwelling catheters, older age (>12 years), malignancy, admission to the ICU, and elevated D-dimer levels.⁴⁸ In a multicenter, Phase 2 trial of enoxaparin thromboprophylaxis in children hospitalized for COVID-19 and MIS-C

(COVAC-TP), children with MIS-C frequently exhibited hyperfibrinogenemia and had significantly elevated D-dimer levels compared to children with primary SARS-CoV-2 infection.⁴⁹

Published data on the risk of bleeding in children with MIS-C who are managed with anticoagulant thromboprophylaxis are limited. Major bleeding events (as defined by the International Society on Thrombosis and Haemostasis) were observed in patients with MIS-C who were treated with anticoagulation in the aforementioned retrospective study⁴⁸ but not in the COVAC-TP trial, which employed prophylactic dosing of enoxaparin and permitted the use of aspirin at a dose of up to 5 mg/kg/day.⁴⁹ However, 5% of patients developed catheter-related thromboembolic events despite the use of enoxaparin thromboprophylaxis in the COVAC-TP trial.

Given the uncertainty regarding the benefit of anticoagulation in the treatment of MIS-C, prophylactic or therapeutic anticoagulation for children with MIS-C who do not have large CAAs or moderate to severe left ventricular dysfunction should be considered on a case-by-case basis, taking into account the risk factors for thrombosis and bleeding.

Antiviral Therapy for MIS-C

The role of SARS-CoV-2 antiviral therapy in treating MIS-C has not been systematically studied. However, it is not expected to be beneficial because MIS-C is considered an immune-mediated phenomenon that occurs weeks after primary SARS-CoV-2 infection. Therefore, the Panel **recommends against** the use of SARS-CoV-2 antiviral therapy for patients with MIS-C (AIII).

Critical Care Management

Shock occurs in approximately 50% of patients with MIS-C and may include elements of distributive, cardiogenic, or hypovolemic shock.^{14,50,51} In general, clinicians should manage shock in patients with MIS-C per the usual critical care standards outlined in the pediatric Surviving Sepsis Campaign guidelines.⁵²

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Care of Critically Ill Adults With COVID-19

Last Updated: February 29, 2024

Summary Recommendations

Hemodynamics

- For adults with COVID-19 and shock, the COVID-19 Treatment Guidelines Panel (the Panel) recommends using dynamic parameters, capillary refill time, and/or lactate levels over static parameters to assess fluid responsiveness (**BIIa**).
- For acute resuscitation in adults who have COVID-19 and shock, there is insufficient evidence for the Panel to recommend either for or against the use of balanced crystalloids, such as Ringer's lactate solution, over normal saline.
- For acute resuscitation in adults with COVID-19 and shock, the Panel **recommends against** the initial use of **albumin (BI)**.
- For adults with COVID-19 and shock, the Panel recommends **norepinephrine** as the first-choice vasopressor (**AI**).
- For adults with COVID-19 and shock, the Panel recommends titrating vasoactive agents and targeting a mean arterial pressure (MAP) of 60 to 65 mm Hg over targeting a higher MAP (**BI**).
- The Panel **recommends against** using hydroxyethyl starches for intravascular volume replacement in adult patients with COVID-19 and sepsis or septic shock (**AI**).
- As a second-line vasopressor, the Panel recommends adding either **vasopressin** (up to 0.03 units/min) (**BIIa**) or **epinephrine (BIIb)** to norepinephrine to raise MAP to the target or adding **vasopressin** (up to 0.03 units/min) (**BIIa**) to decrease the norepinephrine dose.
- The Panel **recommends against** using **low-dose dopamine** for renal protection (**AI**).
- The Panel recommends using **dobutamine** in adult patients with COVID-19 who show evidence of cardiac dysfunction and persistent hypoperfusion despite adequate fluid loading and the use of vasopressor agents (**BIII**).
- The Panel recommends that all adult patients with COVID-19 who require vasopressors have an arterial catheter placed as soon as practical, if resources are available (**BIII**).
- For adult patients with refractory septic shock who have completed a course of corticosteroids to treat COVID-19, the Panel recommends using low-dose corticosteroid therapy ("shock-reversal") over no corticosteroid therapy (**BIIa**).

Oxygenation and Ventilation

- For adults with COVID-19 and acute hypoxemic respiratory failure despite conventional oxygen therapy, the Panel recommends starting therapy with high-flow nasal cannula (HFNC) oxygen; if patients fail to respond, noninvasive ventilation or intubation and mechanical ventilation should be initiated (**BIIa**).
- For adults with COVID-19 and acute hypoxemic respiratory failure despite conventional oxygen therapy who do not have an indication for endotracheal intubation and for whom HFNC oxygen is not available, the Panel recommends performing a closely monitored trial of noninvasive ventilation (**BIIa**).
- For adults with persistent hypoxemia who require HFNC oxygen and for whom endotracheal intubation is not indicated, the Panel recommends a trial of awake prone positioning (**BIIa**).
- The Panel **recommends against** the use of awake prone positioning as a rescue therapy for refractory hypoxemia to avoid intubation in patients who otherwise meet the indications for intubation and mechanical ventilation (**AIII**).
- If intubation becomes necessary, the procedure should be performed by an experienced practitioner in a controlled setting due to the enhanced risk of exposing health care practitioners to SARS-CoV-2 during intubation (**AIII**).
- For mechanically ventilated adults with COVID-19 and acute respiratory distress syndrome (ARDS):
 - The Panel recommends using low tidal volume (VT) ventilation (VT 4–8 mL/kg of predicted body weight) over higher VT ventilation (VT >8 mL/kg) (**AI**).
 - The Panel recommends targeting plateau pressures of <30 cm H₂O (**AIIa**).
 - The Panel recommends using a conservative fluid strategy over a liberal fluid strategy (**BIIa**).
 - The Panel **recommends against** the routine use of inhaled nitric oxide (**AIIa**).

Summary Recommendations, continued

- For mechanically ventilated adults with COVID-19 and moderate to severe ARDS:
 - The Panel recommends using a higher positive end-expiratory pressure (PEEP) strategy over a lower PEEP strategy **(BIIa)**.
 - For mechanically ventilated adults with COVID-19 and refractory hypoxemia despite optimized ventilation, the Panel recommends prone ventilation for 12 to 16 hours per day over no prone ventilation **(BIIa)**.
 - The Panel recommends using, as needed, intermittent boluses of neuromuscular blocking agents or a continuous neuromuscular blocking agent infusion to facilitate protective lung ventilation **(BIIa)**.
- For mechanically ventilated adults with COVID-19, severe ARDS, and hypoxemia despite optimized ventilation and other rescue strategies:
 - The Panel recommends using an inhaled pulmonary vasodilator as a rescue therapy; if rapid improvement in oxygenation is not observed, the treatment should be tapered **(CIII)**.
 - The Panel recommends using recruitment maneuvers rather than not using recruitment maneuvers **(CIIa)**.
 - If recruitment maneuvers are used, the Panel **recommends against** the use of staircase (incremental PEEP) recruitment maneuvers **(AIIa)**.

Pharmacologic Interventions

- In the absence of a proven or suspected secondary infection, the Panel **recommends against** the use of empiric broad-spectrum antimicrobials in patients with severe or critical COVID-19 **(BIII)**.
- As with any hospitalized patient, patients with COVID-19 who receive antimicrobials should be reassessed daily to minimize the adverse consequences of unnecessary antimicrobial therapy **(AIII)**.

Extracorporeal Membrane Oxygenation

- There is insufficient evidence for the Panel to recommend either for or against the use of extracorporeal membrane oxygenation in adults with COVID-19–associated ARDS and refractory hypoxemia.

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

Introduction to Critical Care Management of Adults With COVID-19

Last Updated: February 29, 2024

COVID-19 can progress to critical illness, including hypoxemic respiratory failure, acute respiratory distress syndrome (ARDS), septic shock, thromboembolic disease, renal and hepatic dysfunction, cardiac dysfunction, central nervous system disease, and exacerbation of underlying comorbidities in both adults and children.

In these Guidelines, many of the early recommendations for managing critically ill adults with COVID-19 were extrapolated from experience with other causes of sepsis and respiratory failure.¹ However, the amount of research on the management of these patients has grown, and the COVID-19 Treatment Guidelines Panel's (the Panel) current recommendations have been informed by that research.

Treating patients with COVID-19 in the intensive care unit (ICU) often requires managing underlying illnesses or COVID-19–related morbidities. Clinicians also need to focus on preventing ICU-related complications, as they would for any patient admitted to the ICU.

Selected Clinical Manifestations of COVID-19 Critical Illness

Severe Pulmonary Disease

Most patients who are critically ill with COVID-19 have severe pulmonary disease. Almost all of these patients meet the diagnostic criteria for ARDS. Patients with COVID-19 and severe pulmonary involvement often manifest extrapulmonary disease and exhibit laboratory markers of acute inflammation. Patients typically progress to critical illness 10 to 12 days after the onset of COVID-19 symptoms.

Inflammatory Response

Many critically ill patients with COVID-19 meet the criteria for virus-induced sepsis because they have life-threatening organ dysfunction related to a dysregulated host response to SARS-CoV-2 infection.² Patients with COVID-19 may express increased levels of pro-inflammatory cytokines and anti-inflammatory cytokines, a condition that has been called “cytokine release syndrome” or a “cytokine storm,” but these terms are imprecise and misnomers. The magnitude of cytokine elevation in many critically ill patients with COVID-19 is modest compared to the levels in patients with sepsis and ARDS not related to COVID-19.^{3,4}

Thromboembolic Events

Prothrombotic states and higher rates of venous thromboembolic disease have been observed in adults who are critically ill with COVID-19. In some studies, thromboemboli were diagnosed in patients who received prophylactic doses of heparin.⁵⁻⁷ Autopsy studies provide additional evidence of thromboembolic disease and microvascular thrombosis in patients with COVID-19.⁸ See [Antithrombotic Therapy in Patients With COVID-19](#) for a more detailed discussion.

Renal and Hepatic Dysfunction

Although SARS-CoV-2 is primarily a pulmonary pathogen, renal and hepatic dysfunction are consistently described in adults with severe COVID-19.^{9,10} In a cohort of critically ill adults in Brazil, the development of acute kidney injury that required renal replacement therapy was associated with a poor

prognosis.¹¹ In addition, liver and renal dysfunction may be related to medication side effects or result from shock and poor oxygen delivery.¹⁰

Cardiac Dysfunction, Including Myocarditis

The published literature describes cardiac injury or dysfunction in some (up to 24% in the early years of the pandemic) hospitalized adults with COVID-19.¹² COVID-19 may be associated with an array of cardiovascular complications, including acute coronary syndrome, myocarditis, stress (takotsubo) cardiomyopathy, arrhythmias, and thromboembolic disease.¹³

Central and Peripheral Nervous System Dysfunction

Neurologic manifestations in critically ill patients with acute COVID-19 include thromboembolic or hemorrhagic stroke, cerebral sinus venous thrombosis, seizure, myopathy, and meningoencephalitis.^{14,15} Neurologic manifestations are more common in patients with severe disease.¹⁶ Neuropathologic autopsy studies have reported both macrovascular and microvascular thrombosis with evidence of hypoxic ischemia.¹⁷ Guillain-Barré syndrome has been associated with recent SARS-CoV-2 infection.¹⁸ Critically ill patients with COVID-19 may present with delirium or develop delirium during hospitalization.¹⁹ Risk factors associated with delirium include the use of mechanical ventilation, restraints, benzodiazepines, opioids, vasopressors, and antipsychotics.^{19,20} Adequate management of critically ill patients with COVID-19 includes the use of best practices for sedation and monitoring for stroke.

Multisystem Inflammatory Syndrome in Adults

Case reports have described patients who had minimal respiratory symptoms during a recent or current SARS-CoV-2 infection but who were hospitalized with symptoms such as fever or signs of shock.²¹ Laboratory evidence indicated that these patients had severe inflammation. The patients also had signs of cardiovascular, gastrointestinal, dermatologic, and neurologic disease. This constellation of signs and symptoms has been designated multisystem inflammatory syndrome in adults (MIS-A). The Centers for Disease Control and Prevention has developed a [case definition for MIS-A](#). This syndrome is similar to multisystem inflammatory syndrome in children (MIS-C), which has been well described.

A diagnosis of MIS-A may be made after other causes for the condition (e.g., bacterial sepsis) have been excluded. Although there are currently no controlled clinical trial data from patients with MIS-A to guide treatment of the syndrome, case reports have described the use of intravenous immunoglobulin, corticosteroids, or interleukin-1 inhibitor therapy.²²⁻²⁴ Some observational evidence supports the Panel's recommendations for the therapeutic management of MIS-C, and those recommendations can be applied to MIS-A in most circumstances. See [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#) for more information.

Other Complications Related to Intensive Care Units

When treating patients with COVID-19, clinicians need to minimize the risk of conventional ICU complications. Patients who are critically ill with COVID-19 are at risk for nosocomial infections, such as ventilator-associated pneumonia, hospital-acquired pneumonia, and catheter-related bloodstream infections, and for other complications of critical illness care.

Additional Considerations

Drug-Drug Interactions

All patients in the ICU should be routinely monitored for drug-drug interactions. The potential for drug-drug interactions between investigational medications or any medications used off-label to treat

COVID-19 and concurrent drugs should be considered.

Sedation Management

International guidelines provide recommendations on the prevention, detection, and treatment of pain, sedation, and delirium in ICU patients.^{25,26} Sedation management strategies, such as maintaining a light level of sedation (when appropriate) and minimizing sedative exposure, have shortened the duration of mechanical ventilation and the length of ICU stay among patients without COVID-19.^{27,28}

The Society of Critical Care Medicine (SCCM) ICU Liberation Campaign promotes the ICU Liberation Bundle (A–F) to improve post-ICU patient outcomes.²⁹ The A–F Liberation Bundle includes the following elements:

- A. Assess, prevent, and manage pain
- B. Both spontaneous awakening trials and breathing trials
- C. Choice of analgesia and sedation
- D. Delirium: assess, prevent, and manage
- E. Early mobility and exercise
- F. Family engagement and empowerment

The A–F Liberation Bundle provides frontline staff with practical application strategies for each element, and an interprofessional team model should be used to incorporate the elements.²⁹ This approach helps standardize communication among team members, improves survival, and reduces long-term cognitive dysfunction of patients.³⁰ The elements of the A–F Liberation Bundle represent the steps required to implement the SCCM Clinical Practice Guidelines for the Prevention and Management of Pain, Agitation/Sedation, Delirium, Immobility, and Sleep Disruption in Adult Patients in the ICU.²⁵

Despite the known benefits of the A–F Liberation Bundle,²⁹ its impact has not been directly assessed in patients with COVID-19. However, implementing the elements of the bundle should be encouraged to improve patient outcomes in the ICU.

Some factors may impede routine implementation of the A–F Liberation Bundle and increase the risk of ICU and post-ICU complications. For example, staff workload increases when patients with COVID-19 require prolonged mechanical ventilation, deep sedation, or neuromuscular blockade. In addition, drug shortages, which were common early in the pandemic, may affect the choice of analgesia or sedation. During a drug shortage, older sedatives that have prolonged durations of action and active metabolites are more likely to be prescribed.

Post-Intensive Care Syndrome

Post-intensive care syndrome (PICS) is a spectrum of physical, cognitive, and/or psychiatric disability that affects survivors of critical illness and persists after a patient leaves the ICU.^{31,32} Patients with PICS may present with varying levels of impairment, including profound muscle weakness (ICU-acquired weakness); problems with thinking and judgment (cognitive dysfunction); and mental health problems, such as problems sleeping, post-traumatic stress disorder (PTSD), depression, and anxiety. ICU-acquired weakness affects 33% of all patients who receive mechanical ventilation, 50% of patients with sepsis, and $\leq 50\%$ of patients who remain in the ICU for ≥ 1 week.^{33–35} Cognitive dysfunction affects 30% to 80% of patients discharged from the ICU.^{36–38} About 50% of ICU survivors do not return to work within 1 year after discharge.³⁹

Although no single risk factor has been associated with PICS, there are opportunities to minimize

the risk of PICS through medication management (using the A–F Liberation Bundle), physical rehabilitation, follow-up clinics, family support, and improved education about the syndrome. PICS also affects family members who participate in the care of their loved ones. One study reported that a third of family members who had major decision-making roles experienced mental health problems, such as depression, anxiety, and post-traumatic stress disorder.⁴⁰

Some patients with COVID-19 who have been treated in the ICU express manifestations of PICS.⁴¹ Although specific therapies for PICS induced by COVID-19 are not yet available, health care providers should be aware that cognitive impairment or related problems may develop in patients who have had severe or critical COVID-19.

Advance Care Planning and Goals of Care

The advance care plans and the goals of care for all critically ill patients must be assessed at hospital admission and regularly thereafter. This is an essential element of care for all patients. Information on palliative care for patients with COVID-19 can be found at the [National Coalition for Hospice and Palliative Care](https://www.nationalcoalitionhospiceandpalliativecare.org/) website.

To guide shared decision making in cases of serious illness, advance care planning should include identifying existing advance directives that outline a patient's preferences and values. Values and care preferences should be discussed, documented, and revisited regularly for patients with or without prior directives. Specialty palliative care teams can facilitate communication between clinicians and surrogate decision makers, support frontline clinicians, and provide direct patient care services.

At hospital admission, surrogate decision makers should be identified for all critically ill patients with COVID-19. Early in the pandemic, infection-control policies for COVID-19 often created communication barriers for surrogate decision makers. At that time, most discussions between clinicians and surrogate decision makers about treatment options occurred through telecommunication. However, many of those policies have been rescinded, and health care providers and surrogate decision makers should communicate in person when possible.

Acknowledgments

The Surviving Sepsis Campaign (SSC), an initiative supported by SCCM and the European Society of Intensive Care Medicine, issued the Guidelines on the Management of Critically Ill Adults with Coronavirus Disease 2019 (COVID-19) in March 2020, and a revised version was published in March 2021.¹ The Panel based its recommendations for the care of critically ill adults with COVID-19 on the SSC COVID-19 guidelines with permission, and the Panel gratefully acknowledges the work of the SSC COVID-19 Guidelines panel. The Panel also acknowledges the contributions and expertise of Andrew Rhodes, MBBS, MD, of St. George's University Hospitals in London, England, and Waleed Alhazzani, MBBS, MSc, of McMaster University in Hamilton, Canada.

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Hemodynamics for Adults

Last Updated: February 29, 2024

Most of the hemodynamic recommendations below are similar to those published in Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock 2021.¹ Ultimately, adult patients with COVID-19 who require fluid resuscitation or hemodynamic management of shock should be treated and managed as adult patients with septic shock.

Recommendation

- For adults with COVID-19 and shock, the COVID-19 Treatment Guidelines Panel (the Panel) recommends using dynamic parameters, capillary refill time, and/or lactate levels over static parameters to assess fluid responsiveness (**BIIa**).

Rationale

In a systematic review and meta-analysis of 13 randomized clinical trials in intensive care unit (ICU) patients without COVID-19 ($n = 1,652$), the use of dynamic assessment to guide fluid therapy reduced mortality (risk ratio 0.59; 95% CI, 0.42–0.83), ICU length of stay (weighted mean difference -1.16 days; 95% CI, -1.97 to -0.36), and duration of mechanical ventilation (weighted mean difference -2.98 hours; 95% CI, -5.08 to -0.89).² The dynamic parameters used in these trials included stroke volume variation, pulse pressure variation, and stroke volume change after a passive leg raise or fluid challenge. Passive leg raising followed by assessment of pulse pressure variation and stroke volume variation appears to predict fluid responsiveness with the greatest accuracy.³ The static parameters included some components of early goal-directed therapy (e.g., central venous pressure, mean arterial pressure [MAP]).

In patients who did not have COVID-19, resuscitation therapies for shock, as indicated by serum lactate levels, were summarized in a systematic review and meta-analysis of 7 randomized controlled trials ($n = 1,301$).⁴ When compared with therapy guided by central venous oxygen saturation levels, therapy directed by early lactate clearance was associated with a reduction in mortality (relative ratio 0.68; 95% CI, 0.56–0.82), shorter ICU stay (mean difference -1.64 days; 95% CI, -3.23 to -0.05), and shorter duration of mechanical ventilation (mean difference -10.22 hours; 95% CI, -15.94 to -4.50).

Recommendation

- For acute resuscitation in adults who have COVID-19 and shock, there is insufficient evidence for the Panel to recommend either for or against the use of balanced crystalloids, such as Ringer's lactate solution, over normal saline.

Rationale

The composition of sodium, potassium, and chloride found in balanced crystalloids, such as Ringer's lactate solution, is similar to the composition found in extracellular fluid. The use of balanced crystalloids may prevent hyperchloremic metabolic acidosis, which has been associated with administration of large quantities of normal saline.⁵ Observational data have suggested an association between normal saline and acute kidney injury and higher risk of death,⁶ and many Panel members with experience in acute fluid resuscitation use balanced crystalloids.

A pragmatic randomized trial compared the use of balanced and unbalanced crystalloids for intravenous fluid administration in critically ill adults without COVID-19 ($n = 15,802$).⁷ The rate of the composite

outcome of death, new renal-replacement therapy, or persistent renal dysfunction was lower in the balanced crystalloids arm than in the unbalanced crystalloids arm (OR 0.90; 95% CI, 0.82–0.99; $P = 0.04$). A secondary analysis compared outcomes in a subset of patients with sepsis ($n = 1,641$).⁸ Compared to treatment with unbalanced crystalloids, treatment with balanced crystalloids resulted in fewer deaths (aOR 0.74; 95% CI, 0.59–0.93; $P = 0.01$) and more days free of vasopressors and renal replacements.

A subsequent meta-analysis of 21 non-COVID-19 randomized controlled trials ($n = 20,213$), which included the pragmatic trial cited above,⁷ compared balanced crystalloids to 0.9% saline for resuscitation in critically ill adults and children.⁹ The meta-analysis reported nonsignificant differences between the treatment arms for hospital mortality (OR 0.91; 95% CI, 0.83–1.01) and acute kidney injury (OR 0.92; 95% CI, 0.84–1.00). In a trial of more than 11,000 patients who were in the ICU and required fluid resuscitation, the use of balanced fluids, compared with the use of normal saline, did not reduce mortality at 90 days.¹⁰ Similarly, in a large trial of critically ill adults in Australia and New Zealand, the use of balanced crystalloids for fluid therapy, compared with the use of normal saline, did not reduce the incidence of acute kidney injury or the risk of death at 90 days.¹¹

Recommendation

- For acute resuscitation in adults with COVID-19 and shock, the Panel **recommends against** the initial use of **albumin (BI)**.

Rationale

A meta-analysis of 20 non-COVID-19 randomized controlled trials ($n = 13,047$) that compared the use of albumin or fresh-frozen plasma to crystalloids in critically ill patients found no difference in all-cause mortality between the treatment arms.¹² In contrast, a meta-analysis of 17 non-COVID-19 randomized controlled trials ($n = 1,977$) that compared the use of albumin to crystalloids in patients with sepsis reported a reduction in mortality among the patients who received albumin (OR 0.82; 95% CI, 0.67–1.0; $P = 0.047$).¹³ Given the higher cost of albumin and the lack of a definitive clinical benefit, for acute resuscitation in adults with COVID-19 and shock, the Panel **recommends against** the initial use of **albumin (BI)**.

Recommendation

- For adults with COVID-19 and shock, the Panel recommends **norepinephrine** as the first-choice vasopressor (**AI**).

Rationale

Due to its vasoconstrictive effects, norepinephrine increases MAP with little change to heart rate and less increase in stroke volume than dopamine. Dopamine increases MAP and cardiac output, primarily due to increases in stroke volume and heart rate. Norepinephrine is more potent than dopamine and may be more effective at reversing hypotension in patients with septic shock. Dopamine may be particularly useful in patients with compromised systolic function, but it causes more tachycardia and may be more arrhythmogenic than norepinephrine.¹⁴ Dopamine may also influence the endocrine response via the hypothalamic pituitary axis and have immunosuppressive effects.¹⁵ A systematic review and meta-analysis of 11 non-COVID-19 randomized controlled trials that compared vasopressors used to treat patients with septic shock found that norepinephrine use resulted in lower all-cause mortality (risk ratio 0.89; 95% CI, 0.81–0.98) and a lower risk of arrhythmias (risk ratio 0.48; 95% CI, 0.40–0.58) than dopamine use.¹⁶ Although the beta-1 activity of dopamine would be useful in patients with myocardial dysfunction, the greater risk of arrhythmias limits its use.^{17,18}

Recommendation

- For adults with COVID-19 and shock, the Panel recommends titrating vasoactive agents and targeting a MAP of 60 to 65 mm Hg over targeting a higher MAP (**BI**).

Rationale

A meta-analysis of individual patient data from 2 non-COVID-19 randomized controlled trials (n = 894) compared higher versus lower blood pressure targets for vasopressor therapy in adult patients with shock.¹⁹ The study reported no significant differences between the higher-target and lower-target arms for 28-day mortality (OR 1.15; 95% CI, 0.87–1.52), 90-day mortality (OR 1.08; 95% CI, 0.84–1.44), myocardial injury (OR 1.47; 95% CI, 0.64–3.56), or limb ischemia (OR 0.92; 95% CI, 0.36–2.10). The risk of arrhythmia was increased in the higher-target arm (OR 2.50; 95% CI, 1.35–4.77).

Similarly, the 65 trial, a randomized controlled trial in patients without COVID-19 (n = 2,463), reported no significant difference in mortality between patients that received vasopressor therapy guided by a target MAP of 60 to 65 mm Hg and those that received treatment guided by a higher, standard-of-care target MAP (41% vs. 43.8%; relative risk 0.93; 95% CI, 0.85–1.03).²⁰ Given the indication of improved outcome with lower MAP targets (and no firm indication of harm), the Panel recommends titrating vasoactive agents and targeting a MAP of 60 to 65 over targeting a higher MAP (**BI**).

Additional Recommendations for Adults With COVID-19 and Shock

- The Panel **recommends against** using hydroxyethyl starches for intravascular volume replacement in adult patients with COVID-19 and sepsis or septic shock (**AI**).
- As a second-line vasopressor, the Panel recommends adding either **vasopressin** (up to 0.03 units/min) (**BIIa**) or **epinephrine** (**BIIB**) to norepinephrine to raise MAP to the target or adding **vasopressin** (up to 0.03 units/min) (**BIIa**) to decrease the norepinephrine dose.
- The Panel **recommends against** using **low-dose dopamine** for renal protection (**AI**).
- The Panel recommends using **dobutamine** in adult patients with COVID-19 who show evidence of cardiac dysfunction and persistent hypoperfusion despite adequate fluid loading and the use of vasopressor agents (**BIII**).
- The Panel recommends that all adult patients with COVID-19 who require vasopressors have an arterial catheter placed as soon as practical, if resources are available (**BIII**).
- For adult patients with refractory septic shock who have completed a course of corticosteroids to treat COVID-19, the Panel recommends using low-dose corticosteroid therapy (“shock-reversal”) over no corticosteroid therapy (**BIIa**).
 - A typical corticosteroid regimen in septic shock is intravenous hydrocortisone 200 mg once daily administered either as an infusion or in intermittent doses. The duration of hydrocortisone therapy is usually a clinical decision.
 - Adult patients who are receiving corticosteroids for COVID-19 are receiving sufficient replacement therapy and do not require additional hydrocortisone.

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Oxygenation and Ventilation for Adults

Last Updated: December 20, 2023

The COVID-19 Treatment Guidelines Panel's (the Panel) recommendations in this section were informed by the Surviving Sepsis Campaign [guidelines for managing sepsis](#) and [guidelines for managing COVID-19](#) in adults.

Severe illness in people with COVID-19 typically occurs approximately 1 week after the onset of symptoms. The most common symptom is dyspnea, which is often accompanied by hypoxemia. Patients with severe disease typically require supplemental oxygen and should be monitored closely for worsening respiratory status, because some patients may progress to acute respiratory distress syndrome (ARDS).

Goal of Oxygenation

The optimal oxygen saturation measured by pulse oximetry (SpO_2) in adults with COVID-19 who are receiving supplemental oxygen is unknown. However, a target SpO_2 of 92% to 96% seems logical, considering that indirect evidence from patients without COVID-19 suggests that an SpO_2 <92% or >96% may be harmful.^{1,2} Special care should be taken when assessing SpO_2 in patients with darker skin pigmentation, as recent reports indicate that occult hypoxemia (defined as arterial oxygen saturation [SaO_2] <88% despite an SpO_2 >92%) is more common in these patients.^{3,4} See [Clinical Spectrum of SARS-CoV-2 Infection](#) for more information.

The potential harm of maintaining an SpO_2 <92% was demonstrated during a trial that randomly assigned patients with ARDS who did not have COVID-19 to either a conservative oxygen strategy (target SpO_2 of 88% to 92%) or a liberal oxygen strategy (target SpO_2 $\geq$ 96%).¹ The trial was stopped early due to futility after enrolling 205 patients, but increased mortality was observed at Day 90 in the conservative oxygen strategy arm (between-group risk difference 14%; 95% CI, 0.7% to 27%), and a trend toward increased mortality was observed at Day 28 (between-group risk difference 8%; 95% CI, -5% to 21%).

The results of a meta-analysis of 25 randomized trials that involved patients without COVID-19 demonstrated the potential harm of maintaining an SpO_2 >96%.² This study found that a liberal oxygen supplementation strategy (a median fraction of inspired oxygen [FiO_2] of 0.52) was associated with an increased risk of in-hospital mortality (relative risk 1.21; 95% CI, 1.03–1.43) when compared with a more conservative SpO_2 supplementation strategy (a median FiO_2 of 0.21).

Acute Hypoxemic Respiratory Failure

In adults with COVID-19 and acute hypoxemic respiratory failure, conventional oxygen therapy may be insufficient to meet the oxygen needs of the patient. Options for providing enhanced respiratory support include using high-flow nasal canula (HFNC) oxygen, noninvasive ventilation (NIV), intubation and mechanical ventilation, or extracorporeal membrane oxygenation. In this section, mechanical ventilation refers to the delivery of positive pressure ventilation through an endotracheal or tracheostomy tube. NIV refers to the delivery of either continuous positive airway pressure (CPAP) or bilevel positive airway pressure (e.g., BiPAP) through a noninvasive interface, such as a face mask or nasal mask.

Nonmechanically Ventilated Adults With Acute Hypoxemic Respiratory Failure

High-Flow Nasal Cannula Oxygen and Noninvasive Ventilation

Recommendations

- For adults with COVID-19 and acute hypoxemic respiratory failure despite conventional oxygen

therapy, the Panel recommends starting therapy with HFNC oxygen; if patients fail to respond, NIV or intubation and mechanical ventilation should be initiated (**BIIa**).

- For adults with COVID-19 and acute hypoxemic respiratory failure despite conventional oxygen therapy who do not have an indication for endotracheal intubation and for whom HFNC oxygen is not available, the Panel recommends performing a closely monitored trial of NIV (**BIIa**).

Rationale

Several studies have informed clinical practice on the optimal oxygen delivery system for patients with COVID-19 and acute hypoxemic respiratory failure. A randomized study of 711 patients with COVID-19 in 34 intensive care units (ICUs) in France compared HFNC oxygen delivery to oxygen delivery through a nonrebreather mask.⁵ The patients had acute respiratory failure with a ratio of arterial partial pressure of oxygen to fraction of inspired oxygen ($\text{PaO}_2/\text{FiO}_2$) ≤ 200 mm Hg. The mean FiO_2 was 0.58 in both arms. Although the difference between arms for the primary endpoint of 28-day mortality was not statistically significant (10% in the HFNC oxygen arm vs. 11% in the conventional oxygen arm; absolute difference -1.2%; 95% CI, -5.8% to 3.4%; $P = 0.60$), the intubation rate was significantly lower in the HFNC oxygen arm than in the conventional oxygen arm. Unless a contraindication exists, most Panel members would switch to HFNC oxygen delivery for patients with respiratory failure who do not require mechanical ventilation but have worsening hypoxemia or increased work of breathing despite receiving conventional oxygen at flow rates up to 10 L/min.

For patients with COVID-19 and acute hypoxemic respiratory failure who do not respond to conventional oxygen therapy, HFNC oxygen is preferred over NIV. No studies directly compare HFNC oxygen with mask-delivered NIV in patients with COVID-19; therefore, this guidance is based on an unblinded clinical trial in patients without COVID-19 who had acute hypoxemic respiratory failure.⁶ Study participants were randomized to receive HFNC oxygen, conventional oxygen therapy, or NIV. The patients in the HFNC oxygen arm had more ventilator-free days (mean 24 days) than those in the conventional oxygen therapy arm (mean 22 days) or the NIV arm (mean 19 days; $P = 0.02$). In addition, the conventional oxygen therapy arm (HR 2.01; 95% CI, 1.01–3.99) and the NIV arm (HR 2.50; 95% CI, 1.31–4.78) had higher 90-day mortality than the HFNC oxygen arm. In the subgroup of patients with severe hypoxemia (those with $\text{PaO}_2/\text{FiO}_2 \leq 200$ mm Hg), the HFNC oxygen arm had a lower intubation rate than the conventional oxygen therapy arm (HR 2.07) and the NIV arm (HR 2.57).

The trial's findings were corroborated by a meta-analysis of 8 trials with 1,084 participants that assessed the effectiveness of oxygenation strategies.⁷ Compared to NIV, HFNC oxygen reduced the rate of intubation (OR 0.48; 95% CI, 0.31–0.73) and ICU mortality (OR 0.36; 95% CI, 0.20–0.63).

One small study compared the use of NIV delivered by a helmet device to HFNC oxygen in patients with COVID-19. The HENIVOT trial randomized 109 patients with moderate to severe COVID-19 (defined as those who had $\text{PaO}_2/\text{FiO}_2 \leq 200$ mm Hg) to receive either NIV via a helmet device or HFNC oxygen.⁸ The study found no difference between the arms for the primary outcome of respiratory support-free days. However, only 30% of patients in the NIV arm required endotracheal intubation compared to 51% of patients in the HFNC oxygen arm ($P = 0.03$).

Two larger studies compared the use of NIV with conventional oxygen therapy in patients with COVID-19. The RECOVERY-RS trial was an adaptive randomized controlled trial that was essentially conducted as 2 separate trials that compared NIV and HFNC oxygen to the same conventional oxygen therapy control group.⁹ The trial was stopped early and enrolled fewer than a third of the planned sample size of 4,002 participants. Between April 2020 and May 2021, 1,273 adults with acute hypoxemic respiratory failure related to COVID-19 were randomized to receive NIV ($n = 380$), HFNC oxygen ($n = 418$), or conventional oxygen therapy ($n = 475$). The primary endpoint was a composite of endotracheal intubation or death within 30 days. The proportion of patients who met the primary endpoint was

significantly lower in the NIV arm than in the conventional oxygen therapy arm (36.3% vs. 44.4%; $P = 0.03$). This difference was not due to mortality but was entirely due to a reduction in the number of patients who required intubation. There was no significant difference between the HFNC oxygen arm and the conventional oxygen therapy arm in the occurrence of the primary endpoint (44.3% vs. 45.1%; $P = 0.83$).

There was substantial crossover between the arms, but an inverse probability weighting analysis that corrected for the bias this may have introduced did not change the results.⁹ Adverse events were more common in the NIV arm. Initially, a comparison between NIV and HFNC oxygen was not planned, but a post hoc analysis found that the proportion of patients who required endotracheal intubation or who died was lower in the NIV arm than in the HFNC oxygen arm (34.6% vs. 44.3%; $P = 0.02$).

In contrast to the RECOVERY-RS trial, the HiFlo-COVID trial randomized 220 patients with COVID-19 to receive HFNC oxygen or conventional oxygen therapy and found that a smaller proportion of patients in the HFNC oxygen arm required intubation (34.3% vs. 51.0%; $P = 0.03$).¹⁰ Patients in the HFNC arm also had a shorter median time to recovery (11 vs. 14 days; $P = 0.047$).

The conflicting results of these studies make drawing inferences from the data difficult. Additionally, the RECOVERY-RS trial was stopped long before it reached its planned sample size for reasons not related to futility, efficacy, or harm; inferring benefit in this context is questionable. The Panel recognizes that for patients who need more oxygen support than a conventional nasal cannula can provide, most clinicians will administer oxygen via HFNC and subsequently progress to NIV if needed. Therefore, the pertinent clinical question is whether HFNC oxygen or NIV should be used when a patient does not respond to conventional oxygen therapy. Other than the post hoc analysis in the RECOVERY-RS trial, no study has specifically investigated this question.

NIV is an aerosol-generating procedure, and studies of SARS-CoV show that it may increase the risk of nosocomial transmission.^{11,12} For patients with SARS-CoV-2, it remains unclear whether the use of HFNC oxygen results in a lower risk of nosocomial transmission than the use of NIV.

Awake Prone Positioning in Nonmechanically Ventilated Adults

Recommendations

- For adults with persistent hypoxemia who require HFNC oxygen and for whom endotracheal intubation is not indicated, the Panel recommends a trial of awake prone positioning (**BIIa**).
- The Panel **recommends against** the use of awake prone positioning as a rescue therapy for refractory hypoxemia to avoid intubation in patients who otherwise meet the indications for intubation and mechanical ventilation (**AIII**).

Additional Considerations

- Patients who can adjust their position independently and tolerate lying prone can be considered for awake prone positioning.
- Awake prone positioning is acceptable and feasible for pregnant patients and can be performed in the left lateral decubitus position or the fully prone position.¹³
- Some patients do not tolerate awake prone positioning. Failure rates as high as 63% have been reported in the literature.¹⁴
- Awake prone positioning **should not be used** as a substitute for intubation and mechanical ventilation in patients with refractory hypoxemia who otherwise meet the indications for these interventions.
- Awake prone positioning may be infeasible or impractical in patients with:

- Spinal instability
- Facial or pelvic fractures
- An open chest or unstable chest wall
- Awake prone positioning should be used with caution in patients with confusion, delirium, or hemodynamic instability; patients who cannot independently change position; or patients who have had recent abdominal surgery, nausea, or vomiting.

Rationale

Awake prone positioning, or having a nonintubated patient lie on the stomach, may improve oxygenation and prevent the patient from progressing to requiring intubation and mechanical ventilation. Although prone positioning has been shown to improve oxygenation and outcomes in patients with moderate to severe ARDS who are receiving mechanical ventilation,^{15,16} there is less evidence regarding the benefit of prone positioning in awake patients who require supplemental oxygen without mechanical ventilation. Several case series of patients with COVID-19 who required oxygen or NIV have reported that awake prone positioning improved oxygenation,¹⁷⁻²⁰ and some series have also reported low intubation rates after awake prone positioning.^{19,20}

The Awake Prone Positioning Meta-Trial Group has conducted the largest trial on awake prone positioning.²¹ This study was a prospective, multinational meta-trial of 6 open-label, randomized, controlled, superiority trials that compared awake prone positioning to standard care in adults who required HFNC oxygen for acute hypoxemic respiratory failure due to COVID-19.

The study enrolled 1,126 patients between April 2, 2020, and January 26, 2021, and the intention-to-treat analysis included 1,121 patients.²¹ Of the 564 patients who underwent awake prone positioning, 223 (40%) met the composite primary endpoint of intubation or death within 28 days of enrollment. Among the 557 patients who received standard care, 257 (46%) met the primary endpoint (relative risk 0.86; 95% CI, 0.75–0.98). The incidence of intubation by Day 28 was lower in the awake prone positioning arm than in the standard care arm (HR intubation 0.75; 95% CI, 0.62–0.91). There was no difference in 28-day mortality between the awake prone positioning arm and the standard care arm (HR mortality 0.87; 95% CI, 0.68–1.11).

During the first 14 days of the study, the median daily duration of awake prone positioning was 5.0 hours (IQR 1.6–8.8 hours).²¹ However, the median daily duration varied from 1.6 hours to 8.6 hours across the individual trials. Longer daily durations for awake prone positioning were associated with treatment success by Day 28. This study evaluated the incidences of certain adverse events, including skin breakdown, vomiting, and central or arterial line dislodgment. These events occurred infrequently during the study, and the incidences were similar in each arm. No cardiac arrests occurred during awake prone positioning.

The optimal daily duration of awake prone positioning is unclear. In the meta-trial of awake prone positioning, only 25 of 151 patients (17%) who had an average of ≥ 8 hours of awake prone positioning per day met the primary endpoint of intubation or death when compared with 198 of 413 patients (48%) who remained in awake prone positioning for < 8 hours per day.²¹ This result is consistent with past clinical trials of prone positioning in mechanically ventilated patients with ARDS, in which clinical benefits were observed after longer durations of prone positioning.^{15,16}

Intubation for Mechanical Ventilation

Recommendation

- If intubation becomes necessary, the procedure should be performed by an experienced

practitioner in a controlled setting due to the enhanced risk of exposing health care practitioners to SARS-CoV-2 during intubation (**AIII**).

Rationale

It is essential to closely monitor hypoxemic patients with COVID-19 for signs of respiratory decompensation. To ensure the safety of both patients and health care workers, intubation should be performed in a controlled setting by an experienced practitioner.

Mechanically Ventilated Adults

General Considerations

Recommendations

For mechanically ventilated adults with COVID-19 and ARDS:

- The Panel recommends using low tidal volume (VT) ventilation (VT 4–8 mL/kg of predicted body weight) over higher VT ventilation (VT >8 mL/kg) (**AI**).
- The Panel recommends targeting plateau pressures of <30 cm H₂O (**AIIa**).
- The Panel recommends using a conservative fluid strategy over a liberal fluid strategy (**BIIa**).
- The Panel **recommends against** the routine use of inhaled nitric oxide (**AIIa**).

Rationale

There is no evidence that the ventilator management of patients with hypoxemic respiratory failure due to COVID-19 should differ from the ventilator management of patients with hypoxemic respiratory failure due to other causes.

Positive End-Expiratory Pressure and Prone Positioning in Mechanically Ventilated Adults With Moderate to Severe ARDS

Recommendations

For mechanically ventilated adults with COVID-19 and moderate to severe ARDS:

- The Panel recommends using a higher positive end-expiratory pressure (PEEP) strategy over a lower PEEP strategy (**BIIa**).
- For mechanically ventilated adults with COVID-19 and refractory hypoxemia despite optimized ventilation, the Panel recommends prone ventilation for 12 to 16 hours per day over no prone ventilation (**BIIa**).

Rationale

PEEP is beneficial in patients with ARDS because it prevents alveolar collapse, improves oxygenation, and minimizes atelectrauma, a source of ventilator-induced lung injury. A meta-analysis of individual patient data from the 3 largest trials that compared lower and higher levels of PEEP in patients without COVID-19 found that less ICU mortality and in-hospital mortality was associated with higher levels of PEEP in those with moderate (PaO₂/FiO₂ 100–200 mm Hg) and severe (PaO₂/FiO₂ <100 mm Hg) ARDS.²²

Although there is no clear standard for a high level of PEEP, a conventional threshold is >10 cm H₂O.²³ Recent reports have suggested that, in contrast to patients with ARDS not caused by COVID-19, some patients with moderate or severe ARDS due to COVID-19 have normal static lung compliance. In these patients, high levels of PEEP may cause harm by compromising hemodynamics and cardiovascular performance.^{24,25} Other studies have reported that patients with moderate to severe ARDS due to

COVID-19 had low lung compliance, similar to the lung compliance seen in patients with conventional ARDS.²⁶⁻²⁹ These seemingly contradictory observations suggest that patients with COVID-19 and ARDS are a heterogeneous population, and assessments for responsiveness to high levels of PEEP should be individualized based on oxygenation and lung compliance. Clinicians should monitor patients for known side effects of high levels of PEEP, such as barotrauma and hypotension.

In the prepandemic PROSEVA study of patients with moderate to severe early ARDS ($\text{PaO}_2/\text{FiO}_2 < 150$ mm Hg) who required mechanical ventilation, the patients who were randomized to undergo prone positioning for ≥ 16 hours per day had improved survival compared to those who remained in the supine position throughout the course of mechanical ventilation.¹⁵ A meta-analysis evaluated the results of the PROSEVA study and 7 other randomized controlled trials that investigated the use of prone positioning in people with ARDS.³⁰ A subgroup analysis revealed that mortality was reduced among patients who remained prone for ≥ 12 hours per day when compared with patients who remained in the supine position (risk ratio 0.74; 95% CI, 0.56–0.99). Prone positioning improved oxygenation in all the trials. Patients in the prone positioning arms had higher $\text{PaO}_2/\text{FiO}_2$ on Day 4 than those in the supine positioning arms (mean difference 23.5 mm Hg; 95% CI, 12.4–34.5).

The use of prone positioning may be associated with serious adverse events, including unplanned extubation or central catheter removal. However, the meta-analysis found no differences between the prone positioning and supine positioning arms in the frequency of these events.³⁰ The use of prone positioning was associated with an increased risk of pressure sores (risk ratio 1.22; 95% CI, 1.06–1.41) and endotracheal tube obstruction (risk ratio 1.76; 95% CI, 1.24–2.50) in the 3 studies that evaluated these complications.

Neuromuscular Blockade in Mechanically Ventilated Adults With Moderate to Severe ARDS

Recommendation

For mechanically ventilated adults with COVID-19 and moderate to severe ARDS:

- The Panel recommends using, as needed, intermittent boluses of neuromuscular blocking agents (NMBAs) or a continuous NMBA infusion to facilitate protective lung ventilation (**BIIa**).

Rationale

Although the use of NMBAs in patients with ARDS reduces ventilator dyssynchrony, a large multicenter trial across several ICUs reported no significant difference in mortality between patients who received deep sedation and continuous NMBA infusion and patients who received a usual-care approach of lighter sedation without routine NMBAs.³¹

Rescue Therapies for Mechanically Ventilated Adults With ARDS

Recommendations

For mechanically ventilated adults with COVID-19, severe ARDS, and hypoxemia despite optimized ventilation and other rescue strategies:

- The Panel recommends using an inhaled pulmonary vasodilator as a rescue therapy; if rapid improvement in oxygenation is not observed, the treatment should be tapered (**CIII**).
- The Panel recommends using recruitment maneuvers rather than not using recruitment maneuvers (**CIIa**).
- If recruitment maneuvers are used, the Panel **recommends against** the use of staircase (incremental PEEP) recruitment maneuvers (**AIIa**).

Rationale

A recruitment maneuver refers to a temporary increase in airway pressure during mechanical ventilation to open collapsed alveoli and improve oxygenation. No studies have assessed the effect of recruitment maneuvers on oxygenation in patients with severe ARDS due to COVID-19. However, a systematic review and meta-analysis of 6 trials of recruitment maneuvers in patients with ARDS who did not have COVID-19 found that recruitment maneuvers reduced mortality, improved oxygenation 24 hours after the maneuver, and decreased the need for rescue therapy.³² Because recruitment maneuvers can cause barotrauma or hypotension, patients should be closely monitored during the maneuvers. If a patient decompensates during recruitment maneuvers, the maneuvers should be stopped immediately.

The importance of properly performing recruitment maneuvers was illustrated by an analysis of 8 randomized controlled trials in patients without COVID-19 ($n = 2,544$) that found that recruitment maneuvers did not reduce in-hospital mortality (risk ratio 0.90; 95% CI, 0.78–1.04).²³ However, a subgroup analysis found that traditional recruitment maneuvers significantly reduced in-hospital mortality (risk ratio 0.85; 95% CI, 0.75–0.97). Mortality was higher among patients treated with incremental PEEP titration recruitment maneuvers than among those treated with traditional recruitment maneuvers, but this difference was not statistically significant (risk ratio 1.06; 95% CI, 0.97–1.17).

There are no prospective trials of pulmonary vasodilators in people with COVID-19. However, a meta-analysis of mostly small, retrospective trials did not show improved outcomes.³³ A Cochrane review of 13 trials evaluated the use of inhaled nitric oxide in patients with ARDS who did not have COVID-19 and found no reduction in mortality.³⁴ Because the review showed a transient benefit for oxygenation, it is reasonable to attempt using inhaled nitric oxide as a rescue therapy in patients with COVID-19 and severe ARDS after other options have failed. However, if the use of nitric oxide does not improve a patient's oxygenation, it should be tapered quickly to avoid rebound pulmonary vasoconstriction, which may occur when nitric oxide is discontinued after prolonged use.

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Pharmacologic Interventions for Critically Ill Patients

Last Updated: December 20, 2023

Empiric Broad-Spectrum Antimicrobial Therapy

Recommendations

- In the absence of a proven or suspected secondary infection, the COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** the use of empiric broad-spectrum antimicrobials in patients with severe or critical COVID-19 (**BIII**).
- As with any hospitalized patient, patients with COVID-19 who receive antimicrobials should be reassessed daily to minimize the adverse consequences of unnecessary antimicrobial therapy (**AIII**).

Rationale

Variable rates of community- and hospital-acquired infections have been reported in adult patients with COVID-19. Bacterial coinfection at the time of hospitalization has been reported in 1% to 3.5% of patients with COVID-19.^{1,2} Secondary infections have been reported in 14% to 37% of patients in intensive care units, but the reported rates have been influenced by differences in the severity of illness, duration of hospitalization, method of diagnosis, and time period studied.^{3,4}

No clinical trials have evaluated the use of empiric broad-spectrum antimicrobials in patients with severe or critical COVID-19 or other coronavirus infections. Routine, empiric use of antimicrobials in patients with severe or critical COVID-19 **is not recommended (BIII)**. This recommendation is intended to mitigate the unintended consequences of antimicrobial side effects and resistance. The use of antimicrobials may be considered in specific situations, such as the presence of a lobar infiltrate on a chest X-ray, leukocytosis, an elevated serum lactate level, microbiologic data, or shock.

The use of antimicrobials in patients with severe or critical COVID-19 should follow guidelines established for other hospitalized patients (i.e., for hospital-acquired pneumonia, ventilator-associated pneumonia, or bloodstream infections associated with central lines). It is unclear whether using corticosteroids or other immunomodulatory agents recommended in the Guidelines should alter such approaches.

Therapeutic Management of Hospitalized Adults With COVID-19

For the Panel's recommendations on the use of abatacept, baricitinib, dexamethasone, infliximab, remdesivir, and tocilizumab, see [Therapeutic Management of Hospitalized Adults With COVID-19](#).

Immune-Based Therapy

For recommendations on the use of immunomodulators in patients with COVID-19, see [Immunomodulators](#).

Antithrombotic Therapy

For the Panel's recommendations regarding the use of antithrombotic therapy in critical care settings, see [Antithrombotic Therapy in Patients With COVID-19](#) and [Therapeutic Management of Hospitalized](#)

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Extracorporeal Membrane Oxygenation for Adults

Last Updated: December 20, 2023

Recommendation

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel to recommend either for or against the use of extracorporeal membrane oxygenation (ECMO) in adults with COVID-19–associated acute respiratory distress syndrome (ARDS) and refractory hypoxemia.

Rationale

ECMO has been used as a rescue therapy in patients with ARDS caused by COVID-19 and refractory hypoxemia. However, there is no conclusive evidence that ECMO is responsible for better clinical outcomes, regardless of the cause of hypoxemic respiratory failure.¹⁻⁴

The clinical outcomes for patients with ARDS who are treated with ECMO are variable and depend on multiple factors, including the etiology of hypoxemic respiratory failure, the severity of pulmonary and extrapulmonary illness, the presence of comorbidities, and the ECMO experience of the individual center.⁵⁻⁷ Several multicenter, observational cohort studies from the first half of 2020⁸⁻¹⁰ reported that patients who required ECMO for COVID-19 had similar mortality to patients in a 2018 randomized study who did not have COVID-19 but had ARDS and received ECMO.³

However, subsequent observational studies reported that in patients who required ECMO for COVID-19, outcomes in late 2020 and early 2021 were worse than outcomes in spring 2020.^{11,12} The largest analysis used data from 4,812 patients in the international Extracorporeal Life Support Organization Registry who had COVID-19 and received ECMO in 2020.¹¹ At centers that provided ECMO throughout 2020, patients who started ECMO before May 1, 2020, had a 90-day mortality of 36.9% after ECMO initiation (95% CI, 34.1% to 39.7%). At the same centers, patients who initiated ECMO between May 2 and December 31, 2020, had a 90-day mortality of 51.9% (95% CI, 50.0% to 53.8%). Furthermore, at centers that started using ECMO for patients with COVID-19 after May 1, 2020, the 90-day mortality after ECMO initiation was 58.9% (95% CI, 55.4% to 62.3%). These observational data should be interpreted with caution, as they may reflect a changing case mix of patients with COVID-19 who were referred for ECMO.

Three target emulation trials compared the efficacy of ECMO and conventional mechanical ventilation in patients with severe COVID-19–associated ARDS.^{10,13,14} The largest of these trials included 844 patients with COVID-19 who had hypoxemic respiratory failure and were receiving ECMO.¹⁴ The study reported that the patients who received ECMO had lower 60-day mortality than the patients who received only conventional mechanical ventilation (26% vs. 33.2%; risk difference –7.1%; 95% CI, –8.2% to –6.1%; risk ratio 0.78; 95% CI, 0.75–0.82). Favorable ECMO outcomes were associated with the following factors: aged <65 years, a ratio of arterial partial pressure of oxygen to fraction of inspired oxygen <80 mm Hg, ≤10-day duration of mechanical ventilation, and >15 cm H₂O driving pressure.

Ultimately, the benefits of ECMO cannot be clearly defined for patients with COVID-19 and severe ARDS because no randomized controlled trials have evaluated the use of ECMO in this population.

Clinicians interested in pursuing ECMO for patients with COVID-19 and severe ARDS should consider transferring care to high-volume ECMO centers. These patients should be entered into clinical trials or

registries so more informative data can be obtained. More information on the use of ECMO in patients with COVID-19 can be found on the [Extracorporeal Life Support Organization website](https://www.extracorporeallife.org/).

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Introduction to Critical Care Management of Children With COVID-19

Last Updated: February 29, 2024

COVID-19 may lead to critical illness in children, including hypoxemic respiratory failure, acute respiratory distress syndrome, septic shock, cardiac dysfunction, thromboembolic disease, hepatic or renal dysfunction, central nervous system disease, and exacerbation of underlying comorbidities. In addition, multisystem inflammatory syndrome in children (MIS-C) is a rare, postinfectious complication of SARS-CoV-2 and is frequently associated with critical illness.

Data informing the optimal management of children with acute COVID-19 or MIS-C are limited. In general, care for children with acute COVID-19 or MIS-C should follow the usual principles of pediatric critical care, such as the [2023 Pediatric Acute Lung Injury Consensus Conference \(2023 PALICC-2\) recommendations](#) and the [Surviving Sepsis Campaign guidelines for pediatric sepsis](#). For patients with COVID-19 in the intensive care unit (ICU), treatment often requires managing underlying illnesses other than COVID-19 that may have contributed to the need for ICU admission, as well as managing COVID-19 complications. Finally, prevention of ICU-related complications is critical to achieving optimal clinical outcomes for any patient admitted to the ICU.

Selected Clinical Manifestations of COVID-19 Critical Illness

Inflammatory Response

Patients with COVID-19 may develop a hyperinflammatory state, which appears to be distinct from classic “cytokine storm” syndromes (e.g., macrophage activation syndrome in juvenile idiopathic arthritis, familial hemophagocytic lymphohistiocytosis). This hyperinflammatory state has been well described in adults but not in children.

Multisystem Inflammatory Syndrome in Children

MIS-C is a rare, postinfectious complication of SARS-CoV-2 that is characterized by persistent fever, systemic inflammation, and multisystem organ dysfunction. The majority of children with MIS-C require ICU-level care, primarily for shock and for vasopressor and inotropic support.¹⁻³ For details on the definition of MIS-C, clinical features, and recommended treatments, see [Special Considerations in Children](#) and [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Cardiac Dysfunction, Including Myocarditis

Although cardiac involvement is common in patients with MIS-C,^{2,4} cardiac manifestations have rarely been described in children with acute COVID-19. Myocarditis, cardiac conduction abnormalities, and coronary artery aneurysms have been reported in patients with MIS-C. Myocarditis may also occur after SARS-CoV-2 vaccination, particularly in adolescent males, although the clinical course generally is relatively mild.⁵

Thromboembolic Events

Limited data characterize the prevalence of thromboembolic disease in children with COVID-19 or MIS-C. In a multicenter, retrospective cohort study including 814 hospitalized patients with COVID-19 or MIS-C, thromboembolic events were detected in 2.1% of patients with COVID-19 and 6.5% of patients with MIS-C.⁶ The same study conducted a multivariable analysis and found that the following

variables were associated with an increased risk of thromboembolic events: children aged ≥ 12 years, MIS-C, central venous catheters, and underlying malignancies. See [Antithrombotic Therapy in Patients With COVID-19](#) for additional recommendations.

Acute Kidney Injury

Acute kidney injury is estimated to occur in 12% to 44% of hospitalized children with COVID-19 or MIS-C, but the need for renal replacement therapy is extremely rare.⁷⁻¹⁰

Neurologic Involvement

Neurologic involvement is common in children with COVID-19 or MIS-C and is estimated to occur in approximately 30% to 40% of children hospitalized with these conditions.^{2,11} Severe neurologic manifestations, including severe encephalopathy, stroke, demyelinating conditions, cerebral edema, and Guillain-Barré syndrome, have also been described.¹¹

Additional Considerations

Considerations for the care of children with COVID-19 or MIS-C should generally follow the usual principles of pediatric critical care. Sedation management and considerations related to post-intensive care syndrome–pediatric (PICS-p) are discussed below. See [Oxygenation and Ventilation for Children](#), [Hemodynamic Considerations for Children](#), and [Extracorporeal Membrane Oxygenation for Children](#) for more information on pediatric critical care.

Sedation Management

Guidelines for the management of pain, agitation, neuromuscular blockade, delirium, and early mobility (PANDEM) in infants and children admitted to the pediatric ICU have recently been published.¹² In general, children with COVID-19 or MIS-C who require mechanical ventilation should be managed per the usual principles of critical care for patients with respiratory failure who require mechanical ventilation. The usual care includes sedation with the minimal effective dose required to tolerate mechanical ventilation, optimize gas exchange, and minimize the risk of ventilator-induced lung injury. Using validated pain and sedation scales, the critical care team should set a sedation/pain target based on the phase of ventilation.

Two large randomized controlled trials examined the use of protocols to manage sedation titration in children receiving mechanical ventilation.^{13,14} In both studies, participants received usual care or protocol-driven care implemented by nurses. Use of the protocols did not significantly reduce the duration of ventilation or affect other study outcomes. However, a patient's risk of harm from protocolized sedation is generally low, which led the Society of Critical Care Medicine to issue a conditional recommendation, based on low-level evidence, in its PANDEM clinical practice guidelines for the use of protocolized sedation in children who are critically ill and receiving mechanical ventilation.¹²

Studies evaluating data on the effect of early mobility protocols on critically ill children are limited. One trial evaluated the safety and feasibility of early mobilization in 58 patients who were randomized to receive usual care or early physical therapy, occupational therapy, and speech therapy consultation within 72 hours of admission to the pediatric ICU.¹⁵ Although no differences between the arms were demonstrated for clinical, functional, or quality of life outcomes, the study found that the early rehabilitation consultations were safe and feasible.

Ongoing trials are measuring the effect of early mobilization on patient-centered outcomes in children receiving mechanical ventilation. The PANDEM guideline statement issued by the Society of Critical

Care Medicine conditionally recommends, based on a low quality of evidence, implementing early mobilization strategies in children when feasible, which likely would apply to children with COVID-19 or MIS-C.¹²

Post-Intensive Care Syndrome

In recent years, awareness has been growing that post-intensive care syndrome can occur in pediatric patients. PICS-p has been demonstrated to have a multifaceted effect on the physical, cognitive, emotional, and social health of child survivors of critical illness and their families.¹⁶ Furthermore, many pediatric survivors of sepsis or acute respiratory distress syndrome manifest significant impairments in physical, cognitive, and emotional health.¹⁷⁻¹⁹ Although no clear data characterize the prevalence of PICS-p or long-term morbidity in children with COVID-19 or MIS-C, the prevalence is expected to be similar to that observed in other populations with similar illness severities.

Acknowledgments

For these pediatric recommendations, the COVID-19 Treatment Guidelines Panel integrated the recommendations from pediatric-specific guidelines, including the European Society of Paediatric and Neonatal Intensive Care's recommendations²⁰ for the care of critically ill children with COVID-19 and the Surviving Sepsis Campaign's perspective on managing sepsis in children with COVID-19.²¹ In addition, recommendations from several treatment guidelines not related to COVID-19, such as the Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children,²² the 2023 PALICC-2 recommendations,²³ and the Society of Critical Care Medicine's PANDEM guidelines,¹² were integrated.

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Hemodynamic Considerations for Children

Last Updated: February 29, 2024

Children with acute COVID-19 infrequently experience shock requiring hemodynamic support. However, similar to children with sepsis or septic shock from other causes, children with COVID-19 and shock should be evaluated and managed per the Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children.^{1,2}

Shock occurs in approximately half of the patients with multisystem inflammatory syndrome in children (MIS-C); reported prevalence ranges from 35% to 80%.³⁻⁵ Limited data inform the optimal hemodynamic management for MIS-C. Given that the physiology observed in patients with MIS-C results from a combination of distributive, cardiogenic, and, occasionally, hypovolemic shock, the COVID-19 Treatment Guidelines Panel (the Panel) suggests that clinicians use the management principles outlined in the Surviving Sepsis Campaign's guidelines for children, as well as the principles for clinical management of heart failure and general critical care, as appropriate. The Panel's recommendations apply to the care of children and infants >37 weeks gestational age.

Recommendation

- For children with COVID-19 or MIS-C and shock, the Panel recommends targeting a mean arterial pressure (MAP) between the 5th and 50th percentiles for age or >50th percentile for age (**AIII**).

Rationale

No clinical trials support specific hemodynamic targets for children with septic shock due to COVID-19, MIS-C, or any other etiology. The panel members for the pediatric Surviving Sepsis Campaign guidelines were divided on the most appropriate MAP target and made no specific recommendation for a target MAP.² Therefore, for children with COVID-19 or MIS-C and shock, clinicians should use the same approach used for children without COVID-19 and target a MAP between the 5th and 50th percentiles for age or >50th percentile for age. When MAP cannot be reliably measured, systolic blood pressure is a reasonable alternative.

Recommendation

- The Panel recommends that, when available, a combination of serial clinical assessments; cardiac ultrasound or echocardiography; and laboratory markers, including lactate levels, should be used to monitor the response to resuscitation in children with COVID-19 or MIS-C and shock (**BIII**).

Rationale

Observational data from children with sepsis not related to COVID-19 suggest that using clinical assessment alone limits the ability to classify patients with sepsis as having “warm” (i.e., likely to require fluid or vasopressors) or “cold” (i.e., likely to require inotropes) shock, when compared with assessments that include objective measures of cardiac output/index or systemic vascular resistance.^{6,7} Cardiac ultrasonography can be performed at the bedside and serially, and it may provide additional clinical data on volume responsiveness and cardiac function.⁸ Data from studies evaluating the use of cardiac ultrasound in children with COVID-19 and MIS-C are limited to reports from case series.⁹ However, cardiac ultrasonography may have particular value when patients with MIS-C are being monitored, as these patients can exhibit a wide range of hemodynamic perturbations, and approximately

a third will exhibit left ventricular dysfunction.⁴

Elevated lactate levels are associated with worse outcomes in children with sepsis not related to COVID-19, although the specific threshold is unknown and has varied from 2 to 4 mmol/L across studies.^{10,11} Data on serial lactate measures are limited to a single observational study that demonstrated an association between normalization in lactate and a decreased risk of persistent organ dysfunction in children with sepsis not related to COVID-19 (adjusted relative risk 0.47; 95% CI, 0.29–0.78).¹² The role of serial lactate measures has not been systematically evaluated for COVID-19 or MIS-C. An observational study of 1,080 children with MIS-C demonstrated an association between elevated markers of inflammation (e.g., C-reactive protein, procalcitonin), brain natriuretic peptide (BNP) or N-terminal pro b-type natriuretic peptide (NT-proBNP), and troponin and the presence of cardiac dysfunction, shock, and the need for intensive care unit admission.³ However, the study did not have access to data on the timing of the laboratory values, so the elevated markers may reflect, rather than predict, severe illness.

Recommendation

- The Panel recommends administration of balanced crystalloids rather than 0.9% saline for the initial resuscitation of children with shock due to COVID-19 or MIS-C (**BIIa**).

Rationale

In a randomized trial in India, 708 children with septic shock received either a balanced crystalloid solution (e.g., Plasma-Lyte) or 0.9% saline.¹³ The incidence of the primary outcome of new or progressive acute kidney injury was lower in the balanced crystalloid solution arm than in the saline arm (21% vs. 33%; relative risk 0.62; 95% CI, 0.49–0.80; $P < 0.001$). There was no difference in mortality between the arms. Another randomized trial comparing balanced fluids to 0.9% saline is ongoing (ClinicalTrials.gov Identifier [NCT04102371](https://clinicaltrials.gov/ct2/show/study/NCT04102371)).

Two observational studies used administrative data to compare the use of balanced/buffered crystalloids with 0.9% saline in propensity-matched cohorts of children with severe sepsis or septic shock not related to COVID-19.^{14,15} One of the studies compared patients who received any or only Ringer's lactate solution in the first 3 days of admission with patients who received only normal saline.¹⁴ The study demonstrated no differences between the arms for 30-day mortality or frequency of acute kidney injury. The other study compared patients receiving only balanced fluids with those receiving only 0.9% saline.¹⁵ The study demonstrated that the balanced fluid arm had lower mortality (12.5% vs. 15.9%; OR 0.76; 95% CI, 0.62–0.93; $P = 0.007$), reduced acute kidney injury (16.0% vs. 19.2%; OR 0.82; 95% CI, 0.68–0.98; $P = 0.028$), and fewer days on vasoactive infusions (3.0 days vs. 3.3 days; $P < 0.001$) than the saline arm. No published studies focused on patients with COVID-19 or MIS-C, although hyponatremia is common in patients with MIS-C, and decisions about the type of fluid therapy used should be individualized for this population.

Recommendations

- The Panel recommends the use of **epinephrine** or **norepinephrine** rather than dopamine in children with COVID-19 or MIS-C and shock (**BIIa**).
- There is insufficient evidence to differentiate between norepinephrine and epinephrine as a first-line vasoactive drug in children with COVID-19 or MIS-C. The choice of vasoactive agent should be individualized and based on clinical examination, laboratory data, and data from cardiac ultrasound or echocardiography.

Rationale

Use of vasoactive infusions should be considered for children with shock due to COVID-19 if signs of shock persist after resuscitation with 40 to 60 mL/kg of fluid, or sooner if there is evidence of cardiac dysfunction or signs of fluid overload (e.g., tachypnea, hepatomegaly). Similar principles may be applied to patients with MIS-C, particularly because their clinical presentation overlaps significantly with the clinical presentation of children with septic shock due to other causes. However, given the high prevalence of cardiac dysfunction in patients with MIS-C, clinicians should consider performing echocardiography or cardiac ultrasound early in the initial resuscitation if MIS-C is suspected and consider initiating a vasoactive infusion if cardiac dysfunction is identified.

Data from pediatric studies comparing vasopressors are limited, and there are no data specific to patients with COVID-19 or MIS-C. Two small pediatric trials compared epinephrine with dopamine in patients with fluid-refractory septic shock not related to COVID-19.^{16,17} One study randomized 63 children to receive dopamine 5 to 10 µg/kg/min and 57 children to receive epinephrine 0.1 to 0.3 µg/kg/min.¹⁶ Mortality by Day 28 was 14% in the dopamine arm and 7% in the epinephrine arm (OR 6.5; 95% CI, 1.1–37.8; $P = 0.03$). In the other study, 31 children were randomized to receive incremental doses of dopamine 10 to 20 µg/kg/min, and 29 children were randomized to receive incremental doses of epinephrine 0.1 to 0.3 µg/kg/min.¹⁷ The primary outcome of shock resolution within 1 hour occurred in 4 children (13%) receiving dopamine and 12 children (41%) receiving epinephrine (OR 4.8; 95% CI, 1.3–17.2; $P = 0.019$).

No pediatric trials have compared norepinephrine to other vasoactive agents in patients with sepsis, but based on data from studies of adults, the pharmacologic properties of norepinephrine and dopamine (see [Hemodynamics for Adults](#)), and the Surviving Sepsis Campaign guidelines for children, norepinephrine is suggested over dopamine.²

Collectively, this evidence is insufficient to recommend norepinephrine versus epinephrine as a first-line vasoactive agent in children with COVID-19 or MIS-C. Further, given the varied physiology observed with MIS-C in particular, decisions about which vasopressor to use should be individualized based on clinical and laboratory data and on findings from bedside cardiac ultrasound or echocardiography.

Recommendation

- There is insufficient evidence for the Panel to recommend either for or against the use of inodilators (including dobutamine or milrinone) in children with COVID-19 or MIS-C who show evidence of cardiac dysfunction and persistent hypoperfusion despite adequate fluid loading and the use of vasopressor agents.

Rationale

Data from studies evaluating the use of inodilators in children in COVID-19, MIS-C, and non-COVID-19-related sepsis are limited to reports from case series. However, the majority of the pediatric Surviving Sepsis Campaign guidelines panel (77%) would use an inodilator at least some of the time for patients with sepsis not related to COVID-19, cardiac dysfunction, and persistent hypoperfusion despite adequate fluid loading and the use of vasopressor agents.² Expert consultation from specialists in pediatric cardiology and critical care medicine is recommended in this scenario.

Additional Recommendations

- For the acute resuscitation of children with COVID-19 or MIS-C and shock, the Panel recommends the use of crystalloids rather than albumin (**AIIb**).

- The Panel **recommends against** using hydroxyethyl starches for intravascular volume replacement in children with COVID-19 or MIS-C and sepsis or septic shock (**AIII**).
- For children with refractory shock who have recently completed a course of corticosteroids to treat COVID-19, the Panel recommends using low-dose corticosteroid therapy (“shock-reversal”) over no corticosteroid therapy (**CIII**).
- Children who are currently receiving corticosteroids for COVID-19 or MIS-C are generally receiving sufficient glucocorticoid replacement therapy and do not require additional hydrocortisone for refractory shock.

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Oxygenation and Ventilation for Children

Last Updated: February 29, 2024

The COVID-19 Treatment Guidelines Panel's (the Panel) recommendations in this section were informed by recommendations from the Surviving Sepsis Campaign guidelines for managing [adult sepsis](#), [pediatric sepsis](#), and [COVID-19](#), as well as by recommendations from the [2023 Pediatric Acute Lung Injury Consensus Conference](#) (2023 PALICC-2).

Goal of Oxygenation

Recommendations

- A target oxygen saturation measured by pulse oximetry (SpO_2) of 92% to 97% is recommended for most children with COVID-19 who require supplemental oxygen (**AIIb**).
- For children with severe pediatric acute respiratory distress syndrome (PARDS; i.e., with an oxygenation index ≥ 16 or an oxygen saturation index ≥ 12), an $\text{SpO}_2 < 92\%$ can be considered to minimize exposure to a high fraction of inspired oxygen (FiO_2), but prolonged periods of an $\text{SpO}_2 < 88\%$ should be avoided (**CIII**).

Rationale

The optimal SpO_2 in children with COVID-19 is unknown. However, there is no evidence that the target SpO_2 should differ from the 2023 PALICC-2 recommendation.¹ An SpO_2 of 92% to 97% is recommended for most children with COVID-19 who require supplemental oxygen. The potential harm of hyperoxia in children was demonstrated in a meta-analysis of 11 observational studies of children without COVID-19.² The study demonstrated that critically ill children with hyperoxia had greater odds of mortality than those without hyperoxia (OR 1.59; 95% CI, 1.00–2.51). However, across the included studies, there was significant heterogeneity for populations, definitions of hyperoxia, and the timing of assessments for mortality outcomes. For children with severe PARDS, an $\text{SpO}_2 < 92\%$ can be considered to minimize exposure to a high FiO_2 .¹ Although no evidence clearly identifies a safe minimum SpO_2 in children, prolonged exposure to an $\text{SpO}_2 < 88\%$ should be avoided. When a patient's SpO_2 is $< 92\%$, monitoring oxygen delivery markers, including central venous SpO_2 , is suggested.³

The limitations of currently available measurement devices should be considered when using pulse oximetry to manage children with COVID-19 or PARDS. Observational studies in children have reported that pulse oximetry may be inaccurate, particularly at lower oxygen saturations ($\leq 90\%$) and for children who are Black.^{4,5} These reports are consistent with several observational studies in adults that identified inaccuracies in pulse oximetry measurements, particularly for patients with darker skin pigmentation.^{6–8} See [Clinical Spectrum of SARS-CoV-2 Infection](#) for more information.

Although procedures vary across institutions, the treatment of most patients with PARDS is managed without the use of arterial lines or arterial blood gas testing because arterial line placement in children, especially young children, can result in complications.^{9–11} Clinicians should monitor for adequate delivery of oxygen or consider lowering the threshold for arterial line placement if a patient's SpO_2 measurements could be unreliable (e.g., for children who have darker skin or low SpO_2 levels). Monitoring methods could include observing the patient for altered mentation, measuring venous oxygen saturation, or using near-infrared spectroscopy.

High-Flow Nasal Cannula Oxygen and Noninvasive Ventilation in Children With COVID-19 and Acute Respiratory Failure

Recommendations

- For infants and children with COVID-19 and persistent respiratory failure despite conventional oxygen therapy who have no indicators for endotracheal intubation, a time-limited trial of noninvasive ventilation (NIV) or high-flow nasal cannula (HFNC) oxygen is recommended (**AIIa**).
- There is insufficient evidence for the Panel to recommend either for or against the use of HFNC oxygen over NIV or the use of NIV over HFNC oxygen in infants and children with COVID-19.

Rationale

No high-quality studies have evaluated the use of HFNC oxygen or NIV in children with COVID-19. Therefore, when choosing a mode of respiratory support for children with COVID-19, the principles of management used for patients without COVID-19 should be followed. Both the Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children and the 2023 PALICC-2 recommend the use of NIV for children with respiratory failure who have no indication for intubation.^{12,13}

Furthermore, the response to NIV, particularly in children with more severe hypoxemia or high work of breathing, should be gauged early (within the first several hours).¹³ If the patient does not show improvement, intubation should be considered. To unload respiratory muscles, bilevel modes of NIV (with inspiratory pressure augmentation, such as BiPAP), if tolerated, are preferred over the use of continuous positive airway pressure (CPAP) alone, although CPAP is an alternative for children who cannot achieve an adequate seal with the NIV interface or who have significant patient-ventilator asynchrony.

HFNC oxygen is a relatively new, but increasingly used, mode of respiratory support for infants and children with acute respiratory failure.¹⁴ Data from studies evaluating the effectiveness of HFNC oxygen relative to NIV or conventional oxygen are limited to studies of children with pneumonia in limited-resource settings and to studies of children with bronchiolitis. Two randomized controlled trials of children with pneumonia were conducted in limited-resource settings. One study demonstrated a slightly lower relative risk of mortality with the use of HFNC oxygen when compared with conventional oxygen therapy (aHR 0.79; 95% CI, 0.54–1.16), although the results were not statistically significant.¹⁵ The other trial randomized patients to receive bubble CPAP, low-flow oxygen, or HFNC oxygen.¹⁶ The children who received bubble CPAP demonstrated a lower risk of mortality than the children who received low-flow oxygen (relative risk 0.25; 95% CI, 0.07–0.89; $P = 0.02$). The results also indicated that for the composite outcome of treatment failure, there was no difference between the use of bubble CPAP and HFNC oxygen (relative risk 0.50; 99.7% CI, 0.11–2.29).

A randomized, noninferiority trial compared HFNC oxygen (2 L/kg/min) and nasal CPAP among 142 infants aged <6 months with bronchiolitis not caused by COVID-19.¹⁷ The primary outcome was treatment failure within 24 hours, defined as an increase of ≥ 1 point in the modified Wood's Clinical Asthma Score or Échelle Douleur Inconfort Nouveau-Né (EDIN) score (a neonatal pain and discomfort scale), a respiratory rate >60 breaths/min and an increase of >10 breaths/min from baseline, or >2 severe apnea episodes per hour. Treatment failure occurred more often in the HFNC oxygen arm than in the nasal CPAP arm (51% vs. 31%), a result that failed to meet the prespecified noninferiority margin. Notably, in the HFNC oxygen arm, 72% of the patients who had treatment failure were managed successfully with nasal CPAP, and there were no differences between the arms for intubation rates or length of stay in the pediatric intensive care unit (ICU).

A systematic review of the noninferiority trial and 2 smaller trials that compared HFNC oxygen to nasal CPAP summarized the results of 213 infants and children aged ≤ 2 years with bronchiolitis.¹⁸ Treatment failure in the 2 smaller trials was rare, and no differences were detected between the HFNC oxygen and nasal CPAP arms for any of the clinical outcomes.^{19,20}

In a study that assessed whether higher flow rates of HFNC oxygen improved outcomes, 286 infants aged ≤ 6 months with severe bronchiolitis were randomized to receive HFNC oxygen 2 L/kg/min or HFNC oxygen 3 L/kg/min.²¹ The primary outcome of treatment failure (i.e., an increase of ≥ 1 point in the Wood's Clinical Asthma Score or EDIN score, a respiratory rate > 60 breaths/min and an increase of > 10 breaths/min from baseline, or > 2 severe apnea episodes per hour) occurred in 38.7% of the infants in the 2 L/kg/min arm and in 38.9% of the infants in the 3 L/kg/min arm ($P = 0.98$). Patient discomfort, as measured by EDIN score, occurred more often in the 3 L/kg/min arm than in the 2 L/kg/min arm (43% vs. 16%; $P = 0.002$).

HFNC oxygen is increasingly being used in children. These studies highlight the potential role of an HFNC oxygen trial in the management of children with acute respiratory failure due to COVID-19, particularly infants and young children who may have NIV-related challenges, such as poor mask fit, discomfort, or patient-ventilator asynchrony. For the use of HFNC oxygen in children, consider flow rates of up to 2 L/kg/min, with a maximum of 60 L/min. If patients do not improve within the first few hours of receiving HFNC oxygen, their treatment should be escalated to NIV or intubation.

Awake Prone Positioning in Children Not Receiving Mechanical Ventilation

Recommendations

- There is insufficient evidence for the Panel to recommend either for or against a trial of awake prone positioning in children with persistent hypoxemia who require HFNC oxygen or NIV and do not require endotracheal intubation.
- For patients with refractory hypoxemia who meet the indications for intubation and mechanical ventilation, the Panel **recommends against** the use of awake prone positioning as a rescue therapy to avoid intubation (**AIII**).

Rationale

There are no high-quality pediatric data from studies that evaluated the effect of awake prone positioning on clinical outcomes in children with COVID-19 or in children with illness not related to COVID-19. Awake prone positioning may be considered for older children and adolescents. See [Oxygenation and Ventilation for Adults](#) for more information on the use of awake prone positioning in adults. In addition, pediatric clinicians should consider a child's developmental stage and ability to comply with the protocols for awake prone positioning.

Intubation for Mechanical Ventilation in Children With Acute COVID-19

Recommendations

- If intubation becomes necessary, the Panel recommends that an experienced practitioner perform the procedure in a controlled setting due to the enhanced risk of exposing health care practitioners to SARS-CoV-2 during intubation (**AIII**).
- The Panel recommends using cuffed endotracheal tubes over uncuffed endotracheal tubes in children who require endotracheal intubation (**AIIb**).

Rationale

To optimize the safety of patients and health care workers and maximize first-attempt success, intubation should be performed in a controlled setting by an experienced practitioner. In addition, cuffed endotracheal tubes are preferred for children of all ages to minimize leaks around the endotracheal tube, ensure delivery of ventilator pressure, decrease the risk of aspiration, reduce the need for endotracheal tube exchange, and reduce aerosolization of respiratory secretions during mechanical ventilation.^{3,22-24}

General Considerations for Children With COVID-19 and PARDS Who Require Mechanical Ventilation

Recommendations

For children with COVID-19 and PARDS who require mechanical ventilation:

- The Panel recommends using low tidal volume (VT) ventilation (VT 4–8 mL/kg of predicted body weight) over higher VT ventilation (VT >8 mL/kg) (**AIIb**).
- The Panel recommends targeting plateau pressures of ≤ 28 cm H₂O in children with normal chest wall compliance and ≤ 32 cm H₂O in those with impaired chest wall compliance (**AIII**).
- The Panel recommends using positive end-expiratory pressure (PEEP) at or above the level recommended in the Acute Respiratory Distress Syndrome (ARDS) Network's PEEP/FiO₂ table and titration of PEEP based on observed responses in oxygenation, hemodynamics, and respiratory system compliance (**AIIb**).
- The Panel recommends permissive hypercapnia (i.e., 7.2–7.3 pH) to remain within lung-protective strategies and to minimize ventilator-associated lung injury, provided the patient does not have a coexisting condition that would be worsened by acidosis (e.g., severe pulmonary hypertension, ventricular dysfunction, intracranial hypertension) (**AIII**).
- The Panel recommends limiting driving pressure as part of a lung-protective ventilation strategy (**BIIb**).
- The Panel **recommends against** the routine use of inhaled nitric oxide (**AIII**).

Rationale

There is no evidence that ventilator management in patients with PARDS due to COVID-19 should differ from ventilator management in patients with PARDS due to other causes. The Panel's recommendations are derived from the 2023 PALICC-2 recommendations.¹

A large observational study conducted in 71 international pediatric ICUs reported that for patients with mild to moderate ARDS, less adherence to the recommended VT of 5 to 8 mL/kg (or 3 to 6 mL/kg for patients with severe ARDS) was associated with higher mortality and with more time on ventilation.²⁵ In general, supraphysiologic VT ventilation (>8 mL/kg) should not be used in patients with PARDS, and VT should be adjusted within the acceptable range to maintain other lung-protective ventilation targets (e.g., maintaining ≤ 28 cm H₂O plateau pressure). The use of ultra-low VT ventilation (<4 mL/kg) has not been systematically studied in children, so it should be used with caution.

The ARDS Network established a ventilation protocol that includes suggested low PEEP/high FiO₂ values.²⁶ Two observational studies reported better clinical outcomes associated with use of the suggested (or higher) PEEP levels when compared with lower PEEP levels.^{25,27} The multicenter studies, which included nearly 1,500 pediatric patients with ARDS, demonstrated that PEEP levels lower than those suggested by the ARDS Network were associated with increased mortality.

Driving pressure (i.e., the difference between plateau pressure and PEEP) is a marker for lung strain.

It represents the ratio of delivered VT to respiratory system compliance. An observational study demonstrated that adults with ARDS who were mechanically ventilated and had a driving pressure >15 cm H₂O had increased mortality.²⁸ An observational study in children reported that higher driving pressure was associated with a longer duration of ventilation and fewer ventilator-free days.²⁹ The Panel's recommendation aligns with the 2023 PALICC-2 recommendation to limit driving pressure in patients with PARDS.

Inhaled nitric oxide can be considered as a rescue therapy for children with severe PARDS and COVID-19. In a small, randomized trial, the use of inhaled nitric oxide resulted in reduced use of extracorporeal membrane oxygenation (ECMO).³⁰ However, inhaled nitric oxide has a heterogeneous treatment effect, and many patients do not show improved gas exchange. Although adverse effects are rare, use of inhaled nitric oxide can have a substantial effect on health care costs. Therefore, inhaled nitric oxide should not be considered routine therapy for children with PARDS or COVID-19 who are receiving mechanical ventilation.

Fluid Management in Children With PARDS

Recommendation

- Following an initial resuscitation in children with PARDS due to COVID-19, clinicians should monitor and titrate fluid balance to maintain adequate intravascular volume while aiming to prevent positive fluid balance (**BIIB**).

Rationale

There is no evidence that fluid management in patients with PARDS due to COVID-19 should differ from fluid management in patients with PARDS due to other causes. Therefore, the Panel's recommendation aligns with the 2023 PALICC-2 recommendations.¹ No pediatric randomized trials have directly compared a liberal fluid strategy to a conservative fluid strategy in patients with PARDS of any etiology. Several observational studies have demonstrated an association between greater fluid overload and worse clinical outcomes, including fewer ventilator-free days and increased mortality.³¹⁻³³

In a multicenter study of 168 children with acute lung injury, daily and cumulative fluid balance were measured over the first 7 days after participants met the inclusion criteria.³¹ After adjusting for demographic characteristics, pediatric risk of mortality III (PRISM III) scores, vasopressor use, and the ratio of arterial partial pressure of oxygen to fraction of inspired oxygen (PaO₂/FiO₂), the study found that an increasing cumulative fluid balance on Day 3 was associated with fewer ventilator-free days, but no association with mortality was detected.

A more recent single-center study that included 732 children with acute lung injury demonstrated an association between higher cumulative fluid balance on Days 5 to 7 and increased mortality (OR 1.34 for 100 mL/kg on Day 5; 95% CI, 1.11–1.61) after adjusting for oxygenation index, the number of nonpulmonary organ failures, immunocompromised status, and vasopressor scores.³³ Also, greater cumulative fluid balance on Days 4 to 7 was associated with a lower probability of successful extubation by Day 28.

Collectively, the findings from these pediatric observational studies demonstrate the potential harm of fluid overload in patients with PARDS, particularly after 3 to 4 days of illness. These results are consistent with the findings from FACTT, a trial of conservative versus liberal fluid management strategies in adults.³⁴ In adults, FACTT found no difference between the arms for 60-day mortality, but the conservative strategy arm demonstrated improved oxygenation and less time on mechanical ventilation and in the ICU when compared with the liberal strategy arm. However, no analysis of data from prospective pediatric

trials delineates a causal relationship between a specific, protocolized fluid management strategy, or the timing of such a strategy, and clinical outcomes. Therefore, an individualized fluid management approach that is titrated to maintain intravascular volume while preventing excessive positive fluid balance, as suggested by the 2023 PALICC-2 recommendations, is appropriate.¹

Neuromuscular Blockade in Mechanically Ventilated Children With Severe PARDS

Recommendation

- For mechanically ventilated children with severe PARDS and COVID-19, the Panel recommends minimal yet effective use of neuromuscular blocking agents in conjunction with sedation if sedation alone is inadequate to achieve lung-protective ventilation (**BIII**).

Rationale

There is no evidence that the use of neuromuscular blockade in children with COVID-19 should differ from practices used for severe PARDS from other causes. Therefore, the Panel's recommendation aligns directly with the 2023 PALICC-2 recommendation.¹

Therapies for Mechanically Ventilated Children With Severe PARDS and Refractory Hypoxemia

Recommendations

For children with severe PARDS and refractory hypoxemia after other oxygenation strategies have been optimized:

- The Panel recommends the use of inhaled nitric oxide as a rescue therapy; if a patient's oxygenation does not improve rapidly, the inhaled nitric oxide should be discontinued (**BIIB**).
- The Panel recommends prone positioning for 12 to 16 hours per day over no prone positioning (**BIII**).
- There is insufficient evidence for the Panel to recommend either for or against the use of recruitment maneuvers, but if they are used in children, slow incremental and decremental adjustments in PEEP are preferred over sustained inflation maneuvers.
- There is insufficient evidence for the Panel to recommend either for or against the use of high-frequency oscillatory ventilation (HFOV) in patients with PARDS.

Rationale

There is no evidence that the use of inhaled nitric oxide, prone positioning, or HFOV in children with COVID-19 should differ from practices used for severe PARDS from other causes. Therefore, the Panel's recommendations are based on the 2023 PALICC-2 recommendations.¹

One randomized controlled trial and 2 propensity-matched, observational studies in the past 10 years evaluated the use of inhaled nitric oxide in patients with PARDS.^{30,35,36} The randomized controlled trial included 55 patients and found that the use of inhaled nitric oxide resulted in no statistical difference between the arms for 28-day mortality (8% mortality in the inhaled nitric oxide arm vs. 28% in the placebo arm), although the trial was underpowered for this outcome.³⁰ However, the inhaled nitric oxide arm had approximately 5 more ventilator-free days than the placebo arm, a result that was primarily mediated by avoiding the use of ECMO.

Results from observational studies have shown that patients who received inhaled nitric oxide did not improve or have fewer ventilator-free days.^{35,36} A meta-analysis of randomized controlled trials

in patients with PARDS reported that treatment with inhaled nitric oxide did not decrease mortality, ventilator-free days, or duration of ventilation.³⁷ Although the evidence is insufficient to recommend the use of inhaled nitric oxide for all patients with ARDS, in cases of severe hypoxemia, it can be considered as a rescue therapy to potentially avoid the use of ECMO.

No recent studies have evaluated the role of prone positioning in patients with PARDS, although a large, multicenter trial is ongoing. Therefore, the Panel's recommendation to consider prone positioning in cases of severe PARDS aligns with the 2023 PALICC-2 recommendation and is supported by data from studies in adults, primarily from PROSEVA, a trial on prone positioning in patients with ARDS.³⁸

The 2023 PALICC-2 does not recommend for or against recruitment maneuvers.¹ However, the Panel suggests using careful incremental and decremental adjustments in PEEP if these maneuvers are applied in children with refractory hypoxemia. In children, this approach to recruitment maneuvers is preferred over sustained inflation maneuvers due to the increased risk of harm from barotrauma and hemodynamic compromise in patients with sustained inflation. Clinical trials in adults have highlighted the potential harm of applying these maneuvers to patients who may not have the potential for lung recruitment.^{39,40}

Three small randomized controlled trials examined the use of HFOV for PARDS.⁴¹⁻⁴³ None of these studies found a significant difference for mortality. Several observational studies that used propensity matching reported no difference in outcomes between the HFOV and conventional ventilation arms or reported a potential for higher mortality or longer ventilation time with the use of HFOV when compared with conventional ventilation.⁴⁴⁻⁴⁸ In some of these analyses, residual confounding has been a concern. Therefore, the Panel determined that there is insufficient evidence to recommend either for or against the use of HFOV in patients with PARDS due to COVID-19. Some concerns have been raised about the use of HFOV and the aerosolization of COVID-19; however, adding a filter to the expiratory limb of the HFOV circuit may alleviate these concerns.

Multisystem Inflammatory Syndrome in Children

More than half of the patients with multisystem inflammatory syndrome in children (MIS-C) require mechanical ventilation or NIV.⁴⁹⁻⁵¹ For patients with MIS-C, the indications for mechanical ventilation vary and include shock or cardiac dysfunction, pulmonary edema, procedural preparation (e.g., to facilitate sedation for central venous catheter placement), respiratory failure, or neurologic failure. The management of oxygenation and ventilation in patients with MIS-C should follow the usual principles of shock management outlined in the Surviving Sepsis Campaign guidelines for children, as well as the principles for clinical management of heart failure and general critical care, as appropriate.¹²

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Extracorporeal Membrane Oxygenation for Children

Last Updated: February 29, 2024

Recommendation

- The COVID-19 Treatment Guidelines Panel recommends that the use of extracorporeal membrane oxygenation (ECMO) should be considered for children with acute COVID-19 or multisystem inflammatory syndrome in children (MIS-C) who have refractory hypoxemia or shock when hemodynamic parameters cannot be maintained or lung-protective strategies result in inadequate gas exchange (**III**). Candidacy for ECMO should be determined on a case-by-case basis by the multidisciplinary team.

Rationale

ECMO is used as a rescue therapy for children with refractory hypoxemia or shock. Similar to outcomes for adults, outcomes for children managed with venovenous ECMO are variable and influenced by the etiology and duration of the respiratory failure and by underlying, comorbid medical conditions.^{1,2} In addition, studies have shown that pediatric centers that treat fewer patients with ECMO have worse outcomes than facilities that treat a high volume of patients with ECMO.^{3,4} Other than studies of neonates, no randomized trials have evaluated the efficacy or benefit of ECMO for the treatment of hypoxemic respiratory failure in children without COVID-19. In an observational study of 122 children with severe pediatric acute respiratory distress syndrome (PARDS), children who received ECMO and those supported without ECMO had similar 90-day mortality (25% vs. 30%).⁵

The 2023 Pediatric Acute Lung Injury Consensus Conference suggests that patients with severe PARDS from potentially reversible causes and children who are candidates for lung transplantation be evaluated for management with ECMO if lung-protective strategies result in inadequate ventilation (conditional recommendation, very low quality of evidence).⁶ The Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children issued a weak recommendation, based on very low-quality evidence, for the use of venovenous ECMO in children with PARDS and refractory hypoxemia.⁷

Venoarterial ECMO has been used successfully for the treatment of refractory shock in children, although no trials have evaluated this approach, and the potential benefits must be weighed against risks of bleeding and thromboembolic events.⁸⁻¹⁰ The Surviving Sepsis Campaign guidelines for children issued a weak recommendation, based on very low-quality evidence, for the use of venoarterial ECMO in children with refractory shock who have not improved with any other treatments.⁷ However, a standardized definition for refractory shock in children is not available.

Studies that have evaluated data on the use of ECMO in children with COVID-19 and MIS-C have suggested that these patients have outcomes similar to patients who have received ECMO for illnesses not related to COVID-19.¹¹⁻¹⁵ The Extracorporeal Life Support Organization published guidelines for the use of ECMO in patients with COVID-19.¹⁶ In general, children with COVID-19 or MIS-C who are candidates for ECMO should be assessed using criteria similar to those used for children with severe respiratory failure or shock due to other causes. Cannulation approaches and management principles should follow published [international guidelines](#) and local protocols for patients who do not have COVID-19. Pediatric clinicians should consider entering patients into clinical trials or registries to

inform future recommendations regarding the use of ECMO in children with COVID-19.

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Antiviral Agents, Including Antibody Products

Last Updated: February 29, 2024

Remdesivir and ritonavir-boosted nirmatrelvir (Paxlovid) are approved by the Food and Drug Administration for the treatment of COVID-19.

Molnupiravir and high-titer COVID-19 convalescent plasma (CCP) are available only under Food and Drug Administration Emergency Use Authorizations for the treatment of COVID-19.

Summary Recommendations

Recommendations for Treating Nonhospitalized Adults

- The COVID-19 Treatment Guidelines Panel (the Panel) recommends the following anti-SARS-CoV-2 therapies as preferred treatments for COVID-19. These drugs are listed in order of preference:
 - **Ritonavir-boosted nirmatrelvir (Paxlovid) (AIIa)**
 - **Remdesivir (BIIa)**
- The Panel recommends **molnupiravir** as an alternative therapy when neither of the preferred therapies are available, feasible to use, or clinically appropriate (**CIIa**).
- For more information on using these agents in nonhospitalized adults, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Recommendations for Treating Nonhospitalized Children

- For recommendations on using antiviral therapy in nonhospitalized children, see [Therapeutic Management of Nonhospitalized Children With COVID-19](#).

Recommendations for Treating Hospitalized Adults or Children

- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) and [Therapeutic Management of Hospitalized Children With COVID-19](#) for recommendations on using remdesivir with or without immunomodulators in certain hospitalized patients.

Antiviral Treatments With Insufficient Evidence

- There is insufficient evidence for the Panel to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in hospitalized or nonhospitalized patients who are immunocompromised.
 - Some people who are immunocompromised have prolonged, symptomatic COVID-19 with evidence of ongoing SARS-CoV-2 replication. For the Panel's recommendations for managing these patients, see [Special Considerations in People Who Are Immunocompromised](#).
- There is insufficient evidence for the Panel to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in nonhospitalized patients who are immunocompetent.

Antiviral Treatments That the Panel Recommends Against

- The Panel **recommends against** the use of the following drugs for the treatment of COVID-19, except in a clinical trial:
 - **Interferon alfa or beta** in nonhospitalized patients (**AIIa**)
 - **Systemic interferon alfa** in hospitalized patients (**AIIa**)
 - **Nitazoxanide (BIIa)**
- The Panel **recommends against** the use of the following drugs for the treatment of COVID-19:
 - Anti-SARS-CoV-2 monoclonal antibodies (**AIII**)
 - **Chloroquine or hydroxychloroquine** and/or **azithromycin** in hospitalized patients (**AI**) and nonhospitalized patients (**AIIa**)
 - CCP in hospitalized patients who are immunocompetent (**BIIa**)
 - **Lopinavir/ritonavir** and other HIV protease inhibitors in hospitalized patients (**AI**) and nonhospitalized patients (**AIII**)
 - **Interferon beta** in hospitalized patients (**AI**)

Summary Recommendations, continued

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

Remdesivir

Last Updated: February 29, 2024

Remdesivir is a nucleotide prodrug of an adenosine analog. It binds to the viral RNA-dependent RNA polymerase and inhibits viral replication by terminating RNA transcription prematurely. Remdesivir has demonstrated in vitro and in vivo activity against SARS-CoV-2.¹

Intravenous remdesivir is approved by the Food and Drug Administration (FDA) for the treatment of COVID-19 in adults and pediatric patients aged ≥ 28 days and weighing ≥ 3 kg. In nonhospitalized patients with mild to moderate COVID-19 who are at high risk of progressing to severe disease, remdesivir should be started within 7 days of symptom onset and administered for 3 days. Hospitalized patients should receive remdesivir for 5 days or until hospital discharge, whichever comes first.² The FDA prescribing information for remdesivir indicates that if a patient does not clinically improve within 5 days, clinicians may extend the treatment course for up to 5 additional days (for a total duration of 10 days). See [Table 4e](#) for more information on administering remdesivir.

Remdesivir has been studied in several clinical trials for the treatment of COVID-19. The recommendations from the COVID-19 Treatment Guidelines Panel (the Panel) are based on the results of these studies. See [Table 4a](#) for more information.

Recommendations

- For the Panel's recommendations and information on the clinical efficacy of using remdesivir to treat high-risk, nonhospitalized patients with mild to moderate COVID-19, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).
- For the Panel's recommendations and information on the clinical efficacy of using remdesivir with or without immunomodulators to treat certain hospitalized patients, see [Therapeutic Management of Hospitalized Adults With COVID-19](#).

Monitoring, Adverse Effects, and Drug-Drug Interactions

Remdesivir can cause gastrointestinal symptoms (e.g., nausea), elevated transaminase levels, an increase in prothrombin time without a change in the international normalized ratio, and hypersensitivity reactions. Bradycardia has also been reported.^{3,4}

The FDA approved remdesivir for use without dose adjustment in patients with an estimated glomerular filtration rate (eGFR) of <30 mL/min, including those receiving dialysis.² Remdesivir is formulated with sulfobutylether-beta-cyclodextrin (SBECD) sodium. SBECD is a vehicle that is primarily eliminated through the kidneys. Accumulation of SBECD in patients with renal impairment may result in liver and renal toxicities. The REDPINE trial evaluated the use of remdesivir for 5 days in patients with COVID-19 and an eGFR of <30 mL/min, and a Phase 1 trial evaluated the use of a single dose of remdesivir in individuals with different degrees of renal impairment. Neither trial reported significant safety concerns.

Before starting patients on remdesivir, the FDA recommends performing liver function and prothrombin time tests as clinically appropriate and repeating these tests during treatment as clinically indicated. Remdesivir may need to be discontinued if a patient's alanine transaminase level increases to >10 times the upper limit of normal, and it should be discontinued if increases in alanine transaminase levels and signs or symptoms of liver inflammation are observed.²

Remdesivir should be administered in a setting where severe hypersensitivity reactions, such as anaphylaxis, can be managed. Patients should be monitored during the infusion and observed for at least 1 hour after the infusion as clinically appropriate.

No clinically significant drug-drug interactions are expected with cytochrome P450 3A4 inducers or with inhibitors of P-glycoprotein or organic anion transporting polypeptides 1B1 or 1B3.² See [Table 4e](#) for more information.

Patients Who Are Immunocompromised and Have Prolonged Symptoms and Evidence of Ongoing Viral Replication

Patients who are severely immunocompromised may have a prolonged duration of SARS-CoV-2 replication, which may lead to rapid viral evolution. There is concern that using a single antiviral agent in these patients may result in the emergence of resistant virus.⁵ Additional studies are needed to assess this risk. The data on using combinations of antiviral therapies for the treatment of COVID-19 are limited.⁶ Clinical trials are needed to determine the role of combination therapy in treating patients with COVID-19.

For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication (e.g., those with a low cycle threshold value, as measured by a reverse transcription polymerase chain reaction result or with a positive rapid antigen test result) despite receiving a course of antiviral therapy, the optimal management is unknown. Case reports and case series have documented treating these patients with additional antiviral treatments, prolonged courses of antiviral treatments, high-titer COVID-19 convalescent plasma, or combination therapy.⁷⁻¹¹

For more information on the use of remdesivir in patients who are moderately to severely immunocompromised, see [Special Considerations in People Who Are Immunocompromised](#), [Therapeutic Management of Hospitalized Adults With COVID-19](#), and [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance regarding the use of remdesivir during pregnancy and lactation.

Considerations in Children

See [Therapeutic Management of Nonhospitalized Children With COVID-19](#) and [Therapeutic Management of Hospitalized Children With COVID-19](#) for the Panel's guidance regarding the use of remdesivir in children.

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Table 4a. Remdesivir: Selected Clinical Trial Data

Last Updated: February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for RDV. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
ACTT-1: Double-Blind, Placebo-Controlled Trial of Remdesivir in Hospitalized Patients With COVID-19 in 10 Countries¹		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection ≥1 of the following: - Pulmonary infiltrates - SpO₂ ≤94% on room air - Need for supplemental oxygen, HFNC oxygen, NIV, MV, or ECMO Key Exclusion Criteria - ALT or AST >5 times ULN - eGFR <30 mL/min Interventions - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for up to 9 days (n = 541) - Placebo for up to 10 days (n = 521) Primary Endpoint - Time to clinical recovery Key Secondary Endpoints - Clinical status at Day 15, as measured by an OS - Mortality by Day 29 - Occurrence of SAEs	Participant Characteristics - Mean age 59 years; 64% men; 53% White, 21% Black, 13% Asian, 24% Hispanic/Latinx - Coexisting conditions: 26% with 1; 55% with ≥2 13% not on oxygen; 41% on supplemental oxygen; 18% on HFNC oxygen or NIV; 27% on MV or ECMO - Median of 9 days (IQR 6–12 days) from symptom onset to randomization 23% received corticosteroids during study. Primary Outcomes - Time to clinical recovery: 10 days in RDV arm vs. 15 days in placebo arm (rate ratio for recovery 1.29; 95% CI, 1.12–1.49; <i>P</i> < 0.001) - Benefit of RDV was greatest in patients randomized during first 10 days after symptom onset and in those who required supplemental oxygen at enrollment. - No difference between arms in time to recovery for patients on HFNC oxygen, NIV, MV, or ECMO at enrollment Secondary Outcomes - Improvement in clinical status at Day 15 was more likely in RDV arm (OR 1.5; 95% CI, 1.2–1.9; <i>P</i> < 0.001). - No difference between arms in mortality by Day 29 - Occurrence of SAEs: 25% in RDV arm vs. 32% in placebo arm	Key Limitations - Wide range of disease severity among patients; study not powered to detect differences within subgroups - Study not powered to detect differences in mortality between arms - No data on longer-term morbidity Interpretation - In patients with severe COVID-19, RDV reduced the time to clinical recovery. - The benefit was most apparent in hospitalized patients who were randomized within 10 days of symptom onset and were receiving supplemental oxygen. - There was no observed benefit in those on HFNC oxygen, NIV, MV, or ECMO, but the study was not powered to detect differences within subgroups.

Methods	Results	Limitations and Interpretation
CATCO: Multicenter, Open-Label, Pragmatic RCT of Remdesivir in Hospitalized Patients With COVID-19 in Canada²		
Key Inclusion Criterion - Laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criterion - Already receiving RDV Interventions - RDV 200 mg IV on Day 0, then RDV 100 mg IV once daily on Days 1–9 (n = 634) - Local SOC (n = 647) Primary Endpoint - In-hospital mortality Key Secondary Endpoints - New need for MV - Hospital LOS - Incidence of new hepatic dysfunction, incidence of need for dialysis, and change in SCr at Day 5	Participant Characteristics - Median age 66 years; 60% men; 41% White - Median of 8 days from symptom onset to randomization - At entry: 54% on low-flow oxygen 24% on HFNC oxygen 9% on MV - Rates of comorbidities were similar between arms. 87% in both arms were receiving corticosteroids at baseline. Primary Outcome - In-hospital mortality: 19% in RDV arm vs. 23% in SOC arm (relative risk 0.83; 95% CI, 0.67–1.03) Secondary Outcomes - New need for MV: 8% in RDV arm vs. 15% in SOC arm (relative risk 0.53; 95% CI, 0.38–0.75) - No significant difference between arms in hospital LOS - No difference between arms in incidence of new hepatic dysfunction, incidence of need for dialysis, or change in SCr at Day 5	Key Limitations - Open-label study - Information on comorbidities was not available for 26% of patients. Interpretation - Compared to SOC, RDV did not decrease in-hospital mortality among hospitalized patients with COVID-19. - Patients who received RDV were less likely to require MV than patients who received SOC.

Methods	Results	Limitations and Interpretation
DisCoVeRy: Open-Label, Adaptive RCT of Remdesivir in Hospitalized Patients With Moderate or Severe COVID-19 in Europe³		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - Illness of any duration - SpO₂ ≤94% on room air or use of supplemental oxygen, HFNC oxygen, NIV, or MV Key Exclusion Criteria - ALT or AST >5 times ULN - Severe chronic kidney disease Interventions - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for up to 9 days (n = 429) - SOC (n = 428) Primary Endpoint - Clinical status at Day 15, as measured by an OS Key Secondary Endpoints - Mortality by Day 29 - Occurrence of SAEs	Participant Characteristics - Median age 64 years; 70% men; 69% White 74% with ≥1 coexisting conditions 40% received corticosteroids. - Median of 9 days from symptom onset to randomization in both arms 61% with moderate disease; 39% with severe disease Primary Outcome - No difference between arms in clinical status at Day 15 (OR 0.98; 95% CI, 0.77–1.25; <i>P</i> = 0.85) - A prespecified subgroup analysis based on duration of symptoms found no significant difference in clinical status between arms. Secondary Outcomes - Mortality by Day 29: 8% in RDV arm vs. 9% in SOC arm - Occurrence of SAEs: 33% in RDV arm vs. 31% in SOC arm (<i>P</i> = 0.48)	Key Limitations - Open-label study 440 participants in this study also enrolled in the WHO Solidarity trial. Interpretation - There was no clinical benefit of RDV in hospitalized patients with COVID-19 who were symptomatic for >7 days and who required supplemental oxygen.

Methods	Results	Limitations and Interpretation
WHO Solidarity Trial, Final Report: Open-Label, Adaptive RCT in Hospitalized Patients With COVID-19 in 35 Countries⁴		
Key Inclusion Criterion - Not known to have received any study drug Interventions - RDV 200 mg IV on Day 0, then RDV 100 mg IV once daily on Days 1–9 (n = 4,146) - Local SOC (n = 4,129) Primary Endpoint - In-hospital mortality Key Secondary Endpoint - Initiation of MV	Participant Characteristics 46% aged 50–69 years; 22% aged ≥70 years; 63% men - Rates of comorbidities were similar between arms. - At entry: 71% on supplemental oxygen 9% on MV 68% received corticosteroids during study; 4.6% received IL-6 inhibitors. Primary Outcome - In-hospital mortality: 14.5% in RDV arm vs. 15.6% in SOC arm (rate ratio 0.91; 95% CI, 0.82–1.02; <i>P</i> = 0.12) - On MV: 42.1% vs. 38.6% (rate ratio 1.13; 95% CI, 0.89–1.42; <i>P</i> = 0.32) - Not on MV but receiving oxygen: 14.6% vs. 16.3% (rate ratio 0.87; 95% CI, 0.76–0.99; <i>P</i> = 0.03) - Not on oxygen initially: 2.9% vs. 3.8% (rate ratio 0.76; 95% CI, 0.46–1.28; <i>P</i> = 0.30) Secondary Outcome - Initiation of MV: 14.1% in RDV arm vs. 15.7% in SOC arm (rate ratio 0.88; 95% CI, 0.77–1.00; <i>P</i> = 0.04)	Key Limitations - Open-label study - No data on time from symptom onset to enrollment - Data analysis did not separate receipt of low-flow and high-flow oxygen. Interpretation - There was no benefit of RDV in hospitalized patients with COVID-19 who were on MV at baseline. - Compared to SOC, RDV had a modest but statistically significant effect on reducing the risk of death or progression to MV in hospitalized patients who required oxygen.

Methods	Results	Limitations and Interpretation
GS-US-540-5774 Study: Open-Label RCT of 10 Days or 5 Days of Remdesivir in Hospitalized Patients With Moderate COVID-19 in Asia, Europe, and the United States ⁵		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - Pulmonary infiltrates - SpO₂ >94% on room air Key Exclusion Criteria - ALT or AST >5 times ULN - CrCl <50 mL/min Interventions - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 9 days (n = 193) - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 4 days (n = 191) - Local SOC (n = 200) Primary Endpoint - Clinical status at Day 11, as measured by an OS	Participant Characteristics - Demographic and baseline disease characteristics were similar across arms. - Median age 57 years; 61% men; 58% White 84% required no supplemental oxygen; 15% required low-flow oxygen; 1% required HFNC oxygen or NIV. - Concomitant medication use in the 10-day RDV, 5-day RDV, and SOC arms: - Steroids: 15% vs. 17% vs. 19% - Tocilizumab: 1% vs. 1% vs. 5% - HCQ or CQ: 11% vs. 8% vs. 45% - LPV/RTV: 6% vs. 5% vs. 22% - AZM: 21% vs. 18% vs. 31% - Median duration of therapy: 6 days in 10-day RDV arm vs. 5 days in 5-day RDV arm Primary Outcome - Clinical status at Day 11: - Significantly better in 5-day RDV arm than in SOC arm (OR 1.65; 95% CI, 1.09–2.48; <i>P</i> = 0.02) - No difference between 10-day RDV arm and SOC arm (<i>P</i> = 0.18)	Key Limitations - Open-label design may have affected decisions about concomitant medications (e.g., more patients in SOC arm received AZM, HCQ or CQ, and LPV/RTV) and time of hospital discharge. - No data on time to return to activity for discharged patients Interpretation - Hospitalized patients with moderate COVID-19 who received 5 days of RDV had better clinical status at Day 11 than those who received SOC. - There was no difference in clinical status at Day 11 between patients who received 10 days of RDV and those who received SOC.

Methods	Results	Limitations and Interpretation
GS-US-540-5773 Study: Open-Label RCT of 10 Days or 5 Days of Remdesivir in Hospitalized Patients With Severe COVID-19 in Asia, Europe, and the United States ⁶		
Key Inclusion Criteria • Laboratory-confirmed SARS-CoV-2 infection • Aged ≥12 years • Pulmonary infiltrates and SpO₂ ≤94% on room air or receipt of supplemental oxygen Key Exclusion Criteria • Need for MV or ECMO • Multiorgan failure • ALT or AST >5 times ULN • Estimated CrCl <50 mL/min Interventions • RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 4 days (n = 200) • RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 9 days (n = 197) Primary Endpoint • Clinical status at Day 14, as measured by an OS	Participant Characteristics • Median age: 61 years in 5-day RDV arm vs. 62 years in 10-day RDV arm • 60% men in 5-day RDV arm vs. 68% men in 10-day RDV arm • Oxygen requirements at baseline for 5-day RDV arm vs. 10-day RDV arm: • None: 17% vs. 11% • Low-flow oxygen: 56% vs. 54% • HFNC oxygen or NIV: 24% vs. 30% • MV or ECMO: 2% vs. 5% • Baseline clinical status was worse in 10-day RDV arm than in 5-day RDV arm (<i>P</i> = 0.02). Primary Outcome • After adjusting for baseline clinical status: • Proportion with improved clinical status at Day 14: 65% in 5-day RDV arm vs. 54% in 10-day RDV arm (<i>P</i> = 0.14)	Key Limitations • Open-label study • Lack of placebo arm • Baseline imbalances in clinical status of patients in 5-day RDV and 10-day RDV arms Interpretation • In hospitalized patients with severe COVID-19 who were not receiving MV or ECMO, using RDV for 5 or 10 days had similar clinical benefits.

Methods	Results	Limitations and Interpretation
PINETREE: Double-Blind, Placebo-Controlled Trial of Remdesivir for 3 Days in Nonhospitalized Patients With COVID-19 Who Were at High Risk of Disease Progression in Denmark, Spain, the United Kingdom, and the United States⁷		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection ≤4 days from screening - Aged ≥12 years ≥1 risk factor for disease progression or aged ≥60 years - Symptom onset ≤7 days from randomization ≥1 ongoing COVID-19 symptom Key Exclusion Criteria - COVID-19 vaccination - Receipt of supplemental oxygen - Previous hospitalization or treatment for COVID-19 Interventions - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily on Days 2 and 3 (n = 279) - Placebo (n = 283) Primary Endpoints - COVID-19–related hospitalization or death from any cause by Day 28 - Occurrence of AEs Key Secondary Endpoint - COVID-19–related, medically attended visit or death from any cause by Day 28	Participant Characteristics - Mean age 50 years; 30% aged ≥60 years; 52% men; 80% White, 8% Black 62% with DM; 55% with obesity; 48% with HTN - Median of 5 days (IQR 3–6 days) of symptoms before first infusion - Median of 2 days (IQR 1–4 days) from RT-PCR confirmation to screening for study participation. Primary Outcomes - COVID-19–related hospitalization or death from any cause by Day 28: 2 (0.7%) in RDV arm vs. 15 (5.3%) in placebo arm (HR 0.13; 95% CI, 0.03–0.59; <i>P</i> = 0.008) - Occurrence of AEs: 42% in RDV arm vs. 46% in placebo arm Secondary Outcome - COVID-19–related, medically attended visit or death from any cause by Day 28: 4 (1.6%) in RDV arm vs. 21 (8.3%) in placebo arm (HR 0.19; 95% CI, 0.07–0.56)	Key Limitations - Study halted early due to administrative issues. - Vaccinated individuals were excluded. Interpretation - Among nonhospitalized patients with COVID-19, 3 consecutive days of RDV resulted in an 87% relative reduction in the risk of hospitalization or death when compared to placebo.

Key: AE = adverse event; ALT = alanine transaminase; AST = aspartate aminotransferase; AZM = azithromycin; CQ = chloroquine; CrCl = creatinine clearance; DM = diabetes mellitus; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; HCQ = hydroxychloroquine; HFNC = high-flow nasal cannula; HTN = hypertension; IV = intravenous; IL = interleukin; LOS = length of stay; LPV/RTV = lopinavir/ritonavir; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; RCT = randomized controlled trial; RDV = remdesivir; RT-PCR = reverse transcription polymerase chain reaction; SAE = serious adverse event; SCr = serum creatinine; SOC = standard of care; SpO₂ = oxygen saturation; ULN = upper limit of normal; WHO = World Health Organization

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Ritonavir-Boosted Nirmatrelvir (Paxlovid)

Last Updated: February 29, 2024

Nirmatrelvir is an oral protease inhibitor that is active against M^{PRO}, a viral protease that plays an essential role in viral replication by cleaving the 2 viral polyproteins.¹ It has demonstrated antiviral activity against all coronaviruses that are known to infect humans.² Nirmatrelvir is packaged with ritonavir (as Paxlovid), a strong cytochrome P450 (CYP) 3A4 inhibitor and pharmacokinetic boosting agent that has been used to boost HIV protease inhibitors. Coadministration of ritonavir is required to increase nirmatrelvir concentrations to the target therapeutic range.

Ritonavir-boosted nirmatrelvir is approved by the Food and Drug Administration (FDA) for the treatment of mild to moderate COVID-19 in adults who are at high risk of progressing to severe COVID-19.³ The Emergency Use Authorization (EUA) for ritonavir-boosted nirmatrelvir will continue to authorize the use of the EUA-labeled product for the treatment of nonhospitalized adolescents aged 12 to 17 years and weighing ≥ 40 kg who are at high risk of progressing to severe COVID-19.

Recommendations

- The COVID-19 Treatment Guidelines Panel (the Panel) recommends using **nirmatrelvir 300 mg with ritonavir 100 mg (Paxlovid)** orally (PO) twice daily for 5 days in nonhospitalized adults with mild to moderate COVID-19 who are at high risk of disease progression (**AIIa**). Treatment should be initiated as soon as possible and within 5 days of symptom onset. For information on medical conditions that confer high risk, see the Centers for Disease Control and Prevention webpage [Underlying Medical Conditions Associated With Higher Risk for Severe COVID-19](#).
- Ritonavir-boosted nirmatrelvir is available through an FDA EUA for the treatment of mild to moderate COVID-19 in nonhospitalized adolescents aged 12 to 17 years and weighing ≥ 40 kg.⁴ For recommendations on using ritonavir-boosted nirmatrelvir in nonhospitalized children with COVID-19, see [Therapeutic Management of Nonhospitalized Children With COVID-19](#).
- See the section titled Considerations in Patients With Severe Renal Insufficiency below for information on managing patients with COVID-19 and severe renal insufficiency.
- For a discussion of the treatment of prolonged, symptomatic COVID-19 in patients with evidence of ongoing SARS-CoV-2 replication, see the section titled Patients Who Are Immunocompromised and Have Prolonged COVID-19 Symptoms and Evidence of Ongoing Viral Replication below.

Rationale

The EPIC-HR trial enrolled nonhospitalized adults with mild to moderate COVID-19 who were not vaccinated and who were at high risk of progressing to severe disease.⁵ The trial demonstrated that starting ritonavir-boosted nirmatrelvir within 5 days of symptom onset in these patients reduced the risk of hospitalization or death through Day 28 by 89% compared to placebo. This efficacy is comparable to remdesivir (87% relative reduction)⁶ and greater than the efficacy reported for molnupiravir (31% relative reduction).⁷ However, these agents have not been directly compared in clinical trials.

Although ritonavir-boosted nirmatrelvir demonstrated a clinical benefit during the EPIC-HR trial, the benefits in unvaccinated people who are at low risk of progression to severe disease or in vaccinated people who are at high risk of progression to severe disease are unclear. The EPIC-SR trial, which included both of these populations, found that ritonavir-boosted nirmatrelvir did not reduce the duration of symptoms and did not have a statistically significant effect on the risk of hospitalization or death

compared to placebo, although the event rates were low in this trial.⁸ Some observational studies have shown a benefit of using ritonavir-boosted nirmatrelvir in vaccinated individuals who were at high risk of progressing to severe COVID-19.⁹⁻¹² However, observational studies have inherent limitations. In particular, the results of these studies may be affected by residual confounding. For information on treatment considerations for vaccinated individuals, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

There are no data from randomized clinical trials that support the use of ritonavir-boosted nirmatrelvir in hospitalized patients with COVID-19.

For more information on the use of ritonavir-boosted nirmatrelvir in patients with COVID-19, see [Table 4e](#). Because of the potential for significant drug-drug interactions with concomitant medications, this regimen may not be the optimal choice for all patients. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for details.

Drug-Drug Interactions

The FDA prescribing information for ritonavir-boosted nirmatrelvir includes a boxed warning about significant drug-drug interactions between ritonavir-boosted nirmatrelvir and other medications. These interactions are primarily caused by the ritonavir component of the combination. Ritonavir, a strong CYP3A4 inhibitor and a P-glycoprotein inhibitor, may increase the blood concentration of certain concomitant medications and increase the potential for serious drug toxicities. Before prescribing ritonavir-boosted nirmatrelvir, clinicians should carefully review the patient's concomitant medications, including over-the-counter medications, herbal supplements, and recreational drugs, to evaluate potential drug-drug interactions. Clinicians should consider both the potential benefits of treatment with ritonavir-boosted nirmatrelvir and the potential risks related to drug-drug interactions. Many drug-drug interactions between ritonavir-boosted nirmatrelvir and concomitant medications (e.g., certain statins, calcium channel blockers, or direct oral anticoagulants) **can be safely managed**. For the Panel's recommendations on preferred and alternative antiviral therapies for outpatients with COVID-19, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#). Clinicians should be aware that the drug-drug interaction potential of ritonavir-boosted nirmatrelvir may change if it is used for extended durations (i.e., ≥ 10 days).

The following resources provide information on identifying and managing drug-drug interactions.

- Quick reference lists:
 - [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#). Box 1 lists select outpatient medications that are not expected to have clinically relevant interactions with ritonavir-boosted nirmatrelvir. Box 2 lists select outpatient medications that have clinically relevant drug-drug interactions with ritonavir-boosted nirmatrelvir.
- Web-based drug-drug interaction checker:
 - The [Liverpool COVID-19 Drug Interactions website](#)
- Tables with guidance on managing specific drug-drug interactions:
 - The [University of Waterloo/University of Toronto drug interaction guide](#) for ritonavir-boosted nirmatrelvir
 - The FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir

Patients Who Are Immunocompromised and Have Prolonged COVID-19 Symptoms and Evidence of Ongoing Viral Replication

For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication (e.g., those with a low cycle threshold value, as measured by a reverse transcription polymerase chain reaction result or with a positive rapid antigen test result) despite receiving a course of antiviral therapy, the optimal management is unknown. Case reports and case series have described the treatment of these patients with additional antiviral treatments, prolonged courses of antiviral treatments, high-titer COVID-19 convalescent plasma, or combination therapy.¹³⁻

¹⁹ For information on potential treatment options, see [Special Considerations in People Who Are Immunocompromised](#) and [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Viral Rebound and Symptom Recurrence

Observational studies and the EPIC-HR and MOVE-OUT trials have described SARS-CoV-2 viral rebound and the recurrence of COVID-19 symptoms in some patients who have completed treatment with ritonavir-boosted nirmatrelvir or molnupiravir.^{3,20-23} However, viral rebound can also occur in patients who have not received treatment for COVID-19.²⁴ Some observational studies have reported that patients who were treated with ritonavir-boosted nirmatrelvir had a higher frequency of viral rebound and symptom recurrence than those who did not receive treatment.^{25,26} The re-emergence of culturable SARS-CoV-2 has been reported in some individuals with viral rebound.

To date, virus detection and the recurrence of COVID-19 symptoms following the use of antiviral therapies have not been associated with progression to severe COVID-19.^{27,28} Therefore, concerns about viral rebound or the recurrence of symptoms should not be a reason to avoid using antiviral therapies when their use is indicated.^{23,29-31} There are insufficient data on whether a longer course of ritonavir-boosted nirmatrelvir will prevent viral rebound or symptom recurrence. There are also insufficient data on the efficacy of administering a second course of antiviral therapy to treat viral rebound or symptom recurrence. However, a clinical trial that is evaluating the use of a second course of ritonavir-boosted nirmatrelvir to treat patients with viral rebound and symptom recurrence is underway (ClinicalTrials.gov Identifier [NCT05567952](#)).

SARS-CoV-2 Resistance

Viral mutations that lead to substantial resistance to nirmatrelvir have been selected for in in vitro studies; the fitness of these mutations is unclear. Surveillance for the emergence of significant resistance to nirmatrelvir is critical, particularly in patients who are severely immunocompromised and who experience prolonged replication of SARS-CoV-2.

Additional Considerations

- Nirmatrelvir must be administered with ritonavir to achieve sufficient therapeutic plasma concentrations.
- In patients with moderate renal impairment (i.e., those with an estimated glomerular filtration rate [eGFR] of ≥ 30 to < 60 mL/min), the dose should be reduced to nirmatrelvir 150 mg with ritonavir 100 mg PO twice daily.
- Ritonavir-boosted nirmatrelvir **is not recommended** for patients with known or suspected severe hepatic impairment (i.e., Child-Pugh Class C), and it should be used with caution in patients with pre-existing liver diseases, liver enzyme abnormalities, or hepatitis. No pharmacokinetic or safety data are available for this patient population.
- Patients should complete the 5-day treatment course of ritonavir-boosted nirmatrelvir because

there are concerns that a shorter treatment course may be less effective or may lead to the emergence of drug resistance.

- If a patient requires hospitalization after starting treatment, the full 5-day treatment course of ritonavir-boosted nirmatrelvir should be completed unless there are drug-drug interactions that preclude its use.
- There are very limited data on combining ritonavir-boosted nirmatrelvir with other antiviral therapies to treat nonhospitalized patients with COVID-19. Clinical trials are needed to determine whether combination therapy has a role in the treatment of COVID-19.
- The FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir advises against crushing nirmatrelvir and ritonavir tablets. However, some data indicate that the tablets can be split or crushed if necessary.³²

Monitoring and Adverse Effects

The most common adverse effects of ritonavir-boosted nirmatrelvir are dysgeusia, diarrhea, hypertension, and myalgia. Anaphylaxis, serious skin reactions, and other hypersensitivity reactions have also been reported.

There is no need to check a patient's renal function prior to prescribing ritonavir-boosted nirmatrelvir unless the patient is suspected to have moderate to severe renal impairment (i.e., those with an eGFR of <60 mL/min). For these patients, clinicians may consider checking the patient's renal function to inform the dosing of ritonavir-boosted nirmatrelvir.

Considerations in Patients With Severe Renal Insufficiency

The FDA prescribing information states that ritonavir-boosted nirmatrelvir is not recommended for patients with an eGFR of <30 mL/min until more data are available to establish appropriate dosing.³ Additional information is available in the initial FDA Center for Drug Evaluation and Research review for the EUA of ritonavir-boosted nirmatrelvir.³³ There is limited clinical experience with the use of ritonavir-boosted nirmatrelvir in patients with an eGFR of <30 mL/min and in those who require hemodialysis.^{34,35} Based on limited data, some groups have proposed dosing adjustments for ritonavir-boosted nirmatrelvir in these patients.³⁶⁻³⁸

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance on the use of ritonavir-boosted nirmatrelvir during pregnancy and lactation.

Considerations in Children

Ritonavir-boosted nirmatrelvir is available through an FDA EUA for the treatment of mild to moderate COVID-19 in nonhospitalized adolescents aged 12 to 17 years and weighing ≥ 40 kg. For recommendations on using ritonavir-boosted nirmatrelvir in nonhospitalized children with COVID-19, see [Therapeutic Management of Nonhospitalized Children With COVID-19](#).

Clinical Data

The EPIC-HR study was a multinational randomized trial that compared the use of ritonavir-boosted nirmatrelvir PO twice daily for 5 days to placebo in nonhospitalized patients aged ≥ 18 years with mild to moderate COVID-19 who were at high risk of clinical progression.⁵ Eligible patients were randomized within 5 days of symptom onset, were not vaccinated against COVID-19, and had at least 1 risk factor for progression to severe disease. Patients were excluded if they used medications that were either

highly dependent upon CYP3A4 for clearance or strong inducers of CYP3A4.

A total of 2,246 patients enrolled in the trial. The mean age was 46 years, 51% of the patients were men, and 72% were White. Forty-seven percent of the patients tested negative for SARS-CoV-2 antibodies, and 66% started study treatment within 3 days of symptom onset.

Patients who were randomized within 3 days of symptom onset ($n = 1,379$) were included in the modified intention-to-treat (mITT) analysis. COVID-19–related hospitalizations or all-cause deaths occurred by Day 28 in 5 of 697 patients (0.72%) in the ritonavir-boosted nirmatrelvir arm and in 44 of 682 patients (6.5%) in the placebo arm. Among the 2,085 patients who were randomized within 5 days of symptom onset (mITT1 analysis), COVID-19–related hospitalizations and all-cause deaths occurred in 8 of 1,039 patients (0.77%) in the ritonavir-boosted nirmatrelvir arm and in 66 of 1,046 patients (6.3%) in the placebo arm (89% relative risk reduction; 5.6% estimated absolute reduction; 95% CI, 7.2% to 4.0%; $P < 0.001$). There were no deaths in the ritonavir-boosted nirmatrelvir arm, and 13 deaths occurred in the placebo arm.

A total of 2,224 patients who received at least 1 dose of either ritonavir-boosted nirmatrelvir or placebo were included in the EPIC-HR safety analysis set. Among these patients, dysgeusia and diarrhea occurred more frequently in ritonavir-boosted nirmatrelvir recipients than in placebo recipients (6% vs. 0.3% and 3% vs. 2%, respectively). Fewer ritonavir-boosted nirmatrelvir recipients discontinued the study drug due to an adverse event than placebo recipients (2% vs. 4%).

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Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir (Paxlovid) and Concomitant Medications

Last Updated: February 29, 2024

Ritonavir, a strong cytochrome P450 (CYP) 3A4 inhibitor and a P-glycoprotein (P-gp) inhibitor, is coadministered with nirmatrelvir to increase the blood concentration of nirmatrelvir, thereby making it effective against SARS-CoV-2. Ritonavir may also increase blood concentrations of certain concomitant medications. The Food and Drug Administration (FDA) [prescribing information](#) includes a boxed warning about significant drug-drug interactions between ritonavir-boosted nirmatrelvir (Paxlovid) and other medications.

Before prescribing ritonavir-boosted nirmatrelvir to treat patients with mild to moderate COVID-19, carefully review the patient's concomitant medications, including over-the-counter medicines, herbal supplements, and recreational drugs. Clinicians should consider the potential benefits of treatment with ritonavir-boosted nirmatrelvir, the potential risks of drug-drug interactions, and whether any risks related to drug-drug interactions can be safely managed. Clinicians should be aware that many commonly used medications can be safely coadministered with ritonavir-boosted nirmatrelvir despite its drug-drug interaction potential. Box 1 includes commonly prescribed medications that are not expected to have clinically relevant interactions with ritonavir-boosted nirmatrelvir.

Because ritonavir-boosted nirmatrelvir is the only highly effective oral antiviral for the treatment of COVID-19, drug interactions that can be safely managed should not preclude the use of this medication.

Box 1. Select Outpatient Medications Not Expected to Have Clinically Relevant Interactions With Ritonavir-Boosted Nirmatrelvir (Paxlovid)

This list is primarily based on the most common medication searches by U.S. users on the Liverpool COVID-19 Drug Interactions website.

Medications Without Clinically Relevant Interactions				
These medications do not require dose adjustments when coadministered with ritonavir-boosted nirmatrelvir, and the patients do not require additional monitoring. This list does not include all the noninteracting medications within each drug category.				
Acid Reducers - Famotidine - Omeprazole - Pantoprazole Allergy - Cetirizine - Diphenhydramine - Fexofenadine - Loratadine Anti-Infectives - Azithromycin - Cidofovir - Hydroxychloroquine - Tecovirimat - Valacyclovir	Cardiovascular - Aspirin - Atenolol - Carvedilol - Furosemide - Hydrochlorothiazide - Irbesartan - Isosorbide dinitrate - Lisinopril - Losartan - Metoprolol - Prasugrel Diabetes - Empagliflozin - Insulin - Metformin - Pioglitazone	Immunosuppressants - Abrocitinib - Baricitinib - Methotrexate - Mycophenolate - Prednisone Lipid-Modifiers - Ezetimibe - Pitavastatin - Pravastatin Migraine - Frovatriptan - Naratriptan - Rizatriptan - Sumatriptan - Zavegepant Neuropsychiatric - Amitriptyline - Bupropion	Neuropsychiatric, cont'd - Citalopram - Duloxetine - Escitalopram - Fluoxetine - Gabapentin - Lorazepam - Nortriptyline - Olanzapine - Paroxetine - Sertraline - Venlafaxine Pain - Acetaminophen - Aspirin - Codeine - Ibuprofen - Meloxicam - Naproxen	Respiratory - Corticosteroids (inhaled/nasal) - Formoterol - Montelukast Miscellaneous - Allopurinol - Contraceptives (PO)^a - Cyclobenzaprine - Donepezil - Enoxaparin - Finasteride - Levothyroxine - Most mAb products^b - Ondansetron

Medications Without Clinically Relevant Interactions, continued

^a Coadministering contraceptive products that contain ethinyl estradiol with ritonavir-boosted nirmatrelvir may result in lower ethinyl estradiol concentrations. The FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir suggests that individuals who use these types of contraceptive products should consider using an additional, nonhormonal contraceptive method. However, the lower ethinyl estradiol concentrations are not expected to be clinically significant during the 5 days of therapy. The progestin concentration of a combined hormonal contraceptive is expected to remain similar or increase with coadministration, which would maintain the effectiveness of the PO contraceptive.

^b Ritonavir-boosted nirmatrelvir interacts with certain conjugated mAbs, such as ado-trastuzumab emtansine, mirvetuximab soravtansine, brentuximab vedotin, enfortumab vedotin, polatuzumab vedotin, and tisotumab vedotin. Before coadministering ritonavir-boosted nirmatrelvir and any of these conjugated mAbs, refer to the drug's FDA prescribing information and consult the patient's specialist providers as needed.

Key: FDA = Food and Drug Administration; mAb = monoclonal antibody; PO = oral

Medications That Have Clinically Relevant Drug-Drug Interactions With Ritonavir-Boosted Nirmatrelvir (Paxlovid)

Clinicians should be aware that, in some cases, drug-drug interactions with ritonavir-boosted nirmatrelvir may lead to serious or life-threatening drug toxicities. The recommended treatment course of ritonavir-boosted nirmatrelvir for COVID-19 is 5 days. CYP3A4 inhibition occurs rapidly, with maximum inhibition occurring within 48 hours of ritonavir initiation.¹ After treatment is completed and ritonavir is discontinued, 70% to 90% of CYP3A4 inhibition resolves within 2 to 3 days.² The time to resolution of inhibition varies based on factors such as the patient's age; therefore, resolution may take longer in some individuals, such as in adults of advanced age.

Ritonavir is also an inhibitor of CYP2D6, P-gp, and organic anion transporting polypeptide (OATP) 1B1. When used for longer durations or chronically, ritonavir may induce CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, and uridine diphosphate-glucuronyltransferase (UGT).

Nirmatrelvir and ritonavir are CYP3A4 substrates. Ritonavir-boosted nirmatrelvir should not be given within 2 weeks of administering a strong CYP3A4 inducer (e.g., St. John's wort, rifampin). Ritonavir-boosted nirmatrelvir is **contraindicated** in this setting because the delayed offset of enzyme induction may reduce the concentrations of nirmatrelvir and ritonavir, rendering the treatment ineffective against SARS-CoV-2. An alternative treatment for COVID-19 should be prescribed.

Identifying Drug-Drug Interactions

Consult the following resources for information on identifying and managing drug-drug interactions.

- Quick reference lists:
 - Box 1 above lists select outpatient medications that are not expected to have clinically relevant interactions with ritonavir-boosted nirmatrelvir.
 - Box 2 below lists select outpatient medications that have clinically relevant drug-drug interactions with ritonavir-boosted nirmatrelvir.
- Web-based drug-drug interaction checker:
 - The [Liverpool COVID-19 Drug Interactions website](#)
- Tables with guidance on managing specific drug-drug interactions:
 - The [University of Waterloo/University of Toronto drug interaction guide](#) for ritonavir-boosted nirmatrelvir
 - The FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir

Management Strategies for Drug-Drug Interactions

Consider the magnitude and significance of the potential drug-drug interaction when choosing management strategies for patients who will be receiving ritonavir-boosted nirmatrelvir. Potential strategies include:

- Increasing monitoring for potential adverse effects to the concomitant medication.
- Adjusting the dose of the concomitant medication.
- Temporarily withholding the concomitant medication.
- Using an alternative to the concomitant medication.
- Using alternative COVID-19 therapies (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)).

Use the chosen strategy for the 5-day treatment course of ritonavir-boosted nirmatrelvir and for at least 2 to 3 days after treatment completion. The strategy may need to continue for a longer duration if ritonavir-boosted nirmatrelvir is initiated in an adult of advanced age or if the interacting medication has a long half-life.

Consider consulting with an expert (e.g., a pharmacist or the patient's specialist providers) when treating patients who are receiving highly specialized therapies or drugs that are prone to concentration-dependent toxicities, such as certain anticonvulsant, anticoagulant, immunosuppressant, antiarrhythmic, chemotherapeutic, and neuropsychiatric drugs.

The decision to prescribe ritonavir-boosted nirmatrelvir to patients who are receiving calcineurin and mammalian target of rapamycin inhibitors should always be made in consultation with the patient's specialist providers. Among reports submitted to the FDA Adverse Event Reporting System, the most commonly reported concomitant medications that resulted in serious adverse reactions, including fatal events, were calcineurin inhibitors (e.g., tacrolimus).³ Ritonavir-boosted nirmatrelvir may be prescribed to certain patients who are receiving these medications if an expert in managing the interaction is available and close therapeutic drug monitoring is logistically feasible. Otherwise, an alternative therapy for COVID-19 should be considered. See the [American Society of Transplantation](#) statement for more information.

Interactions between ritonavir-boosted nirmatrelvir and chemotherapeutic agents should also be managed in consultation with the patient's specialist providers. For guidance on managing these interactions, refer to the FDA [prescribing information](#) for ritonavir-boosted nirmatrelvir and the prescribing information for the chemotherapeutic agent. The [University Health Network/Kingston Health Sciences Centre](#) provides an additional resource for evaluating drug-drug interactions between ritonavir-boosted nirmatrelvir and chemotherapeutic agents.

Patients should be counseled about ritonavir-boosted nirmatrelvir's drug-drug interaction potential and the signs and symptoms of potential adverse effects. If ritonavir-boosted nirmatrelvir is prescribed to patients who take certain recreational drugs, those patients will require counseling and careful monitoring for adverse effects.

Box 2. Select Outpatient Medications That Have Clinically Relevant Drug-Drug Interactions With Ritonavir-Boosted Nirmatrelvir (Paxlovid)

The guidance in Box 2 is based on the drug-drug interaction potential of the FDA-approved, 5-day course of ritonavir-boosted nirmatrelvir.

Not all medications that may interact with ritonavir-boosted nirmatrelvir are included in Box 2. Deviation from the recommended strategies may be appropriate in certain clinical scenarios.

Prescribe Alternative COVID-19 Therapy

For these medications, management strategies are not possible or feasible, or the risks outweigh the potential benefits.

Anticonvulsants • Carbamazepine • Phenobarbital • Phenytoin • Primidone Anti-Infectives • Glecaprevir/pibrentasvir • Rifampin • Rifapentine Immunosuppressants • Voclosporin	Cardiovascular • Amiodarone • Clopidogrel^{a,b} • Disopyramide • Dofetilide • Dronedarone • Eplerenone • Flecainide • Ivabradine • Propafenone • Quinidine	Neuropsychiatric • Clozapine • Lurasidone • Midazolam (PO) • Pimozide Pulmonary Hypertension^c • Sildenafil • Tadalafil • Vardenafil	Miscellaneous • Bosentan • Certain chemotherapeutic agents^d • Ergot derivatives • Lumacaftor/ivacaftor • St. John's wort • Tolvaptan
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Temporarily Withhold Concomitant Medication, if Clinically Appropriate

Withhold these medications during ritonavir-boosted nirmatrelvir (Paxlovid) treatment and for at least 2–3 days after treatment completion. They may need to be withheld for longer if the patient is an adult of advanced age or if the interacting medication has a long half-life. If withholding is not clinically appropriate, use an alternative concomitant medication or COVID-19 therapy.

Anticoagulants • Rivaroxaban^e Anti-Infectives • Erythromycin BPH • Alfuzosin • Silodosin Cardiovascular • Aliskiren • Ranolazine • Ticagrelor^b • Vorapaxar	Immunosuppressants^f • Everolimus • Sirolimus • Tacrolimus Lipid-modifiers • Atorvastatin^g • Lomitapide • Lovastatin^g • Rosuvastatin^g • Simvastatin^g	Migraine • Eletriptan • Rimegepant • Ubrogapant Neuropsychiatric • Daridorexant • Lemborexant • Suvorexant • Triazolam^h Erectile Dysfunction • Avanafil	Respiratory • Salmeterol Miscellaneous • Certain chemotherapeutic agents^d • Colchicineⁱ • Finerenone • Flibanserin • Naloxegol
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Adjust Concomitant Medication Dose and Monitor for Adverse Effects

Reduce the dose and/or extend the dosing interval of the concomitant medication. Consult the [Liverpool COVID-19 Drug Interactions website](#) or the [University of Waterloo/University of Toronto drug interaction guide](#) for specific dosing recommendations.^j If the dose of the concomitant medication cannot be adjusted, withhold the medication (if clinically appropriate) or use an alternative concomitant medication or COVID-19 therapy.

Anticoagulants • Apixaban • Dabigatran • Edoxaban Anti-Infectives • Clarithromycin • Itraconazole • Ketoconazole • Maraviroc • Rifabutin BPH • Tamsulosin	Cardiovascular • Amlodipine • Cilostazol • Digoxin • Diltiazem • Felodipine • Nifedipine • Verapamil Diabetes • Saxagliptin Erectile Dysfunction^c • Sildenafil • Tadalafil • Vardenafil	Immunosuppressants • Cyclosporine^f • Dexamethasone^k • Fedratinib • Ruxolitinib • Tofacitinib • Upadacitinib Migraine • Almotriptanⁱ Neuropsychiatric • Alprazolam^h • Aripiprazole • Brexpiprazole	Neuropsychiatric, cont'd • Buspirone • Cariprazine • Chlordiazepoxide^h • Clobazam^h • Clonazepam^h • Clorazepate^h • Diazepam^h • Estazolam^h • Flurazepam^h • Iloperidone • Lumateperone • Pimavanserin • Quetiapine • Trazodone
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Adjust Concomitant Medication Dose and Monitor for Adverse Effects, continued			
Pain - Fentanyl - Hydrocodone - Oxycodone Pulmonary Hypertension - Riociguat	Miscellaneous - Certain chemotherapeutic agents^d - Darifenacin	Miscellaneous, cont'd - Elexacaftor/tezacaftor/ivacaftor - Eluxadoline - Ivacaftor	Miscellaneous, cont'd - Solifenacin - Tezacaftor/ivacaftor
Continue Concomitant Medication and Monitor for Adverse Effects			
There is no need to pre-emptively adjust the doses of these drugs, but dose adjustments may be considered in patients with a high risk of AEs. Educate patients about potential AEs. Consult the Liverpool COVID-19 Drug Interactions website or the University of Waterloo/University of Toronto drug interaction guide for monitoring guidance and dose adjustment information.^j			
Anticoagulants - Warfarin Anti-Infectives - Brincidofovirⁱ - Cobicistat- or ritonavir-boosted ARV drugs - Isavuconazole - Posaconazole - Voriconazole	BPH - Doxazosin - Terazosin Diabetes - Glyburide Cardiovascular - Mexiletine - Sacubitril - Valsartan	Migraine - Zolmitriptan Neuropsychiatric - Haloperidol - Hydroxyzine - Mirtazapine - Risperidone - Ziprasidone - Zolpidem	Pain - Buprenorphine - Hydromorphone - Methadone - Morphine - Tramadol Miscellaneous - Certain chemotherapeutic agents^d - Certain conjugated mAbs^m - Oxybutynin
^a Reduced effectiveness of clopidogrel is likely. It may be acceptable to continue using clopidogrel if the benefits of using ritonavir-boosted nirmatrelvir outweigh the risk of reduced clopidogrel effectiveness. ^b For patients with a very high risk of thrombosis (e.g., those who received a coronary stent within the past 6 weeks), consider prescribing an alternative antiplatelet (e.g., prasugrel, if clinically appropriate) or an alternative COVID-19 therapy. ^c Some PDE5 inhibitors are used to treat both PAH and erectile dysfunction; however, the doses used to treat PAH are higher than those used for erectile dysfunction. Because of this, and because PDE5 inhibitors are used chronically in patients with PAH, coadministration with ritonavir-boosted nirmatrelvir is contraindicated in these patients. PDE5 inhibitors can be coadministered with ritonavir-boosted nirmatrelvir in patients with erectile dysfunction, though the dose of the PDE5 inhibitor should be adjusted. ^d Ritonavir-boosted nirmatrelvir may increase concentrations of some chemotherapeutic agents, leading to an increased potential for drug toxicities. Some chemotherapeutic agents may decrease the effectiveness of ritonavir-boosted nirmatrelvir. Please refer to the FDA prescribing information for ritonavir-boosted nirmatrelvir and the prescribing information for the chemotherapeutic agent and consult the patient's specialist provider. The University Health Network/Kingston Health Sciences Centre is an additional resource for evaluating drug-drug interactions for chemotherapeutic agents. ^e For patients with a high risk of arterial or venous thrombosis (e.g., those who had a stroke within the past 3 months with a CHA₂DS₂-VASc score of 7–9 or a pulmonary embolism within the past month), consult the patient's primary or specialty provider and consider using an alternative anticoagulant (e.g., LMWH) or an alternative COVID-19 therapy. For patients with a lower risk of arterial or venous thrombosis, clinicians may consider administering low-dose aspirin while rivaroxaban is being withheld. ^f The use of another COVID-19 therapy may need to be considered. These immunosuppressants have significant drug-drug interaction potential with ritonavir, and they should not be used if close monitoring, including therapeutic drug monitoring (i.e., measuring drug concentrations), is not feasible. Consult the patient's specialist providers before coadministering these immunosuppressants with ritonavir-boosted nirmatrelvir. See the American Society of Transplantation statement for more information. ^g Withhold lovastatin and simvastatin for at least 12 hours before initiating ritonavir-boosted nirmatrelvir, during treatment, and for 5 days after treatment completion. Withhold atorvastatin and rosuvastatin at the beginning of treatment with ritonavir-boosted nirmatrelvir and resume after completing the 5-day course. If withholding a statin is not clinically appropriate (e.g., because the patient recently had a myocardial infarction), clinicians can reduce the doses of			

Continue Concomitant Medication and Monitor for Adverse Effects, continued

atorvastatin and rosuvastatin and continue treatment. However, lovastatin and simvastatin should be switched to an alternative statin.

^h The guidance on managing drug-drug interactions between certain benzodiazepines and ritonavir-boosted nirmatrelvir can vary significantly between product information resources. Note that abrupt discontinuation or rapid dose reduction of benzodiazepines may precipitate an acute withdrawal reaction.⁴ The risk is greatest for patients who have been using high doses of benzodiazepines over an extended period.

ⁱ Do not coadminister this medication with ritonavir-boosted nirmatrelvir in patients with hepatic or renal impairment.

^j For medications that are not included on the Liverpool COVID-19 Drug Interactions website or in the University of Waterloo/University of Toronto drug interaction guide, refer to the FDA labels for information on coadministering these medications with ritonavir or other strong CYP3A4 and/or P-gp inhibitors (e.g., ketoconazole).

^k Dexamethasone exposure is expected to increase 2.60-fold when dexamethasone is coadministered with ritonavir-boosted nirmatrelvir.⁵ Clinicians should weigh the risks and benefits of continuing the patient's normal dose of dexamethasone (while monitoring for AEs) against the risks and benefits of decreasing the dose. Patients who are receiving higher doses of dexamethasone will be at a greater risk of AEs.

^l Patients should take ritonavir-boosted nirmatrelvir at least 3 hours after taking brincidofovir.

^m Ritonavir-boosted nirmatrelvir interacts with certain conjugated mAbs, such as ado-trastuzumab emtansine, mirvetuximab soravtansine, brentuximab vedotin, enfortumab vedotin, polatuzumab vedotin, and tisotumab vedotin. Before coadministering ritonavir-boosted nirmatrelvir and any of these conjugated mAbs, refer to the drug's FDA prescribing information and consult the patient's specialist providers as needed.

Key: AE = adverse effect; ARV = antiretroviral; BPH = benign prostatic hyperplasia; CHA₂DS₂-VAsC = congestive heart failure, hypertension, age, diabetes, stroke, vascular disease; CYP = cytochrome P450; FDA = Food and Drug Administration; LMWH = low-molecular-weight heparin; mAb = monoclonal antibody; PAH = pulmonary arterial hypertension; PDE5 = phosphodiesterase 5; P-gp = P-glycoprotein; PO = oral

Drug-Drug Interaction Considerations When Using Extended Courses of Ritonavir-Boosted Nirmatrelvir (Paxlovid)

The guidance in this document is based on the drug-drug interaction potential of the FDA-approved, 5-day course of ritonavir-boosted nirmatrelvir. Longer treatment courses may be utilized in certain cases (see [Special Considerations in People Who Are Immunocompromised](#)). Clinicians should be aware that the drug-drug interaction potential of ritonavir may change based on the duration of treatment. Clinicians should also be aware that:

- Induction properties⁶ may become clinically relevant when ritonavir is used for longer durations (i.e., ≥ 10 days) or chronically (e.g., in people who take HIV protease inhibitors).⁷ For example, induction of CYP2C9 and CYP2C19 may decrease warfarin and voriconazole concentrations, and induction of glucuronidation may decrease lamotrigine or valproic acid concentrations.
- The management strategies listed in Box 2 are based on the drug-drug interaction potential of a 5-day treatment course of ritonavir-boosted nirmatrelvir. These strategies may need to be modified when using extended courses. For example, a clinician may need to withhold or reduce the dose of a corticosteroid instead of continuing it as suggested in Box 2. Clinicians may need to adjust monitoring plans for adverse effects or therapeutic drug monitoring in certain patients (e.g., in those receiving tacrolimus). In other cases, the potential risks of withholding certain agents (e.g., chemotherapeutic agents or statins in high-risk individuals) for extended periods to allow for safe coadministration of ritonavir-boosted nirmatrelvir may outweigh the potential benefits of treatment.
- After longer courses of ritonavir-boosted nirmatrelvir are discontinued, drug-drug interactions caused by CYP3A4 inhibition are expected to resolve within 2 to 3 days.² Drug-drug interactions caused by induction (e.g., CYP2C9, CYP2C19, UGT) resolve gradually and variably.^{8,9}

Clinicians should consult with experts (e.g., pharmacists and physicians with HIV expertise) when using extended courses of ritonavir-boosted nirmatrelvir. The Liverpool COVID-19 Drug Interactions website also provides guidance for managing drug-drug interactions during extended courses (i.e., ≥ 10 days) of ritonavir-boosted nirmatrelvir.

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Molnupiravir

Last Updated: February 29, 2023

Molnupiravir is the oral prodrug of beta-D-N4-hydroxycytidine (NHC), a ribonucleoside that has shown antiviral activity against SARS-CoV-2 in vitro and in some clinical trials.^{1,2} NHC uptake by viral RNA-dependent RNA-polymerases results in viral mutations and lethal mutagenesis.^{3,4} On December 23, 2021, the Food and Drug Administration (FDA) issued an Emergency Use Authorization (EUA) for molnupiravir for the treatment of adults with mild to moderate COVID-19 who are within 5 days of symptom onset, who are at high risk of progressing to severe disease, and for whom alternative antiviral therapies are not accessible or clinically appropriate.^{5,6} Molnupiravir is expected to be active against the Omicron variant and its subvariants.⁶

There is a theoretical risk that the molnupiravir metabolite NHC could be incorporated into the host DNA, leading to mutations. The available genotoxicity data and the 5-day duration of treatment led the FDA to conclude that molnupiravir has a low risk for genotoxicity.

Recommendations

- The COVID-19 Treatment Guidelines Panel (the Panel) recommends using **molnupiravir 800 mg** orally (PO) twice daily for 5 days as an alternative therapy in nonhospitalized patients aged ≥ 18 years with mild to moderate COVID-19 who are at high risk of disease progression when ritonavir-boosted nirmatrelvir (Paxlovid) and remdesivir are not available, feasible to use, or clinically appropriate; treatment should be initiated as soon as possible and within 5 days of symptom onset (**CIIa**).
- The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (**AIII**). For more details, see Considerations in Pregnant and Lactating People below.
- People who engage in sexual activity that may result in conception should use effective contraception during and following treatment with molnupiravir. For more details, see Considerations in Sexually Active Individuals below.

Molnupiravir may be used in patients who are hospitalized for a diagnosis other than COVID-19, provided they have mild to moderate COVID-19 and are at high risk of progressing to severe disease. For the Panel's recommendations on preferred and alternative antiviral therapies for outpatients with COVID-19, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Rationale

The MOVE-OUT trial enrolled high-risk, unvaccinated, nonhospitalized adults and reported that molnupiravir reduced the rate of hospitalization or death among these patients by 31% compared to placebo.⁷ A secondary analysis of patients who required hospitalization during the trial found a reduced need for respiratory interventions among those who received molnupiravir compared to those who received placebo.⁸ MOVE-OUT was conducted before the emergence of the Omicron subvariants.

The PANORAMIC trial enrolled nonhospitalized adults with COVID-19 who were at high risk of severe disease during a period when the Omicron variant was circulating.⁹ Ninety-four percent of the patients had received at least 3 doses of a COVID-19 vaccine. The study found that the use of molnupiravir plus usual care did not reduce the occurrence of the primary composite outcome of hospitalization or death

compared to usual care alone. The proportion of patients who met this composite outcome was 1% in both arms. Molnupiravir plus usual care was superior to usual care alone for several secondary clinical endpoints. For example, the time to self-reported recovery was substantially shorter in people who received molnupiravir plus usual care than in people who received usual care alone (median of 9 days vs. 15 days). Because the PANORAMIC trial was an open-label study with self-reported symptoms, the findings are less reliable than those from a placebo-controlled trial.

For more information on the MOVE-Out and PANORAMIC trials, see the Clinical Data section below.

Although the different COVID-19 treatment options have not been directly compared in clinical trials, the Panel recommends using **molnupiravir** only when ritonavir-boosted nirmatrelvir and remdesivir are not available, feasible to use, or clinically appropriate (**CIIa**). Molnupiravir appears to have lower clinical efficacy than these other treatment options.

Some observational studies have evaluated the use of molnupiravir in nonhospitalized or hospitalized adults who are at high risk of progressing to severe disease, including some patients who received COVID-19 vaccines, but these studies have limitations.¹⁰⁻¹² For treatment considerations for vaccinated individuals, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Additional Considerations

- Patients should complete the 5-day treatment course of molnupiravir. It is unknown whether a shorter course is less effective or associated with the emergence of molnupiravir-resistant mutations.
- If a patient requires hospitalization after starting treatment, the full treatment course of molnupiravir can be completed at the health care provider's discretion.
- The FDA EUA for molnupiravir provides instructions for preparing and administering capsule contents through orogastric or nasogastric tubes.⁶
- There are no data on using combinations of antiviral therapies to treat nonhospitalized patients with COVID-19. Clinical trials are needed to determine whether combination therapy has a role in the treatment of SARS-CoV-2 infection.
- Patients who are severely immunocompromised can experience prolonged periods of SARS-CoV-2 replication, which may lead to rapid viral evolution. There are theoretical concerns that using a single antiviral agent in these patients may produce antiviral-resistant viruses. Additional studies are needed to assess this risk. The role of combination antiviral therapy in treating patients who are severely immunocompromised is not yet known. See [Special Considerations in People Who Are Immunocompromised](#) for more information.
- There are limited data on the frequency of SARS-CoV-2 rebound in patients who have completed treatment with molnupiravir.¹³ During the MOVE-OUT trial, rates of symptomatic SARS-CoV-2 rebound were low (approximately 1%) in both those who received molnupiravir and those who received placebo.⁶
- SARS-CoV-2 sequences with molnupiravir-induced mutations have been detected in many countries, but the clinical implications of these mutations remain unclear.¹⁴

Monitoring, Adverse Effects, and Drug-Drug Interactions

The most common adverse effects of molnupiravir are diarrhea, nausea, and dizziness. Based on in vitro studies, neither molnupiravir nor its active metabolite NHC are inhibitors or inducers of major drug-metabolizing enzymes or inhibitors of major drug transporters.

According to the FDA EUA, no drug-drug interactions have been identified for molnupiravir.

Considerations in Sexually Active Individuals

For individuals of childbearing potential, clinicians should assess the patient's pregnancy status before initiating molnupiravir.

Patients of childbearing potential should be counseled about abstaining from sex or using reliable contraception for the duration of therapy and for up to 4 days after taking molnupiravir. Reproductive toxicity has been reported in animal studies of molnupiravir, and molnupiravir may be mutagenic during pregnancy.

The FDA EUA states that men of reproductive potential who are sexually active with individuals of childbearing potential should be counseled to abstain from sex or use a reliable method of contraception for the duration of treatment and for at least 3 months after the last dose of molnupiravir.

Considerations in Pregnant and Lactating People

The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (**AIII**). Because the risk of adverse effects in infants is currently unknown, the FDA EUA fact sheet does not recommend feeding an infant breast milk from a patient who is taking molnupiravir for the duration of the treatment course and until 4 days after the final dose. See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for more information.

Considerations in Children

The MOVE-OUT and PANORAMIC trials excluded participants aged <18 years. There are no data available on the use of molnupiravir in children aged <18 years. Molnupiravir is not authorized for use in those aged <18 years due to potential effects on bone and cartilage growth.

Clinical Data

MOVE-OUT

MOVE-OUT was a Phase 3 trial that evaluated the use of molnupiravir in unvaccinated, nonhospitalized adults with mild to moderate COVID-19 who were at high risk of progressing to severe COVID-19 and who were enrolled within 5 days of symptom onset.⁷ Patients were randomized to receive molnupiravir 800 mg PO every 12 hours for 5 days or placebo. The primary composite endpoint was a hospital stay >24 hours or death by Day 29.

Results

- The final analysis included 1,433 patients.
 - The median age was 43 years (with 17% aged >60 years); 49% of patients were men, 57% were White, 50% were Hispanic/Latinx, and 5% were Black or African American.
 - 4% had a body mass index ≥ 30 , and 16% had diabetes.
- The time from the onset of COVID-19 symptoms to randomization was ≤ 3 days in 48% of patients.
- By Day 29, 48 of 709 patients (6.8%) in the molnupiravir arm and 68 of 699 (9.7%) in the placebo arm experienced hospitalization or death (adjusted difference -3.0%; 95% CI, -5.9% to -0.1%).
- One death occurred in the molnupiravir arm, and 9 deaths occurred in the placebo arm.

- There were no significant differences between the arms in the proportions of patients who experienced adverse events.
- A secondary analysis of data from the patients who were hospitalized during the trial revealed that the use of molnupiravir reduced the risk of requiring respiratory interventions (conventional or high-flow oxygen delivery, noninvasive ventilation, or mechanical ventilation) by 21%.⁸

Limitations

- This study was conducted before the emergence of the Omicron variant and its subvariants.

PANORAMIC

PANORAMIC was a multicenter, open-label, adaptive platform trial conducted in the United Kingdom that evaluated the use of molnupiravir in nonhospitalized adults who were at high risk of progressing to severe COVID-19.⁹ The participants were aged ≥ 50 years or ≥ 18 years with comorbid conditions and were enrolled within 5 days of symptom onset. Patients were randomized to receive molnupiravir 800 mg PO twice daily for 5 days plus usual care or usual care alone. The primary endpoint was a composite of all-cause hospitalization (defined as ≥ 1 overnight hospital stay, ≥ 1 night at home with care and monitoring by hospital clinicians, or an overnight stay in an emergency room) or death within 28 days. The trial was conducted when the Omicron variant was the dominant variant.

Results

- The final analysis included 25,708 patients.
 - The mean age was 56.6 years (26.5% aged ≥ 65 years); 94% were White; 59% were women.
 - 94% received ≥ 3 doses of a COVID-19 vaccine.
 - 69% of patients had comorbidities: 25% with lung disease; 15% with obesity; 12% with diabetes; 8% with heart disease; 8.5% were immunocompromised.
 - 24% were taking inhaled corticosteroids.
- The mean time from symptom onset to starting molnupiravir was 3 days (range 3–5 days). Among the patients who provided information, 95% reported completing the 5-day treatment course.
- The study reported 103 hospitalizations and 3 deaths in the molnupiravir arm and 96 hospitalizations and 5 deaths in the usual care alone arm (aOR 1.06; 95% CrI, 0.81–1.41; probability of superiority 0.33).
- The time to self-reported first recovery among those who received molnupiravir (median of 9 days) was shorter than among those who received usual care alone (median of 15 days).
- No serious adverse events were related to molnupiravir; 145 patients (1.1%) withdrew because of adverse events attributed to molnupiravir.

Limitations

- Because the PANORAMIC trial was an open-label study and the patients knew whether they were receiving molnupiravir or not, this may have affected their reported symptoms. As a result, these findings are less reliable than those from a placebo-controlled trial.

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Anti-SARS-CoV-2 Monoclonal Antibodies

Last Updated: February 29, 2024

Monoclonal antibodies (mAbs) that target the SARS-CoV-2 spike protein have been shown to have clinical benefits in treating SARS-CoV-2 infection. However, laboratory studies have found that the activity of anti-SARS-CoV-2 mAbs against specific variants and subvariants can vary dramatically. The anti-SARS-CoV-2 mAb products that have received Emergency Use Authorizations (EUAs) from the Food and Drug Administration (FDA) are not expected to be effective against the currently circulating SARS-CoV-2 variants and subvariants. As a result, these products are not currently recommended by the COVID-19 Treatment Guidelines Panel (the Panel) for the treatment or prevention of COVID-19.

See [Table 4b](#) for information on the clinical trials that have evaluated the safety and efficacy of using anti-SARS-CoV-2 mAbs in patients with COVID-19.

Recommendation

- The Panel **recommends against** the use of anti-SARS-CoV-2 mAbs for the treatment or prevention of COVID-19 (**AIII**).

Anti-SARS-CoV-2 Monoclonal Antibodies That Have Received Emergency Use Authorizations

Several anti-SARS-CoV-2 mAb products (bamlanivimab plus etesevimab, casirivimab plus imdevimab, sotrovimab, and bebtelovimab) have received EUAs from the FDA for the treatment of outpatients with mild to moderate COVID-19.¹⁻⁴ However, these products are not currently authorized for use in the United States because the dominant Omicron subvariants are not expected to be susceptible to these products. The Centers for Disease Control and Prevention [COVID Data Tracker](#) provides regular updates on the regional proportions of SARS-CoV-2 variants in the United States.

Tixagevimab plus cilgavimab (Evusheld) received an EUA from the FDA for pre-exposure prophylaxis (PrEP) of COVID-19,⁵ and bamlanivimab plus etesevimab and casirivimab plus imdevimab received EUAs for SARS-CoV-2 post-exposure prophylaxis (PEP).^{1,2} However, because many Omicron subvariants, including the dominant Omicron subvariants in the United States, are not expected to be susceptible to these anti-SARS-CoV-2 mAb products, these products are not currently authorized for use as PrEP of COVID-19 or SARS-CoV-2 PEP. See [Prevention of SARS-CoV-2 Infection](#) for more information.

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Table 4b. Anti-SARS-CoV-2 Monoclonal Antibodies: Selected Clinical Trial Data

Last Updated: February 29, 2024

This table describes the clinical trials that have evaluated anti-SARS-CoV-2 mAbs for the treatment of COVID-19.

Methods	Results	Limitations and Interpretation
BLAZE-1: Double-Blind RCT of Bamlanivimab Plus Etesevimab in Nonhospitalized Patients With Mild to Moderate COVID-19 in the United States and Puerto Rico¹		
Key Inclusion Criteria - Aged ≥12 years - At high risk of severe COVID-19 or hospitalization Interventions - Within 3 days of a positive SARS-CoV-2 test result, single IV infusion of: - BAM 700 mg plus ETE 1,400 mg (n = 511) - Placebo (n = 258) Primary Endpoint - Composite of COVID-19–related hospitalization (defined as ≥24 hours of acute care) or death from any cause by Day 29	Participant Characteristics - Median age 56 years; 30% aged ≥65 years; 53% women 87% White, 27% Hispanic/Latinx, 8% Black/African American - Mean of 4 days of symptoms 76% with mild COVID-19, 24% with moderate COVID-19 Primary Outcomes - Composite of COVID-19–related hospitalization or death from any cause by Day 29: 4 (0.8%) in BAM plus ETE arm vs. 15 (5.8%) in placebo arm (change of -5.0%; 95% CI, -8.0% to -2.1%; <i>P</i> < 0.001) - Death from any cause by Day 29: 0 in BAM plus ETE arm vs. 4 (1.6%) in placebo arm	Key Limitation - Conducted before widespread circulation of the Omicron variant Interpretation - Compared to placebo, BAM plus ETE significantly reduced the number of COVID-19–related hospitalizations and deaths from any cause in high-risk patients.
BLAZE-4, Treatment Arms 9–11: Double-Blind RCT of Bamlanivimab Plus Etesevimab Plus Bebtelovimab Versus Bebtelovimab Alone Versus Placebo in Low-Risk, Nonhospitalized Patients With Mild to Moderate COVID-19²		
Key Inclusion Criteria - Aged 18–64 years - No risk factors for progression to severe COVID-19 Key Exclusion Criteria ≥1 of the following: - SpO₂ ≤93% on room air - Respiratory rate ≥30 breaths/min - Heart rate ≥125 bpm	Participant Characteristics - Median age 35 years; 56% women 36% Hispanic/Latinx, 19% Black/African American - Mean of 3.6 days of symptoms prior to enrollment Primary Outcomes - Proportion with PHVL: 13% in BAM plus ETE plus BEB arm vs. 21% in placebo arm (<i>P</i> = 0.098), with a relative reduction of 38% (95% CI, -9% to 65%)	Key Limitations - Only low-risk patients were included. - Not powered to assess hospitalizations and deaths - Conducted before widespread circulation of the Omicron variant Interpretations - There were no differences in the proportion of patients with PHVL across the arms.

Methods	Results	Limitations and Interpretation
BLAZE-4, Treatment Arms 9–11: Double-Blind RCT of Bamlanivimab Plus Etesevimab Plus Bebtelovimab Versus Bebtelovimab Alone Versus Placebo in Low-Risk, Nonhospitalized Patients With Mild to Moderate COVID-19², continued		
Interventions - Within 3 days of a positive SARS-CoV-2 test result, single IV infusion of: - BAM 700 mg plus ETE 1,400 mg plus BEB 175 mg (n = 127) - BEB 175 mg (n = 125) - Placebo (n = 128) Primary Endpoint - Proportion of patients with PHVL (defined as SARS-CoV-2 VL >5.82 log₁₀ by Day 7) Key Secondary Endpoints - Mean change in VL from baseline to Days 3, 5, 7, and 11 - Composite of COVID-19–related hospitalization or death from any cause by Day 29 - Time to sustained symptom resolution	14% in BEB arm vs. 21% in placebo arm ($P = 0.147$), with a relative reduction of 34% (95% CI, -15% to 62%) Secondary Outcomes - Mean decline in VL was greater in mAb arms than in placebo arm at Day 5 but not at Days 3, 7, or 11. - Composite of COVID-19–related hospitalization or death from any cause by Day 29: 3 (2.4%) in BAM plus ETE plus BEB arm, including 1 death 2 (1.6%) in BEB arm 2 (1.6%) in placebo arm - Median time to sustained symptom resolution: 7 days in BAM plus ETE plus BEB arm vs. 8 days in placebo arm ($P = 0.289$) 6 days in BEB arm vs. 8 days in placebo arm ($P = 0.003$)	- Few COVID-19–related hospitalizations or deaths from any cause occurred by Day 29 across the arms, as is expected for a population of individuals who were at low risk of severe COVID-19. - The median time to sustained symptom resolution was shorter in the BEB arm than in the placebo arm.
BLAZE-4, Treatment Arms 12 and 13: Open-Label RCT of Bamlanivimab Plus Etesevimab Plus Bebtelovimab Versus Bebtelovimab Alone in High-Risk, Nonhospitalized Patients With Mild to Moderate COVID-19²		
Key Inclusion Criteria - Aged ≥12 years - Weight ≥40 kg ≥1 risk factors for progression to severe COVID-19 Key Exclusion Criteria ≥1 of the following: - SpO₂ ≤93% on room air - Respiratory rate ≥30 breaths/min - Heart rate ≥125 bpm Interventions - Within 3 days of a positive SARS-CoV-2 test result, single IV infusion of:	Participant Characteristics - Median age 50 years; 52% women 18% Hispanic/Latinx, 18% Black/African American - Mean of 4.7 days of symptoms prior to enrollment 21% received ≥1 doses of a COVID-19 vaccine. Efficacy Outcomes - Composite of COVID-19–related hospitalization or death from any cause by Day 29: 2 (4%) in BAM plus ETE plus BEB arm vs. 3 (3%) in BEB arm - Mean decline in VL was greater in BAM plus ETE plus BEB arm than in BEB arm at Day 5 but not at Days 3, 7, or 11.	Key Limitations - Open-label study - No placebo arm - Not powered to assess hospitalizations and deaths - Conducted before widespread circulation of the Omicron variant Interpretation - There was no difference between the arms in the proportion of patients who were hospitalized or who died.

Methods	Results	Limitations and Interpretation
BLAZE-4, Treatment Arms 12 and 13: Open-Label RCT of Bamlanivimab Plus Etesevimab Plus Bebtelovimab Versus Bebtelovimab Alone in High-Risk, Nonhospitalized Patients With Mild to Moderate COVID-19², continued		
- BAM 700 mg plus ETE 1,400 mg plus BEB 175 mg (n = 50) - BEB 175 mg (n = 100) Efficacy Endpoints - Composite of COVID-19–related hospitalization or death from any cause by Day 29 - Mean change in VL from baseline to Days 3, 5, 7, and 11		
Double-Blind RCT of Casirivimab Plus Imdevimab in Nonhospitalized Patients With Mild to Moderate COVID-19³		
Key Inclusion Criteria - Aged ≥18 years - Laboratory-confirmed SARS-CoV-2 infection - Symptom onset within 7 days of randomization - For patients included in the modified full analysis only: ≥1 risk factors for severe COVID-19 - Positive SARS-CoV-2 RT-PCR result at baseline Interventions - Single IV infusion of: - CAS 600 mg plus IMD 600 mg (n = 736) or placebo (n = 748) - CAS 1,200 mg plus IMD 1,200 mg (n = 1,355) or placebo (n = 1,341) Primary Endpoint - Composite of COVID-19–related hospitalization or death from any cause by Day 29	Participant Characteristics - Median age 50 years 35% Hispanic/Latinx, 5% Black/African American - Median of 3 days of symptoms prior to enrollment Primary Outcomes - COVID-19–related hospitalizations or deaths from any cause by Day 29: 7 (1.0%) in CAS 600 mg plus IMD 600 mg arm vs. 24 (3.2%) in placebo arm ($P = 0.002$) 18 (1.3%) in CAS 1,200 mg plus IMD 1,200 mg arm vs. 62 (4.6%) in placebo arm ($P < 0.001$) - Deaths from any cause by Day 29: 1 (0.1%) in CAS 600 mg plus IMD 600 mg arm vs. 1 (0.1%) in placebo arm 1 (<0.1%) in CAS 1,200 mg plus IMD 1,200 mg arm vs. 3 (0.2%) in placebo arm	Key Limitation - Conducted before widespread circulation of the Omicron variant Interpretation - Compared to placebo, CAS 600 mg plus IMD 600 mg and CAS 1,200 mg plus IMD 1,200 mg reduced the risk of COVID-19–related hospitalization or death from any cause in patients with mild to moderate COVID-19.

Methods	Results	Limitations and Interpretation
COMET-ICE: Double-Blind RCT of Sotrovimab in Nonhospitalized Patients With Mild to Moderate COVID-19 in Brazil, Canada, Peru, Spain, and the United States⁴		
Key Inclusion Criteria • Aged ≥18 years • ≥1 comorbidities or aged ≥55 years • Positive SARS-CoV-2 RT-PCR or antigen test result • Symptom onset ≤5 days before enrollment Key Exclusion Criteria • Hospitalized or required supplemental oxygen • Severely immunocompromised Interventions • SOT 500 mg IV (n = 528) • Placebo (n = 529) Primary Endpoint • Composite of hospitalization or death from any cause by Day 29	Participant Characteristics • Median age 53 years; 20% aged ≥65 years; 54% women • 65% Hispanic/Latinx, 8% Black/African American • 63% with obesity; 22% with DM; 17% with moderate to severe asthma Primary Outcome • Composite of hospitalization or death from any cause by Day 29: 6 (1%) in SOT arm vs. 30 (6%) in placebo arm, including 2 deaths (adjusted relative risk 0.21; 95% CI, 0.09–0.50; absolute difference -4.53%; 95% CI, -6.70% to -2.37%; <i>P</i> < 0.001)	Key Limitation • Conducted before widespread circulation of the Omicron variant Interpretation • Compared to placebo, SOT reduced the incidence of hospitalization and death from any cause among patients with mild to moderate COVID-19.

Key: BAM = bamlanivimab; bpm = beats per minute; BEB = bebtelovimab; CAS = casirivimab; DM = diabetes mellitus; ETE = etesevimab; IMD = imdevimab; IV = intravenous; mAb = monoclonal antibody; PHVL = persistently high viral load; RCT = randomized controlled trial; RT-PCR = reverse transcription polymerase chain reaction; SOT = sotrovimab; SpO₂ = oxygen saturation; VL = viral load

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COVID-19 Convalescent Plasma

Last Updated: February 29, 2024

Plasma from donors who have recovered from COVID-19 (regardless of vaccination status) may contain antibodies to SARS-CoV-2 that could help suppress viral replication.¹ In August 2020, the Food and Drug Administration issued an Emergency Use Authorization (EUA) for COVID-19 convalescent plasma (CCP) for the treatment of hospitalized patients with COVID-19. The EUA was subsequently revised. The current EUA limits the authorization to the use of CCP products that contain high levels of anti-SARS-CoV-2 antibodies (i.e., high-titer products) for the treatment of COVID-19 in outpatients or inpatients who have immunosuppressive disease or who are receiving immunosuppressive treatment. The testing criteria used to identify high-titer CCP products was also revised.²

The use of CCP should be limited to high-titer products. Products that are not labeled “high titer” should not be used.

Recommendations

Patients Who Are Immunocompromised

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in hospitalized or nonhospitalized patients who are immunocompromised.
- Some people who are immunocompromised have prolonged, symptomatic COVID-19 with evidence of ongoing SARS-CoV-2 replication. The data for these approaches are not definitive, but some Panel members would use 1 or more of the following treatment options:
 - Longer and/or additional courses of ritonavir-boosted nirmatrelvir (Paxlovid)
 - Longer and/or additional courses of remdesivir
 - High-titer CCP from a vaccinated donor who recently recovered from COVID-19 likely caused by a SARS-CoV-2 variant similar to the variant causing the patient’s illness

Patients Who Are Immunocompetent

- The Panel **recommends against** the use of CCP for the treatment of COVID-19 in hospitalized patients who are immunocompetent (**BIIa**).
- There is insufficient evidence for the Panel to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in nonhospitalized patients who are immunocompetent.

Rationale

Patients Who Are Immunocompromised

This section pertains to people who are moderately or severely immunocompromised. For examples of moderately or severely immunocompromising conditions and for a broader discussion on the therapeutic management of COVID-19 in people who are immunocompromised, see [Special Considerations in People Who Are Immunocompromised](#).

Patients who are immunocompromised are at risk of having reduced antibody responses to SARS-CoV-2 infection and COVID-19 vaccination, having suboptimal control of viral replication, and progressing to

severe disease.^{3,4} Despite the lack of definitive evidence, there is a physiologic rationale for the use of SARS-CoV-2 antibody-based therapies in these patients.

Under the revised EUA issued on December 27, 2021, CCP is authorized for the treatment of COVID-19 in nonhospitalized or hospitalized patients who have immunosuppressive disease or are receiving immunosuppressive treatment.²

Evidence to support the use of CCP for the treatment of COVID-19 in patients who are immunocompromised is limited. No randomized, adequately powered trials that evaluated the use of CCP for the treatment of COVID-19 in these patients have been published. Some subgroup analyses generated from clinical trials that enrolled patients who were immunocompromised suggested a potential benefit of using CCP in this population.⁵⁻⁷ However, subgroup analyses need to be interpreted with caution. In the overall trial populations, there was no evidence of a benefit of CCP. Other clinical data from case series, retrospective case-control studies, and meta-analyses have been cited as support for the potential benefit of CCP in patients who are immunocompromised. However, the patient populations differed across studies, and the studies had design limitations, making the findings difficult to interpret.⁸⁻¹⁶

The emergence of SARS-CoV-2 variants further complicates the assessment of the benefits of using CCP. Although results from some in vitro studies suggest that CCP collected from vaccinated individuals who recovered from COVID-19 caused by the Omicron variant exhibits neutralizing activity against certain Omicron subvariants,¹⁷⁻²³ extrapolation of these results to the clinical setting is challenging for the following reasons:²⁴

- COVID-19 immune responses across donor populations are heterogeneous; thus, CCP products are variable.
- The tests used to qualify high-titer CCP measure anti-SARS-CoV-2 antibody titers. They do not directly measure neutralizing activity or account for currently circulating subvariants.
- Published in vitro studies that evaluated the virologic activity of CCP against the currently circulating variants used a variety of assays that are difficult to compare and interpret.^{20-23,25-27}
- The pharmacokinetics and pharmacodynamics of individual CCP products are not clearly understood; therefore, determining the clinical relevance of a degree of in vitro neutralization activity is difficult.

In this context, the Panel has concluded that there is insufficient evidence to definitively recommend the use of CCP for the treatment of COVID-19 in people who are immunocompromised. For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication (e.g., those with a low cycle threshold value, as measured by a reverse transcription polymerase chain reaction result or with a positive rapid antigen test result) despite receiving a course of antiviral therapy, the optimal management is unknown. Case reports and case series have described the treatment of these patients with additional antiviral treatments, prolonged courses of antiviral treatments, high-titer CCP, or combination therapy.²⁸⁻³⁴ The data for these approaches are not definitive, but some Panel members would use longer and/or additional courses of ritonavir-boosted nirmatrelvir or remdesivir, high-titer CCP, or combinations of these. If CCP is used, clinicians should attempt to obtain high-titer CCP from a vaccinated donor who recently recovered from COVID-19 likely caused by a SARS-CoV-2 variant similar to the variant causing the patient's illness.

Adequately powered, well-designed, and well-conducted randomized clinical trials are needed to provide more specific, evidence-based guidance on the role of CCP in the treatment of COVID-19 in patients who are immunocompromised.

Hospitalized Patients Who Are Immunocompetent

Under the revised EUA, the use of CCP is not authorized for hospitalized patients with COVID-19 who do not have immunosuppressive disease or who are not receiving immunosuppressive treatments.

Clinical data on the use of CCP for the treatment of COVID-19 in hospitalized patients who are immunocompetent, including data from several randomized trials and the U.S. Expanded Access Program for CCP, are summarized in [Table 4c](#).

Three large randomized controlled trials that evaluated the use of CCP in hospitalized patients—RECOVERY,³⁵ CONCOR-1,³⁶ and REMAP-CAP⁶—reported no benefit of using high-titer CCP in hospitalized patients with COVID-19. All 3 were open-label trials that were stopped early due to futility. However, an open-label randomized trial that enrolled patients with COVID-19–induced acute respiratory distress syndrome who required mechanical ventilation showed a lower mortality among patients who received high-titer CCP than among those who received standard care.³⁷ The benefit was primarily seen among patients who received CCP within 48 hours of mechanical ventilation. Almost all of the patients (98%) received corticosteroids during the trial, but only a few received a second immunomodulator or were previously vaccinated against COVID-19. Concomitant antiviral therapy use was low during the trial (<6% of patients received remdesivir) despite evidence of ongoing viral replication among the participants (i.e., low median cycle threshold values on nasopharyngeal SARS-CoV-2 polymerase chain reaction tests).

The Panel **recommends against** the use of CCP for the treatment of COVID-19 in hospitalized patients who are immunocompetent (**BIIa**). The rationale for this recommendation is based on the evidence from the 4 large randomized controlled trials, the widespread, pre-existing immunity to SARS-CoV-2 in the population, the continuously evolving SARS-CoV-2 variants, and the availability of other antiviral agents (e.g., remdesivir).

Nonhospitalized Patients Who Are Immunocompetent

CCP is not authorized for the treatment of COVID-19 in nonhospitalized patients who do not have immunosuppressive disease or who are not receiving immunosuppressive treatments.

Data from well-designed clinical trials that evaluated the use of CCP for the treatment of nonhospitalized patients with COVID-19 prior to the emergence of the Omicron variants are conflicting. These data are summarized in [Table 4c](#). Differences in patient populations, the placebo used (e.g., some studies used saline; some used non-SARS-CoV-2 plasma), and CCP manufacturing and testing methods may have contributed to the disparate outcomes and the difficulty in reconciling results across these clinical trials. The emergence of SARS-CoV-2 variants further complicates the assessment of the benefits of using CCP.

There is insufficient evidence for the Panel to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in nonhospitalized patients who are immunocompetent.

Monitoring and Adverse Effects

The available data suggest that serious adverse reactions following the administration of CCP are infrequent and consistent with the risks associated with plasma infusions for other indications. These risks include transfusion-transmitted infections (e.g., HIV, hepatitis B, hepatitis C), allergic reactions, anaphylactic reactions, febrile nonhemolytic reactions, transfusion-related acute lung injury, transfusion-associated circulatory overload, and hemolytic reactions. Hypothermia, metabolic complications, and post-transfusion purpura have also been described.^{2,35,38}

Additional risks of CCP transfusion include a theoretical risk of antibody-dependent enhancement

of SARS-CoV-2 infection. In the CONCOR-1 trial, higher levels of full transmembrane spike immunoglobulin G (IgG) were associated with worse outcomes, suggesting that the use of CCP with nonfunctional anti-SARS-CoV-2 antibodies may be harmful.³⁶ A subgroup analysis in the REMAP-CAP trial showed evidence of potential harm in patients who received CCP transfusions more than 7 days after being hospitalized.⁶

When considering the use of CCP in patients with a history of severe allergic or anaphylactic transfusion reactions, consultation with a transfusion medicine specialist is advised.

Considerations in Pregnant People

The safety and efficacy of using CCP during pregnancy have not been evaluated in clinical trials, and published data on its use in pregnant individuals with COVID-19 are limited to case reports.³⁹ Pathogen-specific immunoglobulins are used during pregnancy to prevent infection from varicella zoster virus and rabies virus and have been used in clinical trials of congenital cytomegalovirus infection.^{40,41} Pregnancy is not a reason to withhold CCP from a patient if it is otherwise indicated. The expected physiologic immunomodulation during pregnancy should not affect the decision to use CCP.

Considerations in Children

The safety and efficacy of CCP have not been systematically evaluated in pediatric patients. The published literature on its use in children is limited to case reports and case series. A few clinical trials evaluating the use of CCP in children are ongoing. The use of high-titer CCP may be considered on a case-by-case basis for hospitalized children who are immunocompromised and meet the EUA criteria for its use. CCP is not authorized by the Food and Drug Administration for use in patients who are immunocompetent.

Several other antiviral therapies are available for the treatment of children with COVID-19 who are at high risk of progressing to severe disease. See [Therapeutic Management of Nonhospitalized Children With COVID-19](#) and [Therapeutic Management of Hospitalized Children With COVID-19](#) for more information.

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Table 4c. COVID-19 Convalescent Plasma: Selected Clinical Trial Data

Last Updated: February 29, 2024

The studies described in this table are those that had the greatest impact on the Panel's recommendations. The Panel reviewed other clinical studies of CCP for the treatment of COVID-19.¹⁻⁵ However, those studies have limitations that make them less definitive and informative than the studies summarized in the table.

Methods	Results	Limitations and Interpretation
REMAP-CAP: Multinational, Open-Label RCT of High-Titer CCP in Hospitalized Patients With Critical COVID-19 in Australia, Canada, the United Kingdom, and the United States⁶		
Key Inclusion Criterion - Admitted to ICU while receiving respiratory support (HFNC oxygen, NIV, MV, ECMO) and/or vasopressor or inotrope support Key Exclusion Criteria - CCP contraindicated - Death imminent Interventions - High-titer CCP (550 mL +/- 150 mL) within 48 hours of randomization (n = 1,084) - Usual care (n = 916) Primary Endpoint - Number of organ support-free days by Day 21 Key Secondary Endpoints - In-hospital mortality - Mortality by Day 28 and Day 90 - Number of respiratory support-free days - ICU LOS	Participant Characteristics - Mean age 61 years; 68% men 32% on MV 29% were SARS-CoV-2 antibody negative at baseline. 94% received corticosteroids; 45% received RDV; 39% received IL-6 inhibitors. Primary Outcome - Median number of organ support-free days by Day 21: 0 days in CCP arm vs. 3 days in usual care arm (OR 0.97; 95% CrI, 0.82–1.14) Secondary Outcomes - No difference between arms in: - In-hospital mortality: 37% in CCP arm vs. 38% in usual care arm - Mortality by Day 28 or Day 90 - Median number of respiratory support-free days: 0 days in CCP arm vs. 2 days in usual care arm - Median ICU LOS: 21 days in CCP arm vs. 17 days in usual care arm	Key Limitations - Open-label study - Not all patients in CCP arm received CCP (86% received CCP as per protocol and 95% received some CCP). Interpretation - There was no benefit of CCP in hospitalized patients with critical COVID-19.

Methods	Results	Limitations and Interpretation
CONCOR-1: Multinational, Open-Label RCT of CCP for Hospitalized Patients With COVID-19 in Canada, the United States, and Brazil⁷		
Key Inclusion Criteria • Receipt of supplemental oxygen • Availability of ABO-compatible CCP Key Exclusion Criteria • Imminent or current intubation • >12 days from respiratory symptom onset Interventions • 1–2 units of CCP (approximately 500 mL) from 1–2 donors (n = 625) • SOC (n = 313) Primary Endpoint • Composite of intubation or death by Day 30 Key Secondary Endpoints • Time to intubation or death by Day 30 • Mortality by Day 30 • ICU LOS by Day 30 • Need for renal dialysis by Day 30 • Frequency of SAEs by Day 30	Participant Characteristics • Mean age 68 years; 59% men • 84% were receiving systemic corticosteroids at enrollment. Primary Outcome • Composite of intubation or death by Day 30: 32% in CCP arm vs. 28% in SOC arm (relative risk 1.16; 95% CI, 0.94–1.43; <i>P</i> = 0.18) Secondary Outcomes • By Day 30, no difference between arms in: • Time to intubation or death • Mortality: 23% in CCP arm vs. 21% in SOC arm • Mean ICU LOS: 4.3 days in CCP arm vs. 3.7 days in SOC arm • Need for renal dialysis: 1.6% in CCP arm vs. 2.0% in SOC arm • Frequency of SAEs by Day 30: 33% in CCP arm vs. 26% in SOC arm	Key Limitations • Open-label study • Trial stopped at 78% of planned enrollment after meeting prespecified futility criteria for early termination. Interpretation • There was no benefit of CCP in oxygen-dependent, hospitalized patients with COVID-19 who were within 12 days of symptom onset.
RECOVERY: Open-Label RCT of High-Titer CCP in Hospitalized Patients in the United Kingdom⁸		
Key Inclusion Criterion • Clinically suspected or laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criterion • CCP contraindicated Interventions • 2 units of high-titer CCP (approximately 275 mL/unit) with IgG against SARS-CoV-2 spike protein and sample to cutoff ratio ≥6.0. First unit administered ASAP after randomization; second unit administered ≥12 hours later (n = 5,795).	Participant Characteristics • Mean age 64 years; 64% men • 5% on MV • 92% received corticosteroids. Primary Outcomes • No difference between arms in: • All-cause mortality by Day 28: 24% in each arm • Mortality in patients without detectable SARS-CoV-2 antibodies: 32% in CCP arm vs. 34% in usual care arm	Key Limitation • Open-label study Interpretation • There was no benefit of CCP in hospitalized patients with COVID-19.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of High-Titer CCP in Hospitalized Patients in the United Kingdom⁸, continued		
- Usual care (n = 5,763) Primary Endpoint - All-cause mortality by Day 28 Key Secondary Endpoints - Time to hospital discharge by Day 28 - Among patients not on MV, progression to MV or death by Day 28	Secondary Outcomes - No difference between arms in: - Proportion discharged by Day 28: 66% in both arms - Proportion who progressed to MV or death by Day 28: 29% in CCP arm vs. 29% in usual care arm	
RECOVER: Open-Label RCT of High-Titer CCP in Hospitalized Patients With Severe COVID-19 in 4 Risk Groups in Germany⁹		
Key Inclusion Criteria - PCR-confirmed SARS-CoV-2 infection - Hospitalized with SpO₂ ≤94% on room air or PaO₂/FiO₂ <300 mm Hg ≥1 of the following criteria: - Hematologic cancer and/or receipt of active cancer therapy in past 24 months for any cancer - Chronic immunosuppression due to medications and/or underlying disease - Aged >50 to ≤75 years with ALC <0.8 x 10⁹ cells/L and/or D-dimer >1 µg/mL - Aged >75 years without other listed criteria Key Exclusion Criterion - Required MV or NIV Interventions 2 units (238–337 mL) of high-titer CCP (≥1:80) or vaccinated donor plasma from 2 donors on Days 1 and 2 (n = 68) - SOC (n = 66) Primary Endpoint - Time to 2-point improvement on a 7-point OS or hospital discharge	Participant Characteristics 136 participants were enrolled between September 2020 and January 2022. - Mean age 69 years; 68% men; 97% White - Participants were enrolled from 4 mutually exclusive patient groups: - Patients with cancer (n = 56) - Patients with immunosuppression who did not have cancer (n = 16, including 12 solid organ transplant recipients) - Patients aged >50 to ≤75 years with lymphopenia and/or elevated D-dimer levels (n = 36) - Patients aged >75 years without other criteria (n = 26) 11% were fully vaccinated. 8% received small-molecule antiviral drugs (12% in plasma arm vs. 5% in SOC arm); 37% received anti-inflammatory drugs (40% in plasma arm vs. 33% in SOC arm). 60% received supplemental oxygen via nasal cannula; 21% on HFNC oxygen or NIV - Median 7 days between symptom onset and randomization Primary Outcome - Median time to 2-point improvement on OS or hospital discharge: 13 days in plasma arm vs. 18 days in SOC arm	Key Limitations - Open-label study - The live-virus neutralization assay used to select plasma for this trial may not produce the same results as the assays used to qualify high-titer CCP in the current FDA EUA. - Small sample size - Trial was terminated early because the neutralizing activity of stored plasma against the Omicron variant was not known. - Low proportion of vaccinated participants and limited use of current SOC therapies, such as antiviral or immunomodulatory agents - Subgroup analyses were not adjusted for multiple comparisons. Interpretation - The trial did not demonstrate a benefit of high-titer CCP or vaccinated donor plasma in the overall study population.

Methods	Results	Limitations and Interpretation
RECOVER: Open-Label RCT of High-Titer CCP in Hospitalized Patients With Severe COVID-19 in 4 Risk Groups in Germany ⁴ , continued		
Key Secondary Endpoints 28-day, 56-day, and 84-day overall survival rate	(HR 1.29; 95% CI, 0.86–1.93; $P=0.205$) - Median time to improvement or hospital discharge among patients with cancer: 13 days in plasma arm vs. 31 days in SOC arm (HR 2.50; 95% CI, 1.34–4.79; $P=0.003$) Key Secondary Outcomes - No difference between arms in overall survival; 27 patients (19.9%) died (HR for survival 0.72; 95% CI, 0.33–1.55; $P=0.403$). - Fewer patients with cancer died in plasma arm than in SOC arm (HR 0.28; 95% CI, 0.06–0.96; $P=0.042$).	Results from the predefined subgroup analysis of patients with cancer suggest a potential benefit of CCP or vaccinated donor plasma. However, this analysis was conducted largely before the emergence of Omicron subvariants, so the results should be interpreted with caution.
CONFIDENT: Open-Label RCT of High-Titer CCP in Hospitalized Patients With COVID-19–associated ARDS in Belgium ¹⁰		
Key Inclusion Criteria - PCR-confirmed SARS-CoV-2 infection - Admitted to ICU with COVID-19–associated ARDS and WHO COVID-19 OS score of 7, 8, or 9 - On MV for ≤ 5 days - Clinical Frailty Scale score of <6 Key Exclusion Criteria - Previous transfusion-related side effects - Medical decision to limit therapy Interventions 2 units (400–500 mL total) of high-titer CCP ($\geq 1:320$) from donors who fully recovered from COVID-19 between 28 days and 10 months before study. CCP was administered within 24 hours of study randomization ($n=237$). - SOC ($n=238$) Primary Endpoint - Mortality by Day 28	Participant Characteristics - Median age 64 years; 68% male 10% were vaccinated. 10% with mild ARDS, 58% with moderate ARDS, 32% with severe ARDS - Baseline evidence of ongoing SARS-CoV-2 replication (Ct value of 22 in CCP arm vs. 20 in SOC arm) 98% received corticosteroids; 6% received RDV; 4% received IL-6 inhibitors. Primary Outcome - Mortality by Day 28: 35% in CCP arm vs. 45% in SOC arm ($P=0.03$) - In patients on MV ≤ 48 hours: 33% in CCP arm vs. 47% in SOC arm - In patients on MV >48 hours: 42% in CCP arm vs. 40% in SOC arm - Similar outcomes were seen regardless of whether the original SARS-CoV-2 strain (Wuhan-Hu-1) or the Alpha, Delta, or Omicron variants were in circulation.	Key Limitations - Open-label study - Trial was stopped after recruiting 95% of the target enrollment due to a low ICU admission rate. - Approximately 18% of patients who were assigned to receive CCP received a lower neutralizing titer of 1:160. - The live-virus neutralization assay used to select plasma for this trial may not produce the same results as the assays used to qualify high-titer CCP in the current FDA EUA. - Low proportion of vaccinated patients and limited use of current SOC therapies, such as antiviral agents or a second immunomodulatory agent - There were differences in treatment effects across sites. Interpretation - CCP reduced mortality by Day 28 in patients with COVID-19 and ARDS, and

Methods	Results	Limitations and Interpretation
CONFIDENT: Open-Label RCT of High-Titer CCP in Hospitalized Patients With COVID-19–associated ARDS in Belgium ¹⁰ , continued		
		the effect was most notable in patients who were randomized ≤48 hours after initiating MV.
CSSC-004: RCT of Early Treatment With High-Titer CCP in Outpatients With COVID-19 in the United States ¹¹		
Key Inclusion Criterion - COVID-19 symptoms for <8 days Key Exclusion Criteria - Prior or planned COVID-19–related hospitalization - Receipt of anti-SARS-CoV-2 mAbs Interventions - Approximately 250 mL of CCP with SARS-CoV-2 spike protein IgG (titer ≥1:320) (n = 592) - Non-SARS-CoV-2 plasma (n = 589) Primary Endpoint - Composite of COVID-19–related hospitalization or all-cause death within 28 days	Participant Characteristics - Median age 44 years; 7% aged ≥65 years; 57% women; 79% White 8% with type 2 DM; 2% with CVD; 38% with BMI ≥30 82% were unvaccinated. - Median of 6 days between symptom onset and transfusion Primary Outcomes - COVID-19–related hospitalization within 28 days: 2.9% in CCP arm vs. 6.3% in control arm (absolute risk reduction 3.4 percentage points; 95% CI, 1.0–5.8; <i>P</i> = 0.005) 53 of 54 hospitalizations occurred in unvaccinated individuals. None occurred in fully vaccinated individuals. - All-cause deaths within 28 days: 0 in CCP arm vs. 3 in control arm	Key Limitation - Patients were at relatively low risk for disease progression. Interpretation - This trial demonstrated a benefit of CCP in unvaccinated outpatients with <8 days of COVID-19 symptoms.
CONV-ERT: RCT of High-Titer, Methylene Blue-Treated CCP as an Early Treatment for Outpatients With COVID-19 in Spain ¹²		
Key Inclusion Criteria - Aged ≥50 years - Mild or moderate COVID-19 symptoms for ≤7 days Key Exclusion Criteria - Severe COVID-19 symptoms or need for hospitalization for any reason - Previous SARS-CoV-2 infection - Receipt of ≥1 doses of a COVID-19 vaccine Interventions 250–300 mL of high-titer, methylene blue-treated CCP (n = 188)	Participant Characteristics - Mean age 56 years; 54% men 75% with ≥1 risk factors for COVID-19 progression 97% with mild COVID-19 - Median of 4.4 days of symptoms prior to enrollment - Among 369 patients with available baseline serologic testing, 88% were negative for both IgG anti-SARS-CoV-2 spike and IgM anti-SARS-CoV-2 S1-RBD. Primary Outcomes - Hospitalization within 28 days: 12% in CCP arm vs. 11% in placebo arm (relative risk 1.05; 95% CrI, 0.78–1.41)	Key Limitations - Trial was underpowered because it was terminated early due to rising vaccination rates among the eligible patient population. - Methylene blue, which was used for pathogen inactivation in donor plasma, could have potentially impaired Fc-region functionality of Ig and negatively impacted product efficacy and blinding. Interpretation - This trial did not demonstrate a benefit of CCP in unvaccinated outpatients with

Methods	Results	Limitations and Interpretation
CONV-ERT: RCT of High-Titer, Methylene Blue-Treated CCP as an Early Treatment for Outpatients With COVID-19 in Spain¹², continued		
0.9% saline (n = 188) Primary Endpoints - Hospitalization within 28 days - Mean change in SARS-CoV-2 VL from baseline to Day 7 Key Secondary Endpoints - Death by Day 60 - Time to complete symptom resolution	- Mean change in SARS-CoV-2 VL: -2.41 log₁₀ copies/mL in CCP arm vs. -2.32 log₁₀ copies/mL in placebo arm Key Secondary Outcomes - Death by Day 60: 0 in CCP arm vs. 2 in placebo arm (relative risk 0.20; 95% CI 0.01–4.14) - No difference between arms in median time to symptom resolution: 12.0 days for both arms (HR 1.05; 95% CI, 0.85–1.30)	≤7 days of COVID-19 symptoms.
Double-Blind RCT of Early High-Titer CCP Therapy to Prevent Severe COVID-19 in Nonhospitalized Older Adults in Argentina¹³		
Key Inclusion Criteria - Aged ≥75 years or aged 65–74 years with ≥1 coexisting conditions - Mild COVID-19 symptoms for <72 hours Key Exclusion Criterion - Severe respiratory disease Interventions 250 mL of CCP with SARS-CoV-2 spike protein IgG (titer >1:1,000) (n = 80) - Saline (n = 80) Primary Endpoint - Severe respiratory disease, defined as respiratory rate ≥30 breaths/min and/or SpO₂ <93% on room air, by Day 15	Participant Characteristics - Mean age 77 years; 38% men - Most with comorbidities Primary Outcome - Severe respiratory disease by Day 15: 16% in CCP arm vs. 31% in placebo arm (relative risk 0.52; 95% CI, 0.29–0.94; P = 0.03)	Key Limitations - Small sample size - Trial was terminated early because the number of COVID-19 cases decreased Interpretation - This trial demonstrated a benefit of CCP in older adult outpatients with <72 hours of mild COVID-19 symptoms.

Methods	Results	Limitations and Interpretation
SIREN-C3PO: Multicenter, Single-Blind RCT of High-Titer CCP in Adults With COVID-19 in the United States¹⁴		
Key Inclusion Criteria • ED patient with ≤7 days of symptoms • PCR-confirmed SARS-CoV-2 infection • Aged ≥50 years or aged ≥18 years with ≥1 risk factors for disease progression Key Exclusion Criterion • Need for supplemental oxygen Interventions • 250 mL of high-titer CCP (median titer 1:641) (n = 257) • Saline (n = 254) Primary Endpoint • Disease progression, defined as hospital admission, death, or seeking emergency or urgent care, within 15 days of randomization Key Secondary Endpoints • Severity of illness by Day 30, as measured by an OS • All-cause mortality by Day 30 • Number of hospital-free days by Day 30	Participant Characteristics • Median age 54 years; 46% men • More patients with immunosuppression in CCP arm than in placebo arm (13% vs. 7%) • More patients with ≥3 risk factors in CCP arm than in placebo arm (55% vs. 48%) Primary Outcomes • No difference between arms in proportion with disease progression: 30% in CCP arm vs. 32% in placebo arm (risk difference 1.9%; 95% CrI, -6.0% to 9.8%) • 25 patients (19 in CCP arm vs. 6 in placebo arm) required hospitalization during index visit. In a post hoc analysis that excluded these patients, disease progression occurred in 24% in CCP arm vs. 30% in placebo arm (risk difference 5.8%; 95% CrI, -1.9% to 13.6%). Secondary Outcomes • All-cause mortality by Day 30: 5 (1.9%) in CCP arm vs. 1 (0.4%) in placebo arm • No difference between arms in illness severity or mean number of hospital-free days by Day 30	Key Limitations • In the primary analysis, the number of patients who required hospital admission during the index visit was not balanced across arms. • The CCP arm included more patients with multiple risk factors, including immunosuppression. Interpretation • The use of high-titer CCP within 1 week of symptom onset did not prevent disease progression in outpatients with COVID-19 who were at high risk of severe disease.
CoV-Early: Double-Blind RCT of CCP in Nonhospitalized, High-Risk Adults With COVID-19 in the Netherlands¹⁵		
Key Inclusion Criteria • Aged ≥70 years, aged ≥50 years with a comorbidity, or aged ≥18 years and severely immunocompromised • Positive SARS-CoV-2 RT-PCR or antigen test result • COVID-19 symptoms for ≤7 days Key Exclusion Criteria • Life expectancy <28 days • History of TRALI • IgA deficiency	Participant Characteristics • Median age 60 years; 22% women • Median of 5 days of symptoms • Median of 1 comorbidity • Median SpO₂ of 97% at baseline • 7.9% were SARS-CoV-2 IgG antibody negative at baseline. • 2.9% were fully vaccinated; 5.0% received 1 dose of a COVID-19 vaccine.	Key Limitations • Study was discontinued after 421 of 690 planned participants were enrolled, resulting in decreased power. • The CCP used was selected based on a PRNT50 assay and may not qualify as high-titer CCP per the current FDA EUA.

Methods	Results	Limitations and Interpretation
CoV-Early: Double-Blind RCT of CCP in Nonhospitalized, High-Risk Adults With COVID-19 in the Netherlands¹⁵, continued		
Interventions • 300 mL of CCP with minimum PRNT50 titer of 1:160 (n = 207) • Non-SARS-CoV-2 plasma collected prior to pandemic (n = 209) Primary Endpoint • Improvement based on 5-point OS by Day 28 Secondary Endpoints • Percentage of hospital admissions • Number of days of symptoms	Primary Outcome • Odds of receiving highest score on 5-point OS by Day 28: OR 0.86; 95% CrI, 0.59–1.22 in CCP arm Secondary Outcomes • Percentage of hospital admissions: 4.8% in CCP arm vs. 8.6% in non-SARS-CoV-2 plasma arm (aHR 0.61; 95% CI, 0.28–1.34) • Number of days of symptoms: 13 in CCP arm vs. 12 in non-SARS-CoV-2 plasma arm ($P = 0.99$)	Interpretation • This trial did not demonstrate a benefit of CCP in nonhospitalized, high-risk patients with COVID-19.
Retrospective Evaluation of CCP Antibody Levels and the Risk of Death From COVID-19 in the United States¹⁶		
Key Inclusion Criteria • Severe or life-threatening COVID-19 • Patients for whom samples of transfused CCP were available for retrospective analysis of antibody titer Interventions • High-titer CCP (n = 515), medium-titer CCP (n = 2,006), or low-titer CCP (n = 561), characterized retrospectively Primary Endpoint • Mortality by Day 30 after CCP transfusion	Participant Characteristics • 31% aged ≥ 70 years; 61% men; 48% White, 37% Hispanic/Latinx • 61% in ICU; 33% on MV • 51% received corticosteroids; 31% received RDV. Primary Outcomes • Mortality by Day 30 after transfusion: 22% in high-titer CCP arm vs. 27% in medium-titer CCP arm vs. 30% in low-titer CCP arm • Lower risk of death in high-titer CCP arm than low-titer CCP arm (relative risk 0.75; 95% CI, 0.61–0.93) • Lower mortality among patients not receiving MV before CCP transfusion (relative risk 0.66; 95% CI, 0.48–0.91) • No difference in mortality between high-titer and low-titer arms among patients on MV before CCP transfusion (relative risk 1.02; 95% CI, 0.78–1.32)	Key Limitation • Lack of untreated control arm Interpretation • The study data are not sufficient to establish the efficacy or safety of CCP.

Key: Key: ALC = absolute lymphocyte count; ARDS = acute respiratory distress syndrome; ASAP = as soon as possible; BMI = body mass index; CCP = COVID-19 convalescent plasma; Ct = cycle threshold; CVD = cardiovascular disease; DM = diabetes mellitus; ECMO = extracorporeal membrane oxygenation; ED = emergency department; EUA = Emergency Use Authorization; Fc = fragment crystallizable; FDA = Food and Drug Administration; HFNC = high-flow nasal cannula; ICU = intensive care unit; Ig = immunoglobulin; IL = interleukin; LOS = length of stay; mAb = monoclonal antibody; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PaO₂/FiO₂ = ratio of arterial partial pressure of oxygen to fraction of inspired oxygen; PCR = polymerase chain reaction; PRNT50 = 50% plaque reduction neutralization test; RBD = receptor-binding domain; RCT = randomized controlled trial; RDV = remdesivir; RT-PCR = reverse transcription polymerase chain reaction; SAE = serious adverse event; SOC = standard of care; SpO₂ = oxygen saturation; TRALI = transfusion-related acute lung injury; VL = viral load; WHO = World Health Organization

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Interferons

Last Updated: December 20, 2023

Interferons are a family of cytokines with in vitro and in vivo antiviral properties. Interferon beta-1a has been approved by the Food and Drug Administration (FDA) to treat relapsing forms of multiple sclerosis, and pegylated formulations of interferon alfa-2a and interferon alfa-2b have been approved by the FDA to treat hepatitis B and hepatitis C virus infections. Several interferons, including interferon alfa, beta, and lambda, have been evaluated for the treatment of COVID-19. Interferon lambda is not currently approved or authorized by the FDA for any use.

Recommendations

- For nonhospitalized patients with mild to moderate COVID-19, the COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** the use of **interferon alfa** or **beta**, except in a clinical trial (**AIIa**).
- For hospitalized patients with COVID-19, the Panel **recommends against** the use of **systemic interferon alfa**, except in a clinical trial (**AIIa**).
- For hospitalized patients with COVID-19, the Panel **recommends against** the use of **systemic interferon beta** (**AI**).
- The Panel is unable to recommend either for or against the use of interferon lambda because this product is not currently available for clinical use.

Rationale

Interferon Alfa and Beta

Many of the studies that evaluated the use of systemic interferons for the treatment of hospitalized adults with COVID-19 were conducted in early 2020, before the widespread use of remdesivir or corticosteroids and other immunomodulators. In addition, these studies administered interferons with other drugs that have since been shown to have no clinical benefit in people with COVID-19, such as lopinavir/ritonavir and hydroxychloroquine.¹⁻³

More recent studies have shown no benefit of using interferon beta-1a to treat patients with COVID-19, and some of the trials have suggested that interferon beta-1a can cause harm in patients with severe disease, such as those who require high-flow oxygen, noninvasive ventilation, or mechanical ventilation.^{4,5} In a large randomized controlled trial of hospitalized patients with COVID-19, the combination of interferon beta-1a plus remdesivir showed no clinical benefit when compared to remdesivir alone.⁴ Similarly, the World Health Organization Solidarity trial did not show a benefit of administering interferon beta-1a to hospitalized patients, approximately 50% of whom were on corticosteroids.⁵

Systemic interferon alfa and inhaled interferons have also been evaluated in patients with COVID-19. The trials that have evaluated the use of interferon alfa have generally been small or moderate in size and have not been adequately powered to assess whether this agent provides a clinical benefit for patients with COVID-19.⁶⁻⁸

Interferon Lambda

Pegylated interferon lambda was studied in a randomized, double-blind, adaptive clinical trial that

enrolled nonhospitalized patients with COVID-19 in Brazil and Canada.⁹ A total of 1,941 patients with risk factors for severe COVID-19 were randomized to receive either a single subcutaneous injection of pegylated interferon lambda 180 µg or placebo. Eighty-three percent of these patients had received at least 1 dose of a COVID-19 vaccine. The primary outcome was a composite of observation in an emergency department for >6 hours or hospitalization, and 1 of the secondary outcomes was a composite of hospitalization or death. By Day 28 after randomization, the use of interferon lambda was associated with a 51% decrease in the occurrence of the primary outcome and a 39% decrease in the occurrence of this secondary outcome. Patients with a high baseline SARS-CoV-2 viral load who received interferon lambda were more likely to have cleared the virus by Day 7 than those who received placebo.

The drug was generally well tolerated. However, since pegylated interferon lambda is an investigational agent that is not currently available for clinical use, the Panel cannot make a recommendation for its use at this time.

Summaries of the studies that informed the Panel's recommendations can be found in [Table 4d](#).

Considerations in Pregnant People

According to analyses of data from several large pregnancy registries, exposure to interferon beta-1b prior to conception or during pregnancy does not lead to an increased risk of adverse birth outcomes (e.g., spontaneous abortion, congenital anomaly).^{10,11} In a study that used data from pregnancy registries in Sweden and Finland, women who were exposed to interferon beta during pregnancy did not report significant changes in the birth weight, height, or head circumference of their infants.¹²

Considerations in Children

There are insufficient data on the use of interferons to treat respiratory viral infections in children to make any recommendations for treating children with COVID-19.

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Table 4d. Interferons: Selected Clinical Trial Data

Last Updated: December 20, 2023

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for interferons. The studies summarized below are the randomized controlled trials that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
ACTT-3: Multinational, Double-Blind RCT of Interferon Beta-1a and Remdesivir in Hospitalized Adults With COVID-19¹		
Key Inclusion Criteria - Evidence of pneumonia (radiographic infiltrates, SpO₂ ≤94% on room air, or supplemental oxygen) - No MV required Key Exclusion Criteria - AST or ALT >5 times ULN - Impaired renal function - Hospital discharge or transfer anticipated within 72 hours Interventions - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 9 days plus IFN beta-1a 44 µg SUBQ every other day for up to 4 doses (n = 487) - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily for 9 days plus placebo (n = 482) Primary Endpoint - Time to recovery by Day 28 Key Secondary Endpoints - Clinical status at Day 14, as measured by an OS - Mortality by Day 28	Participant Characteristics - Mean age 59 years; 38% were aged ≥65 years 58% men; 32% Latinx, 60% White, 17% Black - Mean of 8.6 days of symptoms before enrollment 90% had ≥1 comorbidity; 58% with HTN; 58% with obesity; 37% with DM Primary Outcome - Median time to recovery: 5 days in both arms (rate ratio 0.99; 95% CI, 0.87–1.13; <i>P</i> = 0.88) - In patients on high-flow oxygen or NIV (OS6) at baseline, median time to recovery: >28 days in IFN beta-1a arm vs. 9 days in placebo arm (rate ratio 0.40; 95% CI, 0.22–0.75; <i>P</i> = 0.0031) Secondary Outcomes - No difference between arms in clinical status at Day 14 (OR 1.01; 95% CI, 0.79–1.28) - No difference between IFN beta-1a arm and placebo arm in mortality by Day 28 in: - All patients: 5% vs. 3% (HR 1.33; 95% CI, 0.69–2.55) - Patients who were OS6 at baseline: 21% vs. 12% (HR 1.74; 95% CI, 0.51–5.93)	Key Limitation - After 270 patients were enrolled, OS6 patients were excluded because of an increased frequency of AEs in this group. Interpretation - There was no clinical benefit of adding IFN beta-1a to RDV in hospitalized patients with COVID-19. - The use of IFN beta-1a was associated with worse outcomes among patients who were OS6 at baseline.

Methods	Results	Limitations and Interpretation
WHO Solidarity Trial: Multinational, Open-Label, Adaptive RCT of IV or SUBQ Interferon Beta-1a or Other Repurposed Drugs in Hospitalized Adults With COVID-19²		
Key Inclusion Criteria • Diagnosis of COVID-19 • Not expected to be transferred elsewhere within 72 hours Interventions • IFN beta-1a 44 µg SUBQ on day of randomization, Day 3, and Day 6 (n = 1,656) • IFN beta-1a 10 µg IV daily for 6 days for patients on high-flow oxygen, ventilation, or ECMO (n = 394) • IFN beta-1a (either SUBQ or IV) and LPV/RTV 400 mg/50 mg twice daily for 14 days (n = 651) • Local SOC (n = 2,050) Primary Endpoint • In-hospital mortality Key Secondary Endpoint • Initiation of ventilation	Participant Characteristics • 35% aged <50 years; 19% aged ≥70 years; 63% men • 70% on supplemental oxygen; 7% on ventilation • Approximately 50% received corticosteroids during the study. Primary Outcome • In-hospital mortality: 11.9% in combined IFN beta-1a arms vs. 10.5% in SOC arm (rate ratio 1.16; 95% CI, 0.96–1.39) • For IFN beta-1a only (without LPV/RTV) recipients vs. SOC recipients, rate ratio was 1.12 (95% CI, 0.83–1.51). • Among those on ventilation at baseline, age-stratified rate ratio for in-hospital mortality was 1.40 (95% CI, 0.93–2.11). Secondary Outcome • 10% initiated ventilation in the combined IFN beta-1a arms and SOC arm.	Key Limitations • Open-label study • IFN beta-1a given as IV or SUBQ formulations at different doses. Interpretation • IFN beta-1a did not reduce in-hospital mortality in hospitalized patients with COVID-19.

Methods	Results	Limitations and Interpretation
DisCoVeRy Solidarity Trial Add-On: Open-Label, Adaptive RCT of Interferon Beta-1a Plus Lopinavir/Ritonavir, Lopinavir/Ritonavir, or Hydroxychloroquine in Hospitalized Adults With COVID-19 in France³		
Key Inclusion Criteria • Positive SARS-CoV-2 PCR result • Patients had pulmonary rales or crackles with SpO₂ ≤94% on room air or required supplemental oxygen Interventions • IFN beta-1a 44 µg SUBQ on Days 1, 3, and 6 plus LPV/RTV 400 mg/100 mg PO twice daily for 14 days plus SOC (n = 145) • LPV/RTV 400 mg/100 mg PO twice daily for 14 days plus SOC (n = 145) • HCQ 400 mg twice on Day 1, then HCQ 400 mg daily for 9 days plus SOC (n = 145) • SOC alone, which included corticosteroids, anticoagulants, or immunomodulatory agents but not antivirals (n = 148) Primary Endpoint • Clinical status at Day 15, as measured by an OS Key Secondary Endpoints • Clinical status at Day 29 • Rate of SARS-CoV-2 viral clearance • Time to SARS-CoV-2 viral clearance by Day 29 • Time to improvement of 2 OS categories by Day 29 • Time to hospital discharge by Day 29	Participant Characteristics • Median age 63 years; 72% men • 29% with obesity; 26% with chronic cardiac disease; 22% with DM • 36% had severe disease • Median of 9 days of symptoms before randomization • 30% received steroids during the study. Primary Outcome • No difference in clinical status at Day 15 for any intervention compared to SOC: • IFN beta-1a plus LPV/RTV: aOR 0.69 (95% CI, 0.45–1.04; <i>P</i> = 0.08) • LPV/RTV: aOR 0.83 (95% CI, 0.55–1.26; <i>P</i> = 0.39) • HCQ: aOR 0.93 (95% CI, 0.62–1.41; <i>P</i> = 0.75) Secondary Outcomes • No difference between arms in clinical status at Day 29 • No difference between arms in rate or time to SARS-CoV-2 viral clearance • Time to improvement of 2 OS categories and hospital discharge by Day 29 was longer in LPV/RTV plus IFN beta-1a and LPV/RTV arms than in SOC arm.	Key Limitations • Open-label study • Most patients had moderate disease. • No IFN beta-1a arm without LPV/RTV • Study stopped early for futility. Interpretation • Compared to SOC alone, the use of IFN-beta-1a plus LPV/RTV did not improve clinical status, rate of viral clearance, or time to viral clearance in hospitalized patients with COVID-19.

Methods	Results	Limitations and Interpretation
TOGETHER: Double-Blind, Adaptive RCT of Pegylated Interferon Lambda in Nonhospitalized Patients With COVID-19 in Brazil and Canada⁴		
Key Inclusion Criteria • Positive SARS-CoV-2 antigen test result • Within 7 days of symptom onset • ≥1 high-risk factor for disease progression (e.g., age ≥50 years, comorbidities, immunosuppression) • Up to 25% of patients could have no high-risk factors. Key Exclusion Criteria • Need for hospitalization • SpO₂ ≤93% on room air Interventions • Single dose of PEG-IFN lambda 180 µg SUBQ (n = 931) • Placebo (n = 1,018; 825 received single SUBQ injection, 193 received PO placebo) Primary Endpoint • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints • Composite of COVID-19–related hospitalization or death by Day 28 • SARS-CoV-2 viral clearance at Day 7 • Occurrence of AEs	Participant Characteristics • Median age 43 years; 57.1% women; 95.1% self-identified as mixed race • 1,919 (98.5%) from Brazil, 30 (1.5%) from Canada • 50% with obesity • 59.4% were randomized within 3 days of symptom onset. • 83% received ≥1 COVID-19 vaccine dose. Primary Outcome • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 (ITT): 25 (2.7%) in PEG-IFN lambda arm vs. 57 (5.6%) in placebo arm (relative risk 0.49; 95% Bayesian CrI, 0.30–0.76) • 61 events (74%) were hospitalizations (ITT). Secondary Outcomes • Composite of COVID-19–related hospitalization or death by Day 28: 22 (2.4%) in PEG-IFN lambda arm vs. 40 (3.9%) in placebo arm (relative risk 0.61; 95% CrI, 0.36–0.99) • SARS-CoV-2 viral clearance at Day 7 among the 15% of patients with VL >192 million copies/mL at baseline: 50.5% in PEG-IFN lambda arm vs. 32.9% in placebo arm (OR 2.13; 95% CrI, 1.14–4.00) • Occurrence of AEs: 141 (15.1%) in PEG-IFN lambda arm vs. 172 (16.9%) in placebo arm (relative risk 0.90; 95% CrI, 0.73–1.10)	Key Limitations • Health care facility capacity may have influenced the number and duration of ED observations. • As this was an adaptive platform trial where multiple investigational treatments or placebos were being evaluated simultaneously, not all patients in the placebo arm received a placebo that was matched to PEG-IFN lambda. Interpretation • In outpatients with COVID-19 who were within 7 days of symptom onset, PEG-IFN lambda reduced the need for ED observations >6 hours or hospitalization when compared with placebo.

Methods	Results	Limitations and Interpretation
Single-Blind RCT of Pegylated Interferon Lambda-1a for Treatment of Outpatients With Uncomplicated COVID-19 in the United States⁵		
Key Inclusion Criteria • Aged 18–65 years • Asymptomatic or symptomatic • Positive SARS-CoV-2 RT-PCR result within 72 hours of enrollment Key Exclusion Criteria • Current or imminent hospitalization • Respiratory rate >20 breaths/min • SpO₂ <94% on room air • Decompensated liver disease Interventions • Single dose of PEG-IFN lambda-1a 180 µg SUBQ (n = 60) • Placebo (n = 60) Primary Endpoint • Time to first negative SARS-CoV-2 RT-PCR result Key Secondary Endpoints • Hospitalization by Day 28 • Time to complete symptom resolution	Participant Characteristics • Median age 36 years; 42% women; 63% Latinx, 28% White • 7% were asymptomatic. • Median of 5 days of symptoms before randomization Primary Outcome • Median time to cessation of viral shedding: 7 days in both arms (aHR 0.81; 95% CI, 0.56–1.19; <i>P</i> = 0.29) Secondary Outcomes • No difference between PEG-IFN lambda-1a and placebo arms in: • Proportion of patients hospitalized by Day 28: 3.3% for each arm • Time to resolution of symptoms: 8 days vs. 9 days (HR 0.94; 95% CI, 0.64–1.39) Other Outcome • Patients who received PEG-IFN lambda-1a were more likely to have elevations of transaminase concentrations than patients who received placebo (25% vs. 8%; <i>P</i> = 0.027).	Key Limitation • Small sample size Interpretation • PEG-IFN lambda-1a provided no virologic or clinical benefit compared to placebo among outpatients with uncomplicated COVID-19.

Methods	Results	Limitations and Interpretation
Double-Blind RCT of Pegylated Interferon Lambda in Outpatients With Laboratory-Confirmed COVID-19 in Canada⁶		
Key Inclusion Criteria - Positive SARS-CoV-2 PCR result - Patients were within 7 days of symptom onset, or, if asymptomatic, were within 7 days of first positive SARS-CoV-2 test result. Key Exclusion Criterion - Immunosuppression or condition that could be worsened by PEG-IFN lambda Interventions - Single dose of PEG-IFN lambda 180 µg SUBQ (n = 30) - Placebo (n = 30) Primary Endpoint - Proportion of patients with negative SARS-CoV-2 test result on nasal mid-turbinate swab at Day 7 Key Secondary Endpoints - Quantitative change in SARS-CoV-2 RNA over time - Hospitalization by Day 14	Participant Characteristics - Median age 46 years; 58% women; 52% White 19% were asymptomatic. - Mean of 4.5 days of symptoms before randomization Primary Outcome 80% in PEG-IFN lambda arm vs. 63% in placebo arm were negative for SARS-CoV-2 RNA at Day 7 ($P = 0.15$). Secondary Outcomes - VL decline by Day 7 was greater in PEG-IFN lambda arm than in placebo arm ($P = 0.0041$). 1 participant in each arm hospitalized by Day 14 Other Outcome 3 participants in each arm had mild elevations of aminotransferase concentrations. Increase was greater in PEG-IFN lambda arm.	Key Limitation - Small sample size Interpretation - PEG-IFN lambda may accelerate VL decline and clearance in outpatients with COVID-19; however, the clinical significance of this finding is unclear.

Key: AE = adverse event; ALT = alanine transaminase; AST = aspartate aminotransferase; DM = diabetes mellitus; ECMO = extracorporeal membrane oxygenation; ED = emergency department; HCQ = hydroxychloroquine; HTN = hypertension; IFN = interferon; ITT = intention-to-treat; IV = intravenous; LPV/RTV = lopinavir/ritonavir; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PCR = polymerase chain reaction; PEG-IFN = pegylated interferon; PO = oral; RCT = randomized controlled trial; RDV = remdesivir; RT-PCR = reverse transcription polymerase chain reaction; SOC = standard of care; SpO₂ = oxygen saturation; SUBQ = subcutaneous; ULN = upper limit of normal; VL = viral load; WHO = World Health Organization

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Table 4e. Characteristics of Antiviral Agents, Including Antibody Products

Last Updated: February 29, 2024

- This table contains drugs and products that have shown antiviral activity against SARS-CoV-2, including small-molecule antiviral drugs, CCP, and IFNs.
- RDV and RTV-boosted nirmatrelvir (Paxlovid) are approved by the FDA for the treatment of COVID-19.
- MOV and CCP have received EUAs from the FDA for the treatment of COVID-19.
- For drug-drug interaction information, please refer to product labels, EUA fact sheets, and [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#).
- For the Panel's recommendations on using the drugs listed in this table, refer to [Antiviral Agents, Including Antibody Products](#); [Therapeutic Management of Nonhospitalized Adults With COVID-19](#); [Therapeutic Management of Hospitalized Adults With COVID-19](#); [Therapeutic Management of Nonhospitalized Children With COVID-19](#); [Therapeutic Management of Hospitalized Children With COVID-19](#); and [Pregnancy, Lactation, and COVID-19 Therapeutics](#).

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Anti-SARS-CoV-2 Antiviral Drugs (Small-Molecule Antivirals)					
Ritonavir-Boosted Nirmatrelvir (Paxlovid) <i>Approved by the FDA for use in adults and authorized under an FDA EUA for use in adolescents (aged ≥12 years and weighing ≥40 kg) for the treatment of mild to moderate COVID-19 in high-risk individuals.</i>	FDA Prescribing Information/EUA Dose for COVID-19^{1,2} <i>eGFR ≥60 mL/min</i> • Nirmatrelvir 300 mg (two 150-mg tablets) with RTV 100 mg (one 100-mg tablet) twice daily for 5 days <i>eGFR ≥30 to <60 mL/min</i> • Nirmatrelvir 150 mg (one 150-mg tablet) with RTV 100 mg (one 100-mg tablet) twice daily for 5 days <i>eGFR <30 mL/min</i> • Not recommended (see comments) <i>Severe Hepatic Impairment (Child-Pugh Class C)</i> • Not recommended	• Dysgeusia • Diarrhea • Anaphylaxis, serious skin reactions, and other HSRs	• Boxed warning: Monitor for potential AEs due to drug-drug interactions with concomitant medications. Weigh potential benefits of treatment against potential risks of drug-drug interactions. • Use with caution in patients with pre-existing liver diseases, liver enzyme abnormalities, or hepatitis. • Monitor for HSRs.	• RTV-boosted nirmatrelvir has significant drug-drug interactions. Before prescribing RTV-boosted nirmatrelvir, carefully review concomitant medications, including OTC medicines, herbal supplements, and recreational drugs. • See Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir (Paxlovid) and Concomitant Medications for additional guidance and resources to assist with identifying drug-drug interactions.	• The FDA prescribing information and the EUA do not recommend using RTV-boosted nirmatrelvir in patients with eGFR <30 mL/min. See Ritonavir-Boosted Nirmatrelvir (Paxlovid) for more information. • Both nirmatrelvir and RTV tablets can be taken with or without food. • The FDA prescribing information and the EUA advise against crushing nirmatrelvir and RTV tablets. However, some data indicate that the tablets can be split or crushed if necessary.³

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Anti-SARS-CoV-2 Antiviral Drugs (Small-Molecule Antivirals), continued					
Remdesivir <i>Approved by the FDA for the treatment of COVID-19 in individuals aged ≥28 days and weighing ≥3 kg.</i>	Dose for Adults and Children Weighing ≥40 kg - RDV 200 mg IV on Day 1, then RDV 100 mg IV once daily from Day 2 Dose for Children Aged ≥28 Days and Weighing 3 kg to <40 kg - RDV 5 mg/kg IV on Day 1, then RDV 2.5 mg/kg IV once daily from Day 2 Total Treatment Duration <i>Nonhospitalized Patients or Patients Hospitalized for Reasons Other than COVID-19</i> 3 days <i>Hospitalized Patients</i> 5 days or until hospital discharge - If a patient does not clinically improve, clinicians may extend the treatment course for ≤5 additional days, for a total duration of 10 days.	- Nausea - ALT and AST elevations - HSRs - Increases in prothrombin time - Bradycardia	- Monitor patients for infusion-related reactions during the infusion and observe them for ≥1 hour after the infusion as clinically appropriate. - Monitor renal function, hepatic function, and prothrombin time as clinically indicated. - Monitor heart rate.	No clinically significant drug-drug interactions are expected with CYP3A4 inducers or inhibitors of OATP1B1, OATP1B3, or P-gp.	- Administer each infusion over 30–120 minutes. - RDV may be used without dose adjustment in patients with renal impairment, including those receiving dialysis.⁴

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Anti-SARS-CoV-2 Antiviral Drugs (Small-Molecule Antivirals), continued					
Molnupiravir <i>Authorized under an FDA EUA for the treatment of mild to moderate COVID-19 in high-risk individuals aged ≥18 years.</i>	Dose Recommended in FDA EUA - MOV 800 mg (four 200-mg capsules) PO every 12 hours for 5 days - MOV is not authorized for use in people aged <18 years due to potential effects on bone and cartilage growth.	- Diarrhea - Nausea - Dizziness - Per the EUA, the 5-day course of MOV has a low risk for genotoxicity.⁵ See Molnupiravir for details.	- Before initiating MOV, assess the patient's pregnancy status as clinically indicated. - Monitor for potential AEs.	- Clinical drug-drug interaction studies of MOV have not been conducted. - Drug-drug interactions related to hepatic metabolism are not expected.	- People of reproductive potential who are sexually active should use effective contraception during and after treatment with MOV. - Pregnant patients should also be offered the opportunity to participate in the COVID-19 International Drug Pregnancy Registry. - Breastfeeding is not recommended while a patient is taking MOV and for 4 days after the last dose. - MOV can be taken with or without food. - The EUA provides instructions for preparing and administering MOV capsule contents through OG or NG tubes.⁵

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
COVID-19 Convalescent Plasma					
High-Titer COVID-19 Convalescent Plasma <i>Authorized under an FDA EUA for the treatment of COVID-19 in patients who are immunocompromised or who are receiving immunosuppressive treatment.</i>	Dose Recommended in FDA EUA Administer 1 high-titer CCP unit (about 200 mL) IV. Administer an additional CCP unit IV based on the prescribing provider's judgment and the patient's clinical response.	- TRALI - TACO - Allergic reactions - Anaphylactic reactions - Febrile nonhemolytic reactions - Hemolytic reactions - Hypothermia - Metabolic complications - Transfusion-transmitted infections⁶ - Thrombotic events - Theoretical risk of antibody-mediated enhancement of infection and suppressed, long-term immunity	- Before administering CCP to patients with a history of severe allergic or anaphylactic transfusion reactions, consult a transfusion medicine specialist who is associated with the hospital's blood bank. - Monitor for transfusion-related reactions. - Monitor vital signs at baseline and during and after transfusion.	Drug products should not be added to the IV infusion line for the blood product.	In patients with impaired cardiac function and heart failure, it may be necessary to reduce the CCP volume or decrease the transfusion rate.
Interferons					
IFN Beta <i>Not approved by the FDA and not recommended by the Panel for the treatment of COVID-19.</i>	Various doses and durations for IFN beta-1a and IFN beta-1b are being studied in clinical trials.	- Flu-like symptoms (e.g., fever, fatigue, myalgia) - Leukopenia, neutropenia, thrombocytopenia, lymphopenia - Liver function abnormalities (ALT > AST) - Injection site reactions - Headache - Hypertonia - Pain - Rash - Worsening depression - Induction of autoimmunity	- Monitor CBC with differential and liver enzymes. - Monitor for worsening CHF. - Monitor for signs of depression and suicidal ideation.	- Low potential for drug-drug interactions Use with caution with other hepatotoxic agents. - Reduce dose if ALT is >5 times ULN.	Inhaled IFN beta-1a is not approved by the FDA for use in the United States.

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Interferons, continued					
PEG-IFN Lambda <i>Not approved by the FDA and not recommended by the Panel for the treatment of COVID-19.</i>	Dose for COVID-19 in Clinical Trials Single dose of PEG-IFN lambda 180 µg SUBQ	- Liver function abnormalities (ALT > AST) - Injection site reactions	- CBC with differential - Liver enzymes - Monitor for potential AEs.	- Low potential for drug-drug interactions Use with caution with other hepatotoxic agents.	PEG-IFN lambda is not approved by the FDA for use in the United States.

Key: AE = adverse event; ALT = alanine transaminase; AST = aspartate aminotransferase; CBC = complete blood count; CCP = COVID-19 convalescent plasma; CHF = congestive heart failure; CYP = cytochrome P450; eGFR = estimated glomerular filtration rate; EUA = Emergency Use Authorization; FDA = Food and Drug Administration; HSR = hypersensitivity reaction; IFN = interferon; IV = intravenous; MOV = molnupiravir; NG = nasogastric; OATP = organic anion transporting polypeptide; OG = orogastric; OTC = over-the-counter; the Panel = the COVID-19 Treatment Guidelines Panel; PEG-IFN = pegylated interferon; P-gp = P-glycoprotein; PO = oral; RDV = remdesivir; RTV = ritonavir; SUBQ = subcutaneous; TACO = transfusion-associated circulatory overload; TRALI = transfusion-related acute lung injury; ULN = upper limit of normal

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Immunomodulators

Last Updated: February 29, 2024

Summary Recommendations
• The hyperactive inflammatory response to SARS-CoV-2 infection plays a central role in the pathogenesis of COVID-19. See Therapeutic Management of Hospitalized Adults With COVID-19 and Therapeutic Management of Hospitalized Children With COVID-19 for the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations on the use of the following immunomodulators in hospitalized patients with COVID-19 according to their disease severity (listed in alphabetical order): • Abatacept • Baricitinib (or tofacitinib) • Dexamethasone • Infliximab • Tocilizumab (or sarilumab) • There is insufficient evidence for the Panel to recommend either for or against the use of the following immunomodulators for the treatment of COVID-19: • Anakinra • Inhaled corticosteroids • Vilobelimab • The Panel recommends against the use of canakinumab for the treatment of COVID-19, except in a clinical trial (BIIa). Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Systemic Corticosteroids

Last Updated: February 29, 2024

The results of several randomized trials indicate that systemic corticosteroid therapy improves clinical outcomes and reduces mortality in hospitalized patients with COVID-19 who require supplemental oxygen,^{1,2} presumably by mitigating the COVID-19–induced systemic inflammatory response that can lead to lung injury and multisystem organ dysfunction. In contrast, in hospitalized patients with COVID-19 who do not require supplemental oxygen, the use of systemic corticosteroids provided no benefit and increased mortality.^{3,4} The COVID-19 Treatment Guidelines Panel’s (the Panel) recommendations for the use of systemic corticosteroids in hospitalized patients with COVID-19 are based on results from these clinical trials (see [Table 5a](#)). There are no data to support the use of systemic corticosteroids in nonhospitalized patients with COVID-19.

Recommendations

- The Panel **recommends against** the use of **dexamethasone** or other systemic corticosteroids to treat outpatients with mild to moderate COVID-19 who do not require hospitalization or supplemental oxygen (**AIIb**).
- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for the Panel’s recommendations on the use of dexamethasone or other systemic corticosteroids in certain hospitalized patients with COVID-19.
- Patients with COVID-19 who are receiving **dexamethasone** or another corticosteroid for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Rationale

Nonhospitalized Adults

There are no data to support the use of systemic corticosteroids in nonhospitalized patients with COVID-19. Therefore, the safety and efficacy of using systemic corticosteroids in this population have not been established. Generally, the use of systemic corticosteroids is associated with adverse events (e.g., hyperglycemia, neuropsychiatric symptoms, secondary infections), which may be difficult to detect and monitor in an outpatient setting. For more information, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#).

Hospitalized Adults

The RECOVERY trial was a multicenter, open-label trial in the United Kingdom that randomly assigned 6,425 hospitalized patients to receive up to 10 days of dexamethasone 6 mg once daily plus standard care or standard care alone.³ Mortality at 28 days was lower among the patients who received dexamethasone than among those who received standard care alone. This benefit of dexamethasone was observed in patients who were mechanically ventilated or who required supplemental oxygen at enrollment. In contrast, no benefit was seen in patients who did not require supplemental oxygen at enrollment.

Several clinical trials have evaluated the use of systemic corticosteroids in critically ill patients with COVID-19 who were on supplemental oxygen with or without mechanical ventilation. These trials reported that all-cause mortality at 28 days was lower in those who received systemic corticosteroids

than in those who received standard of care or placebo; however, some of these trials were terminated early.¹

In addition to the randomized controlled trials, a large observational study evaluated the use of systemic corticosteroids in 15,404 hospitalized patients with positive SARS-CoV-2 polymerase chain reaction or antigen test results from a Department of Veterans Affairs database.⁴ Corticosteroids were administered to 60% of the patients within 48 hours of admission, and 95% of the patients who received corticosteroids received dexamethasone. A total of 9,450 patients did not receive supplemental oxygen during the study. Of these patients, 3,514 (37%) received dexamethasone for a median duration of 5 days (IQR 3–8 days). Using average treatment effect estimates, patients who received dexamethasone without supplemental oxygen had an increased risk of death within 90 days (HR 1.76; 95% CI, 1.47–2.12). Patients who received dexamethasone either without supplemental oxygen or with low-flow nasal cannula oxygen had a 60% higher risk of death. Although this study was observational, the investigators employed several statistical techniques to minimize potential bias, including propensity scoring and weighted analyses. Additionally, several subgroup and sensitivity analyses in this study confirmed the overall results.

Dexamethasone Dose

The RECOVERY platform trial studied the use of dexamethasone 6 mg once daily for up to 10 days,³ which is currently the recommended dose for hospitalized adults with COVID-19 who require supplemental oxygen. Several other randomized controlled trials evaluated the role of higher doses of dexamethasone or other corticosteroids in hospitalized patients with different levels of respiratory support. The results of some key studies are summarized below.

Patients Who Received Conventional Oxygen or No Supplemental Oxygen

The RECOVERY platform trial included an additional study in which patients with COVID-19 and evidence of hypoxemia (i.e., they were receiving conventional supplemental oxygen or had an oxygen saturation <92% on room air) were randomized to receive usual care plus high-dose dexamethasone (20 mg once daily for 5 days, then 10 mg once daily for 5 days or until hospital discharge, whichever came first) or usual care alone, which included low-dose dexamethasone (usually 6 mg once daily for 10 days).⁵ On May 11, 2022, the trial's independent data monitoring committee stopped enrolling patients receiving conventional oxygen therapy and those not receiving any supplemental oxygen. Among the 1,272 patients enrolled, 28-day mortality was higher in the high-dose dexamethasone arm than in the usual care arm (19% vs. 12%; rate ratio 1.59; 95% CI, 1.20–2.10; $P = 0.0012$).

Patients Who Received Noninvasive or Mechanical Ventilation

The COVID STEROID 2 trial investigated the use of different doses of corticosteroids in patients with COVID-19 and severe hypoxemia.⁶ In this multicenter trial, hospitalized patients who required at least 10 L/min of oxygen or mechanical ventilation were randomized to receive up to 10 days of dexamethasone 6 mg once daily ($n = 485$) or dexamethasone 12 mg once daily ($n = 497$). The median number of days alive without life support at 28 days after randomization was 20.5 days in the dexamethasone 6 mg arm and 22.0 days in the dexamethasone 12 mg arm, yielding an adjusted mean difference of 1.3 days (95% CI, 0–2.6; $P = 0.07$). No differences between the arms were found for 28- or 90-day mortality. Although these conventional analyses did not quite reach statistical significance, a preplanned Bayesian analysis found that dexamethasone 12 mg had a higher probability of benefit and a lower probability of harm than dexamethasone 6 mg.⁷

In the COVIDICUS trial, patients with COVID-19 and acute hypoxemic respiratory failure were randomized to receive dexamethasone 6 mg once daily for 10 days ($n = 276$, of which 37 received

placebo prior to release of results from the RECOVERY trial)³ or high-dose dexamethasone (i.e., 20 mg once daily for 5 days, then 10 mg once daily for 5 days; n = 270).⁸ At baseline, 98 patients were receiving mechanical ventilation, 114 were receiving continuous positive airway pressure, 10 were receiving noninvasive ventilation (NIV), 199 were receiving high-flow nasal cannula oxygen, and 125 were receiving standard oxygen therapy through a nonrebreather mask. There was no difference in 60-day mortality between the arms (HR 0.96; 95% CI, 0.69–1.33; *P* = 0.79).

The mixed results from these studies have led the Panel to continue to recommend 6 mg once daily as the preferred dose of dexamethasone in hospitalized patients with COVID-19 who require supplemental oxygen, including patients receiving NIV or mechanical ventilation. However, the Panel notes that both the conventional and Bayesian analyses conducted during the COVID STEROID 2 trial suggest that a dose of 12 mg might confer a benefit in patients who require NIV or mechanical ventilation.^{6,7} Most patients in the COVID STEROID 2 trial did not receive additional immunomodulators beyond corticosteroids.⁶ Currently, there are no data from clinical trials that evaluated the safety and efficacy of using doses of dexamethasone that were larger or smaller than 6 mg once daily in combination with other immunomodulators to treat hospitalized adults with COVID-19.

Combination Immunomodulator Therapy

Using systemic corticosteroids in combination with other agents, including tocilizumab (see [Interleukin-6 Inhibitors](#)),^{9,10} baricitinib (see [Janus Kinase Inhibitors](#)),¹¹ abatacept, or infliximab,¹² has been shown to have a clinical benefit in subsets of hospitalized patients with COVID-19, especially those who are in the early stages of critical illness and those with signs of systemic inflammation. For the Panel's recommendations on when to use dexamethasone with another immunomodulator, see [Therapeutic Management of Hospitalized Adults With COVID-19](#).

See [Table 5a](#) for a summary of the data from clinical trials that have evaluated the use of systemic corticosteroids in patients with COVID-19.

Systemic Corticosteroids Other Than Dexamethasone

Systemic corticosteroids other than dexamethasone, including hydrocortisone^{13,14} and methylprednisolone,^{15,16} have been studied for the treatment of COVID-19 in several randomized trials. Some of these trials were stopped early due to low enrollment following the release of the RECOVERY trial results. Consequently, the sample size of these trials was insufficient to assess efficacy (i.e., there were too few events to definitively confirm or exclude an effect, although many point estimates suggested a beneficial effect). Therefore, the evidence supporting the use of hydrocortisone or methylprednisolone for the treatment of COVID-19 is not as strong as the evidence supporting the use of dexamethasone. Based on the available evidence, the Panel has concluded the following:

- If dexamethasone is not available, alternative corticosteroids (e.g., **prednisone, methylprednisolone, hydrocortisone**) can be used (**BIII**).
- For these drugs, the total daily dose equivalencies to dexamethasone 6 mg (orally or intravenously)¹⁷ are:
 - Prednisone 40 mg
 - Methylprednisolone 32 mg
 - Hydrocortisone 160 mg
- Half-life, duration of action, and frequency of administration vary among corticosteroids.
 - *Long-acting corticosteroid: Dexamethasone*; half-life 36 to 72 hours; administer once daily.

- *Intermediate-acting corticosteroids*: Prednisone and methylprednisolone; half-lives 12 to 36 hours; administer once daily or in 2 divided doses daily.
- *Short-acting corticosteroid*: Hydrocortisone; half-life 8 to 12 hours; administer in 2 to 4 divided doses daily.
- Hydrocortisone is commonly used to manage septic shock in patients with COVID-19; see [Hemodynamics for Adults](#) for more information. Unlike other corticosteroids that have previously been studied in patients with acute respiratory distress syndrome, dexamethasone lacks mineralocorticoid activity and, thus, its effects on sodium balance and fluid volume are minimal.¹⁸

Monitoring, Adverse Effects, and Drug-Drug Interactions

Clinicians should closely monitor patients with COVID-19 who are receiving dexamethasone for certain adverse effects (e.g., hyperglycemia, secondary infections, psychiatric effects, avascular necrosis). The use of systemic corticosteroids may increase the risk of opportunistic fungal infections (e.g., aspergillosis, mucormycosis) and reactivation of latent infections (e.g., hepatitis B virus infection, herpes simplex virus and varicella zoster virus infections, strongyloidiasis, tuberculosis).¹⁹⁻²⁶ Cases of severe and disseminated strongyloidiasis have been reported in patients with COVID-19 during treatment with tocilizumab and corticosteroids.^{24,27} Many clinicians would initiate empiric treatment for strongyloidiasis (e.g., with the antiparasitic drug ivermectin), with or without serologic testing, in patients who currently reside or who have previously resided in areas where *Strongyloides* is endemic (i.e., tropical, subtropical, or warm temperate areas).²⁸

Using systemic corticosteroids with other immunosuppressants, such as tocilizumab or baricitinib, could theoretically increase the risk of secondary infections. However, clinical trials have reported no difference in the rates of secondary infections between patients who received corticosteroids in combination with another immunomodulatory agent and those who received corticosteroids alone. Dexamethasone is a moderate cytochrome P450 3A4 inducer. Therefore, it could reduce the concentration and potential efficacy of concomitant medications that are cytochrome P450 3A4 substrates. Clinicians should carefully review a patient's concomitant medications to assess the potential for drug-drug interactions.

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance regarding the use of dexamethasone during pregnancy and lactation.

Considerations in Children

Dexamethasone is recommended for hospitalized children with COVID-19 who require supplemental oxygen. See [Therapeutic Management of Hospitalized Children With COVID-19](#) for the Panel's recommendations. Methylprednisolone or another corticosteroid is recommended for the treatment of multisystem inflammatory syndrome in children (MIS-C). See [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#) for the Panel's recommendations.

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Table 5a. Systemic Corticosteroids: Selected Clinical Trial Data

Last Updated: February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for systemic corticosteroids. The studies summarized below are those that have had the greatest impact on the Panel's recommendations. Unless stated otherwise, the clinical trials listed below only included participants aged ≥ 18 years.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of Dexamethasone in Hospitalized Patients With COVID-19 in the United Kingdom¹		
Key Inclusion Criterion - Hospitalized with suspected or laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criteria - Physician determination, based on patient's medical history, that risk of participation was too great - An indication for corticosteroid therapy outside of the study Interventions - DEX 6 mg IV or PO once daily plus SOC for up to 10 days or until hospital discharge, whichever came first (n = 2,104) - SOC alone (n = 4,321) Primary Endpoint - All-cause mortality at 28 days	Participant Characteristics - Mean age 66 years; 64% men; 73% White 56% had ≥ 1 comorbidities; 24% with DM 89% had laboratory-confirmed SARS-CoV-2 infection. - Median of 7 days of DEX therapy - At randomization: 16% received MV or ECMO. 60% required supplemental oxygen but not MV. 24% required no supplemental oxygen. <1% in both arms received RDV. - Received tocilizumab or sarilumab: 2% in DEX arm vs. 3% in SOC arm Primary Outcome - All-cause mortality at 28 days in DEX arm vs. SOC arm: - All patients: 23% vs. 26% (age-adjusted rate ratio 0.83; 95% CI, 0.75–0.93; $P < 0.001$) - Patients who required MV or ECMO at randomization: 29% vs. 41% (rate ratio 0.64; 95% CI, 0.51–0.81) - Patients who required supplemental oxygen but not MV at randomization: 23% vs. 26% (rate ratio 0.82; 95% CI, 0.72–0.94) - Patients who did not require supplemental oxygen at randomization: 18% vs. 14% (rate ratio 1.19; 95% CI, 0.92–1.55)	Key Limitations - Open-label study - Published data did not include results for key secondary endpoints (e.g., cause-specific mortality, need for renal replacement), AEs, and key subgroups (e.g., patients with comorbidities). - Patients who required supplemental oxygen (but not MV) had a range of disease severity. It is unclear whether all patients in this group benefited from DEX or whether benefit was restricted to those who required higher levels of supplemental oxygen. - Patients aged >80 years were preferentially assigned to receive supplemental oxygen therapy (and not MV). - The high mortality in this study may limit the generalizability of results to populations with a lower baseline mortality. Interpretation - In hospitalized patients with severe COVID-19 who required supplemental oxygen, the use of DEX reduced mortality at 28 days. The greatest benefit was seen in those who were receiving MV at randomization. - There was no survival benefit for DEX in patients who did not require supplemental oxygen at randomization.

Methods	Results	Limitations and Interpretation
CoDEX: Open-Label RCT of Dexamethasone in Patients With Moderate to Severe ARDS and COVID-19 in Brazil²		
Key Inclusion Criteria • Suspected or laboratory-confirmed SARS-CoV-2 infection • Received MV within 48 hours of meeting criteria for moderate to severe ARDS (PaO₂/FiO₂ ≤200 mm Hg) Key Exclusion Criteria • Received immunosuppressive drugs in past 21 days • Death expected within 24 hours Interventions • DEX 20 mg IV once daily for 5 days, then DEX 10 mg IV once daily for 5 days or until ICU discharge (n = 151) • SOC alone (n = 148) Primary Endpoint • Number of days alive and free from MV by Day 28 Key Secondary Endpoints • All-cause mortality by Day 28 • Number of ICU-free days by Day 28 • Duration of MV by Day 28 • Score on 6-point OS at Day 15 • SOFA score at Day 7 Key Exploratory Analysis • Death or MV at Day 15	Participant Characteristics • Mean age 61 years; 63% men • Comorbidities in DEX arm vs. SOC arm: • Obesity: 31% vs. 24% • DM: 38% vs. 47% • Vasopressor use: 66% in DEX arm vs. 68% in SOC arm • Mean PaO₂/FiO₂: 131 mm Hg in DEX arm vs. 133 mm Hg in SOC arm • Median of 10 days of DEX therapy • No patients received RDV or tocilizumab. • 35% in SOC arm received corticosteroids for indications such as bronchospasm or septic shock. Primary Outcome • Mean number of days alive and free from MV by Day 28: 7 in DEX arm vs. 4 in SOC arm (<i>P</i> = 0.04) Secondary Outcomes • No differences between arms by Day 28 in all-cause mortality (56% in DEX arm vs. 62% in SOC arm), number of ICU-free days, or duration of MV • No difference between arms at Day 15 in score on 6-point OS • Mean SOFA score at Day 7: 6.1 in DEX arm vs. 7.5 in SOC arm (<i>P</i> = 0.004) Other Outcome • Post hoc analysis of probability of death or MV at Day 15: 68% in DEX arm vs. 80% in SOC arm (OR 0.46; <i>P</i> = 0.01)	Key Limitations • Open-label study • Underpowered; enrollment stopped after release of data from the RECOVERY trial. • Since no follow-up data were collected after hospital discharge for patients who were discharged before Day 28, no data on deaths or rehospitalization were available for these patients between day of discharge and Day 28. • The high mortality in this study may limit the generalizability of results to populations with a lower baseline mortality. • More than a third of those randomized to receive SOC also received corticosteroids. Interpretation • Compared with SOC alone, DEX increased the mean number of days alive and free of MV by Day 28 in patients with COVID-19 and moderate to severe ARDS.

Methods	Results	Limitations and Interpretation
Observational Cohort Study of Dexamethasone in Hospitalized Patients With COVID-19 Who Were Not on Intensive Respiratory Support in the United States³		
Key Inclusion Criterion • Within 14 days of a positive SARS-CoV-2 test result Key Exclusion Criteria • Recent receipt of corticosteroids • Receipt of IRS (defined as HFNC oxygen, NIV, or MV) within 48 hours • Hospital LOS <48 hours Interventions • Corticosteroids (95% received DEX) administered within 48 hours of admission (n = 7,507) • No corticosteroids administered (n = 7,433) Primary Endpoint • All-cause mortality at 90 days	Participant Characteristics • Mean age 71 years; 95% men; 27% Black, 55% White • 77% did not receive IRS within 48 hours. • 83% were hospitalized within 1 day of a positive SARS-CoV-2 test result. • Median duration of DEX for patients who did not receive IRS: 5 days for those not on supplemental oxygen at baseline vs. 6 days for those on low-flow nasal cannula oxygen • Received RDV: 43% of those who received DEX vs. 13% of those who did not • Received anticoagulants: 46% of those who received DEX vs. 10% of those who did not Primary Outcome • Risk of all-cause mortality at 90 days was higher in those who received DEX. • Combination of those not on supplemental oxygen and those on low-flow nasal cannula oxygen: HR 1.59; 95% CI, 1.39–1.81 • Those not on supplemental oxygen: HR 1.76; 95% CI, 1.47–2.12 • Those on low-flow nasal cannula oxygen: HR 1.08; 95% CI, 0.86–1.36	Key Limitations • Retrospective observational study • Because nearly all patients on MV or HFNC oxygen received DEX, analysis was restricted to patients who did not receive IRS (i.e., those who received no supplemental oxygen or only low-flow nasal cannula oxygen). • There were differences between the arms in other therapies received. The investigators attempted to account for this using different approaches (e.g., propensity scoring, weighted analyses, subgroup/sensitivity analyses). Interpretation • In hospitalized patients with COVID-19, the use of DEX was not associated with a reduction in mortality among those who received low-flow nasal cannula oxygen during the first 48 hours after hospital admission, but it was associated with increased mortality among those who received no supplemental oxygen during the first 48 hours after admission.

Methods	Results	Limitations and Interpretation
COVID STEROID 2: Blinded RCT of Dexamethasone 12 mg Versus 6 mg in Hospitalized Adults With COVID-19 and Severe Hypoxemia in Denmark, India, Sweden, and Switzerland^{4,5}		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - Required oxygen ≥ 10 L/min, NIV, CPAP, or MV Key Exclusion Criteria - Treated with DEX >6 mg (or equivalent) - Treated with corticosteroid within past 5 days - Invasive fungal infection or active TB Interventions - DEX 12 mg IV once daily for up to 10 days (n = 497) - DEX 6 mg IV once daily for up to 10 days (n = 485) Primary Endpoint - Number of days alive without life support (MV, circulatory support, or kidney replacement therapy) at 28 days Key Secondary Endpoints - Number of days alive without life support at 90 days - Number of days alive and out of hospital at 90 days - Mortality at 90 days - Mortality at 28 days - SAEs at 28 days	Participant Characteristics - Median age 65 years; 31% women - DM: 27% in 12 mg arm vs. 34% in 6 mg arm - Median of 7 days from symptom onset to hospitalization in both arms - Received ICU care: 78% in 12 mg arm vs. 81% in 6 mg arm - Oxygen requirements: 54% on oxygen via nasal cannula or face mask (median flow rate 23 L/min) 25% on NIV 21% on MV 63% received RDV; 12% received IL-6 inhibitors or JAK inhibitors. - Median of 7 days of DEX therapy in both arms Primary Outcome - Median number of days alive without life support at 28 days: 22.0 in 12 mg arm vs. 20.5 in 6 mg arm (adjusted mean difference 1.3 days; 95% CI, 0.0–2.6; $P = 0.07$) 63.9% Bayesian probability of clinically important benefit and 0.3% Bayesian probability of clinically important harm for DEX 12 mg Secondary Outcomes - At 90 days: - Median number of days alive without life support: 84 in 12 mg arm vs. 80 in 6 mg arm ($P = 0.15$) - Median number of days alive and out of hospital: 62 in 12 mg arm vs. 48 in 6 mg arm ($P = 0.09$) - Mortality: 32% in 12 mg arm vs. 38% in 6 mg arm (adjusted relative risk 0.87; 99% CI, 0.70–1.07; $P = 0.09$) - At 28 days: - Mortality: 27% in 12 mg arm vs. 32% in 6 mg arm (adjusted relative risk 0.86; 99% CI, 0.68–1.08; $P = 0.10$) - SAEs, including septic shock and invasive fungal infections: 11% in 12 mg arm vs. 13% in 6 mg arm (adjusted relative risk 0.83; 99% CI, 0.54–1.29; $P = 0.27$)	Key Limitation - The randomized intervention period was <10 days for some patients because the trial allowed up to 4 days of DEX before enrollment. Interpretation - Among patients with COVID-19 and severe hypoxemia, the use of DEX 12 mg once daily did not result in more days alive without life support at 28 days than the use of DEX 6 mg once daily. - A preplanned Bayesian analysis showed that DEX 12 mg had a higher probability of benefit and a lower probability of harm than DEX 6 mg.⁵

Methods	Results	Limitations and Interpretation
CAPE COVID: Double-Blind RCT of Hydrocortisone In Critically Ill Patients With COVID-19 in France⁶		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection or radiographically suspected COVID-19 with ≥ 1 of the following: - MV with PEEP ≥ 5 cm H₂O - PaO₂/FiO₂ < 300 mm Hg and FiO₂ $\geq 50\%$ on HFNC - PaO₂/FiO₂ < 300 mm Hg on reservoir mask oxygen - Pulmonary severity index score > 130 Key Exclusion Criteria - Septic shock - Do-not-intubate orders Interventions - Continuous IV infusion of hydrocortisone 200 mg per day for 7 days, then 100 mg per day for 4 days, then 50 mg per day for 3 days; if patient improved by Day 4, then IV infusion of hydrocortisone 200 mg per day for 4 days, then 100 mg per day for 2 days, then 50 mg per day for 2 days (n = 76) - Placebo (n = 73) Primary Endpoint - Treatment failure (death or dependency on MV or high-flow oxygen) by Day 21 Key Secondary Endpoints - Need for intubation, prone positioning, ECMO, or inhaled nitric oxide - Nosocomial infection by Day 28 - Clinical status by Day 21, as measured by a 5-item scale: - Death - In ICU and on MV - Required high-flow oxygen therapy - Discharged from ICU	Participant Characteristics - Mean age 62 years; 70% men; median BMI 28 96% had laboratory-confirmed SARS-CoV-2 infection. - Median symptom duration of 9–10 days 81% required MV at baseline. - Received vasopressors: 24% in hydrocortisone arm vs. 18% in placebo arm $< 5\%$ received RDV or tocilizumab. - Median duration of treatment with study drug: 11 days in hydrocortisone arm vs. 13 days in placebo arm ($P = 0.25$) Primary Outcome - Treatment failure by Day 21: 42% in hydrocortisone arm vs. 51% in placebo arm ($P = 0.29$) Secondary Outcomes - No difference between arms in need for intubation or prone positioning (too few received ECMO or inhaled nitric oxide for comparison) - Need for intubation in those not on MV at baseline: 50% in hydrocortisone arm vs. 75% in placebo arm - No difference between arms in proportion of patients with nosocomial infection by Day 28 - No difference between arms in clinical status by Day 21, but 15% died in hydrocortisone arm vs. 27% in placebo arm ($P = 0.06$) - Discharged from ICU by Day 21: 57% in hydrocortisone arm vs. 44% in placebo arm; 23% in both arms still required MV	Key Limitations - Underpowered; enrollment stopped after release of data from the RECOVERY trial. - Limited information about comorbidities Interpretation - The use of hydrocortisone did not reduce the proportion of patients with COVID-19 and acute respiratory failure who experienced treatment failure by Day 21.

Methods	Results	Limitations and Interpretation
REMAP-CAP: Randomized, Open-Label, Adaptive Trial of Hydrocortisone in Patients With Severe COVID-19⁷		
Key Inclusion Criteria • Presumed or laboratory-confirmed SARS-CoV-2 infection • ICU admission for respiratory or cardiovascular support Key Exclusion Criteria • Presumed imminent death • Systemic corticosteroid use • >36 hours since ICU admission Interventions • Hydrocortisone 50 mg IV every 6 hours for 7 days (n = 137) • Shock-dependent hydrocortisone 50 mg IV every 6 hours for duration of shock for up to 28 days (n = 146) • No hydrocortisone (n = 101) Primary Endpoint • Number of days free of respiratory and cardiovascular organ support by Day 21 Key Secondary Endpoint • In-hospital mortality	Participant Characteristics • Mean age 60 years; 71% men; 53% White • Mean BMI range of 29.7–30.9 for the 3 arms • 50% to 64% required MV. Primary Outcome • Median number of days free of organ support by Day 21: 0 in both arms • Median adjusted ORs for hydrocortisone arms vs. no hydrocortisone arm: • OR 1.43 (95% CrI, 0.91–2.27) with 93% Bayesian probability of superiority for fixed-dose hydrocortisone arm • OR 1.22 (95% CrI, 0.76–1.94) with 80% Bayesian probability of superiority for shock-dependent hydrocortisone arm Key Secondary Outcome • No difference between arms in in-hospital mortality (30% in fixed-dose hydrocortisone arm vs. 26% in shock-dependent hydrocortisone arm vs. 33% in no hydrocortisone arm)	Key Limitations • Open-label study • Enrollment stopped after release of data from the RECOVERY trial. Interpretation • The use of hydrocortisone did not increase the median number of days free of organ support in either the fixed-dose or the shock-dependent hydrocortisone arms; however, early termination limited the power to detect differences between the arms.

Methods	Results	Limitations and Interpretation
Single-Blind RCT of Methylprednisolone in Hospitalized Patients With COVID-19 Pneumonia in China⁸		
Key Inclusion Criteria • Laboratory-confirmed SARS-CoV-2 infection • Pneumonia confirmed by chest CT scan • Hospitalized on general ward for <72 hours Key Exclusion Criteria • Severe immunosuppression • Corticosteroid use for other diseases Interventions • Methylprednisolone 1 mg/kg IV per day for 7 days (n = 43) • Saline (n = 43) Primary Endpoint • Clinical deterioration at 14 days Key Secondary Endpoints • Clinical cure at 14 days • Time to clinical cure • ICU admission • In-hospital mortality • Number of days hospitalized	Participant Characteristics • Mean age 56 years; 48% men • Median of 8 days from symptom onset to randomization • At randomization, 71% were receiving oxygen via nasal cannula. Primary Outcome • Clinical deterioration at 14 days: 4.8% in both arms (OR 1.0; 95% CI, 0.134–7.442; $P = 1.00$) Secondary Outcomes • No difference (all $P > 0.05$) between methylprednisolone arm and placebo arm for: • Clinical cure at 14 days: 51% vs. 58% • Median number of days to clinical cure: 14 vs. 12 • ICU admission: 4.8% in both arms • In-hospital mortality: 0% vs. 2.3% • Median number of days hospitalized: 17 vs. 13	Key Limitations • Small sample size • Terminated early because of decreasing incidence of COVID-19 pneumonia at study sites Interpretation • The use of methylprednisolone did not reduce the incidence of clinical deterioration among hospitalized patients with COVID-19.

Methods	Results	Limitations and Interpretation
COVIDICUS: RCT of High-Dose Dexamethasone Versus Standard of Care Dexamethasone in Patients With COVID-19–Related Respiratory Failure in the Intensive Care Unit in France⁹		
Key Inclusion Criteria • Suspected or laboratory-confirmed SARS-CoV-2 infection • ICU admission in past 48 hours • Respiratory failure (defined as PaO₂ <70 mm Hg, SpO₂ <90% on room air, >30 breaths/min, labored breathing, respiratory distress, or need for oxygen ≥6 L/min) Key Exclusion Criteria • Decision to limit life-sustaining treatment • Therapy with ≥0.5 mg/kg per day of prednisone equivalent for ≥3 weeks • Active and untreated bacterial, fungal, or parasitic infection Interventions • High dose: DEX 20 mg IV once daily for 5 days, then DEX 10 mg IV once daily for 5 days (n = 270) • SOC: DEX 6 mg IV once daily for 10 days (n = 239) or placebo (n = 37) Primary Endpoint • All-cause mortality by Day 60	Participant Characteristics • Median age 67 years; 76% men • Median of 9 days from symptom onset to randomization • 81% had ≥1 comorbidities. • 17% received RDV; <1% received tocilizumab. Primary Outcome • All-cause mortality by Day 60: 26% in high-dose arm vs. 27% in SOC arm (HR 0.96; 95% CI, 0.69–1.33; <i>P</i> = 0.79)	Key Limitation • Comparator arm was initially a placebo but was changed to a standard dose of DEX after the RECOVERY trial results were released. Interpretation • Among ICU patients with COVID-19–related respiratory failure, high-dose DEX did not significantly improve 60-day survival.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of 2 Doses of Dexamethasone in Hospitalized Patients With COVID-19 in the United Kingdom, Asia, and Africa¹⁰		
Key Inclusion Criteria - Hospitalized with suspected or laboratory-confirmed SARS-CoV-2 infection - SpO₂ <92% on room air Key Exclusion Criteria - Physician determination, based on patient's medical history, that risk of participation was too great - Contraindication for short-term corticosteroids - Suspected or confirmed influenza - Current use of ritonavir-boosted nirmatrelvir (Paxlovid), ritonavir, or other potent CYP3A inhibitor Interventions - High dose: DEX 20 mg once daily plus SOC for 5 days followed by DEX 10 mg once daily for 5 days or until hospital discharge, whichever came first (n = 659) - SOC: DEX 6 mg once daily plus SOC for 10 days or until hospital discharge, whichever came first (n = 613) Primary Endpoint - All-cause mortality at 28 days Key Secondary Endpoints - Time to discharge from hospital - Composite of MV (including ECMO) or death Key Safety Endpoints - Infections other than COVID-19 - Metabolic complications	Participant Characteristics - Enrollment for the subgroup of patients who received conventional oxygen or did not receive supplemental oxygen was stopped prematurely due to safety concerns. The results reported for this analysis only include patients from this subgroup. - Mean age 61 years; 60% men; 54% Asian, 36% White 51% had ≥1 comorbidities; 19% with DM. 53% received ≥1 COVID-19 vaccine doses. 34% received RDV; 12% had received tocilizumab or were going to receive tocilizumab within 24 hours of randomization. Primary Outcome - All-cause mortality at 28 days: 19% in high-dose arm vs. 12% in SOC arm (rate ratio 1.59; 95% CI, 1.20–2.10; P = 0.0012) Secondary Outcomes - Time to discharge from hospital: 9 days in both arms - Composite of MV or death: 20% in high-dose arm vs. 13% in SOC arm (risk ratio 1.52; 95% CI, 1.18–1.97) Safety Outcomes - Pneumonia not due to COVID-19: 10% in high-dose arm vs. 6% in SOC arm (absolute difference 3.7%; 95% CI, 0.7–6.6) - Hyperglycemia requiring new or increased insulin dose: 22% in high-dose arm vs. 14% in SOC arm (absolute difference 7.4%; 95% CI, 3.2–11.5)	Key Limitations - Open-label study - The larger RECOVERY trial stopped enrollment of patients in this subgroup (i.e., those who received conventional oxygen or did not receive supplemental oxygen) due to safety concerns. Interpretation - In patients hospitalized with COVID-19 who had clinical hypoxemia (SpO₂ <92%) and did not require supplemental oxygen or required only conventional oxygen, the use of high-dose DEX increased the risk of death and hyperglycemia when compared with the use of standard doses of corticosteroids.

Key: AE = adverse event; ARDS = acute respiratory distress syndrome; BMI = body mass index; CPAP = continuous positive airway pressure; CT = computed tomography; CYP = cytochrome P450; DEX = dexamethasone; DM = diabetes mellitus; ECMO = extracorporeal membrane oxygenation; FiO₂ = fraction of inspired oxygen; HFNC = high-flow nasal cannula; ICU = intensive care unit; IL = interleukin; IRS = intensive respiratory support; IV = intravenous; JAK = Janus kinase; LOS = length of stay; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PaO₂ = arterial partial pressure of oxygen; PEEP = positive end-expiratory pressure; PO = oral; RCT = randomized controlled trial; RDV = remdesivir; SAE = serious adverse event; SOC = standard of care; SOFA = sequential organ failure assessment; SpO₂ = oxygen saturation; TB = tuberculosis

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Inhaled Corticosteroids

Last Updated: December 20, 2023

Inhaled corticosteroids have been identified as potential COVID-19 therapeutic agents because of their targeted anti-inflammatory effects on the lungs. In addition, certain inhaled corticosteroids have been shown to impair viral replication of SARS-CoV-2¹ and downregulate the expression of the receptors used for cell entry.^{2,3} Several trials provide additional insights regarding the role of inhaled corticosteroids in treating outpatients with COVID-19. These trials are described below and in [Table 5b](#).

Recommendations

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of inhaled corticosteroids for the treatment of COVID-19.
- There is insufficient evidence for the Panel to recommend either for or against the use of the combination of inhaled budesonide plus fluvoxamine for the treatment of COVID-19 in nonhospitalized patients.
- Patients with COVID-19 who are receiving an inhaled corticosteroid for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Rationale

Compared to usual care, inhaled corticosteroid therapy decreased the time to recovery in 2 open-label randomized controlled trials in outpatients with mild symptoms of COVID-19.^{4,5} However, subsequent placebo-controlled, double-blind trials have shown that corticosteroid therapy did not reduce the duration of COVID-19 symptoms.⁶⁻⁸ The available evidence does not show that inhaled corticosteroid therapy reduces the risk of hospitalization or death due to COVID-19. However, the Panel acknowledges that there are areas of uncertainty. Studies conducted predominantly among unvaccinated patients have reported mixed results.

ACTIV-6 is the only randomized controlled trial of inhaled corticosteroid monotherapy that was conducted in a predominantly vaccinated population.⁸ In this study, treatment with inhaled fluticasone did not reduce the number of hospitalizations or health care visits or the time to sustained recovery. However, this study included patients who were at modest risk for complications from COVID-19. The median age of the patients was 45 years, and patients were not required to have a comorbidity to be included in the study.

The mixed results from these studies make it difficult to draw definitive conclusions about the benefit of using inhaled corticosteroids in people who are at high risk of disease progression. See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for recommendations regarding therapies for high-risk outpatients.

The combination of inhaled budesonide plus oral fluvoxamine was studied in a large, double-blind, placebo-controlled, adaptive randomized trial in Brazil.⁹ Over 90% of the patients had received at least 2 doses of a COVID-19 vaccine. Treatment with this combination significantly reduced the incidence of the primary outcome, which was a composite of hospitalization or retention in an emergency setting for >6 hours. The proportion of patients who were hospitalized was the same in the treatment and placebo arms (0.9% vs. 1.1%), and the treatment did not significantly impact secondary outcomes such as health care attendance or the need for an emergency setting visit. It is unclear how the >6-hour emergency

setting outcome translates to other settings. In addition, the treatment with budesonide plus fluvoxamine was associated with significantly more adverse events.

For more information on these trials, see [Table 5b](#).

No clinical trials have assessed the role of inhaled corticosteroids for the treatment of COVID-19 in hospitalized patients.

Monitoring, Adverse Effects, and Drug-Drug Interactions

Patients who are receiving inhaled corticosteroids may develop oral candidiasis.

Using a cytochrome P450 3A4 inhibitor, such as ritonavir-boosted nirmatrelvir (Paxlovid), with inhaled budesonide or fluticasone may lead to increased systemic absorption of the corticosteroid, which may result in systemic adverse effects from the corticosteroid.

Considerations in Pregnant People

There is insufficient evidence for the Panel to recommend either for or against the use of inhaled corticosteroids for the treatment of COVID-19 in people who are pregnant. Pregnant patients with COVID-19 who are receiving an inhaled corticosteroid for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Considerations in Children

There is insufficient evidence for the Panel to recommend either for or against the use of inhaled corticosteroids for the treatment of COVID-19 in children. Children with COVID-19 who are receiving an inhaled corticosteroid for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

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Table 5b. Inhaled Corticosteroids: Selected Clinical Trial Data

Last Updated: December 20, 2023

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for inhaled corticosteroids. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
PRINCIPLE: Open-Label RCT of Inhaled Budesonide in Nonhospitalized Patients With COVID-19 in the United Kingdom¹		
Key Inclusion Criteria • Aged ≥65 years or aged ≥50 years with comorbidities • PCR-confirmed or suspected COVID-19 • ≤14 days of COVID-19 symptoms Key Exclusion Criteria • Already taking inhaled or systemic corticosteroids • Unable to use an inhaler • Contraindication for inhaled budesonide Interventions • Usual care plus inhaled budesonide 800 µg twice daily for 14 days (n = 1,069) • Usual care (n = 787) Primary Endpoints • COVID-19–related hospitalization or death by Day 28 • Time to reported recovery up to 28 days from randomization	Participant Characteristics • Mean age 64.2 years; 52% women; 92% White • 81% with comorbidities • Median of 6 days from symptom onset to randomization Primary Outcomes • COVID-19–related hospitalization or death by Day 28: 6.8% in budesonide arm vs. 8.8% in usual care arm (OR 0.75; 95% CrI, 0.55–1.03) • Median time to reported recovery: 11.8 days in budesonide arm vs. 14.7 days in usual care arm (HR 1.21; 95% CrI, 1.08–1.36)	Key Limitations • Open-label trial • Primary endpoint of time to recovery was based on patient self-report. Interpretation • Inhaled budesonide reduced the time to reported recovery but not the incidence of COVID-19–related hospitalization or death. • The clinical significance of self-reported time to recovery in an open-label study is unclear.
STOIC: Open-Label, Phase 2 RCT of Inhaled Budesonide in Nonhospitalized Adults With Early COVID-19 in the United Kingdom²		
Key Inclusion Criteria • Aged ≥18 years • ≤7 days of COVID-19 symptoms Key Exclusion Criteria • Use of inhaled or systemic glucocorticoids in past 7 days • Known allergy or contraindication to budesonide	Participant Characteristics • Mean age 45 years; 58% women • 9% with CVD; 5% with DM • 95% with positive SARS-CoV-2 RT-PCR result • Median of 3 days from symptom onset to randomization	Key Limitations • Small, open-label trial • Trial was terminated early after statistical analysis determined that additional patients would not alter study outcome.

Methods	Results	Limitations and Interpretation
STOIC: Open-Label, Phase 2 RCT of Inhaled Budesonide in Nonhospitalized Adults With Early COVID-19 in the United Kingdom², continued		
Interventions - Usual care plus inhaled budesonide 800 µg twice daily until symptom resolution (n = 70) - Usual care (n = 69) Primary Endpoint - COVID-19–related urgent care visit, including ED visit or hospitalization Key Secondary Endpoint - Time to clinical recovery	Primary Outcome - COVID-19–related urgent care visit: 1% in budesonide arm vs. 14% in usual care arm (difference in proportion 0.131; 95% CI, 0.043–0.218; <i>P</i> = 0.004) Secondary Outcome - Median time to clinical recovery: 7 days in budesonide arm vs. 8 days in usual care arm	- Secondary endpoint of time to recovery was based on patient self-report. Interpretation - In adult outpatients with mild COVID-19, inhaled budesonide may reduce the need for urgent care, ED visit, or hospitalization. - The clinical significance of self-reported time to recovery in an open-label study is unclear.
Phase 3, Double-Blind, Placebo-Controlled RCT of Inhaled Ciclesonide in Nonhospitalized Patients With COVID-19 in the United States³		
Key Inclusion Criteria - Aged ≥12 years - Positive SARS-CoV-2 molecular or antigen diagnostic test result in previous 72 hours ≥1 symptoms of COVID-19 (i.e., fever, cough, dyspnea) Key Exclusion Criteria - Use of inhaled or intranasal corticosteroid within 14 days of enrollment or systemic corticosteroid within 90 days of enrollment - Unable to use an inhaler Interventions - Ciclesonide MDI 160 µg/actuation, administered as 2 actuations twice daily for 30 days (n = 197) - Placebo MDI twice daily for 30 days (n = 203) Primary Endpoint - Time to alleviation of all COVID-19–related symptoms by Day 30 Key Secondary Endpoints - Alleviation of COVID-19–related symptoms by Day 30	Participant Characteristics - Mean age 43.3 years; 55.3% women; 86.3% White - Mean BMI 29.4 22.3% with HTN; 7.5% with type 2 DM - Higher rates of DM and asthma in ciclesonide arm Primary Outcome - Median of 19 days in both arms for alleviation of all COVID-19–related symptoms (HR 1.08; 95% CI, 0.84–1.38) Secondary Outcomes - Alleviation of COVID-19–related symptoms by Day 30: 70.6% in ciclesonide arm vs. 63.5% in placebo arm (OR 1.28; 95% CI, 0.84–19.7) - ED visit or hospital admission for COVID-19 by Day 30: 1.0% in ciclesonide arm vs. 5.4% in placebo arm (OR 0.18; 95% CI, 0.04–0.85) - Hospital admission or death by Day 30: 1.5% in ciclesonide arm vs. 3.4% in placebo arm (OR 0.45; 95% CI, 0.11–1.84) - No deaths by Day 30 in either arm	Key Limitations - ED visit or hospital admission outcome was based on a small number of events. - Primary endpoint of time to alleviation of all symptoms was based on patient self-report. Interpretation - Inhaled ciclesonide did not reduce the time to reported recovery in nonhospitalized patients with COVID-19. - The robustness of the conclusion that inhaled ciclesonide reduced COVID-19–related ED visits or hospital admissions is uncertain. The small number of events is most likely due to the relatively low rate of comorbidities in the study population.

Methods	Results	Limitations and Interpretation
Phase 3, Double-Blind, Placebo-Controlled RCT of Inhaled Ciclesonide in Nonhospitalized Patients With COVID-19 in the United States³ , continued		
• ED visit or hospital admission for COVID-19 by Day 30 • Hospital admission or death by Day 30		
CONTAIN: Double-Blind RCT of Inhaled Ciclesonide Plus Intranasal Ciclesonide in Nonhospitalized Patients With COVID-19 in Canada⁴		
Key Inclusion Criteria • Aged ≥18 years • Positive SARS-CoV-2 molecular diagnostic test result • ≥1 symptoms of COVID-19 (i.e., fever, cough, shortness of breath) • ≤5 days of COVID-19 symptoms Key Exclusion Criteria • Receiving an inhaled corticosteroid or received a PO or IM corticosteroid within 7 days of enrollment • Unable to use an inhaler • Has only nonrespiratory symptoms • Use of oxygen at home • Vaccinated against COVID-19 Interventions • Ciclesonide MDI 600 µg/actuation plus intranasal ciclesonide 100 µg, both twice daily for 14 days (n = 105) • Saline placebo MDI plus intranasal saline, both twice daily for 14 days (n = 98) Primary Endpoint • Resolution of fever and all respiratory symptoms at Day 7 Key Secondary Endpoints • Resolution of fever and all respiratory symptoms at Day 14 • Hospital admission by Day 14	Participant Characteristics • Median age 35 years; 54% women; 61% White • 20% with comorbidities Primary Outcome • Resolution of fever and all respiratory symptoms at Day 7: 40% in ciclesonide arm vs. 35% in placebo arm (adjusted risk difference 5.5%; 95% CI, -7.8% to 18.8%) Secondary Outcomes • Resolution of fever and all respiratory symptoms at Day 14: 66% in ciclesonide arm vs. 58% in placebo arm (adjusted risk difference 7.5%; 95% CI, -5.9% to 20.8%) • Hospital admission by Day 14: 6% in ciclesonide arm vs. 3% in placebo arm (adjusted risk difference 2.3%; 95% CI, -3.0% to 7.6%)	Key Limitation • Small study with a relatively young, healthy population Interpretation • The use of inhaled ciclesonide plus intranasal ciclesonide did not improve resolution of fever and respiratory symptoms in nonhospitalized patients with COVID-19.

Methods	Results	Limitations and Interpretation
COVERAGE: Open-Label RCT of Inhaled Ciclesonide in Nonhospitalized Adults With COVID-19 in France ⁵		
Key Inclusion Criteria • Aged ≥60 years or aged ≥50 years with comorbidities • Positive SARS-CoV-2 nasopharyngeal RT-PCR result or antigen test result • ≤7 days of COVID-19 symptoms Key Exclusion Criteria • Chronic use of inhaled corticosteroid therapy • Unable to use an inhalation chamber • Ongoing therapy with a potent CYP3A4 inhibitor Interventions • Ciclesonide 160 µg via inhalation chamber, 2 puffs twice daily for 10 days (n = 110) • Vitamin and trace element supplement, 2 capsules PO once or twice daily for 10 days (n = 107) Primary Endpoint • Composite of hospitalization from any cause, need for COVID-19–related oxygen therapy at home, or death by Day 14 Key Secondary Endpoint • Sustained alleviation of symptoms by Day 14	Participant Characteristics • Median age 63 years; 51% women • 72% with ≥1 comorbidities • 14% received ≥1 COVID-19 vaccine doses. Primary Outcome • Composite of hospitalization from any cause, need for COVID-19–related oxygen therapy at home, or death by Day 14: 16% in ciclesonide arm vs. 12% in control arm Secondary Outcome • Sustained alleviation of symptoms by Day 14: 54% in ciclesonide arm vs. 57% in control arm	Key Limitation • Small, open-label study Interpretation • In adult outpatients with mild COVID-19, inhaled ciclesonide did not reduce the proportion of patients who died, were hospitalized, or required COVID-19–related oxygen therapy at home.

Methods	Results	Limitations and Interpretation
ACTIV-6: Decentralized, Placebo-Controlled, Platform RCT of Inhaled Fluticasone in Outpatients With COVID-19 in the United States⁶		
Key Inclusion Criteria • Aged ≥30 years • Positive SARS-CoV-2 nasopharyngeal RT-PCR result or antigen test result • ≤7 days of ≥2 COVID-19 symptoms Key Exclusion Criterion • Use of inhaled or systemic corticosteroids in preceding 30 days Interventions • Inhaled fluticasone 200 µg once daily for 14 days (n = 656) • Matching inhaled placebo (n = 350) or placebo from a different study (n = 271) Primary Endpoint • Time to sustained recovery (i.e., the last of 3 consecutive days without symptoms) Key Secondary Endpoints • Hospitalization or death by Day 28 • Urgent care visit, ED visit, or hospitalization by Day 28 • Number of days unwell with ongoing symptoms	Participant Characteristics • Median age 45 years; 63% women • 39% with BMI >30; 26% with HTN • 65% received ≥2 COVID-19 vaccine doses. Primary Outcome • No difference between arms in time to sustained recovery (HR 1.01; 95% CrI, 0.91–1.12) Secondary Outcomes • Hospitalization or death by Day 28: 0.5% in fluticasone arm vs. 0.5% in placebo arm • Urgent care visit, ED visit, or hospitalization by Day 28: 3.7% in fluticasone arm vs. in 2.1% placebo arm (HR 1.9; 95% CrI, 0.8–3.5) • Mean number of days unwell with ongoing symptoms: 11.2 in fluticasone arm vs. 11.3 in placebo arm	Key Limitations • Low numbers of some clinical endpoints limited the ability to assess the effect of inhaled fluticasone on the key secondary endpoints. • Not all patients in the placebo arm received a matched placebo. Interpretation • In adult outpatients with mild COVID-19, inhaled fluticasone did not reduce the time to sustained symptom recovery or the occurrence of urgent care visits, ED visits, or hospitalizations.

Methods	Results	Limitations and Interpretation
TOGETHER: Placebo-Controlled, Platform RCT of Oral Fluvoxamine and Inhaled Budesonide in Adults With Early-Onset COVID-19 in Brazil⁷		
Key Inclusion Criteria • Aged ≥50 years or aged ≥18 years with comorbidities • Laboratory-confirmed SARS-CoV-2 infection • ≤7 days of COVID-19 symptoms Key Exclusion Criteria • Use of an SSRI • Severe mental illness • Cirrhosis, recent seizures, or severe ventricular cardiac arrhythmia Interventions • Fluvoxamine 100 mg PO twice daily plus inhaled budesonide 800 mcg twice daily for 10 days (n = 738) • Placebo (n = 738; route, dosing frequency, and duration may have differed from fluvoxamine arm) Primary Endpoint • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints • Hospitalization by Day 28 • Health care attendance by Day 28 • Any ED visit by Day 28 • Occurrence of treatment-emergent AEs	Participant Characteristics • Median age 51 years; 61% women • 42% with BMI >30 • 44% with HTN; 68% with ≥2 comorbidities • 94% received ≥2 COVID-19 vaccine doses. Primary Outcome • Composite of ED observation >6 hours or hospitalization by Day 28: 1.8% in fluvoxamine and budesonide arm vs. 3.7% in placebo arm (relative risk 0.50; 95% CrI, 0.25–0.92) Secondary Outcomes • Hospitalization by Day 28: 0.9% in fluvoxamine plus budesonide arm vs. 1.1% in placebo arm • Health care attendance by Day 28: 2.6% in fluvoxamine plus budesonide arm vs. 4.1% in placebo arm (relative risk 0.64; 95% CrI, 0.36–1.11) • Any ED visit by Day 28: 12.2% in fluvoxamine plus budesonide arm vs. 13.0% in placebo arm • Occurrence of treatment-emergent AEs: 17.6% in fluvoxamine plus budesonide arm vs. 12.9% in placebo arm (relative risk 1.37; 95% CrI, 1.07–1.75) • Most AEs were grade 2 events.	Key Limitation • Multiple investigational treatments or placebos were evaluated simultaneously. Not all patients in the placebo arm received a matched placebo. Interpretation • Adult outpatients with mild COVID-19 who received a combination of fluvoxamine and inhaled budesonide had fewer ED observations >6 hours or hospitalizations for COVID-19 by Day 28 than those who received placebo. • The use of fluvoxamine plus inhaled budesonide did not reduce the risk of hospitalization, health care attendance, or ED visits. • It is difficult to define the clinical relevance of the >6-hour ED observation endpoint and apply it to practice settings in different countries. • More AEs occurred with the use of fluvoxamine plus inhaled budesonide than with placebo.

Key: AE = adverse event; BMI = body mass index; CVD = cardiovascular disease; CYP = cytochrome P450; DM = diabetes mellitus; ED = emergency department; HTN = hypertension; IM = intramuscular; MDI = metered dose inhaler; the Panel = the COVID-19 Treatment Guidelines Panel; PCR = polymerase chain reaction; PO = oral; RCT = randomized controlled trial; RT-PCR = reverse transcription polymerase chain reaction; SSRI = selective serotonin reuptake inhibitor

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Interleukin-6 Inhibitors

Last Updated: February 29, 2024

Interleukin (IL)-6 is a pleiotropic, proinflammatory cytokine produced by a variety of cell types, including lymphocytes, monocytes, and fibroblasts. Infection by SARS-CoV induces a dose-dependent production of IL-6 from bronchial epithelial cells.¹ COVID-19–associated systemic inflammation and hypoxemic respiratory failure can be associated with heightened cytokine release, as indicated by elevated levels of IL-6, C-reactive protein (CRP), D-dimer, and ferritin in the blood.²⁻⁴

The recombinant humanized anti-IL-6 receptor monoclonal antibodies (mAbs) tocilizumab and sarilumab have been evaluated in hospitalized patients with COVID-19 who had systemic inflammation. Tocilizumab is approved by the Food and Drug Administration (FDA) for use in patients with rheumatologic disorders and in patients with cytokine release syndrome induced by chimeric antigen receptor T cell therapy. On December 21, 2022, the FDA approved the use of intravenous (IV) tocilizumab for the treatment of COVID-19 in hospitalized adults who are receiving systemic corticosteroids and require supplemental oxygen, noninvasive ventilation (NIV), mechanical ventilation, or extracorporeal membrane oxygenation (ECMO).^{5,6} Tocilizumab can be administered as an IV infusion or a subcutaneous (SUBQ) injection; however, only the IV formulation of tocilizumab should be used for the treatment of COVID-19.

Sarilumab is approved by the FDA for use in patients with rheumatoid arthritis. It is available as a SUBQ formulation and is not approved for the treatment of cytokine release syndrome or COVID-19.

Recommendations

- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations on the use of tocilizumab in combination with dexamethasone in hospitalized patients who require conventional oxygen, high-flow nasal cannula (HFNC) oxygen, NIV, or mechanical ventilation.

Additional Considerations

- If none of the recommended immunomodulatory therapies discussed in [Therapeutic Management of Hospitalized Adults With COVID-19](#) are available or feasible to use, IV sarilumab can be used in combination with dexamethasone (**CIIa**). Sarilumab is only commercially available as a SUBQ injection; see [Table 5e](#) for information regarding the preparation of an IV infusion using the SUBQ product.
- Tocilizumab and sarilumab should be used with caution in patients with COVID-19 who belong to populations that have not been adequately represented in clinical trials. This includes patients who are significantly immunosuppressed, such as those who have recently received other biologic immunomodulators, and patients with any of the following:
 - Alanine transaminase levels >5 times the upper limit of normal
 - A high risk for gastrointestinal perforation
 - An uncontrolled serious bacterial, fungal, or non-SARS-CoV-2 viral infection
 - Absolute neutrophil counts <500 cells/ μ L
 - Platelet counts <50,000 cells/ μ L

- Known hypersensitivity to tocilizumab or sarilumab
- In both the REMAP-CAP and RECOVERY trials, 29% of patients received a second dose of tocilizumab at the discretion of their treating physician.^{7,8} However, there is insufficient evidence for the Panel to recommend either for or against the use of a second dose of tocilizumab for the treatment of COVID-19.
- Cases of severe and disseminated strongyloidiasis have been reported in patients with COVID-19 during treatment with tocilizumab and corticosteroids.^{9,10} Many clinicians would initiate empiric treatment for strongyloidiasis (e.g., with the antiparasitic drug ivermectin), with or without serologic testing, in patients from areas where *Strongyloides* is endemic (i.e., tropical, subtropical, or warm temperate areas).¹¹

Rationale

The results of the RECOVERY and REMAP-CAP trials provide consistent evidence that tocilizumab, when administered as a second immunomodulatory agent in combination with a corticosteroid, offers a survival benefit in certain patients with COVID-19.^{7,8} Specifically, the patients who may benefit are those who are severely ill and require HFNC oxygen or NIV, those who are rapidly deteriorating and have increasing oxygen needs, or those who are having a significant inflammatory response. In the REMAP-CAP trial, a long-term follow-up through 180 days confirmed that treatment with an anti-IL-6 receptor mAb improved survival among patients with severe to critical COVID-19.¹² However, the Panel found it challenging to determine which patients with COVID-19 who are receiving low-flow oxygen would benefit from receiving tocilizumab or sarilumab plus dexamethasone.

If none of the recommended immunomodulatory therapies are available or feasible to use, sarilumab may be used because the REMAP-CAP trial demonstrated that the use of tocilizumab and the use of sarilumab improved survival and reduced the duration of organ support.^{12,13} Sarilumab is currently only approved for use in the United States as a SUBQ injection.

Monitoring and Adverse Effects

The primary laboratory abnormalities reported in people receiving tocilizumab and sarilumab are elevated liver enzyme levels that appear to be dose dependent and rare occurrences of neutropenia or thrombocytopenia.^{5,14} In randomized trials, no excess secondary infections were seen among patients who received tocilizumab plus corticosteroids when compared with control patients.¹⁵⁻¹⁷ Additional adverse effects of tocilizumab, such as serious infections (e.g., tuberculosis, bacterial or fungal infections) and bowel perforation, have been reported. These adverse effects may occur with long-term use of sarilumab as well.

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance regarding the use of tocilizumab during pregnancy and lactation.

There are insufficient data to determine whether the use of sarilumab is associated with an increased risk for major birth defects or miscarriage. As pregnancy progresses, mAbs are actively transported across the placenta (with the greatest transfer occurring during the third trimester), and immune responses in the exposed fetus may be affected.

Considerations in Children

See [Therapeutic Management of Hospitalized Children With COVID-19](#) for the Panel's

recommendations regarding the use of tocilizumab and sarilumab in children.

Drug Availability

On December 21, 2022, the FDA approved the use of IV tocilizumab for the treatment of COVID-19 in hospitalized adults who are receiving systemic corticosteroids and require supplemental oxygen, NIV, mechanical ventilation, or ECMO.⁵ In June 2021, the FDA issued an Emergency Use Authorization for the use of tocilizumab in combination with corticosteroids for the treatment of COVID-19 in hospitalized children aged ≥ 2 years who require supplemental oxygen, NIV, mechanical ventilation, or ECMO.⁶ If a patient's clinical signs or symptoms worsen or do not improve after the first dose of tocilizumab, 1 additional IV infusion of tocilizumab may be administered at least 8 hours after the initial infusion.

The IV administration of sarilumab is not approved by the FDA, but in clinical trials, single SUBQ sarilumab doses were modified to enable IV administration. See [Table 5e](#) for additional details.

Clinical Data

Two large randomized controlled trials, REMAP-CAP and RECOVERY, evaluated the use of tocilizumab in combination with standard of care corticosteroids.^{7,8} Both studies reported a statistically significant survival benefit from the use of tocilizumab in certain patients, including patients who exhibited rapid respiratory decompensation associated with an inflammatory response.

REMAP-CAP enrolled critically ill patients who were within 24 hours of receiving respiratory support in an intensive care unit.⁷ At baseline, 29% of these patients were receiving HFNC oxygen, 42% were receiving NIV, and 29% were receiving mechanical ventilation. The patients were randomized to receive open-label tocilizumab or usual care. In-hospital mortality was 28% in the tocilizumab arm and 36% in the usual care arm. A follow-up analysis confirmed these findings.¹² At 180 days, mortality was 36% in the tocilizumab arm and 40% in the usual care arm.

The RECOVERY trial enrolled hospitalized patients with COVID-19 into an open-label platform trial that included several treatment options.⁸ A subset of all trial participants who had hypoxemia and CRP levels ≥ 75 mg/L were offered enrollment into a second randomization that compared the use of tocilizumab to usual care. In this subgroup, the 28-day mortality was 31% in the tocilizumab arm and 35% in the usual care arm.

In contrast to the REMAP-CAP and RECOVERY trials, other randomized trials, including the REMDACTA and EMPACTA trials, found that tocilizumab did not reduce all-cause mortality.^{18,19} In those trials, $>80\%$ of participants received corticosteroids as part of standard care, and most participants in the REMDACTA trial required NIV or HFNC oxygen.¹⁸

In the REMAP-CAP trial, the efficacy results for sarilumab were similar to those for tocilizumab.¹³ When compared with patients in the standard of care arm ($n = 406$), patients in the sarilumab arm ($n = 485$) had more organ support-free days (OR 1.50; 95% CrI, 1.13–2.00) and a greater likelihood of survival while hospitalized (OR 1.51; 95% CrI, 1.06–2.20). In-hospital mortality was 33% for the sarilumab arm and 37% for the standard of care arm, and mortality at 180 days was 33% for the sarilumab arm and 40% for the standard of care arm.¹² A notable limitation to the sarilumab findings in the REMAP-CAP trial is that patients in the standard of care arm were enrolled earlier in the pandemic than those in the sarilumab arm.¹³ Randomization closed in November 2020 for the standard of care arm and continued through April 2021 for the sarilumab arm.

An adaptive, multinational, double-blind, randomized (2:2:1) trial compared the efficacy and safety of sarilumab 400 mg IV and sarilumab 200 mg IV to placebo in hospitalized patients with COVID-19.²⁰

This trial did not show a clinical benefit of sarilumab in hospitalized patients who were receiving supplemental oxygen.

A similar adaptive study conducted in the United States in patients with severe and critical COVID-19 also failed to show a benefit of sarilumab.²¹ In this placebo-controlled trial, mortality by Day 22 was reduced among the sarilumab recipients with critical COVID-19 pneumonia who required mechanical ventilation and received corticosteroids at baseline. However, due to the small sample size, this result was not statistically significant.

The clinical data on the use of tocilizumab and sarilumab for the treatment of COVID-19, including data from several randomized trials and large observational studies, are summarized in [Table 5c](#).

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Table 5c. Interleukin-6 Inhibitors: Selected Clinical Trial Data

Last Updated February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for IL-6 inhibitors. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of Tocilizumab and Usual Care in Hospitalized Adults With COVID-19 in the United Kingdom¹		
Key Inclusion Criteria - Evidence of COVID-19 progression ≤ 21 days after initial randomization to an intervention within the RECOVERY protocol, defined as: - SpO₂ <92% on room air or receipt of supplemental oxygen; and - CRP ≥ 75 mg/L Key Exclusion Criterion - Presence of non-SARS-CoV-2 infection Interventions 1 weight-based dose of tocilizumab (maximum 800 mg) with possible second dose (n = 2,022) - Usual care (n = 2,094) Primary Endpoint 28-day all-cause mortality Key Secondary Endpoints - Time to hospital discharge within 28 days - Among those not on MV at baseline, death or receipt of MV (including ECMO) within 28 days	Participant Characteristics - Mean age 64 years; 67% men; 76% White 95% with PCR-confirmed SARS-CoV-2 infection - At baseline: 45% on conventional oxygen 41% on HFNC oxygen or NIV 14% on MV 82% receiving corticosteroids Primary Outcomes 28-day all-cause mortality: 31% in tocilizumab arm vs. 35% in usual care arm (rate ratio 0.85; 95% CI, 0.76–0.94; $P = 0.003$) 28-day all-cause mortality among those who required MV at baseline: 49% in tocilizumab arm vs. 51% in usual care arm (risk ratio 0.93; 95% CI, 0.74–1.18) Secondary Outcomes - Proportion discharged from hospital within 28 days: 57% in tocilizumab arm vs. 50% in usual care arm (rate ratio 1.22; 95% CI, 1.12–1.33; $P < 0.0001$) - Median time to hospital discharge: 19 days in tocilizumab arm vs. 28 days in usual care arm - Proportion not on MV at baseline who died or required MV within 28 days: 35% in tocilizumab arm vs. 42% in usual care arm (rate ratio 0.84; 95% CI, 0.77–0.92; $P < 0.0001$)	Key Limitations - Arbitrary CRP ≥ 75 mg/L cutoff for enrollment - Difficult to define exact subset of patients in RECOVERY cohort who were subsequently selected for secondary randomization/tocilizumab trial Interpretation - Among hospitalized patients with COVID-19, hypoxemia, and elevated CRP levels, the use of tocilizumab was associated with a reduction in all-cause mortality and a shorter time to hospital discharge.

Methods	Results	Limitations and Interpretation
REMAP-CAP: Open-Label, Adaptive-Platform RCT of Tocilizumab and Sarilumab in Adults With COVID-19 in 21 Countries in Europe and North America^{2,4}		
Key Inclusion Criteria • ICU admission • Suspected or laboratory-confirmed SARS-CoV-2 infection • Receipt of MV, NIV, or cardiovascular support Key Exclusion Criteria • >24 hours after ICU admission • Death imminent • Immunosuppression • ALT >5 times ULN Interventions • SOC plus 1 of the following (drug selection based on provider preference, availability, or adaptive probability): • 1 dose of tocilizumab 8 mg/kg IV with possible second dose in 12–24 hours (n = 952) • Single dose of sarilumab 400 mg IV (n = 485) • SOC alone (n = 406) Primary Endpoint • Composite of in-hospital mortality or number of organ support-free days by Day 21, as measured by an OS Key Secondary Endpoints • In-hospital survival • All-cause mortality at 180 days	Participant Characteristics • Mean age 60 years; 69% men; 75% White • 86% with PCR-confirmed SARS-CoV-2 infection • Median of 14 hours between ICU admission and enrollment • At baseline: • 68% on HFNC oxygen or NIV • 32% on MV • Receiving corticosteroids: 67% in SOC arm, 82% in tocilizumab arm, 89% in sarilumab arm Primary Outcomes <i>Tocilizumab vs. SOC</i> • Median number of organ support-free days: 7 in tocilizumab arm vs. 0 in SOC arm • Improvement in OS score by Day 21 for composite outcome was more likely in tocilizumab arm (median aOR 1.46; 95% CrI, 1.13–1.87). • Highest CRP tercile: aOR 1.87 (95% CrI, 1.35–2.59) • Outcomes were consistent across subgroups according to oxygen requirement at baseline. <i>Sarilumab vs. SOC</i> • Median number of organ support-free days: 9 in sarilumab arm vs. 0 in SOC arm • Improvement in OS score by Day 21 for composite outcome was more likely in sarilumab arm (median aOR 1.50; 95% CrI, 1.13–2.00). • Highest CRP tercile: aOR 1.85 (95% CrI, 1.24–2.69) • Outcomes were consistent across subgroups according to oxygen requirement at baseline. Secondary Outcomes <i>Tocilizumab vs. SOC</i> • In-hospital survival: 66% in tocilizumab arm vs. 63% in SOC arm (aOR 1.42; 95% CrI, 1.05–1.93)	Key Limitation • The SOC arm closed in November 2020, after which patients were randomized to active drug arms only; enrollment in the tocilizumab and sarilumab arms was partially nonconcurrent with the SOC arm. Although comparisons to the SOC arm were adjusted for this time period, there is a possibility of bias. Interpretation • Among patients with respiratory failure who were within 24 hours of ICU admission, the tocilizumab and sarilumab arms had higher rates of in-hospital survival and shorter durations of organ support than the SOC arm. • The use of IL-6 receptor antagonists reduced all-cause mortality at 180 days. • The treatment effect appeared to be strongest in the highest CRP tercile. • Tocilizumab and sarilumab were similarly effective in these patients.

Methods	Results	Limitations and Interpretation
REMAP-CAP: Open-Label, Adaptive-Platform RCT of Tocilizumab and Sarilumab in Adults With COVID-19 in 21 Countries in Europe and North America^{2,4}, cont'd		
	- All-cause mortality at 180 days: 36% in tocilizumab arm vs. 40% in SOC arm (aHR 0.76; 95% CrI, 0.61–0.93) <i>Sarilumab vs. SOC</i> - In-hospital survival: 67% in sarilumab arm vs. 63% in SOC arm (aOR 1.51; 95% CrI, 1.06–2.20) - All-cause mortality at 180 days: 33% in sarilumab arm vs. 40% in SOC arm (aHR 0.72; 95% CrI, 0.56–0.91) <i>Pooled Tocilizumab and Sarilumab Arms vs. SOC Arm</i> - All-cause mortality at 180 days: 35% in pooled arms vs. 40% in SOC arm (aHR 0.74; 95% CrI, 0.61–0.90)	
COVACTA: Double-Blind RCT of Tocilizumab in Hospitalized Adults With COVID-19 in 9 Countries in Europe and North America⁵		
Key Inclusion Criteria - PCR-confirmed SARS-CoV-2 infection - Hypoxemia - Bilateral chest infiltrates Key Exclusion Criteria - Death imminent - Presence of active non-SARS-CoV-2 infection Interventions 1 dose of tocilizumab 8 mg/kg with possible second dose, plus SOC (n = 294) - Placebo plus SOC (n = 144) Primary Endpoint - Clinical status at Day 28, as measured by an OS Key Secondary Endpoints - Time to hospital discharge - ICU LOS - Mortality by Day 28	Participant Characteristics - Mean age 61 years; 70% men; 58% White 30% on HFNC oxygen or NIV 38% on MV 25% with multiorgan failure - Received corticosteroids at entry or during follow-up: 36% in tocilizumab arm vs. 55% in placebo arm Primary Outcome - No significant difference between arms in clinical status at Day 28 ($P = 0.31$) Secondary Outcomes - Median time to hospital discharge: 20 days in tocilizumab arm vs. 28 days in placebo arm (HR 1.35; 95% CI, 1.02–1.79) - Median ICU LOS: 9.8 days in tocilizumab arm vs. 15.5 days in placebo arm (difference 5.8 days; 95% CI, -15.0 to 2.9) - Mortality by Day 28: 20% in tocilizumab arm vs. 19% in placebo arm ($P = 0.94$)	Key Limitations - Modest power to detect differences in clinical status at Day 28 - More patients received corticosteroids in placebo arm than in tocilizumab arm. Interpretation - There was no difference between the tocilizumab and placebo recipients in clinical status at Day 28 or survival. - The median time to hospital discharge was significantly shorter in the tocilizumab arm than in the placebo arm. - Although the result was not statistically significant, the tocilizumab arm had a shorter ICU LOS than the placebo arm.

Methods	Results	Limitations and Interpretation
EMPACTA: Double-Blind RCT of Tocilizumab in Hospitalized Adults With COVID-19 in 6 Countries in North America, South America, and Africa⁶		
Key Inclusion Criteria • PCR-confirmed SARS-CoV-2 infection • COVID-19 pneumonia Key Exclusion Criteria • Death imminent • Receiving NIV or MV Interventions • 1 dose of tocilizumab 8 mg/kg with possible second dose, plus SOC (n = 249) • Placebo plus SOC (n = 128) Primary Endpoint • Progression to MV, ECMO, or death by Day 28 Key Secondary Endpoints • Time to hospital discharge or readiness for discharge, as measured by an OS • All-cause mortality by Day 28	Participant Characteristics • Mean age 56 years; 59% men; 56% Hispanic/Latinx, 15% Black/African American, 13% American Indian/Alaska Native • 84% with elevated CRP • Concomitant medications: • Corticosteroids: 80% in tocilizumab arm vs. 88% in placebo arm • RDV: 53% in tocilizumab arm vs. 59% in placebo arm Primary Outcome • Proportion who progressed to MV, ECMO, or death by Day 28: 12% in tocilizumab arm vs. 19% in placebo arm (HR 0.56; 95% CI, 0.33–0.97; <i>P</i> = 0.04) Secondary Outcomes • Median time to hospital discharge or readiness for discharge: 6.0 days in tocilizumab arm vs. 7.5 days in placebo arm (HR 1.16; 95% CI, 0.91–1.48) • All-cause mortality by Day 28: 10.4% in tocilizumab arm vs. 8.6% in placebo arm (95% CI, -5.2 to 7.8)	Key Limitation • Moderate sample size Interpretation • In patients with COVID-19 pneumonia, tocilizumab reduced the likelihood of progression to MV, ECMO, or death by Day 28 but did not reduce all-cause mortality by Day 28.

Methods	Results	Limitations and Interpretation
BACC Bay: Double-Blind RCT of Tocilizumab in Hospitalized Adults With COVID-19 in the United States⁷		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection ≥2 of the following conditions: - Fever >38°C - Pulmonary infiltrates - Need for supplemental oxygen ≥1 of the following laboratory criteria: - CRP ≥50 mg/L - D-dimer >1,000 ng/mL - LDH ≥250 U/L - Ferritin >500 ng/mL Key Exclusion Criteria - Receipt of supplemental oxygen at rate >10 L/min - Recent use of biologic agents or small-molecule immunosuppressive therapy - Receipt of immunosuppressive therapy that increased risk for infection Interventions - Tocilizumab 8 mg/kg plus usual care (n = 161) - Placebo plus usual care (n = 81) Primary Endpoint - Progression to MV or death by Day 28 Key Secondary Endpoints - Clinical worsening of disease by Day 28, as measured by an OS - Discontinuation of supplemental oxygen among patients receiving it at baseline	Participant Characteristics - Median age 60 years; 58% men; 45% Hispanic/Latinx, 43% White 50% with BMI ≥30; 49% with HTN; 31% with DM 80% receiving oxygen ≤6 L/min; 4% on HFNC oxygen; 16% receiving no supplemental oxygen - Concomitant medications: - Corticosteroids: 11% in tocilizumab arm vs. 6% in placebo arm - RDV: 33% in tocilizumab arm vs. 29% in placebo arm Primary Outcome - Progression to MV or death by Day 28: 11% in tocilizumab arm vs. 12% in placebo arm (HR 0.83; 95% CI, 0.38–1.81; <i>P</i> = 0.64) Secondary Outcomes - Proportion with clinical worsening of disease by Day 28: 19% in tocilizumab arm vs. 17% in placebo arm (HR 1.11; 95% CI, 0.59–2.10; <i>P</i> = 0.73) - Median time to discontinuation of supplemental oxygen: 5.0 days in tocilizumab arm vs. 4.9 days in placebo arm (<i>P</i> = 0.69)	Key Limitations - Wide confidence intervals due to small sample size and low event rates - Few patients received RDV or corticosteroids. Interpretation - The use of tocilizumab did not prevent MV or death, reduce the risk of clinical worsening, or reduce the time to discontinuation of oxygen. This could be due to the low rate of concomitant corticosteroid use among the study participants.

Methods	Results	Limitations and Interpretation
Double-Blind RCT of Sarilumab in Hospitalized Adults With Severe to Critical COVID-19 in 11 Countries in Europe, North America, South America, and Asia⁸		
Key Inclusion Criteria • COVID-19 pneumonia • Need for supplemental oxygen or intensive care Key Exclusion Criteria • Low probability of surviving or remaining at study site • Dysfunction of ≥ 2 organ systems and need for ECMO or renal replacement therapy Interventions • Sarilumab 200 mg IV (n = 159) • Sarilumab 400 mg IV (n = 173) • Placebo (n = 84) Primary Endpoint • Time to clinical improvement of ≥ 2 points on a 7-point OS Key Secondary Endpoint • Survival to Day 29	Participant Characteristics • Median age 59 years; 63% men; 77% White, 36% Hispanic/Latinx • 39% on HFNC oxygen, MV, or NIV • 42% with BMI ≥ 30; 43% with HTN; 26% with type 2 DM • 20% received systemic corticosteroids before receiving intervention; 63% received ≥ 1 doses of corticosteroids during the study. Primary Outcomes • Median time to clinical improvement: 10 days in sarilumab 200 mg arm vs. 10 days in sarilumab 400 mg arm vs. 12 days in placebo arm • Sarilumab 200 mg arm vs. placebo arm: HR 1.03; 95% CI, 0.75–1.40; $P = 0.96$ • Sarilumab 400 mg arm vs. placebo arm: HR 1.14; 95% CI, 0.84–1.54; $P = 0.34$ Secondary Outcome • Survival to Day 29: 92% in placebo arm; 90% in sarilumab 200 mg arm ($P = 0.63$ vs. placebo); 92% in sarilumab 400 mg arm ($P = 0.85$ vs. placebo)	Key Limitation • Moderate sample size Interpretation • The use of sarilumab did not reduce mortality or time to clinical improvement in hospitalized adults with COVID-19.

Methods	Results	Limitations and Interpretation
REMDACTA: Double-Blind RCT of Tocilizumab and Remdesivir in Hospitalized Patients With Severe COVID-19 Pneumonia in Brazil, Russia, Spain, and the United States⁹		
Key Inclusion Criteria • Aged ≥12 years • PCR-confirmed SARS-CoV-2 infection • Hospitalized with pneumonia confirmed by CXR or CT scan • Required supplemental oxygen >6 L/min Key Exclusion Criteria • eGFR <30 mL/min • ALT or AST >5 times ULN • Presence of non-SARS-CoV-2 infection • Treatment with antivirals, CCP, CQ, HCQ, or JAK inhibitors Interventions • Up to 10 days of RDV plus: • Tocilizumab 8 mg/kg IV with possible second dose in 8–24 hours (n = 434) • Placebo (n = 215) Primary Endpoint • Time to hospital discharge or readiness for discharge by Day 28 Key Secondary Endpoints • Time to MV or death by Day 28 • Clinical status at Day 14, as measured by an OS • Time to death by Day 28	Participant Characteristics • Mean age 59 years; 40% in tocilizumab arm and 34% in placebo arm aged ≥65 years; 63% men; 67% White • Respiratory support: • HFNC oxygen or NIV: 78% in tocilizumab arm vs. 83% in placebo arm • MV or ECMO: 15% in tocilizumab arm vs. 11% in placebo arm • Corticosteroid use: • At baseline: 83% in tocilizumab arm vs. 86% in placebo arm • During trial: 88% in each arm Primary Outcome • Median time to hospital discharge or readiness for discharge by Day 28: 14 days in each arm (HR 0.97; 95% CI, 0.78–1.19; <i>P</i> = 0.74) Secondary Outcomes • No difference between arms in: • Proportion who required MV or died by Day 28: 29% in each arm; time to death was not evaluable (HR 0.98; 95% CI, 0.72–1.34; <i>P</i> = 0.90). • Mean OS score for clinical status at Day 14: 2.8 in tocilizumab arm vs. 2.9 in placebo arm (<i>P</i> = 0.72) • Proportion who died by Day 28: 18% in tocilizumab arm vs. 20% in placebo arm; time to death was not evaluable (HR 0.95; 95% CI, 0.65–1.39; <i>P</i> = 0.79).	Key Limitations • During the trial, primary outcome changed from clinical status at Day 28 to time to hospital discharge or readiness for discharge by Day 28. • Imbalances in patient characteristics at baseline between arms • Possible underrepresentation of patients with rapidly progressive disease Interpretation • Compared with placebo plus RDV, tocilizumab plus RDV did not shorten the time to discharge or readiness for discharge in patients with severe COVID-19 pneumonia. • In these patients, the use of tocilizumab plus RDV did not reduce mortality when compared with RDV alone.

Key: ALT = alanine transaminase; AST = aspartate aminotransferase; BMI = body mass index; CCP = COVID-19 convalescent plasma; CQ = chloroquine; CRP = C-reactive protein; CT = computed tomography; CXR = chest X-ray; DM = diabetes mellitus; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; HCQ = hydroxychloroquine; HFNC = high-flow nasal cannula; HTN = hypertension; ICU = intensive care unit; IL = interleukin; IV = intravenous; JAK = Janus kinase; LDH = lactate dehydrogenase; LOS = length of stay; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PCR = polymerase chain reaction; RCT = randomized controlled trial; RDV = remdesivir; SOC = standard of care; SpO₂ = oxygen saturation; ULN = upper limit of normal

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Janus Kinase Inhibitors

Last Updated: February 29, 2024

Janus kinase (JAK) inhibitors interfere with phosphorylation of the signal transducer and activator of transcription (STAT) proteins^{1,2} that are involved in vital cellular functions, including signaling, growth, and survival. JAK inhibitors are used to treat COVID-19 because they can prevent phosphorylation of key proteins involved in the signal transduction that leads to immune activation and inflammation (e.g., the cellular response to proinflammatory cytokines such as interleukin [IL]-6).³ Several JAK inhibitors are available for clinical use, but only baricitinib and tofacitinib have been studied for the treatment of COVID-19.

In May 2022, the Food and Drug Administration (FDA) approved the use of baricitinib for the treatment of COVID-19 in hospitalized adults who require supplemental oxygen, noninvasive ventilation (NIV), mechanical ventilation, or extracorporeal membrane oxygenation.⁴ Baricitinib is an oral JAK inhibitor that is selective for JAK1 and JAK2. It can modulate downstream inflammatory responses via JAK1/JAK2 inhibition and has exhibited dose-dependent inhibition of IL-6–induced STAT3 phosphorylation.⁵

Tofacitinib is a JAK inhibitor that is predominantly selective for JAK1 and JAK3. However, it also has modest activity against JAK2; thus, it can block signaling from gamma-chain cytokines (e.g., IL-2, IL-4) and glycoprotein 130 proteins (e.g., IL-6, IL-11, interferons). It is an oral agent that was first approved by the FDA for the treatment of rheumatoid arthritis and has been shown to decrease levels of IL-6 in patients with this disease.⁶ Tofacitinib is also approved by the FDA for the treatment of psoriatic arthritis, juvenile idiopathic arthritis, and ulcerative colitis.⁷

Recommendation

- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations on the use of baricitinib in combination with dexamethasone in hospitalized patients who require conventional oxygen, high-flow nasal cannula oxygen, NIV, or mechanical ventilation.

Additional Consideration

- If none of the recommended immunomodulatory therapies discussed in [Therapeutic Management of Hospitalized Adults With COVID-19](#) are available or feasible to use, **oral tofacitinib** can be used in combination with dexamethasone (**CIIa**).

Rationale

Several large randomized controlled trials have demonstrated that some patients who require supplemental oxygen and most patients who require oxygen through a high-flow device, NIV, or mechanical ventilation benefit from the use of dexamethasone in combination with a JAK inhibitor.

In the RECOVERY trial, baricitinib was associated with a survival benefit among hospitalized patients, with a treatment effect that was most pronounced among patients who were receiving NIV or oxygen supplementation through a high-flow device.⁸ The COV-BARRIER trial also demonstrated a survival benefit for baricitinib that was most pronounced among patients who were receiving high-flow oxygen or NIV.⁹ In the addendum to the COV-BARRIER trial, the benefit extended to patients who were receiving mechanical ventilation.¹⁰ Data from the ACTT-2¹¹ and ACTT-4¹² trials support the overall

safety of baricitinib and the potential for benefit, but neither trial studied baricitinib use in combination with dexamethasone as standard care.

The STOP-COVID study examined the use of tofacitinib in people with COVID-19 pneumonia who were not receiving NIV, mechanical ventilation, or extracorporeal membrane oxygenation at the time of enrollment.¹³ The study demonstrated a survival benefit in patients who received tofacitinib; nearly all of these patients also received corticosteroids. If none of the other recommended immunomodulatory therapies for the treatment of COVID-19 are available or feasible to use, tofacitinib may be used as a substitute based on the findings from the STOP-COVID study.

The clinical trial data on the use of baricitinib and tofacitinib in patients with COVID-19 are summarized in [Table 5d](#).

Monitoring, Adverse Effects, and Drug-Drug Interactions

The FDA reviewed the data from a large, randomized, safety clinical trial that compared the use of tofacitinib to tumor necrosis factor inhibitors in people with rheumatoid arthritis over 4 years.¹⁴ The FDA review reported that the use of tofacitinib was associated with a higher incidence of serious adverse events, including heart attack, stroke, cancer, blood clots, and death. Therefore, the FDA now requires new and updated warnings for drugs in the JAK inhibitor class, including baricitinib and tofacitinib. The data from randomized trials that have evaluated the safety of short-term use of JAK inhibitors in patients with COVID-19 have not revealed significant safety signals, such as thrombosis.^{8,9,11-13} Because JAK inhibitors have immunosuppressive effects, all patients who are receiving either baricitinib or tofacitinib should be monitored for new infections.

No clinically significant drug-drug interactions are expected between baricitinib and concomitant drugs. Tofacitinib is a cytochrome P450 (CYP) 3A4 substrate. Dose modifications are required when the drug is coadministered with strong CYP3A4 inhibitors, or when it is used with a moderate CYP3A4 inhibitor that is coadministered with a strong CYP2C19 inhibitor. Coadministering tofacitinib with a strong CYP3A4 inducer is not recommended. See [Table 5e](#) for JAK inhibitor drug characteristics and dosing information.

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance regarding the use of baricitinib during pregnancy and lactation. Pregnancy registries provide some outcome data on tofacitinib use during pregnancy for other conditions (e.g., ulcerative colitis, rheumatoid arthritis, psoriasis). Among the cases reported, pregnancy outcomes were similar to those among the general population.¹⁵⁻¹⁷

Considerations in Children

See [Therapeutic Management of Hospitalized Children With COVID-19](#) for the Panel's recommendations regarding the use of baricitinib or tofacitinib in children with COVID-19.

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Table 5d. Janus Kinase Inhibitors: Selected Clinical Trial Data

Last Updated: February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for JAK inhibitors. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of Baricitinib Versus Usual Care in the United Kingdom¹		
Key Inclusion Criterion - Hospitalized with suspected or laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criteria - eGFR <15 mL/min/1.73m² - ANC <500 cells/mm³ - Evidence of active TB Interventions - BAR 4 mg PO daily for 10 days or until hospital discharge, whichever came first (n = 4,148) - SOC (n = 4,008) Primary Endpoint 28-day all-cause mortality Key Secondary Endpoints - Time to discharge from hospital - Composite of MV, ECMO, or death within 28 days	Participant Characteristics - Mean age 58 years; 66% men; 80% White - Median of 9 days of symptoms at enrollment 91% with laboratory-confirmed SARS-CoV-2 infection - At baseline: 42% received ≥1 doses of a COVID-19 vaccine. 95% received corticosteroids; 23% received tocilizumab; 20% received RDV. 6% not on supplemental oxygen; 68% on conventional oxygen; 24% on NIV; 3% on MV Primary Outcome 28-day all-cause mortality: 12% in BAR arm vs. 14% in SOC arm (age-adjusted rate ratio 0.87; 95% CI, 0.77–0.98; <i>P</i> = 0.028) Secondary Outcomes - Hospital discharge within 28 days: 80% in BAR arm vs. 78% in SOC arm (age-adjusted rate ratio 1.10; 95% CI, 1.04–1.15; <i>P</i> = 0.002) - Median time to discharge: 8 days in both arms - Composite of MV, ECMO, or death within 28 days: 16% in BAR arm vs. 17% in SOC arm (age-adjusted risk ratio 0.89; 95% CI, 0.81–0.98; <i>P</i> = 0.016)	Key Limitation - Open-label study Interpretation - In patients who were hospitalized for COVID-19, the use of BAR reduced the risk of death.

Methods	Results	Limitations and Interpretation
COV-BARRIER: Double-Blind, Placebo-Controlled, Randomized Trial of Baricitinib in Hospitalized Adults in 12 Countries in Asia, Europe, North America, and South America²		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - Evidence of pneumonia or active, symptomatic COVID-19 ≥1 elevated inflammatory markers (CRP, D-dimer, LDH, or ferritin) Key Exclusion Criteria - Required MV or ECMO - Receiving immunosuppressants (including high-dose corticosteroids) - Previously received CCP or IVIG - ANC <1,000 cells/μL - ALC <200 cells/μL - ALT or AST >5 times ULN - eGFR <30 mL/min Interventions - BAR 4 mg PO once daily for up to 14 days (n = 764) - Placebo (n = 761) Primary Endpoint - Clinical progression or death by Day 28 Key Secondary Endpoint - Mortality by Day 28	Participant Characteristics - Mean age 58 years; 63% men 79% received corticosteroids; 19% received RDV; 13% received oxygen but no steroids. Primary Outcome - Clinical progression or death by Day 28: 28% in BAR arm vs. 31% in placebo arm (OR 0.85; 95% CI, 0.67–1.08; <i>P</i> = 0.18) Secondary Outcomes - Mortality by Day 28: 8% in BAR arm vs. 13% in placebo arm (HR 0.57; 95% CI, 0.41–0.78; <i>P</i> = 0.0018) - Mortality by Day 28 for those receiving corticosteroids at baseline: 9% in BAR arm vs. 14% in placebo arm (HR 0.63; 95% CI, 0.45–0.89)	Key Limitation - Results from the ACTT-2 trial prompted a protocol amendment that limited enrollment to patients who required oxygen at baseline. Interpretation - Although the primary outcome of clinical progression or death was not significantly different between the arms, treatment with BAR was associated with reduced mortality in hospitalized adults with COVID-19 who were not receiving MV (see the addendum below for results in patients who required MV or ECMO). - For patients who were receiving oxygen but not corticosteroids at baseline, the primary and secondary outcomes were similar to the outcomes for the overall study population.

Methods	Results	Limitations and Interpretation
COV-BARRIER Addendum: Double-Blind, Placebo-Controlled, Randomized Trial of Baricitinib in Hospitalized Adults on Mechanical Ventilation or Extracorporeal Membrane Oxygenation in Argentina, Brazil, Mexico, and the United States³		
Key Inclusion Criteria • Laboratory-confirmed SARS-CoV-2 infection • Evidence of pneumonia or active, symptomatic COVID-19 • ≥ 1 elevated inflammatory markers (CRP, D-dimer, LDH, or ferritin) • Receiving MV or ECMO at baseline Key Exclusion Criteria • Receiving immunosuppressants (including high-dose corticosteroids) • Previously received CCP or IVIG • ANC $< 1,000$ cells/μL • ALC < 200 cells/μL • ALT or AST > 5 times ULN • eGFR < 30 mL/min Interventions • BAR 4 mg PO once daily for up to 14 days (n = 51) • Placebo (n = 50) Key Endpoints • Mortality by Day 28 • Number of ventilator-free days • Duration of hospitalization	Participant Characteristics • Mean age 59 years; 55% men • 86% received corticosteroids; 2% received RDV. Outcomes • Mortality by Day 28: 39% in BAR arm vs. 58% in placebo arm (HR 0.54; 95% CI, 0.31–0.96; $P = 0.030$) • No significant difference between arms in number of ventilator-free days or duration of hospitalization	Key Limitations • Very small sample size • Exploratory analysis • High mortality in placebo arm Interpretation • In critically ill patients with COVID-19 who were receiving MV or ECMO, treatment with BAR decreased mortality.

Methods	Results	Limitations and Interpretation
ACTT-2: Double-Blind, Placebo-Controlled, Randomized Trial of Baricitinib Plus Remdesivir in Hospitalized Adults With COVID-19 in 8 Countries in Europe, North America, and Asia⁴		
Key Inclusion Criteria • Positive SARS-CoV-2 PCR result • Radiographic infiltrates; SpO₂ ≤94% on room air; or required supplemental oxygen, MV, or ECMO Key Exclusion Criteria • Receiving glucocorticoids for COVID-19 indications • ALT or AST >5 times ULN • Impaired renal function Interventions • BAR 4 mg PO once daily for 14 days or until hospital discharge, plus RDV IV for 10 days or until hospital discharge (n = 515) • Placebo PO once daily for 14 days or until hospital discharge, plus RDV IV for 10 days or until hospital discharge (n = 518) Primary Endpoint • Time to recovery by Day 28 Key Secondary Endpoints • Clinical status at Day 15, as measured by an OS • Mortality by Day 28	Participant Characteristics • Mean age 55 years; 63% men; 48% White, 15% Black, 10% Asian • At baseline: • 13% not on supplemental oxygen • 55% on conventional oxygen • 21% on HFNC oxygen or NIV • 11% on MV or ECMO Primary Outcomes • Median time to recovery by Day 28: 7 days in BAR arm vs. 8 days in placebo arm (rate ratio 1.16; 95% CI, 1.01–1.32; <i>P</i> = 0.03) • Median time to recovery for those receiving HFNC oxygen or NIV: 10 days in BAR arm vs. 18 days in placebo arm (rate ratio for recovery 1.51; 95% CI, 1.10–2.08) Secondary Outcomes • Odds of improvement in clinical status at Day 15 were greater in BAR arm than in placebo arm (OR 1.3; 95% CI, 1.0–1.6). • Mortality by Day 28: 5% in BAR arm vs. 8% in placebo arm (HR 0.65; 95% CI, 0.39–1.09)	Key Limitations • Not powered to detect difference in mortality between arms • Steroids not part of SOC Interpretation • Compared with RDV alone, BAR plus RDV reduced recovery time and improved clinical status, particularly for patients who were receiving HFNC oxygen or NIV at baseline.

Methods	Results	Limitations and Interpretation
ACTT-4: Double-Blind, Placebo-Controlled, Randomized Trial of Remdesivir With Baricitinib Versus Dexamethasone for Hospitalized Patients Requiring Supplemental Oxygen in Japan, Mexico, Singapore, South Korea, and the United States⁵		
Key Inclusion Criteria - Hospitalized and required conventional oxygen, HFNC oxygen, or NIV - Laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criterion - Receipt of CCP or >1 dose of DEX 6 mg (or equivalent) or BAR before enrollment Interventions - RDV IV for ≤10 days plus BAR 4 mg PO daily for ≤14 days plus DEX placebo IV (n = 516) - RDV IV for ≤10 days plus BAR placebo PO plus DEX 6 mg IV daily for ≤10 days (n = 494) Primary Endpoint - MV-free survival by Day 29 Key Secondary Endpoints - Clinical status at Day 15, as measured by an OS - Time to recovery Key Safety Endpoints - Occurrence of treatment-related AEs - Occurrence of SAEs	Participant Characteristics - Median age 58 years; 58% men; 58% White, 34% Hispanic/Latinx - At baseline: 85% on low-flow oxygen 15% on HFNC oxygen or NIV - Mean of 8 days of symptoms at enrollment Primary Outcome - MV-free survival by Day 29: 87% in BAR arm vs. 88% in DEX arm (risk difference 0.6%; 95% CI, -3.6% to 4.8%; <i>P</i> = 0.91) Secondary Outcomes - Odds of improvement in clinical status at Day 15 were similar between arms (OR 1.01; 95% CI, 0.80–1.27). - For those on low-flow oxygen at baseline: OR 0.91; 95% CI, 0.70–1.17 - For those on HFNC oxygen or NIV at baseline: OR 1.64; 95% CI, 0.92–2.90 - Median time to recovery: 6 days in BAR arm vs. 5 days in DEX arm (rate ratio 1.04; 95% CI, 0.91–1.19) Safety Outcomes - Occurrence of treatment-related AEs: 4% in BAR arm vs. 10% in DEX arm (risk difference 6.0%; 95% CI, 2.8%–9.3%; <i>P</i> = 0.0004) - Occurrence of SAEs: 28% in BAR arm vs. 36% in DEX arm (risk difference 7.7%; 95% CI, 1.8%–13.4%; <i>P</i> = 0.012) - Most SAEs and treatment-related AEs were laboratory abnormalities.	Key Limitations - Study closed before enrolling 1,500 patients, as it was unlikely to show a difference between arms. - Study was not powered to detect differences between BAR and DEX in the subgroup of patients on HFNC oxygen or NIV at baseline. - Few patients died or required MV, which may have decreased the power to detect a difference between arms for MV-free survival. - Treatment-related differences in AEs for BAR arm vs. DEX arm were mainly related to laboratory abnormalities, not clinical events. The clinical relevance of these differences in laboratory abnormalities is unclear. Interpretation - In hospitalized patients who required conventional oxygen, HFNC oxygen, or NIV, the use of BAR or DEX resulted in similar MV-free survival by Day 29.

Methods	Results	Limitations and Interpretation
STOP-COVID: Double-Blind, Placebo-Controlled, Randomized Trial of Tofacitinib in Hospitalized Patients With COVID-19 Pneumonia in Brazil⁶		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - COVID-19 pneumonia on CXR or CT - Hospitalized for <72 hours Key Exclusion Criteria - Receiving NIV, MV, or ECMO at baseline - History of or current thrombosis - Immunosuppression or active cancer treatment Interventions - Tofacitinib 10 mg PO twice daily for up to 14 days or until hospital discharge (n = 144) - Placebo (n = 145) Primary Endpoint - Mortality or respiratory failure by Day 28 Key Secondary Endpoint - Mortality by Day 28	Participant Characteristics - Mean age 56 years; 35% women - Median of 10 days from symptom onset to randomization - At baseline: 75% on supplemental oxygen 13% on HFNC oxygen - Use of glucocorticoids: 79% at baseline, 89% during hospitalization Primary Outcome - Mortality or respiratory failure by Day 28: 18% in tofacitinib arm vs. 29% in placebo arm (risk ratio 0.63; 95% CI, 0.41–0.97; <i>P</i> = 0.04) Secondary Outcome - Mortality by Day 28: 2.8% in tofacitinib arm vs. 5.5% in placebo arm (HR 0.49; 95% CI, 0.15–1.63)	Key Limitations - Small sample size - RDV not available during trial Interpretation - When compared with placebo, use of tofacitinib led to a lower risk of mortality or respiratory failure among hospitalized adults with COVID-19 pneumonia, most of whom received glucocorticoids.

Key: AE = adverse event; ALC = absolute lymphocyte count; ALT = alanine transaminase; ANC = absolute neutrophil count; AST = aspartate aminotransferase; BAR = baricitinib; CCP = COVID-19 convalescent plasma; CRP = C-reactive protein; CT = computed tomography; CXR = chest X-ray; DEX = dexamethasone; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; HFNC = high-flow nasal cannula; IV = intravenous; IVIG = intravenous immunoglobulin; JAK = Janus kinase; LDH = lactate dehydrogenase; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PCR = polymerase chain reaction; PO = oral; RCT = randomized controlled trial; RDV = remdesivir; SAE = serious adverse event; SOC = standard of care; SpO₂ = oxygen saturation; TB = tuberculosis; ULN = upper limit of normal

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Abatacept

Last Updated: February 29, 2024

Cytotoxic T-lymphocyte–associated antigen 4 (CTLA-4) is a protein receptor that is expressed by activated T cells. By mediating inhibitory signals, this receptor can decrease T cell proliferation and cytokine production.^{1,2} Abatacept (CTLA-4-Ig) is a soluble fusion protein that contains CTLA-4 linked to human immunoglobulin, and it is used to block T cell activation. Because excessive T cell stimulation and proliferation is thought to propagate the pathogenesis of COVID-19,³ modulating this response may be a potential option for treating COVID-19.⁴

Abatacept is approved by the Food and Drug Administration for the treatment of inflammatory arthritis and for the prophylaxis of acute graft-versus-host disease.⁵ It is currently not approved for the treatment of COVID-19, but it has been evaluated in clinical trials for the treatment of hospitalized patients with moderate to severe COVID-19.

Recommendation

- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for the COVID-19 Treatment Guidelines Panel's (the Panel) recommendations on the use of abatacept in hospitalized patients who require conventional oxygen, high-flow nasal cannula (HFNC) oxygen, or noninvasive ventilation (NIV).

Rationale

The ACTIV-1 immunomodulator trial was a double-blind, multi-arm, randomized trial in moderately to severely ill adults who were hospitalized with COVID-19.⁶ The trial separately evaluated treatment with abatacept, cenicriviroc, and infliximab versus placebo. All arms received standard care, and the separate analyses included data from a shared placebo arm. One substudy compared the use of a single dose of intravenous abatacept 10 mg/kg to placebo. The primary endpoint was time to recovery by Day 28. Key secondary endpoints included clinical status at Day 14 and mortality through Days 28 and 60.

The study concluded that use of abatacept in patients with COVID-19 did not have a significant effect on the time to recovery. A reduction in 28-day mortality, a secondary endpoint, was found. Patients who required mechanical ventilation or extracorporeal membrane oxygenation did not benefit from the use of abatacept.

Monitoring and Adverse Effects

Concomitant use of abatacept and other immunomodulatory agents may increase the risk of serious infections. Due to its immunosuppressive effects, all patients who are receiving abatacept should also be monitored for new infections. In the ACTIV-1 trial, the use of a single dose of abatacept in patients with COVID-19 did not reveal significant safety concerns.

Most of the data on the adverse effects of abatacept come from the chronic use of the agent for the treatment of autoimmune diseases and graft-versus-host disease. When abatacept is used for the prevention of acute graft-versus-host disease, the most commonly reported adverse effects include fever, anemia, hypertension, cytomegalovirus infection (or reactivation), pneumonia, epistaxis, CD4 lymphopenia, and acute kidney injury.⁵

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance regarding the use of

abatacept during pregnancy and lactation.

Considerations in Children

The intravenous formulation of abatacept is approved by the Food and Drug Administration for the treatment of juvenile idiopathic arthritis and acute graft-versus-host disease in children aged ≥ 2 years. It is not approved for the treatment of COVID-19 in children, and there are no published reports on the efficacy of using abatacept in this population. No patients aged <18 years were included in the ACTIV-1 trial.

Clinical Data

In the ACTIV-1 trial, the modified intention-to-treat analysis for the abatacept substudy included 509 patients in the abatacept arm and 510 patients in the placebo arm. At baseline, 53% of patients required conventional oxygen supplementation, and 33% required HFNC oxygen or NIV. As part of their standard care before or during the study, 93% of patients received remdesivir, and 91% received corticosteroids.

Results

- The median time to recovery was 9 days in both the infliximab and placebo arms (recovery rate ratio 1.12; 95% CI, 0.98–1.28; $P = 0.09$), and there was no differential effect across subgroups based on disease severity (interaction $P = 0.66$).
- Mortality by Day 28 was lower among patients who received abatacept (56 of 509 patients [11.0%]) than among those who received placebo (77 of 510 patients [15.1%]; OR 0.62; 95% CI, 0.41–0.94).
- Subgroup analyses showed reduced mortality only among patients in the abatacept arm who required HFNC oxygen or NIV (OR 0.48; 95% CI, 0.28–0.84).
- Among patients who required mechanical ventilation or extracorporeal membrane oxygenation, there was no difference in mortality by Day 28 (OR 1.63; 95% CI, 0.66–4.05).
- There were no differences in secondary infections or in the number or severity of serious adverse events between the abatacept and placebo arms.

Limitations

- Each of the 3 active agents was compared to a shared placebo group without adjusting for multiple comparisons.
- Mortality was a secondary endpoint. Although the treatment difference for mortality by Day 28 was nominally significant, no adjustment was made to the analysis to assess multiple outcomes (primary outcome and mortality).
- The study was not powered to analyze differences within disease severity subgroups.

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Infliximab

Last Updated: February 29, 2024

Tumor necrosis factor–alpha (TNF-alpha) is a pleiotropic proinflammatory cytokine that is mainly generated by activated macrophages, lymphocytes, and natural killer cells. TNF-alpha plays a significant role in immune-mediated inflammatory diseases. Early in the COVID-19 pandemic, increased levels of interleukin-6 and TNF-alpha were identified as independent predictors of disease severity and death.¹ Furthermore, several cohort studies and registries noted that people with immune-mediated inflammatory diseases who were receiving TNF-alpha inhibitors were at lower risk for COVID-19–related hospitalizations and severe disease than people with immune-mediated inflammatory diseases who were receiving non–TNF-alpha biologic products.^{2,3} It has been hypothesized that modulating levels of TNF-alpha or its effects may reduce the duration or severity of COVID-19. Infliximab is a TNF-alpha inhibitor that has been evaluated for the treatment of hospitalized patients with moderate to severe COVID-19.

Recommendation

- See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for the COVID-19 Treatment Guidelines Panel’s (the Panel) recommendations on the use of infliximab in hospitalized patients who require conventional oxygen, high-flow nasal cannula (HFNC) oxygen, or noninvasive ventilation (NIV).

Rationale

The ACTIV-1 immunomodulator trial was a double-blind, multi-arm, randomized trial in moderately to severely ill adults who were hospitalized with COVID-19.⁴ The trial separately evaluated treatment with abatacept, cenicriviroc, and infliximab versus placebo. All arms received standard care, and the separate analyses included data from a shared placebo arm. One substudy compared the use of a single dose of intravenous infliximab 5 mg/kg to placebo. The primary endpoint was time to recovery by Day 28. Key secondary endpoints included clinical status at Day 14 and mortality through Days 28 and 60.

The study concluded that use of infliximab in patients with COVID-19 did not have a significant effect on the time to recovery. A reduction in 28-day mortality, a secondary endpoint, was found. Patients who required mechanical ventilation or extracorporeal membrane oxygenation did not benefit from the use of infliximab.

Monitoring and Adverse Effects

Because of infliximab’s immunosuppressive effects, all patients who receive it should be monitored for new infections. In the ACTIV-1 trial, the use of a single dose of infliximab in patients with COVID-19 did not reveal significant safety concerns.

Most of the data on the adverse effects of infliximab come from the chronic use of the agent for the treatment of autoimmune diseases. Adverse effects include serious infections (including invasive fungal infections), infusion-related reactions and hypersensitivity, cytopenias, hepatotoxicity, and, rarely, cardiovascular and cerebrovascular events.⁵

Considerations in Pregnant and Lactating People

See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel’s guidance regarding the use of

infliximab during pregnancy and lactation.

Considerations in Children

Infliximab is approved for the treatment of inflammatory bowel disease in children and is often used to treat juvenile idiopathic arthritis. The Food and Drug Administration has not approved the use of infliximab for the treatment of COVID-19 in children, and there are no published reports on the efficacy of using infliximab in this population. No patients aged <18 years were included in the ACTIV-1 trial.

See [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#) for the Panel's recommendations regarding the use of infliximab in pediatric patients with multisystem inflammatory syndrome in children (MIS-C).

Clinical Data

In the ACTIV-1 trial, the modified intention-to-treat analysis for the infliximab substudy included 517 patients in the infliximab arm and 516 patients in the placebo arm. At baseline, 52% of patients required conventional oxygen supplementation, and 33% required HFNC oxygen or NIV. As part of their standard care before or during the study, 93% of patients received remdesivir, and 92% received corticosteroids.

Results

- The median time to recovery was 8 days in the infliximab arm versus 9 days in the placebo arm (recovery rate ratio 1.12; 95% CI, 0.99–1.28; $P = 0.08$), and there was no differential effect across subgroups based on disease severity (interaction $P = 0.36$).
- Mortality by Day 28 was lower among patients who received infliximab (52 of 517 patients [10.1%]) than among those who received placebo (75 of 516 patients [14%]; OR 0.59; 95% CI, 0.39–0.90).
- Subgroup analyses showed reduced mortality only among patients in the infliximab arm who required HFNC oxygen or NIV (OR 0.52; 95% CI, 0.29–0.91).
- Among patients who required mechanical ventilation or extracorporeal membrane oxygenation, there was no difference in mortality by Day 28 (OR 1.11; 95% CI, 0.45–2.72).
- There were no differences in secondary infections or in the number or severity of serious adverse events between the infliximab and placebo arms.

Limitations

- Each of the 3 active agents was compared to a shared placebo group without adjusting for multiple comparisons.
- Mortality was a secondary endpoint. Although the treatment difference for mortality by Day 28 was nominally significant, no adjustment was made to the analysis to assess multiple outcomes (primary outcome and mortality).
- The study was not powered to analyze differences within disease severity subgroups.

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Interleukin-1 Inhibitors

Last Updated: February 29, 2024

Endogenous interleukin (IL)-1 is elevated in patients with COVID-19.¹⁻³ In addition, SARS-CoV-2 infection causes epithelial damage that leads to the release of IL-1 beta, which recruits inflammatory cells and induces the release of IL-1 beta in monocytes. This in turn leads to the release of more IL-1 to recruit and activate additional innate immune cells. Drugs that block the IL-1 receptor (e.g., anakinra) or drugs that block IL-1 signaling (e.g., canakinumab) can potentially interrupt this autoinflammatory loop. These drugs have been investigated as potential treatments for COVID-19.

Anakinra is a recombinant human IL-1 receptor antagonist. It is approved by the Food and Drug Administration (FDA) to treat rheumatoid arthritis and cryopyrin-associated periodic syndromes, specifically neonatal-onset multisystem inflammatory disease.⁴ It is used off-label to treat severe chimeric antigen receptor T cell-mediated cytokine release syndrome and macrophage activation syndrome/secondary hemophagocytic lymphohistiocytosis. On November 8, 2022, the FDA issued an Emergency Use Authorization for anakinra. The Emergency Use Authorization allows the use of anakinra to treat COVID-19 in certain hospitalized adults with pneumonia. These patients must have laboratory-confirmed SARS-CoV-2 infection, require supplemental oxygen (either low- or high-flow oxygen), be at risk of progressing to severe respiratory failure, and be likely to have elevated plasma levels of soluble urokinase plasminogen activator receptor (suPAR), a marker of inflammation.⁵

Canakinumab is a human monoclonal antibody that targets the beta subunit of IL-1 and is approved by the FDA for the treatment of systemic juvenile idiopathic arthritis, Still's disease, and gout.⁶

Recommendations

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of anakinra for the treatment of COVID-19.
- The Panel **recommends against** the use of **canakinumab** for the treatment of COVID-19, except in a clinical trial (BIIa).

Rationale

In the SAVE-MORE trial, 594 hospitalized patients who had moderate or severe COVID-19 pneumonia and plasma suPAR levels ≥ 6 ng/mL were randomized to receive either anakinra or placebo.⁷ The study found that patients who received anakinra had a lower risk of clinical progression of COVID-19 than those who received placebo. REMAP-CAP, an open-label, adaptive platform trial that evaluated the use of several immunomodulators in patients with COVID-19 who required organ support, found no clinical benefit of anakinra in these patients. In addition, among patients who received anakinra, no reduction in mortality was observed during a 180-day follow up.⁸ Several smaller trials that evaluated the use of anakinra in people with COVID-19 were either stopped early due to futility or did not detect a benefit of anakinra in these patients.⁹⁻¹¹ Meta-analyses of the available data have not detected a benefit of using anakinra to treat COVID-19.^{12,13}

The SAVE-MORE study population was restricted to participants with high levels of suPAR (≥ 6 ng/mL) based on the hypothesis that this group is most likely to benefit from IL-1 inhibition. However, the laboratory assay that is used to assess suPAR levels is not currently available in many countries, including the United States. Using data from the SAVE-MORE and SAVE trials (both a priori, open-label, single-arm prospective studies), the FDA developed a scoring system that uses common

clinical and laboratory factors to identify patients who are likely to have suPAR levels ≥ 6 ng/mL.

The Panel has concluded that there is insufficient evidence to recommend either for or against the use of anakinra for the treatment of COVID-19. After reviewing the available evidence, the Panel notes the following:

- Data from randomized trials have not consistently demonstrated a benefit of using anakinra to treat COVID-19.
- The suPAR assays that were used to identify patients for participation in the SAVE-MORE trial are not available in the United States.
- The scoring system that the FDA developed to identify patients who are likely to have high suPAR levels requires further validation.

The Panel **recommends against** the use of **canakinumab** for the treatment of COVID-19, except in a clinical trial (**BIIa**). CAN-COVID, a randomized controlled trial that evaluated the use of canakinumab in hospitalized patients with COVID-19 who were hypoxemic but did not require ventilatory support, reported that the use of canakinumab did not improve the likelihood of survival without mechanical ventilation.¹⁴ CanCovDia was a randomized controlled trial that compared the use of canakinumab to placebo in hospitalized patients with COVID-19, type 2 diabetes, and a body mass index >25 . This trial did not find a difference between the arms in the occurrence of a composite outcome that included length of survival, ventilation, intensive care unit (ICU) stay, and hospitalization through Day 29.¹⁵

Monitoring and Adverse Effects

Headache, nausea, vomiting, and liver enzyme elevations can occur with both anakinra and canakinumab.

Anakinra was not associated with any significant safety concerns when used in clinical trials for the treatment of sepsis.¹⁶⁻¹⁸ Increased rates of infection were reported with prolonged anakinra use in combination with tumor necrosis factor–alpha blockade but not with short-term use.¹⁹

Considerations in Pregnant People

The data on using IL-1 inhibitors to treat COVID-19 in pregnant patients are currently limited. The American College of Rheumatology recommends against the use of anakinra during pregnancy.²⁰ Unintentional first-trimester exposure to anakinra is unlikely to be harmful, given the minimal transfer of monoclonal antibodies across the placenta early in pregnancy.²¹

Considerations in Children

Anakinra has been used to treat severely ill children with rheumatologic conditions, including systemic juvenile idiopathic arthritis and macrophage activation syndrome. The data on the use of anakinra in pediatric patients with acute respiratory distress syndrome or sepsis are limited. Information on the use of anakinra in pediatric patients with acute COVID-19 is limited to small case series.^{22,23} However, anakinra has been used in approximately 10% of cases of multisystem inflammatory syndrome in children (MIS-C).²⁴⁻²⁹ Anakinra is often included in institutional protocols for the treatment of MIS-C in the United States, and it is an option for second-line therapy for refractory MIS-C in national consensus guidelines, including the COVID-19 Treatment Guidelines.³⁰⁻³² For more information, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Data on using canakinumab in pediatric patients are limited to use in patients with periodic fever syndromes and systemic juvenile idiopathic arthritis. There are no data on its use in pediatric patients

with acute COVID-19 or MIS-C.

Clinical Data

SAVE-MORE

SAVE-MORE was a randomized controlled trial in 594 hospitalized patients with moderate to severe COVID-19 pneumonia and plasma suPAR levels ≥ 6 ng/mL.⁷ Patients who required noninvasive ventilation (NIV) or mechanical ventilation were excluded from the study. Patients were randomized 2:1 to receive anakinra 100 mg subcutaneously once daily for 10 days or placebo. The primary endpoint was clinical status at Day 28 on the 11-point World Health Organization Clinical Progression Scale (WHO-CPS). Additional analyses assessed outcomes at Days 60 and 90.³³

Results

- Patients who were randomized to receive anakinra had lower odds of a worse WHO-CPS score at Day 28 (OR 0.36; 95% CI, 0.26–0.50; $P < 0.0001$).
- The secondary endpoints also favored anakinra, including the absolute decrease in WHO-CPS scores from baseline at Days 14 and 28, the absolute decrease in sequential organ failure assessment (SOFA) scores from baseline at Day 7, the median time to hospital discharge, and the median duration of ICU stays.
- A smaller proportion of patients in the anakinra arm experienced secondary infections, including ventilator-associated pneumonias, than in the placebo arm (8.4% vs. 15.9%; $P = 0.01$).
- Twenty-eight-day mortality was lower among patients who received anakinra than among those who received placebo (3.2% vs. 6.9%; HR 0.45; 95% CI, 0.21–0.98; $P = 0.045$).
- Additional analyses performed at Days 60 and 90 showed a sustained survival benefit for anakinra.

Limitations

The laboratory assay that is used to assess suPAR levels is not currently available in many countries, including the United States. The FDA worked with the SAVE-MORE investigators to develop a scoring system that predicts whether a patient has suPAR levels ≥ 6 ng/mL using baseline data from patients who were randomized during the trial and a subset of patients who were screened but not randomized. The FDA's surrogate for suPAR levels ≥ 6 ng/mL is called SCORE 2, and it includes the following patient characteristics:

- Aged ≥ 75 years
- Severe pneumonia, as determined by WHO criteria
- Current or past smoker
- SOFA score ≥ 3
- Neutrophil to lymphocyte ratio ≥ 7
- Hemoglobin ≤ 10.5 g/dL
- Medical history of ischemic stroke
- Blood urea ≥ 50 mg/dL and/or medical history of renal disease

Patients who met ≥ 3 of these criteria were considered positive for SCORE 2 and likely to have a suPAR level ≥ 6 ng/mL. SCORE 2 had a positive predictive value of 0.95, a sensitivity of 0.41, and specificity of 0.96 when retrospectively applied to the SAVE-MORE trial, and it had similar characteristics when applied to the SAVE trial, an open-label, single-arm prospective study that served as an external

validation dataset. In the SAVE-MORE trial, a greater proportion of patients who were positive for SCORE 2 developed severe respiratory failure at Day 14 compared with those who met ≤ 2 of the SCORE 2 criteria (41.4% vs. 8.0%).^{4,34}

REMAP-CAP

The REMAP-CAP trial is an open-label, adaptive platform trial in which eligible participants are randomized to several domains, including the Immune Modulation Therapy domain, which consists of 2 IL-6 inhibitors, anakinra, interferon beta-1a, and a control group.⁸ Participants are eligible for enrollment if they are within 24 hours of receiving respiratory or cardiovascular organ support in the ICU and they have suspected or microbiologically confirmed COVID-19. This population had more advanced disease than the population enrolled in the SAVE-MORE trial.

Anakinra 300 mg was given intravenously (IV) as a loading dose, followed by anakinra 100 mg IV every 6 hours for 14 days until patients were either free from mechanical ventilation for >24 hours or discharged from the ICU. The primary outcome was measured using an ordinal scale that included a composite of in-hospital mortality and duration of respiratory and cardiovascular organ support at 21 days; all deaths up to 90 days were assigned the worst outcome. The trial used a Bayesian design that allowed the authors to compare nonconcurrently randomized interventions across time periods. Additional analyses assessed outcomes at 180 days.^{3,8}

Results

- Of the 2,274 participants who were randomized to 1 of the arms in the Immune Modulation Therapy domain, 365 individuals were assigned to receive anakinra and included in the analysis, 406 were assigned to the usual care (control) arm, 943 were assigned to receive tocilizumab, and 483 were assigned to receive sarilumab.
- Of those assigned to receive anakinra, 37% were receiving mechanical ventilation at study entry compared with 32% of patients in the other arms. The other patients received oxygen through a high-flow nasal cannula or NIV, with a few exceptions.
- The median number of organ support-free days was similar for patients who received anakinra and those who received usual care (0 days [IQR -1 to 15 days] vs. 0 days [IQR -1 to 15 days]). The aOR for organ support-free days was 0.99 for anakinra (95% CrI, 0.74–1.35), with a 47% posterior probability of superiority to control. Sixty percent of those who were assigned to receive anakinra survived compared with 63% of those who were assigned to the control arm, with a 44% posterior probability that anakinra was superior to usual care.
- Additional analyses performed at 180 days showed no reduction in mortality among patients who received anakinra.^{3,8}
- The risk of experiencing serious adverse events was similar between the arms.

Limitations

Patients were not randomized contemporaneously to receive anakinra or usual care; the treatment effect was estimated from an overarching model that mostly included patients who were randomized to receive an IL-6 inhibitor (tocilizumab or sarilumab) or usual care, and patients who were randomized to receive an IL-6 inhibitor or anakinra. Thus, the estimate of the treatment effect is not fully protected by randomization. This study also had an open-label design.

CORIMUNO-ANA-1

The CORIMUNO-ANA-1 trial randomized 116 hospitalized patients with COVID-19 pneumonia 1:1 to receive either usual care plus anakinra (200 mg IV twice daily on Days 1–3, 100 mg IV twice on Day

4, and 100 mg IV once on Day 5) or usual care alone.⁹ Patients were eligible for enrollment if they had laboratory-confirmed SARS-CoV-2 infection with COVID-19 pneumonia and they required >3 L/min of supplemental oxygen. Patients who required high-flow oxygen, ventilation, or ICU admission were excluded. The 2 coprimary outcomes were the proportion of patients who had died or who needed NIV or mechanical ventilation by Day 4 (score of >5 on the WHO-CPS) and the proportion who survived without the need for NIV or mechanical ventilation (including high-flow oxygen) by Day 14.

Results

- There was no difference between the arms in the occurrence of the coprimary outcomes: by Day 4, 36% of patients in the anakinra arm had died or required high-flow oxygen or ventilation compared with 38% in the usual care arm (90% CrI, -17.1 to 12.0; posterior probability of benefit 61%). By Day 14, 47% of patients in the anakinra arm had died or required NIV or mechanical ventilation compared with 51% in the usual care arm (median HR 0.97; 90% CrI, 0.62–1.52; posterior probability of benefit 55%).
- Fifty-two percent of patients received corticosteroids at study entry.
- Serious adverse events occurred in 46% of patients in the anakinra arm compared with 38% in the usual care arm; 11 of 59 patients (18.6%) in the anakinra arm experienced bacterial or fungal infections compared with 4 of 55 patients (7.3%) who received usual care.

Limitations

The limitations of this study include the small sample size, narrow eligibility criteria, and the fact that many patients did not receive current standard of care therapy (e.g., corticosteroids, remdesivir).

COV-AID

The COV-AID trial enrolled 342 hospitalized patients with COVID-19, hypoxia, and signs of hyperinflammation.¹⁰ This trial had an open-label, 2 x 2 factorial design to compare IL-1 inhibition to no IL-1 inhibition and IL-6 inhibition to no IL-6 inhibition. The primary outcome was the time to clinical improvement, which was defined as an increase of 2 or more points on a 6-point ordinal scale or discharge from the hospital.

Results

- There was no difference between the anakinra arm and the usual care arm in the occurrence of the primary outcome. The estimated median time to clinical improvement was 12 days (95% CI, 10–16 days) in the anakinra arm and 12 days (95% CI, 10–15 days) in the usual care arm (HR 0.94; 95% CI, 0.73–1.21).
- Fifty-five patients died during the study, and no statistically significant differences in mortality were found between the study arms.
- The risk of experiencing serious adverse events was similar between the arms.

Limitations

The limitations of this study include the open-label design and the fact that many patients did not receive current standard of care therapy (e.g., corticosteroids, remdesivir). In addition, the 2 x 2 factorial structure was underpowered to detect interactions between treatment arms, because some patients received both an IL-1 inhibitor and an IL-6 inhibitor.

ANA-COVID-GEAS

ANA-COVID-GEAS was a multicenter, randomized, open-label, Phase 2/3 clinical trial of 179 hospitalized patients with severe COVID-19 pneumonia and hyperinflammation.¹¹ Patients were

randomized 1:1 to receive anakinra 100 mg IV 4 times daily plus standard of care or standard of care alone for up to 15 days. The length of treatment was based on the patient's clinical response per the protocol-defined criteria. The primary outcome was the proportion of patients who did not require mechanical ventilation up to 15 days after treatment initiation.

Results

- There was no statistical difference between the anakinra arm and standard of care arm in the proportion of patients who did not require mechanical ventilation up to 15 days after treatment initiation (77.1% vs. 85.9%; relative risk ratio 0.90; 95% CI, 0.77–1.04)
- The secondary outcomes were also not statistically different between groups. These included the time to mechanical ventilation (HR 1.72; 95% CI, 0.82–3.62) and the number of deaths by Day 28 (4 vs. 5; relative risk ratio 0.77; 95% CI, 0.21–2.77).

Limitations

Key limitations of this study were its open-label design, modest sample size, the fact that only half of the patients received dexamethasone, and the fact that the proportion of patients who required oxygen supplementation at baseline was significantly higher in the anakinra arm.

CAN-COVID

CAN-COVID was a double-blind, placebo-controlled randomized trial of 454 hospitalized patients with COVID-19 who were hypoxemic but not mechanically ventilated and had elevated levels of C-reactive protein (≥ 20 mg/L) or ferritin (≥ 600 μ g/L).¹⁴ Patients were randomized 1:1 to receive a single dose of IV canakinumab (450 mg for a body weight of 40 kg to <60 kg, 600 mg for 60–80 kg, and 750 mg for >80 kg) or placebo. The primary outcome was survival without the need for mechanical ventilation from Days 3 through Day 29.

Results

- There was no statistical difference between the canakinumab arm and placebo arm in the proportion of patients who survived without mechanical ventilation (88.8% vs. 85.7%; $P = 0.29$).
- The number of COVID-19–related deaths at 4 weeks was similar for the 2 arms (11 of 223 patients [4.9%] in the canakinumab arm vs. 16 of 222 patients [7.2%] in the placebo arm; OR 0.67; 95% CI, 0.30–1.50).
- Forty-one percent of patients in the canakinumab arm and 32% in the placebo arm received dexamethasone.
- Serious adverse events occurred in 16% of patients who received canakinumab and in 21% of patients who received placebo.

Limitations

The use of corticosteroids was unbalanced in this study, with more patients receiving dexamethasone at baseline in the canakinumab arm than in the placebo arm. More patients received dexamethasone after the trial was underway in the placebo arm than in the canakinumab arm (22.5% vs. 14.5%), and more patients received tocilizumab in the placebo arm than in the canakinumab arm (8.8% vs. 2.2%).

CanCovDia

CanCovDia was a double-blind, placebo-controlled, randomized trial of 116 hospitalized patients with type 2 diabetes, a body mass index >25 , and COVID-19.¹⁵ Most patients (65.8%) required oxygen supplementation. Patients were randomized 1:1 to receive a single dose of canakinumab (using weight-adapted dosing between 450 and 750 mg) or placebo. The primary outcome was reported as a win

ratio, which was calculated by dividing the number of winners by the number of losers in a sequence of hierarchical comparisons based on ordered components. This hierarchy included:

1. Longer survival time
2. Longer ventilation-free time
3. Longer ICU-free time
4. Shorter hospitalization time within 29 days after treatment with canakinumab compared with placebo

Results

- The win ratio for canakinumab versus placebo was 1.08 (95% CI, 0.69–1.69; $P = 0.75$).
- At 4 weeks, there was no statistically significant difference between the canakinumab arm and placebo arm in the number of deaths (4 vs. 7; OR 0.54; 95% CI, 0.13–1.90).

Limitations

At baseline, patients in the canakinumab arm had poorer kidney function and higher levels of ferritin and suPAR compared to those in the placebo arm.

Other small cohort studies, case-control studies, and case series have reported mixed findings with regard to improvement in outcomes among patients who received anakinra for the treatment of COVID-19.³⁵⁻³⁸ The clinical implication of these findings is uncertain due to small sample sizes and unmeasured confounding factors. Therefore, these studies did not substantially influence the Panel's current recommendations for using IL-1 inhibitors.

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Vilobelimab

Last Updated: December 20, 2023

Vilobelimab is an anti-C5a monoclonal antibody. High concentrations of C5a have been reported in patients with severe COVID-19.¹ C5a activates innate immune system responses, including inflammation and the release of histamines, and can increase damage to local tissues.² A study in mice demonstrated that an anti-C5a monoclonal antibody reduced immune system activation and inhibited lung injury.³ Vilobelimab targets C5a, which is a product of complement activation, and preserves membrane attack complex function.⁴ Vilobelimab is not approved by the Food and Drug Administration (FDA) for any indication.

On April 4, 2023, the FDA issued an Emergency Use Authorization for the use of vilobelimab for the treatment of COVID-19 in hospitalized adults when it is administered within 48 hours of mechanical ventilation or extracorporeal membrane oxygenation.⁵

Recommendation

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of vilobelimab for the treatment of COVID-19.

Rationale

Results from the PANAMO trial were used to support the FDA Emergency Use Authorization.⁵ However, the prespecified analysis that stratified by study site showed that 28-day mortality among patients with COVID-19 who received vilobelimab was not significantly different from 28-day mortality among those who received placebo. The initially proposed primary study analysis did not stratify by study site. In the second phase of the study, the primary analysis was changed to stratify by site based on a recommendation from the FDA. The analysis that did not stratify by site demonstrated that all-cause mortality through Day 28 was significantly lower in the vilobelimab arm than in the placebo arm. Concomitant use of corticosteroids (97%) and antithrombotic agents (98%) was high in this study population. Prior or concomitant use of additional immunomodulators, such as tocilizumab (17% in the vilobelimab arm, 16% in the placebo arm) and baricitinib (3% in each arm), was low. The Panel determined that the results from the PANAMO trial were insufficient to recommend either for or against the use of vilobelimab for the treatment of COVID-19.

Monitoring, Adverse Effects, and Drug-Drug Interactions

Reports of adverse effects of vilobelimab are limited to a Phase 3 trial that included critically ill adult patients with COVID-19 who received intravenous vilobelimab 800 mg for up to 6 doses.^{5,6} Common adverse reactions (i.e., those with an incidence $\geq 3\%$ and that were observed at least 1% more frequently in the vilobelimab arm than in the placebo arm through Day 60) were pneumonia, sepsis, delirium, pulmonary embolism, hypertension, pneumothorax, deep vein thrombosis, herpes simplex, enterococcal infection, bronchopulmonary aspergillosis, increased hepatic enzymes, urinary tract infection, hypoxemia, thrombocytopenia, pneumomediastinum, respiratory tract infection, supraventricular tachycardia, constipation, and rash.

Vilobelimab is not expected to be associated with any pharmacokinetic drug-drug interactions.

Considerations in Pregnant People

There are no data on the use of vilobelimab during pregnancy, as pregnant individuals were excluded from the PANAMO trial.

Considerations in Children

There are no data on the use of vilobelimab in children. Vilobelimab is not authorized by the FDA for the treatment of COVID-19 in pediatric patients.

Clinical Data

The small (n = 30) Phase 2 portion of the Phase 2/3 PANAMO trial was too underpowered to draw any conclusions about study outcomes, including physiologic improvement at 5 days and mortality.⁷

The Phase 3 portion of the trial was a double-blind, randomized trial performed at 46 hospitals in Western Europe (i.e., Netherlands, France, Germany, Belgium), Brazil, Mexico, Russia, Peru, and South Africa from October 1, 2020, to October 4, 2021.⁶ The trial compared the use of vilobelimab plus standard of care with placebo plus standard of care in patients aged ≥ 18 years who had laboratory-confirmed SARS-CoV-2 infection, were receiving mechanical ventilation (and were within 48 hours of intubation), and had a ratio of arterial partial pressure of oxygen to fraction of inspired oxygen of 60 to 200 mm Hg at study entry. Vilobelimab 800 mg was administered intravenously on Days 1, 2, 4, 8, 15, and 22, if the patient remained hospitalized, for a maximum of 6 doses.

The primary outcome was all-cause mortality at 28 days. Secondary outcomes included all-cause mortality at 60 days, the proportion of patients who improved on a World Health Organization 8-point ordinal scale, the proportion of patients who developed acute kidney failure by Day 28, and the proportion of patients free from renal replacement therapy at Day 28.

Results

- The trial enrolled 369 patients; 368 patients were included in the analysis that did not stratify by study site (177 in the vilobelimab arm, 191 in the placebo arm).
- In the prespecified analysis that stratified by study site (n = 307), 28-day mortality was not significantly different between the vilobelimab and placebo arms (HR 0.73; 95% CI, 0.50–1.06; $P = 0.094$). The analysis for 28-day mortality that stratified by study site excluded the 61 patients (16.6%) from sites that had no deaths or had only 1 treatment group.
- In the analysis that did not stratify by study site (n = 368), 28-day mortality was lower in the vilobelimab arm than in the placebo arm (54 of 177 patients [31%] vs. 77 of 191 patients [44%]), and the difference between arms was statistically significant (HR 0.67; 95% CI, 0.48–0.96; $P = 0.027$).
- Prespecified subgroup analyses identified a significant reduction in 28-day mortality in the vilobelimab arm for subgroups of patients with severe acute respiratory distress syndrome (HR 0.55; 95% CI, 0.30–0.98; $P = 0.044$), patients with an estimated glomerular filtration rate of <60 mL/min (HR 0.55; 95% CI, 0.31–0.96; $P = 0.036$), and patients receiving mechanical ventilation and additional organ support (category 7 on the World Health Organization 8-point ordinal scale; HR 0.62; 95% CI, 0.40–0.95; $P = 0.028$).
- In a prespecified analysis of the Western Europe subgroup (i.e., Netherlands, France, Germany, Belgium), the vilobelimab arm had significantly lower 28-day mortality than the placebo arm (HR 0.51; 95% CI, 0.30–0.87; $P = 0.014$).
- For the secondary outcomes:
 - The analysis that stratified by study site showed no significant difference between the arms for

all-cause mortality at 60 days (HR 0.74; 95% CI, 0.52–1.04; $P = 0.082$).

- The vilobelimab arm had significantly fewer patients who required renal replacement therapy at Day 28 than the placebo arm (age-adjusted HR 0.54; 95% CI, 0.30–0.98; $P = 0.042$).

Limitations

- The results for the study's site-stratified, prespecified analysis were not significant.
- The analysis for 28-day mortality that stratified by study site excluded the 61 patients (16.6%) from sites that had no deaths or had only 1 treatment group.
- Very few patients received a second immunomodulator (tocilizumab or baricitinib), which makes the study results difficult to apply to current practice.
- Compared to other studies that have evaluated the use of immunomodulators for the treatment of COVID-19, Phase 3 of the PANAMO trial had a relatively small sample size.

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Table 5e. Characteristics of Immunomodulators

Last Updated: February 29, 2024

- The information in this table is derived from data on the use of these drugs for FDA-approved indications or from clinical trials that evaluated their use in patients with COVID-19.
- For dose modifications for patients with organ failure or those who require extracorporeal devices, please refer to product labels or EUAs, when available.
- There are currently not enough data to determine whether certain medications can be safely coadministered with therapies for the treatment of COVID-19. When using concomitant medications with similar toxicity profiles, consider performing additional safety monitoring.
- The potential additive, antagonistic, or synergistic effects and the safety of using certain combination therapies for the treatment of COVID-19 are unknown. Clinicians are encouraged to report AEs to the [FDA MedWatch program](#).
- For drug-drug interaction information, please refer to product labels and visit the [Liverpool COVID-19 Drug Interactions website](#).
- For the Panel's recommendations on using the drugs listed in this table, please refer to the drug-specific sections of the Guidelines; [Therapeutic Management of Nonhospitalized Adults With COVID-19](#); [Therapeutic Management of Hospitalized Adults With COVID-19](#); [Therapeutic Management of Hospitalized Children With COVID-19](#); and [Pregnancy, Lactation, and COVID-19 Therapeutics](#).

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Corticosteroid (Systemic) <i>Recommended by the Panel for the treatment of COVID-19 in certain hospitalized patients.</i>					
Dexamethasone	Dose for Adults With COVID-19 • DEX 6 mg IV or PO once daily for up to 10 days or until hospital discharge, whichever comes first¹	• Hyperglycemia • Secondary infections • Reactivation of latent infections (e.g., HBV, HSV, strongyloidiasis, TB) • Psychiatric disturbances • Avascular necrosis • Adrenal insufficiency • Increased BP • Peripheral edema • Myopathy (particularly if used with NMBAs)	• Blood glucose • BP • Signs and symptoms of new infection	• Moderate CYP3A4 inducer • CYP3A4 substrate	• If DEX is not available, an alternative corticosteroid (e.g., prednisone, methylprednisolone, hydrocortisone) can be used. • For these drugs, the total daily dose equivalencies to DEX 6 mg (IV or PO) are: • Prednisone 40 mg • Methylprednisolone 32 mg • Hydrocortisone 160 mg

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Janus Kinase Inhibitors <i>Recommended by the Panel for the treatment of COVID-19 in certain hospitalized patients.</i>					
Baricitinib	FDA-Approved Doses for COVID-19 in Adults Aged ≥18 Years, per eGFR² ≥60 mL/min/1.73 m² • BAR 4 mg PO once daily for 14 days or until hospital discharge, whichever comes first 30 to <60 mL/min/1.73 m² • BAR 2 mg PO once daily for 14 days or until hospital discharge, whichever comes first 15 to <30 mL/min/1.73 m² • BAR 1 mg PO once daily for 14 days or until hospital discharge, whichever comes first <15 mL/min/1.73 m² • Not recommended FDA EUA Dose for COVID-19 in Children Aged 9–17 Years³ • Same as adults FDA EUA Doses for COVID-19 in Children Aged 2 to <9 Years, per eGFR³ ≥60 mL/min/1.73 m² • BAR 2 mg PO once daily for 14 days or until hospital discharge, whichever comes first <30 mL/min/1.73 m² • Not recommended	• Lymphoma and other malignancies • Thrombotic events (e.g., PE, DVT, arterial thrombosis) • GI perforation • Treatment-related changes in lymphocytes, neutrophils, Hgb, liver enzymes • HSV reactivation • Herpes zoster • Secondary infections • Serious cardiac-related events (e.g., MI, stroke)	• CBC with differential • Renal function • Liver enzymes • Signs and symptoms of new infections	• Dose modification recommended when coadministering BAR with a strong OAT3 inhibitor.	• See the FDA label² and EUA³ for dosing guidance in patients with: • ALC <200 cells/μL • ANC <500 cells/μL • If increases in ALT or AST are observed and DILI is suspected, interrupt BAR treatment until the diagnosis of DILI is excluded. • BAR tablets can be taken PO or crushed, dispersed in water, and given via gastrostomy tube.² Availability • BAR is approved by the FDA for the treatment of COVID-19 in adults aged ≥18 years.² • BAR is available through an FDA EUA for children aged 2–17 years who require supplemental oxygen, NIV, MV, or ECMO.³

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Janus Kinase Inhibitors, continued					
Tofacitinib	Dose for COVID-19 in Clinical Trials Tofacitinib 10 mg PO twice daily for up to 14 days or until hospital discharge, whichever comes first⁴	- Thrombotic events (e.g., PE, DVT, arterial thrombosis) - Anemia - Secondary infections - GI perforation - Diarrhea - Headache - Herpes zoster - Lipid elevations - Liver enzyme elevations - Lymphoma and other malignancies - Serious cardiac-related events (e.g., MI, stroke)	- CBC with differential - Liver enzymes - Signs and symptoms of new infections	- Requires dose modification when administered with strong CYP3A4 inhibitors or when used with a moderate CYP3A4 inhibitor that is coadministered with a strong CYP2C19 inhibitor - Coadministration with strong CYP3A4 inducers is not recommended.	- Avoid use in patients with ALC <500 cells/μL, ANC <1,000 cells/μL, or Hgb <9 g/dL. - May require dose modification in patients with moderate to severe renal impairment or moderate hepatic impairment
Interleukin-6 Inhibitors (Anti-Interleukin-6 Receptor Monoclonal Antibodies) <i>Recommended by the Panel for the treatment of COVID-19 in certain hospitalized patients.</i>					
Sarilumab	Dose for COVID-19 in Clinical Trials 1 dose of sarilumab 400 mg by IV infusion over 1 hour^{5,6}	- Neutropenia - Thrombocytopenia - GI perforation - HSRs - Liver enzyme elevations - HBV reactivation - Infusion-related reactions	- HSRs - Infusion-related reactions - CBC with differential - Liver enzymes	- Elevated IL-6 may downregulate CYP enzymes; thus, use of sarilumab may lead to increased metabolism of coadministered drugs that are CYP substrates. - The effects of sarilumab on CYP enzymes may persist for weeks after the drug is stopped.	- Sarilumab is not recommended in patients with ALT or AST >1.5 times the upper limit of the reference range, ANC <2,000 cells/μL, or PLT <150,000 cells/μL.⁷ Availability - The IV formulation of sarilumab is not approved by the FDA, but in clinical trials, a single SUBQ dose (using the prefilled syringes, not the prefilled pen) of sarilumab 400 mg was reconstituted in 100 cc 0.9% NaCl and given as an IV infusion over 1 hour.^{6,8}

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Interleukin-6 Inhibitors (Anti-Interleukin-6 Receptor Monoclonal Antibodies) , continued					
					IV infusion of sarilumab should occur within 4 hours of its preparation; it can be stored at room temperature until administered.
Tocilizumab	FDA-Approved Dose for COVID-19 in Hospitalized Adults 1 dose of tocilizumab 8 mg/kg actual body weight (up to 800 mg) by IV infusion over 1 hour FDA EUA Doses for COVID-19 in Hospitalized Children <i>Body Weight ≥30 kg</i> - Tocilizumab 8 mg/kg by IV infusion over 1 hour <i>Body Weight <30 kg</i> - Tocilizumab 12 mg/kg by IV infusion over 1 hour For All Doses - If clinical signs or symptoms worsen or do not improve following the first IV infusion, 1 additional dose may be administered at least 8 hours after the first dose.	- HSRs - Infusion-related reactions - GI perforation - Hepatotoxicity - Treatment-related changes on laboratory tests for neutrophils, platelets, lipids, and liver enzymes - HBV reactivation - Secondary infections - Cases of disseminated strongyloidiasis have been reported in patients with COVID-19 during treatment with tocilizumab and corticosteroids.	- HSRs - Infusion-related reactions - CBC with differential - Liver enzymes	- Inhibition of IL-6 may lead to increased metabolism of coadministered drugs that are CYP450 substrates. - The effects of tocilizumab on CYP enzymes may persist for weeks after the drug is stopped.	- Tocilizumab is not recommended in patients with ALT or AST >10 times the upper limit of the reference range, ANC <1,000 cells/μL, or PLT <50,000 cells/μL.⁹ - SUBQ formulation of tocilizumab is not intended for IV administration. Availability - IV tocilizumab is approved by the FDA for the treatment of COVID-19 in hospitalized adults aged 18 years.¹⁰ - Tocilizumab is available through an FDA EUA for the treatment of COVID-19 in certain hospitalized children aged 2–17 years.⁹

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Cytotoxic T-Lymphocyte-Associated Antigen 4 Agonist <i>Recommended by the Panel for the treatment of COVID-19 in certain hospitalized patients.</i>					
Abatacept	Dose for COVID-19 in Clinical Trials 1 dose of abatacept 10 mg/kg actual body weight (up to 1,000 mg) by IV infusion over 30 minutes¹¹	- HSRs, including anaphylaxis - Infusion-related reactions - HBV reactivation - Secondary infections - Patients with COPD may develop more frequent respiratory AEs. - Headache - Upper respiratory infection, nasopharyngitis - Nausea - Anemia - HTN - Decrease in CD4 count - Hypermagnesemia - Acute kidney injury¹²	- HSRs - Infusion-related reactions - CBC with differential - Electrolytes - Renal function	Drug-drug interactions are unlikely between abatacept and medications that are CYP substrates, inhibitors, or inducers.	- The IV formulation of abatacept includes maltose, which may give falsely elevated blood glucose readings with certain blood glucose monitors (e.g., GDH-PQQ–based monitoring systems) on the day of infusion. - In the ACTIV-1 trial, 1 case of anaphylaxis and 2 infusion-related reactions were reported among abatacept recipients.¹¹ Availability - The IV formulation of abatacept is commercially available.
Tumor Necrosis Factor–Alpha Inhibitor <i>Recommended by the Panel for the treatment of COVID-19 in certain hospitalized patients.</i>					
Infliximab	Dose for COVID-19 in Clinical Trials 1 dose of infliximab 5 mg/kg actual body weight by IV infusion over 2 hours¹¹	- HSRs, including anaphylaxis - Infusion-related reactions - The following AEs are associated with chronic use of infliximab: - Hepatotoxicity - Cytopenia (e.g., leukopenia, neutropenia, thrombocytopenia, pancytopenia)	- HSRs - Infusion-related reactions - CBC with differential - PLT - Liver enzymes - If infliximab is administered to patients with heart failure, they should be closely monitored.	Inhibition of cytokine activity may lead to increased metabolism of coadministered drugs that are CYP450 substrates.	Availability Infliximab is available as an originator biologic or a biosimilar.

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Tumor Necrosis Factor–Alpha Inhibitor , continued					
		• HBV reactivation • Secondary infections (e.g., invasive fungal infections, reactivation of latent TB) • Heart failure • CVA, MI, hypotension, hypertension, arrhythmias • Transient vision loss • Demyelinating disease • Lupus-like syndrome • Headache • Abdominal pain¹³			
Anti-C5a Monoclonal Antibody <i>Received an FDA EUA for the treatment of COVID-19 when it is administered within 48 hours of MV or ECMO. There is insufficient evidence for the Panel to recommend either for or against its use..</i>					
Vilobelimab	FDA EUA Dose for COVID-19 in Hospitalized Adults Receiving MV or ECMO • Vilobelimab 800 mg by IV infusion after dilution, up to 6 doses; start treatment within 48 hours of intubation (Day 1) followed by administration on Days 2, 4, 8, 15, and 22 if patient is still hospitalized (even if discharged from ICU)	• Secondary infections • Infusion-related reactions • Delirium • PE • HTN • Pneumothorax • DVT • Liver enzyme elevations • Hypoxemia • Thrombocytopenia • Pneumomediastinum • Supraventricular tachycardia • Constipation • Rash	• Signs and symptoms of new infections • Infusion-related reactions • CBC with differential • Liver enzymes	• None expected	Availability • Vilobelimab is not approved by the FDA, but it is commercially available for use in hospitalized adults with COVID-19, as authorized by the EUA.¹⁴

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Interleukin-1 Inhibitors <i>Anakinra: Received an FDA EUA for the treatment of COVID-19 in certain hospitalized adults. There is insufficient evidence for the Panel to recommend either for or against its use.</i> <i>Canakinumab: The Panel recommends against the use of canakinumab for the treatment of COVID-19, except in a clinical trial.</i>					
Anakinra	FDA EUA Dose for COVID-19 in Hospitalized Adults Aged ≥18 Years - Anakinra 100 mg SUBQ once daily for 10 days Dose for Patients With CrCl <30 mL/min - Anakinra 100 mg SUBQ every other day for 5 total doses over 10 days¹⁵	- Neutropenia, particularly when used concomitantly with other agents that can cause neutropenia - HSRs, including anaphylaxis and angioedema - Secondary infections - Injection site reactions - Liver enzyme elevations - Hyperkalemia - Hyponatremia - Rash	- CBC with differential; assess neutrophils before starting treatment and during therapy. - BMP - Liver enzymes - Renal function	Use with TNF-blocking agents is not recommended due to potential increased risk of infection.	Contraindicated in patients with known hypersensitivity to proteins derived from <i>Escherichia coli</i>, anakinra, or any component of the product¹⁵
Canakinumab	Dose for COVID-19 in Clinical Trials - Canakinumab 450–750 mg (based on body weight) by IV infusion over 2 hours^{16,17} FDA-Approved Dose for Systemic JIA - Canakinumab 4 mg/kg (up to 300 mg) SUBQ every 4 weeks¹⁸	- HSRs - Neutropenia - Nasopharyngitis - Diarrhea - Respiratory tract infections - Bronchitis - Gastroenteritis - Pharyngitis - Musculoskeletal pain - Vertigo - Abdominal pain - Injection site reactions - Liver enzyme elevations	- HSRs - CBC with differential - Liver enzymes	- Binding of canakinumab to IL-1 may increase formation of CYP enzymes and alter metabolism of drugs that are CYP substrates. - Use with TNF-blocking agents is not recommended due to potential increased risk of infection.	Availability The IV formulation of canakinumab is not approved by the FDA for use in the United States.¹⁸

Drug Name	Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Corticosteroids (Inhaled) <i>There is insufficient evidence for the Panel to recommend either for or against the use of inhaled corticosteroids for the treatment of COVID-19.</i>					
Budesonide (Inhaled)	Dose for COVID-19 in Clinical Trials Budesonide 800 µg oral inhalation twice daily until symptom resolution or for up to 14 days^{19,20}	- Secondary infections - Oral thrush - Systemic AEs are not common, but they may occur when budesonide is coadministered with a strong CYP3A4 inhibitor.	- Signs of AEs involving the oral mucosa or throat, including thrush - Signs of systemic corticosteroid effects (e.g., adrenal suppression)	- CYP3A4 substrate Do not use with strong CYP3A4 inhibitors.	No comments
Ciclesonide (Inhaled)	Dose for COVID-19 in Clinical Trials Ciclesonide 160 µg as 2 MDI inhalations twice daily for 30 days²¹	- Secondary infections - Oral thrush - Systemic AEs (less common)	- Signs of AEs involving the oral mucosa or throat, including thrush - Signs of systemic corticosteroid effects (e.g., adrenal suppression)	- CYP3A4 substrate - Strong CYP3A4 inhibitors are expected to have less effect on ciclesonide exposure than on budesonide or fluticasone exposure.	No comments
Fluticasone (Inhaled)	Dose for COVID-19 in Clinical Trials Fluticasone 200 µg as 1 MDI inhalation once daily for 14 days²²	- Secondary infections - Oral thrush - Systemic AEs are not common, but they may occur when fluticasone is coadministered with a strong CYP3A4 inhibitor.	- Signs of AEs involving the oral mucosa or throat, including thrush - Signs of systemic corticosteroid effects (e.g., adrenal suppression)	- CYP3A4 substrate Do not use with strong CYP3A4 inhibitors.	No comments

Key: AE = adverse event; ALC = absolute lymphocyte count; ALT = alanine transaminase; ANC = absolute neutrophil count; AST = aspartate aminotransferase; BAR = baricitinib; BMP = basic metabolic panel; BP = blood pressure; CBC = complete blood count; CD4 = CD4 T lymphocyte; COPD = chronic obstructive pulmonary disease; CrCl = creatinine clearance; CVA = cerebral vascular accident; CYP = cytochrome P450; DEX = dexamethasone; DILI = drug-induced liver injury; DVT = deep vein thrombosis; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; EUA = Emergency Use Authorization; FDA = Food and Drug Administration; GDH-PQQ = glucose dehydrogenase pyrroloquinoline quinone; GI = gastrointestinal; HBV = hepatitis B virus; Hgb = hemoglobin; HSR = hypersensitivity reaction; HSV = herpes simplex virus; HTN = hypertension; ICU = intensive care unit; IL = interleukin; IV = intravenous; JIA = juvenile idiopathic arthritis; MDI = metered dose inhaler; MI = myocardial infarction; MV = mechanical ventilation; NaCl = sodium chloride; NIV = noninvasive ventilation; NMBA = neuromuscular blocking agent; OAT = organic anion transporter; the Panel = the COVID-19 Treatment Guidelines Panel; PE = pulmonary embolism; PLT = platelet count; PO = oral; SUBQ = subcutaneous; TB = tuberculosis; TNF = tumor necrosis factor

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Antithrombotic Therapy in Patients With COVID-19

Last Updated: February 29, 2024

Summary Recommendations

Chronic Anticoagulant and Antiplatelet Therapy

- The COVID-19 Treatment Guidelines Panel (the Panel) recommends that patients with COVID-19 who are receiving anticoagulant or antiplatelet therapies for underlying conditions continue these medications unless significant bleeding develops or other contraindications are present **(AIII)**.
- Before prescribing ritonavir-boosted nirmatrelvir (Paxlovid) to patients who are receiving anticoagulant or antiplatelet therapy, clinicians should carefully review the patient's concomitant medications to evaluate potential drug-drug interactions. It may be necessary to modify the dosage of the antithrombotic agent, switch to another antithrombotic agent, or prescribe an alternative COVID-19 therapy. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

Screening and Evaluation for Venous Thromboembolism

- There is insufficient evidence for the Panel to recommend either for or against routine screening for venous thromboembolism (VTE) in patients with COVID-19 who do not have signs or symptoms of VTE, regardless of the status of their coagulation markers.
- For hospitalized patients with COVID-19 who experience rapid deterioration of pulmonary, cardiac, or neurological function or sudden, localized loss of peripheral perfusion, the Panel recommends evaluating the patients for thromboembolic disease **(AIII)**.

Antithrombotic Therapy for Nonhospitalized Adults Without Evidence of Venous Thromboembolism

- In nonhospitalized patients with COVID-19, the Panel **recommends against** the use of anticoagulant and antiplatelet therapy (i.e., **aspirin, P2Y₁₂ inhibitors**) for the prevention of VTE or arterial thrombosis, except in a clinical trial **(AIIa)**. This recommendation does not apply to patients with other indications for antithrombotic therapy.

Antithrombotic Therapy for Hospitalized, Nonpregnant Adults Without Evidence of Venous Thromboembolism

- The Panel **recommends against** using anticoagulant or antiplatelet therapy to prevent arterial thrombosis outside of the usual standard of care for patients without COVID-19 **(AIII)**.
- In hospitalized patients, **low-molecular-weight heparin (LMWH)** or **unfractionated heparin (UFH)** is preferred over oral anticoagulants **(AIII)**. Because these types of heparin have shorter half-lives, their effects can be reversed quickly. They can also be administered intravenously or subcutaneously, and they have fewer drug-drug interactions than oral anticoagulants.
- When heparin is used, LMWH is preferred over UFH.

For adults who require low-flow oxygen and do not require intensive care unit (ICU)-level care:

- The Panel recommends the use of a **therapeutic dose of heparin** for patients with D-dimer levels above the upper limit of normal who require low-flow oxygen and who do not have an increased risk of bleeding **(CIIa)**.
 - Contraindications for the use of therapeutic anticoagulation in patients with COVID-19 include a platelet count <50,000 cells/ μ L, hemoglobin <8 g/dL, the need for dual antiplatelet therapy, bleeding within the past 30 days that required an emergency department visit or hospitalization, a history of a bleeding disorder, or an inherited or active acquired bleeding disorder. This list is based on the exclusion criteria from clinical trials; patients with these conditions have an increased risk of bleeding
- In patients without VTE who have started treatment with therapeutic doses of heparin, treatment should continue until 1 of the following occurs, whichever comes first:
 - The patient receives 14 days of treatment, at which time, they should be switched to prophylactic anticoagulation until hospital discharge;
 - The patient is transferred to the ICU, and prophylactic anticoagulation should be administered for the remainder of the hospitalization period; *or*

Summary Recommendations, continued

- The patient is discharged from the hospital.
- The Panel recommends the use of a **prophylactic dose of heparin** for patients who do not meet the criteria for receiving therapeutic heparin or are not receiving a therapeutic dose of heparin for other reasons, unless a contraindication exists (**AI**).
- There is insufficient evidence for the Panel to recommend either for or against the use of a therapeutic dose of apixaban for VTE prophylaxis or the prevention of COVID-19 progression.
- The Panel **recommends against** the use of a **therapeutic dose of rivaroxaban** for VTE prophylaxis or the prevention of COVID-19 progression (**AIIa**).
- The Panel **recommends against** the use of antiplatelet therapy to prevent COVID-19 progression or death in noncritically ill patients (**BIIa**).

For adults who require ICU-level care, including those receiving high-flow oxygen:

- The Panel recommends using a **prophylactic dose of heparin** as VTE prophylaxis, unless a contraindication exists (**AI**).
- The Panel **recommends against** the use of a therapeutic dose of anticoagulation for VTE prophylaxis (**BI**).
- For patients who start on a therapeutic dose of heparin in a non-ICU setting due to COVID-19 and then transfer to the ICU, the Panel recommends switching from the therapeutic dose to a **prophylactic dose of heparin**, unless VTE is confirmed (**BIII**).
- There is insufficient evidence for the Panel to recommend either for or against the use of an intermediate dose of anticoagulation for VTE prophylaxis.
- There is insufficient evidence for the Panel to recommend either for or against the use of antiplatelet therapy in critically ill patients with COVID-19.

Antithrombotic Therapy for Patients Discharged From the Hospital

- The Panel **recommends against** routinely continuing VTE prophylaxis in patients with COVID-19 after hospital discharge unless they have another indication for anticoagulation (**AIIa**).

Pregnant and Lactating Patients

- The Panel recommends that pregnant patients who are receiving anticoagulant or antiplatelet therapies for underlying conditions continue these medications after they receive a diagnosis of COVID-19 (**AIII**).
- The Panel recommends the use of a prophylactic dose of anticoagulation for pregnant patients who are hospitalized for manifestations of COVID-19, unless a contraindication exists (**BIII**).
- Because pregnant patients were not included in most of the clinical trials that evaluated the use of therapeutic anticoagulation in people with COVID-19, there is insufficient evidence for the Panel to recommend either for or against the use of therapeutic anticoagulation in pregnant patients with COVID-19 who do not have evidence of VTE.
- As in nonpregnant patients with COVID-19, VTE prophylaxis after hospital discharge is not routinely recommended for pregnant patients (**BIII**). Clinicians should consider an individual patient's VTE risk factors when making decisions about continuing VTE prophylaxis after discharge in pregnant or postpartum patients.
- The use of anticoagulant therapy during labor and delivery requires specialized care and planning. The management of anticoagulant therapy in pregnant patients with COVID-19 should be similar to the management used for pregnant patients with other conditions (**AIII**).
- UFH, LMWH, and warfarin do not accumulate in breast milk and do not induce an anticoagulant effect in the newborn; therefore, they can be used by breastfeeding individuals who require VTE prophylaxis or treatment (**AIII**).

Children With COVID-19 or MIS-C

- For the Panel's recommendations on the use of antithrombotic therapy in children, see [Therapeutic Management of Hospitalized Children With COVID-19](#) and [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

Association Between COVID-19 and Thromboembolism

COVID-19 has been associated with inflammation and a prothrombotic state, with increases in levels of fibrin, fibrin degradation products, fibrinogen, and D-dimer.^{1,2} In some studies, elevations in these markers have been associated with worse clinical outcomes.^{3,4}

Studies have reported varying incidences of venous thromboembolism (VTE) in patients with COVID-19. A meta-analysis of studies of hospitalized patients with COVID-19 who received VTE prophylaxis found an overall VTE prevalence of 14.1% (95% CI, 11.6–16.9).⁵ The VTE prevalence was higher in studies that used ultrasound screening (40.3%; 95% CI, 27.0–54.3) than in studies that did not (9.5%; 95% CI, 7.5–11.7). In randomized controlled trials conducted prior to the pandemic, the incidence of VTE in hospitalized patients who received VTE prophylaxis ranged from 0.3% to 1% for symptomatic VTE and from 2.8% to 5.6% for VTE overall.^{6–8} In randomized trials, the VTE incidence among critically ill patients without COVID-19 who received a prophylactic dose of anticoagulants ranged from 5% to 16%, and a prospective cohort study of critically ill patients with sepsis reported a VTE incidence of 37%.^{9–12}

Guidelines for the use of antithrombotic therapy in patients with COVID-19 have been released by multiple organizations, including the American College of Chest Physicians,¹³ the American Society of Hematology,¹⁴ the Anticoagulation Forum,¹⁵ the International Society on Thrombosis and Haemostasis,¹⁶ the Italian Society on Thrombosis and Haemostasis,¹⁷ the National Institute for Health and Care Excellence,¹⁸ and the Royal College of Physicians.¹⁹ The American College of Chest Physicians also has guidance on the use of antithrombotic therapy to treat arterial thrombosis in people with COVID-19.²⁰

The guidelines referenced above agree that hospitalized, nonpregnant patients with COVID-19 should receive, at a minimum, a prophylactic dose of anticoagulation to prevent VTE. The [National Institute for Health and Care Excellence guidelines](#) state: “Consider a treatment dose of a low-molecular-weight heparin (LMWH) for young people and adults with COVID-19 who need low-flow oxygen and who do not have an increased bleeding risk.” Results from clinical trials have provided further information on the safety and efficacy of different antithrombotic strategies for patients with COVID-19.

Chronic Anticoagulant or Antiplatelet Therapy

The COVID-19 Treatment Guidelines Panel (the Panel) recommends that patients with COVID-19 who are receiving anticoagulant or antiplatelet therapies for underlying conditions continue these medications unless significant bleeding develops or other contraindications are present (**AIII**). Outpatients with COVID-19 who are receiving warfarin and are in isolation and unable to have international normalized ratio monitoring may be candidates for switching to direct oral anticoagulant therapy.²¹ Patients with a mechanical heart valve, a ventricular assist device, valvular atrial fibrillation, or antiphospholipid antibody syndrome and patients who are lactating should not discontinue treatment with warfarin (**AIII**).

Before prescribing ritonavir-boosted nirmatrelvir (Paxlovid) to patients who are receiving anticoagulant or antiplatelet therapy, clinicians should carefully review the patient’s concomitant medications to evaluate potential drug-drug interactions. It may be necessary to modify the dosage of the antithrombotic agent, switch to another antithrombotic agent, or prescribe an alternative COVID-19 therapy. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

Screening and Evaluation for Venous Thromboembolism

VTE guidelines for patients without COVID-19 have recommended against performing routine screening ultrasounds in critically ill patients because no study has shown that this strategy reduces

the rate of subsequent symptomatic thromboembolic complications.²² Although the incidence of thromboembolic events, especially pulmonary embolism, can be high among hospitalized patients with COVID-19, no published data demonstrate the clinical utility of using lower extremity ultrasounds as routine surveillance for deep vein thrombosis in this population.

There is insufficient evidence for the Panel to recommend either for or against routine screening for VTE in patients with COVID-19 who do not have signs or symptoms of VTE, regardless of the status of their coagulation markers. For hospitalized patients with COVID-19 who experience rapid deterioration of pulmonary, cardiac, or neurological function or sudden, localized loss of peripheral perfusion, the Panel recommends evaluating the patients for thromboembolic disease (**AIII**).

Selection of Anticoagulant or Antiplatelet Drugs

Whenever anticoagulant or antiplatelet therapy is used, potential drug-drug interactions with other concomitant medications must be considered. The [Liverpool COVID-19 Drug Interactions website](#) provides a list of drug-drug interactions. In hospitalized patients, **LMWH** or **unfractionated heparin** (UFH) is preferred over oral anticoagulants (**AIII**). Because these types of heparin have shorter half-lives, their effects can be reversed quickly. They can also be administered intravenously or subcutaneously (SUBQ), and they have fewer drug-drug interactions than oral anticoagulants.

Management of Nonhospitalized Adults

ACTIV-4b was a placebo-controlled, randomized trial that evaluated the efficacy of using aspirin or prophylactic doses (2.5 mg) or therapeutic doses (5 mg) of apixaban in outpatients with COVID-19 aged >40 years.²³ After 657 outpatients were randomized, the trial was stopped in June 2021 due to a low event rate for the composite outcome of thromboembolic events, hospitalization, or death (1 patient each in the placebo, aspirin, and apixaban 2.5 mg arms and 2 patients in the apixaban 5 mg arm). The median time from randomization to receipt of treatment was 3 days, and 22 patients were hospitalized for COVID-19 prior to initiation of the study drugs.

Two trials evaluated the use of LMWH and its impact on hospitalization and mortality in outpatients with COVID-19. The ETHIC trial was a multicenter, open-label randomized controlled trial of unvaccinated outpatients with COVID-19.²⁴ Adults with at least 1 risk factor for severe disease were randomized to receive enoxaparin 40 mg SUBQ once daily (if they weighed <100 kg) or enoxaparin 40 mg SUBQ twice daily (if they weighed >100 kg) for 21 days or standard of care. The study was terminated early due to a low event rate and slow accrual of participants. There was no difference between the arms in the number of patients who met the composite endpoint of all-cause mortality or all-cause hospitalization (12 of 105 patients [11%] in the enoxaparin arm vs. 12 of 114 patients [11%] in the standard of care arm). Four of the 12 patients in the enoxaparin arm who were admitted to the hospital required acute medical care or intensive care unit (ICU) admission (3 patients required mechanical ventilation or extracorporeal membrane oxygenation [ECMO]). There were no hospitalizations in the standard of care arm. Bleeding events occurred in 2 patients who received enoxaparin and in 1 patient who received standard of care.

The OVID trial was a multicenter, open-label randomized controlled trial of 472 outpatients with COVID-19 aged >50 years who were randomized to receive enoxaparin 40 mg SUBQ once daily for 14 days or standard of care.²⁵ The study was terminated after recruiting 50% of the planned number of participants due to a low probability that enoxaparin would be superior to standard of care for the primary outcome. There was no difference between the arms in the number of patients who met the primary composite endpoint of all-cause hospitalization or mortality (8 of 234 patients [3%] in the enoxaparin arm vs. 8 of 238 patients [3%] in the standard of care arm). No major bleeding events

occurred during the study.

The clinical data for these trials are summarized in [Table 6a](#).

In nonhospitalized patients with COVID-19, the Panel **recommends against** the use of anticoagulant and antiplatelet therapy (i.e., **aspirin, P2Y12 inhibitors**) for the prevention of VTE or arterial thrombosis, except in a clinical trial (**AIIa**). This recommendation does not apply to patients with other indications for antithrombotic therapy.

Management of Hospitalized Adults

Several studies have evaluated the risks and benefits of using prophylactic or therapeutic doses of anticoagulants in patients with COVID-19. Observational studies and clinical trials have examined the effects of anticoagulation on mortality, progression of COVID-19, thrombosis, and bleeding. Some of these studies are outlined below. Observational studies are included here only when evidence from clinical trials is not available. The clinical data for these trials are summarized in [Table 6a](#).

Prophylactic Dose of Anticoagulation Versus No Anticoagulation

An observational study of 4,297 veterans hospitalized with COVID-19 evaluated the use of prophylactic anticoagulation.²⁶ A prophylactic dose of anticoagulation was administered to 3,627 patients with COVID-19 within 24 hours of hospital admission. An inverse probability of treatment weighted analysis showed a cumulative 30-day mortality of 14% among patients who received prophylactic anticoagulation and 19% among patients who were not treated with anticoagulation (HR 0.73; 95% CI, 0.66–0.81). Patients treated with the prophylactic dose did not have a significant difference in the risk of bleeding that required transfusion when compared with patients who were not treated with anticoagulation (HR 0.87; 95% CI, 0.71–1.05). Overall, the study demonstrated that patients with COVID-19 may benefit from a prophylactic dose of anticoagulation.

Therapeutic Versus Prophylactic Doses of Heparin in Hospitalized Patients Who Do Not Require Intensive Care Unit-Level Care

Four open-label randomized controlled trials (the large ATTACC/ACTIV-4a/REMAP-CAP multiplatform trial and the FREEDOM trial, and the smaller RAPID and HEP-COVID trials) compared therapeutic doses of heparin to prophylactic or intermediate doses of the anticoagulant in selected hospitalized patients who did not require intensive care. The inclusion and exclusion criteria for these studies varied, but most of the studies included patients who required supplemental oxygen and had no risk of a major bleeding event. In the larger multiplatform trial, therapeutic doses of heparin increased the number of organ support-free days but did not significantly affect mortality or length of hospitalization when compared with prophylactic doses of heparin.²⁷ In the FREEDOM trial, there was no difference between the therapeutic and prophylactic anticoagulation arms in the occurrence of the 30-day primary composite outcome of all-cause mortality, need for ICU-level care, systemic thromboembolism, or ischemic stroke. In a secondary analysis, 30-day mortality was significantly lower in patients who received therapeutic enoxaparin than in patients who received prophylactic enoxaparin.²⁸ However, only a small proportion of patients received concomitant corticosteroids or remdesivir as standard of care, and the trial was stopped early due to slow recruitment.

The RAPID trial enrolled patients with elevated D-dimer levels and hypoxemia.²⁹ The patients were randomized to receive therapeutic or prophylactic doses of heparin. There was no statistically significant difference between the arms for the primary endpoint, which was a composite of ICU admission, noninvasive ventilation (NIV) or mechanical ventilation, or death by Day 28. However, the therapeutic dose of heparin reduced the risk of all-cause death, a secondary outcome.

The HEP-COVID trial enrolled patients who required supplemental oxygen and had a D-dimer value >4 times the upper limit of normal (ULN) or a sepsis-induced coagulopathy score of ≥ 4 .³⁰ There were significantly fewer occurrences of the primary endpoint of VTE, arterial thromboembolism, or all-cause death within 32 days of randomization in the therapeutic LMWH arm than in the prophylactic LMWH arm, but there was no difference between arms for the outcome of death within 32 days.

Given the results of the ATTACC/ACTIV-4a/REMAP-CAP, FREEDOM, RAPID, and HEP-COVID trials, for hospitalized, nonpregnant adults with COVID-19 who do not require ICU-level care and have no evidence of VTE:

- The Panel recommends the use of a **therapeutic dose of heparin** for patients with D-dimer levels above the ULN who require low-flow oxygen and who do not have an increased risk of bleeding (**CIIa**).
 - Contraindications for the use of therapeutic anticoagulation in patients with COVID-19 include a platelet count $<50,000$ cells/ μ L, hemoglobin <8 g/dL, the need for dual antiplatelet therapy, bleeding within the past 30 days that required an emergency department visit or hospitalization, a history of a bleeding disorder, or an inherited or active acquired bleeding disorder. This list is based on the exclusion criteria from clinical trials; patients with these conditions have an increased risk of bleeding.
- LMWH is preferred over UFH because of its ease of administration and because LMWH was the predominant form of heparin used in the clinical trials for COVID-19.
- In patients without VTE who have started treatment with therapeutic doses of heparin, treatment should continue until 1 of the following occurs, whichever comes first:
 - The patient receives 14 days of treatment, at which time, they should be switched to prophylactic anticoagulation until hospital discharge;
 - The patient is transferred to the ICU, and prophylactic anticoagulation should be administered for the remainder of the hospitalization period; *or*
 - The patient is discharged from the hospital.
- Patients with predicted hospitalizations of <72 hours were excluded from the multiplatform ATTACC/ACTIV-4a/REMAP-CAP trial. It is currently unknown whether the benefits of using therapeutic doses of anticoagulation for short hospital stays outweigh the risks.
- The Panel recommends the use of a **prophylactic dose of heparin** for patients who do not meet the criteria for receiving therapeutic heparin or are not receiving a therapeutic dose of heparin for other reasons, unless a contraindication exists (**AI**).

Prophylactic Versus Intermediate or Therapeutic Doses of Heparin in Hospitalized Patients Who Require Intensive Care Unit-Level Care

Several randomized controlled trials have evaluated the role of therapeutic doses of heparin in reducing the incidence of VTE events or death in patients in the ICU setting. The clinical data for these trials are summarized in [Table 6a](#).

The INSPIRATION trial compared the use of an intermediate dose of enoxaparin (1 mg/kg SUBQ once daily) to a prophylactic dose of enoxaparin (40 mg/kg SUBQ once daily) in patients with COVID-19 who were in the ICU.³¹ The study reported no difference between the arms in the occurrence of the composite endpoint of adjudicated VTE, arterial thrombosis, ECMO, or all-cause mortality. Major bleeding occurred in 2.5% of patients in the intermediate-dose anticoagulation arm and in 1.4% of patients who received the prophylactic dose. Overall, there was no significant benefit of receiving an

intermediate dose of anticoagulation for patients with COVID-19 who were in the ICU.

The ANTICOVID trial was an open-label study of hospitalized patients with COVID-19 who required oxygen therapy.³² Patients were randomized to receive a prophylactic dose of LMWH (n = 114), an intermediate dose of LMWH (n = 110), or a therapeutic dose of LMWH (n = 110). Patients in the study received either enoxaparin or tinzaparin. Patients underwent a computed tomography scan at baseline to ensure they did not have a pulmonary embolism. The study excluded patients weighing <40 kg or >100 kg.

The primary hierarchical outcome for this study was all-cause mortality or time to clinical improvement by Day 28. There was no difference between the arms for this outcome. The study also evaluated net clinical outcome, which was defined as a composite of venous and arterial thrombosis, major bleeding events (as defined by the International Society on Thrombosis and Haemostasis), or all-cause mortality by Day 28. A smaller percentage of patients who received the intermediate dose of anticoagulation met the net clinical outcome criteria compared with those who received the prophylactic dose of anticoagulation (16.4% vs. 29.8%; absolute difference -13.5%; $P = 0.02$). There was no statistically significant difference in the occurrence of the net clinical outcome between the therapeutic-dose anticoagulation arm and the prophylactic-dose or intermediate-dose arms. No difference in the occurrence of major bleeding events was seen among the study arms.

Tinzaparin is not available in the United States. This lack of availability, combined with the conflicting results of the INSPIRATION and ANTICOVID trials, has led the Panel to conclude that there is insufficient evidence to recommend either for or against the use of an intermediate dose of anticoagulation for VTE prophylaxis.

The multiplatform ATTACC/ACTIV-4a/REMAP-CAP trial compared the effectiveness of a therapeutic dose of heparin or LMWH with usual care in reducing the number of organ support-free days among critically ill patients with COVID-19.²⁷ All 3 trials were stopped for futility. The doses of heparin that were administered to patients in the usual care arm varied. The median number of organ support-free days and likelihood of survival to hospital discharge did not differ between the arms. Major bleeding occurred in 4% of patients who received therapeutic anticoagulation and in 2% of patients who received usual care. Therapeutic doses of heparin showed no significant benefit in patients with COVID-19 who were admitted to the ICU.

The COVID-PACT trial was a multicenter trial with a 2 x 2 factorial design.³³ Critically ill patients with COVID-19 were randomized to receive a therapeutic dose or a prophylactic dose of anticoagulation. They were also randomized to receive either clopidogrel or no antiplatelet therapy. The trial was stopped early because the decreasing number of ICU admissions for patients with COVID-19 made recruitment difficult. There was no difference between the arms in the occurrence of the primary endpoint (a composite of VTE or arterial thrombotic events at hospital discharge or Day 28). More moderate to severe bleeding events occurred among patients who were treated with therapeutic anticoagulation than among those who received prophylactic anticoagulation.

Given the results from the studies discussed above, for hospitalized, nonpregnant adults with COVID-19 who require ICU-level care and who do not have documented or suspected VTE:

- The Panel recommends using a **prophylactic dose of heparin** as VTE prophylaxis, unless a contraindication exists (**AI**).
- The Panel **recommends against** the use of a therapeutic dose of anticoagulation for VTE prophylaxis (**BI**).
- For patients who start on a therapeutic dose of heparin in a non-ICU setting due to COVID-19

and then transfer to the ICU, the Panel recommends switching from the therapeutic dose to a **prophylactic dose of heparin**, unless VTE is confirmed (**BIII**).

- There is insufficient evidence for the Panel to recommend either for or against the use of an intermediate dose of anticoagulation for VTE prophylaxis.

Apixaban or Rivaroxaban in Hospitalized Patients

The FREEDOM trial randomized patients 1:1:1 to receive a therapeutic dose of apixaban, a therapeutic dose of enoxaparin, or a prophylactic dose of enoxaparin.²⁸ The trial showed no difference in the occurrence of the primary composite endpoint between the therapeutic and prophylactic anticoagulation arms. In a secondary analysis, fewer deaths were reported at 30 days among patients who were treated with a therapeutic dose of apixaban than among those who received prophylactic enoxaparin (5% vs. 7%; HR 0.7; 95% CI, 0.49–0.99). Only a small proportion of patients were treated with dexamethasone or remdesivir as part of usual care; both of these drugs have been shown to have a benefit in this population. This open-label trial was also stopped early due to slow recruitment.

The FREEDOM trial is the only study that has evaluated the use of therapeutic apixaban in patients with COVID-19; in contrast, 4 trials have evaluated the use of therapeutic heparin. Additionally, oral anticoagulants have the potential for drug-drug interactions and present unique challenges for managing hemorrhages. Due to these limitations, there is insufficient evidence for the Panel to recommend either for or against the use of a therapeutic dose of apixaban for VTE prophylaxis or the prevention of COVID-19 progression.

The ACTION trial randomized adults who were hospitalized with COVID-19 and elevated D-dimer levels (defined as levels that were above the laboratory ULN) to receive rivaroxaban 20 mg once daily for 30 days (n = 311) or usual care (n = 304).³⁴ A heterogeneous population was included; 25% of patients did not require oxygen, 60% were treated with low-flow oxygen, and 15% needed high-flow oxygen, NIV, or mechanical ventilation. No statistical difference was found between the arms for the composite endpoint of time to death, hospitalization duration, or oxygen use duration (hierarchical analysis; win ratio 0.86; 95% CI, 0.59–1.22) or for the individual components of the composite endpoint. The probability of clinically relevant, nonmajor bleeding was greater in the rivaroxaban arm (5% in the rivaroxaban arm vs. 1% in the usual care arm; relative risk 5.23; 95% CI, 1.54–17.77), but for major bleeding events, the difference in probability between the arms was not significant (3% in the rivaroxaban arm vs. 1% in the usual care arm; relative risk 2.45; 95% CI, 0.78–7.73). Given the lack of benefit and the increased risk of bleeding events, the Panel **recommends against** the use of a **therapeutic dose of rivaroxaban** for VTE prophylaxis or the prevention of COVID-19 progression (**AIIa**).

Antiplatelet Therapy Versus Usual Care in Hospitalized Patients

The RECOVERY trial randomized hospitalized adults with COVID-19 to receive usual care plus aspirin 150 mg once daily (n = 7,351) or usual care only (n = 7,541).³⁵ At enrollment, 38% of the patients required NIV or mechanical ventilation. Mortality at 28 days was 17% in both arms (rate ratio 0.96; 95% CI, 0.89–1.04). Among patients who were not receiving mechanical ventilation at baseline, there was no difference between the arms in the proportion of patients who progressed to requiring mechanical ventilation or who died (21% in the aspirin arm vs. 22% in the usual care arm; rate ratio 0.96; 95% CI, 0.90–1.03). Among those treated with aspirin, the incidence of thrombotic events was lower (4.6% vs. 5.3%; absolute difference 0.6%; SE 0.4%), and the incidence of major bleeding events was higher (1.6% vs. 1.0%; absolute difference 0.6%; SE 0.2%). Overall, the use of aspirin was associated with an increase in the incidence of major bleeding events and did not reduce the risk of death.

The ACTIV-4a trial compared the use of P2Y12 inhibitor therapy plus a therapeutic dose of heparin to a therapeutic dose of heparin alone in hospitalized patients with COVID-19.³⁶ In this study, enrollment of noncritically ill patients was stopped early due to futility; the combination therapy did not increase the number of organ support-free days. The limitations of this study include the open-label design, the use of different P2Y12 inhibitors, and the small sample size.

After reviewing the results of the ACTIV-4a and RECOVERY trials, the Panel **recommends against** the use of antiplatelet therapy to prevent COVID-19 progression or death in noncritically ill patients (**BIIa**).

The REMAP-CAP study team randomized critically ill patients with COVID-19 to receive aspirin (n = 565), a P2Y12 inhibitor (n = 455), or no antiplatelet therapy (n = 529).³⁷ Treatment continued for 14 days or until hospital discharge, whichever came first. The aspirin and P2Y12 inhibitor arms were pooled for analysis because the criteria for equivalence were met. The trial was stopped early due to futility, as the median number of organ support-free days did not differ between the pooled antiplatelet arm and the control arm (7 days; IQR 1–16 days; 95.7% posterior probability of futility). There was no statistically significant difference between the arms in the number of patients who survived to hospital discharge (723 of 1,011 patients [71.5%] in the pooled antiplatelet arm vs. 354 of 521 patients [67.9%] in the control arm; median-adjusted OR 1.27; 95% CrI, 0.99–1.62). The pooled antiplatelet arm had improved survival by 90 days (median aHR 1.22; 95% CrI, 1.06–1.40). The use of antiplatelet therapy was associated with an increased incidence of major bleeding (2.1% in the pooled antiplatelet arm vs. 0.4% in the control arm; aOR 2.97; 95% CrI, 1.23–8.28; adjusted absolute risk difference of 0.8%; 95% CrI, 0.1% to 2.7%).

In the RECOVERY trial, the use of aspirin therapy was not associated with a reduction in mortality at 28 days in the subgroups of patients who required NIV or mechanical ventilation at baseline. In the REMAP-CAP trial, administering antiplatelet therapy to critically ill patients with COVID-19 improved 90-day survival but did not increase the number of organ support-free days. In both studies, the use of antiplatelet therapy was associated with an increased risk of bleeding. The COVID-PACT trial randomized 292 adult patients with COVID-19 who required ICU-level care to receive either clopidogrel or no antiplatelet therapy.³³ There was no difference between the arms in the incidence of VTE, arterial thrombotic events, or bleeding.

Given the inconsistent results of these trials, there is insufficient evidence for the Panel to recommend either for or against the use of antiplatelet therapy in critically ill patients with COVID-19. Eligible patients should be encouraged to participate in clinical trials that are evaluating the use of antiplatelet therapy.

The clinical data for the trials discussed above are summarized in [Table 6b](#).

Patients Discharged From the Hospital

For patients with a high risk of VTE who do not have COVID-19, post-discharge prophylaxis has been shown to be beneficial. The Food and Drug Administration approved the use of rivaroxaban 10 mg once daily for 31 to 39 days in these patients.³⁸ Inclusion criteria for the trials^{39,40} that studied post-discharge VTE prophylaxis included:

- A VTE risk score of ≥ 4 on the modified International Medical Prevention Registry on Venous Thromboembolism (IMPROVE) tool;⁴¹ *or*
- A VTE risk score ≥ 2 on the modified IMPROVE tool⁴² and a D-dimer level >2 times ULN.³⁹

The MICHELLE trial randomized 320 patients with COVID-19 and an IMPROVE score of ≥ 4 or an IMPROVE score of 2 to 3 with a D-dimer level >500 ng/mL to receive rivaroxaban 10 mg orally

once daily or no anticoagulation for 35 days.⁴³ The primary outcome was a composite of symptomatic VTE, fatal pulmonary embolism, symptomatic arterial thromboembolism, cardiovascular death, or asymptomatic VTE detected on screening imaging at Day 35. Five patients (3%) who were treated with rivaroxaban and 15 patients (9%) who did not receive anticoagulation experienced a thrombotic event (relative risk 0.33; 95% CI, 0.13–0.9). One patient who received rivaroxaban and 10 patients who did not receive anticoagulation experienced symptomatic events. No major bleeding events occurred, and 2 patients in each arm had clinically relevant, nonmajor bleeding. The open-label design and the inclusion of asymptomatic events that were detected on screening ultrasounds and computed tomography scans may have biased the results. Additionally, two-thirds of the screened patients did not meet the eligibility criteria for the trial, which limits the generalizability of the results.

The ACTIV-4c trial randomized 1,217 patients who were hospitalized for symptomatic COVID-19 for >48 hours to receive apixaban 2.5 mg orally twice daily or placebo at hospital discharge.⁴⁴ The 30-day composite endpoint of all-cause mortality, venous thrombosis, or arterial thrombosis occurred in 2.13% of patients in the apixaban arm and in 2.31% of patients in the placebo arm. Major bleeding events were infrequent, occurring in 2 patients (0.4%) in the apixaban arm and in 1 patient (0.2%) in the placebo arm. The trial's leadership and sponsors stopped the trial early because the event rate for the composite endpoint was lower than expected, and the decreasing number of hospitalizations for people with COVID-19 made recruitment difficult.

After reviewing the results of the MICHELLE and ACTIV-4c trials, the Panel **recommends against** routinely continuing VTE prophylaxis in patients with COVID-19 after hospital discharge unless they have another indication for anticoagulation (**AIIa**).

Although there is no clear benefit of administering anticoagulation after hospital discharge in all patients with COVID-19, results from the MICHELLE trial, which evaluated patients with COVID-19, and the MARINER trial, which evaluated patients who were hospitalized for other conditions and who had risk factors for VTE, suggest a possible benefit of using anticoagulation after discharge in patients who are at high risk of VTE. The need for VTE prophylaxis after a COVID-19–related hospital discharge should be assessed on a case-by-case basis. The criteria for assessing the risk of VTE in these patients are the same as the criteria used for patients who are hospitalized for other acute illnesses.

Pregnant and Lactating Patients

Because pregnancy is a hypercoagulable state, the risk of thromboembolism is greater in pregnant individuals than in nonpregnant individuals.⁴⁵ It is not yet known whether COVID-19 increases this risk, though some data do suggest that there is an increased risk. A cohort study in California compared perinatal outcomes among almost 44,000 pregnant people with and without COVID-19.⁴⁶ After adjusting for demographic factors and comorbidities, those with COVID-19 had a higher risk of severe maternal morbidity, preterm birth, and VTE.

In several other cohort studies of pregnant women with COVID-19 in the United States and Europe, VTE was not reported as a complication even among women with severe disease, although the receipt of prophylactic or therapeutic anticoagulation varied across the studies.^{47–49} Although there are currently not enough data to recommend either for or against the use of VTE prophylaxis, the [American College of Obstetricians and Gynecologists](#) notes that VTE prophylaxis can reasonably be considered for pregnant individuals who are hospitalized with COVID-19, particularly those who have severe disease. If there are no contraindications, the Society for Maternal-Fetal Medicine recommends using heparin or LMWH in pregnant patients who are critically ill or receiving mechanical ventilation.⁵⁰ Several professional societies, including the American Society of Hematology and the American College of Obstetricians and Gynecologists, have guidelines that specifically address the management of VTE in the context of pregnancy.^{51,52} If delivery is imminent, or if there are other risks for bleeding, the risk of bleeding may

outweigh the potential benefit of using VTE prophylaxis in pregnant individuals.

In nonpregnant people, D-dimer levels have been used to stratify VTE risk. However, physiologic increases in D-dimer levels may occur during pregnancy, making elevated D-dimer values an unreliable predictor that should not be used to evaluate VTE risk in pregnant people with COVID-19.⁵³⁻⁵⁵

In general, heparin compounds are the preferred anticoagulants to use during pregnancy. Because of its reliability and ease of administration, LMWH is recommended rather than UFH for the prevention and treatment of VTE in pregnant people.⁵² Direct-acting anticoagulants are not routinely recommended for use during pregnancy because of a lack of safety data for pregnant individuals.⁵¹ The use of warfarin to prevent or treat VTE should be avoided in pregnant individuals regardless of their COVID-19 status, especially during the first trimester, due to the concern for teratogenicity.

Specific recommendations for pregnant or lactating individuals with COVID-19 include:

- The Panel recommends that pregnant patients who are receiving anticoagulant or antiplatelet therapies for underlying conditions continue these medications after they receive a diagnosis of COVID-19 (**AIII**).
- The Panel recommends the use of a prophylactic dose of anticoagulation for pregnant patients who are hospitalized for manifestations of COVID-19, unless a contraindication exists (**BIII**).
- Because pregnant patients were not included in most of the clinical trials that evaluated the use of therapeutic anticoagulation in people with COVID-19, there is insufficient evidence for the Panel to recommend either for or against the use of therapeutic anticoagulation in pregnant patients with COVID-19 who do not have evidence of VTE.
- As in nonpregnant patients with COVID-19, VTE prophylaxis after hospital discharge is not routinely recommended for pregnant patients (**BIII**). Clinicians should consider an individual patient's VTE risk factors when making decisions about continuing VTE prophylaxis after discharge in pregnant or postpartum patients.
- The use of anticoagulant therapy during labor and delivery requires specialized care and planning. The management of anticoagulant therapy in pregnant patients with COVID-19 should be similar to the management used for pregnant patients with other conditions (**AIII**).
- UFH, LMWH, and warfarin do not accumulate in breast milk and do not induce an anticoagulant effect in the newborn; therefore, they can be used by breastfeeding individuals who require VTE prophylaxis or treatment (**AIII**).

Children With COVID-19 or MIS-C

For the Panel's recommendations on the use of antithrombotic therapy in children, see [Therapeutic Management of Hospitalized Children With COVID-19](#) and [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

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Table 6a. Anticoagulant Therapy: Selected Clinical Trial Data

Last Updated: February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for anticoagulant therapy. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
ATTACC/ACTIV-4a/REMAP-CAP: Multiplatform, Open-Label RCT of Therapeutic Anticoagulation in Noncritically Ill, Hospitalized Patients With COVID-19 in 9 Countries¹		
Key Inclusion Criterion - Hospitalized with laboratory-confirmed SARS-CoV-2 infection without need for HFNC oxygen, NIV, MV, vasopressors, or inotropes Key Exclusion Criteria - Hospital discharge expected in ≤72 hours - Need for therapeutic anticoagulation or dual antiplatelet therapy - High bleeding risk Interventions - Therapeutic UFH or LMWH for 14 days or until hospital discharge, whichever came first (n = 1,190) - SOC, which included prophylactic UFH or LMWH (n = 1,054) Primary Endpoint - Organ support-free days by Day 21, as measured by an OS Key Secondary Endpoints - Survival until hospital discharge - Hospital LOS - Thrombosis or major bleeding events	Participant Characteristics - Median age 59 years; 59% men; median BMI 30 52% with HTN; 30% with DM; 11% with CVD 66% required low-flow oxygen. - D-dimer level: 48.4% <2 times ULN 28.4% ≥2 times ULN 23.1% unknown 62% on corticosteroids; 36% on RDV Primary Outcomes - Therapeutic anticoagulation was superior to SOC for organ support-free days (aOR 1.27; 95% CrI, 1.03–1.58; 99% posterior probability). 4% absolute difference in survival until hospital discharge without organ support that favored therapeutic arm (95% CrI, 0.5–7.2) - Outcome was consistent across D-dimer stratum. Secondary Outcomes - Survival until hospital discharge: 92% in both arms - No difference between arms in hospital LOS (aOR 1.03; 95% CrI, 0.94–1.13) - Thrombosis: 1% in therapeutic arm vs. 2% in SOC arm - Major bleeding events: 2% in therapeutic arm vs. 1% in SOC arm	Key Limitations - Open-label study - Dose for anticoagulation varied in SOC arm (27% received intermediate dose of thromboprophylaxis). - Inclusion criteria for hospital LOS and ICU-level care differed across trials. - Trial only enrolled 17% of screened patients. Interpretation - Therapeutic heparin increased the number of organ support-free days and decreased the number of patients requiring organ support. - Therapeutic heparin did not significantly affect hospital LOS or the number of major thrombosis events or deaths. - Major bleeds occurred 1% more frequently in the therapeutic arm than in the SOC arm.

Methods	Results	Limitations and Interpretation
RAPID: Open-Label RCT of Therapeutic Heparin in Moderately Ill, Hospitalized Patients With COVID-19 in 6 Countries²		
Key Inclusion Criteria - Hospitalized with COVID-19 and D-dimer level ≥ 2 times ULN or any elevated D-dimer level and $\text{SpO}_2 \leq 93\%$ on room air - Hospitalized < 5 days Key Exclusion Criteria - Need for therapeutic anticoagulation - Receiving dual antiplatelet therapy - High bleeding risk Interventions - Therapeutic UFH or LMWH for 28 days or until hospital discharge or death ($n = 228$) - Prophylactic UFH or LMWH for 28 days or until hospital discharge or death ($n = 237$) Primary Endpoint - Composite of ICU admission, NIV or MV, or death at 28 days Key Secondary Endpoints - All-cause death at 28 days - Mean number of organ support-free days - VTE - Major bleeding events - Mean number of hospital-free days alive	Participant Characteristics - Median age 60 years; 57% men; mean BMI 30 48% with HTN; 34% with DM; 7% with CVD 91% with hypoxia; 6% received HFNC oxygen. - D-dimer level: 49% < 2 times ULN 51% ≥ 2 times ULN 69% on corticosteroids Primary Outcome - Composite of ICU admission, NIV or MV, or death at 28 days: 16% in therapeutic arm vs. 22% in prophylactic arm (OR 0.69; 95% CI, 0.43–1.10) Secondary Outcomes - All-cause death at 28 days: 2% in therapeutic arm vs. 8% in prophylactic arm (OR 0.22; 95% CI, 0.07–0.65) - Mean number of organ support-free days: 26 in therapeutic arm vs. 24 in prophylactic arm (OR 1.41; 95% CI, 0.9–2.21) - No difference between arms for VTE (1% in therapeutic arm vs. 3% in prophylactic arm) or major bleeding events (1% in therapeutic arm vs. 2% in prophylactic arm) - Mean number of hospital-free days alive: 20 in therapeutic arm vs. 18 in prophylactic arm (OR 1.09; 95% CI, 0.79–1.50)	Key Limitations - Open-label study - Trial only enrolled 12% of screened patients. Interpretation - Compared to prophylactic heparin, therapeutic heparin reduced mortality (a secondary endpoint) but had no effect on the primary composite endpoint of ICU admission, the need for NIV or MV, or death up to 28 days. - There were no differences between the arms in the percentages of patients who experienced VTE or major bleeding events.

Methods	Results	Limitations and Interpretation
HEP-COVID: Open-Label RCT of Therapeutic Heparin in High-Risk, Hospitalized Patients With COVID-19 in the United States³		
Key Inclusion Criteria • Hospitalized with COVID-19 and required supplemental oxygen • D-dimer level >4 times ULN or sepsis-induced coagulopathy score of ≥ 4 • Hospitalized <72 hours Key Exclusion Criteria • Need for therapeutic anticoagulation • Receiving dual antiplatelet therapy • High bleeding risk • CrCl <15 mL/min Interventions • Therapeutic LMWH until hospital discharge or primary endpoint met (n = 129) • Usual care of prophylactic or intermediate-dose LMWH until hospital discharge or primary endpoint met (n = 124) Primary Endpoint • Composite of VTE, ATE, or death from any cause within 32 days of randomization Key Safety Endpoint • Major bleeding events within 32 days	Participant Characteristics • Median age 67 years; 54% men; mean BMI 30 • 60% with HTN; 37% with DM; 75% with CVD • 64% received oxygen via nasal cannula; 15% received HFNC oxygen or NIV; 5% received MV. • 80% on corticosteroids Primary Outcomes • Composite of VTE, ATE, or death within 32 days: 29% in therapeutic arm vs. 42% in usual care arm (relative risk 0.68; 95% CI, 0.49–0.96) • Thrombotic events: 11% in therapeutic arm vs. 29% in usual care arm (relative risk 0.37; 95% CI, 0.21–0.66) • Death: 19% in therapeutic arm vs. 25% in usual care arm (relative risk 0.78; 95% CI, 0.49–1.23) • Non-ICU stratum composite of VTE, ATE, or death within 32 days: 17% in therapeutic arm vs. 36% in usual care arm (relative risk 0.46; 95% CI, 0.27–0.81) Safety Outcomes • Major bleeding events within 32 days: 5% in therapeutic arm vs. 2% in usual care arm (relative risk 2.88; 95% CI, 0.59–14.02) • Non-ICU stratum major bleeding events within 32 days: 2% in both arms	Key Limitations • Open-label study • Trial only enrolled 2% of screened patients. Interpretation • Compared to usual care, therapeutic LMWH reduced the incidence of VTE, ATE, and death. • Among patients who were not in the ICU, therapeutic LMWH significantly reduced the percentage of patients who experienced thrombotic events and did not increase the percentage of patients who experienced major bleeding events.

Methods	Results	Limitations and Interpretation
ACTION: Open-Label RCT of Therapeutic Rivaroxaban in Hospitalized Patients With COVID-19 in Brazil⁴		
Key Inclusion Criteria - Hospitalized with COVID-19 and elevated D-dimer level - Symptoms for ≤14 days Key Exclusion Criteria - Need for therapeutic anticoagulation - CrCl <30 mL/min - Receiving P2Y12 inhibitor therapy or aspirin >100 mg - High bleeding risk Interventions - Therapeutic anticoagulation for 30 days: rivaroxaban 15 mg or 20 mg once daily; if clinically unstable, enoxaparin 1 mg/kg twice daily or UFH (n = 311) - Usual care prophylactic anticoagulation with enoxaparin or UFH during hospitalization (n = 304) Primary Endpoint - Hierarchical composite of time to death, hospital duration, or oxygen use duration by Day 30 Key Secondary Endpoints - Thrombosis, with and without all-cause death - Death by Day 30 - Bleeding events	Participant Characteristics - Median age 57 years; 60% men; mean BMI 30 49% with HTN; 24% with DM; 5% with CAD - Critically ill: 7% in therapeutic arm vs. 5% in usual care arm 75% required oxygen: 60% received low-flow oxygen; 8% received HFNC oxygen; 1% received NIV; 6% received MV. 83% on corticosteroids Primary Outcome - No difference between arms in the composite of time to death, hospital duration, or oxygen use duration by Day 30 (win ratio 0.86; 95% CI, 0.59–1.22) Secondary Outcomes - No difference between therapeutic and usual care arms in: - Thrombosis: 7% vs. 10% - Death by Day 30: 11% vs. 8% - Any bleeding events: 12% in therapeutic arm vs. 3% in usual care arm - Major bleeding events: 3% in therapeutic arm vs. 1% in usual care arm - Clinically relevant, nonmajor bleeding events: 5% in therapeutic arm vs. 1% in usual care arm	Key Limitations - Open-label study - Trial only enrolled 18% of screened patients. - Therapeutic rivaroxaban was administered for a longer duration than prophylactic anticoagulation (30 days vs. a mean duration of 8 days). Interpretation - When compared with usual care, therapeutic rivaroxaban did not reduce mortality, hospital duration, oxygen use duration, or the percentage of patients who experienced thrombosis. - Patients who received therapeutic rivaroxaban had more clinically relevant, nonmajor bleeding events than those who received usual care. - The longer duration of therapy in the therapeutic arm may have influenced the difference in bleeding events.

Methods	Results	Limitations and Interpretation
FREEDOM: RCT of Anticoagulation Strategies in Noncritically Ill Patients Who Were Hospitalized With COVID-19 in 10 Countries⁵		
Key Inclusion Criterion - Hospitalized with symptomatic COVID-19 for <48 hours Key Exclusion Criteria - Need for therapeutic anticoagulation - CrCl <30 mL/min - Receiving P2Y12 inhibitor therapy or aspirin >100 mg per day - Anticipated hospitalization for <72 hours Interventions - Therapeutic apixaban 5 mg twice daily (n = 1,121) - Therapeutic enoxaparin 1 mg/kg twice daily (n = 1,136) - Usual care prophylactic enoxaparin (n = 1,141) Primary Endpoint 30-day composite of all-cause mortality, need for ICU-level care, systemic thromboembolism, or ischemic stroke. Endpoint assessed for the combined therapeutic arms vs. the prophylactic arm. Key Secondary Endpoints 30-day all-cause mortality - BARC type 3 or 5 bleeding events	Participant Characteristics - Median age 52 years; 59% men; mean BMI 26 32% with HTN; 19% with DM 22% on corticosteroids; 10% on RDV Primary Outcome 30-day composite outcome: 11.3% in combined therapeutic arms vs. 13.2% in prophylactic arm (HR 0.85; 95% CI, 0.69–1.04; <i>P</i> = 0.11) - Primary endpoint was not statistically significant when therapeutic enoxaparin or apixaban were compared to prophylactic enoxaparin. Secondary Outcomes 30-day all-cause mortality: 4.9% in therapeutic enoxaparin arm vs. 7.0% in prophylactic enoxaparin arm (HR 0.69; 95% CI, 0.49–0.99) 30-day all-cause mortality: 5.0% in therapeutic apixaban arm vs. 7.0% in prophylactic enoxaparin arm (HR 0.7; 95% CI, 0.49–0.99) - BARC type 3 or 5 bleeding events: 0.4% in combined therapeutic arms vs. 0.1% in prophylactic arm (IRR 3.96; 95% CI, 0.50–31.27)	Key Limitations - Open-label study - Trial was terminated early due to slow recruitment (3,452 of 3,600 planned patients recruited). - Minimal treatment with RDV or DEX as SOC for COVID-19 Interpretation - When compared with prophylactic enoxaparin, therapeutic apixaban and therapeutic enoxaparin did not reduce 30-day mortality, the need for ICU-level care, or the occurrence of thromboembolism or ischemic stroke. - Fewer patients died in the therapeutic enoxaparin and therapeutic apixaban arms than in the prophylactic enoxaparin arm. - There were no statistically significant differences between the arms in the percentages of patients who experienced severe bleeding events.

Methods	Results	Limitations and Interpretation
COVID-PACT: Open-Label RCT of Full-Dose Versus Prophylactic-Dose Anticoagulation in Adults With COVID-19 Who Were Receiving Intensive Care Unit-Level Care in the United States⁶		
Key Inclusion Criteria • Aged ≥18 years • Acute SARS-CoV-2 infection • Required ICU-level care for ≤96 hours prior to randomization Key Exclusion Criteria • Ongoing or planned use of full-dose anticoagulation or dual antiplatelet therapy • High bleeding risk • History of HIT • Ischemic stroke within 2 weeks Interventions • Full-dose anticoagulation until Day 28 or hospital discharge, whichever came first (n = 197) • Prophylactic anticoagulation (n = 193) • Eligible patients were also randomized 1:1 to receive clopidogrel or no antiplatelet therapy (n = 292) Primary Endpoint • Composite of VTE or ATE events by hospital discharge or Day 28. Events included death due to VTE or ATE, PE, clinically evident DVT, MI, ischemic stroke, systemic embolic event or acute limb ischemia, and clinically silent DVT. Key Secondary Endpoint • Individual outcomes listed above, with the exception of clinically silent DVT Key Safety Endpoints • Fatal or life-threatening bleeding events • Moderate or severe bleeding events	Participant Characteristics • Median age 61 years; 41% women; 71% White • 99% received HFNC oxygen, NIV, or MV; 15% received MV. • 41% received MV during the study. • 31% to 37% crossed over to an alternative study treatment during the study. Primary Outcome • Composite of VTE or ATE events by hospital discharge or Day 28: 12% in full-dose anticoagulation arm vs. 6% in prophylactic anticoagulation arm (win ratio 1.95; 95% CI, 1.08–3.55; <i>P</i> = 0.028) Secondary Outcome • Composite of clinically evident VTE or ATE events by hospital discharge or Day 28: 10% in full-dose anticoagulation arm vs. 6% in prophylactic anticoagulation arm (win ratio 1.79; 95% CI, 0.92–3.47; <i>P</i> = 0.087) Safety Outcomes • No fatal bleeding events occurred. • Life-threatening bleeding events: 4 (2.1%) in full-dose anticoagulation arm vs. 1 (0.5%) in prophylactic anticoagulation arm (<i>P</i> = 0.19) • Moderate or severe bleeding events: 15 (7.9%) in full-dose anticoagulation arm vs. 1 (0.5%) in prophylactic anticoagulation arm (<i>P</i> = 0.002)	Key Limitations • Open-label study (adjudication committee members were blinded to the study arms). • Trial was stopped early because the decreasing number of ICU admissions for patients with COVID-19 made recruitment difficult. • There was an unequal crossover between the arms, with a greater crossover from the prophylactic anticoagulation arm to the full-dose anticoagulation arm. Interpretation • Among patients with COVID-19 who required ICU-level care, patients who received full-dose anticoagulation had fewer VTE or ATE events but no survival benefit compared to those who received prophylactic anticoagulation. • The prevalence of moderate or severe bleeding events was higher among patients who received full-dose anticoagulation than among those who received prophylactic anticoagulation.

Methods	Results	Limitations and Interpretation
REMAP-CAP/ACTIV-4a/ATTACC: Multiplatform, Open-Label RCT of Therapeutic Anticoagulation in Critically Ill, Hospitalized Patients With COVID-19 in 20 Countries⁷		
Key Inclusion Criteria - Hospitalized with severe COVID-19 and receiving HFNC oxygen, NIV, MV, ECMO, vasopressors, or inotropes - Hospitalized <72 hours (ACTIV-4a, ATTACC) or <14 days (REMAP-CAP) Key Exclusion Criteria - Hospital discharge expected in ≤72 hours - Need for therapeutic anticoagulation or dual antiplatelet therapy - High bleeding risk Interventions - Therapeutic UFH or LMWH for 14 days or until hospital discharge, whichever came first (n = 534) - Usual care thromboprophylaxis (n = 564) Primary Endpoint - Number of organ support-free days by Day 21 Key Secondary Endpoints - Survival to hospital discharge - Any thrombosis - Composite of major thrombotic events or death - Bleeding events	Participant Characteristics - Median age 60 years; 70% men; median BMI 30 24% with chronic respiratory disease; 33% with DM; 10% with chronic kidney disease; 8% with severe CVD 32% received HFNC oxygen; 38% received NIV; 29% received MV. 18% on vasopressors; 82% on corticosteroids; 32% on RDV Primary Outcome - Median number of organ support-free days by Day 21: 4 in therapeutic arm vs. 5 in usual care arm (aOR 0.83; 95% CrI, 0.67–1.03; 99.9% posterior probability of futility; OR < 1.2) Secondary Outcomes - No difference between therapeutic and usual care arms in: - Survival to hospital discharge: 63% vs. 65% (aOR 0.84; 95% CrI, 0.64–1.11) - Thrombosis: 6% vs. 10% - Composite of major thrombotic events or death: 41% in both arms - Major bleeding events: 4% vs. 2% (aOR 1.48; 95% CrI, 0.75–3.04)	Key Limitations - Open-label study - Dose of thromboprophylaxis varied in usual care arm (51% received intermediate dose; 2% received subtherapeutic dose; 5% received therapeutic dose). - Inclusion criteria for hospital LOS and ICU-level care differed across trials. - Trial stopped for futility. Interpretation - In patients who required ICU-level care, therapeutic heparin did not reduce the duration of organ support or mortality. - Although the differences were not significant, patients who received therapeutic anticoagulation had more bleeding events and fewer thrombotic events than patients who received usual care.

Methods	Results	Limitations and Interpretation
INSPIRATION: Open-Label RCT of Intermediate-Dose Versus Prophylactic-Dose Anticoagulation in Patients With COVID-19 in Intensive Care Units in Iran⁸		
Key Inclusion Criteria - Admitted to ICU - Hospitalized <7 days Key Exclusion Criteria - Life expectancy <24 hours - Need for therapeutic anticoagulation - Bleeding or high bleeding risk Interventions - Intermediate-dose anticoagulation: enoxaparin 1 mg/kg once daily (n = 276) - Prophylactic-dose anticoagulation (n = 286) Primary Endpoint - Composite of adjudicated acute VTE, ATE, the need for ECMO, or all-cause mortality at 30 days Key Secondary Endpoints - All-cause mortality at 30 days - VTE - Bleeding events	Participant Characteristics - Median age 62 years; 58% men; median BMI 27 44% with HTN; 28% with DM; 14% with CAD 32% received NIV; 20% received MV. 23% on vasopressors; 93% on corticosteroids; 60% on RDV Primary Outcome - Composite of adjudicated acute VTE, ATE, the need for ECMO, or all-cause mortality at 30 days: 46% in intermediate-dose arm vs. 44% in prophylactic arm (OR 1.06; 95% CI, 0.76–1.48) Secondary Outcomes - No difference between intermediate-dose arm and prophylactic arm in: - All-cause mortality at 30 days: 43% vs. 41% - VTE: 3% in both arms - Major bleeding events and clinically relevant, nonmajor bleeding events: 6.3% vs. 3.1% (OR 2.02; 95% CI, 0.89–4.61)	Key Limitations - Open-label study - Not all patients received ICU-level care. Interpretation - Intermediate-dose anticoagulation did not significantly reduce the occurrence of VTE and ATE, the need for ECMO, or mortality. - Although the difference was not significant, patients who received intermediate-dose anticoagulation had more bleeding events than patients who received prophylactic-dose anticoagulation.

Methods	Results	Limitations and Interpretation
ANTICOVID: Open-Label RCT of Therapeutic-Dose Versus Intermediate-Dose Versus Prophylactic-Dose Anticoagulation in Patients With COVID-19 in Intensive Care Units in France⁹		
Key Inclusion Criterion - Hospitalized for <72 hours with hypoxemic COVID-19 pneumonia Key Exclusion Criteria - Weight <40 kg or >100 kg - Indication or contraindication for therapeutic anticoagulation - Bleeding or high bleeding risk Interventions - Therapeutic-dose anticoagulation: tinzaparin 175 IU/kg once daily or enoxaparin 100 IU/kg twice daily (n = 110) - Intermediate-dose anticoagulation: tinzaparin 7,000 IU once daily or enoxaparin 4,000 IU twice daily (n = 110) - Prophylactic-dose anticoagulation: tinzaparin 3,500 IU once daily or enoxaparin 4,000 IU once daily (n = 114) Primary Endpoint - Hierarchical outcome of all-cause mortality or time to clinical improvement of 2 points on a WHO scale by Day 28	Participant Characteristics - Mean age 58 years; 67% men; median BMI 27–28 31% with HTN; 18% with DM; 4% with CAD 23% received conventional oxygen; 61% received HFNC oxygen; 7% received NIV; 10% received MV. 92% on corticosteroids; 0.6% on RDV; 34% on tocilizumab; 3% on vasopressors Primary Outcome - No difference between arms for hierarchical outcome of all-cause mortality or time to clinical improvement by Day 28 Secondary Outcomes - Net clinical outcome by Day 28: 20.0% in therapeutic-dose arm vs. 16.4% in intermediate-dose arm vs. 29.8% in prophylactic-dose arm - Venous or arterial thrombosis: 5% in therapeutic-dose arm vs. 5% in intermediate-dose arm vs. 20% in prophylactic-dose arm - Major bleeding events: 4% in therapeutic-dose arm vs. 4% in intermediate-dose arm vs. 3% in prophylactic-dose arm	Key Limitations - Open-label study - Not all patients received ICU-level care. - Study excluded patients weighing >100 kg. - Tinzaparin is not available in the United States. Interpretation - The use of intermediate doses of anticoagulants improved the net clinical outcome by reducing the number of thrombosis events. - There was no difference between the arms in the occurrence of major bleeding events.

Methods	Results	Limitations and Interpretation
ACTIV-4B: Double-Blind RCT of Anticoagulant and Antiplatelet Therapy in Symptomatic, Nonhospitalized Patients With COVID-19 in the United States¹⁰		
Key Inclusion Criteria • Aged 40–80 years • Symptomatic SARS-CoV-2 infection • CrCl >30 mL/min • PLT >100,000 cells/μL Key Exclusion Criteria • Previously hospitalized with COVID-19 • Acute leukemia, recent major bleeding events, or indication or contraindication for anticoagulant or antiplatelet therapy Interventions • 558 of 657 randomized patients received study drugs. • Aspirin 81 mg PO once daily for 45 days (n = 144) • Prophylactic-dose anticoagulation: apixaban 2.5 mg PO twice daily for 45 days (n = 135) • Therapeutic-dose anticoagulation: apixaban 5 mg PO twice daily for 45 days (n = 143) • Placebo (n = 136) Primary Endpoint • Composite of all-cause mortality, symptomatic VTE, ATE, MI, stroke, or hospitalization for cardiovascular or pulmonary cause at 45 days Key Secondary and Safety Endpoints • Component events of primary endpoint • Major bleeding	Participant Characteristics • Listed characteristics are for all 657 randomized patients. • Median age 54 years; 59% women; median BMI 30; 13% Black, 28% Hispanic • 18% with DM; 20% with a history of smoking; 35% with HTN Primary Outcome • Composite of all-cause mortality, symptomatic VTE, ATE, MI, stroke, or hospitalization at 45 days: 1 (0.7%) in aspirin arm vs. 1 (0.7%) in apixaban 2.5 mg arm vs. 2 (1.4%) in apixaban 5 mg arm vs. 1 (0.7%) in placebo arm • Risk differences compared with placebo: • 0.0% (95% CI, not calculable) in aspirin arm • 0.7% (95% CI, -2.1% to 4.1%) in 2.5 mg apixaban arm • 1.4% (95% CI, -1.5% to 5.0%) in 5 mg apixaban arm • Cumulative incidence of primary outcomes did not significantly differ across treatment arms (log-rank $P = 0.78$). Secondary Outcome • No differences between all treatment arms and placebo arm in occurrence of individual components of the primary endpoint Safety Outcome • No major bleeding events occurred.	Key Limitation • Initial target sample size was 7,000 patients, but trial was terminated after enrolling only 9% of target because of a low event rate. Interpretation • Among symptomatic outpatients with COVID-19 who were clinically stable, treatment with aspirin or a therapeutic or prophylactic dose of apixaban did not reduce the risk of death, symptomatic VTE or ATE, or hospitalization for cardiovascular or pulmonary causes compared to placebo.

Methods	Results	Limitations and Interpretation
OVID: Open-Label RCT of Low-Molecular-Weight Heparin for Thromboprophylaxis in Symptomatic, Nonhospitalized Patients With COVID-19 in Germany and Switzerland¹¹		
Key Inclusion Criteria • Aged ≥50 years • Positive SARS-CoV-2 test result within past 5 days • Respiratory symptoms or temperature ≥37.5 °C Key Exclusion Criteria • Severe renal or hepatic dysfunction • Severe anemia or recent major bleeding events • Receiving dual antiplatelet therapy Interventions • Enoxaparin 40 mg SUBQ once daily for 14 days (n = 234) • SOC (n = 238) Primary Endpoint • Composite of any untoward hospitalization or all-cause death by Day 30 Key Secondary Endpoint • Composite of major arterial and venous cardiovascular events by Day 30 • Bleeding events	Participant Characteristics • Median age 57 years; 46% women; 96% White • Median of 3 days from COVID-19 diagnosis to randomization • 24% with HTN; 8% with DM; 5% with CVD • 9.5% received ≥1 COVID-19 vaccine doses. Primary Outcome • Composite of any untoward hospitalization or all-cause death by Day 30: 8 (3%) in enoxaparin arm vs. 8 (3%) in SOC arm (adjusted relative risk 0.98; 95% CI, 0.37–2.56; <i>P</i> = 0.96) Secondary Outcomes • Composite of major arterial and venous cardiovascular events by Day 30: 2 (1%) in enoxaparin arm vs. 4 (2%) in SOC arm (relative risk 0.51; 95% CI, 0.09–2.74) • No major or clinically relevant, nonmajor bleeding events occurred.	Key Limitations • Open-label study • Trial terminated early due to a low probability that enoxaparin would be superior to the standard of care for the primary outcome. Interpretation • Thromboprophylaxis with enoxaparin did not reduce the risk of hospitalization or death among nonhospitalized, symptomatic patients with COVID-19.

Methods	Results	Limitations and Interpretation
ETHIC: Open-Label RCT of Low-Molecular-Weight Heparin for Thromboprophylaxis in Symptomatic Outpatients With COVID-19 in Belgium, Brazil, India, South Africa, Spain, and the United Kingdom¹²		
Key Inclusion Criteria • Aged ≥30 years • RT-PCR-confirmed SARS-CoV-2 infection, with symptoms for ≤9 days • ≥1 risk factors for severe disease Key Exclusion Criteria • Receipt of a COVID-19 vaccine • eGFR <30 mL/min • Receiving anticoagulant or antiplatelet therapy, except low-dose aspirin Interventions • Enoxaparin 40 mg SUBQ once daily (for patients weighing <100 kg) or enoxaparin 40 mg SUBQ twice daily (for patients weighing ≥100 kg), self-administered for 21 days (n = 105) • SOC (n = 114) Primary Endpoint • Composite of all-cause hospitalization or all-cause mortality by Day 21 Key Secondary Endpoints • VTE by Day 90 • Bleeding events by Day 50	Participant Characteristics • Median age 59 years; 56% men • Median of 5 days from first symptom to randomization Primary Outcomes • Composite of all-cause hospitalization or all-cause mortality by Day 21: 12 (11%) in enoxaparin arm vs. 12 (11%) in SOC arm (HR 1.09; 95% CI, 0.49–2.43; <i>P</i> = 0.83) • Patients who required hospitalization: 12 in enoxaparin arm vs. 12 in SOC arm • Hospitalized patients who required acute medical care or ICU admission: 4 in enoxaparin arm vs. 0 in SOC arm Secondary Outcomes • VTE by Day 90: 1 (1%) in enoxaparin arm vs. 2 (2%) in SOC arm • Bleeding events by Day 50: 2 (2%) in enoxaparin arm vs. 2 (2%) in SOC arm	Key Limitations • Open-label study • Study terminated early because of low event rate and lack of efficacy. Interpretation • This study demonstrated no benefit of prophylaxis with LMWH in outpatients with COVID-19 who were at risk of progressing to severe disease.

Methods	Results	Limitations and Interpretation
ACTIV-4C: Double-Blind RCT of 30 Days of Apixaban After Hospital Discharge in Patients With COVID-19 in the United States¹³		
Key Inclusion Criteria • Hospitalized >48 hours with confirmed SARS-CoV-2 infection within 2 weeks of admission • PLT >50,000 cells/μL and Hgb >8 g/dL Key Exclusion Criteria • Need for therapeutic or prophylactic anticoagulation at hospital discharge • Ischemic stroke, intracranial bleed, or neurosurgery within 3 months • Bleeding events within past 30 days • Major surgery within 14 days • Inherited or active acquired bleeding disorder Interventions • Apixaban 2.5 mg PO twice daily for 30 days, starting at hospital discharge (n = 610) • Placebo (n = 607) Primary Endpoint • Composite of death, ATE, or VTE by Day 30 Key Safety Endpoint • Bleeding events	Participant Characteristics • Median age 54 years; 50% men; 27% Black, 17% Hispanic • 15% on antiplatelet therapy • At hospital discharge, 16% were prescribed antiplatelet therapy; 93% received aspirin. Primary Outcome • Composite of death, ATE, or VTE by Day 30: 13 (2.1%) in apixaban arm vs. 14 (2.3%) in placebo arm (relative risk 0.92; 95% CI, 0.44–1.95; <i>P</i> = 0.85) Safety Outcomes • Major bleeding events: 2 (0.4%) in apixaban arm vs. 1 (0.2%) in placebo arm (relative risk 2.00; 95% CI, 0.18–22.03) • Clinically relevant, nonmajor bleeding events: 3 (0.6%) in apixaban arm vs. 6 (1.1%) in placebo arm (relative risk 0.50; 95% CI, 0.13–1.99)	Key Limitation • Trial was terminated early due to a low event rate and because the decreasing number of hospitalizations for people with COVID-19 made recruitment difficult. Interpretation • Incidence of death or thromboembolism was low in this cohort of patients. • Because the trial was terminated early, the results were imprecise, and the study was inconclusive.

Methods	Results	Limitations and Interpretation
MICHELLE: Open-Label RCT of Using Rivaroxaban After Hospital Discharge in Patients With COVID-19 Who Were at High Risk of Venous Thromboembolism in Brazil¹⁴		
Key Inclusion Criteria - Hospitalized for ≥3 days with confirmed SARS-CoV-2 infection - Increased risk of VTE, defined as an IMPROVE VTE score at hospital discharge of >4 or 2–3 with D-dimer >500 ng/mL Key Exclusion Criterion - Suspicion or confirmation of a thrombotic event Interventions - Rivaroxaban 10 mg PO once daily for 35 days, starting at hospital discharge (n = 159) - No anticoagulation (n = 159) Primary Endpoint - Composite of symptomatic or fatal VTE, asymptomatic VTE on bilateral lower-limb venous ultrasound and CTPA, symptomatic ATE, or cardiovascular death by Day 35 Key Secondary Endpoints - Symptomatic or fatal VTE - Composite of symptomatic VTE, MI, non-hemorrhagic stroke, or cardiovascular death Key Safety Endpoint - Bleeding events	Participant Characteristics - Median age 57 years; 60% men - While hospitalized, 86% received thromboprophylaxis with enoxaparin, 14% received unfractionated heparin, and 5% received antiplatelet therapy. Primary Outcome - Primary composite outcome by Day 35: 5 (3%) in rivaroxaban arm vs. 15 (9%) in no anticoagulation arm (relative risk 0.33; 95% CI, 0.12–0.90; <i>P</i> = 0.03) - Difference driven mainly by incidence of PE (2 in rivaroxaban arm vs. 10 in no anticoagulation arm). Secondary Outcomes - Symptomatic or fatal VTE: 1 (0.6%) in rivaroxaban arm vs. 8 (5.0%) in no anticoagulation arm (relative risk 0.13; 95% CI, 0.02–0.99; <i>P</i> = 0.049) - Composite of symptomatic VTE, MI, non-hemorrhagic stroke, or cardiovascular death: 1 (0.6%) in rivaroxaban arm vs. 9 (5.7%) in no anticoagulation arm (relative risk 0.11; 95% CI, 0.01–0.87; <i>P</i> = 0.036) Safety Outcome - No major bleeding events occurred.	Key Limitations - Open-label study with no placebo - Not all patients had the protocol-specified CTPA or Doppler ultrasound during the study. However, a higher number of imaging evaluations occurred among the patients in the rivaroxaban arm. Interpretation - In patients who were at high risk of VTE, the use of thromboprophylaxis with rivaroxaban 10 mg PO once daily for 35 days improved clinical outcomes when compared with no anticoagulation.

Key: ATE = arterial thromboembolism; BARC = Bleeding Academic Research Consortium; BMI = body mass index; CAD = coronary artery disease; CrCl = creatinine clearance; CTPA = computed tomography pulmonary angiogram; CVD = cardiovascular disease; DEX = dexamethasone; DM = diabetes mellitus; DVT = deep vein thrombosis; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; HFNC = high-flow nasal cannula; Hgb = hemoglobin; HIT = heparin-induced thrombocytopenia; HTN = hypertension; ICU = intensive care unit; IMPROVE = International Medical Prevention Registry on Venous Thromboembolism; ISTH = International Society on Thrombosis and Haemostasis; LMWH = low-molecular-weight heparin; LOS = length of stay; MI = myocardial infarction; MV = mechanical ventilation; NIV = noninvasive ventilation; OS = ordinal scale; the Panel = the COVID-19 Treatment Guidelines Panel; PE = pulmonary embolism; PLT = platelet count; PO = oral; RCT = randomized controlled trial; RDV = remdesivir; RT-PCR = reverse transcription polymerase chain reaction; SOC = standard of care; SpO₂ = oxygen saturation; SUBQ = subcutaneous; UFH = unfractionated heparin; ULN = upper limit of normal; VTE = venous thromboembolism; WHO = World Health Organization

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Table 6b. Antiplatelet Therapy: Selected Clinical Trial Data

Last Updated: February 29, 2024

The clinical trials described in this table do not represent all the trials that the Panel reviewed while developing the recommendations for antiplatelet therapy. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Methods	Results	Limitations and Interpretation
ACTIV-4a: Open-Label, Adaptive RCT of Adding a P2Y12 Inhibitor to Anticoagulant Therapy in Noncritically Ill Hospitalized Patients With COVID-19 in Brazil, Italy, Spain, and the United States¹		
Key Inclusion Criteria - Laboratory-confirmed SARS-CoV-2 infection - Any 1 of the following: - D-dimer level ≥ 2 times ULN - Aged 60–84 years - Aged < 60 years with oxygen requirement > 2 L/min, HTN, DM, eGFR < 60 mL/min, CVD, or BMI ≥ 35 Key Exclusion Criteria - Required HFNC oxygen ≥ 20 L/min, NIV, MV, ECMO, vasopressors, or inotropes ≥ 72 hours since hospital admission Interventions - Therapeutic dose of heparin plus P2Y12 inhibitor for 14 days or until hospital discharge, whichever came first (n = 293) - Therapeutic dose of heparin (usual care arm; n = 269) Primary Endpoints - Number of organ support-free days by Day 21 - Major bleeding events by Day 28 Key Secondary Endpoint - Composite of major thrombotic events or death by Day 28	Participant Characteristics - Mean age 53 years; 42% women; 62% White - HTN: 43% in P2Y12 inhibitor arm vs. 55% in usual care arm 65% received glucocorticoids; 52% received RDV; 3% received IL-6 inhibitors; 14% received aspirin. - Median duration of P2Y12 inhibitor treatment was 6 days. 63% received ticagrelor; 37% received clopidogrel. Primary Outcomes - Median number of organ support-free days by Day 21: 21 in both arms (aOR 0.83; 95% CrI, 0.55–1.25; posterior probability of futility 96%) - Major bleeding events by Day 28: 2.0% in P2Y12 inhibitor arm vs. 0.7% in usual care arm (aOR 3.31; 95% CI, 0.64–17.2; $P = 0.15$) Secondary Outcome - Composite of major thrombotic events or death by Day 28: 6.1% in P2Y12 inhibitor arm vs. 4.5% in usual care arm (aOR 1.42; 95% CI, 0.64–3.13)	Key Limitations - Open-label study - Study was stopped early for futility. - Different P2Y12 inhibitors were used. - Median duration of P2Y12 inhibitor treatment was 6 days, which may not be sufficient to observe effects. Interpretation - Among hospitalized patients with COVID-19 who were not critically ill, adding a P2Y12 inhibitor to a therapeutic dose of heparin did not increase the number of organ support-free days. - Major bleeding events occurred infrequently during the study. The number of patients who experienced a major bleeding event was not significantly different between the arms.

Methods	Results	Limitations and Interpretation
RECOVERY: Open-Label RCT of Aspirin in Hospitalized Patients With COVID-19 in Indonesia, Nepal, and the United Kingdom²		
Key Inclusion Criterion - Clinically suspected or laboratory-confirmed SARS-CoV-2 infection Key Exclusion Criteria - Hypersensitivity to aspirin - Recent history of major bleeding events - Currently receiving aspirin or another antiplatelet treatment Interventions - Aspirin 150 mg once daily until hospital discharge (n = 7,351) - SOC alone (n = 7,541) Primary Endpoint - All-cause mortality at 28 days Key Secondary Endpoints - Composite of progression to MV or death at 28 days - Major bleeding events at 28 days - Thrombotic events at 28 days	Participant Characteristics - Mean age 59 years; 62% men; 75% White 97% had laboratory-confirmed SARS-CoV-2 infection - At baseline: 5% on MV 28% on NIV 34% received intermediate- or therapeutic-dose LMWH. 60% received standard-dose LMWH. 7% received no thromboprophylaxis. 94% received corticosteroids; 26% received RDV; 13% received tocilizumab; 6% received baricitinib. Primary Outcome - All-cause mortality at 28 days: 17% in both arms (rate ratio 0.96; 95% CI, 0.89–1.04; <i>P</i> = 0.35) Secondary Outcomes - Composite of progression to MV or death at 28 days: 21% in aspirin arm vs. 22% in SOC arm (risk ratio 0.96; 95% CI, 0.90–1.03) - Major bleeding events at 28 days: 1.6% in aspirin arm vs. 1.0% in SOC arm (<i>P</i> = 0.0028) - Thrombotic events at 28 days: 4.6% in aspirin arm vs. 5.3% in SOC arm (<i>P</i> = 0.07)	Key Limitation - Because of the open-label design, reporting of major bleeding and thrombotic events may have been influenced by the treatment allocation. Interpretation - In hospitalized patients with COVID-19, the use of aspirin was not associated with reductions in 28-day mortality or the risk of progressing to MV or death.

Methods	Results	Limitations and Interpretation
REMAP-CAP: Open-Label, Adaptive RCT of Antiplatelet Therapy in Critically Ill Patients With COVID-19 in 8 Countries in Europe and Asia³		
Key Inclusion Criteria - Clinically suspected or laboratory-confirmed SARS-CoV-2 infection - Within 48 hours of ICU admission Key Exclusion Criteria - Bleeding risk sufficient to contraindicate antiplatelet therapy - CrCl <30 mL/min - Receiving antiplatelet therapy or NSAIDs Interventions 1 of the following plus anticoagulation for 14 days or until hospital discharge, whichever came first: - Aspirin 75–100 mg once daily (n = 565) - P2Y12 inhibitor (n = 455) - No antiplatelet therapy (control arm; n = 529) Primary Endpoint - Number of organ support-free days by Day 21 Key Secondary Endpoints - Survival to hospital discharge - Survival to Day 90 - Major bleeding events by Day 14	Participant Characteristics - Mean age 57 years; 34% women; 77% White - At baseline, 98% were receiving LMWH: 19% received low-dose LMWH. 59% received intermediate-dose LMWH. 12% received therapeutic-dose LMWH. 98% received steroids; 21% received RDV; 44% received tocilizumab; 11% received sarilumab. - In P2Y12 inhibitor arm, 88.5% received clopidogrel, 1.3% received ticagrelor, 1.3% received prasugrel, and 8.8% received an unknown P2Y12 inhibitor. Primary Outcome - Data from aspirin and P2Y12 inhibitor arms were pooled and reported as “pooled antiplatelet arm” in final analysis: - Median number of organ support-free days by Day 21: 7 in pooled antiplatelet arm vs. 7 in control arm (aOR 1.02; 95% CrI, 0.86–1.23; posterior probability of futility 96%) Secondary Outcomes - Survival to hospital discharge: 71.5% in pooled antiplatelet arm vs. 67.9% in control arm (median-adjusted OR 1.27; 95% CrI, 0.99–1.62; adjusted absolute difference 5%; 95% CrI, -0.2% to 9.5%; 97% posterior probability of efficacy) - Survival to Day 90: 72% in pooled antiplatelet arm vs. 68% in control arm (HR with pooled antiplatelets 1.22; 95% CrI, 1.06–1.40; 99.7% posterior probability of efficacy) - Major bleeding events by Day 14: 2.1% in pooled antiplatelet arm vs. 0.4% in control arm (aOR 2.97; 95% CrI, 1.23–8.28; posterior probability of harm 99.4%)	Key Limitations - Open-label study - Different P2Y12 inhibitors were used. - Trial was stopped for futility. Because equivalence for aspirin and P2Y12 inhibitor arms was reached, these arms were pooled for analyses. Interpretation - In critically ill patients with COVID-19, the use of aspirin or a P2Y12 inhibitor did not reduce the number of organ support-free days or in-hospital mortality. - Patients in the pooled antiplatelet arm had more major bleeding events than those in the control arm, but they had improved survival over 90 days.

Methods	Results	Limitations and Interpretation
COVID-PACT: Open-Label RCT of Clopidogrel in Adults With COVID-19 Who Were Receiving Intensive Care Unit-Level Care in the United States⁴		
Key Inclusion Criteria • Aged ≥18 years • Acute SARS-CoV-2 infection • Required ICU-level care for ≤96 hours prior to randomization Key Exclusion Criteria • Ongoing or planned use of a therapeutic dose of anticoagulation or dual antiplatelet therapy • High risk of bleeding • History of HIT • Ischemic stroke within 2 weeks Interventions • Clopidogrel 300 mg at randomization, then clopidogrel 75 mg once daily until hospital discharge or Day 28, whichever came first (n = 152) • No clopidogrel therapy (n = 140) • Some patients were also randomized to receive a therapeutic or prophylactic dose of anticoagulation (n = 290) Primary Endpoint • Composite of VTE or ATE events by hospital discharge or Day 28. Events included death due to VTE or ATE, PE, clinically evident DVT, MI, ischemic stroke, systemic embolic events or acute limb ischemia, and clinically silent DVT. Secondary Endpoint • Composite of clinically evident VTE or ATE events by hospital discharge or Day 28 Safety Endpoints • Fatal or life-threatening bleeding events • Moderate or severe bleeding events	Participant Characteristics • Median age 58 years; 41% women; 71% White • At baseline, 99% on HFNC oxygen, NIV, or MV; 15% on MV • 37% required MV during the study. Primary Outcome • Composite of VTE or ATE events by hospital discharge or Day 28: 10% in both arms (win ratio 1.04; 95% CI, 0.54–2.01; <i>P</i> = 0.90) Secondary Outcome • Composite of clinically evident VTE or ATE events by hospital discharge or Day 28: 7% in clopidogrel arm vs. 9% in no clopidogrel arm (win ratio 0.79; 95% CI, 0.38–1.65; <i>P</i> = 0.53) Safety Outcomes • Fatal or life-threatening bleeding events: 1.3% in clopidogrel arm vs. 1.4% in no clopidogrel arm (<i>P</i> = 1.00) • Moderate or severe bleeding events: 4.0% in clopidogrel arm vs. 6.4% in no clopidogrel arm (<i>P</i> = 0.83)	Key Limitations • Open-label study (adjudication committee members were blinded to the study arms). • Trial was stopped early because the decreasing number of ICU admissions for patients with COVID-19 made recruitment difficult. • 31% discontinued clopidogrel. Interpretation • In patients with COVID-19 who required ICU-level care, clopidogrel did not reduce the incidence of thrombotic complications.

Key: ATE = arterial thromboembolism; BMI = body mass index; CrCl = creatinine clearance; CVD = cardiovascular disease; DM = diabetes mellitus; DVT = deep vein thrombosis; ECMO = extracorporeal membrane oxygenation; eGFR = estimated glomerular filtration rate; HFNC = high-flow nasal cannula; HIT = heparin-induced thrombocytopenia; HTN = hypertension; ICU = intensive care unit; IL = interleukin; LMWH = low-molecular-weight heparin; MI = myocardial infarction; MV = mechanical ventilation; NIV = noninvasive ventilation; NSAID = nonsteroidal anti-inflammatory drug; the Panel = the COVID-19 Treatment Guidelines Panel; PE = pulmonary embolism; RCT = randomized controlled trial; RDV = remdesivir; SOC = standard of care; ULN = upper limit of normal; VTE = venous thromboembolism

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Miscellaneous Drugs

Last Updated: December 20, 2023

Summary Recommendations
Fluvoxamine • The COVID-19 Treatment Guidelines Panel (the Panel) recommends against the use of fluvoxamine for the treatment of COVID-19 in nonhospitalized patients (AIIa). • There is insufficient evidence for the Panel to recommend either for or against the use of the combination of inhaled budesonide plus fluvoxamine for the treatment of COVID-19 in nonhospitalized patients. • Patients with COVID-19 who are receiving fluvoxamine for an underlying condition should continue this therapy as directed by their health care provider (AIII). Intravenous Immunoglobulin • The Panel recommends against the use of intravenous immunoglobulin (IVIG) for the treatment of acute COVID-19 in adults and children, except in a clinical trial (AIII). This recommendation should not preclude the use of IVIG when it is otherwise indicated for the treatment of underlying conditions or complications that arise during the course of COVID-19. • For the Panel's recommendations on the use of IVIG in people with multisystem inflammatory syndrome in children (MIS-C) or multisystem inflammatory syndrome in adults (MIS-A) and a discussion of the clinical data that support those recommendations, see Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A. Ivermectin • The Panel recommends against the use of ivermectin for the treatment of COVID-19 (AIIa). Metformin • There is insufficient evidence for the Panel to recommend either for or against the use of metformin for the treatment of COVID-19 in nonhospitalized patients. • The Panel recommends against the use of metformin for the treatment of COVID-19 in hospitalized patients, except in a clinical trial (BIII). • Patients with COVID-19 who are receiving metformin for an underlying condition should continue this therapy as directed by their health care provider (AIII). The Panel reviewed clinical trials that evaluated the use of the anti-inflammatory drug colchicine for the treatment of COVID-19; however, these trials failed to show a benefit of using colchicine in patients with COVID-19. Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Fluvoxamine

Last Updated: December 20, 2023

Fluvoxamine is a selective serotonin reuptake inhibitor (SSRI) that is approved by the Food and Drug Administration (FDA) for the treatment of obsessive-compulsive disorder and is used for other conditions, including depression. Fluvoxamine is not approved by the FDA for the treatment of any infection.

In a murine sepsis model, fluvoxamine was found to bind to the sigma-1 receptor on immune cells, resulting in reduced production of inflammatory cytokines.¹ In an in vitro study of human endothelial cells and macrophages, fluvoxamine reduced the expression of inflammatory genes.²

Recommendations

- The COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** the use of **fluvoxamine** for the treatment of COVID-19 in nonhospitalized patients (**AIIa**).
- There is insufficient evidence for the Panel to recommend either for or against the use of the combination of inhaled budesonide plus fluvoxamine for the treatment of COVID-19 in nonhospitalized patients.
- Patients with COVID-19 who are receiving **fluvoxamine** for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Rationale

Six randomized trials have studied the use of fluvoxamine for the treatment of nonhospitalized patients with COVID-19.³⁻⁸ The TOGETHER and STOP COVID 2 trials enrolled unvaccinated patients with COVID-19 who had at least 1 risk factor for disease progression.^{3,5} These studies did not identify a consistent benefit of using fluvoxamine in these patients, although STOP COVID 2 was stopped early due to low primary outcome rates. Other outpatient therapies (i.e., ritonavir-boosted nirmatrelvir [Paxlovid], remdesivir) have been shown to be effective in preventing hospitalization or death in unvaccinated patients who are at high risk of disease progression. In subsequent trials where the majority of enrolled patients were vaccinated against COVID-19, fluvoxamine did not significantly reduce the risk of hospitalization or death, the time to recovery, or health care utilization.⁶⁻⁸ In several of these studies, fluvoxamine was associated with decreased adherence and/or an increase in the occurrence of nonserious adverse effects, primarily gastrointestinal symptoms.^{3,5,6}

The TOGETHER trial was a large, double-blind, placebo-controlled, adaptive randomized trial in Brazil that evaluated the use of inhaled budesonide plus oral fluvoxamine in patients with COVID-19.⁹ Over 90% of the patients had received at least 2 doses of a COVID-19 vaccine. Treatment with this combination significantly reduced the incidence of the primary outcome, which was a composite of hospitalization or retention in an emergency setting for >6 hours. The proportion of patients who were hospitalized was the same in the treatment and placebo arms (0.9% vs. 1.1%), and the treatment did not significantly impact secondary outcomes such as health care attendance or the need for an emergency setting visit. It is unclear how the >6-hour emergency setting outcome translates to other settings. In addition, treatment with budesonide plus fluvoxamine was associated with significantly more adverse events.

Summaries of the studies that informed the Panel's recommendations can be found in [Table 7a](#).

Monitoring, Adverse Effects, and Drug-Drug Interactions

When fluvoxamine is used to treat psychiatric conditions, the most common adverse effect is nausea, but adverse effects can include other gastrointestinal effects (e.g., diarrhea, indigestion), neurologic effects (e.g., asthenia, insomnia, somnolence, anxiety, headache), and, rarely, suicidal ideation.

Fluvoxamine is a cytochrome P450 (CYP) 2D6 substrate, a potent inhibitor of CYP1A2 and CYP2C19, and a moderate inhibitor of CYP2C9, CYP2D6, and CYP3A4.¹⁰ Fluvoxamine can enhance the serotonergic effects of other SSRIs or monoamine oxidase inhibitors, resulting in serotonin syndrome; therefore, it should not be used within 2 weeks of receiving other SSRIs or monoamine oxidase inhibitors. Fluvoxamine may enhance the anticoagulant effects of antiplatelets and anticoagulants. Patients who are receiving these drugs should be closely monitored.

Considerations in Pregnant People

Fluvoxamine is not thought to increase the risk of congenital abnormalities; however, the data on its use in pregnant individuals are limited.^{11,12} An association between SSRI use in the late third trimester and a small increase in the risk of primary persistent pulmonary hypertension in newborns has not been excluded, although the absolute risk is likely low.¹³

Considerations in Children

Fluvoxamine is approved by the FDA for the treatment of obsessive-compulsive disorder in children aged ≥ 8 years.¹⁴ The adverse effects of SSRI use seen in children are similar to those seen in adults, although children and adolescents appear to have higher rates of activation and vomiting than adults.¹⁵ There are no data on the use of fluvoxamine for the prevention or treatment of COVID-19 in children.

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Table 7a. Fluvoxamine: Selected Clinical Trial Data

Last Updated: December 20, 2023

Methods	Results	Limitations and Interpretation
ACTIV-6: Decentralized, Randomized, Placebo-Controlled, Platform Trial of Low-Dose Fluvoxamine in Patients With Mild to Moderate COVID-19¹		
Key Inclusion Criteria - Aged ≥30 years - Positive SARS-CoV-2 nasopharyngeal RT-PCR result or antigen test result ≥2 COVID-19 symptoms for ≤7 days Key Exclusion Criterion - Receipt of fluvoxamine in past 14 days Interventions - Fluvoxamine 50 mg PO twice daily for 10 days (n = 674) - Placebo (n = 614; 326 received matching placebo, 288 received placebo from another study arm) Primary Endpoint - Time to recovery, defined as time to third day of 3 consecutive days without symptoms Key Secondary Endpoints - Hospitalization or death by Day 28 - Urgent care visit, ED visit, or hospitalization by Day 28	Participant Characteristics - Mean age 47 years; 57% women; 81% White 36% with BMI ≥30; 24% with HTN 67% received ≥2 doses of a SARS-CoV-2 vaccine. - Median of 5 days from symptom onset to receipt of study drug Primary Outcome - Median time to recovery: 12 days in fluvoxamine arm vs. 13 days in placebo arm (HR 0.96; 95% CrI, 0.86–1.06) Secondary Outcomes - Hospitalization or death by Day 28: 0.2% in fluvoxamine arm vs. 0.3% in placebo arm (3 events total) - Urgent care visit, ED visit, or hospitalization by Day 28: 3.9% in fluvoxamine arm vs. 3.8% in placebo arm (HR 1.1; 95% CrI, 0.5–1.8)	Key Limitation - Low number of some clinical events, such as hospitalization Interpretation - In outpatients with mild to moderate COVID-19, fluvoxamine 50 mg twice daily for 10 days did not reduce the time to recovery or the incidence of clinical events such as hospitalization, urgent care visits, or ED visits.

Methods	Results	Limitations and Interpretation
ACTIV-6: Decentralized, Randomized, Placebo-Controlled, Platform Trial of High-Dose Fluvoxamine in Patients With Mild to Moderate COVID-19²		
Key Inclusion Criteria • Aged ≥30 years • Positive SARS-CoV-2 nasopharyngeal RT-PCR result or antigen test result • ≥2 COVID-19 symptoms for ≤7 days Key Exclusion Criterion • Receipt of fluvoxamine or other selective serotonin or norepinephrine reuptake inhibitors in past 14 days Interventions • Fluvoxamine 50 mg PO twice daily for 1 day, then fluvoxamine 100 mg PO twice daily for 12 days (n = 589) • Placebo (n = 586) Primary Endpoint • Time to recovery, defined as time to third day of 3 consecutive days without symptoms Key Secondary Endpoints • Hospitalization or death by Day 28 • Urgent care visit, ED visit, or hospitalization by Day 28	Participant Characteristics • Median age 50 years; 66% women; 73% White • 36% with BMI ≥30; 26% with HTN • 77% received ≥2 doses of a SARS-CoV-2 vaccine. • Median of 5 days from symptom onset to receipt of study drug Primary Outcome • Median time to recovery: 10 days in fluvoxamine arm vs. 10 days in placebo arm (HR 0.99; 95% CrI, 0.89–1.09) Secondary Outcomes • Hospitalization or death by Day 28: 0.2% in fluvoxamine arm vs. 0.3% in placebo arm (3 events total) • Urgent care visit, ED visit, or hospitalization by Day 28: 2.4% in fluvoxamine arm vs. 3.6% in placebo arm (HR 0.69; 95% CrI, 0.27–1.21)	Key Limitation • Low number of some clinical events, such as hospitalization Interpretation • In outpatients with mild to moderate COVID-19, fluvoxamine 100 mg twice daily did not reduce the time to symptom recovery or the incidence of clinical events such as hospitalization, urgent care visits, or ED visits.

Methods	Results	Limitations and Interpretation
COVID-OUT: Randomized Trial of Metformin, Ivermectin, and Fluvoxamine in Patients With COVID-19³		
Key Inclusion Criteria • Aged 30–85 years • BMI ≥ 25 or ≥ 23 if Asian or Latinx • Laboratory-confirmed SARS-CoV-2 infection within 3 days of randomization • < 7 days of symptoms Key Exclusion Criteria • Immunocompromised • Hepatic impairment, severe kidney disease Interventions • Fluvoxamine 50 mg PO twice daily for 14 days (n = 334) • Control (n = 327) Primary Endpoint • Composite of hypoxemia (as measured by a home pulse oximeter), ED visit, hospitalization, or death by Day 14 Key Secondary Endpoints • Individual components of the composite endpoint • Total symptom severity score • Drug interruption or discontinuation	Participant Characteristics • Median age 43–46 years; 54% women; 82% White • 27% with CVD; 47% with BMI ≥ 30 • 56% received primary vaccination series. • Mean of 5 days from symptom onset to randomization Primary Outcome • Composite of hypoxemia, ED visit, hospitalization, or death by Day 14: 24% in fluvoxamine arm vs. 25% in control arm (aOR 0.94; 95% CI, 0.66–1.36) Secondary Outcomes • Hospitalization by Day 14: 1.8% in fluvoxamine arm vs. 1.5% in control arm (aOR 1.11; 95% CI, 0.33–3.76). • Composite of ED visit, hospitalization, or death: 5.5% in fluvoxamine arm vs. 4.6% in control arm (aOR 1.17; 95% CI, 0.57–2.40) • No deaths occurred in either arm. • No difference between arms in total symptom severity score over 14 days • Drug interruption or discontinuation: 30% in those who only received fluvoxamine vs. 25% in those who only received placebo	Key Limitation • In this trial, the study arms that did not include metformin were underpowered to detect differences in the primary endpoint. Interpretation • Fluvoxamine did not impact the incidence of COVID-19–related complications such as hospitalization. • Fluvoxamine did not impact symptom severity.

Methods	Results	Limitations and Interpretation
TOGETHER: Double-Blind, Adaptive RCT of Fluvoxamine in Nonhospitalized Patients With COVID-19 in Brazil⁴		
Key Inclusion Criteria • Aged ≥50 years or aged ≥18 years with comorbidities • Laboratory-confirmed SARS-CoV-2 infection • ≤7 days of symptoms Key Exclusion Criteria • Use of an SSRI • Severe mental illness • Cirrhosis, recent seizures, severe ventricular cardiac arrhythmia Interventions • Fluvoxamine 100 mg PO twice daily for 10 days (n = 741) • Placebo (n = 756; route, dosing frequency, and duration of placebo may have differed from fluvoxamine for some patients) Primary Endpoint • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints • COVID-19–related hospitalization by Day 28 • Composite of hospitalization or ED observation >24 hours • Time to symptom resolution • Adherence to study drugs, defined as receiving >80% of possible doses • Mortality in both the primary ITT population and a PP population that included patients who took >80% of the study medication doses	Participant Characteristics • Median age 50 years; 58% women; 95% self-identified as mixed race • 13% with uncontrolled HTN; 13% with type 2 DM; 50% with BMI ≥30 • Mean of 3.8 days from symptom onset to randomization Primary Outcome • Composite of ED observation >6 hours or hospitalization by Day 28: 11% in fluvoxamine arm vs. 16% in placebo arm (relative risk 0.68; 95% CrI, 0.52–0.88) Secondary Outcomes • 87% of clinical events were hospitalizations. • No difference between arms in COVID-19–related hospitalizations: 10% in fluvoxamine arm vs. 13% in placebo arm (OR 0.77; 95% CI, 0.55–1.05) • Lower risk of hospitalization or ED observation >24 hours in fluvoxamine arm than in placebo arm (relative risk 0.74; 95% CI, 0.56–0.98)⁵ • No difference between arms in time to symptom resolution • Adherence: 74% in fluvoxamine arm vs. 82% in placebo arm (OR 0.62; 95% CI, 0.48–0.81) • 11% in fluvoxamine arm vs. 8% in placebo arm stopped drug due to issues of tolerability. • Mortality (ITT): 2% in fluvoxamine arm vs. 3% in placebo arm (OR 0.68; 95% CI, 0.36–1.27) • Mortality (PP): <1% in fluvoxamine arm vs. 2% in placebo arm (OR 0.09; 95% CI, 0.01–0.47)	Key Limitations • The >6-hour ED observation endpoint has not been used in other studies of interventions for nonhospitalized patients who are at high risk of hospitalization and death. • Hospitalization or ED observation for >24 hours was analyzed in a post hoc analysis. • As this was an adaptive platform trial where multiple investigational treatments or placebos were being evaluated simultaneously, not all patients in the placebo arm received a placebo that was matched to fluvoxamine by route of administration, dosing frequency, or duration of therapy. • PP analyses are not randomized comparisons, and they introduce bias when adherence is associated with factors that influence the outcome. • Adherence was self-reported and not verified. Interpretation • Fluvoxamine reduced the proportion of patients who met the composite endpoint of COVID-19–related hospitalization or retention in an ED for >6 hours. • The use of fluvoxamine did not impact the incidence of COVID-19-related hospitalizations but did reduce the need for hospitalization or ED observations >24 hours. • It is difficult to define the clinical relevance of the >6-hour ED observation endpoint and apply it to practice settings in different countries. • Fluvoxamine did not have a consistent impact on mortality. • Fluvoxamine did not impact the time to symptom resolution.

Methods	Results	Limitations and Interpretation
STOP COVID 2: Fully Remote RCT of Fluvoxamine Versus Placebo in Outpatients With Symptomatic COVID-19⁶		
Key Inclusion Criteria • Aged ≥30 years • Positive SARS-CoV-2 PCR result per patient self-report • ≤7 days of symptoms • ≥1 risk factor for clinical deterioration Key Exclusion Criteria • Unstable medical comorbidities • Significant interacting medications Interventions • Fluvoxamine 50 mg PO for 1 dose, then fluvoxamine 100 mg PO twice daily through Day 15 (n = 272) • Placebo (n = 275) Primary Endpoint • Clinical deterioration within 15 days of randomization. Clinical deterioration was defined as: • Having dyspnea or being hospitalized for dyspnea or pneumonia; <i>and</i> • Having SpO₂ <92% on room air or requiring supplemental oxygen to attain SpO₂ ≥92% Key Secondary Endpoints • Occurrence of AEs	Participant Characteristics • Median age 47 years; 62% women; 27% non-White • 44% with obesity; 21% with HTN • Median of 5 days from symptom onset to randomization Primary Outcome • Clinical deterioration: 4.8% in fluvoxamine arm vs. 5.5% in placebo arm (absolute difference 0.68%; 95% CI, -3.0 to 4.4) Secondary Outcomes • GI AEs were significantly more common in fluvoxamine arm	Key Limitations • Small sample size compared to other trials • Short follow-up period Interpretation • Fluvoxamine did not reduce the proportion of patients who experienced clinical deterioration by Day 15.

Methods	Results	Limitations and Interpretation
STOP COVID: Double-Blind RCT of Fluvoxamine in Nonhospitalized Patients With COVID-19 in the United States⁷		
Key Inclusion Criteria • Aged ≥18 years • Positive SARS-CoV-2 PCR result • ≤7 days of symptoms Key Exclusion Criteria • Immunocompromised • Unstable medical comorbidities Interventions • Fluvoxamine 50 mg PO for 1 dose, then fluvoxamine 100 mg PO twice daily, then fluvoxamine 100 mg PO 3 times daily through Day 15 (n = 80) • Placebo (n = 72) Primary Endpoint • Clinical deterioration within 15 days of randomization. Clinical deterioration was defined as: • Having dyspnea or being hospitalized for dyspnea or pneumonia; <i>and</i> • Having SpO₂ <92% on room air or requiring supplemental oxygen to attain SpO₂ ≥92% Key Secondary Endpoint • Hospitalization by Day 15	Participant Characteristics • Mean age 46 years; 72% women; 25% Black • 56% with obesity; 20% with HTN; 17% with asthma • Median of 4 days from symptom onset to randomization Primary Outcome • Clinical deterioration: 0% in fluvoxamine arm vs. 8.3% in placebo arm (absolute difference 8.7%; 95% CI, 1.8% to 16.4%) Secondary Outcome • No patients in fluvoxamine arm and 4 patients in placebo arm were hospitalized.	Key Limitations • Small sample size • Short follow-up period • Ascertaining clinical deterioration was challenging because all assessments were done remotely. • 24% of patients stopped responding to follow-up prior to Day 15 but were included in the final analysis. Interpretation • Fluvoxamine reduced the proportion of patients who experienced clinical deterioration. • Due to significant limitations, it is difficult to draw definitive conclusions about the efficacy of using fluvoxamine to treat COVID-19.

Methods	Results	Limitations and Interpretation
TOGETHER: Randomized Platform Trial of Oral Fluvoxamine Plus Inhaled Budesonide for the Treatment of Early Onset COVID-19⁸		
Key Inclusion Criteria • Aged ≥50 years or aged ≥18 years with comorbidities • Laboratory-confirmed SARS-CoV-2 infection • ≤7 days of symptoms Key Exclusion Criteria • Use of an SSRI • Severe mental illness • Cirrhosis, recent seizures, severe ventricular cardiac arrhythmia Interventions • Fluvoxamine 100 mg PO twice daily plus budesonide 800 µg inhaled twice daily for 10 days (n = 738) • Placebo (n = 738; route, dosing frequency, and duration for some patients may have differed from treatment group) Primary Endpoint • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints • Hospitalization by Day 28 • Health care attendance by Day 28 • Any ED visit by Day 28 • Occurrence of AEs	Participant Characteristics • Median age 51 years; 61% women • 42% with BMI >30; 44% with HTN; 68% with multiple comorbidities • 94% received ≥2 doses of a COVID-19 vaccine. Primary Outcome • Composite of ED observation >6 hours or hospitalization by Day 28: 1.8% in fluvoxamine plus inhaled budesonide arm vs. 3.7% in placebo arm (relative risk 0.50; 95% CrI, 0.25–0.92) Secondary Outcomes • Hospitalization by Day 28: 0.9% in fluvoxamine plus inhaled budesonide arm vs. 1.1% in placebo arm • Health care attendance by Day 28: 2.6% in fluvoxamine plus inhaled budesonide arm vs. 4.1% in placebo arm (relative risk 0.64; 95% CrI, 0.36–1.11) • Any ED visit by Day 28: 12.2% in fluvoxamine plus inhaled budesonide arm vs. 13.0% in placebo arm • Treatment-emergent AEs: 17.6% in fluvoxamine plus inhaled budesonide arm vs. 12.9% in placebo arm (relative risk 1.37; 95% CrI, 1.07–1.75) • Most AEs were grade 2.	Key Limitation • Multiple investigational treatments or placebos were evaluated simultaneously. Not all patients in the placebo arm received a matched placebo. Interpretation • In adult outpatients with mild COVID-19, fluvoxamine plus inhaled budesonide reduced the need for ED observations >6 hours or hospitalization when compared with placebo. • The use of fluvoxamine plus inhaled budesonide did not reduce hospitalization, health care attendance, or the occurrence of any ED visit. • It is difficult to define the clinical relevance of the >6-hour ED observation endpoint and apply it to practice settings in different countries. • The use of fluvoxamine plus inhaled budesonide resulted in more AEs than placebo.

Key: AE = adverse event; BMI = body mass index; CVD = cardiovascular disease; DM = diabetes mellitus; ED = emergency department; GI = gastrointestinal; HTN = hypertension; ITT = intention-to-treat; the Panel = the COVID-19 Treatment Guidelines Panel; PCR = polymerase chain reaction; PO = oral; PP = per protocol; RCT = randomized controlled trial; RT-PCR = reverse transcription polymerase chain reaction; SpO₂ = oxygen saturation; SSRI = selective serotonin reuptake inhibitor

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Intravenous Immunoglobulin

Last Updated: December 20, 2023

Recommendations

- The COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** the use of **intravenous immunoglobulin (IVIG)** for the treatment of acute COVID-19 in adults and children, except in a clinical trial (**AIII**). This recommendation should not preclude the use of IVIG when it is otherwise indicated for the treatment of underlying conditions or complications that arise during the course of COVID-19.
- For the Panel's recommendations on the use of IVIG in people with multisystem inflammatory syndrome in children (MIS-C) or multisystem inflammatory syndrome in adults (MIS-A) and a discussion of the clinical data that support those recommendations, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Rationale

It is unknown whether IVIG products derived from pooled donor plasma contain high titers of antibodies that neutralize SARS-CoV-2. Information on SARS-CoV-2 antibody titer was not reported in the clinical trials that evaluated the use of IVIG for the treatment of COVID-19. The levels of SARS-CoV-2 antibodies in IVIG products likely vary depending on which SARS-CoV-2 variant was dominant when the plasma products were collected, and different lots of IVIG may have different titers of antibodies. Although IVIG preparations may have general immunomodulatory effects, these theoretical effects do not appear to benefit patients with COVID-19.¹

Considerations in Pregnant People

IVIG is commonly used during pregnancy for indications such as alloimmune thrombocytopenia.² However, because there is no clear evidence that IVIG is an effective treatment for acute COVID-19 in nonpregnant adults, the Panel **recommends against** the use of **IVIG** for the treatment of acute COVID-19 in pregnant individuals, except in a clinical trial (**AIII**). This recommendation should not preclude the use of IVIG when it is otherwise indicated for the treatment of underlying conditions or complications that arise during the course of COVID-19.

Considerations in Children

No comparative studies have evaluated the use of IVIG in pediatric patients with acute COVID-19. IVIG is used in combination with glucocorticoids to treat MIS-C in pediatric patients.³⁻⁶ However, because there is no clear evidence that IVIG is an effective treatment for acute COVID-19 in adults, the Panel **recommends against** the use of **IVIG** for the treatment of acute COVID-19 in children, except in a clinical trial (**AIII**). This recommendation should not preclude the use of IVIG when it is otherwise indicated.

For the Panel's recommendations for children with MIS-C, see [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#).

Clinical Data

In a meta-analysis of 6 randomized controlled trials that enrolled hospitalized patients with COVID-19, the use of non-SARS-CoV-2-specific IVIG was not associated with a survival benefit.¹ All of the

included trials were conducted in 2020, when the presence of SARS-CoV-2 antibodies in blood donors was likely uncommon. None of the studies measured the titers of anti-SARS-CoV-2 antibodies. Blood supplies collected since that time likely have a higher level of these antibodies, and the IVIG derived from those supplies could be expected to have a higher level of SARS-specific antibodies. A British study performed in 2022 evaluated serum anti-SARS-CoV-2 spike antibody titers before and after IVIG infusion in 35 patients with primary immunodeficiencies who were receiving regular immunoglobulin replacement therapy.⁷ The study found that anti-SARS-CoV-2 spike antibody titers and the neutralization capacity of serum increased after IVIG infusion in most patients.

Different brands of commercially available IVIG products exhibit different levels of neutralizing activity against SARS-CoV-2 variants (e.g., BA.1, BA.4, BA.5, BQ.1.1, XBB). A study compared the anti-SARS-CoV-2 antibody levels in U.S. IVIG products that had expiration dates from 2020 to 2025.⁸ The study found that products with expiration dates in 2023 and 2024 were more likely to have higher levels of anti-SARS-CoV-2 antibodies than those with earlier expiration dates. In addition, the study reported an association between later expiration dates and increased inhibition of angiotensin-converting enzyme 2 binding activity. Preparations that were intended for intravenous administration had higher titers than those intended for subcutaneous administration. However, the neutralizing activity against the Omicron variant was lower than the activity against prior variants, and the efficacy of using IVIG for the treatment of COVID-19 remains uncertain.

These data do not provide clear evidence for a clinical benefit of administering IVIG to people with COVID-19. Randomized controlled trials are needed to further define the role of IVIG in the treatment of COVID-19. The use of non-SARS-CoV-2–specific IVIG for the treatment of COVID-19 should be limited to clinical trials.

Concentrated antibody preparations derived from pooled plasma collected from individuals who have recovered from COVID-19 can be manufactured as SARS-CoV-2 hyperimmunoglobulin (hIVIG). Treatment with SARS-CoV-2 hIVIG did not alter patient outcomes in a large randomized controlled trial of hospitalized patients with COVID-19, and hIVIG is not currently available for clinical use in the United States.⁹

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Ivermectin

Last Updated: December 20, 2023

Ivermectin is a Food and Drug Administration (FDA)-approved antiparasitic drug used to treat several neglected tropical diseases, including onchocerciasis, helminthiasis, and scabies.¹ For these indications, ivermectin has been widely used and is generally well tolerated.^{1,2} Ivermectin is not approved by the FDA for the treatment of any viral infection, including COVID-19. See the FDA webpage [Why You Should Not Use Ivermectin to Treat or Prevent COVID-19](#) for more information.

Ivermectin has been shown to inhibit the replication of SARS-CoV-2 in cell cultures.³ However, pharmacokinetic and pharmacodynamic studies suggest that achieving the plasma concentrations necessary for the antiviral efficacy detected in vitro would require administration of doses up to 100-fold higher than those approved for use in humans.^{4,5}

The safety and efficacy of ivermectin for the prevention and treatment of COVID-19 have been evaluated in clinical trials and observational cohorts. Summaries of the studies that informed the COVID-19 Treatment Guidelines Panel's (the Panel) recommendation can be found in [Table 7b](#). The Panel reviewed additional studies, but these studies are not summarized in Table 7b because they have study design limitations or results that make them less definitive and informative.

Recommendation

- The Panel **recommends against** the use of **ivermectin** for the treatment of COVID-19 (**AIIa**).

Rationale

The Panel's recommendation is primarily informed by adequately powered, randomized trials of ivermectin that reported clinical outcomes. Studies that randomized participants to receive ivermectin or a matched placebo had the greatest impact on the Panel's recommendation.⁶⁻¹³

Trials have failed to find a clinical benefit of using ivermectin to treat COVID-19 in outpatients. In TOGETHER, an adaptive platform trial conducted in Brazil, there was no apparent difference between the ivermectin and placebo arms for the primary outcome of risk of emergency department visits or hospitalization (14.7% vs. 16.4%).¹⁴ In addition, there was no statistically significant difference between the ivermectin and placebo arms in mortality (3.1% vs. 3.5%). In COVID-OUT, a randomized factorial trial, the use of ivermectin did not reduce the occurrence of a composite outcome of emergency department visits, hospitalization, or death when compared with a matched control (5.7% vs. 4.1%).⁶

The ACTIV-6 trial was an adaptive platform trial conducted in outpatients with mild to moderate COVID-19 in the United States.^{15,16} Participants were randomized to receive an ivermectin regimen (either 400 µg/kg for 3 days or 600 µg/kg for 6 days) or a matching placebo. In the 400 µg/kg phase of the study, the median time to sustained recovery was 12 days for the ivermectin arm and 13 days for the placebo arm. In the 600 µg/kg phase of the study, the median time to sustained recovery was 11 days for both arms.

I-TECH, an open-label trial conducted in Malaysia, found no difference between the ivermectin and standard of care arms in the occurrence of the primary outcome of risk of progression to severe COVID-19 (21.6% vs. 17.3%).¹⁷ Patients in the ivermectin arm had a lower risk of mortality than those in the standard of care arm (relative risk 0.31; 95% CI, 0.09–1.11; *P* = 0.09), but this difference was not statistically significant.

The study populations in most of the reviewed trials were patients with mild to moderate COVID-19 who had a relatively low risk of disease progression, and the number of deaths was low (as expected). In these randomized trials, completely excluding an effect of ivermectin on COVID-19 disease progression is difficult because the trials were not powered to detect differences in secondary outcomes, such as death. However, data from these trials do not provide evidence that the use of ivermectin is effective for the treatment of COVID-19. For this reason, and because other medications now have demonstrated clear clinical benefits for the treatment of COVID-19, the Panel **recommends against** the use of **ivermectin** for the treatment of COVID-19 (**AIIa**).

See [Table 7b](#) for summaries of key studies that informed the Panel’s recommendation.

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Table 7b. Ivermectin: Selected Clinical Trial Data

Last Updated: December 20, 2023

The clinical trials described in this table are the RCTs that had the greatest impact on the Panel's recommendation. The Panel reviewed other clinical studies that evaluated the use of IVM for the treatment of COVID-19.¹⁻²⁶ However, those studies have limitations that make them less definitive and informative than the studies summarized in this table.

Methods	Results	Limitations and Interpretation
ACTIV-6: Double-Blind RCT of Ivermectin 600 µg/kg in Outpatients With Mild to Moderate COVID-19 in the United States²⁷		
Key Inclusion Criteria • Aged ≥30 years • Not hospitalized • Positive SARS-CoV-2 test result within past 10 days • ≥2 COVID-19 symptoms for ≤7 days Key Exclusion Criteria • End-stage kidney disease • Liver failure or decompensated cirrhosis Interventions • IVM 600 µg/kg PO once daily for 6 days (n = 602) • Placebo (n = 604) Primary Endpoint • Time to sustained recovery (i.e., ≥3 consecutive days without symptoms) Key Secondary Endpoint • Hospitalization or death by Day 28 Safety Endpoint • Occurrence of AEs and SAEs	Participant Characteristics • Median age 48 years; 59.1% women • 38.1% with BMI >30; 9.2% with DM; 26.8% with HTN • 83.6% received ≥2 COVID-19 vaccine doses. • Median of 5 days from symptom onset to receipt of study drug Primary Outcome • Median time to sustained recovery: 11 days in IVM arm vs. 11 days in placebo arm (HR 1.02; 95% CrI, 0.92–1.13) Secondary Outcome • Hospitalization or death by Day 28: 5 (0.8%) in IVM arm vs. 2 (0.3%) in placebo arm Safety Outcomes • Occurrence of AEs: 52 of 566 patients (9.2%) in IVM arm vs. 41 of 576 patients (7.1%) in placebo arm • Occurrence of SAEs: 5 of 566 patients (0.9%) in IVM arm vs. 3 of 576 patients (0.5%) in placebo arm	Key Limitation • The low number of events limited the power to determine an effect on hospitalization and death. Interpretation • Among outpatients with COVID-19, IVM 600 µg/kg PO once daily for 6 days did not shorten time to sustained recovery or reduce incidence of hospitalization or death.

Methods	Results	Limitations and Interpretation
ACTIV-6: Double-Blind RCT of Ivermectin 400 µg/kg Once Daily in Outpatients With Mild to Moderate COVID-19 in the United States²⁸		
Key Inclusion Criteria • Aged ≥30 years • Not hospitalized • Positive SARS-CoV-2 test result within past 10 days • ≥2 COVID-19 symptoms for ≤7 days Key Exclusion Criteria • End-stage kidney disease • Liver failure or decompensated cirrhosis Interventions • IVM 400 µg/kg PO once daily for 3 days (n = 817) • Placebo (n = 774) Primary Endpoint • Time to sustained recovery (i.e., ≥3 consecutive days without symptoms) Key Secondary Endpoint • Hospitalization or death by Day 28 Safety Endpoint • Occurrence of AEs and SAEs	Participant Characteristics • Mean age 48 years; 59% women • 41% with BMI >30; 11.5% with DM; 26% with HTN • 47% received ≥2 COVID-19 vaccine doses. • Median of 6 days from symptom onset to receipt of study drug Primary Outcome • Median time to sustained recovery: 12 days in IVM arm vs. 13 days in placebo arm (HR 1.07; 95% CrI, 0.96–1.17) Secondary Outcome • Hospitalization or death by Day 28: 10 (1.2%) in IVM arm vs. 9 (1.2%) in placebo arm Safety Outcomes • Occurrence of AEs: 24 of 766 patients (3.1%) in IVM arm vs. 27 of 724 patients (3.7%) in placebo arm • Occurrence of SAEs: 9 of 766 patients (1.2%) in IVM arm vs. 9 of 724 patients (1.2%) in placebo arm	Key Limitation • The low number of events limited the power to determine an effect on hospitalization and death. Interpretation • Among outpatients with COVID-19, IVM 400 µg/kg PO once daily for 3 days did not shorten time to sustained recovery or reduce incidence of hospitalization or death.

Methods	Results	Limitations and Interpretation
TOGETHER: Double-Blind, Adaptive RCT of Ivermectin in Nonhospitalized Patients With COVID-19 in Brazil²⁹		
Key Inclusion Criteria • Positive SARS-CoV-2 antigen test result • Within 7 days of symptom onset • ≥1 high-risk factor for disease progression (e.g., aged >50 years, comorbidities, immunosuppression) Interventions • IVM 400 µg/kg PO once daily for 3 days (n = 679) • Placebo (n = 679; not all patients received IVM placebo) Primary Endpoint • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints • Viral clearance at Day 7 • All-cause mortality • Occurrence of AEs	Participant Characteristics • Median age 49 years; 46% aged ≥50 years; 58% women; 95% self-identified as mixed race • Most prevalent risk factor: 50% with obesity • 44% within 3 days of symptom onset at enrollment Primary Outcome • Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 (ITT): 100 (14.7%) in IVM arm vs. 111 (16.4%) in placebo arm (relative risk 0.90; 95% CrI, 0.70–1.16) • 171 (81%) of events were hospitalizations (ITT) Secondary Outcomes • No difference between IVM arm and placebo arm in: • Viral clearance at Day 7 (relative risk 1.00; 95% CrI, 0.68–1.46) • All-cause mortality: 21 (3.1%) vs. 24 (3.5%) (relative risk 0.88; CrI, 0.49–1.55) • Occurrence of AEs	Key Limitations • Health care facility capacity may have influenced the number and duration of ED visits and hospitalizations. • No details on safety outcomes (e.g., type of treatment-emergent AEs) other than grading were reported. Interpretation • In outpatients with recent SARS-CoV-2 infection, IVM did not reduce the need for ED visits or hospitalization when compared with placebo.

Methods	Results	Limitations and Interpretation
COVID-OUT: RCT of Metformin, Ivermectin, and Fluvoxamine in Nonhospitalized Adults With COVID-19 in the United States³⁰		
Key Inclusion Criteria • Aged 30–85 years • BMI ≥ 25 or ≥ 23 if Asian or Latinx • Laboratory-confirmed SARS-CoV-2 infection within 3 days of randomization • ≤ 7 days of COVID-19 symptoms Key Exclusion Criteria • Immunocompromised • Hepatic impairment • Stage 4–5 chronic kidney disease or eGFR < 45 mL/min/1.73 m² Interventions • IVM 390–470 ug/kg PO once daily for 3 days (n = 410) in the following arms: • IVM alone (n = 206) • Metformin plus IVM (n = 204) • IVM control (n = 398), which included the following arms: • Placebo alone (n = 203) • Metformin alone (n = 195) Primary Endpoints • Composite of hypoxemia (SpO₂ $\leq 93\%$, as measured by a home pulse oximeter), ED visit, hospitalization, or death by Day 14 • A prespecified secondary analysis evaluated the occurrence of ED visits, hospitalization, or death by Day 14. Key Secondary Endpoints • Total symptom severity score by Day 14, as measured by a symptom severity scale • Drug discontinuation or interruption	Participant Characteristics • Median age 46 years; 56% women; 82% White • Median BMI 30 • 27% with CVD • 52% received primary COVID-19 vaccination series. • Mean of 4.8 days of symptoms • Approximately 68% enrolled while Delta was the dominant variant; approximately 29% enrolled while Omicron was dominant. Primary Outcomes • Composite of hypoxemia, ED visit, hospitalization, or death by Day 14: 105 (25.8%) in IVM arm vs. 96 (24.6%) in control arm (aOR 1.05; 95% CI, 0.76–1.45, $P = 0.78$) • No difference between IVM alone arm and placebo alone arm in occurrence of primary endpoint (aOR 1.06; 95% CI, 0.67–1.67) • ED visit, hospitalization, or death by Day 14 in a prespecified secondary analysis: 23 (5.7%) in IVM arm vs. 16 (4.1%) in control arm (aOR 1.39; 95% CI, 0.72–2.69) • Hospitalization or death by Day 14 in a prespecified secondary analysis: 4 (1.0%) in IVM arm vs. 5 (1.3%) in control arm (aOR 0.73; 95% CI, 0.19–2.77); 1 death in IVM arm vs. 0 deaths in control arm Secondary Outcomes • No difference between arms in total symptom severity score by Day 14 • Drug discontinuation or interruption: 20% in IVM arm vs. 25% in placebo alone arm	Key Limitations • Study included SpO₂ measurements using home pulse oximeters as 1 of the composite measures of the primary endpoint. However, the FDA has issued a statement concerning the accuracy of these home pulse oximeters, making this study endpoint less reliable. • SpO₂ data were incomplete or missing for 30% of the patients. • The low number of events limited the power to determine the effect on hospitalization and death. Interpretations • IVM did not prevent the composite endpoint of hypoxemia, ED visit, hospitalization, or death. • No primary, secondary, or subgroup analysis demonstrated a benefit for the use of IVM over placebo.

Methods	Results	Limitations and Interpretation
IVERCOR-COVID19: Double-Blind, Placebo-Controlled, Randomized Trial of Ivermectin in Nonhospitalized Patients With COVID-19 in Argentina³¹		
Key Inclusion Criterion - Positive SARS-CoV-2 RT-PCR result within 48 hours of screening Key Exclusion Criteria - Required supplemental oxygen or hospitalization - Concomitant use of CQ or HCQ Interventions - Weight-based dose of IVM PO at enrollment and 24 hours later for a maximum total dose of 48 mg (n = 250) - Placebo (n = 251) Primary Endpoint - Hospitalization for any reason Key Secondary Endpoints - Need for MV - All-cause mortality - Occurrence of AEs	Participant Characteristics - Mean age 42 years; 8% aged ≥65 years; 47% women 24% with HTN; 10% with DM; 58% with ≥1 comorbidity - Median of 4 days from symptom onset Primary Outcome - Hospitalization for any reason: 5.6% in IVM arm vs. 8.3% in placebo arm (OR 0.65; 95% CI, 0.32–1.31; $P = 0.23$) Secondary Outcomes - Need for MV: 2% in IVM arm vs. 1% in placebo arm ($P = 0.7$) - All-cause mortality: 2% in IVM arm vs. 1% in placebo arm ($P = 0.7$) - Occurrence of AEs: 18% in IVM arm vs. 21% in placebo arm ($P = 0.6$)	Key Limitation - Study enrolled a young population with few of the comorbidities that predict disease progression. Interpretation - Among patients who had recently acquired SARS-CoV-2 infection, there was no evidence that IVM provided any clinical benefit.

Methods	Results	Limitations and Interpretation
Double-Blind, Placebo-Controlled, Randomized Trial of Ivermectin in Patients With Mild COVID-19 in Colombia ³²		
Key Inclusion Criteria • Positive SARS-CoV-2 RT-PCR or antigen test result • ≤7 days of COVID-19 symptoms • Mild disease Key Exclusion Criteria • Asymptomatic disease • Severe pneumonia • Hepatic dysfunction Interventions • IVM 300 µg/kg PO once daily for 5 days (n = 200) • Placebo PO (n = 198) Primary Endpoint • Time to symptom resolution within 21 days Key Secondary Endpoints • Clinical deterioration • Escalation of care • Occurrence of AEs	Participant Characteristics • Median age 37 years; 4% aged ≥65 years in IVM arm, 8% in placebo arm; 39% men in IVM arm, 45% in placebo arm • 79% with no known comorbidities • Median of 5 days from symptom onset to randomization Primary Outcome • Median time to symptom resolution: 10 days in IVM arm vs. 12 days in placebo arm (HR 1.07; <i>P</i> = 0.53) • Symptoms resolved by Day 21: 82% in IVM arm vs. 79% in placebo arm Secondary Outcomes • No difference between arms in proportion of patients who showed clinical deterioration or required escalation of care • Occurrence of AEs: • Discontinued treatment due to AEs: 8% in IVM arm vs. 3% in placebo arm • No SAEs related to intervention	Key Limitations • Due to low event rates, the primary endpoint changed from the proportion of patients with clinical deterioration to the time to symptom resolution during the trial. • The study enrolled younger, healthier patients, a population that does not typically develop severe COVID-19. Interpretation • In patients with mild COVID-19, IVM 300 µg/kg once daily for 5 days did not improve the time to symptom resolution.

Methods	Results	Limitations and Interpretation
I-TECH: Open-Label RCT of Ivermectin in Patients With Mild to Moderate COVID-19 in Malaysia³³		
Key Inclusion Criteria • Positive SARS-CoV-2 RT-PCR or antigen test result within 7 days of symptom onset • Aged ≥50 years • ≥1 comorbidities Key Exclusion Criteria • Required supplemental oxygen • Severe hepatic impairment (ALT >10 times the ULN) Interventions • IVM 400 µg/kg PO once daily for 5 days plus SOC (n = 241) • SOC (n = 249) Primary Endpoint • Progression to severe COVID-19 (i.e., hypoxemia requiring supplemental oxygen to maintain SpO₂ ≥95%) Key Secondary Endpoints • In-hospital, all-cause mortality by Day 28 • MV or ICU admission • Occurrence of AEs	Participant Characteristics • Mean age 63 years; 55% women • 68% received ≥1 COVID-19 vaccine dose; 52% received 2 doses. • Most common comorbidities: 75% with HTN; 54% with DM; 24% with dyslipidemia • Mean of 5 days symptom duration Primary Outcome • Progression to severe COVID-19 (mITT): 52 (21.6%) in IVM plus SOC arm vs. 43 (17.3%) in SOC alone arm (relative risk 1.25; 95% CI, 0.87–1.80; <i>P</i> = 0.25) Secondary Outcomes • No difference between IVM plus SOC arm and SOC alone arm in: • In-hospital, all-cause mortality by Day 28: 3 (1.2%) vs. 10 (4.0%) (relative risk 0.31; 95% CI, 0.09–1.11; <i>P</i> = 0.09) • MV: 4 (1.7%) vs. 10 (4.0%) (relative risk 0.41; 95% CI, 0.13–1.30; <i>P</i> = 0.17) • ICU admission: 6 (2.5%) vs. 8 (3.2%) (relative risk 0.78; 95% CI, 0.27–2.20; <i>P</i> = 0.79) • Occurrence of AEs: 33 (13.7%) in IVM plus SOC arm vs. 11 (4.4%) in SOC alone arm; most with diarrhea (14 vs. 4)	Key Limitation • Open-label study Interpretation • In patients with mild to moderate COVID-19, there was no evidence that IVM provided any clinical benefit, including no evidence that IVM reduced the risk of progression to severe disease.

Methods	Results	Limitations and Interpretation
COVER: Phase 2, Double-Blind RCT of Ivermectin in Nonhospitalized Patients With COVID-19 in Italy³⁴		
Key Inclusion Criteria - Asymptomatic or oligosymptomatic disease - SARS-CoV-2 infection confirmed by RT-PCR result - Not hospitalized or receiving supplemental oxygen Key Exclusion Criteria - CNS disease - Receiving dialysis - Severe medical condition with <6 months survival prognosis - Use of warfarin, antiviral agents, CQ, or HCQ Interventions - IVM 1,200 µg/kg PO once daily for 5 days (n = 32) - IVM 600 µg/kg plus placebo PO once daily for 5 days (n = 29) - Placebo PO (n = 32) Primary Endpoints - Number of SAEs - Change in VL at Day 7 Other Endpoint - Drug discontinuation or interruption	Participant Characteristics - Median age 47 years; 58% men 86% with COVID-19 symptoms 2.2% received a COVID-19 vaccine. Primary Outcomes - No SAEs related to intervention - Mean log₁₀ reduction in VL at Day 7: 2.9 in IVM 1,200 µg/kg arm vs. 2.5 in IVM 600 µg/kg arm vs. 2.0 in placebo arm (IVM 1,200 µg/kg vs. placebo, <i>P</i> = 0.099; IVM 600 µg/kg vs. placebo, <i>P</i> = 0.122) Other Outcomes 14 (15.1%) discontinued treatment: 11 (34.4%) in IVM 1,200 µg/kg arm vs. 2 (6.9%) in IVM 600 µg/kg arm vs. 1 (3.1%) in placebo arm - All discontinuations in IVM 1,200 µg/kg arm were due to tolerability	Key Limitations - Small, Phase 2 study 90% of subjects screened were not enrolled for various reasons. - Recruitment stopped early because of a decline in the number of COVID-19 cases. Interpretations - A high dose of IVM (1,200 µg/kg) appears to be safe but not well tolerated; 34% of patients discontinued therapy due to AEs. - There was no significant difference in reduction of VL between IVM and placebo arms.

Methods	Results	Limitations and Interpretation
Open-Label RCT of Ivermectin in Hospitalized Patients With COVID-19 in Egypt³⁵		
Key Inclusion Criteria • RT-PCR-confirmed SARS-CoV-2 infection by pharyngeal swab • Hospitalized with mild to moderate COVID-19 Key Exclusion Criterion • Cardiac problems Interventions • IVM 12 mg PO once daily for 3 days (n = 82) • SOC (n = 82) Primary Endpoint • All-cause mortality by 28 days Key Secondary Endpoints • Hospital LOS • Need for MV	Participant Characteristics • Mean age 42 years for IVM arm, 39 years for SOC arm; 50% men • 49% with ≥1 comorbidities Primary Outcome • All-cause mortality by 28 days: 3 (3.7%) in IVM arm vs. 4 (4.9%) in SOC arm ($P = 1.00$) Secondary Outcomes • Mean hospital LOS: 9 days in IVM arm vs. 11 days in SOC arm ($P = 0.085$) • Need for MV: 3 (3.7%) in each arm ($P = 1.00$)	Key Limitation • Small, open-label study Interpretation • The use of IVM did not reduce all-cause mortality, hospital LOS, or the need for MV among patients with mild to moderate COVID-19.

Methods	Results	Limitations and Interpretation
Double-Blind, Placebo-Controlled, Randomized Trial of Ivermectin in Patients With Mild to Moderate COVID-19 in India ³⁶		
Key Inclusion Criteria • Positive SARS-CoV-2 RT-PCR or antigen test result • Hospitalized with mild to moderate COVID-19 Interventions • IVM 12 mg PO once daily for 2 days (n = 55) • Placebo PO (n = 57) Primary Endpoint • Negative SARS-CoV-2 RT-PCR result on Day 6 Key Secondary Endpoints • Symptom resolution by Day 6 • Discharge by Day 10 • Need for ICU admission or MV • In-hospital mortality	Participant Characteristics • Mean age 53 years; 28% women • 35% with HTN; 36% with DM • 79% with mild COVID-19 • Mean of 6.9 days from symptom onset • 100% received HCQ, steroids, and antibiotics; 21% received RDV; 6% received tocilizumab. Primary Outcome • Negative SARS-CoV-2 RT-PCR result on Day 6: 24% in IVM arm vs. 32% in placebo arm (rate ratio 0.8; $P = 0.348$) Secondary Outcomes • Symptom resolution by Day 6: 84% in IVM arm vs. 90% in placebo arm (rate ratio 0.9; $P = 0.36$) • Discharge by Day 10: 80% in IVM arm vs. 74% in placebo arm (rate ratio 1.1; $P = 0.43$) • No difference between arms in need for ICU admission or MV • In-hospital mortality: 0 in IVM arm (0%) vs. 4 in placebo arm (7%)	Key Limitations • Although the primary endpoint was a negative SARS-CoV-2 RT-PCR result on Day 6, no RT-PCR result or an inconclusive RT-PCR result was reported for 42% of patients in the IVM arm and 23% in the placebo arm. • The time to discharge was not reported, and outcomes after discharge were not evaluated. Interpretation • IVM provided no significant virologic or clinical benefit for patients with mild to moderate COVID-19.

Methods	Results	Limitations and Interpretation
RIVET-COV: Double-Blind, Placebo-Controlled, Randomized Trial of Ivermectin in Patients With Mild to Moderate COVID-19 in India³⁷		
Key Inclusion Criteria • Positive SARS-CoV-2 RT-PCR or antigen test result • Nonsevere COVID-19 Key Exclusion Criteria • CrCl <30 mL/min • Transaminases >5 times ULN • MI, heart failure, QTc interval prolongation • Severe comorbidity Interventions • Single dose of IVM 24 mg PO (n = 51) • Single dose of IVM 12 mg PO (n = 49) • Placebo (n = 52) Primary Endpoints • Negative SARS-CoV-2 RT-PCR result at Day 5 • Decline of VL at Day 5 Key Secondary Endpoints • Time to symptom resolution • Clinical worsening at Day 14 • Number of hospital-free days at Day 28 • Frequency of AEs	Participant Characteristics • Mean age 35 years; 89% men • 60% to 68% with mild COVID-19 (including asymptomatic patients); 33% to 40% with moderate COVID-19 • Median of 4–5 days symptom duration; similar across arms • 10% in each arm received concurrent antivirals (RDV, favipiravir, or HCQ). Primary Outcomes • Negative SARS-CoV-2 RT-PCR result at Day 5: 48% in IVM 24 mg arm vs. 35% in IVM 12 mg arm vs. 31% in placebo arm ($P = 0.30$) • No significant difference between arms in decline of VL at Day 5 Secondary Outcomes • No difference between arms in time to symptom resolution • Clinical worsening at Day 14: 8% in IVM 24 mg arm vs. 5% in IVM 12 mg arm vs. 11% in placebo arm ($P = 0.65$) • No difference between arms in number of hospital-free days at Day 28 • No difference between arms in frequency of AEs; no SAEs were reported	Key Limitation • Small sample size Interpretation • The use of IVM did not affect the proportion of patients with negative SARS-CoV-2 RT-PCR results at Day 5 or the clinical outcomes.

Methods	Results	Limitations and Interpretation
Double-Blind RCT of Ivermectin, Chloroquine, or Hydroxychloroquine in Hospitalized Adults With Severe COVID-19 in Brazil³⁸		
Key Inclusion Criteria - Hospitalized with laboratory-confirmed SARS-CoV-2 infection ≥1 of the following severity criteria: - Dyspnea - Tachypnea (>30 breaths/min) - SpO₂ <93% - PaO₂/FiO₂ <300 mm Hg - Involvement of >50% of lungs confirmed by CXR or CT scan Key Exclusion Criterion - Cardiac arrhythmia Interventions - IVM 14 mg once daily for 3 days (n = 53) - CQ 450 mg twice daily on Day 0, then once daily for 4 days (n = 61) - HCQ 400 mg twice daily on Day 0, then once daily for 4 days (n = 54) Endpoints - Need for supplemental oxygen, MV, or ICU admission - Occurrence of AEs - Mortality	Participant Characteristics - Mean age 53 years; 58% men - Most common comorbidities: 43% with HTN; 28% with DM; 38% with BMI >30 76% with respiratory failure on admission Outcomes - No difference between IVM, CQ, and HCQ arms in: - Need for supplemental oxygen: 88% vs. 89% vs. 90% - Need for MV: 24% vs. 21% vs. 21% - ICU admission: 28% vs. 22% vs. 21% - Mortality: 23% vs. 21% vs. 22% - Mean number of days of supplemental oxygen: 8 days in each arm - No difference between arms in occurrence of AEs - Baseline characteristics significantly associated with mortality: - Aged >60 years (HR 2.4) - DM (HR 1.9) - BMI >33 (HR 2.0) - SpO₂ <90% (HR 5.8)	Key Limitations - Small sample size - No clearly defined primary endpoint Interpretation - Compared to CQ or HCQ, IVM did not reduce the proportion of hospitalized patients with severe COVID-19 who died or who required supplemental oxygen, ICU admission, or MV

Key: AE = adverse event; ALT = alanine transaminase; BMI = body mass index; CNS = central nervous system; CQ = chloroquine; CrCl = creatinine clearance; CT = computed tomography; CVD = cardiovascular disease; CXR = chest X-ray; DM = diabetes mellitus; ED = emergency department; eGFR = estimated glomerular filtration rate; FDA = Food and Drug Administration; HCQ = hydroxychloroquine; HTN = hypertension; ICU = intensive care unit; ITT = intention-to-treat; IVM = ivermectin; LOS = length of stay; MI = myocardial infarction; mITT = modified intention-to-treat; MV = mechanical ventilation; the Panel = the COVID-19 Treatment Guidelines Panel; PaO₂/FiO₂ = ratio of arterial partial pressure of oxygen to fraction of inspired oxygen; PO = oral; RCT = randomized controlled trial; RDV = remdesivir; RT-PCR = reverse transcriptase polymerase chain reaction; SAE = severe adverse event; SOC = standard of care; SpO₂ = oxygen saturation; ULN = upper limit of normal; VL = viral load

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Metformin

Last Updated: December 20, 2023

Metformin has been identified as a potential COVID-19 therapeutic agent because of its possible action against the proteins that are involved in translation, its antiviral activity in vitro, and its anti-inflammatory and antithrombotic activities.¹⁻⁴ Data from observational studies have suggested that patients who were receiving metformin as treatment for diabetes at the time of their COVID-19 diagnosis had a lower risk of progressing to severe COVID-19.⁵⁻⁷ Randomized controlled trials have provided insight into the role of metformin in treating nonhospitalized patients with COVID-19. These trials are described below and in [Table 7c](#).

Recommendations

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of metformin for the treatment of COVID-19 in nonhospitalized patients.
- The Panel **recommends against** the use of **metformin** for the treatment of COVID-19 in hospitalized patients, except in a clinical trial (**BIII**).
- Patients with COVID-19 who are receiving **metformin** for an underlying condition should continue this therapy as directed by their health care provider (**AIII**).

Rationale

Two randomized controlled trials (the TOGETHER and COVID-OUT trials) assessed the efficacy of using metformin in nonhospitalized patients with COVID-19. In these trials, the use of metformin did not reduce the risk of hospitalization or death in these patients. The Panel's recommendations are based on the results of these trials.

Other outpatient therapies (i.e., ritonavir-boosted nirmatrelvir [Paxlovid], remdesivir, molnupiravir) have been shown to be effective in preventing hospitalization or death in unvaccinated patients who are at high risk of disease progression.

Monitoring, Adverse Effects, and Drug-Drug Interactions

The most common adverse effects of metformin are nausea, vomiting, diarrhea, and headache. In rare cases, lactic acidosis may occur. The risk factors associated with lactic acidosis include older age, impaired renal or hepatic function, the use of iodinated contrast dye, cardiac dysfunction, metabolic disturbances, and excessive alcohol consumption. Metformin is not recommended for patients with an estimated glomerular filtration rate of <30 mL/min/1.73m².

Metformin is a substrate of the human organic cation transporters OCT1 and OCT2. Drugs that inhibit these transporters may increase the systemic exposure of metformin and increase the risk of metformin-related adverse effects.

Considerations in Pregnant People

Metformin is commonly used in pregnant people with type 2 diabetes mellitus. However, because clinical trials have not demonstrated a clear clinical benefit of using metformin in nonpregnant adults with COVID-19, there is no justification for administering it to pregnant people to treat COVID-19.

outside of a clinical trial.

Considerations in Children

Although metformin is approved by the Food and Drug Administration for the treatment of type 2 diabetes mellitus in children aged >10 years, clinical trials that have evaluated its use for the treatment of COVID-19 have not included people aged <18 years. Given the lack of clear evidence of efficacy in adults, the Panel **recommends against** the use of **metformin** for the treatment of COVID-19 in pediatric patients, except in a clinical trial (**AIII**).

Clinical Data

TOGETHER Trial

The TOGETHER trial was a placebo-controlled platform clinical trial that was conducted in Brazil.⁸ The study enrolled nonhospitalized patients who had symptomatic SARS-CoV-2 infection for ≤7 days, no history of COVID-19 vaccination, and an increased risk of progressing to severe disease. Patients were randomized to receive extended-release metformin 750 mg (n = 215) or placebo (n = 203) twice daily for 10 days.

The primary endpoint was a composite of retention in an emergency setting for >6 hours or hospitalization within 28 days of randomization. Secondary endpoints included viral clearance at Days 3 and 7, clinical improvement at Day 28, time to hospitalization or death, and the occurrence of adverse events. The study was stopped by the data and safety monitoring board for futility, as there was a low probability of demonstrating a difference between the study arms. Overall, there was no difference between the arms in the number of adverse events; however, the proportion of patients who experienced grade 3 events was higher in the metformin arm (9.8%) than in the placebo arm (4.4%).

COVID-OUT Trial

The COVID-OUT trial was a Phase 3, double-blind, placebo-controlled 2 x 3 factorial trial that evaluated the effectiveness of metformin, ivermectin, or fluvoxamine in patients with COVID-19.⁹ Patients were randomized to receive metformin or placebo in 1 factor and ivermectin, fluvoxamine, or placebo in the other factor. The study enrolled nonhospitalized adults within 3 days of a confirmed diagnosis of COVID-19 and ≤7 days from symptom onset. Patients were aged 30 to 85 years and overweight. Those with stage 4 or 5 chronic kidney disease or an estimated glomerular filtration rate of <45 mL/min/1.73 m² were excluded. The metformin arm included those assigned to receive immediate-release oral metformin (titrated over several days to a final daily dose of 1,500 mg) alone or in combination with ivermectin or fluvoxamine. The control arm included those who received placebo with or without ivermectin or fluvoxamine.

The primary endpoint was a composite of development of hypoxemia (defined as oxygen saturation ≤93%, as measured by a home pulse oximeter), emergency department visit, hospitalization, or death by Day 14. While this study was underway, the Food and Drug Administration raised concerns about the accuracy of home pulse oximeters. Approximately 50% of the patients received a primary COVID-19 vaccine series. The analyses showed no benefit for any of the 3 investigational agents in preventing the primary endpoint. In addition, the use of these agents did not lower the severity of COVID-19 symptoms over 14 days. A prespecified secondary analysis determined that, over 14 days of follow-up, those who received metformin had a lower risk of an emergency department visit, hospitalization, or death than those who did not receive metformin (adjusted OR 0.58; 95% CI, 0.35–0.94). A key secondary endpoint in the analysis was a composite of hospitalization or death by Day 28. Eight of 596 patients (1.3%) who received metformin met this endpoint compared with 19 of 601 patients (3.2%) who did not receive

metformin.

A secondary endpoint in the COVID-OUT trial assessed the impact of metformin on the development of long COVID. Since there is no standardized definition for long COVID, the endpoint was based on whether the patient had been given this diagnosis by a health care provider during the 10 months of follow-up. The study reported lower rates of long COVID in the metformin arm than in the control arm.¹⁰

Although a secondary analysis of the COVID-OUT trial data demonstrated a benefit of metformin in patients with COVID-19, the results of the TOGETHER and COVID-OUT trials did not show a consistent benefit of metformin in these patients. Therefore, the Panel believes there is insufficient evidence to recommend either for or against the use of metformin for the treatment of COVID-19 in nonhospitalized patients. For more information on these trials, see [Table 7c](#).

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Table 7c. Metformin: Selected Clinical Trial Data

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The Panel's recommendations for metformin are based on data from the clinical trials described in this table.

Methods	Results	Limitations and Interpretation
TOGETHER: RCT of Metformin in Nonhospitalized Patients With COVID-19 in Brazil¹		
Key Inclusion Criteria - Aged ≥50 years or aged ≥18 years with ≥1 comorbidities - Positive rapid antigen test result for SARS-CoV-2 infection ≤7 days of COVID-19 symptoms Key Exclusion Criteria - Acute respiratory symptoms that required hospitalization - Receipt of a COVID-19 vaccine Interventions - Extended-release metformin 750 mg PO twice daily for 10 days (n = 215) - Placebo PO twice daily for 10 days (n = 203) Primary Endpoint - Composite of ED observation >6 hours or hospitalization for COVID-19 by Day 28 Key Secondary Endpoints - Clinical improvement by Day 28 - Viral clearance by Day 7 - Time to hospitalization or death - Occurrence of AEs - Study adherence	Participant Characteristics - Median age 52 years; 57% women; 91% self-identified as mixed race 45% with BMI ≥30; 40% with HTN; 15% with DM 44% had COVID-19 symptoms for 0–3 days at enrollment Primary Outcome - Study was stopped early by DSMB for futility. At the time the study was stopped, primary endpoint had occurred in 16% in metformin arm vs. 14% in placebo arm (relative risk 1.14; 95% CI, 0.73–1.81; probability of superiority 28%). Secondary Outcomes - No difference between arms in: - Clinical improvement by Day 28 (OR 1.05; 95% CI, 0.71–1.56) - Viral clearance by Day 7 (OR 0.99; 95% CI, 0.88–1.11) - Time to hospitalization or death ($P = 0.53$) - Occurrence of treatment-emergent, grade 3 AEs: 9.8% in metformin arm vs. 4.4% in placebo arm (relative risk 2.11; 95% CI, 1.05–4.61) - Did not complete all phases of the study: 22% in metformin arm vs. 12% in placebo arm	Key Limitations - The >6-hour ED observation endpoint has not been used in other studies of interventions for nonhospitalized patients who are at high risk of hospitalization and death. - Study was stopped early for futility. - Vaccinated individuals were excluded from trial. Interpretation - This trial demonstrated no clinical benefit of metformin in nonhospitalized patients with COVID-19. - The use of metformin was associated with more grade 3 AEs than placebo.

Methods	Results	Limitations and Interpretation
COVID-OUT: RCT of Metformin, Ivermectin, and Fluvoxamine in Nonhospitalized Adults With COVID-19 in the United States²		
Key Inclusion Criteria • Aged 30–85 years • BMI ≥ 25 or ≥ 23 if Asian or Latinx • Laboratory-confirmed SARS-CoV-2 infection within 3 days of randomization • ≤ 7 days of COVID-19 symptoms Key Exclusion Criteria • Immunocompromised • Hepatic impairment • Stage 4–5 chronic kidney disease or eGFR of < 45 mL/min/1.73m² Interventions • Immediate-release metformin 500 mg PO on Day 1, 500 mg twice daily on Days 2–5, and 500 mg in morning and 1,000 mg in evening on Days 6–14 (n = 663) in the following arms: • Metformin alone (n = 284) • Metformin plus IVM 390–470 μg/kg PO once daily for 3 days (n = 204) • Metformin plus fluvoxamine 50 mg PO twice daily for 14 days (n = 175) • Control (n = 655), which included the following arms: • Placebo alone (n = 293) • IVM or fluvoxamine alone (n = 362) Primary Endpoints • Composite of hypoxemia (SpO₂ $\leq 93\%$, as measured by a home pulse oximeter), ED visit, hospitalization, or death by Day 14 • A prespecified secondary analysis evaluated the occurrence of ED visits, hospitalization, or death by Day 14.	Participant Characteristics • Median age 46 years; 56% women; 82% White • Median BMI 30 • 27% with CVD • 52% received primary COVID-19 vaccination series • Mean duration of symptoms was 4.8 days • Approximately 66% enrolled while Delta was the dominant variant; approximately 22% enrolled while Omicron was dominant Primary Outcomes • Composite of hypoxemia, ED visit, hospitalization, or death by Day 14: 154 (24%) in metformin arm vs. 179 (27%) in control arm (aOR 0.84; 95% CI, 0.66–1.09; <i>P</i> = 0.19) • No difference between metformin alone arm and placebo alone arm in occurrence of primary endpoint (aOR 0.91; 95% CI, 0.62–1.33) • ED visit, hospitalization, or death by Day 14 in a prespecified secondary analysis: 27 (4.1%) in metformin arm vs. 48 (7.3%) in control arm (aOR 0.58; 95% CI, 0.35–0.94) • Hospitalization or death by Day 14 in a prespecified secondary analysis: 8 (1.2%) in metformin arm vs. 18 (2.7%) in control arm (aOR 0.47; 95% CI, 0.20–1.11) Secondary Outcomes • No difference between arms in total symptom severity score by Day 14 • Drug discontinuation or interruption: 29% in metformin arm vs. 25% in control arm • Hospitalization or death by Day 28: 8 of 596 (1.3%) in metformin arm vs. 19 of 601 (3.2%) in control arm	Key Limitations • Analyses of secondary endpoints were not adjusted for multiple comparisons. • Study included SpO₂ measurements using home pulse oximeters as 1 of the composite measures of the primary endpoint. However, the FDA has issued a statement concerning the accuracy of these home pulse oximeters, making this study endpoint less reliable. Interpretation • The use of metformin did not prevent the occurrence of the primary composite endpoint of hypoxemia, ED visit, hospitalization, or death by Day 14. • Although the results of the prespecified secondary analyses of ED visits, hospitalization, or death by Day 14 and the secondary endpoint of hospitalization or death by Day 28 suggest a potential benefit of metformin, these results are not considered definitive.

Methods	Results	Limitations and Interpretation
COVID-OUT: RCT of Metformin, Ivermectin, and Fluvoxamine in Nonhospitalized Adults With COVID-19 in the United States², continued		
Key Secondary Endpoints • Total symptom severity score by Day 14, as measured by a symptom severity scale • Drug discontinuation or interruption • Hospitalization or death by Day 28		

Key: AE = adverse event; BMI = body mass index; CVD = cardiovascular disease; DM = diabetes mellitus; DSMB = data and safety monitoring board; ED = emergency department; eGFR = estimated glomerular filtration rate; FDA = Food and Drug Administration; HTN = hypertension; IVM = ivermectin; the Panel = the COVID-19 Treatment Guidelines Panel; PO = oral; RCT = randomized controlled trial; SpO₂ = oxygen saturation

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Table 7d. Characteristics of Miscellaneous Drugs

Last Updated: December 20, 2023

- The information in this table is derived from data on the use of these drugs for FDA-approved indications or in investigational trials. It is supplemented with data on the use of these drugs in patients with COVID-19 or MIS-C, when available.
- For dose modifications for patients with organ failure or those who require extracorporeal devices, please refer to product labels, when available.
- There are currently not enough data to determine whether certain medications can be safely coadministered with therapies for the treatment of COVID-19. When using concomitant medications with similar toxicity profiles, consider performing additional safety monitoring.
- The potential additive, antagonistic, or synergistic effects and the safety of using certain combination therapies for the treatment of COVID-19 are unknown. Clinicians are encouraged to report AEs to the [FDA MedWatch program](#).
- For drug-drug interaction information, please refer to product labels and visit the [Liverpool COVID-19 Drug Interactions website](#).

Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Fluvoxamine <i>Not recommended by the Panel for the treatment of COVID-19 in nonhospitalized patients.</i>				
Doses for COVID-19 in Clinical Trials • Fluvoxamine 50 mg PO twice daily • Fluvoxamine 100 mg PO twice daily • Fluvoxamine 100 mg PO 3 times daily	• Nausea • Diarrhea • Dyspepsia • Asthenia • Insomnia • Somnolence • Sweating • Suicidal ideation (rare)	• Hepatic function • Drug-drug interactions • Withdrawal symptoms during dose tapering	• CYP2D6 substrate • Fluvoxamine inhibits several CYP isoenzymes (CYP1A2, CYP2C9, CYP3A4, CYP2C19, CYP2D6). • Coadministration of tizanidine, thioridazine, alosetron, or pimozide with fluvoxamine is contraindicated.	• Fluvoxamine may enhance the anticoagulant effects of antiplatelets and anticoagulants. Consider additional monitoring when these drugs are used concomitantly with fluvoxamine. • The use of MAOIs concomitantly with fluvoxamine or within 14 days of treatment with fluvoxamine is contraindicated.

Dosing Regimens	Adverse Events	Monitoring Parameters	Drug-Drug Interaction Potential	Comments
Intravenous Immunoglobulin <i>Primarily used for the treatment of MIS-C. Currently under investigation in clinical trials.</i>				
Dose for MIS-C • 1 dose of IVIG 2 g/kg IBW IV, up to a maximum total dose of 100 g • In the event of cardiac dysfunction or fluid overload, consider administering IVIG in divided doses (i.e., IVIG 1 g/kg IBW IV, up to 50 g daily for 2 doses).	• Allergic reactions, including anaphylaxis • Renal failure • Thromboembolic events • Aseptic meningitis syndrome • Hemolysis • TRALI • Transmission of infectious pathogens • AEs may vary by formulation. • Risk and severity of AEs may increase with high dose or rapid infusion.	• Transfusion-related reactions • Vital signs at baseline and during and after infusion • Renal function; discontinue treatment if renal function deteriorates.	• Not a CYP substrate; no drug-drug interactions expected	• Rapid infusion should be avoided in patients with renal dysfunction or those who are at risk of thromboembolic events.
Metformin <i>There is insufficient evidence for the Panel to recommend either for or against the use of metformin in nonhospitalized patients. Not recommended by the Panel for the treatment of COVID-19 in hospitalized patients, except in a clinical trial.</i>				
Doses for COVID-19 in Clinical Trials • Immediate-release metformin 500 mg PO on Day 1; 500 mg twice daily on Days 2–5; and 500 mg in morning and 1,000 mg in evening on Days 6–14 • Extended-release metformin 750 mg PO twice daily for 10 days	• Diarrhea • Nausea and vomiting • Headache • Lactic acidosis	• Renal function • Hepatic function • Drug-drug interactions • Alcohol use disorder	• OCT1 and OCT2 substrate • Drugs that interfere with OCT systems (e.g., cimetidine, dolutegravir, ranolazine, vandetanib) could increase systemic exposure to metformin. • Concomitant use with carbonic anhydrase inhibitors (e.g., acetazolamide, topiramate, zonisamide) may increase the risk of lactic acidosis.	• Alcohol intake may increase the risk of lactic acidosis.

Key: AE = adverse event; CYP = cytochrome P450; FDA = Food and Drug Administration; IBW = ideal body weight; IV = intravenous; IVIG = intravenous immunoglobulin; MAOI = monoamine oxidase inhibitor; MIS-C = multisystem inflammatory syndrome in children; OCT = organic cation transporter; the Panel = the COVID-19 Treatment Guidelines Panel; PO = oral; TRALI = transfusion-related acute lung injury

Supplements

Last Updated: December 20, 2023

Summary Recommendations
Vitamin C • There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of vitamin C for the treatment of COVID-19 in nonhospitalized patients. • The Panel recommends against the use of vitamin C for the treatment of COVID-19 in hospitalized patients (AIIa). Vitamin D • There is insufficient evidence for the Panel to recommend either for or against the use of vitamin D for the prevention or treatment of COVID-19. Zinc • There is insufficient evidence for the Panel to recommend either for or against the use of zinc for the treatment of COVID-19. • The Panel recommends against using zinc supplementation above the recommended dietary allowance (i.e., zinc 11 mg daily for men, zinc 8 mg daily for nonpregnant women) for the prevention of COVID-19, except in a clinical trial (BIII). Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Vitamin C

Last Updated: December 20, 2023

Vitamin C (ascorbic acid) is a water-soluble vitamin that has been considered for potential beneficial effects in patients with varying degrees of illness severity. It is an antioxidant and free radical scavenger that has anti-inflammatory properties, influences cellular immunity and vascular integrity, serves as a cofactor in endogenous catecholamine generation, and has been studied in many disease states, including COVID-19.^{1,2}

Recommendation for Nonhospitalized Patients With COVID-19

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of vitamin C for the treatment of COVID-19 in nonhospitalized patients.

Rationale

Because patients who are not critically ill with COVID-19 are less likely to experience oxidative stress or severe inflammation, the role of vitamin C in this setting is unknown.

Clinical Data for Nonhospitalized Patients With COVID-19

In an open-label trial conducted at 2 sites in the United States, outpatients with laboratory-confirmed SARS-CoV-2 infection were randomized to receive either 10 days of oral ascorbic acid 8,000 mg, zinc gluconate 50 mg, both agents, or standard of care.³ The primary endpoint was the number of days required to reach a 50% reduction in the patient's symptom severity score. The study was stopped early by an operational and safety monitoring board due to futility after 214 of the planned 520 participants were enrolled.

Patients who received standard of care achieved a 50% reduction in their symptom severity scores at a mean of 6.7 days (SD 4.4 days) compared with 5.5 days (SD 3.7 days) in the ascorbic acid arm, 5.9 days (SD 4.9 days) in the zinc gluconate arm, and 5.5 days (SD 3.4 days) in the arm that received both agents (overall $P = 0.45$).³ No serious adverse events related to the treatments were reported. Nonserious adverse events were experienced by 39.5% of patients in the ascorbic acid arm, 18.5% in the zinc gluconate arm, and 32.1% in the arm that received both agents, compared with 0% of patients in the standard of care arm (overall $P < 0.001$). The most common nonserious adverse effects in this study were gastrointestinal events.

The limitations of this study include the small sample size and the lack of a placebo control. In outpatients with COVID-19, treatment with high-dose zinc gluconate, ascorbic acid, or a combination of the 2 supplements, when compared with standard care, did not significantly decrease the number of days required to reach a 50% reduction in a symptom severity score.

Recommendation for Hospitalized Patients With COVID-19

- The Panel **recommends against** the use of vitamin C for the treatment of COVID-19 in hospitalized patients (**AIIa**).

Rationale

Randomized clinical trials have failed to demonstrate benefit from vitamin C as a therapeutic

intervention for hospitalized patients with COVID-19. The data from these trials are summarized below.

Clinical Data for Hospitalized Patients With COVID-19

Two harmonized, randomized trials (LOVIT-COVID and REMAP-CAP) evaluated intravenous (IV) vitamin C versus a control in hospitalized patients with COVID-19 between July 2020 and July 2022.⁴ The studies enrolled patients from Asia, Australia, Europe, and North America, and data from the 2 studies were analyzed together. Patients in intensive care units who were critically ill and receiving organ support (1,568 patients from 90 sites) and hospitalized patients who were not critically ill (1,022 patients from 40 sites) were randomized to a vitamin C arm or a control arm. Patients in the intervention arm received IV vitamin C every 6 hours for 96 hours, for a maximum of 16 doses. Patients in the control arm received either no vitamin C or placebo. The composite primary outcome was a measure for days free of organ support up to 21 days and survival to hospital discharge. The study terminated enrollment after meeting criteria for harm and futility.

Among patients who were critically ill, the vitamin C arm ($n = 1,037$) had a median of 7 days free of organ support versus 10 days in the control arm ($n = 531$), with posterior probabilities of 8.6% for vitamin C efficacy and 99.9% for futility.⁴ Among patients who were not critically ill, both the vitamin C arm ($n = 456$) and the control arm ($n = 566$) had a median of 22 days free of organ support, with posterior probabilities of 2.9% for vitamin C efficacy and >99.9% for futility.

This study was limited by its use of combined data from 2 trials. The majority of patients enrolled were from an open-label study that used response-adaptive randomization.⁴ In addition, the precision of the treatment effect estimate in critically ill patients was limited because enrollment was stopped for harm. Data on individual vaccination status and the vitamin C product administered were unavailable. The study authors concluded that, in hospitalized patients with COVID-19, the probability that the use of vitamin C would increase the number of days free of organ support was low.

In a small, prospective, open-label randomized trial of hospitalized patients with severe COVID-19 in Pakistan, patients were randomized to receive vitamin C 50 mg/kg IV daily plus standard therapy ($n = 75$) or standard therapy alone ($n = 75$).⁵ Standard therapy included antipyretics, dexamethasone, and prophylactic antibiotics. Vitamin C recipients became symptom-free earlier (7.1 days vs. 9.6 days; $P < 0.0001$) and had a shorter duration of hospitalization (8.1 days vs. 10.7 days; $P < 0.0001$) than patients who received standard therapy alone. There were no significant differences between the arms for the outcomes of mortality and the need for mechanical ventilation. Limitations of this study include a small sample size, enrollment from only 1 hospital, and no clear method for recording symptoms.

In a pilot trial in China, 56 adults with COVID-19 who were in the intensive care unit were randomized to receive vitamin C 24 g IV daily for 7 days or placebo. The study was terminated early due to a reduction of cases of COVID-19 in China.⁶ Overall, the study found no differences between the arms for the outcomes of mortality, duration of mechanical ventilation, or change in median sequential organ failure assessment (SOFA) scores. The study reported improvements in oxygenation (as measured by the ratio of arterial partial pressure of oxygen to fraction of inspired oxygen) from baseline to Day 7 in the treatment arm that were statistically greater than those observed in the placebo arm (+20.0 vs. -51.9; $P = 0.04$).

In a randomized trial of 66 hospitalized patients with COVID-19 who required supplemental oxygen, treatment with vitamin C at doses escalating from 0.3 to 0.9 g/kg IV over 6 days ($n = 44$) was compared to standard of care ($n = 22$).⁷ The vitamin C did not improve the primary outcome of clinical status (defined as a composite of a 50% reduction in oxygen use, a 50% reduction in bronchodilator use, or

hospital discharge) at 72 hours after randomization.

Other Consideration

High concentrations of circulating vitamin C may affect the accuracy of point-of-care glucometers.^{8,9}

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Vitamin D

Last Updated: December 20, 2023

Vitamin D is critical for bone and mineral metabolism. Because the vitamin D receptor is present on immune cells such as B cells, T cells, and antigen-presenting cells, and because these cells can synthesize the active vitamin D metabolite, vitamin D also has the potential to modulate innate and adaptive immune responses.¹ It is postulated that these immunomodulatory effects of vitamin D could potentially protect against SARS-CoV-2 infection or decrease the severity of COVID-19.

Vitamin D deficiency (defined as a serum concentration of 25-hydroxyvitamin D ≤ 20 ng/mL) is common in the United States, particularly among people who identified as Hispanic or non-Hispanic Black.² These groups are overrepresented among cases of COVID-19 in the United States.³ Vitamin D deficiency is also more common in older patients and patients with obesity and hypertension; these factors have been associated with worse outcomes in patients with COVID-19.⁴ High levels of vitamin D may cause hypercalcemia and nephrocalcinosis.⁵

Recommendation

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of vitamin D for the prevention or treatment of COVID-19.

Rationale

The results from several cohort studies, clinical trials, and meta-analyses on the use of vitamin D for the prevention or treatment of COVID-19 have been published in peer-reviewed journals or have been made available as manuscripts ahead of peer review. However, most of these studies had significant limitations, such as small sample sizes or a lack of randomization and/or blinding. In addition, these studies used varying doses and formulations of vitamin D, enrolled participants with a range of COVID-19 severities, included different concomitant medications, and measured different study outcomes. All these factors make it difficult to compare results across studies. The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Although multiple observational cohort studies suggest that people with low vitamin D levels are at increased risk of SARS-CoV-2 infection and worse clinical outcomes after infection (e.g., higher mortality), clear evidence that vitamin D supplementation provides protection against infection or improves outcomes in patients with COVID-19 is still lacking.^{6,7}

Clinical Data on Vitamin D for Prevention

In a double-blind trial conducted at 4 hospitals in Mexico, frontline health care workers were randomized to receive vitamin D₃ 4,000 IU or placebo for 30 days.⁸ Participants were enrolled before COVID-19 vaccines became available. Over one-third of the enrolled participants dropped out before study completion. Of the 192 participants who completed follow-up, 6.4% of participants in the vitamin D₃ arm and 24.5% in the placebo arm acquired SARS-CoV-2 infection (relative risk 0.22; 95% CI, 0.08–0.59). At baseline, approximately 67% of participants had vitamin D deficiency, but this was not found to be an independent predictor of acquiring SARS-CoV-2 infection. The frequency of SARS-CoV-2 infection was very high in the placebo group, and it is unclear how these results translate to the use of vitamin D in vaccinated health care workers.

Clinical Data on Vitamin D for Treatment

In a double-blind trial conducted from June to October 2020 at 2 sites in Brazil, 240 hospitalized patients with moderate to severe COVID-19 were randomized to receive a single dose of vitamin D₃ 200,000 IU or placebo.⁹ Patients were considered to have moderate to severe COVID-19 if they had a positive polymerase chain reaction (PCR) result for SARS-CoV-2 or compatible computed tomography scan findings and a respiratory rate >24 breaths/min or oxygen saturation <93% on room air. The primary outcome was length of hospital stay. The study found no significant difference in the median length of stay between the vitamin D₃ arm (7.0 days; IQR 4.0–10.0 days) and the placebo arm (7.0 days; IQR 5.0–13.0 days; log-rank $P=0.59$). No significant differences were observed between the arms in the proportion of patients who were admitted to the intensive care unit (ICU), the need for mechanical ventilation, or mortality. There were no significant safety concerns.

A randomized, double-blind, placebo-controlled study conducted in Argentina included 218 adult patients with COVID-19 who had been admitted to the hospital during the preceding 24 hours and who had oxygen saturation $\geq 90\%$ on room air and a risk factor for disease progression.¹⁰ Patients were randomized to receive a single oral dose of vitamin D₃ 500,000 IU or placebo. The primary outcome was the change in the respiratory sepsis-related organ failure assessment (rSOFA) score between baseline and the highest value recorded up to Day 7. There was no significant difference between the arms for this outcome, with a median change of 0 in both arms ($P = 0.925$). There were also no significant differences between the arms in the median length of hospital stay, the number of patients admitted to the ICU, or in-hospital mortality.

A randomized, open-label study conducted in France compared the effect of a high dose of vitamin D₃ (400,000 IU) to the standard dose of vitamin D₃ (50,000 IU) on mortality in 254 patients who were either hospitalized or living in nursing facilities near the study hospital sites.¹¹ Patients were aged ≥ 65 years, had been diagnosed with SARS-CoV-2 infection within the preceding 3 days, and had at least 1 risk factor for disease progression (i.e., aged ≥ 75 years, hypoxemia). Mortality was significantly different between the arms at 14 days, with 7 deaths (6%) among patients in the high-dose arm and 14 deaths (11%) among patients in the standard-dose arm (adjusted HR 0.33; 95% CI, 0.12–0.86; $P = 0.02$). However, mortality was not significantly different between the arms at 28 days (adjusted HR 0.70; 95% CI, 0.36–1.36; $P = 0.29$).

In an open-label pilot study, 50 hospitalized adults in New York with PCR-confirmed SARS-CoV-2 infection were randomized to receive calcitriol 0.5 μg daily for 14 days or no treatment.¹² Calcitriol is the active metabolite of cholecalciferol or vitamin D₃ and is more commonly used to treat parathyroid disease. The study evaluated the change in oxygen saturation between patient admission and discharge. Additional outcomes were the length of hospital stay; mortality; and the need for endotracheal intubation, ICU admission, or hospital readmission within 30 days. Oxygen saturation was calculated using the ratio of peripheral oxygen saturation (measured by pulse oximetry) to fraction of inspired oxygen ($\text{SpO}_2/\text{FiO}_2$) as a surrogate for the ratio of arterial partial pressure of oxygen to fraction of inspired oxygen ($\text{PaO}_2/\text{FiO}_2$). Between admission and discharge, the patients who received no treatment had an average increase of 13.2 (SD 127.7) in the ratio, and those who received calcitriol had an increase of 91.04 (SD 119.08; $P = 0.0305$), implying an improvement in oxygenation.¹² There were no differences between the arms in the length of hospital stay, mortality, or the need for ICU admission or hospital readmission.

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Zinc

Last Updated: December 20, 2023

Increased intracellular zinc concentrations efficiently impair the replication of a number of RNA viruses.¹ Zinc has been shown to enhance cytotoxicity and induce apoptosis when used in vitro with a zinc ionophore (e.g., chloroquine). Chloroquine has also been shown to enhance intracellular zinc uptake in vitro.² Zinc levels are difficult to measure accurately, as zinc is distributed as a component of various proteins and nucleic acids.³

The recommended dietary allowance for elemental zinc is 11 mg daily for men and 8 mg daily for nonpregnant women.⁴ Long-term zinc supplementation can cause copper deficiency with subsequent reversible hematologic defects (i.e., anemia, leukopenia) and potentially irreversible neurologic manifestations (i.e., myelopathy, paresthesia, ataxia, spasticity).⁵⁻⁷ The use of zinc supplementation for durations as short as 10 months has been associated with copper deficiency.³ In addition, oral zinc can decrease the absorption of medications that bind with polyvalent cations (e.g., fluoroquinolones, HIV integrase inhibitors, tetracyclines).⁴

Recommendations

- There is insufficient evidence for the COVID-19 Treatment Guidelines Panel (the Panel) to recommend either for or against the use of zinc for the treatment of COVID-19.
- The Panel **recommends against** using zinc supplementation above the recommended dietary allowance (i.e., zinc 11 mg daily for men, zinc 8 mg daily for nonpregnant women) for the prevention of COVID-19, except in a clinical trial (**BIII**).

Rationale

The results from some cohort studies and clinical trials that evaluated the use of zinc in patients with COVID-19 have been published in peer-reviewed journals or have been made available as manuscripts ahead of peer review. However, most of these studies have significant limitations, such as small sample sizes or a lack of randomization or blinding. In addition, these studies used varying doses and formulations of zinc, enrolled participants with a range of COVID-19 severities, included different concomitant medications, and measured different study outcomes. All of these factors make it difficult to compare results across studies. Because zinc has not been shown to have a clinical benefit and may be harmful, the Panel **recommends against** using zinc supplementation above the recommended dietary allowance for the prevention of COVID-19, except in a clinical trial (**BIII**).

The studies summarized below are those that have had the greatest impact on the Panel's recommendations.

Clinical Data

In a double-blind, multicenter trial in Tunisia, nonhospitalized and hospitalized adults with COVID-19 were randomized within 7 days of symptom onset to receive elemental zinc 25 mg orally twice daily (n = 231) or matching placebo (n = 239) for 15 days.⁸ Approximately 20% of these patients had received a COVID-19 vaccine prior to enrollment. During the study, none of the patients received antiviral drugs, and <40% received corticosteroids.

The primary outcome in the study was a composite of death due to COVID-19 or intensive care unit

admission within 30 days of randomization.⁸ This study has several limitations. The study enrolled nonhospitalized and hospitalized patients, and comparing the results for these populations is difficult. In addition, only some patients received standard of care treatments. The data presented in the published paper had numerous and substantial inconsistencies.^{9,10} Together, these limitations make it difficult to interpret the results of this study or apply these findings to the current U.S. population with COVID-19.

In an open-label trial conducted at 2 sites in the United States, outpatients with laboratory-confirmed SARS-CoV-2 infection were randomized to receive either 10 days of zinc gluconate 50 mg, ascorbic acid 8,000 mg, both agents, or standard of care.¹¹ The primary endpoint was the number of days required to reach a 50% reduction in the patient's symptom severity score. The study was stopped early by an operational and safety monitoring board due to futility after 214 of the planned 520 participants were enrolled. Compared with standard of care, treatment with high-dose zinc gluconate, ascorbic acid, or a combination of the 2 supplements did not significantly decrease the number of days required to reach a 50% reduction in a symptom severity score. Patients who received standard of care achieved a 50% reduction in their symptom severity scores at a mean of 6.7 days (SD 4.4 days) compared with 5.5 days (SD 3.7 days) for the ascorbic acid arm, 5.9 days (SD 4.9 days) for the zinc gluconate arm, and 5.5 days (SD 3.4 days) for the arm that received both agents (overall $P = 0.45$).

Nonserious adverse effects occurred more frequently in patients who received supplements than in those who did not.¹¹ Nonserious adverse effects were experienced by 39.5% of patients in the ascorbic acid arm, 18.5% in the zinc gluconate arm, and 32.1% in the arm that received both agents, compared with 0% of patients in the standard of care arm (overall $P < 0.001$). The most common nonserious adverse effects in this study were gastrointestinal events.

In a randomized clinical trial conducted at 3 academic medical centers in Egypt, 191 patients with laboratory-confirmed SARS-CoV-2 infection were randomized to receive either zinc 220 mg twice daily plus hydroxychloroquine or hydroxychloroquine alone for a 5-day course.¹² The primary endpoints were recovery within 28 days, the need for mechanical ventilation, and death. The 2 arms were matched for age and gender. There were no significant differences between the arms in the percentages of patients who recovered within 28 days (79.2% in the zinc plus hydroxychloroquine arm vs. 77.9% in the hydroxychloroquine alone arm; $P = 0.969$), the number of patients who required mechanical ventilation (4 in the zinc plus hydroxychloroquine arm vs. 6 in the hydroxychloroquine alone arm; $P = 0.537$), or overall mortality (2 patients in each arm; $P = 0.986$). The only risk factors for mortality were age and the need for mechanical ventilation.

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Considerations for Using Concomitant Medications in Patients With COVID-19

Last Updated: December 20, 2023

Summary Recommendations
- Patients with COVID-19 who are receiving concomitant medications (e.g., angiotensin-converting enzyme [ACE] inhibitors, angiotensin receptor blockers [ARBs], HMG-CoA reductase inhibitors [statins], systemic or inhaled corticosteroids, nonsteroidal anti-inflammatory drugs, acid-suppressive therapy) for underlying medical conditions should not discontinue ACE inhibitors and ARBs (AIIa) or other medications (AIII) unless discontinuation is otherwise warranted by their clinical condition. - The COVID-19 Treatment Guidelines Panel recommends against using medications off-label to treat COVID-19 if they have not been shown to be safe and effective for this indication in a clinical trial (AIII).
Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Individuals with underlying medical conditions, such as cardiovascular disease, pulmonary disease, diabetes, and malignancy, and those who receive chronic immunosuppressive therapy are at higher risk of severe illness with COVID-19. These patients are often prescribed medications to treat their underlying medical conditions.

Early in the pandemic, some of these medications, such as angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs),¹ HMG-CoA reductase inhibitors (statins),^{2,3} and histamine-2 receptor antagonists,⁴ were hypothesized to offer potential as COVID-19 therapeutic agents. Others, such as nonsteroidal anti-inflammatory agents, were postulated to have negative impacts.⁵ Currently, there is no evidence that discontinuing medication for underlying medical conditions offers a clinical benefit for patients with COVID-19.⁶⁻⁸ For example, the Food and Drug Administration stated that there is no evidence linking the use of nonsteroidal anti-inflammatory agents with worsening of COVID-19 and advised patients to use them as directed.⁹ Additionally, the American Heart Association, the Heart Failure Society of America, and the American College of Cardiology issued a joint statement that renin-angiotensin-aldosterone system antagonists, such as ACE inhibitors and ARBs, should be continued as prescribed in those with COVID-19.¹⁰

Therefore, patients with COVID-19 who are treated with concomitant medications for an underlying medical condition **should not discontinue** ACE inhibitors and ARBs (**AIIa**) or other medications (**AIII**) unless discontinuation is otherwise warranted by their clinical condition. For patients with COVID-19 who require nebulized medications, precautions should be taken to minimize the potential for transmission of SARS-CoV-2 in the home and in health care settings.^{11,12}

The COVID-19 Treatment Guidelines Panel **recommends against** using medications off-label to treat COVID-19 if they have not been shown to be safe and effective for this indication in a clinical trial (**AIII**). Clinicians should refer to the [Therapies](#) section of the Guidelines for information on the medications that have been studied as potential therapeutic options for patients with COVID-19.

When prescribing medications to treat COVID-19, clinicians should always assess the patient's current medications for potential drug-drug interactions and additive adverse effects. The decision to continue or change a patient's medications should be individualized based on their specific clinical condition. Clinicians can refer to product labels and visit the [Liverpool COVID-19 Drug Interactions](#) website for guidance on identifying and managing drug-drug interactions. It is also worth noting that

ritonavir-boosted nirmatrelvir (Paxlovid), which is approved by the Food and Drug Administration for the treatment of mild to moderate COVID-19 in adults who are at high risk of progressing to severe COVID-19, has significant drug-drug interactions. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

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Special Considerations in People Who Are Immunocompromised

Last Updated: February 29, 2024

Summary Recommendations
Prevention of COVID-19 • The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible, including those who are immunocompromised, according to the Centers for Disease Control and Prevention's Advisory Committee on Immunization Practices (AI). • Vaccine response rates may be lower in patients who are moderately or severely immunocompromised. Specific guidance on administering vaccines to these individuals is provided by the Centers for Disease Control and Prevention. • Clinicians should strongly encourage all household members and close contacts of patients who are immunocompromised to be vaccinated against COVID-19 (AI). • Currently, antibody tests are not recommended for assessing SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated. Management of Patients With COVID-19 Who Are Immunocompromised • Clinicians should consult with the appropriate specialists when making decisions about stopping or adjusting the doses of immunosuppressive drugs in patients with COVID-19. • When selecting treatments for COVID-19, clinicians should consider factors such as the underlying disease; the specific immunosuppressants being used; the severity of COVID-19; and the potential for drug-drug interactions, overlapping toxicities, and secondary infections. • For nonhospitalized patients with mild to moderate COVID-19 who are immunocompromised, the Panel recommends prompt treatment with antiviral drugs at the doses and durations recommended for the general population (AIII). For more information, see Therapeutic Management of Nonhospitalized Adults With COVID-19. • For most hospitalized patients with severe or critical COVID-19 who are immunocompromised, the Panel recommends using antiviral drugs and immunomodulatory therapies at the doses and durations recommended for the general population (AIII). For more information, see Therapeutic Management of Hospitalized Adults With COVID-19. • For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication, the optimal management is unknown. The data for these approaches are not definitive, but some Panel members would use 1 or more of the following treatment options: • Longer and/or additional courses of ritonavir-boosted nirmatrelvir (Paxlovid) • Longer and/or additional courses of remdesivir • High-titer COVID-19 convalescent plasma from a vaccinated donor who recently recovered from COVID-19 likely caused by a SARS-CoV-2 variant similar to the variant causing the patient's illness Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Introduction

Approximately 3% of people in the United States have immunocompromising conditions.¹ People who are immunocompromised are a heterogeneous population, and the severity of COVID-19 can vary significantly in this group. Some individuals who are immunocompromised may have a higher risk of hospitalization, complications, or death, and some may have outcomes that are comparable to those in the general population.

This section pertains to people who are moderately or severely immunocompromised, which includes those who:

- Are receiving active treatment for solid tumor and hematologic malignancies.
- Have hematologic malignancies (e.g., chronic lymphocytic lymphoma, non-Hodgkin lymphoma, multiple myeloma, acute leukemia) and are known to have poor responses to COVID-19 vaccines, regardless of the treatment status for the hematologic malignancy.
- Received a solid-organ or islet transplant and are receiving immunosuppressive therapy.
- Received chimeric antigen receptor T cell (CAR T-cell) therapy or a hematopoietic cell transplant (HCT) and are within 2 years of transplantation or are receiving immunosuppressive therapy.
- Have a moderate or severe primary immunodeficiency (e.g., severe combined immunodeficiency, DiGeorge syndrome, Wiskott-Aldrich syndrome, common variable immunodeficiency disease).
- Have advanced or untreated HIV infection (defined as people with HIV and CD4 T lymphocyte cell counts <200 cells/mm³, a history of an AIDS-defining illness without immune reconstitution, or clinical manifestations of symptomatic HIV).
- Are receiving active treatment with high-dose corticosteroids (i.e., ≥ 20 mg prednisone or equivalent per day for ≥ 2 weeks), alkylating agents, antimetabolites, transplant-related immunosuppressive drugs, cancer chemotherapeutic agents classified as severely immunosuppressive, or immunosuppressive or immunomodulatory biologic agents (e.g., B cell-depleting agents).

Analyses have found a higher risk of hospitalization or death from COVID-19 in those with a variety of immunocompromising conditions, including rheumatic diseases, hematological malignancies, solid organ transplants, and HIV.²⁻⁷ Factors that increase the risk of severe COVID-19 in the general population, such as older age, chronic kidney disease, cardiovascular disease, and other comorbidities, are significant drivers of risk in people who are immunocompromised. Moreover, some immunodeficiencies seem to increase the risk above and beyond traditional risk factors. For example, there is evidence that individuals who make autoantibodies to type I interferons (proteins that are critical to the protective immune response against viral infections) have a higher risk of severe COVID-19.⁸ Similarly, certain classes of medications, such as T cell-depleting or T cell-suppressing agents (e.g., antithymocyte globulin, calcineurin inhibitors, mycophenolate mofetil, belatacept) and B cell-depleting agents (e.g., rituximab, ocrelizumab, obinutuzumab), have been associated with more severe disease and death.^{7,9,10}

Prolonged shedding of SARS-CoV-2 has been reported in patients who are immunocompromised. A systematic review found that replication-competent virus could be detected for a median of 20 days in these patients, compared to 11 days in the general population.¹¹ Prolonged viral shedding may affect SARS-CoV-2 testing strategies and isolation durations for this group of patients. Moreover, case reports suggest that prolonged infections can create evolutionary pressure for the emergence of variants that resist therapies or evade vaccine-induced immunity.¹²⁻¹⁴

For any person who is eligible, clinicians should prescribe therapies for the treatment of COVID-19 as recommended in these Guidelines. However, if logistical constraints limit the availability of therapies, the COVID-19 Treatment Guidelines Panel (the Panel) suggests prioritizing the treatment of patients with COVID-19 who are at the highest risk of clinical progression (see [Prioritization of Anti-SARS-CoV-2 Therapies for the Treatment of COVID-19 in Nonhospitalized Patients When There Are Logistical Constraints](#)). Providers should use their clinical judgment when prioritizing patients for treatment and assess a patient's immunocompromised status, age, comorbidities, and vaccination status.

The sections below outline the Panel's rationale for the recommendations on preventing and managing COVID-19 in people who are immunocompromised. Some of the special considerations for patients who are immunocompromised include the timing of COVID-19 vaccination, the management of immunosuppressive medications, and the strategies for treating COVID-19.

Prevention of COVID-19

Vaccination

COVID-19 vaccination remains the most effective way to prevent serious outcomes and death from SARS-CoV-2 infection. The Panel recommends COVID-19 vaccination for everyone who is eligible, including those who are immunocompromised, according to the Centers for Disease Control and Prevention's (CDC) Advisory Committee on Immunization Practices (AI).

Authorized or approved COVID-19 vaccines in the United States are not live-virus vaccines and can be safely administered to patients who are immunocompromised. However, in people who are immunocompromised, the immune response to vaccination may be blunted, and the timing of vaccination requires special consideration. Nevertheless, vaccination is still recommended, as it may confer partial protection, including the protection provided by vaccine-induced, cell-mediated immunity.¹⁵

The Panel recommends following the [CDC's COVID-19 vaccination guidance](#) for people who are moderately or severely immunocompromised. This guidance includes information on the use of the updated 2023–2024 mRNA vaccines, which target the SARS-CoV-2 Omicron variant lineage XBB.1.5. Current CDC guidance allows for the use of additional vaccine doses in people who are moderately or severely immunocompromised.¹⁶ Data on the optimal timing for repeat vaccination in people who are immunocompromised are lacking; the CDC recommends an interval of at least 2 months after the last dose. Other considerations include the patient's current or expected level of immunosuppression, their age, comorbidities, and the time since their last vaccine dose. Clinicians should also account for the prevalence of SARS-CoV-2 infection in the community and whether the patient intends to travel.

A preprint of a large observational study from Israel suggests a potential benefit from administering COVID-19 boosters every 6 months in groups with the highest risk of COVID-19–related hospitalization or death.¹⁷ The CDC-funded VISION Network evaluated the effectiveness of bivalent vaccines between September 13, 2022, and April 21, 2023, at 5 sites in 7 states.¹⁸ Among adults who were immunocompromised, a lower vaccine effectiveness was observed for the bivalent booster, but vaccine effectiveness was sustained against critical COVID-19–associated outcomes, including intensive care unit admission and death. Vaccine effectiveness against hospitalization was 28% during the first 7 to 59 days after receipt of the bivalent dose and declined to 13% by 120 to 179 days; this indirectly supports using a 6-month interval for repeat vaccination.

The pivotal clinical trials that evaluated the safety and efficacy of the COVID-19 vaccines excluded people who were severely immunocompromised; therefore, the data for this population are less robust.^{19,20} Observational data suggest that serological responses to vaccines may be blunted in patients who are immunocompromised.^{21,22} However, the MELODY trial reported detectable immunoglobulin G spike protein antibodies in approximately 80% of a large cohort of individuals in the United Kingdom who were immunocompromised and had received at least 3 doses of COVID-19 vaccines.²³ Those who had received anti-CD20 therapies within the past year were less likely than other groups in the study to have detectable anti-spike protein antibodies. In another study conducted during the Omicron era, a fourth dose of an mRNA COVID-19 vaccine reduced the risk of SARS-CoV-2 infection and severe COVID-19 among patients receiving treatment for systemic autoimmune rheumatic diseases.²⁴

Vaccination of Close Contacts

Clinicians should strongly encourage all household members and close contacts of patients who are immunocompromised to be vaccinated against COVID-19 (AI). Before Omicron became the dominant circulating variant, a large cohort study of health care workers in Finland reported that COVID-19 vaccines were associated with a reduction in SARS-CoV-2 infections not only among vaccinated

individuals but also among unvaccinated adult household members.²⁵ A 2022 systematic review and meta-analysis of 96 studies reported that people who received a complete primary COVID-19 vaccine series had reduced susceptibility to SARS-CoV-2 infection and were less infectious if they become infected.²⁶

Vaccination Timing and Immunosuppressive Therapies

If possible, COVID-19 vaccines should be administered at least 2 weeks before initiating or resuming immunosuppressive therapies. The timing of the vaccination should be determined based on the patient's current or planned immunosuppressive therapies, as well as the patient's medical condition and predicted response to the vaccine. Guidance about the timing of COVID-19 vaccination in solid organ transplant, HCT, and cellular immunotherapy candidates can be found in [Special Considerations in Solid Organ Transplant, Hematopoietic Cell Transplant, and Cellular Immunotherapy Candidates, Donors, and Recipients](#). The CDC's guidance allows the use of additional vaccine doses in people who are immunocompromised. Each additional dose should be administered at least 2 months after the last dose.

The CDC recommends that HCT and CAR T-cell recipients who received doses of COVID-19 vaccines before or during treatment with HCT or CAR T-cell therapy should be revaccinated with the currently recommended primary vaccine series at least 3 months after the transplant or CAR T-cell therapy.¹⁶ The American College of Rheumatology also provides guidance for temporarily stopping immunosuppressive regimens during vaccination.²⁷

Polyethylene Glycol Allergies

The BNT162b2 (Pfizer-BioNTech) and mRNA-1273 (Moderna) mRNA vaccines contain polyethylene glycol (PEG), whereas the NVX-CoV2373 (Novavax) vaccine contains polysorbate 80. PEG and polysorbate are used in many products, including in agents used for cancer chemotherapy (e.g., PEG-asparaginase). PEG and polysorbate are structurally related, and cross-reactive hypersensitivity between these compounds might occur. The detection of PEG antibodies after vaccination was not associated with increased adverse reactions (such as delayed-onset reactions, including injection site rashes, or severe allergic reactions) to the mRNA COVID-19 vaccines.²⁸ Therefore, testing for anti-PEG antibodies should not be used as a screening tool to assess the risk of allergic reactions²⁹ and should not replace an assessment by a specialist in those rare individuals with a history of anaphylaxis.³⁰ The [CDC has issued guidance](#) on triaging people with a history of allergies or allergic reactions to the components of COVID-19 vaccines.

Pre-Exposure Prophylaxis

As of February 2024, no biomedical intervention other than vaccines prevents COVID-19 disease. Previously, the Food and Drug Administration (FDA) authorized the use of the anti-SARS-CoV-2 monoclonal antibodies tixagevimab plus cilgavimab (Evusheld) as pre-exposure prophylaxis (PrEP) of COVID-19 in certain people who were not expected to mount an adequate immune response to COVID-19 vaccination and in people with COVID-19 vaccine contraindications. Because the current Omicron subvariants are not susceptible to tixagevimab plus cilgavimab, this combination is not currently authorized by the FDA for use as PrEP of COVID-19.³¹

Serologic Testing to Guide Vaccination Strategies

Currently, antibody tests are not recommended for assessing SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated. More than 80 SARS-CoV-2 serologic tests, including quantitative, semiquantitative, neutralizing antibody, and point-of-care tests, have been issued Emergency Use Authorizations by the FDA to aid in detecting antibodies to SARS-CoV-2.³² However, these tests are not currently authorized for routine use in making

individual medical decisions, and their ability to assess a person's level of immunity or protection from SARS-CoV-2 infection has not been evaluated.³³ Most of these tests have not been calibrated to a reference standard, limiting the ability to compare and reproduce results from different tests.

Management of Patients With COVID-19 Who Are Immunocompromised

Adjusting Chronic Immunosuppressive Therapies

Clinicians should consult with the appropriate specialists when making decisions about stopping or adjusting the doses of immunosuppressive drugs in patients with COVID-19. When selecting treatments for COVID-19, clinicians should consider factors such as the underlying disease; the specific immunosuppressants being used; the severity of COVID-19; and the potential for drug-drug interactions, overlapping toxicities, and secondary infections.

Early in the clinical course of COVID-19, the disease is primarily driven by the replication of SARS-CoV-2. Immunosuppressive medications can reduce the host immune responses that suppress viral replication, increasing the risk of prolonged viral shedding and infection.^{34,35} Clinicians should consider adjusting the doses of immunosuppressive medications or substituting certain immunosuppressive medications, if possible, to improve the patient's immune response to infection. When making decisions about stopping or reducing the dose of immunosuppressive drugs, clinicians should balance the potential benefit of enhancing the patient's immune response to COVID-19 with the risk of exacerbating the underlying condition. They should also consider the role of immunomodulation in the treatment of COVID-19.

Clinicians should be aware that many immunosuppressive drugs, particularly biologic agents, have long half-lives or prolonged periods of biologic activity. Patients may remain immunosuppressed long after the drugs are stopped. Care should be taken to not stop glucocorticoids abruptly, since this may result in adrenal insufficiency. For medications other than glucocorticoids, decisions about dose adjustments should be made on a case-by-case basis. For example, for some autoimmune diseases, temporary cessation of immunosuppression is often possible, and restarting medications 7 to 14 days after symptom resolution may be appropriate.^{27,36} Observational data suggest that patients receiving tumor necrosis factor- α (TNF- α) inhibitors may be at lower risk of progressing to severe COVID-19 than those receiving treatment with other immunomodulators.^{37,38} Therefore, some patients with mild COVID-19 may safely continue receiving treatment with TNF- α inhibitors.

For solid organ transplant recipients, adjustments to immunosuppressive regimens should be individualized based on disease severity, the risk of graft rejection, the specific immunosuppressants being used, the type of transplant, the time since transplantation, the concentration of immunosuppressants, and the potential for drug-drug interactions.³⁹ See [Special Considerations in Solid Organ Transplant, Hematopoietic Cell Transplant, and Cellular Immunotherapy Candidates, Donors, and Recipients](#) for more information.

Therapeutic Management of Nonhospitalized Patients With COVID-19

For nonhospitalized patients with mild to moderate COVID-19 who are immunocompromised, the Panel recommends prompt treatment with antiviral drugs at the doses and durations recommended for the general population (AIII). See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) to review the Panel's recommendations. Some special considerations for using these therapies in people who are immunocompromised are outlined below.

Ritonavir-Boosted Nirmatrelvir (Paxlovid)

In the EPIC-HR trial, the use of ritonavir-boosted nirmatrelvir reduced the risk of hospitalization

or death when compared with placebo in nonhospitalized, unvaccinated adults who had laboratory-confirmed SARS-CoV-2 infection and a high risk of progressing to severe COVID-19.⁴⁰ Because the trial did not enroll many participants who were immunocompromised, the efficacy of ritonavir-boosted nirmatrelvir was not established for this population. In subsequent retrospective studies, some potential benefits of using ritonavir-boosted nirmatrelvir in people with various immunocompromising conditions have been observed.^{41,42}

A retrospective study investigated the use of ritonavir-boosted nirmatrelvir for the treatment of COVID-19 in patients who were moderately or severely immunocompromised.⁴³ Among 3,188 patients, the use of ritonavir-boosted nirmatrelvir reduced the risk of death or hospitalization in extremely vulnerable patients, such as those with solid organ, bone marrow, or stem cell transplants; those with severe primary immunodeficiencies; and those receiving B cell-depletion therapy or treatment for hematologic malignancies.

Because ritonavir-boosted nirmatrelvir is the only highly effective oral antiviral therapy for COVID-19, it should be considered for patients who are immunocompromised if there are no potentially significant drug-drug interactions or if the interactions can be safely managed. Clinicians should be aware of drug-drug interactions that may be life- or organ-threatening (see [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \[Paxlovid\] and Concomitant Medications](#)).⁴⁴ Notably, calcineurin inhibitors (e.g., tacrolimus, cyclosporine A) and mammalian target of rapamycin drugs (e.g., sirolimus, everolimus) have important drug-drug interactions with ritonavir. For this reason, the American Society of Transplantation recommends preferentially using other therapies, such as remdesivir, over ritonavir-boosted nirmatrelvir in people who are taking calcineurin inhibitors or mammalian target of rapamycin inhibitors.³⁹ Ritonavir can inhibit the metabolism of many cancer-directed therapies and should only be given after consulting with specialty pharmacists and other appropriate specialists. Remdesivir and molnupiravir are other antiviral options for individuals who cannot receive ritonavir-boosted nirmatrelvir because of drug-drug interactions.

Observational studies and the EPIC-HR and MOVE-OUT trials have described SARS-CoV-2 viral rebound and the recurrence of COVID-19 symptoms in some patients who have completed treatment with ritonavir-boosted nirmatrelvir or molnupiravir.⁴⁵⁻⁴⁹ However, viral rebound can also occur in patients who have not received treatment for COVID-19.⁵⁰ Some observational studies have reported that patients who were treated with ritonavir-boosted nirmatrelvir had a higher frequency of viral rebound and symptom recurrence than those who did not receive treatment.^{51,52}

To date, virus detection and the recurrence of COVID-19 symptoms following the use of antiviral therapies have not been associated with progression to severe COVID-19.^{53,54} Therefore, concerns about viral rebound or the recurrence of symptoms should not be a reason to avoid using antiviral therapies when their use is indicated.^{49,55-57} A clinical trial that is evaluating the use of a second course of ritonavir-boosted nirmatrelvir to treat patients with viral rebound and symptom recurrence is underway (ClinicalTrials.gov Identifier [NCT05567952](#)). See [Ritonavir-Boosted Nirmatrelvir \(Paxlovid\)](#) for more information on rebound.

Remdesivir

Remdesivir was studied in nonhospitalized patients with mild to moderate COVID-19 who were at high risk of progressing to severe disease, and it was shown to be highly effective in reducing the risk of hospitalization and death.⁵⁸ However, this trial only included a small number of participants who were immunocompromised. Because remdesivir treatment for nonhospitalized patients requires an intravenous infusion for 3 consecutive days, there may be logistical constraints to administering this drug in many settings. It can be considered for patients who are immunocompromised if other options, such as ritonavir-boosted nirmatrelvir, are not appropriate or available.

Molnupiravir

In the MOVe-OUT trial, molnupiravir reduced the rate of hospitalization or death when compared with placebo in nonhospitalized patients with COVID-19.⁵⁹ However, this trial only enrolled a small number of participants who were immunocompromised. In a post hoc analysis of data from 55 patients who were immunocompromised, 2 of 24 patients (8%) who received molnupiravir were hospitalized or died through Day 29 compared with 7 of 31 patients (23%) who received placebo.⁶⁰ The PANORAMIC trial enrolled a larger population of people who were immunocompromised, but this population was heterogeneous and the results of the study were inconclusive.⁶¹ Although the different treatment options have not been directly compared in clinical trials, the available evidence suggests that molnupiravir has a lower efficacy than ritonavir-boosted nirmatrelvir or remdesivir (see [Molnupiravir](#)). Other COVID-19 therapies should be prioritized over molnupiravir in patients who are immunocompromised.

COVID-19 Convalescent Plasma

There is insufficient evidence for the Panel to recommend either for or against the use of high-titer COVID-19 convalescent plasma (CCP) for the treatment of COVID-19 in nonhospitalized patients who are immunocompromised.

The FDA issued an Emergency Use Authorization that allows the use of high-titer CCP for the treatment of COVID-19 in nonhospitalized or hospitalized patients who have immunosuppressive disease or are receiving immunosuppressive treatment.⁶² However, the evidence generated from well-designed clinical trials that evaluated the use of CCP for the treatment of nonhospitalized patients with COVID-19 is conflicting; these trials only enrolled a small number of patients who were immunocompromised.⁶³⁻⁶⁶

For a discussion on the potential use of high-titer CCP in patients who are immunocompromised and have prolonged viral replication, see the Therapeutic Management of Patients With COVID-19 Symptoms and Prolonged SARS-CoV-2 Replication section below.

Intravenous Immunoglobulin

The Panel **recommends against** the use of **intravenous immunoglobulin (IVIG)** for the prevention or treatment of acute COVID-19 in adults and children, except in a clinical trial (**AIII**).

Some individuals who are immunocompromised and have hypogammaglobulinemia are candidates for receiving supplemental antibodies in the form of IVIG for the prevention of a variety of infections and in the setting of acute infections, including COVID-19. IVIG can be administered as outpatient or inpatient therapy. However, outside these specific circumstances, the Panel's recommendation should not preclude the use of IVIG when it is otherwise indicated for underlying conditions. See [Intravenous Immunoglobulin](#) for more information.

Therapeutic Management of Patients Who Are Hospitalized for COVID-19

For most hospitalized patients with severe or critical COVID-19 who are immunocompromised, the Panel recommends using antiviral drugs and immunomodulatory therapies at the doses and durations recommended for the general population (**AIII**). See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for more information. The optimal management strategies and treatments for COVID-19 in hospitalized patients who are immunocompromised are unknown since these individuals were either excluded from or poorly represented in major clinical trials. Nevertheless, clinical experience and retrospective data suggest that many patients who are immunocompromised have the expected responses to standard therapies for COVID-19.

Remdesivir

Case reports suggest that remdesivir can suppress, but does not always eliminate, viral replication in this

population.^{67,68} In a large retrospective study of hospitalized patients who were immunocompromised, including patients who did not require supplemental oxygen, patients who received remdesivir had a lower risk of mortality at 14 days and 28 days than patients who did not receive remdesivir.⁶⁹ The optimal duration of treatment with remdesivir in patients who are immunocompromised is unknown. Given the risk of prolonged viral replication in patients who are immunocompromised, some clinicians may choose to extend the course of antiviral therapy past 5 to 10 days. For patients receiving immunomodulatory therapy who have severe respiratory impairment due to COVID-19, clinicians may consider adding remdesivir treatment, although remdesivir has not been adequately studied in prospective clinical trials to determine whether there is a benefit in these patients.

Corticosteroids

The RECOVERY trial reported a survival benefit for dexamethasone in inpatients with COVID-19 who were receiving oxygen, high-flow nasal cannula oxygen, noninvasive ventilation, or mechanical ventilation; however, specific data regarding the subgroup of patients who were immunocompromised are not available.⁷⁰ Corticosteroids should not be used for the treatment of COVID-19 in patients who are not receiving oxygen. In the RECOVERY trial, no survival benefit was observed for dexamethasone among the subset of patients with COVID-19 who did not require supplemental oxygen at enrollment. In an observational cohort study of U.S. veterans, the use of dexamethasone was associated with higher mortality in hospitalized patients with COVID-19 who did not require supplemental oxygen.^{70,71}

Patients who are immunocompromised may experience delayed development of favorable adaptive responses and a prolonged period of viral replication, as discussed above. For patients who are immunocompromised, are receiving minimal levels of conventional oxygen, and are earlier in the course of COVID-19 (e.g., those with <10 days of symptoms), the preferred approach may be emphasizing supportive care, using antiviral therapy, and avoiding corticosteroids. This strategy may reduce the duration of viral replication and the risk of secondary infections. Dexamethasone should be added if the patient has escalating oxygen requirements.

For patients who are immunocompromised and who were on chronic corticosteroids prior to hospitalization, the optimal dose of dexamethasone for the treatment of COVID-19 is unknown. The recommended dose of dexamethasone is 6 mg, which is equivalent to 40 mg of prednisone. This is the minimum dose of a steroid that should be used. Maintenance doses of corticosteroids should be discontinued while a patient is receiving dexamethasone, and the doses should be resumed as soon as possible after the patient recovers from COVID-19 or completes the course of dexamethasone.

Other Immunomodulators

Several randomized trials have shown that adding baricitinib or tocilizumab as a second immunomodulator to dexamethasone improves clinical outcomes in patients with severe or critical COVID-19.⁷²⁻⁷⁴ Another randomized trial that examined the use of abatacept, cenicriviroc, or infliximab in combination with dexamethasone in hospitalized adults with COVID-19 reported no differences between the study arms in the primary endpoint of time to recovery.⁷⁵ However, patients who received infliximab or abatacept had a lower risk of mortality at 28 days. These trials generally excluded patients who were immunocompromised or only included small numbers of these patients. For patients who are immunocompromised, the use of these agents may provide a clinical benefit similar to the benefit seen in the general population. However, it is not clear whether augmenting immunomodulation in this population increases the risk of serious bacterial, invasive fungal, or parasitic infections.

The Panel currently recommends adding another immunomodulator to dexamethasone in hospitalized patients with COVID-19 who are hypoxemic and experiencing clinical progression (see [Therapeutic Management of Hospitalized Adults With COVID-19](#)). This approach can also be used for most patients

with COVID-19 who are immunocompromised. However, clinicians should consult with specialists to ensure that the risks of using additional immunosuppressive medications, including the risks of serious infections, do not outweigh the benefits. The patient should be closely monitored for secondary infections.

COVID-19 Convalescent Plasma

There is insufficient evidence for the Panel to recommend either for or against the use of high-titer CCP for the treatment of COVID-19 in hospitalized patients who are immunocompromised.

Three key randomized trials that evaluated the use of CCP for the treatment of COVID-19—RECOVERY, CONCOR-1, and REMAP-CAP—reported no evidence of a benefit of CCP in hospitalized patients with COVID-19. However, most of the patients enrolled in these trials were not immunocompromised.⁷⁶⁻⁷⁸ Some of the subgroup analyses generated from clinical trials that enrolled patients who were immunocompromised suggested a potential benefit of CCP in this population,⁷⁸⁻⁸⁰ but subgroup analyses need to be interpreted with caution (see [Table 4c](#)). Other clinical data from case series, retrospective case-control studies, and meta-analyses have been cited as support for the potential benefit of CCP in patients who are immunocompromised.⁸¹⁻⁸⁹ However, the patient populations differed across studies, and the studies had design limitations, making the findings difficult to interpret.

The RECOVER trial was a small, randomized trial that evaluated the use of plasma from donors who were vaccinated against COVID-19 or convalescent after SARS-CoV-2 infection as a treatment for COVID-19 in hospitalized people with cancer, people with immunosuppression, people with lymphopenia and D-dimer levels >1 µg/mL, and people aged >75 years.⁷⁹ Only the subgroup of patients with cancer who received plasma treatment experienced a shorter median time to improvement and lower mortality when compared with the control arm.

For a discussion on the potential use of high-titer CCP in patients who are immunocompromised and have prolonged viral replication, see *Therapeutic Management of Patients With COVID-19 Symptoms and Prolonged SARS-CoV-2 Replication* below.

Therapeutic Management of Patients Who Are Hospitalized for Reasons Other Than COVID-19

People who are immunocompromised and have COVID-19 but were hospitalized for conditions other than COVID-19 should receive the same treatments as nonhospitalized patients (AIII). See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for more information.

Therapeutic Management of Patients With COVID-19 Symptoms and Prolonged SARS-CoV-2 Replication

For patients who are immunocompromised and have prolonged COVID-19 symptoms and evidence of ongoing viral replication (e.g., those with a low cycle threshold value, as measured by a reverse transcription polymerase chain reaction result or with a positive rapid antigen test result) despite receiving a course of antiviral therapy, the optimal management is unknown. Case reports and case series have described the treatment of these patients with additional antiviral treatments, prolonged courses of antiviral treatments, high-titer CCP, or combination therapy.⁹⁰⁻⁹⁶ The data for these approaches are not definitive, but some Panel members would use 1 or more of the following treatment options:

- Longer and/or additional courses of ritonavir-boosted nirmatrelvir
- Longer and/or additional courses of remdesivir
- High-titer CCP from a vaccinated donor who recently recovered from COVID-19 likely caused by

a SARS-CoV-2 variant similar to the variant causing the patient's illness

Clinicians should be aware that the drug-drug interaction potential of ritonavir may change based on the duration of treatment. Although the primary concern with the use of a 5-day course of ritonavir is cytochrome P450 (CYP) 3A4 inhibition, the induction properties of ritonavir may become clinically relevant when it is used for ≥ 10 days.⁹⁷

After longer courses of ritonavir-boosted nirmatrelvir are discontinued, drug-drug interactions caused by CYP3A4 inhibition are expected to resolve within 2 to 3 days.⁹⁸ Drug-drug interactions that are caused by induction (e.g., CYP2C9, CYP2C19, uridine diphosphate-glucuronyltransferase) resolve gradually and variably.

Clinicians should consult experts (e.g., pharmacists and physicians with HIV expertise) for guidance on drug-drug interactions when using extended courses of ritonavir-boosted nirmatrelvir. For more information, see [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#). The [Liverpool COVID-19 Drug Interactions website](#) provides guidance on managing drug-drug interactions in patients who are receiving for extended courses (i.e., ≥ 10 days) of ritonavir-boosted nirmatrelvir.

Considerations in Pregnant and Lactating People

Multiple studies have found that pregnant individuals have an increased risk of severe COVID-19 compared to age-matched controls, with increased rates of intensive care unit admission, mechanical ventilation, extracorporeal membrane oxygenation, and death.⁹⁹⁻¹⁰¹ Although hormonally mediated immunomodulation occurs during pregnancy, pregnancy is not a state of systemic immunosuppression. Changes in the immune response to certain infectious pathogens during pregnancy may increase the severity of respiratory illness in pregnant individuals. Physiologic changes, such as reduced pulmonary residual capacity, may also contribute to respiratory disease severity.¹⁰²⁻¹⁰⁵ Pregnant people who have underlying immunocompromising conditions or are receiving immunosuppressive medications likely have an even higher risk of severe disease. This patient group should be prioritized for the prevention and treatment of COVID-19.

Prevention

The Panel recommends COVID-19 vaccination for everyone who is eligible, including pregnant and lactating individuals, according to the CDC's Advisory Committee on Immunization Practices (AI). COVID-19 vaccination is strongly recommended for pregnant individuals due to their increased risk for severe disease.^{106,107} Vaccination is especially important for pregnant people with concomitant risk factors such as underlying immunocompromising conditions (including those who are receiving immunosuppressive medications), as the risk for severe disease is likely additive.¹⁰⁰

Treatment

Although pregnant patients have been excluded from the majority of the clinical trials that evaluated the use of COVID-19 therapeutics, pregnant patients with COVID-19 can be treated the same as nonpregnant patients, with a few exceptions. Pregnant patients who are immunocompromised or who have other risk factors likely have an even higher risk of severe COVID-19 and should be prioritized for treatment. Providers should refer to [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for the Panel's guidance on treating COVID-19 in pregnant and lactating patients. Pregnant people who are immunocompromised are a heterogeneous group of patients, ranging from those who are mildly immunocompromised to those who are severely immunocompromised. Evaluating and managing pregnant patients require collaboration from a multidisciplinary team. This team should include a

transplant or specialty provider, an obstetrician or maternal-fetal medicine specialist, a pediatrician or neonatology specialist, and a pharmacist.

Considerations in Children

Although the overall risk of critical illness and death related to COVID-19 is lower in children than in adults, severe disease does occur, particularly in children with risk factors such as moderate or severe immunocompromising conditions. See [Special Considerations in Children](#) for a discussion of the risk factors for severe COVID-19 in children, and see [Therapeutic Management of Nonhospitalized Children With COVID-19](#) for the Panel's framework for assessing a child's risk of progression to severe COVID-19 based on vaccination status, comorbidities, and age.

Prevention

The Panel recommends COVID-19 vaccination for everyone who is eligible, including children, according to the CDC's Advisory Committee on Immunization Practices (AI).

Treatment

The majority of children with mild to moderate COVID-19 will not progress to more severe illness; therefore, the Panel recommends managing these patients with supportive care alone (AIII). Few children, if any, have been enrolled in clinical trials of treatments for COVID-19. Among the children who were enrolled, very few were immunocompromised. Therefore, clinicians should be cautious when applying recommendations based on adult data to children. Clinicians need to consider the potential risks and benefits of therapy, the severity of the patient's disease, and underlying risk factors. See [Therapeutic Management of Hospitalized Children With COVID-19](#) and [Therapeutic Management of Nonhospitalized Children With COVID-19](#) for the Panel's treatment recommendations in these scenarios.

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Last Updated: February 29, 2024

Summary Recommendations
• COVID-19 vaccination remains the most effective way to prevent serious outcomes and death from SARS-CoV-2 infection. The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible, including patients with active cancer and patients receiving treatment for cancer, according to the Centers for Disease Control and Prevention's (CDC) Advisory Committee on Immunization Practices (AI). • See the CDC webpage COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised for the current COVID-19 vaccination schedule. • Vaccinating household members, close contacts, and health care providers who provide care to patients with cancer is important to protect these patients from infection. Clinicians should strongly encourage all household members and close contacts of patients with cancer to be vaccinated against COVID-19 (AI). • The Panel defers to CDC recommendations for diagnostic molecular or antigen testing for SARS-CoV-2 infection in patients with cancer who develop signs and symptoms that suggest acute COVID-19. The Panel also defers to CDC recommendations for testing of asymptomatic patients before procedures and hospital admissions. • For patients with cancer and COVID-19, clinicians should follow COVID-19 evaluation and management guidelines for patients who do not have cancer. See Therapeutic Management of Nonhospitalized Adults With COVID-19 and Therapeutic Management of Hospitalized Adults With COVID-19 for more information. • Decisions about administering cancer-directed therapy to patients with acute COVID-19 and those who are recovering from COVID-19 should be made on a case-by-case basis; clinicians should consider the indication for chemotherapy, the goals of care, and the patient's history of tolerance to the cancer treatment (BIII). • Clinicians should consult with a hematologist or oncologist when making decisions about stopping or adjusting the doses of cancer-directed medications in patients with cancer and COVID-19. • Clinicians should pay careful attention to potential overlapping toxicities and drug-drug interactions between drugs used to treat COVID-19 (e.g., ritonavir-boosted nirmatrelvir [Paxlovid], dexamethasone) and cancer-directed therapies, prophylactic antimicrobials, and other medications (AIII). Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

People being treated for cancer may be at increased risk of severe COVID-19, and clinical outcomes of COVID-19 are generally worse in people with cancer than in people without cancer.¹⁻⁴ A meta-analysis of 46,499 patients with COVID-19 showed that all-cause mortality was higher in patients with cancer (risk ratio 1.66; 95% CI, 1.33–2.07), and that patients with cancer were more likely to be admitted to intensive care units (risk ratio 1.56; 95% CI, 1.31–1.87).⁵ A patient's risk of immunosuppression and susceptibility to SARS-CoV-2 infection depend on the type of cancer, the treatments administered, and the stage of disease (e.g., patients actively being treated compared to those in remission). In a study that used data from the COVID-19 and Cancer Consortium registry, patients with cancer who were in remission or who had no evidence of disease had a lower risk of death from COVID-19 than those who were receiving active treatment.⁶ Cancer survivors may also have an increased risk of severe COVID-19.⁷

This section of the COVID-19 Treatment Guidelines focuses on testing for SARS-CoV-2 and managing COVID-19 and cancer-directed therapies in people with cancer and COVID-19.

COVID-19 Vaccination

The clinical trials that evaluated the COVID-19 vaccines that received Emergency Use Authorizations or approvals from the Food and Drug Administration excluded patients who were severely immunocompromised. The COVID-19 vaccines authorized for use in the United States are not live vaccines; therefore, they can be safely administered to people who are immunocompromised.

COVID-19 vaccination remains the most effective way to prevent serious outcomes and death from SARS-CoV-2 infection. The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible, including patients with active cancer and patients receiving treatment for cancer, according to the Centers for Disease Control and Prevention's (CDC) Advisory Committee on Immunization Practices (AI). See the CDC webpage [COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised](#) for the current COVID-19 vaccination schedule.

Observational data suggest that serological responses to vaccines may be blunted in patients who are immunocompromised.^{8,9} However, vaccination is still recommended for these patients because it may provide partial protection, including protection from vaccine-induced, cell-mediated immunity.

Vaccinating household members, close contacts, and health care providers who provide care to patients with cancer is important to protect these patients from infection. Clinicians should strongly encourage all household members and close contacts of patients with cancer to be vaccinated against COVID-19 (AI). There is evidence that vaccinated individuals infected with SARS-CoV-2 have lower viral loads than unvaccinated individuals^{10,11} and that COVID-19 vaccines reduce the incidence of SARS-CoV-2 infections not only among vaccinated individuals but also among their household contacts.¹²⁻¹⁴

When determining the timing of COVID-19 vaccination in patients with cancer, clinicians should consider the following factors:

- If possible, patients planning to receive chemotherapy should receive vaccinations for COVID-19 at least 2 weeks before starting chemotherapy.^{15,16}
- COVID-19 vaccines can be offered as early as 3 months after a patient receives hematopoietic cell or chimeric antigen receptor T cell therapy.¹⁶
- Graft-versus-host disease symptoms may flare after COVID-19 vaccination.¹⁷ No immune-related adverse events were reported after COVID-19 vaccination in 2 studies of patients with cancer who received immune checkpoint inhibitors.^{18,19}

Decreased immunologic responses to COVID-19 vaccination have been reported in patients receiving treatment for solid tumors and hematologic malignancies.^{9,20} The type of therapy has been shown to influence the patient's response to vaccination. For example, people with chronic lymphocytic leukemia who were treated with Bruton's tyrosine kinase inhibitors or venetoclax with or without anti-CD20 antibodies had extremely low response rates (16.0% and 13.6%, respectively).²⁰ In comparison, approximately 80% to 95% of patients with solid tumors showed immunologic responses.^{9,21,22} Several observational studies support the use of a third vaccine dose in patients with cancer, even though vaccine failure may still occur.²³⁻²⁵ See the [CDC website COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised](#) for guidance on vaccine dosing.

Polyethylene Glycol Allergies

The BNT162b2 (Pfizer-BioNTech) and mRNA-1273 (Moderna) mRNA vaccines contain polyethylene glycol (PEG), whereas the NVX-CoV2373 (Novavax) vaccine contains polysorbate 80.²⁶ PEG and polysorbate are used in many products, including in agents used for cancer chemotherapy (e.g., PEG-

asparaginase). PEG and polysorbate are structurally related, and cross-reactive hypersensitivity between these compounds might occur.²⁷

The detection of PEG antibodies after vaccination was not associated with increased adverse reactions (such as delayed-onset reactions, including injection site rashes, or severe allergic reactions) to the mRNA vaccines.²⁷ Therefore, testing for anti-PEG antibodies should not be used as a screening tool to assess the risk of allergic reactions²⁸ and should not replace an assessment by a specialist in those rare individuals with a history of anaphylaxis.²⁹ The [CDC has issued guidance](#) on triaging people with a history of allergies or allergic reactions to the components of COVID-19 vaccines.

Testing for SARS-CoV-2

The Panel defers to CDC recommendations for diagnostic molecular or antigen testing for SARS-CoV-2 infection in patients with cancer who develop signs and symptoms that suggest acute COVID-19. The Panel also defers to CDC recommendations for testing of asymptomatic patients before procedures and hospital admissions.

Patients with cancer who are receiving chemotherapy are at risk of developing neutropenia. The National Comprehensive Cancer Network Guidelines for Hematopoietic Growth Factors categorize cancer treatment regimens based on the patient's risk of developing neutropenia.³⁰ A retrospective study suggests that patients with cancer and neutropenia have an increased risk of mortality if they develop COVID-19.³¹ Studies have reported an increased risk of poor clinical outcomes for patients with COVID-19 in the setting of neutropenia and during the perioperative period.^{32,33}

General Guidance for Patients With Cancer

Patients with cancer frequently engage with the health care system to receive treatment and supportive care for cancer or treatment-related complications. Nosocomial transmission of SARS-CoV-2 to patients and health care workers has been reported.³⁴⁻³⁶ Health care providers and patients should take precautions to reduce the risk of SARS-CoV-2 exposure and infection, including wearing a mask and practicing good hand hygiene.³⁷ Telemedicine can minimize the need for in-person services and reduce the risk of SARS-CoV-2 exposure. For people who have difficulty accessing health care, telemedicine may improve access to providers, but it could worsen disparities if these populations have limited access to technology.

Decisions about treatment regimens, surgery, and radiation therapy for the underlying malignancy should be made on a case-by-case basis, and clinicians should consider the biology of the cancer, the need for hospitalization, the number of clinic visits required, and the anticipated degree of immunosuppression. Cancer treatment regimens that do not affect the outcomes of COVID-19 in patients with cancer may not need to be altered. In a prospective observational study, receipt of immunotherapy, hormonal therapy, or radiotherapy in the month prior to SARS-CoV-2 infection was not associated with an increased risk of mortality among patients with cancer and COVID-19.³⁸

Febrile Neutropenia

Patients with cancer and febrile neutropenia should undergo diagnostic molecular or antigen testing for SARS-CoV-2 and evaluation for other infectious agents. They should also be given empiric antimicrobial therapy per the standard of care.³⁹

Treating COVID-19 and Managing Chemotherapy

Retrospective studies suggest that patients with cancer who were admitted to the hospital with

SARS-CoV-2 infection have a high case-fatality rate, with higher rates observed in patients with hematologic malignancies than in those with solid tumors.^{40,41}

For patients with cancer and COVID-19, clinicians should follow COVID-19 evaluation and management guidelines for patients who do not have cancer. Patients with cancer are at high risk of progressing to severe COVID-19 and are eligible to receive anti-SARS-CoV-2 therapies in the outpatient setting if they develop mild to moderate COVID-19. See [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for details.

The Panel also provides recommendations for treating COVID-19 in hospitalized patients. See [Therapeutic Management of Hospitalized Adults With COVID-19](#) for more information. In patients with COVID-19 who required supplemental oxygen or mechanical ventilation, the use of dexamethasone has been associated with lower mortality than standard of care treatment alone.⁴² In patients with cancer, dexamethasone is commonly used to prevent chemotherapy-induced nausea, as a part of tumor-directed therapy, and to treat inflammation associated with brain metastasis. The side effects of dexamethasone are expected to be the same in patients with cancer as in those without cancer.

The immunomodulators baricitinib, tocilizumab, abatacept, or inflixamab used in combination with dexamethasone are recommended for some patients with severe or critical COVID-19 who exhibit rapid respiratory decompensation (see [Therapeutic Management of Hospitalized Adults With COVID-19](#)).⁴³⁻
⁴⁶ The risks and benefits of using dexamethasone in combination with another immunomodulator in patients with cancer who recently received chemotherapy is unknown. Because dexamethasone and the other immunomodulators are immunosuppressive agents, patients who receive these medications should be closely monitored for secondary infections.

Therapeutic anticoagulation for patients with cancer who are hospitalized for COVID-19 should be managed similarly to anticoagulation for other hospitalized patients. Patients with platelet counts <50,000 cells/ μ L should not receive therapeutic anticoagulation for COVID-19. Clinicians should follow hospital protocols for managing anticoagulation in patients with thrombocytopenia.

Decisions about administering cancer-directed therapy to patients with acute COVID-19 and those who are recovering from COVID-19 should be made on a case-by-case basis; clinicians should consider the indication for chemotherapy, the goals of care, and the patient's history of tolerance to the cancer treatment (**BIII**). The optimal time to initiate or restart cancer-directed therapies after the infection has resolved is unclear. If possible, clinicians should withhold treatment until COVID-19 symptoms have resolved. Prolonged viral shedding may occur in patients with cancer,² although it is unknown how this relates to infectious virus or outcomes. The decision to restart cancer treatments in this setting should be made on a case-by-case basis. Clinicians should consult with a hematologist or oncologist when making decisions about stopping or adjusting the doses of cancer-directed medications in patients with cancer and COVID-19.

Drug-Drug Interactions

The use of antiviral or immune-based therapies to treat COVID-19 can present additional challenges in patients with cancer. Clinicians should pay careful attention to potential overlapping toxicities and drug-drug interactions between drugs used to treat COVID-19 (e.g., ritonavir-boosted nirmatrelvir [Paxlovid], dexamethasone) and cancer-directed therapies, prophylactic antimicrobials, and other medications (**AIII**).

A 5-day course of ritonavir-boosted nirmatrelvir is 1 of the preferred therapies for treating mild to moderate COVID-19 in nonhospitalized patients who are at risk for disease progression. However, this regimen has the potential for significant and complex drug-drug interactions with concomitant

medications, primarily due to the ritonavir component of the combination. Boosting with ritonavir, a strong cytochrome P450 (CYP) 3A inhibitor, is required to increase the exposure of nirmatrelvir to a concentration that is effective against SARS-CoV-2. Ritonavir may also increase concentrations of certain concomitant medications, including some chemotherapeutic agents and immunotherapies used to treat cancer. Significant increases in the concentrations of these drugs may lead to serious or life-threatening drug toxicities. Additionally, ritonavir is an inhibitor, inducer, and substrate of various other drug-metabolizing enzymes and drug transporters.

Before prescribing ritonavir-boosted nirmatrelvir, clinicians should carefully review the patient's concomitant medications, including over-the-counter medications, herbal supplements, and recreational drugs, to evaluate potential drug-drug interactions. Clinicians should refer to resources such as the [Liverpool COVID-19 Drug Interactions website](#), [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#), and the Food and Drug Administration's [prescribing information](#) for ritonavir-boosted nirmatrelvir for guidance on identifying and managing potential drug-drug interactions. If significant interactions prohibit the concomitant use of ritonavir-boosted nirmatrelvir, another antiviral treatment option should be used (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)).

Dexamethasone is commonly used as an antiemetic for patients with cancer and is recommended for the treatment of certain patients with COVID-19 (see [Therapeutic Management of Hospitalized Adults With COVID-19](#)). Dexamethasone is a weak to moderate CYP3A4 inducer; therefore, interactions with any CYP3A4 substrates need to be considered.

Special Considerations in Children

Preliminary published reports suggest that pediatric patients with cancer may have milder manifestations of COVID-19 than adult patients with cancer, although larger studies are needed.⁴⁷⁻⁴⁹ Guidance on managing children with cancer and COVID-19 is available from an international group that received input from the International Society of Paediatric Oncology, the Children's Oncology Group, St. Jude Global, and Childhood Cancer International.⁵⁰ Guidance on managing specific malignancies and supportive care and weblinks relevant to the care of children with cancer and COVID-19 are available in 2 publications authored by groups of experts.^{50,51} Special considerations for the treatment of children with COVID-19, including those who are immunocompromised, can be found in [Therapeutic Management of Hospitalized Children With COVID-19](#) and [Therapeutic Management of Nonhospitalized Children With COVID-19](#).

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Special Considerations in Solid Organ Transplant, Hematopoietic Cell Transplant, and Cellular Immunotherapy Candidates, Donors, and Recipients

Last Updated: February 29, 2024

Summary Recommendations
COVID-19 Vaccination • COVID-19 vaccination remains the most effective way to prevent serious outcomes and death from SARS-CoV-2 infection. The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible, including transplant and cellular immunotherapy candidates and recipients, according to the Centers for Disease Control and Prevention's Advisory Committee on Immunization Practices (AI). • See the Centers for Disease Control and Prevention webpage COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised for the current COVID-19 vaccination schedule for transplant and cellular immunotherapy recipients. • Clinicians should strongly encourage all household members, close contacts, and health care providers who provide care to transplant and cellular immunotherapy candidates and recipients to be vaccinated against COVID-19 (AI). • Clinicians should strongly encourage all potential organ and hematopoietic cell donors to get vaccinated against COVID-19 (AI). Potential Transplant and Cellular Immunotherapy Candidates • The Panel recommends performing diagnostic molecular or antigen testing for SARS-CoV-2 in all potential solid organ transplant, hematopoietic cell transplant, and cellular immunotherapy candidates with signs and symptoms that suggest acute COVID-19 (AIII). Additional guidance is available from medical professional organizations. See the text below for more information. • If SARS-CoV-2 is detected or infection is strongly suspected in a potential transplant or cellular immunotherapy candidate, transplantation or immunotherapy should be deferred, if possible (BIII). • The optimal management and therapeutic approach to COVID-19 in these populations is unknown. At this time, the procedures for evaluating and managing COVID-19 in transplant candidates are the same as those for nontransplant patients (AIII). Potential Transplant Donors • The Panel recommends performing diagnostic molecular testing for SARS-CoV-2 and assessing for symptoms of COVID-19 in all potential solid organ transplant and hematopoietic cell transplant donors prior to donation (AIII). Additional guidance is available from medical professional organizations. See the text below for more information. Transplant and Cellular Immunotherapy Recipients With COVID-19 • For transplant and cellular immunotherapy recipients with COVID-19, clinicians should follow COVID-19 evaluation and management guidelines for nontransplant patients. See Therapeutic Management of Hospitalized Adults With COVID-19, Therapeutic Management of Nonhospitalized Adults With COVID-19, and Special Considerations in People Who Are Immunocompromised for more information. • For nonhospitalized patients with mild to moderate COVID-19 who are transplant or cellular immunotherapy recipients, the Panel recommends prompt treatment with antiviral drugs at the doses and durations recommended for the general population (AIII). • Clinicians should consult with a transplant specialist when making decisions about stopping or adjusting the doses of immunosuppressive drugs in transplant or cellular immunotherapy recipients with COVID-19. • Clinicians should pay special attention to the potential for drug-drug interactions and overlapping toxicities between treatments for COVID-19 and concomitant medications, such as immunosuppressants used to prevent or treat allograft rejection and antimicrobials used to prevent or treat opportunistic infections. Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Introduction

Treating COVID-19 in solid organ transplant, hematopoietic cell transplant (HCT), and cellular immunotherapy recipients can be challenging due to the presence of coexisting medical conditions, the potential for transplant-related cytopenias, and the need for chronic immunosuppressive therapy to prevent graft rejection and graft-versus-host disease. Transplant recipients may also have a higher risk of exposure to SARS-CoV-2 given their frequent contact with the health care system. Since immunosuppressive agents modulate several aspects of immune response, the severity of COVID-19 could potentially be affected by the type and intensity of the immunosuppressive effect of the agent, as well as by specific combinations of immunosuppressive agents. Some transplant recipients have medical comorbidities that have been associated with more severe cases of COVID-19 and a greater risk of mortality, which makes the impact of transplantation on disease severity difficult to assess.

The [International Society for Heart and Lung Transplantation](#), the [American Society of Transplantation](#), the [American Society for Transplantation and Cellular Therapy](#), and the [European Society for Blood and Marrow Transplantation](#) provide guidance for clinicians who are caring for transplant recipients with COVID-19 and guidance on screening potential donors and transplant or cellular immunotherapy candidates. In addition, the American Society of Hematology offers guidance regarding [COVID-19 vaccination](#) for [transplant and cellular immunotherapy recipients](#).

This section of the COVID-19 Treatment Guidelines complements these sources and focuses on considerations for managing COVID-19 in solid organ transplant, HCT, and cellular immunotherapy recipients. The optimal management and therapeutic approach to COVID-19 in these populations is unknown. At this time, the procedures for evaluating and managing COVID-19 in transplant recipients are the same as those for nontransplant patients (AIII). See [Therapeutic Management of Hospitalized Adults With COVID-19](#) and [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) for more information. The risks and benefits of each medication used to treat COVID-19 may be different for transplant patients and nontransplant patients.

COVID-19 Vaccination

The clinical trials that evaluated the safety and efficacy of the COVID-19 vaccines excluded patients who were severely immunocompromised.^{1,2} Authorized and approved COVID-19 vaccines in the United States are not live-virus vaccines and can be safely administered to patients who are immunocompromised. However, solid organ transplant recipients have reduced immunological antibody responses following a primary 2-dose or 3-dose series of the mRNA COVID-19 vaccines.³⁻⁶

COVID-19 vaccination remains the most effective way to prevent serious outcomes and death from SARS-CoV-2 infection. The COVID-19 Treatment Guidelines Panel (the Panel) recommends COVID-19 vaccination for everyone who is eligible, including transplant and cellular immunotherapy candidates and recipients, according to the Centers for Disease Control and Prevention's (CDC) Advisory Committee on Immunization Practices (AI). See the CDC webpage [COVID-19 Vaccines for People Who Are Moderately or Severely Immunocompromised](#) for the current COVID-19 vaccination schedule for transplant and cellular immunotherapy recipients.

When determining the timing of COVID-19 vaccination in solid organ transplant, HCT, and cellular immunotherapy recipients, clinicians should consider the following factors:

- Ideally, solid organ transplant candidates should receive COVID-19 vaccines while they wait for transplant.
- In general, vaccination should be completed at least 2 weeks before a solid organ transplant, or vaccination should be started 1 month after a solid organ transplant.

- In certain situations, such as when T cell– or B cell–ablative therapy (with antithymocyte globulin or rituximab) is used at the time of transplant, delaying vaccination until 3 months after a solid organ transplant may be appropriate.⁷
- Reducing the dose of immunosuppressants and withholding immunosuppressants before vaccination **are not recommended**.
- COVID-19 vaccines can be offered as early as 3 months after a patient receives HCT or chimeric antigen receptor T cell therapy, although the vaccines may be less effective in these patients than in the general population.⁸⁻¹⁰
- If possible, patients who are scheduled to receive cytotoxic or B cell–depleting therapies should receive their COVID-19 vaccination before initiating these therapies or between cycles of these therapies. The suggested interval before resuming vaccination is about 6 months after completion of the B cell–depleting therapy.¹¹
- Graft-versus-host disease symptoms may flare after COVID-19 vaccination.¹²
- After receiving a vaccination, people who are immunocompromised should be advised to continue to exercise precautions to reduce their risk of SARS-CoV-2 exposure and infection (e.g., they should wear a mask and avoid crowds and poorly ventilated spaces).⁸

Currently, antibody tests are not recommended for assessing SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated. For people who received COVID-19 vaccines during treatment with immunosuppressive drugs, it is currently unknown whether revaccination offers a clinical benefit.

Clinicians should strongly encourage all household members, close contacts, and health care providers who provide care to transplant and cellular immunotherapy candidates and recipients to be vaccinated against COVID-19 (**AI**). There is evidence that vaccinated individuals who are infected with SARS-CoV-2 have lower viral loads than unvaccinated individuals^{13,14} and that COVID-19 vaccines reduce the incidence of SARS-CoV-2 infections not only among vaccinated individuals but also among their household contacts.¹⁵⁻¹⁷ Clinicians should strongly encourage all potential organ and hematopoietic cell donors to get vaccinated against COVID-19 (**AI**).

Assessing SARS-CoV-2 Infection

The risk of transmission of SARS-CoV-2 from donors to candidates is unknown. The probability that a donor or candidate may have SARS-CoV-2 infection can be estimated by considering the epidemiologic risk, obtaining a clinical history, and testing with molecular techniques. No current testing strategy is sensitive enough or specific enough to totally exclude active infection.

Assessing Transplant and Cellular Immunotherapy Candidates

The Panel recommends performing diagnostic molecular or antigen testing for SARS-CoV-2 in all potential solid organ transplant, HCT, and cellular immunotherapy candidates with signs and symptoms that suggest acute COVID-19 (**AIII**). The CDC testing algorithm recommends performing additional confirmatory testing with a laboratory-based nucleic acid amplification test (NAAT) when a person who is strongly suspected of having SARS-CoV-2 infection (i.e., a person who is symptomatic) receives a negative result on an antigen test.¹⁸ Shortly before solid organ transplant, HCT, or cellular immunotherapy, all candidates should undergo diagnostic molecular testing for SARS-CoV-2 and assessment for symptoms of COVID-19 (**AIII**).

Assessing Donors

The Panel recommends performing diagnostic molecular testing for SARS-CoV-2 and assessing for symptoms of COVID-19 in all potential solid organ transplant and HCT donors prior to donation **(AIII)**. Additional guidance is available from medical professional organizations, such as the [Organ Procurement and Transplantation Network](#) and the [American Society of Transplantation](#).

Living donors should undergo a SARS-CoV-2 NAAT using a specimen collected from the respiratory tract within 3 days of donation. Deceased donors should be tested for SARS-CoV-2 infection using a NAAT with a specimen taken from the upper respiratory tract within 72 hours of death; ideally, the test should be performed as close to organ recovery as possible. Lower respiratory sampling for COVID-19 testing is required for potential lung transplant donors by the United Network for Organ Sharing.¹⁹ The Organ Procurement and Transplantation Network and American Society of Transplantation provide information to help guide the decision-making process when managing solid organ transplant donors with a history of COVID-19.

If SARS-CoV-2 Infection Is Detected or Strongly Suspected in Transplant and Cellular Immunotherapy Candidates

If SARS-CoV-2 is detected or infection is strongly suspected in a potential transplant or cellular immunotherapy candidate, transplantation or immunotherapy should be deferred, if possible **(BIII)**. The optimal disease-free interval before transplantation or immunotherapy is not known. In this situation, decisions about the appropriate timing for transplantation or cellular immunotherapy should be made on a case-by-case basis. Clinicians should consider both the risk of viral transmission and the risks of delaying or altering therapy, which may include progression of the underlying disease or death.

Transplant Recipients With COVID-19

Solid organ transplant recipients receiving immunosuppressive therapy should be considered at increased risk of severe COVID-19.^{20,21} Initial reports of transplant recipients hospitalized with COVID-19 suggested mortality of up to 28%.²²⁻²⁶ However, COVID-19 vaccines and better treatments have improved clinical outcomes.

Risk of Graft Rejection

There are concerns that COVID-19 itself may increase the risk of acute rejection. In solid organ transplant recipients with or without COVID-19, acute cellular rejection should not be presumed without biopsy confirmation.²⁰ Similarly, in recipients with or without COVID-19 who have had rejection confirmed by a biopsy, immunosuppressive therapy should be initiated.

Data on the incidence and clinical characteristics of SARS-CoV-2 infection in HCT²⁷ and cellular immunotherapy recipients are limited.²⁸ Data from the Center for International Blood and Marrow Transplant Research demonstrated that approximately 30% of a cohort of 318 HCT recipients died within 30 days of COVID-19 diagnosis.²⁷ This probability of mortality was observed in both allogeneic and autologous recipients. Older age (≥ 50 years), male sex, and receipt of a COVID-19 diagnosis within 12 months of transplantation were associated with a higher risk of mortality among allogeneic recipients. In autologous recipients, patients with lymphoma had a higher risk of mortality than patients who had plasma cell disorder or myeloma.

A smaller study demonstrated slightly lower mortality among HCT and cellular immunotherapy recipients than earlier reports. This study found that the number of comorbidities, the presence of infiltrates on initial chest imaging, and neutropenia were predictors for increased disease severity.²⁹ Additional factors that have been used to determine the clinical severity of other respiratory viral

infections include the degree of cytopenia, the intensity of the conditioning regimen, the graft source, the degree of mismatch, and the need for further immunosuppression to manage graft-versus-host disease. Prolonged viral shedding has been described in solid organ transplant and HCT recipients;³⁰ this can have implications for preventing infection and for the timing of therapeutic interventions.

Treating COVID-19 in Transplant Recipients

For transplant and cellular immunotherapy recipients with COVID-19, clinicians should follow COVID-19 evaluation and management guidelines for nontransplant patients. See [Therapeutic Management of Hospitalized Adults With COVID-19](#), [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) and [Special Considerations in People Who Are Immunocompromised](#) for more information.

For nonhospitalized patients with mild to moderate COVID-19 who are transplant or cellular immunotherapy recipients, the Panel recommends prompt treatment with antiviral drugs at the doses and durations recommended for the general population (AIII). In hospitalized people with severe COVID-19 who required supplemental oxygen or mechanical ventilation, data from a large randomized controlled trial found that a short course of dexamethasone (6 mg once daily for up to 10 days) improved survival.³¹ A second immunomodulator (e.g., baricitinib, tocilizumab, abatacept, infliximab) used in combination with dexamethasone is recommended for some patients with severe or critical COVID-19.³¹⁻³⁴ Because dexamethasone and the other immunomodulators are immunosuppressive agents, patients who receive these medications should be closely monitored for secondary infections.

Therapeutic anticoagulation for transplant recipients who are hospitalized for COVID-19 should be managed similarly to anticoagulation for other hospitalized patients. Patients with platelet counts <50,000 cells/ μ L should not receive therapeutic anticoagulation for COVID-19. Clinicians should follow hospital protocols for managing anticoagulation in patients with thrombocytopenia.

The Panel's recommendations for the treatment of COVID-19 in hospitalized patients with COVID-19 can be found in [Therapeutic Management of Hospitalized Adults With COVID-19](#).

Concomitant Medications

Clinicians should pay special attention to the potential for drug-drug interactions and overlapping toxicities between treatments for COVID-19 and concomitant medications, such as immunosuppressants used to prevent or treat allograft rejection and antimicrobials used to prevent or treat opportunistic infections. Dose modifications may be necessary for drugs used to treat COVID-19 in transplant recipients with pre-existing organ dysfunction. Adjustments to the immunosuppressive regimen should be individualized based on disease severity, the specific immunosuppressants used, the type of transplant, the time since transplantation, the drug concentration, and the risk of graft rejection.²³ Clinicians should consult with a transplant specialist when making decisions about stopping or adjusting the doses of immunosuppressive drugs in transplant or cellular immunotherapy recipients with COVID-19.

Drug-Drug Interactions

Calcineurin inhibitors (e.g., cyclosporine, tacrolimus) and mammalian target of rapamycin (mTOR) inhibitors (e.g., everolimus, sirolimus), which are commonly used to prevent allograft rejection, have narrow therapeutic indices. Medications that inhibit or induce cytochrome P450 (CYP) enzymes or P-glycoprotein may put patients who receive these drugs at risk of clinically significant drug-drug interactions, increasing the need for therapeutic drug monitoring and the need to assess for signs of toxicity or rejection.³⁵

Some clinicians prefer to administer a 3-day course of intravenous remdesivir to nonhospitalized transplant recipients who are receiving immunosuppressive therapy to avoid significant drug-drug interactions. If remdesivir is not available or feasible to use, a 5-day course of ritonavir-boosted nirmatrelvir (Paxlovid) may be used with caution and only when close therapeutic drug monitoring of the antirejection therapy is possible. Clinicians should consult with transplant specialists throughout the treatment. Boosting with ritonavir, a strong CYP3A inhibitor, is required to increase the exposure of nirmatrelvir to a concentration that is effective against SARS-CoV-2. Ritonavir may also increase concentrations of certain concomitant medications, including calcineurin and mTOR inhibitors, during the treatment course and for ≥ 3 days after ritonavir is discontinued. Significant increases in the concentrations of these drugs may lead to serious or life-threatening drug toxicities.

General guidance for coadministering ritonavir-boosted nirmatrelvir with concomitant medications includes temporarily withholding certain immunosuppressive agents (e.g., tacrolimus, everolimus, sirolimus) or reducing the dosage of certain immunosuppressive agents (e.g., cyclosporine), monitoring the patient closely for toxicities, and performing therapeutic drug monitoring during and after the 5-day treatment course of ritonavir-boosted nirmatrelvir.^{36,37}

Some small case series have reported success using these recommendations to manage patients.^{38,39} However, cases of significant toxicities due to supratherapeutic tacrolimus concentrations have also been reported.⁴⁰ Therapeutic drug monitoring should be used to guide the process of reintroducing or modifying the doses of calcineurin and mTOR inhibitors in patients who have completed a course of ritonavir-boosted nirmatrelvir. Clinicians should also consult with a specialist who has experience with dose management. Clinicians should take additional precautions when treating transplant recipients who are also receiving other concomitant medications (e.g., certain triazole antifungals) that may interact with ritonavir, the immunosuppressants, or both. The extent and significance of multiple drug-drug interactions are much more complex and unpredictable.

Clinicians should refer to resources such as the [Liverpool COVID-19 Drug Interactions website](#), [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#), and the Food and Drug Administration [prescribing information](#) for ritonavir-boosted nirmatrelvir for guidance on identifying and managing potential drug-drug interactions. If significant interactions prohibit the concomitant use of ritonavir-boosted nirmatrelvir, another COVID-19 treatment option should be used (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)).

Among the drugs commonly used to treat hospitalized patients with COVID-19, dexamethasone is a moderate inducer of CYP3A4, and interleukin-6 inhibitors may lead to increased metabolism of CYP substrates. Clinicians should closely monitor the serum concentrations of calcineurin and mTOR inhibitors when these drugs are used.

Additional details about the adverse effects and drug-drug interactions of antiviral medications and immune-based therapies for COVID-19 are noted in [Tables 4e](#) and [5e](#).

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Special Considerations During Pregnancy and After Delivery

Last Updated: February 29, 2024

Summary Recommendations
• Pregnant people should be counseled about the increased risk for severe disease from SARS-CoV-2 infection and the measures they can take to protect themselves and their families from infection. • The COVID-19 Treatment Guidelines Panel (the Panel) recommends against withholding COVID-19 vaccination from pregnant or lactating individuals specifically because of pregnancy or lactation (AIII). • If hospitalization for COVID-19 is indicated for a pregnant patient, care should be provided in a facility that can conduct maternal and fetal monitoring, when appropriate. • General management of COVID-19 in pregnant patients should include: • Fetal and uterine contraction monitoring based on gestational age, when appropriate • Individualized delivery planning • A multispecialty, team-based approach that may include consultation with obstetric, maternal-fetal medicine, infectious disease, pulmonary-critical care, and pediatric specialists, as appropriate • The Panel recommends against withholding COVID-19 treatment from pregnant or lactating individuals specifically because of pregnancy or lactation (AIII). • In general, the therapeutic management of pregnant patients with COVID-19 should be the same as for nonpregnant patients, with a few exceptions (AIII). Notable exceptions include: • The Panel recommends against the use of molnupiravir for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (AIII). • There is insufficient evidence for the Panel to recommend either for or against the use of therapeutic anticoagulation in pregnant patients with COVID-19 who do not have evidence of venous thromboembolism. See Antithrombotic Therapy in Patients With COVID-19 for more information. • For details regarding therapeutic recommendations and pregnancy considerations, see Therapeutic Management of Nonhospitalized Adults With COVID-19; Therapeutic Management of Hospitalized Adults With COVID-19; Pregnancy, Lactation, and COVID-19 Therapeutics; and the individual drug sections. • Data on the use of COVID-19 therapeutic agents in pregnant and lactating people are limited. When making decisions about treatment, pregnant or lactating people and their clinical teams should use a shared decision-making process and consider several factors, including the severity of COVID-19, the risk of disease progression, and the safety of specific medications for the fetus, infant, or pregnant or lactating individual. For detailed guidance on using the Panel-recommended COVID-19 therapeutic agents during pregnancy, see Pregnancy, Lactation, and COVID-19 Therapeutics. • The decision to feed the infant breast milk while the lactating patient is receiving therapeutic agents for COVID-19 should be a collaborative effort between the patient and the clinical team, including infant care providers. The patient and the clinical team should discuss the potential benefits of the therapeutic agent and evaluate the potential risk of pausing lactation on future breast milk delivery to the infant. For more information, see Pregnancy, Lactation, and COVID-19 Therapeutics. Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Epidemiology of COVID-19 in Pregnancy

Although the overall risk of severe illness is low, pregnant people with COVID-19 are at a higher risk of severe disease than nonpregnant people.¹ After adjustments have been made for age, race/ethnicity, and underlying medical conditions, pregnant women have significantly higher rates of intensive care unit (ICU) admission (10.5 vs. 3.9 cases per 1,000 cases; adjusted risk ratio [aRR] 3.0; 95% CI, 2.6–3.4),

mechanical ventilation (2.9 vs. 1.1 cases per 1,000 cases; aRR 2.9; 95% CI, 2.2–3.8), extracorporeal membrane oxygenation (0.7 vs. 0.3 cases per 1,000 cases; aRR 2.4; 95% CI, 1.5–4.0), and death (1.5 vs. 1.2 cases per 1,000 cases; aRR 1.7; 95% CI, 1.2–2.4).

An ongoing systematic review and meta-analysis of 435 studies also described increased odds of ICU admission and mechanical ventilation among pregnant and recently pregnant patients with COVID-19 when compared with nonpregnant patients of reproductive age.^{2,3} Compared with pregnant women and recently pregnant women without COVID-19, pregnant women with COVID-19 were at a higher risk of preterm birth and stillbirth.

A national cohort study in the United States evaluated whether severe maternal morbidity, as defined by the Centers for Disease Control and Prevention (CDC), was associated with SARS-CoV-2 infection during pregnancy.⁴ The study reported that among the 93,624 pregnant individuals who delivered, 4.8% were infected with SARS-CoV-2 at some point during pregnancy. Among the cases of SARS-CoV-2 infection, 59.5% occurred early in pregnancy, 13.5% occurred 7 to 30 days before delivery, and 27% occurred <7 days before delivery. The adjusted risk of severe maternal morbidity was 2.22 times higher for pregnant individuals who were infected <7 days before delivery and 1.67 times higher for those who were infected 7 to 30 days before delivery, when compared with the risk for pregnant individuals who delivered and had not been infected with SARS-CoV-2 during pregnancy. The study also reported that at the time of hospitalization for delivery, severe maternal morbidity was associated with younger (15–25 years) and older maternal (>35 years) ages, non-Hispanic Black race, lower household income and educational attainment, pre-existing medical conditions, and diagnosed pregnancy complications.

Obstetric and Perinatal Outcomes in Patients With COVID-19

An observational cohort study at 33 U.S. hospitals evaluated the maternal characteristics and outcomes across disease severity for all of the pregnant patients who had a singleton gestation and a positive result on a SARS-CoV-2 virologic test.⁵ The data suggested that adverse perinatal outcomes were more common in patients with severe or critical disease than in asymptomatic patients with SARS-CoV-2 infection, including an increased incidence of cesarean delivery (59.6% vs. 34.0% of patients; aRR 1.57; 95% CI, 1.30–1.90), hypertensive disorders of pregnancy (40.4% vs. 18.8%; aRR 1.61; 95% CI, 1.18–2.20), and preterm birth (41.8% vs. 11.9%; aRR 3.53; 95% CI, 2.42–5.14). The perinatal outcomes for those with mild to moderate illness were similar to those observed among asymptomatic patients with SARS-CoV-2 infection.

Among 1,249,634 delivery hospitalizations in the United States from March 2020 through September 2021, women with COVID-19 had an increased risk of stillbirth, which was defined as fetal death at >20 weeks' gestation (aRR 1.90; 95% CI, 1.69–2.15).⁶ The risk of stillbirth was higher during the time period that Delta was the dominant variant in the United States (aRR 4.04; 95% CI, 3.28–4.97) than during the pre-Delta period (aRR 1.47; 95% CI, 1.27–1.71).

A retrospective cohort analysis collected data from 14,104 pregnant or recently postpartum individuals who delivered at U.S. hospitals that participated in the Gestational Research Assessments for COVID-19 (GRAVID) study.⁷ Compared with pregnant individuals who did not have SARS-CoV-2 infection, patients with COVID-19 during pregnancy had an increased risk of meeting the composite endpoint of maternal death or severe morbidity related to hypertensive disorders of pregnancy, postpartum hemorrhage, or infection. Eighty percent of the patients in this cohort tested positive for SARS-CoV-2 infection during the third trimester. The primary composite endpoint occurred in 13.4% of patients who had COVID-19 during pregnancy or within 6 weeks postpartum and in 9.2% of those who did not have COVID-19 (aRR 1.41; 95% CI, 1.23–1.61).

When compared with those who did not have a positive SARS-CoV-2 test result, pregnant patients who had SARS-CoV-2 infection prior to 28 weeks' gestation had a subsequent increased risk of fetal/neonatal death (aRR 1.97; 95% CI, 1.01–3.85), preterm birth at <37 weeks (aRR 1.29; 95% CI, 1.02–1.63), and hypertensive disorders of pregnancy with delivery at <37 weeks' gestation (aRR 1.74; 95% CI, 1.19–2.55).⁸ There were no significant differences between these groups of patients in the risk of preterm birth at <34 weeks, any major congenital abnormalities, or a size for gestational age of less than the 5th or 10th percentiles. There were also no significant differences between these groups in the rates of gestational hypertension overall or preeclampsia with severe features. These data suggest that those with SARS-CoV-2 infection early in gestation may also have an increased risk of subsequent adverse pregnancy outcomes.

Vertical Transmission of COVID-19

Although vertical transmission of SARS-CoV-2 is possible, current data suggest that it is rare.⁹ A review of 101 infants born to 100 women with SARS-CoV-2 infection at a single U.S. academic medical center found that 2 infants (2%) had indeterminate SARS-CoV-2 polymerase chain reaction (PCR) results, which were presumed to be positive. However, the infants exhibited no evidence of clinical disease. It is reassuring that the majority of the infants received negative PCR results after rooming with their mothers and breastfeeding directly (the mothers in this study practiced appropriate hand and breast hygiene).

Data collected by the CDC as part of the Surveillance for Emerging Threats to Mothers and Babies Network showed that among 4,038 infants born to people with COVID-19, for whom laboratory testing information was available and who were tested during the delivery hospitalization, 227 infants (5.6%) had positive PCR results for SARS-CoV-2.¹⁰

The published data to date were largely collected prior to the emergence of the Omicron variants. The risk of vertical transmission may vary based on viral dynamics and the transmissibility of the circulating variants in a community. However, the variant-specific factors associated with vertical transmission have not been determined. For additional information on vertical transmission and infants born to people with SARS-CoV-2 infection, see [Special Considerations in Children](#).

Racial and Ethnic Disparities Among Pregnant People With COVID-19

Between January 22 and June 7, 2020, the CDC received reports of 8,207 women who were pregnant at the time of their COVID-19 diagnosis. Among these women, 46% were reported to be Hispanic, and 22% were reported to be Black.¹¹ Those proportions were higher than the proportions of Hispanic (24%) and Black (15%) women who gave birth in 2019, suggesting that pregnant people who are Hispanic or Black may be disproportionately affected by SARS-CoV-2 infection. It is important to note that these disparities are related to social determinants of health, current and historic inequities in access to health care and other resources, and structural racism. The American College of Obstetricians and Gynecologists (ACOG) has [published guidance](#) on addressing health equity during the COVID-19 pandemic.

Prevention of COVID-19 in Pregnancy

Pregnant people should be counseled about the increased risk for severe disease from SARS-CoV-2 infection and the measures they can take to protect themselves and their families from infection. Nonpharmacologic measures include practicing physical distancing, washing hands regularly, and wearing a face covering as per guidance from the CDC.

COVID-19 Vaccines

The COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** withholding COVID-19 vaccination from pregnant or lactating individuals specifically because of pregnancy or lactation (**AIII**).

Pregnant people should be counseled about the benefits of COVID-19 vaccination, which include a decreased risk of severe disease and hospitalization for the pregnant person and a decreased risk of hospitalization for the infant in the first 6 months of life.¹² The Society for Maternal-Fetal Medicine, the ACOG, and the CDC recommend that all eligible people, including pregnant and lactating individuals and those planning to become pregnant, receive COVID-19 vaccines as recommended.¹³⁻¹⁵ The CDC has published up-to-date guidance regarding COVID-19 vaccination, including guidance for administering vaccines to pregnant and lactating individuals.¹⁶ COVID-19 vaccines can be administered regardless of trimester and in concert with other vaccines recommended during pregnancy.¹⁴

Pregnant people were not included in the initial COVID-19 vaccine studies. However, a growing body of observational data supports the efficacy and safety of administering COVID-19 vaccines to this population. At this time, the mRNA vaccines BNT162b2 (Pfizer-BioNTech) and mRNA-1273 (Moderna) are recommended for pregnant or lactating individuals. The adjuvanted vaccine NVX-CoV2373 (Novavax) can also be used.^{14,15} For the most up-to-date clinical recommendations, see the [CDC guidelines on using COVID-19 vaccines](#). The ACOG and the Society for Maternal-Fetal Medicine provide guidance for counseling pregnant and lactating patients about COVID-19 vaccination.^{13,14}

Efficacy

A prospective cohort study of 131 subjects at 2 academic medical centers compared the immunogenicity and reactogenicity of the mRNA COVID-19 vaccines in pregnant and lactating women and nonpregnant controls.¹⁷ The study also compared vaccine-generated immunity to the immune response to natural SARS-CoV-2 infection among pregnant participants. Maternal immunoglobulin G antibody levels were similar after vaccination in pregnant and lactating women and in nonpregnant controls, and the antibody response did not differ by trimester of vaccination. Vaccinated pregnant women had significantly higher levels of antibodies than pregnant women who had had natural SARS-CoV-2 infection during the previous 4 to 12 weeks.

Antibody Transfer to the Neonate

The available data indicate that vaccine-derived antibodies are passively transferred to the neonate during pregnancy and lactation.¹⁸ A case control study that was conducted at 20 pediatric hospitals in 17 states in the United States from July 1, 2021, to January 17, 2022, assessed the relationship between maternal vaccination with a 2-dose mRNA COVID-19 vaccine during pregnancy and pediatric hospitalization for COVID-19.¹² In this study, 379 infants aged <6 months were hospitalized. Among these infants, 176 had COVID-19 and were considered case infants. The remaining 203 infants did not have COVID-19 and were considered control infants. Sixteen percent of the mothers of the case infants had received 2 doses of COVID-19 vaccine during pregnancy compared with 32% of the mothers of control infants.

Maternal completion of a 2-dose primary mRNA COVID-19 vaccination series during pregnancy led to a decrease in the number of infant hospitalizations for COVID-19 during the first 6 months of life (61% decrease; 95% CI, 31% to 78%).¹² There were no statistically significant differences between the case infants and control infants in the presence of underlying medical conditions or the occurrence of premature birth. Of the 43 case infants who were admitted to the ICU, 88% had mothers who were unvaccinated. These data further support the CDC's recommendation for COVID-19 vaccination in people who are pregnant, breastfeeding, or trying to become pregnant or who might become pregnant in the future.¹⁶

Safety

A study that used data from 3 vaccine safety reporting systems in the United States reported that the frequency of adverse events among 35,691 vaccine recipients who identified as pregnant was similar to the frequency observed among nonpregnant patients.¹⁹ Local injection site pain, nausea, and vomiting were reported slightly more frequently in pregnant people than in nonpregnant people. Other systemic reactions were reported more frequently among nonpregnant vaccine recipients, but the overall

reactogenicity profile was similar for pregnant and nonpregnant patients.

From December 2020 to June 2021, a total of 22,953 pregnant people in the United States were enrolled in the COVID-19 Vaccine Pregnancy Registry.²⁰ An analysis that compared historic cohorts with people who received a COVID-19 vaccine during pregnancy found no increased risk of spontaneous abortion; major birth defects; or pregnancy-associated outcomes, including stillbirth, preterm birth, hypertensive disorders of pregnancy, neonatal ICU admission, or maternal ICU admission.²¹

Managing COVID-19 in Pregnancy

In pregnant patients, as in nonpregnant patients, SARS-CoV-2 infection can present as asymptomatic or presymptomatic disease or with a wide range of clinical manifestations, from mild symptoms to severe disease and respiratory failure that requires ICU admission. The illness severity, underlying comorbidities, and clinical status of pregnant patients who have symptoms compatible with COVID-19 should be assessed to determine whether in-person evaluation for potential hospitalization is needed.

If hospitalization for COVID-19 is indicated, care should be provided in a facility that can conduct maternal and fetal monitoring, when appropriate. General management of COVID-19 in pregnant patients should include:

- Fetal and uterine contraction monitoring based on gestational age, when appropriate
- Individualized delivery planning
- A multispecialty, team-based approach that may include consultation with obstetric, maternal-fetal medicine, infectious disease, pulmonary-critical care, and pediatric specialists, as appropriate

In general, the recommendations for managing COVID-19 in nonpregnant patients also apply to pregnant patients.

Therapeutic Management of COVID-19 in the Setting of Pregnancy

To date, most SARS-CoV-2-related clinical trials have excluded individuals who are pregnant or lactating. In cases where lactating and pregnant individuals have been included in studies, only a small number have been enrolled. This makes providing evidence-based recommendations on the use of anti-SARS-CoV-2 therapies in these vulnerable patients difficult and potentially limits their treatment options. When possible, pregnant and lactating individuals should not be excluded from clinical trials of COVID-19 therapeutic agents or vaccines.

The Panel **recommends against** withholding COVID-19 treatments from pregnant or lactating individuals specifically because of pregnancy or lactation (**AIII**).

Data on the use of COVID-19 therapeutic agents in pregnant and lactating people are limited. Pregnancy is a risk factor for severe COVID-19. Studies have demonstrated that pregnant individuals are at increased risk of ICU admission, mechanical ventilation, and death, as well as poor obstetric and neonatal outcomes.¹⁻⁴ When making decisions about treatment, pregnant or lactating people and their clinical teams should use a shared decision-making process and consider several factors, including the severity of COVID-19, the risk of disease progression based on the person's comorbidities, and the safety of specific medications for the fetus, infant, or pregnant or lactating individual.

For details regarding therapeutic recommendations and pregnancy and lactation considerations, see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#); [Therapeutic Management of Hospitalized Adults With COVID-19](#); [Pregnancy, Lactation, and COVID-19 Therapeutics](#); and the individual drug sections. In general, the therapeutic management of pregnant patients with COVID-19

should be the same as for nonpregnant patients, with a few exceptions (AIII). Notable exceptions include:

- The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (AIII). For more information regarding the use of molnupiravir in pregnant patients, see [Pregnancy, Lactation, and COVID-19 Therapeutics](#).
- Pregnant patients were not included in most of the clinical trials that evaluated therapeutic anticoagulation in the setting of COVID-19, and there is a potential for increased maternal risks if bleeding occurs during pregnancy. Therefore, there is insufficient evidence for the Panel to recommend either for or against the use of therapeutic anticoagulation in pregnant patients with COVID-19 who do not have evidence of venous thromboembolism.

Timing of Delivery

The ACOG provides [detailed guidance](#) on the timing of delivery and the risk of vertical transmission of SARS-CoV-2.

In most cases, the timing of delivery should be dictated by obstetric indications rather than maternal diagnosis of COVID-19. For people who had suspected or confirmed COVID-19 early in pregnancy and who recovered, no alteration to the usual timing of delivery is indicated.

After Delivery

Therapeutic management in postpartum patients should follow guidelines for nonpregnant patients. However, the use of anticoagulant therapy in the immediate postpartum period should be individualized, as there may be an increased risk of bleeding, especially after an operative delivery.

The majority of studies have not demonstrated the presence of SARS-CoV-2 in breast milk; therefore, breastfeeding is not contraindicated for people with laboratory-confirmed or suspected SARS-CoV-2 infection.²² Precautions should be taken to avoid transmission to the infant, including practicing appropriate hand hygiene, wearing face coverings, and performing proper pump cleaning before and after breast milk expression.

The decision to feed the infant breast milk while the lactating patient is receiving therapeutic agents for COVID-19 should be a collaborative effort between the patient and the clinical team, including infant care providers. The patient and the clinical team should discuss the potential benefits of the therapeutic agent and evaluate the potential risk of pausing lactation on future breast milk delivery to the infant.

Specific guidance on the postdelivery management of infants born to individuals with known or suspected SARS-CoV-2 infection, including breastfeeding recommendations, is provided by the [American Academy of Pediatrics](#).

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Pregnancy, Lactation, and COVID-19 Therapeutics

Last Updated: February 29, 2024

General Considerations

The COVID-19 Treatment Guidelines Panel (the Panel) **recommends against** withholding COVID-19 treatments or vaccination from pregnant or lactating individuals specifically because of pregnancy or lactation (**AIII**). Pregnant patients should be offered the opportunity to participate in the [COVID-19 International Drug Pregnancy Registry](#) or other pregnancy registries.

The decision to feed the infant breast milk while the lactating patient is receiving therapeutic agents for COVID-19 should be a collaborative effort between the patient and the clinical team, including infant care providers. The patient and the clinical team should consider the benefits of breastfeeding, the postnatal age of the infant, the need for the medication, any underlying risks of exposing the infant to the drug, and the potential adverse outcomes of COVID-19.

If a patient is receiving a treatment for COVID-19 that prevents them from feeding breast milk to their infant for a limited time, the clinical team should still encourage pumping breast milk and provide lactation support. This ensures that lactation can continue after the patient stops receiving the treatment.

While a person with COVID-19 is breastfeeding, prevention measures should be taken to avoid transmitting SARS-CoV-2 to the infant. These measures include practicing appropriate hand hygiene, wearing face coverings, and performing proper pump cleaning before and after breast milk expression.

Table A: Recommendations for the Use of COVID-19 Therapeutics in Pregnant and Lactating People

For the Panel's recommendations on when to use the medications listed below, refer to [Therapeutic Management of Nonhospitalized Adults With COVID-19](#) and [Therapeutic Management of Hospitalized Adults With COVID-19](#). Pregnant patients should be offered the opportunity to participate in the [COVID-19 International Drug Pregnancy Registry](#) or other pregnancy registries. For additional information on the use of these medications during pregnancy and lactation, see the section text below.

Drug Name (in alphabetical order)	Pregnancy	Lactation
Abatacept	Recommended in hospitalized patients, if indicated. Pregnant patients and their health care providers should jointly decide whether to use abatacept during pregnancy, and the decision-making process should include a discussion of the potential risks and benefits.	Should be offered to patients who qualify for this therapy. There is minimal data on the transfer of abatacept to breast milk. Breastfeeding may be considered while a patient receives abatacept.
Baricitinib	Recommended in hospitalized patients, if indicated. Pregnant patients and their health care providers should jointly decide whether to use baricitinib during pregnancy, and the decision-making process should include a discussion of the potential risks and benefits.	Feeding breast milk should be avoided while taking baricitinib and for 4 days after the last dose. Lactation support should be provided during this time. ^a
Dexamethasone	Recommended in hospitalized patients, if indicated.	Should be offered to patients who qualify for this therapy. Breastfeeding can continue while a patient receives dexamethasone.

Drug Name (in alphabetical order)	Pregnancy	Lactation
Heparin (LMWH and UFH)	Recommended in hospitalized patients if indicated and if the patient does not have an obstetric-related bleeding risk (e.g., imminent delivery, bleeding complications of pregnancy) that would preclude use. See Antithrombotic Therapy in Patients With COVID-19 for more information.	Should be offered to patients who qualify for this therapy. Breastfeeding can continue while a patient receives LMWH or UFH.
Infliximab	Recommended in hospitalized patients, if indicated. Pregnant patients and their health care providers should jointly decide whether to use infliximab during pregnancy, and the decision-making process should include a discussion of the potential risks and benefits.	Should be offered to patients who qualify for this therapy. The available data show that the amount of infliximab that transfers through breast milk is negligible. Breastfeeding can continue while a patient receives infliximab.
Molnupiravir	Recommended against , unless there are no other options and therapy is clearly indicated.	Breastfeeding is not recommended while a patient is taking molnupiravir and for 4 days after the last dose. ¹ Lactation support should be provided during this time. ^a
Remdesivir	Recommended , if indicated.	Should be offered to patients who qualify for this therapy. Breastfeeding can continue while a patient receives remdesivir.
Ritonavir-Boosted Nirmatrelvir (Paxlovid)	Recommended , if indicated.	Should be offered to patients who qualify for this therapy. Breastfeeding can continue while a patient receives ritonavir-boosted nirmatrelvir.
Tocilizumab	Recommended in hospitalized patients, if indicated. Pregnant patients and their health care providers should jointly decide whether to use tocilizumab during pregnancy, and the decision-making process should include a discussion of the potential risks and benefits.	Should be offered to patients who qualify for this therapy. Breastfeeding can continue while a patient receives tocilizumab.

^a If a patient is receiving a treatment for COVID-19 that prevents them from feeding breast milk for a limited time, the clinical team should still encourage pumping breast milk and provide lactation support. This ensures that lactation can continue after the patient stops receiving the treatment.

Key: LMWH = low-molecular-weight heparin; the Panel = the COVID-19 Treatment Guidelines Panel; UFH = unfractionated heparin

Rationale

Abatacept

Pregnancy

As there are no data on the use of abatacept during pregnancy in hospitalized patients with COVID-19, this drug should be used only if baricitinib and tocilizumab are not available or feasible to use. When deciding whether to prescribe abatacept to a pregnant individual, clinicians need to consider the severity of the patient's COVID-19, the patient's comorbidities, and the gestational age of the fetus.

There is a paucity of data on the use of abatacept in pregnant individuals. It is currently not known whether abatacept can cross the human placenta; however, abatacept has crossed the placenta in animal studies. One study reported alterations to the immune systems of the offspring of animals that received supratherapeutic doses of abatacept throughout pregnancy.² It is not known whether the immune systems of infants who were exposed to a single dose of abatacept in utero might be impacted. Abatacept should only be used during pregnancy if the benefits clearly outweigh the potential risks.

Lactation

Abatacept should be offered to patients who qualify for this therapy. It is not known whether abatacept is transferred to breast milk during lactation or whether it is absorbed systemically by the infant. Because abatacept is a large molecule, only small amounts are thought to be transferred to breast milk. Patients who are receiving abatacept may consider breastfeeding.

Baricitinib

Pregnancy

When deciding whether to prescribe baricitinib to a pregnant individual, clinicians need to consider the severity of the patient's COVID-19, the patient's comorbidities, and the gestational age of the fetus.

Baricitinib is a Janus kinase inhibitor. As a small-molecule drug, baricitinib is likely to pass through the placenta; therefore, fetal risk cannot be ruled out.³ In animal studies, baricitinib doses that exceeded the therapeutic human dose were associated with embryofetal developmental abnormalities. Pregnancy registries provide some data on the use of tofacitinib, another Janus kinase inhibitor, during pregnancy for other conditions (e.g., ulcerative colitis, rheumatoid arthritis, psoriasis).⁴⁻⁶ Pregnancy outcomes among the participants who received tofacitinib were similar to those among the general population.

Lactation

There is no information on the use of baricitinib in lactating people or on the effects of baricitinib on breastfed infants; however, baricitinib has been detected in the breast milk of lactating rats.⁷ Feeding breast milk should be avoided for 4 days (approximately 5–6 elimination half-lives) after baricitinib is discontinued.

Dexamethasone

Pregnancy

A short course of betamethasone or dexamethasone, which are both known to cross the placenta, is routinely used to decrease neonatal complications of prematurity in people who are at risk of imminent preterm birth.^{8,9} Treating COVID-19 with a short course of dexamethasone can lower the risk of death in pregnant individuals. In addition, dexamethasone carries a low risk of fetal adverse effects.

Lactation

Dexamethasone should be offered to lactating patients with COVID-19 who qualify for this therapy. Breast milk can be fed to the infant while the lactating patient is receiving dexamethasone. Although there are limited data on the use of dexamethasone in lactating patients, some published reports about a related antenatal corticosteroid (betamethasone) reported a time-limited decrease in the volume of breast milk production.^{10,11} Given the benefits of breast milk, additional lactation support has been recommended if needed.

Heparin (Low-Molecular-Weight and Unfractionated)

Pregnancy

In general, the preferred anticoagulants during pregnancy are heparin compounds. Because of its reliability and ease of administration, low-molecular-weight heparin is recommended rather than unfractionated heparin for the prevention and treatment of venous thromboembolism in pregnant people.

The use of anticoagulant therapy during labor and delivery requires specialized care and planning. The management of anticoagulant therapy in pregnant patients with COVID-19 should be similar to the management used for pregnant patients with other conditions.

Lactation

Low-molecular-weight heparin, unfractionated heparin, and warfarin do not accumulate in breast milk and do not induce an anticoagulant effect in the newborn. Therefore, they can be used by breastfeeding individuals who require venous thromboembolism prophylaxis or treatment.

Infliximab

Pregnancy

As there are no data on the use of infliximab during pregnancy in hospitalized patients with COVID-19, infliximab should be used only if baricitinib and tocilizumab are not available or feasible to use. When deciding whether to prescribe infliximab to a pregnant individual, clinicians need to consider the severity of the patient's COVID-19, the patient's comorbidities, and the gestational age of the fetus.

There are limited data on the use of infliximab to treat COVID-19 in pregnant patients. It has been used to treat autoimmune diseases in pregnant individuals when the benefits outweigh the potential risks. Infliximab crosses the placenta and has been detected in the serum of infants born to patients treated with infliximab during pregnancy. No adverse effects have been reported in these infants.

Lactation

Infants who are breastfed by people receiving infliximab show minimal absorption of this agent.¹² No adverse effects have been reported in these infants. Therefore, infliximab should be offered to patients who qualify. Breastfeeding can continue while a patient receives infliximab.

Molnupiravir

Pregnancy

The Panel **recommends against** the use of **molnupiravir** for the treatment of COVID-19 in pregnant patients unless there are no other options and therapy is clearly indicated (**AIII**).

The Food and Drug Administration (FDA) Emergency Use Authorization states that molnupiravir is not recommended for use in pregnant patients because fetal toxicity has been reported in animal studies of molnupiravir.¹ However, when other therapies are not available, pregnant people with COVID-19 who are at high risk of progressing to severe disease may reasonably choose molnupiravir therapy after being fully informed of the potential risks, particularly if they are beyond the time of embryogenesis (i.e., >10 weeks' gestation).

Lactation

There is no data on the use of molnupiravir in lactating people; however, molnupiravir has been detected in the offspring of lactating rats. Molnupiravir is not authorized for use in children aged <18 years. Because the risk of adverse effects in infants is currently unknown, the FDA Emergency Use Authorization fact sheet does not recommend feeding an infant breast milk from a patient who is taking molnupiravir for the duration of the treatment course and until 4 days after the final dose.¹

Remdesivir

Pregnancy

While pregnant individuals were excluded from the clinical trials that evaluated the safety and efficacy of remdesivir for the treatment of COVID-19, subsequent reports on the use of remdesivir in pregnant individuals have been reassuring. Among 95 pregnant patients with moderate, severe, or critical COVID-19 who were included in a secondary analysis of data from a COVID-19 pregnancy registry in Texas, the composite maternal and neonatal outcomes were similar between those who received

remdesivir (n = 39) and those who did not.¹³

A systematic review of 13 observational studies that included 113 pregnant people also reported few adverse effects of remdesivir in pregnant patients with COVID-19.¹⁴ The most common adverse effect was a mild elevation in transaminase levels.

Lactation

Remdesivir is approved by the FDA for use in pediatric patients aged ≥ 28 days and weighing ≥ 3 kg.¹⁵ Limited data have suggested that the drug is poorly absorbed via the oral route; therefore, the levels of the drug that are absorbed when the infant ingests breast milk are low.^{16,17} One case report described a patient with COVID-19 who received remdesivir during the immediate postpartum period.¹⁷ Based on the concentration of remdesivir in the maternal serum and breast milk, the calculated milk-to-serum ratio was low. Therefore, the levels of remdesivir that would have reached a breastfed infant were estimated to be low.

Ritonavir-Boosted Nirmatrelvir (Paxlovid)

Pregnancy

Ritonavir has been used extensively during pregnancy in people with HIV and has a favorable safety profile during pregnancy. The mechanisms of action for both nirmatrelvir and ritonavir and the results of animal studies and case series suggest that this regimen can be used safely in pregnant individuals.

Two descriptive case series evaluated outcomes among pregnant patients with COVID-19 who received ritonavir-boosted nirmatrelvir.^{18,19} One case series included 47 patients with COVID-19 and a median gestational age of 28.4 weeks.¹⁸ These patients started taking ritonavir-boosted nirmatrelvir after a median duration of 1 day of COVID-19 symptoms. Thirty patients (64%) in the cohort had clinical characteristics in addition to pregnancy that increased their risk of progressing to severe COVID-19. The patients tolerated ritonavir-boosted nirmatrelvir well, with no serious adverse effects noted in either the pregnant patients or the neonates during the study period.

The other case series included 7 patients with a mean gestational age of 26.4 weeks who initiated ritonavir-boosted nirmatrelvir after approximately 2 days of COVID-19 symptoms.¹⁹ One patient developed dysgeusia and stopped treatment, but the remaining 6 patients completed 5 days of treatment. Six of the patients were fully vaccinated, and 4 of these patients had also received a booster dose. All the patients reported resolution of their COVID-19 symptoms, and no fetal or neonatal adverse effects were observed during the study period.

Ritonavir-boosted nirmatrelvir should be offered to pregnant and recently pregnant patients with COVID-19 who qualify for this therapy based on the results of a risk-benefit assessment. The risk-benefit assessment may include factors such as medical comorbidities, body mass index, vaccination status, and the number and severity of the risk factors for severe disease.

Obstetricians should be aware of potential drug-drug interactions when prescribing this agent. See [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#) for more information.

Lactation

Studies of infants who were exposed to ritonavir through breast milk suggest that the amount of ritonavir that transfers through breast milk is negligible and not considered clinically significant.²⁰⁻²³

There are no data on the use of nirmatrelvir in lactating people. However, a prebirth-to-lactation study performed in rats reported an 8% decrease in body weight on Postnatal Day 17 in the offspring of

rats that received nirmatrelvir and had systemic exposures that were 9 times higher than the clinical exposures at the authorized human dose.²⁴ This reduction in body weight was not seen in the offspring of rats that had exposures that were 5 times higher than the clinical exposures at the authorized human dose. Because the overall oral absorption of nirmatrelvir is poor, it is unlikely that the levels of nirmatrelvir absorbed from breast milk ingestion would be clinically relevant or expected to cause adverse effects in an infant.²⁵

Tocilizumab

Pregnancy

Pregnant individuals have been excluded from clinical trials that evaluated the use of the anti-interleukin-6 receptor monoclonal antibody tocilizumab for the treatment of COVID-19. An analysis of data from a global safety database reported pregnancy outcomes from 288 women who were exposed to tocilizumab during their pregnancies.²⁶ Eighty-nine percent of these women received tocilizumab as ongoing treatment for rheumatoid arthritis, and most were exposed to tocilizumab during their first trimester. The rates of congenital abnormalities among the infants born to these women were not higher than the rates seen in the general population. However, an increased rate of preterm birth was observed among these individuals when compared with the general population. A retrospective report of 61 pregnant women who were exposed to tocilizumab at conception or during their first trimesters showed no increased rates of congenital abnormalities or spontaneous abortion.²⁷

As pregnancy progresses, monoclonal antibodies are actively transported across the placenta, with the greatest transfer occurring during the third trimester. This may affect immune responses in the exposed fetus. If a pregnant patient receives tocilizumab after 20 weeks' gestation, clinicians should delay administering live viral vaccines to the infant for at least 6 months.

Lactation

There is limited information on the use of tocilizumab in lactating patients. Based on case report data, the amount of tocilizumab transferred to the infant via breast milk appears to be very low, with no reports of adverse effects.²⁸

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Influenza and COVID-19

Last Updated: December 20, 2023

Summary Recommendations
Influenza Vaccination • People with acute COVID-19 who have not received an influenza vaccine during influenza season should be vaccinated after they recover from acute illness and are no longer in isolation (BIII). • Patients may be vaccinated while they are still in isolation if they are in a health care setting. • An influenza vaccine and a COVID-19 vaccine may be administered concurrently at different injection sites. The Advisory Committee on Immunization Practices and the Centers for Disease Control and Prevention (CDC) provide more information on COVID-19 and influenza vaccines. Diagnosis of Influenza and COVID-19 When Influenza Viruses and SARS-CoV-2 Are Cocirculating • Only testing can distinguish between SARS-CoV-2 and influenza virus infections and identify SARS-CoV-2 and influenza virus coinfection. • The COVID-19 Treatment Guidelines Panel (the Panel) recommends performing influenza testing in addition to SARS-CoV-2 testing in outpatients with acute respiratory illness if the results will change the clinical management strategy for the patient (e.g., administering antiviral treatment for influenza) (BIII). • The Panel recommends testing for both viruses in all hospitalized patients with acute respiratory illness (AIII). • Clinicians should consider performing additional testing in specific clinical circumstances. Secondary bacterial infection is more common with influenza than with COVID-19, so additional testing for bacterial pathogens is important in patients with influenza who have clinical signs that suggest bacterial superinfection, especially for those who are immunocompromised or intubated. • See the CDC webpage Information for Clinicians on Influenza Virus Testing and the Infectious Diseases Society of America (IDSA) clinical practice guidelines for more information. Antiviral Treatment of Influenza When Influenza Viruses and SARS-CoV-2 Are Cocirculating • Antiviral treatment for influenza is the same for all patients regardless of SARS-CoV-2 coinfection (AIII). • For information on using antiviral drugs to treat influenza in hospitalized and nonhospitalized patients, see the CDC and IDSA recommendations. • There are no clinically significant drug-drug interactions between the antiviral agents used to treat influenza and the antiviral agents or immunomodulators used to treat COVID-19. • The Panel recommends starting hospitalized patients who are suspected of having influenza on empiric treatment for influenza with oseltamivir as soon as possible regardless of their COVID-19 status and without waiting for influenza test results (AIIb). • Oseltamivir treatment should be continued until nucleic acid detection assay results rule out influenza. For patients who are not intubated, assays should be performed on upper respiratory tract specimens. For patients who are intubated, assays should be performed on both upper and lower respiratory tract specimens. Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See Guidelines Development for more information.

Introduction

Clinicians should monitor local influenza and SARS-CoV-2 activities during influenza season to inform the evaluation and management of patients with acute respiratory illness. This can be done by tracking local and state public health surveillance data, assessing the results of testing performed at health care facilities, and reviewing the Centers for Disease Control and Prevention (CDC) [Weekly U.S. Influenza Surveillance Report](#).

Influenza Vaccination

For Patients With Acute COVID-19 or Those Recovering From COVID-19

The Advisory Committee on Immunization Practices (ACIP) recommends offering an influenza vaccine by the end of October to all people aged ≥ 6 months in the United States.¹ Unvaccinated persons can still benefit from influenza vaccination after October as long as influenza viruses are still circulating in the community. People with acute COVID-19 who have not received an influenza vaccine should be vaccinated after they recover from acute illness and are no longer in isolation (**BIII**). Patients may be vaccinated while they are still in isolation if they are in a health care setting.

There are currently no data on the safety, immunogenicity, or efficacy of administering influenza vaccines to patients with acute COVID-19 or those who are recovering from COVID-19. Vaccination in people who have mild illness is safe and effective.² Clinicians should consider deferring influenza vaccination for symptomatic patients with moderate or severe COVID-19 until they have recovered and completed their COVID-19 isolation period. It is not known whether administering dexamethasone or other immunomodulatory therapies to patients with severe COVID-19 will affect the immune response to the influenza vaccine. People with asymptomatic SARS-CoV-2 infection or mild COVID-19 should seek influenza vaccination when they no longer require isolation. They may be vaccinated sooner if they are in a health care setting for other reasons. See the influenza vaccine recommendations from the [CDC](#) and the [American Academy of Pediatrics](#).

Coadministration of COVID-19 Vaccines and Influenza Vaccines

Coadministration of a COVID-19 vaccine and an influenza vaccine at different injection sites has been shown to be safe.³⁻⁶ Providers and patients should be aware of a potential increase in reactogenicity when both vaccines are administered concurrently. The [CDC](#) and [ACIP](#) provide more information on coadministering influenza and COVID-19 vaccines.

Clinical Presentation of Influenza Versus COVID-19

The signs and symptoms of uncomplicated, clinically mild influenza overlap with those of mild COVID-19. Loss of taste and smell can occur with both diseases, but these symptoms are more common with COVID-19 than with influenza. Fever is not always present in patients with either disease, particularly in young infants, adults of advanced age, and patients who are immunosuppressed. Complications of influenza and COVID-19 can be similar, but the onset of influenza complications and severe disease typically occurs within a week of illness, whereas the onset of severe COVID-19 usually occurs in the second week of illness.

Because of the overlap in signs and symptoms, when SARS-CoV-2 and influenza viruses are cocirculating, diagnostic testing for both viruses is needed to distinguish between SARS-CoV-2 and influenza virus infection and to identify coinfection in people with an acute respiratory illness. Coinfection with influenza virus and SARS-CoV-2 has been described in case reports and case series,⁷⁻¹¹ but it is uncommon.¹² Observational studies have reported greater disease severity in adult patients with influenza virus and SARS-CoV-2 coinfection than in those with SARS-CoV-2 infection alone.^{13,14} In pediatric patients, coinfection with the 2 viruses was associated with greater disease severity than infection with influenza virus alone.¹⁵

Testing for SARS-CoV-2 and Influenza

The COVID-19 Treatment Guidelines Panel (the Panel) recommends performing influenza testing in addition to SARS-CoV-2 testing in outpatients with acute respiratory illness if the results will change the clinical management strategy for the patient (e.g., administering antiviral treatment for influenza) (**BIII**).

The Panel recommends testing for both viruses in all hospitalized patients with acute respiratory illness (AIII).

Several multiplex molecular assays and multiplex antigen assays that detect SARS-CoV-2 and influenza A and B viruses have received Food and Drug Administration Emergency Use Authorizations or De Novo classifications and can provide results in 15 minutes to 8 hours using a single respiratory specimen.¹⁶⁻¹⁸ For more information, see the CDC webpage [Information for Clinicians on Influenza Virus Testing](#) and the recommendations from the [Infectious Diseases Society of America](#) (IDSA) on the use of influenza tests and the interpretation of test results.

Treating Influenza With Antiviral Agents

Antiviral treatment for influenza is the same for all patients regardless of SARS-CoV-2 coinfection (AIII). There are no clinically significant drug-drug interactions between the antiviral agents used to treat influenza and the antiviral agents or immunomodulators used to treat COVID-19. The IDSA recommends administering antiviral treatment for influenza to all hospitalized patients with influenza.¹⁹

The Panel recommends starting hospitalized patients who are suspected of having influenza on empiric treatment for influenza with **oseltamivir** as soon as possible regardless of their COVID-19 status and without waiting for influenza test results (AIIb). Oseltamivir has no activity against SARS-CoV-2.²⁰ The standard dose of oseltamivir is absorbed well, even in critically ill patients. For patients who cannot tolerate oral or enterically administered oseltamivir (e.g., because of gastric stasis, malabsorption, or gastrointestinal bleeding), intravenous peramivir is an option.¹⁹ There are no data on the activity of peramivir against SARS-CoV-2.

See the CDC webpage [Influenza Antiviral Medications: Summary for Clinicians](#) for clinical algorithms for using antiviral agents in patients with suspected or laboratory-confirmed influenza, including pregnant people and other people who are at high risk for influenza complications. The IDSA clinical practice guidelines also provide recommendations on using antiviral agents to treat influenza,¹⁹ and the American Academy of Pediatrics provides recommendations on the antiviral treatment of influenza in children.²¹

When the result of an influenza nucleic acid detection assay from an upper respiratory tract specimen is negative in a patient who is receiving antiviral treatment for influenza:

- *In a patient who is not intubated:* Antiviral treatment for influenza can be stopped.
- *In a patient who is intubated:* Antiviral treatment for influenza should be continued, and a lower respiratory tract specimen (e.g., endotracheal aspirate) should be collected and tested using an influenza nucleic acid detection assay. If the lower respiratory tract specimen is also negative, antiviral treatment for influenza can be stopped.

COVID-19 Treatment Considerations for Hospitalized Patients With Suspected or Confirmed Influenza Virus Coinfection

Corticosteroids, which are used to treat patients with severe COVID-19, may prolong influenza viral replication and may be associated with poor outcomes for influenza.^{19,22} Currently, no data are available on the use of corticosteroids in patients with SARS-CoV-2 and influenza virus coinfection. However, because dexamethasone has demonstrated substantial benefits in patients with COVID-19 who require supplemental oxygen, the benefits of using corticosteroids in patients with severe SARS-CoV-2 and influenza virus coinfection likely outweigh any potential harm.

Although severe influenza may be associated with a dysregulated innate immune response, there are

no data on the use of immunomodulatory therapies, such as interleukin-6 inhibitors (e.g., tocilizumab, sarilumab) or Janus kinase inhibitors (e.g., baricitinib, tofacitinib), for the treatment of severe influenza. There are also no data on the effects these therapies may have on influenza virus infection, such as potentially prolonging viral replication. These immunomodulators have demonstrated a clinical benefit in certain patients with COVID-19. When considering using these drugs in patients with COVID-19 who have suspected or laboratory-confirmed influenza, clinicians should carefully weigh the known benefits for treatment of severe COVID-19 against the unknown theoretical risks for patients with influenza.

Observational studies have reported that co-occurrence of community-acquired secondary bacterial pneumonia appears to be infrequent in people with COVID-19; it is more common in people who have influenza.²³⁻²⁸ Typical bacterial causes of community-acquired pneumonia with severe influenza are *Staphylococcus aureus* (both methicillin-resistant *S. aureus* [MRSA] and methicillin-susceptible *S. aureus* [MSSA]), *Streptococcus pneumoniae*, and group A *Streptococcus*.¹⁹

Patients with COVID-19 who develop new respiratory symptoms with or without fever or respiratory distress and who do not have a clear diagnosis should be evaluated for the possibility of nosocomial influenza.

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Special Considerations in People With HIV

Last Updated: February 29, 2024

Summary Recommendations

COVID-19 Vaccination

- The COVID-19 Treatment Guidelines Panel (the Panel) recommends that people with HIV receive COVID-19 vaccines, regardless of their CD4 T lymphocyte (CD4) cell count or HIV viral load, because the potential benefits outweigh the potential risks **(AIIb)**.
- For people with untreated or advanced HIV, the Panel recommends following the most recent [COVID-19 vaccination schedule](#) from the Centers for Disease Control and Prevention (CDC) for people who are moderately or severely immunocompromised. The CDC defines advanced HIV as CD4 counts <200 cells/mm³, a history of an AIDS-defining illness without immune reconstitution, or clinical manifestations of symptomatic HIV.

Diagnosis of SARS-CoV-2 Infection

- The Panel defers to CDC recommendations for diagnostic molecular or antigen testing for SARS-CoV-2 infection in people with HIV who develop signs and symptoms that suggest acute COVID-19.

Managing COVID-19 in People With HIV

- The recommendations for the triage, management, and treatment of COVID-19 in people with HIV are the same as those for the general population **(AIII)**.
- Nonhospitalized people with HIV and mild to moderate COVID-19 may be eligible to receive the therapies that are currently recommended for treatment (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)). However, in situations where there are logistical constraints for administering these therapies, priority should be given to those with untreated or advanced HIV **(AIII)**. See [Prioritization of Anti-SARS-CoV-2 Therapies for the Treatment of COVID-19 in Nonhospitalized Patients When There Are Logistical Constraints](#) for details.
- People with HIV who are receiving a 5-day course of ritonavir-boosted nirmatrelvir (Paxlovid) to treat COVID-19 can continue using their antiretroviral therapy (ART) doses of ritonavir or cobicistat without alteration or interruption.
- In patients with advanced HIV who have suspected or laboratory-confirmed SARS-CoV-2 infection, clinicians should consider HIV-associated opportunistic infections in the differential diagnosis of clinical symptoms and consider consulting an HIV specialist.
- When starting treatment for COVID-19 in patients with HIV, clinicians should pay careful attention to potential drug-drug interactions and overlapping toxicities among COVID-19 treatments, antiretroviral (ARV) medications, antimicrobial therapies, and other medications **(AIII)**.

Managing HIV in People With COVID-19

- People with HIV who develop COVID-19, including those who require hospitalization, should continue their ART and their medications for the treatment or prevention of opportunistic infections whenever possible.
- Clinicians treating COVID-19 in people with HIV should consult an HIV specialist before adjusting or switching a patient's ARV medications.
- An ARV regimen should not be modified for the purpose of preventing or treating SARS-CoV-2 infection.
- Clinicians should consult with an HIV specialist about the timing of ART in people who present with COVID-19 and untreated HIV.

Each recommendation in the Guidelines receives a rating for the strength of the recommendation (A, B, or C) and a rating for the evidence that supports it (I, IIa, IIb, or III). See [Guidelines Development](#) for more information.

Introduction

Approximately 1.2 million people in the United States are living with HIV. Most of these individuals are in care, and many are receiving antiretroviral therapy (ART) and have well-controlled disease.¹ Similar to COVID-19, HIV disproportionately affects racial and ethnic minorities and people living in

low-income settings in the United States; these demographic groups also appear to have a higher risk of poor outcomes for COVID-19.² Many people with HIV have 1 or more comorbidities or conditions that may put them at higher risk of severe COVID-19.³

Clinical Outcomes of COVID-19

In some of the initial case series of people with COVID-19 in Europe and the United States, no significant differences were observed in the clinical outcomes of COVID-19 between people with HIV and people who did not have HIV.⁴⁻¹¹ Several subsequent studies have reported worse outcomes for patients with HIV and COVID-19, especially in patients with CD4 T lymphocyte (CD4) cell counts <200 cells/mm³.¹²⁻¹⁸ Many of these studies were done before the widespread use of COVID-19 vaccines; however, people with advanced HIV may have a suboptimal response to vaccines.^{19,20}

Prevention of COVID-19

The COVID-19 Treatment Guidelines Panel (the Panel) recommends that people with HIV receive COVID-19 vaccines, regardless of their CD4 count or HIV viral load, because the potential benefits outweigh the potential risks (**AIIb**). People with HIV were included in the clinical trials of the 2 mRNA vaccines (Pfizer and Moderna) and the glycoprotein vaccine (Novavax) that are currently available through Emergency Use Authorizations and/or approval from the Food and Drug Administration.²¹⁻²³ Typically, people with HIV who are receiving ART and who have achieved virologic suppression respond well to licensed vaccines. Data from studies that used COVID-19 vaccines in people with HIV confirm that people who are receiving ART and have normal CD4 counts have good immunologic responses to the vaccines.²⁴⁻²⁶ However, vaccine response rates are generally lower in people with lower CD4 counts (e.g., <200 cells/mm³).^{19,20,27}

For people with untreated or advanced HIV, the Panel recommends following the most recent [COVID-19 vaccination schedule](#) from the Centers for Disease Control and Prevention (CDC) for people who are moderately or severely immunocompromised. The CDC defines advanced HIV as CD4 counts <200 cells/mm³, a history of an AIDS-defining illness without immune reconstitution, or clinical manifestations of symptomatic HIV. Patients who have poor adherence or who experience virologic failure while on ART may have a similar risk of severe COVID-19 as those with untreated HIV. For additional considerations regarding vaccination in people who are immunocompromised, see [Special Considerations in People Who Are Immunocompromised](#).

There is currently no clear evidence that antiretroviral (ARV) medications can prevent SARS-CoV-2 infection. Some studies suggested that tenofovir disoproxil fumarate/emtricitabine may play a role in preventing SARS-CoV-2 acquisition or hospitalization or death associated with COVID-19; however, the significance of these findings is unclear.²⁸⁻³⁰ These studies may not have adequately controlled for confounding variables such as age and comorbidities. In addition, most of these studies were conducted in unvaccinated patients.

Diagnostic and Laboratory Testing for COVID-19

Diagnosis of SARS-CoV-2 Infection in People With HIV

The Panel defers to CDC recommendations for diagnostic molecular or antigen testing for SARS-CoV-2 infection in people with HIV who develop signs and symptoms that suggest acute COVID-19. See [Testing for SARS-CoV-2 Infection](#) for more information. There is currently no evidence that the performance characteristics of nucleic acid amplification tests (NAATs) and antigen tests differ in people with or without HIV when diagnosing acute SARS-CoV-2 infection. Antibody tests should not be used to diagnose current SARS-CoV-2 infection. Currently, antibody tests are not recommended for assessing

SARS-CoV-2 immunity following COVID-19 vaccination or for assessing the need for vaccination in a person who is unvaccinated. However, if serologic testing is performed in a patient with HIV, the results should be interpreted with caution because cross-reactivity between antibodies to SARS-CoV-2 and HIV has been reported.³¹

Correlation of CD4 Count in People With HIV and COVID-19

The normal range for CD4 counts in healthy adults is about 500 to 1,600 cells/mm³. People with HIV who have a CD4 count of ≥ 500 cells/mm³ have similar cellular immune function to those without HIV. In people with HIV, a CD4 count < 200 cells/mm³ meets the definition for AIDS. For patients receiving ART, the hallmark of treatment success is a plasma HIV RNA measurement below the level of detection by a polymerase chain reaction assay. Lymphopenia is a common laboratory finding in patients with COVID-19. Therefore, in patients with HIV, clinicians should note that CD4 counts obtained during acute COVID-19 may not accurately reflect the patient's HIV disease stage.

There have been some reports of people with advanced HIV who have presented with COVID-19 and another coinfection, including *Pneumocystis jirovecii* pneumonia and other opportunistic infections.³²⁻³⁶ In patients with advanced HIV who have suspected or laboratory-confirmed SARS-CoV-2 infection, clinicians should consider HIV-associated opportunistic infections in the differential diagnosis of clinical symptoms and consider consulting an HIV specialist.

Clinical Presentation of COVID-19

It is currently unknown whether people with HIV have a higher incidence of SARS-CoV-2 infection or a higher rate of progression to symptomatic disease than the general population. Approximately 50% of people with HIV in the United States are aged > 50 years,³⁷ and many have comorbidities that are associated with more severe COVID-19. These comorbidities include hypertension, diabetes mellitus, cardiovascular disease, a history of smoking, chronic lung disease, chronic liver disease, and cancer.³⁸

There are a number of case reports and case series that describe the clinical presentation of COVID-19 in people with HIV.^{4-11,28,39} These studies indicate that the clinical presentation of COVID-19 is similar in people with and without HIV. Most of the published reports describe populations in which the majority of individuals with HIV are receiving ART and have achieved virologic suppression. Consequently, the current understanding of the impact of COVID-19 in people with advanced HIV and low CD4 counts or persistent HIV viremia is limited.

Managing COVID-19 in People With HIV

The recommendations for the triage, management, and treatment of COVID-19 in people with HIV are the same as those for the general population (**AIII**). Nonhospitalized people with HIV and mild to moderate COVID-19 may be eligible to receive the therapies that are currently recommended for treatment (see [Therapeutic Management of Nonhospitalized Adults With COVID-19](#)). However, in situations where there are logistical constraints for administering these therapies, priority should be given to those with untreated or advanced HIV (**AIII**).

When starting treatment for COVID-19 in patients with HIV, clinicians should pay careful attention to potential drug-drug interactions and overlapping toxicities among COVID-19 treatments, ARV medications, antimicrobial therapies, and other medications (**AIII**).

Among the antiviral drugs recommended for nonhospitalized patients with COVID-19, drug-drug interactions are a special concern with the use of ritonavir-boosted nirmatrelvir (Paxlovid). People with HIV who are receiving a 5-day course of ritonavir-boosted nirmatrelvir to treat COVID-19 can continue

using their ART doses of ritonavir or cobicistat without alteration or interruption. Before prescribing ritonavir-boosted nirmatrelvir to a patient who is not already on a ritonavir-based or cobicistat-based regimen, clinicians should carefully review the patient's concomitant medications, including over-the-counter medications, herbal supplements, and recreational drugs, to evaluate potential drug-drug interactions. Clinicians should use resources such as [Drug-Drug Interactions Between Ritonavir-Boosted Nirmatrelvir \(Paxlovid\) and Concomitant Medications](#), the Food and Drug Administration [prescribing information](#) for ritonavir-boosted nirmatrelvir, and the [Liverpool COVID-19 Drug Interactions website](#) for additional guidance on identifying and managing drug-drug interactions.

In hospitalized patients, the appropriate treatment strategy depends on disease severity (see [Therapeutic Management of Hospitalized Adults With COVID-19](#)). Dexamethasone, which is recommended for use in combination with baricitinib or tocilizumab for some patients with severe or critical COVID-19, is an immunosuppressive agent. The safety of using this drug in patients who are immunocompromised, including those with advanced HIV, has not been studied. Therefore, patients with advanced HIV who are receiving dexamethasone should be closely monitored for secondary infections.

Dexamethasone is a dose-dependent inducer of cytochrome P450 3A4 and could potentially lower the levels of certain coadministered ARV drugs. More than a single dose of dexamethasone **is not recommended** for patients receiving rilpivirine as part of their ARV regimen. Clinicians should consult an HIV specialist before administering dexamethasone to these patients. It is currently unknown whether administering ≤ 10 days of dexamethasone impacts the clinical efficacy of other ARV drugs. Patients with HIV who are receiving dexamethasone as treatment for COVID-19 should follow up with their HIV care providers to assess their virologic response.

Although some ARV drugs were studied early in the pandemic for the treatment of COVID-19, none of these agents have been shown to be effective.

Managing HIV in People With COVID-19

People with HIV who develop COVID-19, including those who require hospitalization, should continue their ART and their medications for the treatment or prevention of opportunistic infections whenever possible. If a patient with HIV needs to receive the next dose of the long-acting injectables cabotegravir/rilpivirine, ibalizumab, or lenacapavir while hospitalized for COVID-19, clinicians should make arrangements with the patient's hospital provider to continue administering the medication without interruption. ART interruption may lead to rebound viremia, and, in some cases, the emergence of drug resistance. If the appropriate ARV drugs are not on the hospital's formulary, administer medications from the patient's home supplies, if available.

Clinicians treating COVID-19 in people with HIV should consult an HIV specialist before adjusting or switching a patient's ARV medications. An ARV regimen should not be modified for the purpose of preventing or treating SARS-CoV-2 infection. Many drugs, including some ARV agents (e.g., lopinavir/ritonavir, boosted darunavir, tenofovir disoproxil fumarate/emtricitabine), were evaluated in clinical trials or have been prescribed off-label to treat or prevent SARS-CoV-2 infection. Lopinavir/ritonavir and darunavir/cobicistat have not been found to be effective for the treatment of COVID-19.^{40,41}

For patients receiving an investigational ARV medication as part of their ARV regimen, arrangements should be made with the investigational study team to continue the medication, if possible.

For critically ill patients who require tube feeding, some ARV medications are available in liquid formulations, and some ARV pills may be crushed. Clinicians should consult an HIV specialist and/or pharmacist to assess the best way to continue an effective ARV regimen for a patient with a feeding tube. Information may be available in the drug product label or from the [Toronto General Hospital](#).

For people who present with COVID-19 and have either a new diagnosis of HIV or a history of HIV but are not receiving ART, the optimal time to start or restart ART is currently unknown. For people with HIV who have not initiated ART or who have been off therapy for >2 weeks before presenting with COVID-19, an HIV specialist should be consulted about initiating or reinitiating ART as soon as clinically feasible. If ART is initiated, maintaining treatment and linking patients to HIV care upon hospital discharge is critical. If an HIV specialist is not available, clinical consultation is available by phone through the [National Clinician Consultation Center](#).

Considerations in Pregnant and Lactating People

Pregnant or recently pregnant individuals are at a higher risk of severe illness and death from COVID-19 than nonpregnant individuals (see [Special Considerations During Pregnancy and After Delivery](#)). Although the data on pregnancy and maternal outcomes in individuals who have COVID-19 and HIV are limited, a prospective meta-analysis demonstrated that pregnant individuals with COVID-19 and HIV had a 67% greater risk of being admitted to the intensive care unit and a 72% greater risk of needing critical care.⁴² An observational study from Botswana found that offspring who were exposed to both HIV and SARS-CoV-2 had a high prevalence of adverse birth outcomes.⁴³

Given the severity of COVID-19 in pregnant or recently pregnant individuals, COVID-19 vaccines should be offered to all pregnant and lactating individuals and to those who are planning to become pregnant, including those who are also living with HIV. Pregnant individuals with HIV who have COVID-19 should be triaged, managed, and treated the same way as pregnant individuals without HIV. Clinicians should consider any additional comorbidities when assessing the risk of severe COVID-19 in these patients. See [Pregnancy, Lactation, and COVID-19 Therapeutics](#) for information regarding the therapies recommended for the treatment of COVID-19.

Pregnant individuals with HIV who are hospitalized for COVID-19 should continue their ART and opportunistic infection treatment and prophylaxis. Clinicians should consult an HIV specialist if any changes to ARV regimens are needed.

Considerations in Children

In general, children appear less likely to become severely ill with COVID-19 than adults. In the few publications that have described cases of COVID-19 among children or adolescents with HIV, most cases were mild, and HIV did not appear to be an independent predictor of severe COVID-19.⁴⁴⁻⁴⁷ Children with HIV who are eligible should receive COVID-19 vaccines and booster doses regardless of their CD4 count or viral load. Children with HIV and COVID-19 or multisystem inflammatory syndrome in children (MIS-C) should receive the same treatment as children without HIV. See [Therapeutic Management of Hospitalized Children With COVID-19](#), [Therapeutic Management of Nonhospitalized Children With COVID-19](#), and [Therapeutic Management of Hospitalized Children With MIS-C, Plus a Discussion on MIS-A](#) for more information.

Parents of children with HIV and COVID-19 should be advised to continue their child's ART without interruption if the child is being managed at home. For children with HIV who are hospitalized for COVID-19, ART should be continued for the duration of hospitalization.

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Appendix A, Table 1. COVID-19 Treatment Guidelines Panel Roster

Last Updated: February 29, 2024

Name	Affiliation
Co-Chairs	
Roy M. Gulick, MD, MPH	Weill Cornell Medicine, New York, NY
H. Clifford Lane, MD	National Institutes of Health, Bethesda, MD
Henry Masur, MD	National Institutes of Health, Bethesda, MD
Executive Secretary	
Alice K. Pau, PharmD	National Institutes of Health, Bethesda, MD
Members	
Judith Aberg, MD	Icahn School of Medicine at Mount Sinai, New York, NY
Adaora Adimora, MD, MPH	University of North Carolina School of Medicine, Chapel Hill, NC
Jason Baker, MD, MS	Hennepin Healthcare and University of Minnesota, Minneapolis, MN
Lisa Baumann Kreuziger, MD, MS	Versiti and Medical College of Wisconsin, Milwaukee, WI
Roger Bedimo, MD, MS	University of Texas Southwestern and Veterans Affairs North Texas Health Care System, Dallas, TX
Pamela S. Belperio, PharmD	Department of Veterans Affairs, Los Angeles, CA
Anoopindar Bhalla, MD	Children's Hospital Los Angeles and University of Southern California, Los Angeles, CA
Stephen V. Cantrill, MD	Denver Health, Denver, CO
Kara Chew, MD, MS	University of California, Los Angeles, Los Angeles, CA
Kathleen Chiotos, MD, MSCE	Children's Hospital of Philadelphia and University of Pennsylvania, Philadelphia, PA
Craig Coopersmith, MD	Emory University School of Medicine, Atlanta, GA
Eric Daar, MD	Harbor-UCLA Medical Center, Torrance, CA
Amy L. Dzierba, PharmD	New York-Presbyterian Hospital, New York, NY
Gregory Eschenauer, PharmD	University of Michigan, Ann Arbor, MI
Laura Evans, MD, MSc	University of Washington, Seattle, WA
John J. Gallagher, DNP, RN	Case Western Reserve University School of Nursing, Cleveland, OH
Rajesh Gandhi, MD	Massachusetts General Hospital and Harvard Medical School, Boston, MA
David V. Glidden, PhD	University of California, San Francisco, San Francisco, CA
Neil Goldenberg, MD, PhD	Johns Hopkins All Children's Institute for Clinical and Translational Research, St. Petersburg, FL
Birgit Grund, PhD	University of Minnesota, Minneapolis, MN
Erica J. Hardy, MD, MMSc	Warren Alpert Medical School of Brown University, Providence, RI
Lauren Henderson, MD, MMSc	Boston Children's Hospital and Harvard Medical School, Boston, MA
Carl Hinkson, MSRC	Providence Health & Services, Everett, WA
Brenna L. Hughes, MD, MSc	Duke University School of Medicine, Durham, NC
Steven Johnson, MD	University of Colorado School of Medicine, Aurora, CO
Marla J. Keller, MD	Albert Einstein College of Medicine and Montefiore Medical Center, Bronx, NY
Arthur Kim, MD	Massachusetts General Hospital and Harvard Medical School, Boston, MA
Jeffrey L. Lennox, MD	Emory University School of Medicine, Atlanta, GA
Mitchell M. Levy, MD	Warren Alpert Medical School of Brown University, Providence, RI
Jonathan Li, MD, MMSc	Brigham and Women's Hospital and Harvard Medical School, Boston, MA

Name	Affiliation
Members, continued	
Christine MacBrayne, PharmD, MSCS	Children's Hospital Colorado, Aurora, CO
Gregory Martin, MD, MSc	Emory University School of Medicine, Atlanta, GA
Nandita R. Nadig, MD	Northwestern University Feinberg School of Medicine, Chicago, IL
Andrew T. Pavia, MD	University of Utah School of Medicine, Salt Lake City, UT
Grant Schulert, MD, PhD	Cincinnati Children's Hospital Medical Center and University of Cincinnati College of Medicine, Cincinnati, OH
Nitin Seam, MD	National Institutes of Health, Bethesda, MD
Steven Q. Simpson, MD	University of Kansas Medical Center, Kansas City, KS
Susan Swindells, MBBS	University of Nebraska Medical Center, Omaha, NE
Pablo Tebas, MD	University of Pennsylvania, Philadelphia, PA
Phyllis Tien, MD, MSc	University of California, San Francisco and San Francisco Veterans Affairs Health Care System, San Francisco, CA
Alpana A. Waghmare, MD	Seattle Children's Hospital, Seattle, WA
Cameron R. Wolfe, MBBS	Duke University School of Medicine, Durham, NC
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Community Members	
Danielle M. Campbell, MPH	University of California, Los Angeles, Los Angeles, CA
Carly Harrison	LupusChat, New York, NY
Richard Knight, MBA	American Association of Kidney Patients, Bowie, MD
Pharmacology Consultants	
Sarita Boyd, PharmD	Food and Drug Administration, Silver Spring, MD
Jomy George, PharmD	National Institutes of Health, Bethesda, MD
Kimberly Scarsi, PharmD	University of Nebraska Medical Center, Omaha, NE
Ex Officio Members, U.S. Government Representatives	
Timothy Burgess, MD	Department of Defense, Bethesda, MD
Derek Eisnor, MD	Biomedical Advanced Research and Development Authority, Washington, DC
Joseph Francis, MD, MPH	Department of Veterans Affairs, Washington, DC
Pragna Patel, MD, MPH, DTM&H	Centers for Disease Control and Prevention, Atlanta, GA
Virginia Sheikh, MD, MHS	Food and Drug Administration, Silver Spring, MD
Timothy M. Uyeki, MD, MPH	Centers for Disease Control and Prevention, Atlanta, GA
U.S. Government Support Team	
Richard T. Davey, Jr., MD	National Institutes of Health, Bethesda, MD
Alison Han, MD (Co-Team Coordinator)	National Institutes of Health, Bethesda, MD
Elizabeth S. Higgs, MD, DTM&H, MIA	National Institutes of Health, Bethesda, MD
Martha C. Nason, PhD (Biostatistics Support)	National Institutes of Health, Bethesda, MD
Michael Proschan, PhD (Biostatistics Support)	National Institutes of Health, Bethesda, MD
Renee Ridzon, MD (Co-Team Coordinator)	National Institutes of Health, Bethesda, MD
Kanal Singh, MD, MPH (Co-Team Coordinator)	National Institutes of Health, Bethesda, MD
Assistant Executive Secretaries	
Safia Kuriakose, PharmD	Frederick National Laboratory for Cancer Research, in support of NIAID, Frederick, MD
Andrea M. Lerner, MD, MS	National Institutes of Health, Bethesda, MD

Appendix A, Table 2. COVID-19 Treatment Guidelines Panel Financial Disclosure for Companies Related to COVID-19 Treatment or Diagnostics

Last Updated: February 29, 2024

Reporting Period: January 1, 2023, to December 31, 2023

Panel Member	Financial Disclosure	
	Company	Relationship
Judith Aberg, MD	Emergent	Research support
	Gilead	Research support
	GSK/ViiV Healthcare	Advisory board, research support
	Janssen	Research support
	Kintor	DSMB chair/member
	Merck	Advisory board, research support
	Pfizer	Research support
	Regeneron	Research support, protocol reviewer
Adaora Adimora, MD, MPH	Gilead	Consultant, honoraria
	Merck	Advisory board, consultant, honoraria, research support
Jason Baker, MD, MS	None	N/A
Lisa Baumann Kreuziger, MD, MS	3M	Spouse is an employee, stockholder
	CSL Behring	Research support
	Takeda	Research support
	Versiti	Employee
Roger Bedimo, MD, MS	Gilead	Advisory board
	GSK/ViiV Healthcare	Advisory board
	Janssen	Advisory board
	Merck	Advisory board, research support
	Shionogi	Advisory board
	Theratechnologies	Advisory board
Pamela S. Belperio, PharmD	None	N/A
Anoopindar Bhalla, MD	None	N/A
Timothy Burgess, MD	AstraZeneca	Research support
Danielle M. Campbell, MPH	Gilead	Advisory board
	GSK/ViiV Healthcare	Attendee at a community stakeholder meeting, honoraria
Stephen V. Cantrill, MD	None	N/A
Kara Chew, MD, MS	Pardes Biosciences	Consultant
Kathleen Chiotos, MD, MSCE	None	N/A
Craig Coopersmith, MD	None	N/A
Eric Daar, MD	Gilead	Consultant, research support
	GSK/ViiV Healthcare	Consultant, research support
	Merck	Consultant
	Theratechnologies	Consultant
Richard T. Davey, Jr., MD	None	N/A

Panel Member	Financial Disclosure	
	Company	Relationship
Amy L. Dzierba, PharmD	None	N/A
Derek Eisnor, MD	None	N/A
Gregory Eschenauer, PharmD	None	N/A
Laura Evans, MD, MSc	None	N/A
Joseph Francis, MD, MPH	None	N/A
John J. Gallagher, DNP, RN	None	N/A
Rajesh Gandhi, MD	None	N/A
David V. Glidden, PhD	Gilead	Consultant
Neil Goldenberg, MD, PhD	Anthos Therapeutics	Advisory board, research support
	Bayer	Advisory board, research support
	Boehringer Ingelheim	Research support
	CPC Clinical Research	DSMB chair/member
	Daiichi Sankyo	Steering committee chair, research support
	Janssen	Advisory board, research support
	Novartis	DSMB chair/member
Birgit Grund, PhD	None	N/A
Roy M. Gulick, MD, MPH	None	N/A
Alison Han, MD	None	N/A
Erica J. Hardy, MD, MMSc	None	N/A
Carly Harrison	Novartis	Advisory board
Lauren Henderson, MD, MMSc	Bristol Myers Squibb	Research support
	Sobi	Consultant, research support, travel support
	SkyGenic	Consultant, spouse is an employee
Elizabeth S. Higgs, MD, DTM&H, MIA	None	N/A
Carl Hinkson, MSRC	None	N/A
Brenna L. Hughes, MD, MSc	None	N/A
Steven Johnson, MD	None	N/A
Marla J. Keller, MD	None	N/A
Arthur Kim, MD	None	N/A
Richard Knight, MBA	None	N/A
Safia Kuriakose, PharmD	None	N/A
H. Clifford Lane, MD	None	N/A
Jeffrey L. Lennox, MD	Pfizer	Spouse is an employee of Pfizer contract
Andrea M. Lerner, MD, MS	None	N/A
Mitchell M. Levy, MD	None	N/A
Jonathan Li, MD, MMSc	Merck	Research support
Christine MacBrayne, PharmD, MSCS	None	N/A
Gregory Martin, MD, MSc	Eagle Pharmaceuticals	Research grants review panel
	Grifols	Research grants review panel

Panel Member	Financial Disclosure	
	Company	Relationship
Henry Masur, MD	None	N/A
Nandita R. Nadig, MD	None	N/A
Martha C. Nason, PhD	Bristol Myers Squibb	Stockholder
	Medtronic	Stockholder
Pragna Patel, MD, MPH, DTM&H	None	N/A
Alice K. Pau, PharmD	None	N/A
Andrew T. Pavia, MD	AstraZeneca	Consultant
	Haleon	Honoraria
Michael Proschan, PhD	None	N/A
Renee Ridzon, MD	None	N/A
Grant Schulert, MD, PhD	Boehringer Ingelheim	Consultant
Nitin Seam, MD	None	N/A
Virginia Sheikh, MD, MHS	None	N/A
Steven Q. Simpson, MD	None	N/A
Kanal Singh, MD, MPH	None	N/A
Susan Swindells, MBBS	GSK/ViiV Healthcare	Research support
Pablo Tebas, MD	Shionogi	Consultant
Phyllis Tien, MD, MSc	Merck	Research support
Timothy M. Uyeki, MD, MPH	None	N/A
Alpana A. Waghmare, MD	AlloVir	Research support
	Ansun Biopharma	Research support
	GSK	Research support
	Pfizer	Research support
	Vir Biotechnology	Consultant, research support
Cameron R. Wolfe, MBBS	AbbVie	Consultant
	AstraZeneca	Research support
	Atea Pharmaceuticals	DSMB chair/member
	Biogen	DSMB chair/member
	Gilead	Advisory board
	Janssen	DSMB chair/member
Jinoos Yazdany, MD, MPH	AstraZeneca	Consultant, research support
	Aurinia	Research support
	Bristol Myers Squibb	Research support
	Gilead	Research support
	Pfizer	Consultant

Key: DSMB = data and safety monitoring board; N/A = not applicable